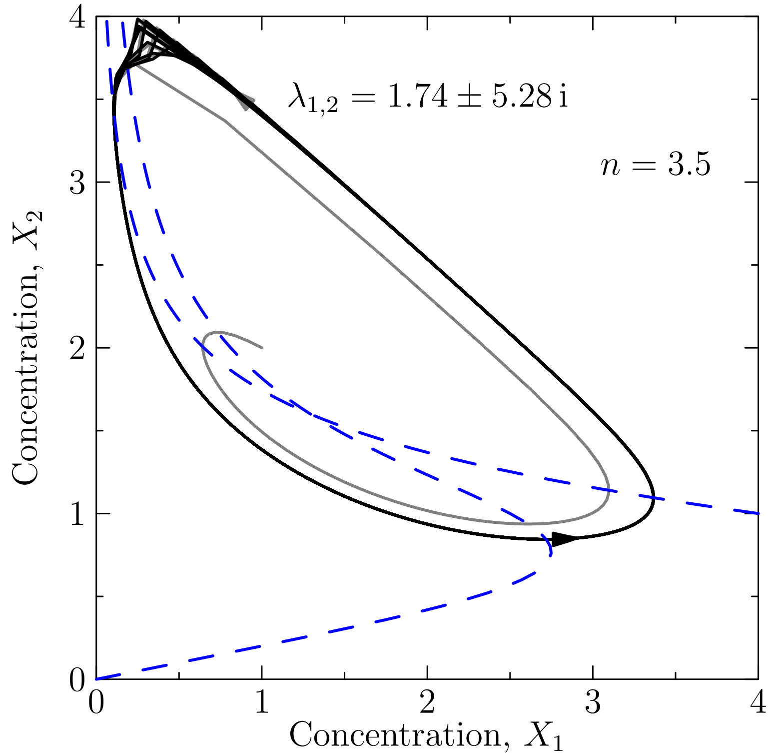
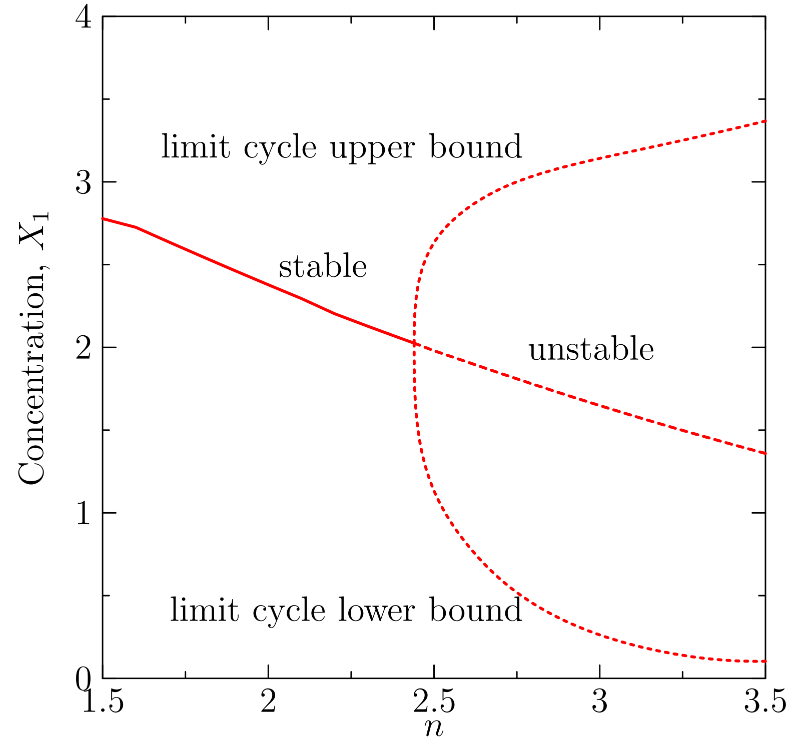
Computational Biology

Lecture 4: *Dynamical Systems*

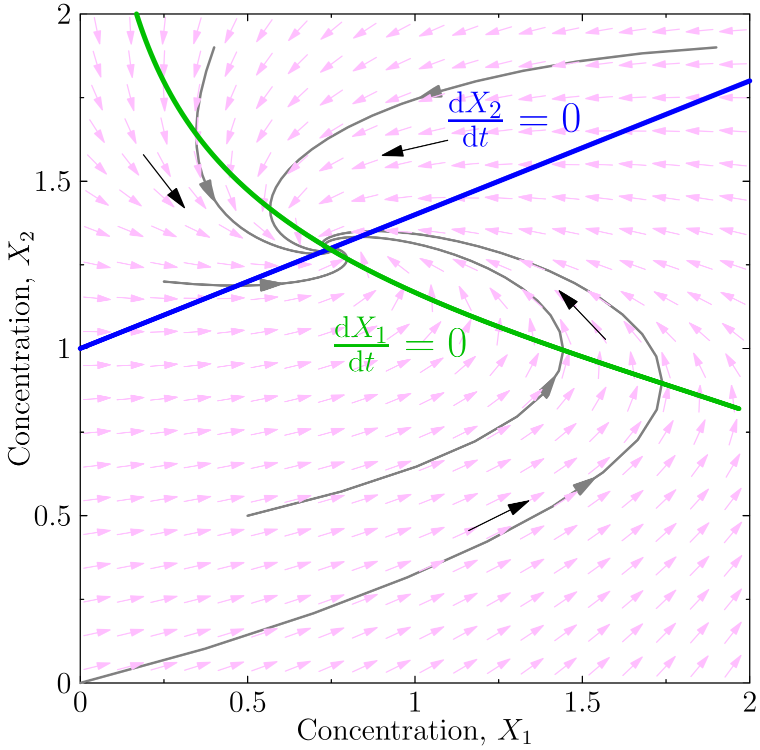




ODEs, nullclines, fixed points, stability analysis, bifications

Outline

1. **Understanding ODEs**
2. Stability
3. Bifications



Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X})$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}) \qquad \text{or} \qquad \frac{\mathrm{d}[X_i]}{\mathrm{d}t} = f_i([\boldsymbol{X}])$$

- These model the mean behaviour of a set of chemical equations

- Provide the number of molecules is sufficiently large this model is accurate

- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}) \qquad \text{or} \qquad \frac{\mathrm{d}[X_i]}{\mathrm{d}t} = f_i([\boldsymbol{X}])$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}) \qquad \text{or} \qquad \frac{\mathrm{d}[X_i]}{\mathrm{d}t} = f_i([\boldsymbol{X}])$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\boldsymbol{X} = (X_i | i = 1, 2, \ldots, n)$ in terms of differential equations

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}) \qquad \text{or} \qquad \frac{\mathrm{d}[X_i]}{\mathrm{d}t} = f_i([\boldsymbol{X}])$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

Two Species

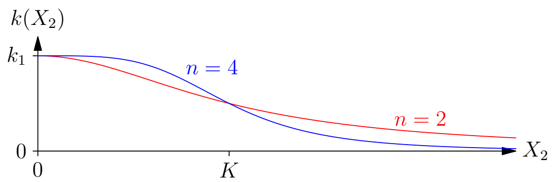
- Consider the simple system

$$\varnothing \xrightarrow{k(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$k(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$



- Leading to the coupled ODEs

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(X_1, X_2) = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(X_1, X_2) = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

Two Species

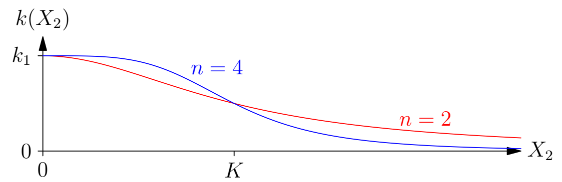
- Consider the simple system

$$\varnothing \xrightarrow{k(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$k(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$



- Leading to the coupled ODEs

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(X_1, X_2) = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(X_1, X_2) = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

Two Species

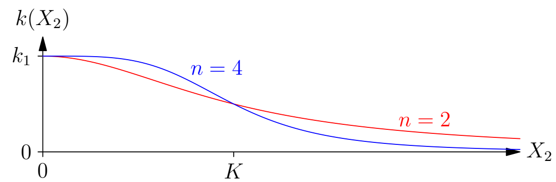
- Consider the simple system

$$\varnothing \xrightarrow{k(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$k(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$

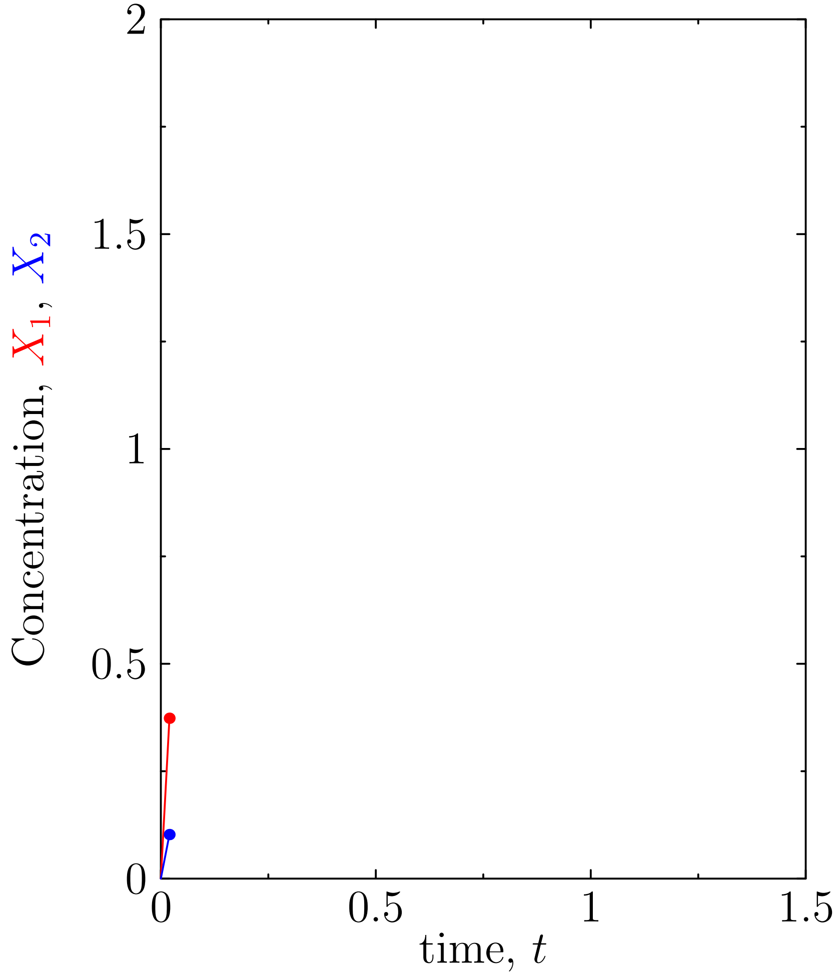


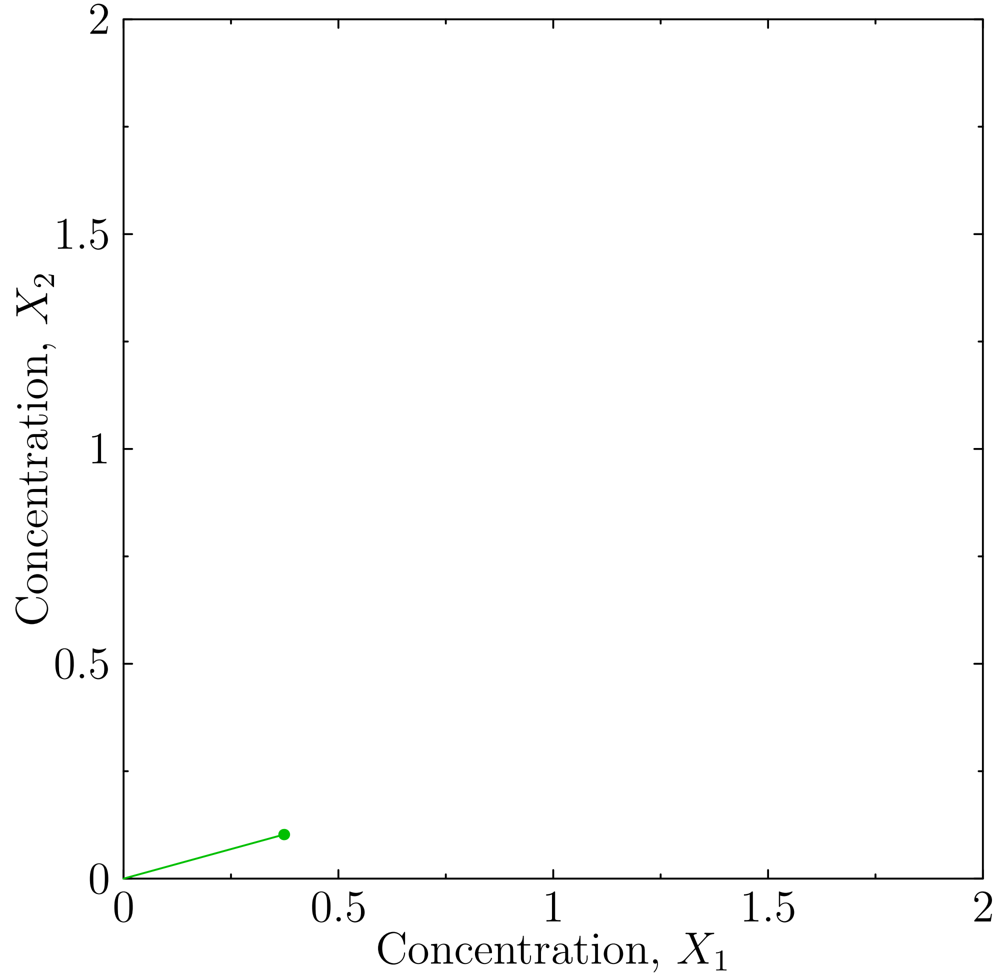
- Leading to the coupled ODEs

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(X_1, X_2) = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(X_1, X_2) = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

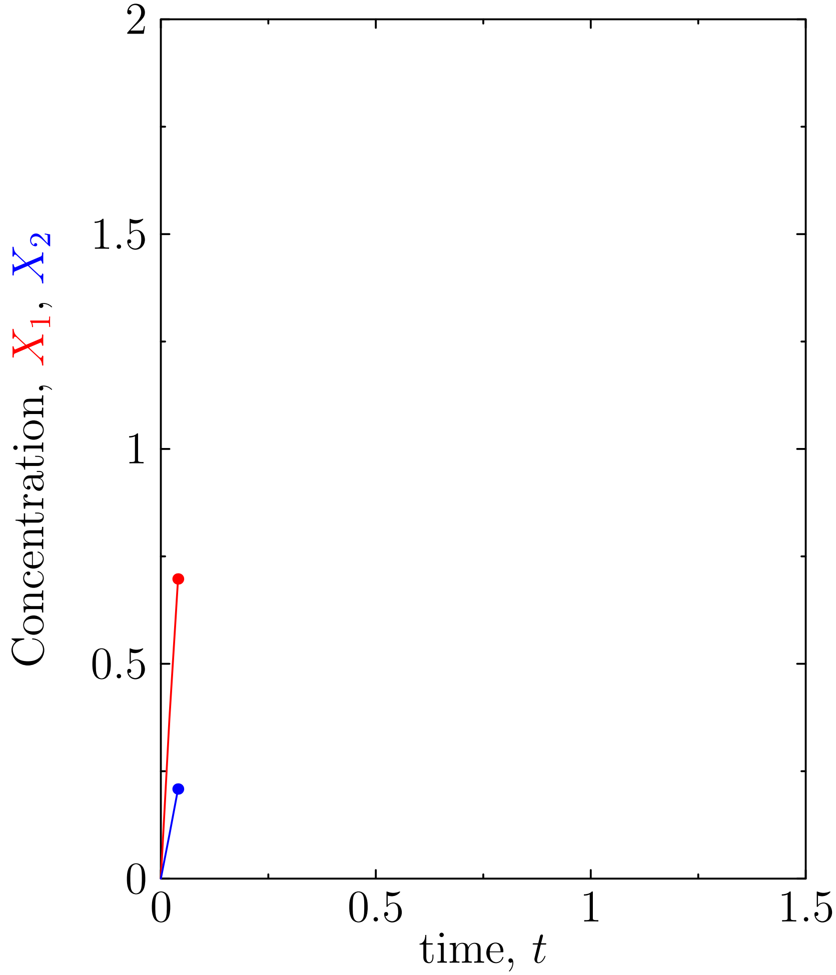
Solving the Dynamics

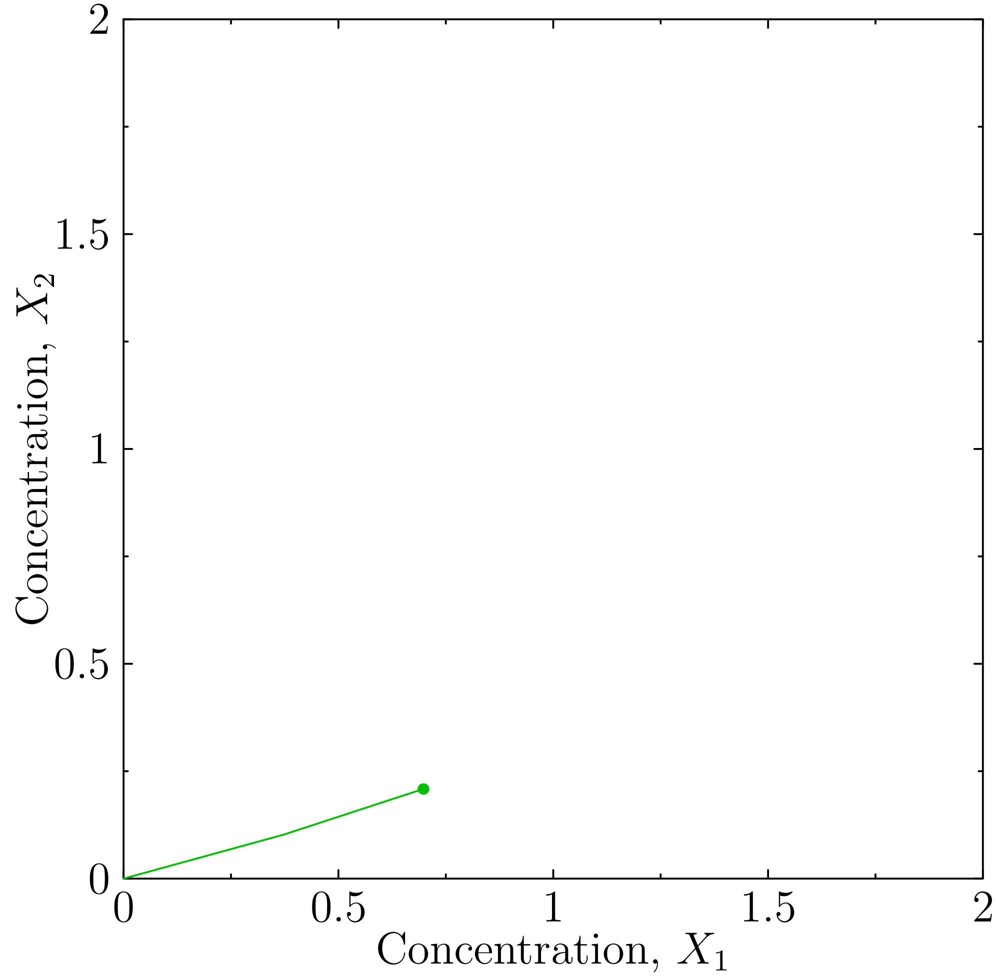




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

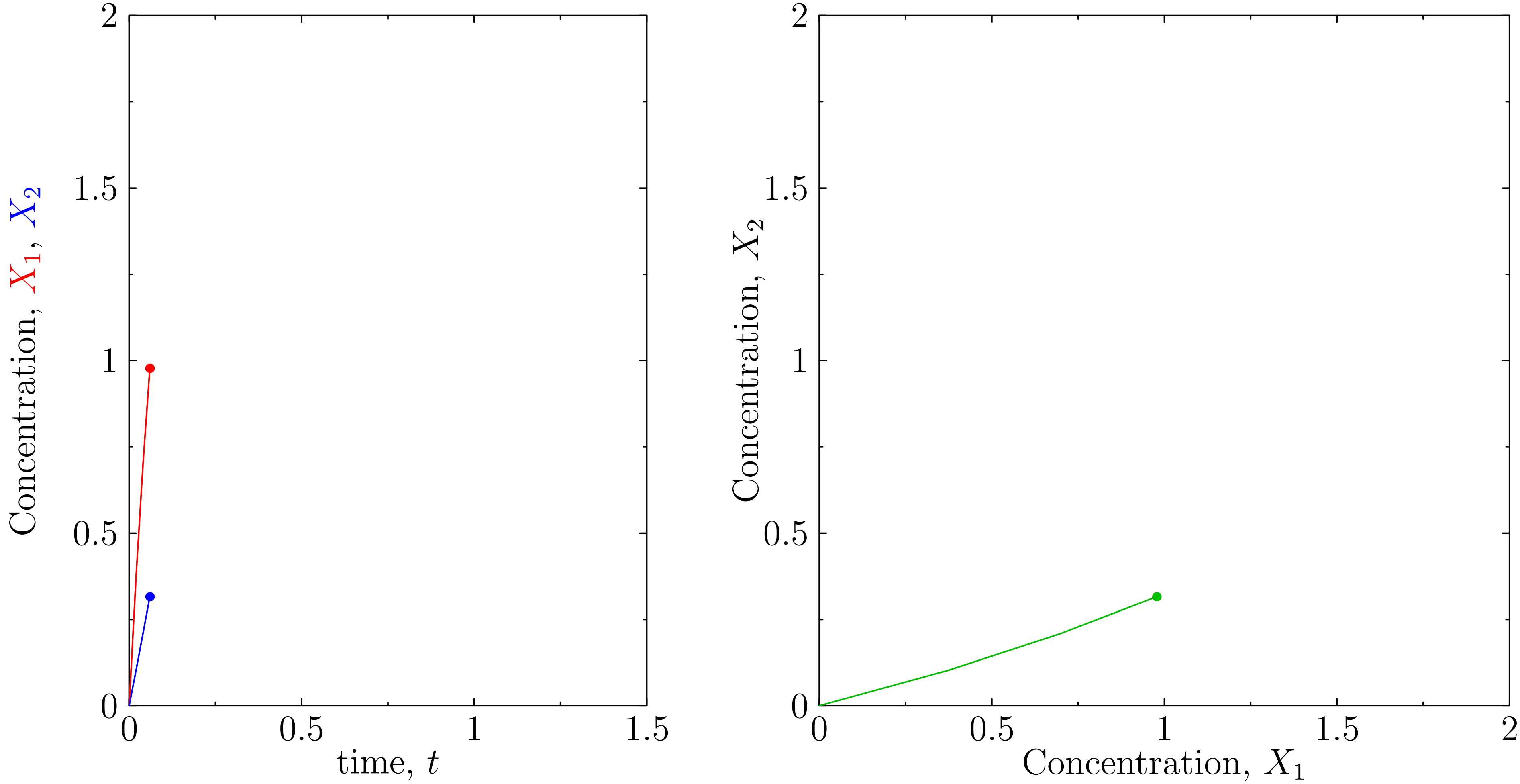
Solving the Dynamics





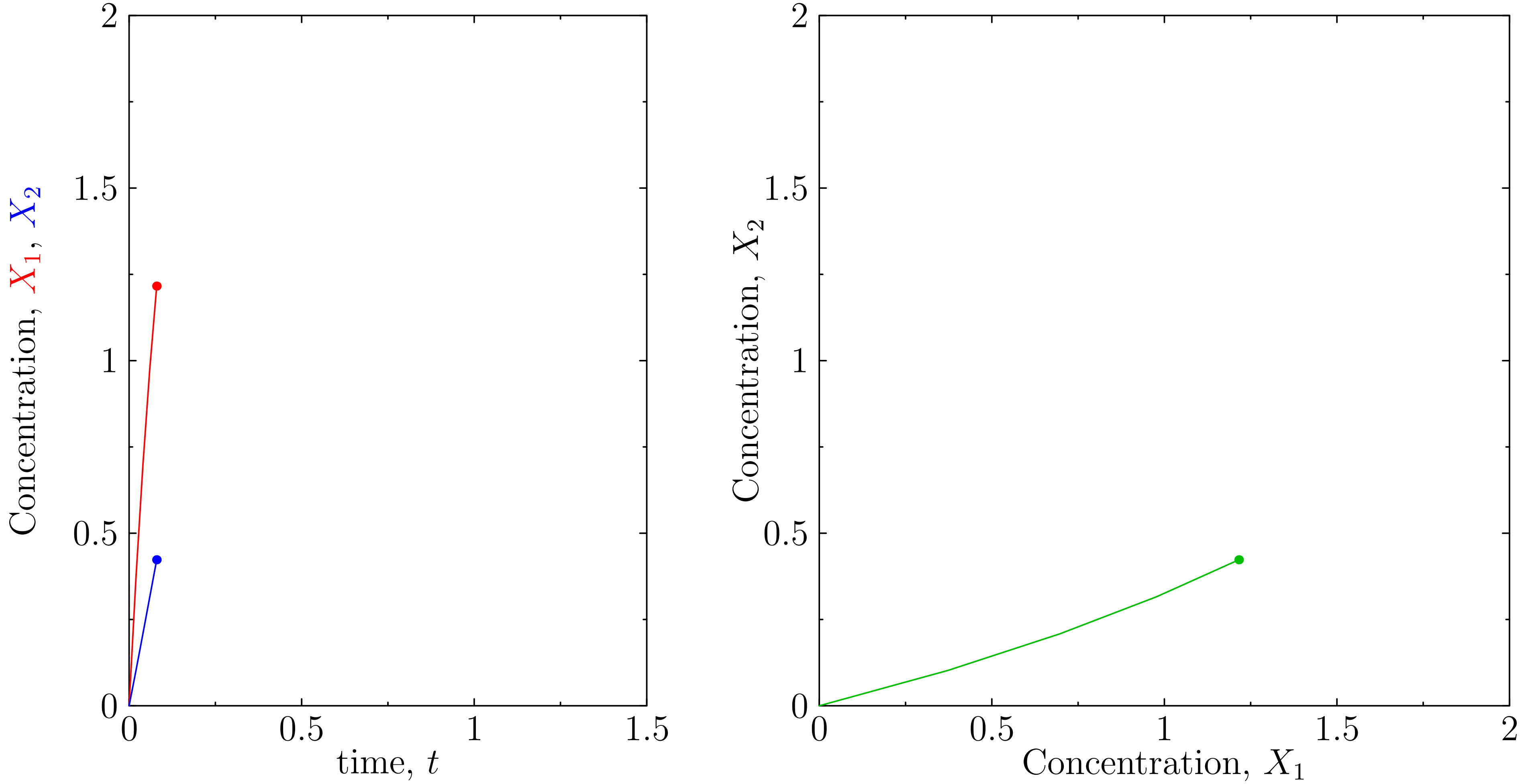
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



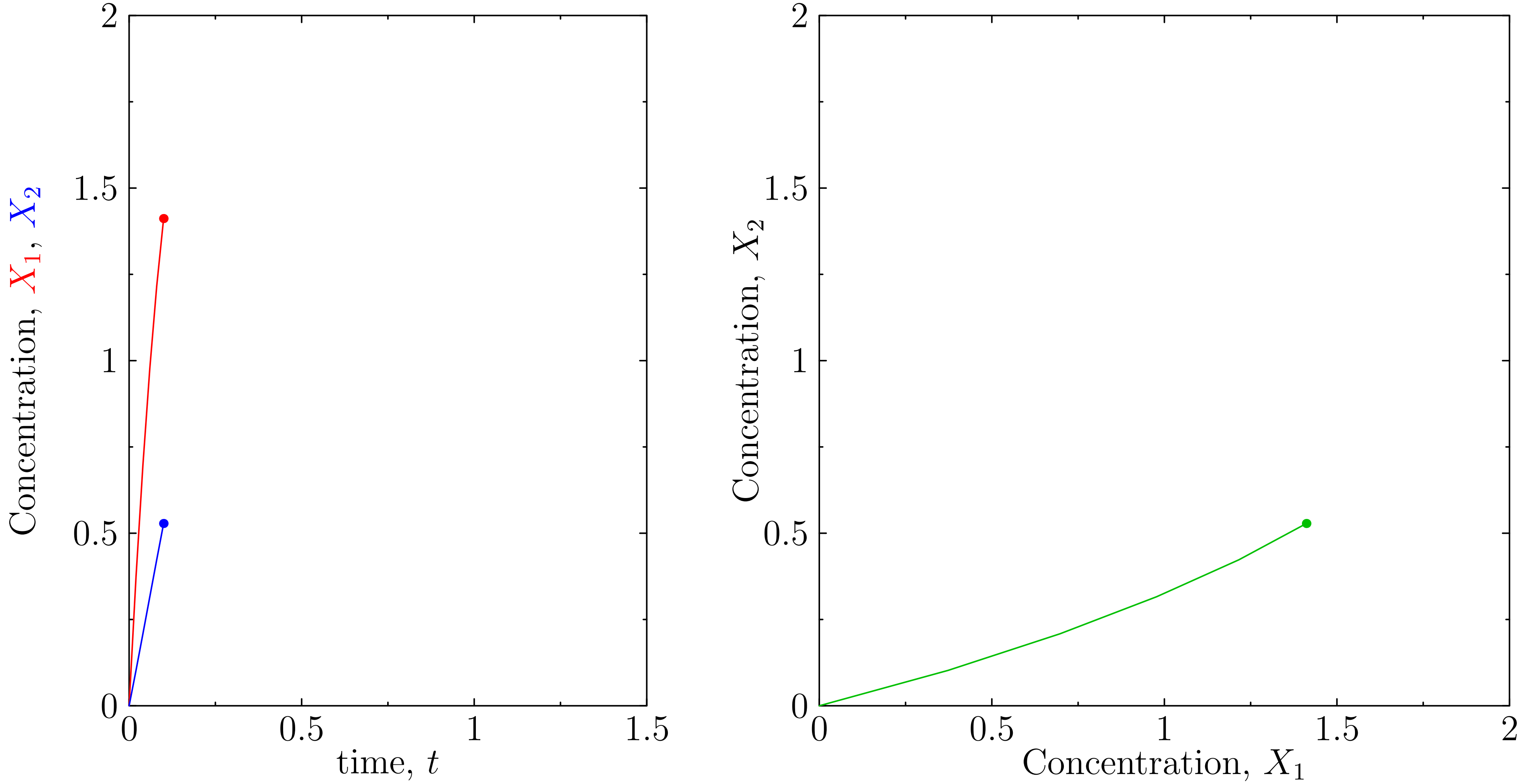
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



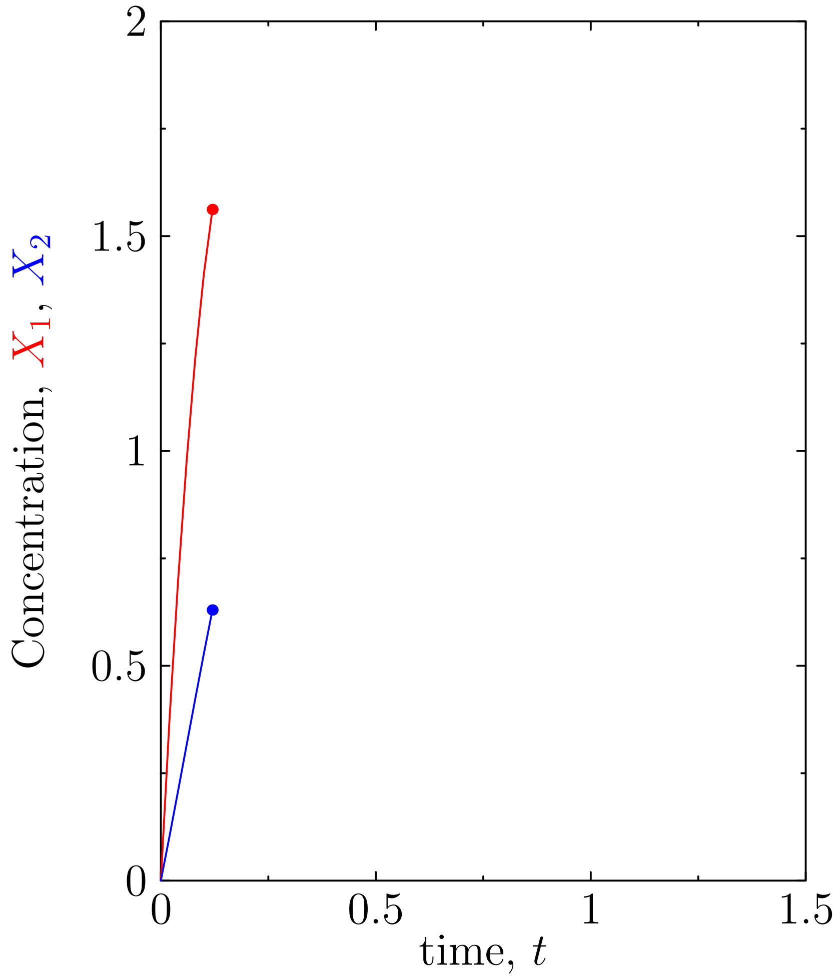
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

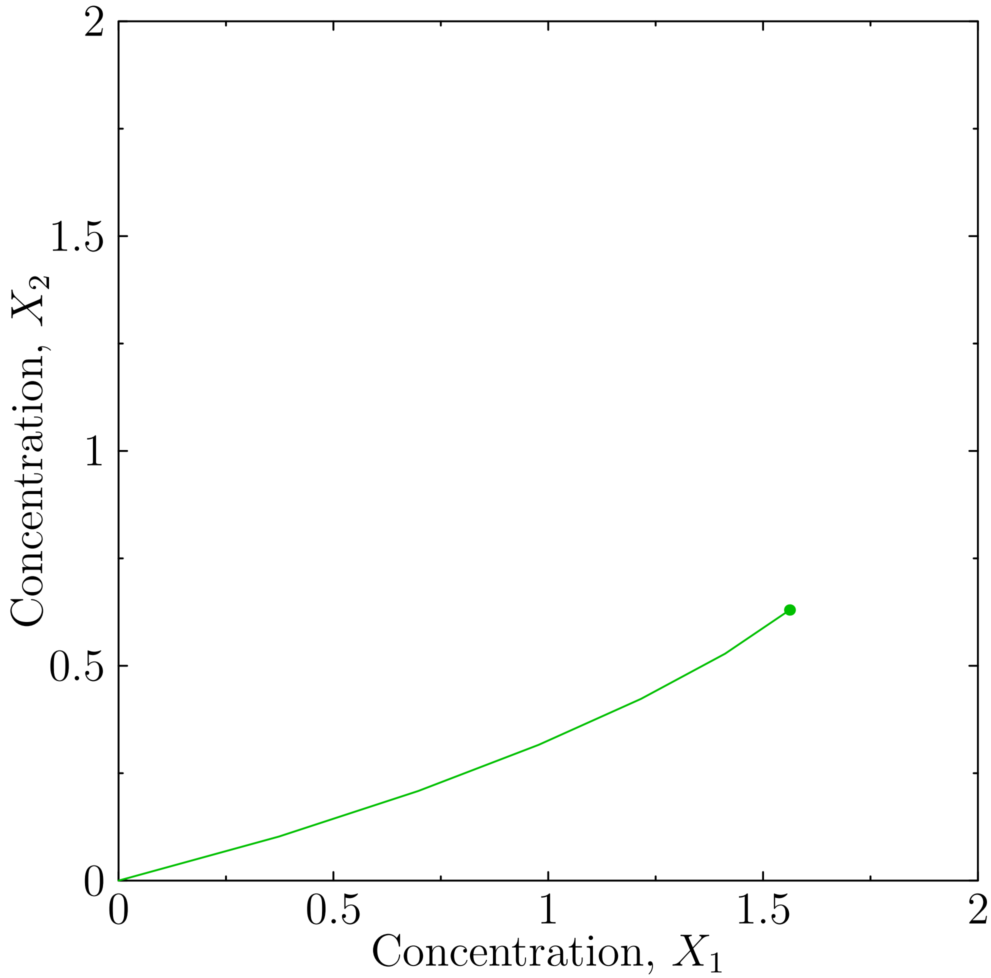
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

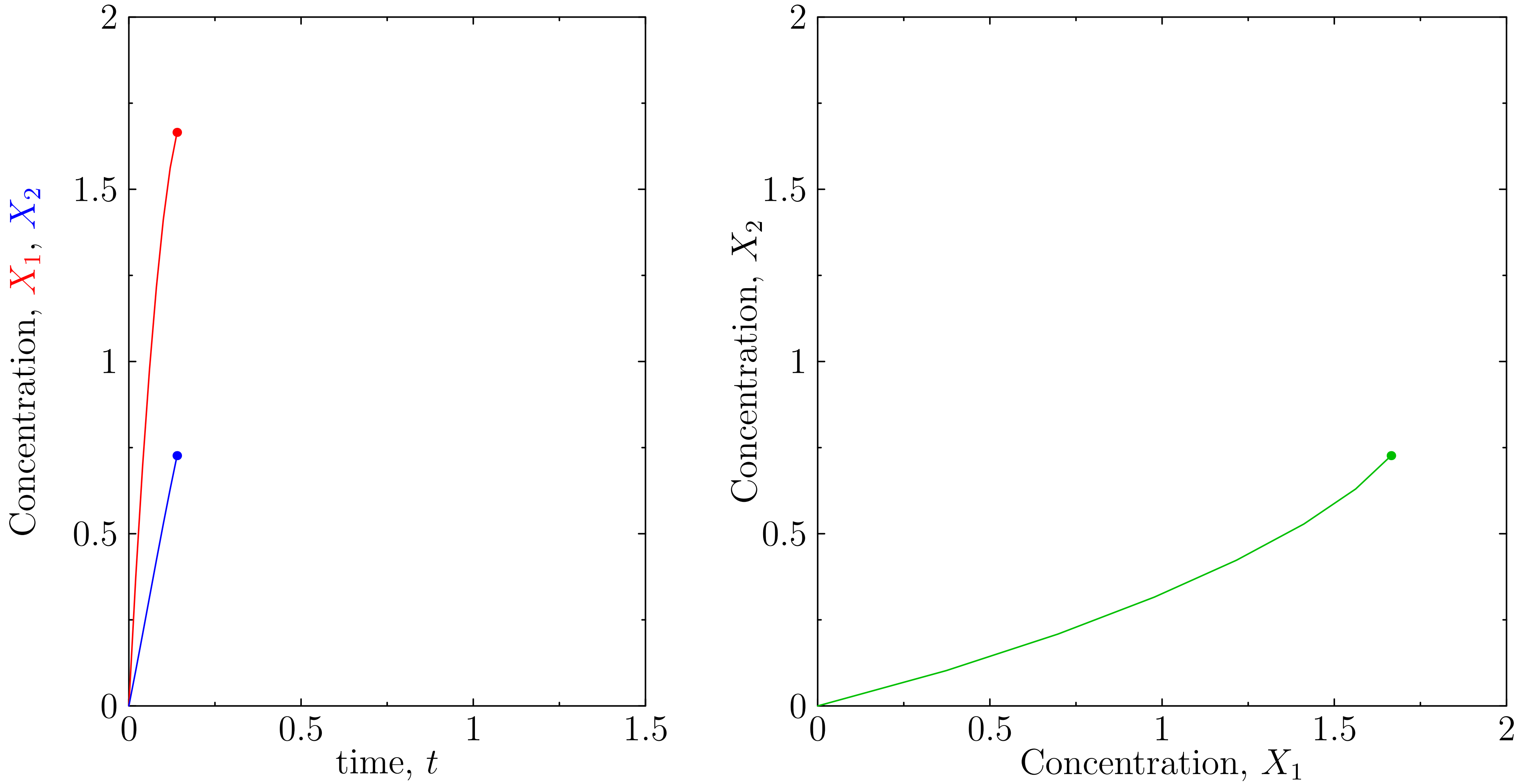
Solving the Dynamics





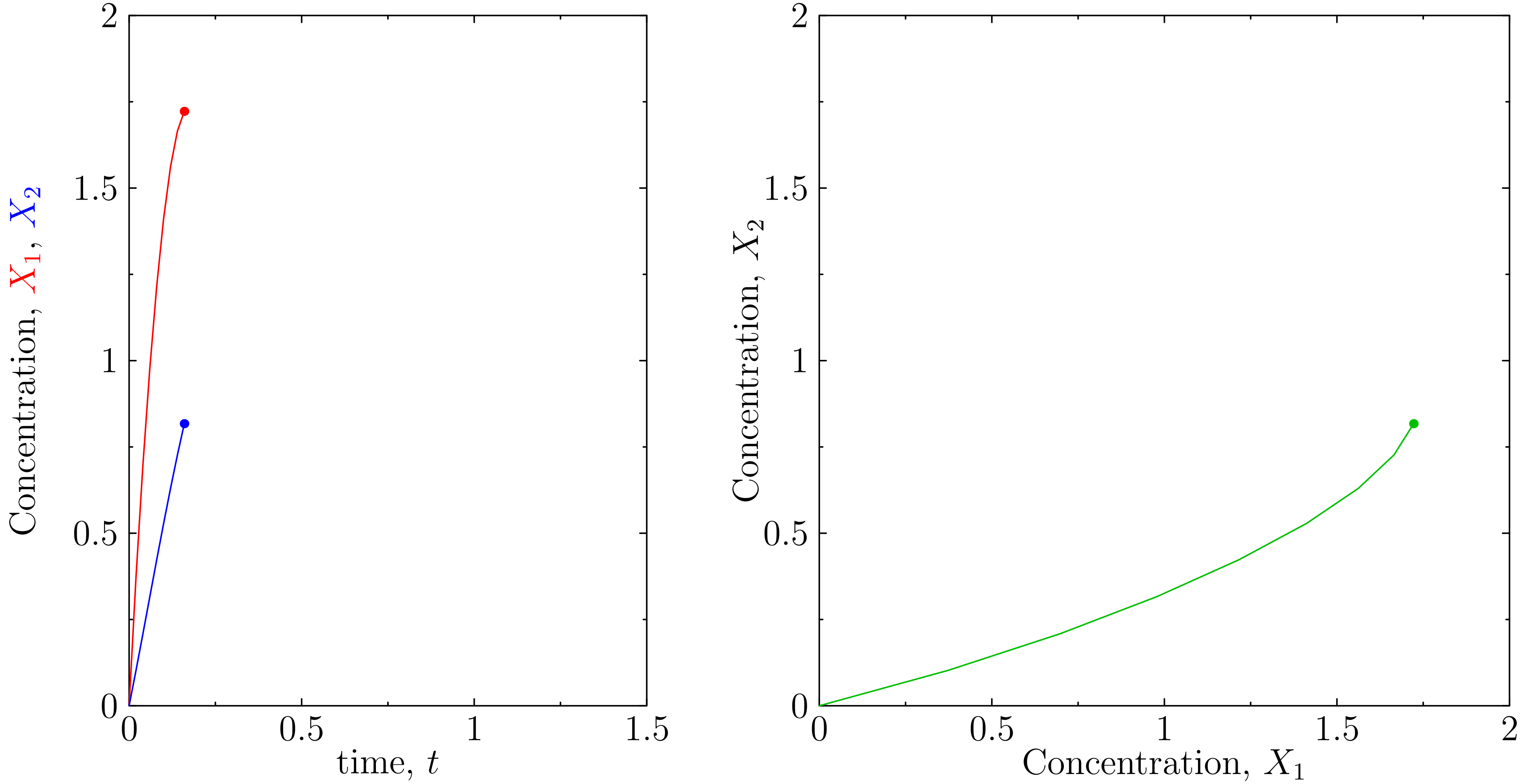
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



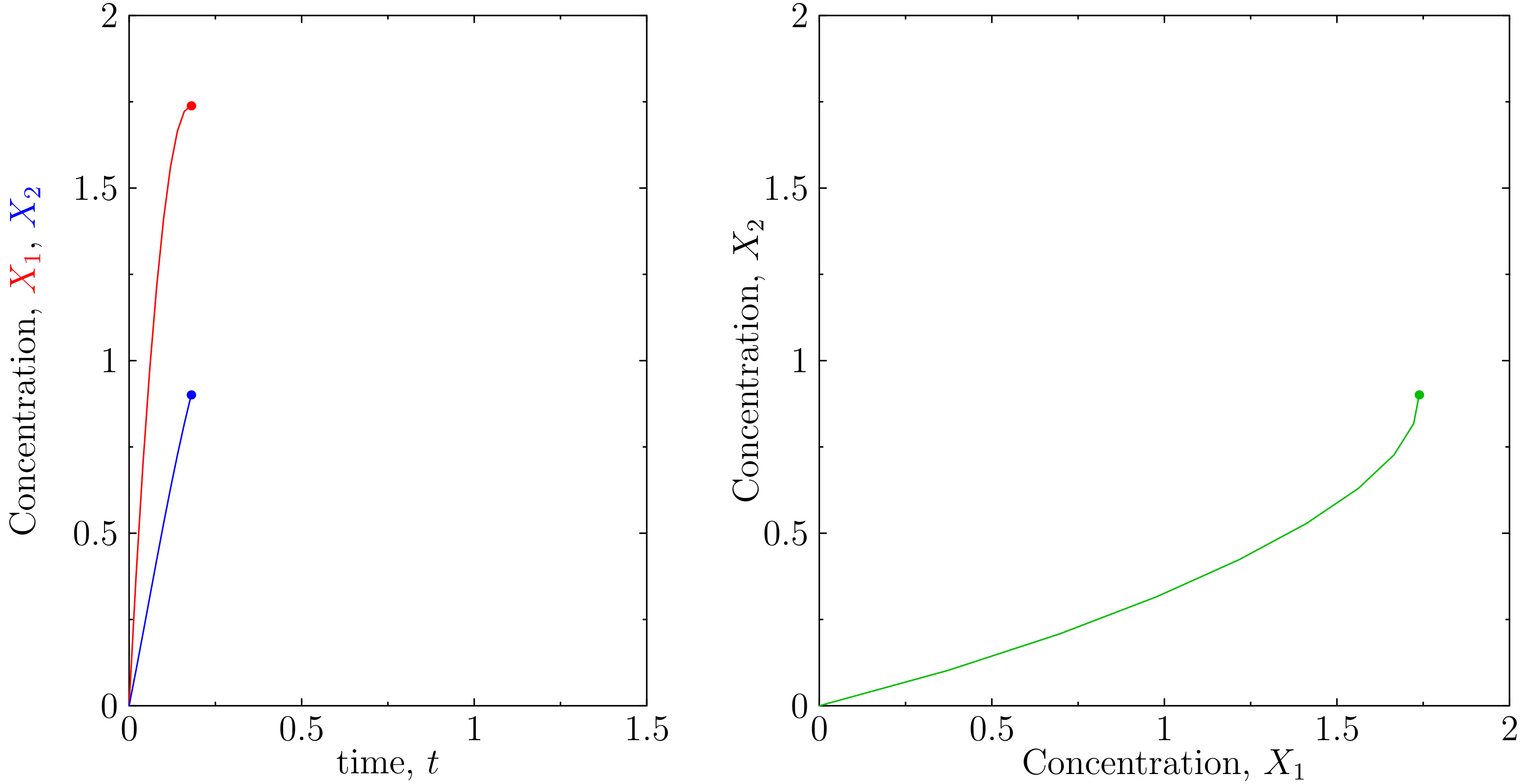
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



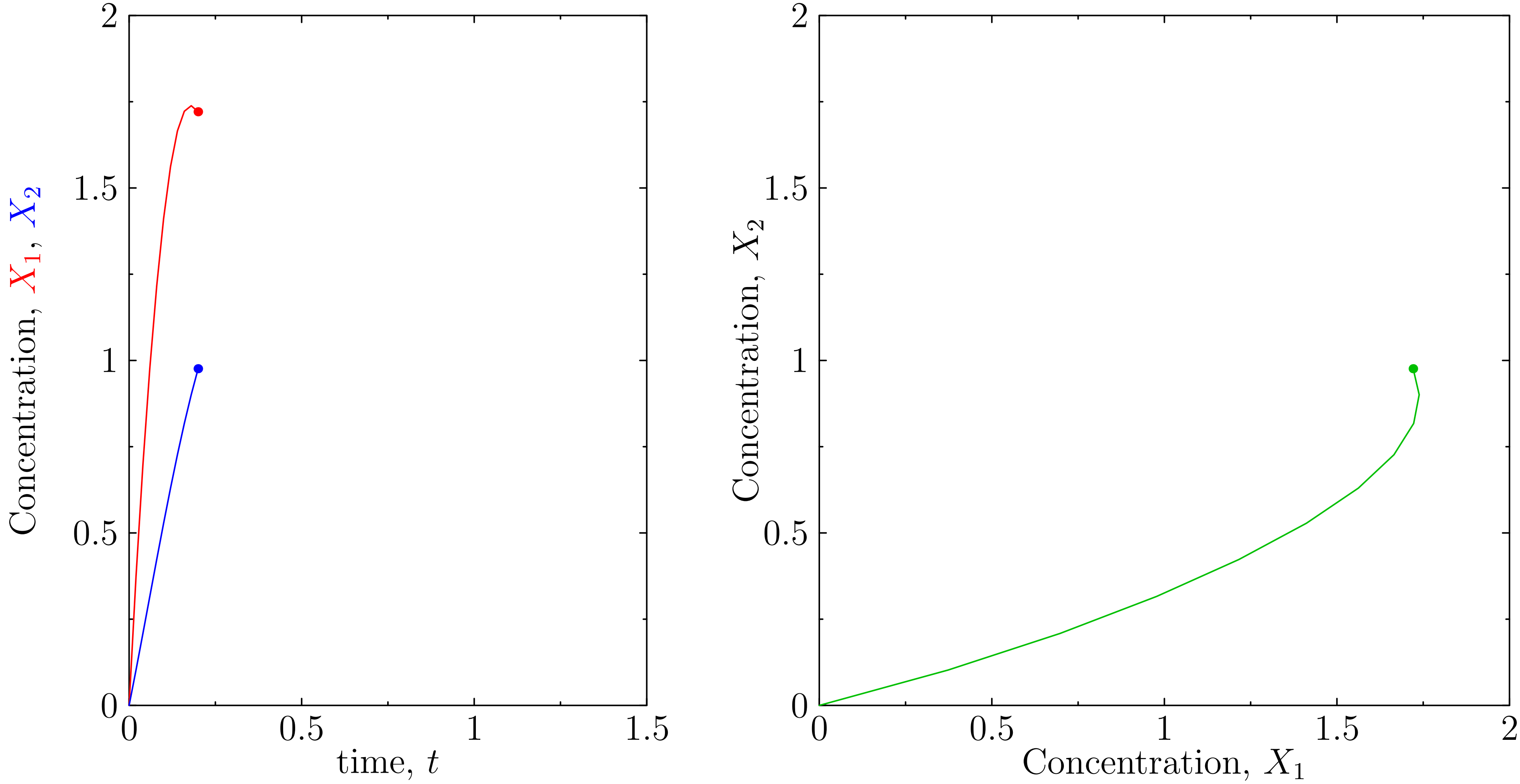
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



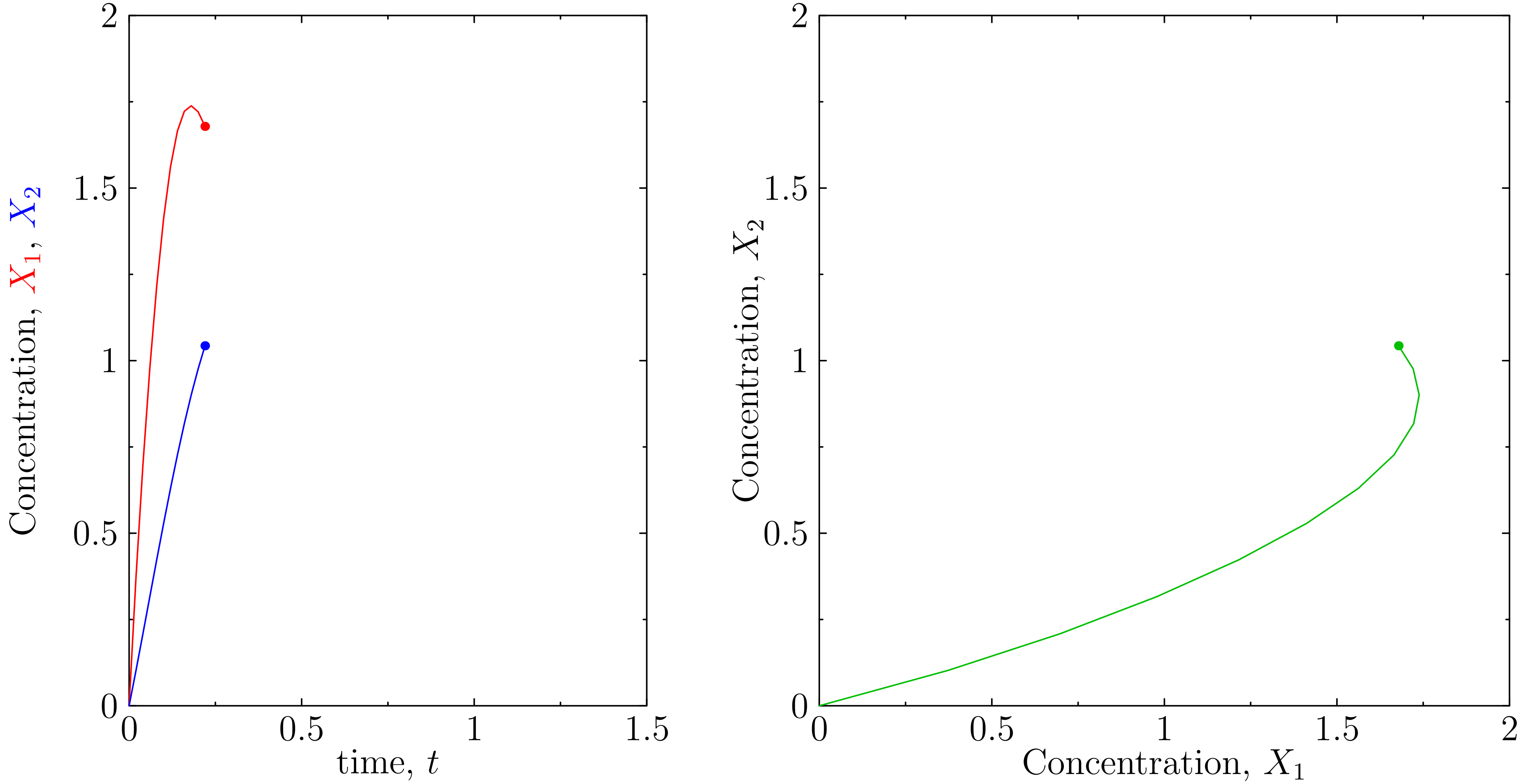
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



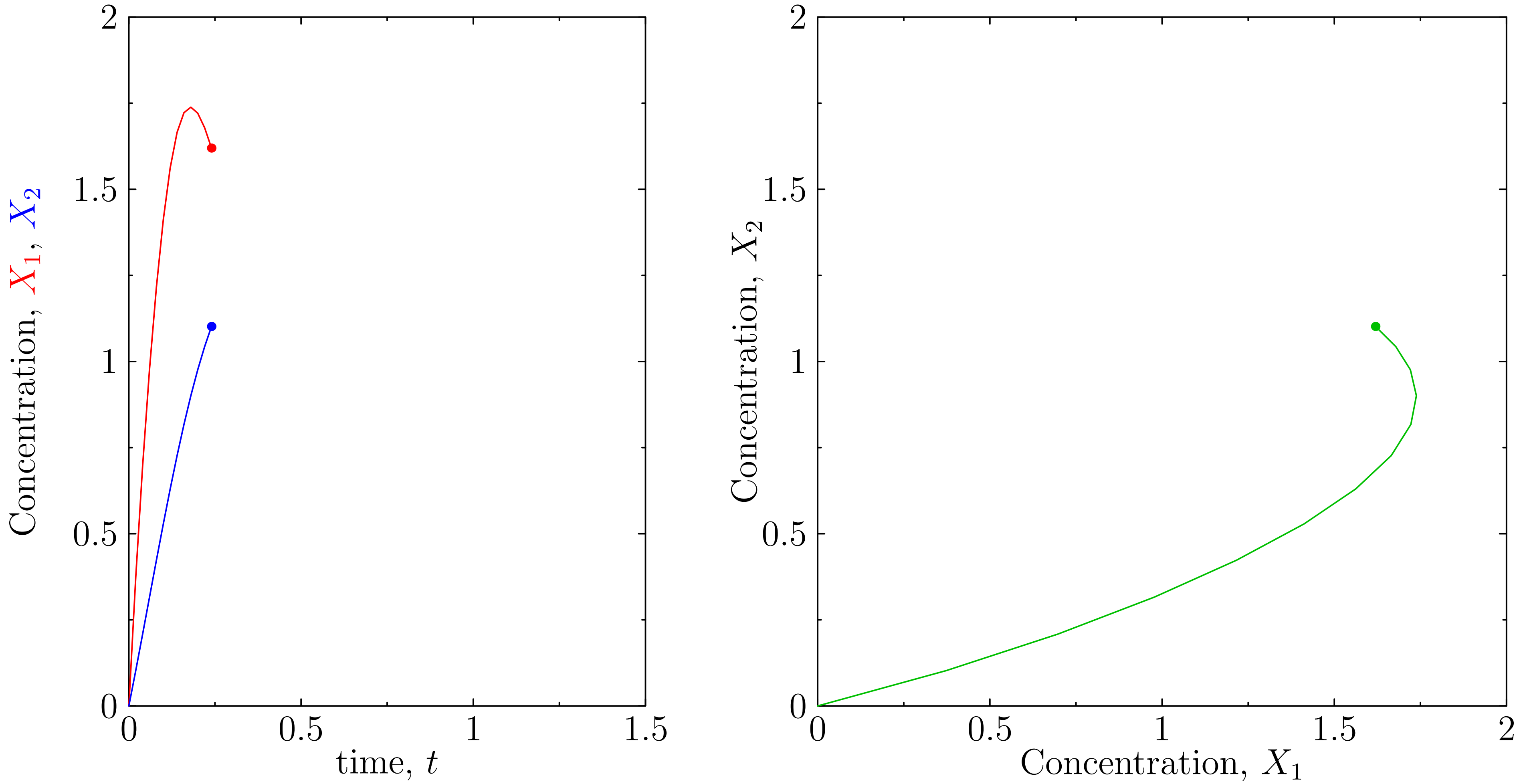
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



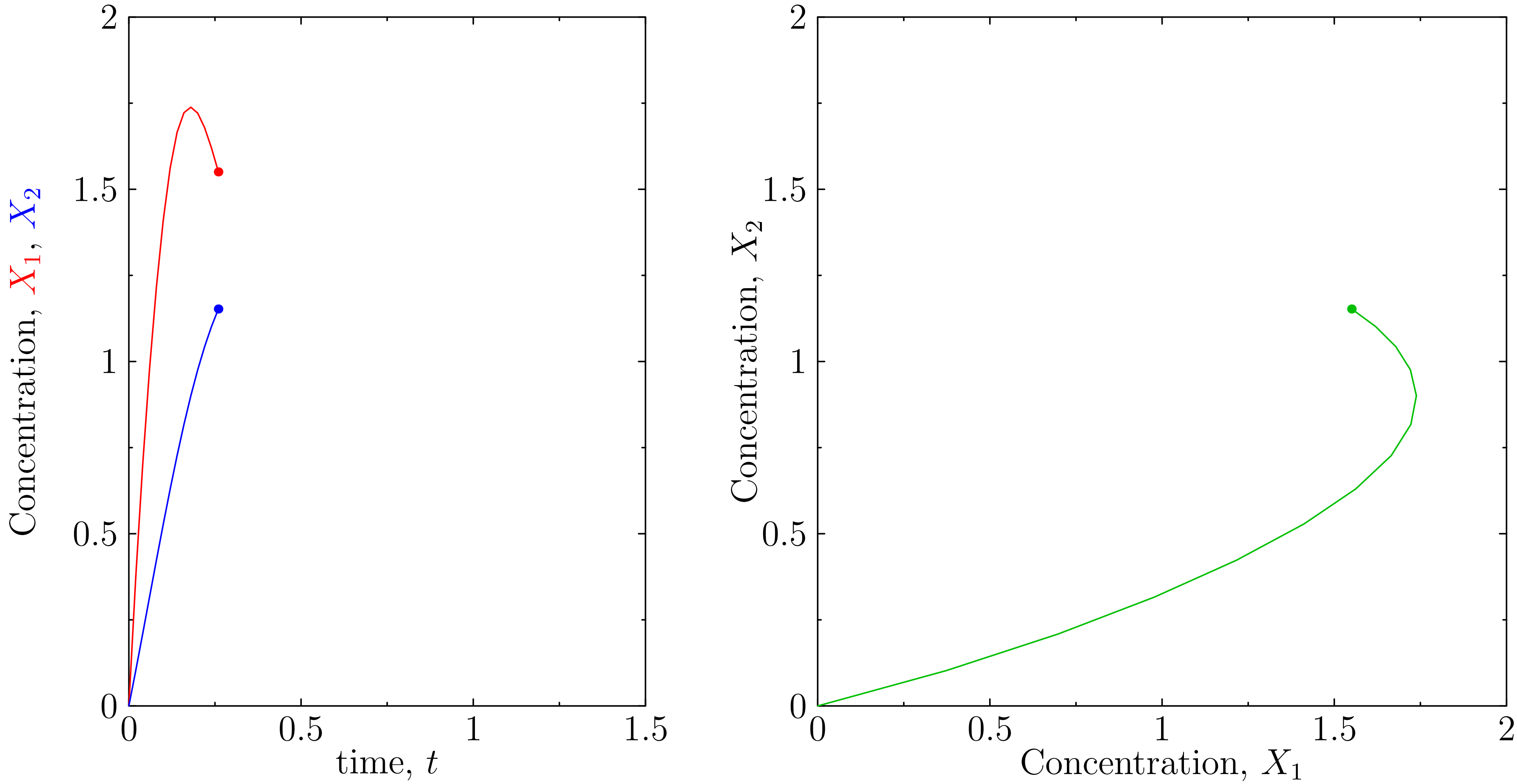
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



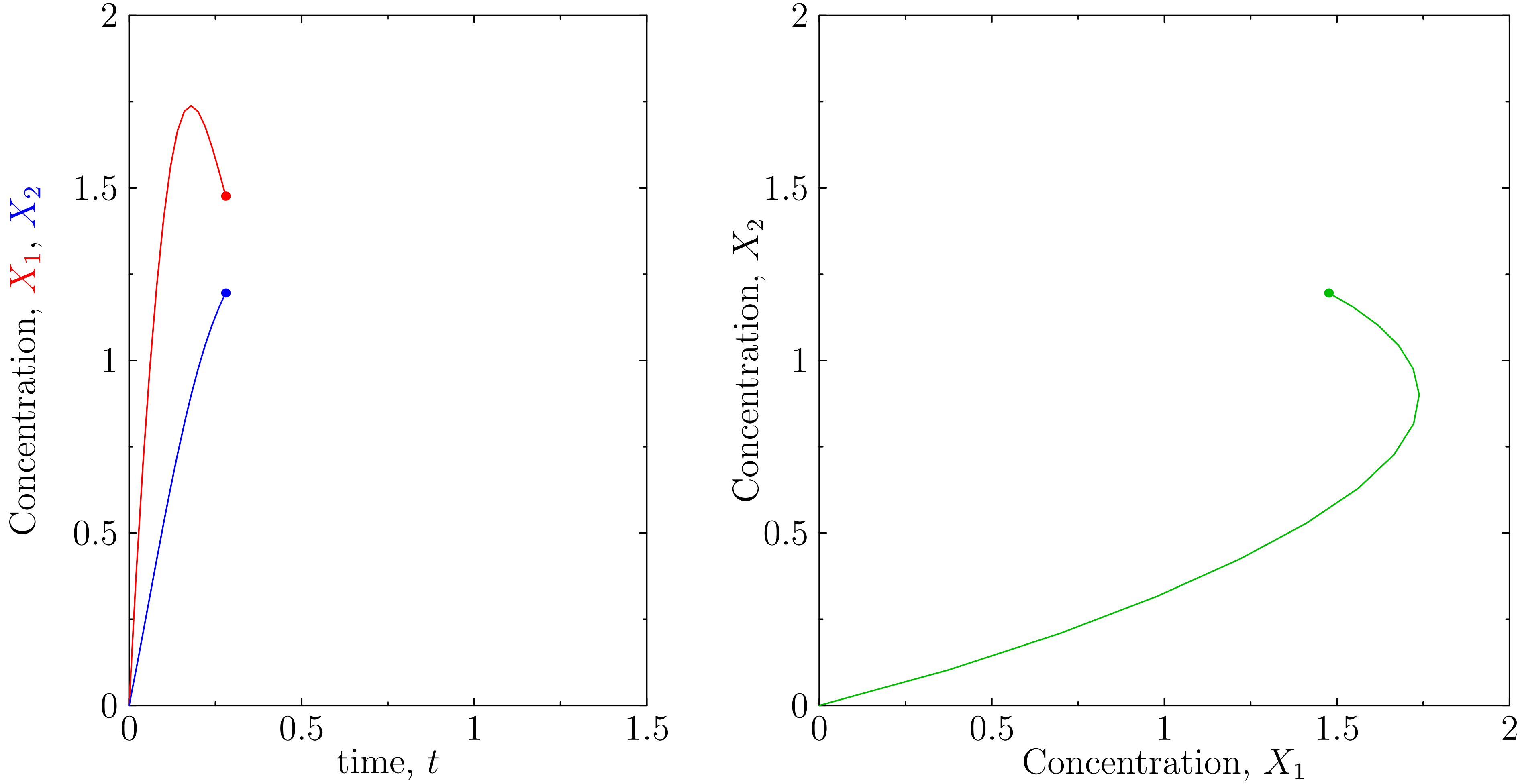
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



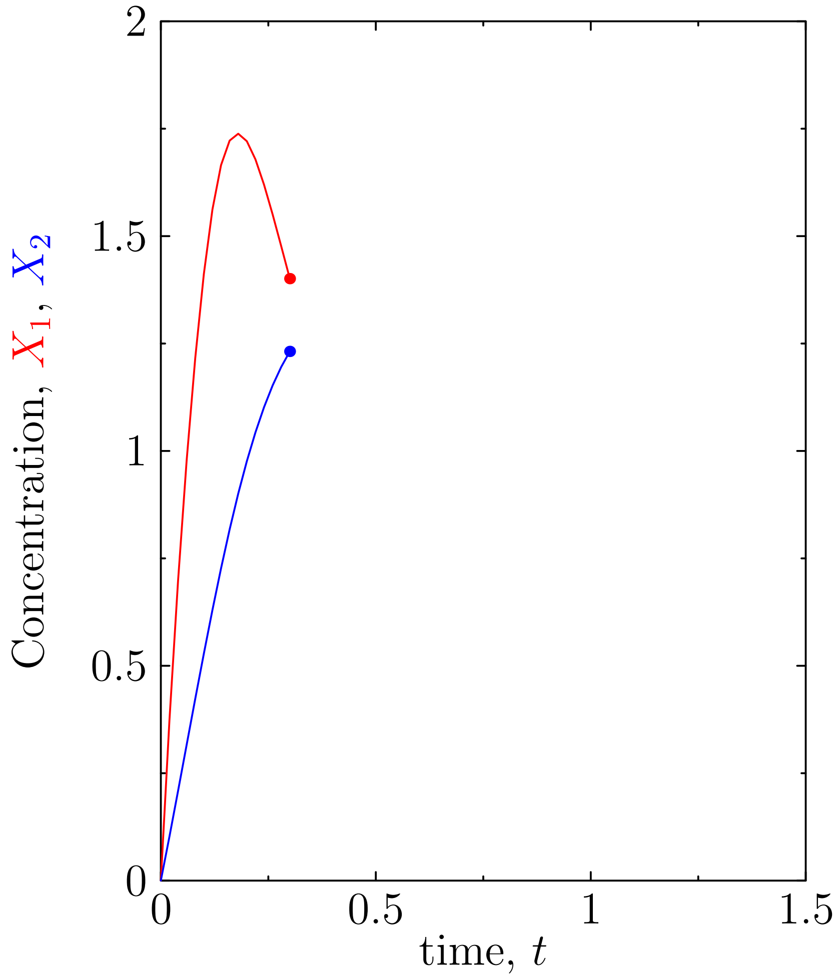
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

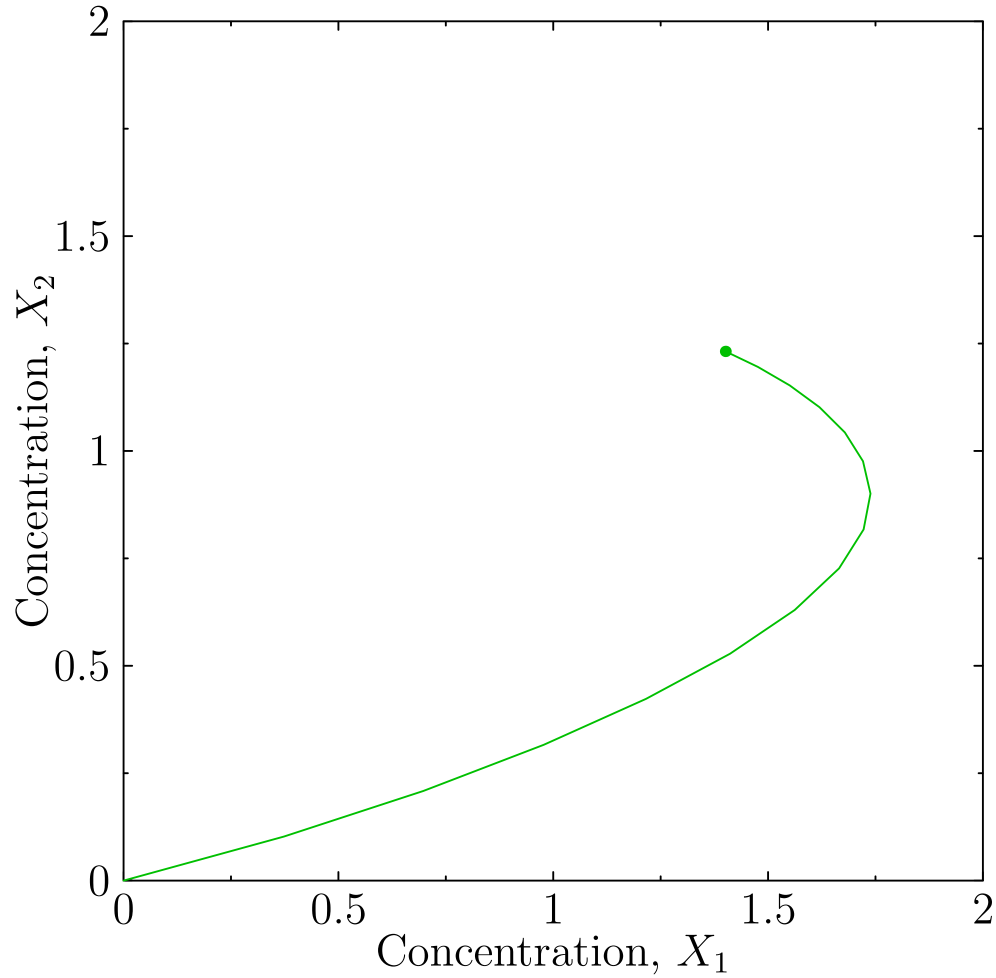
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

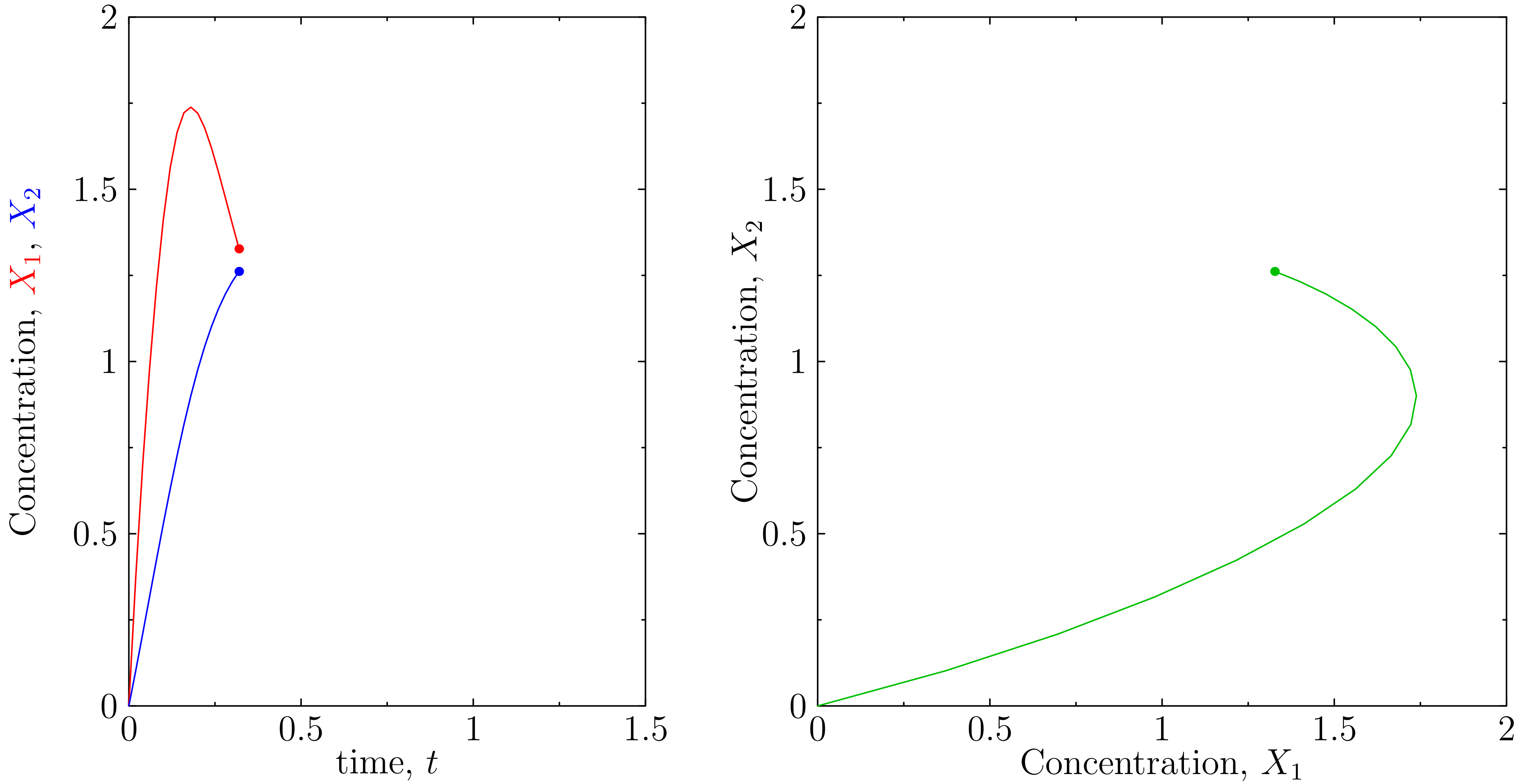
Solving the Dynamics





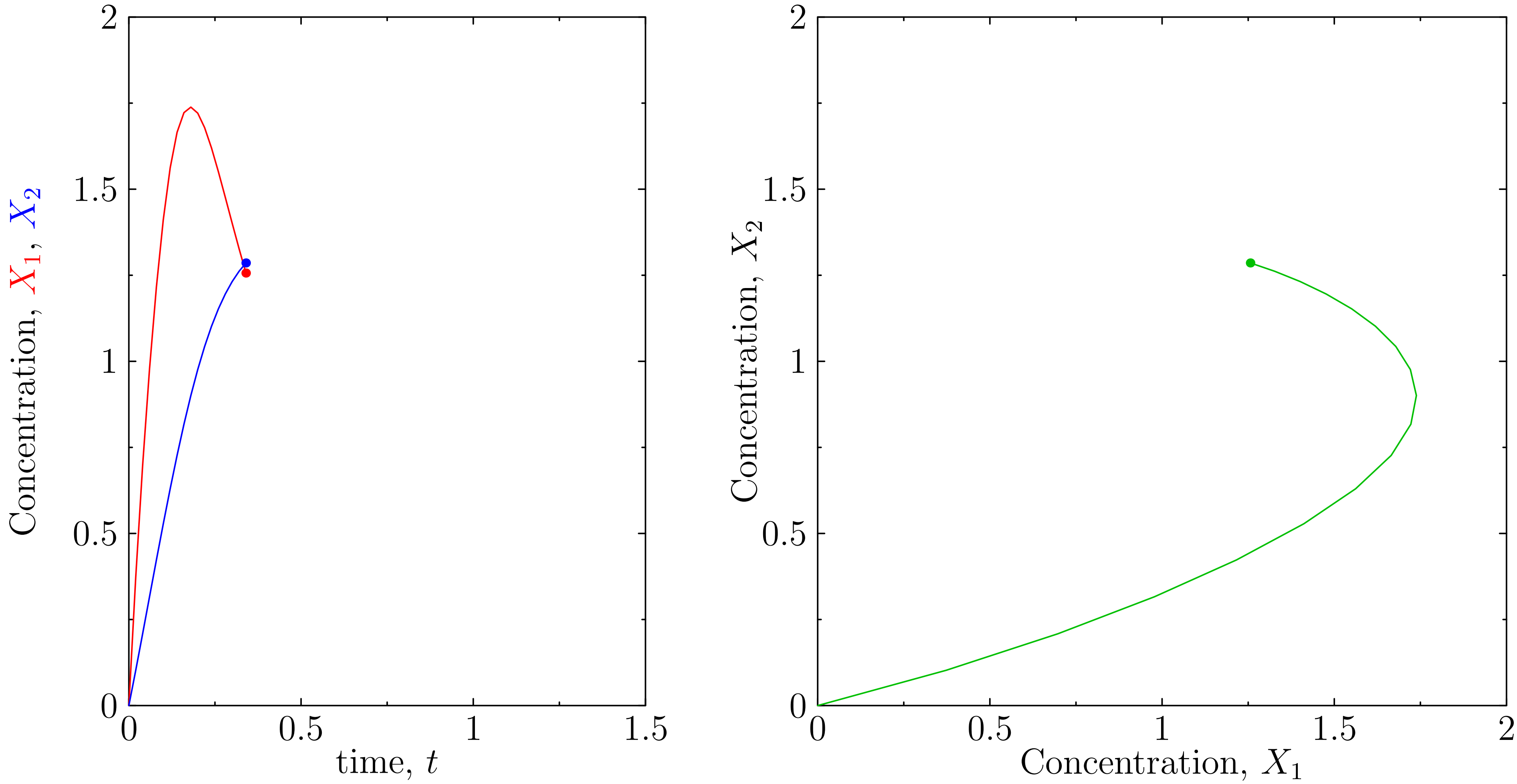
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



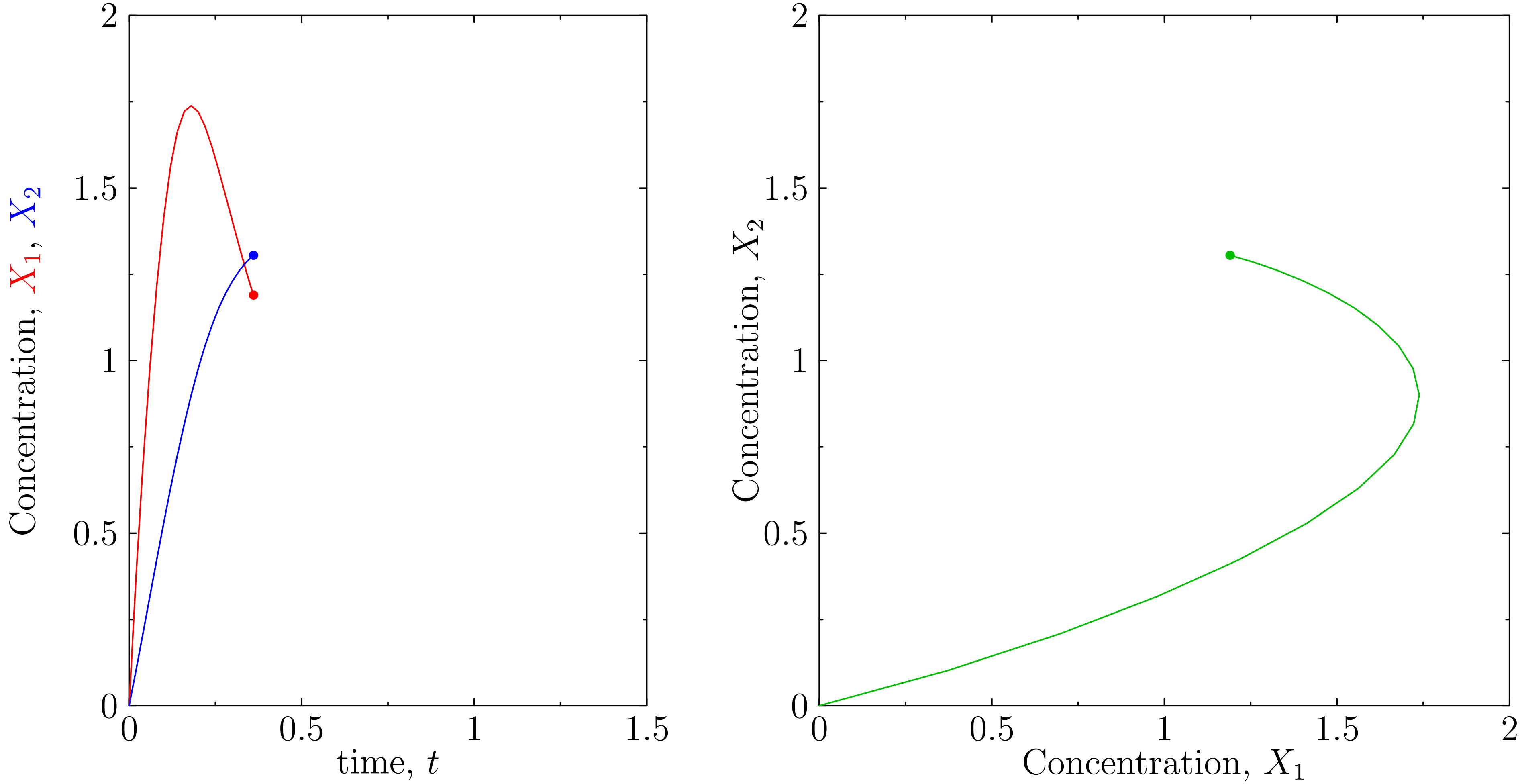
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



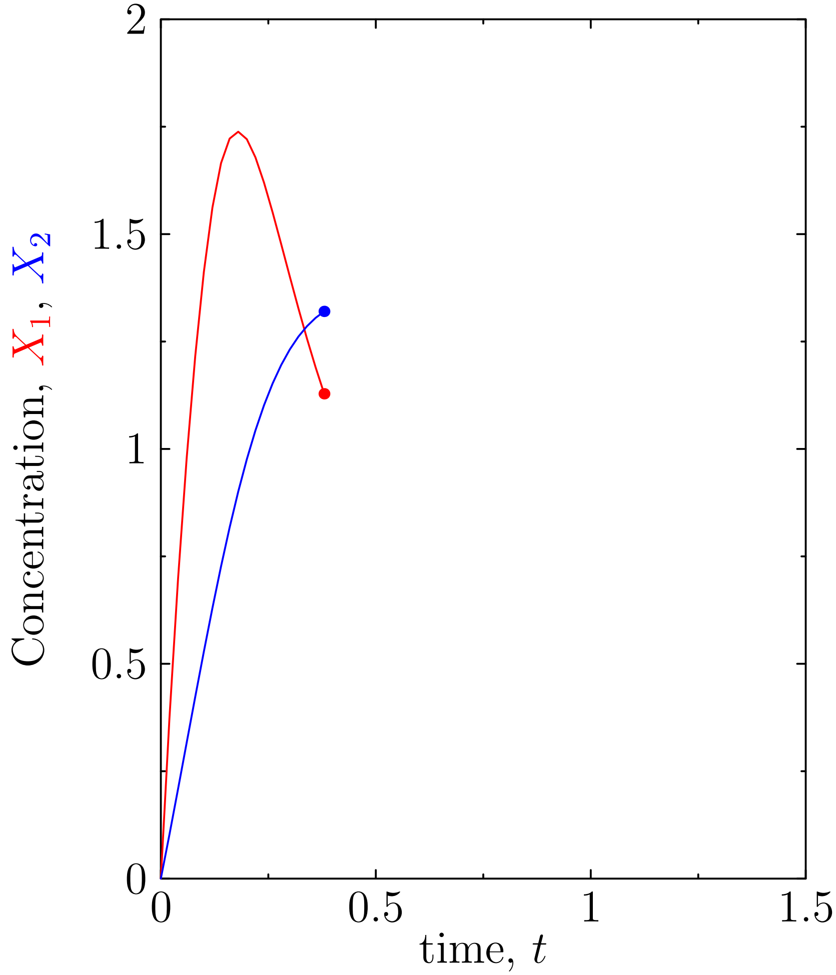
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

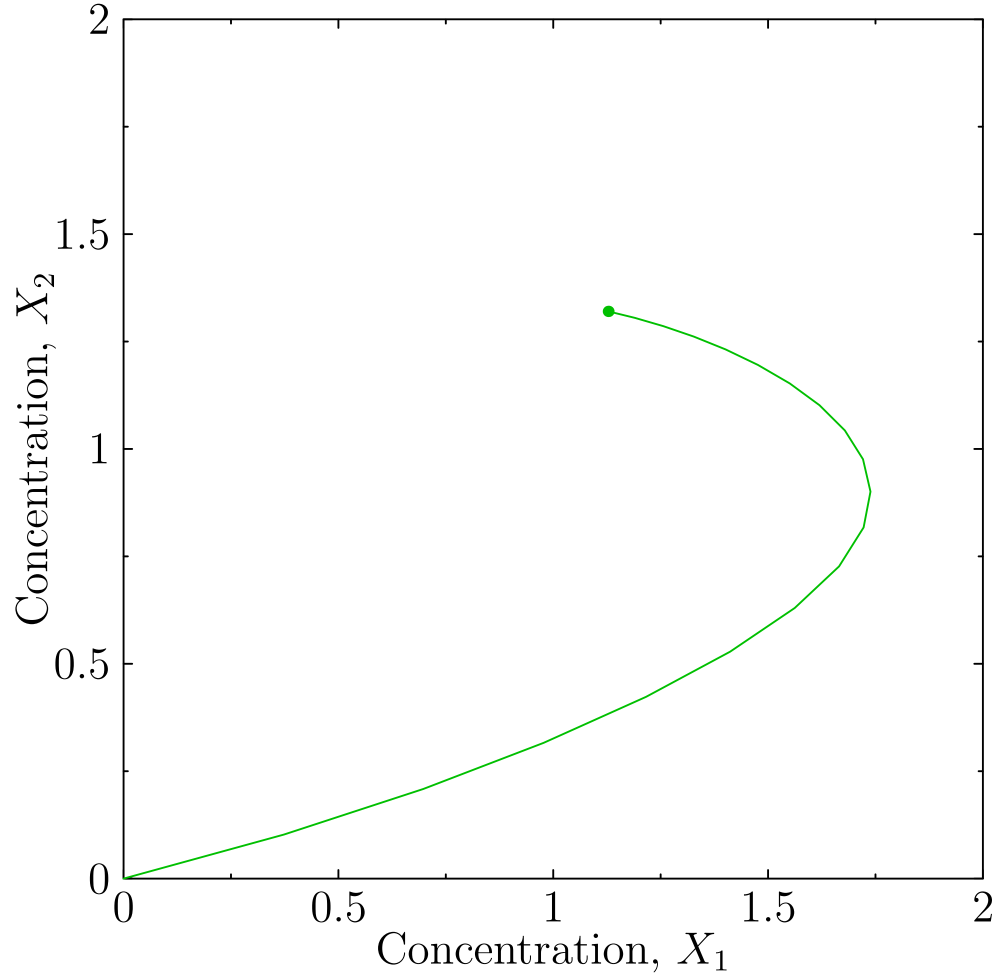
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

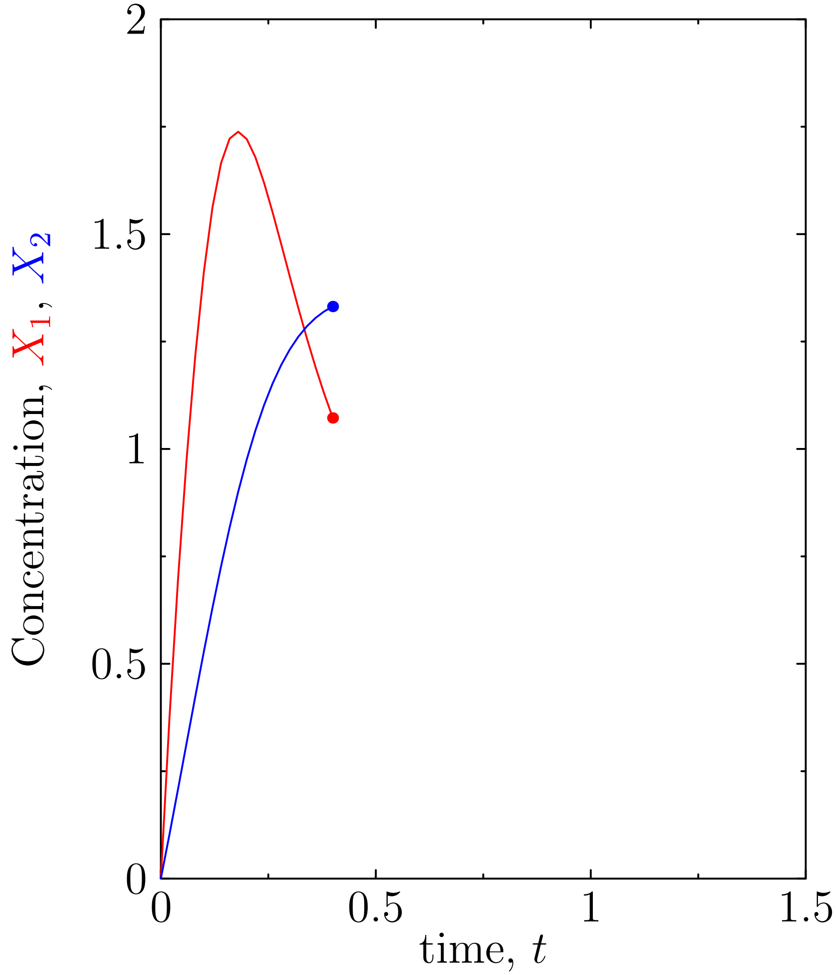
Solving the Dynamics

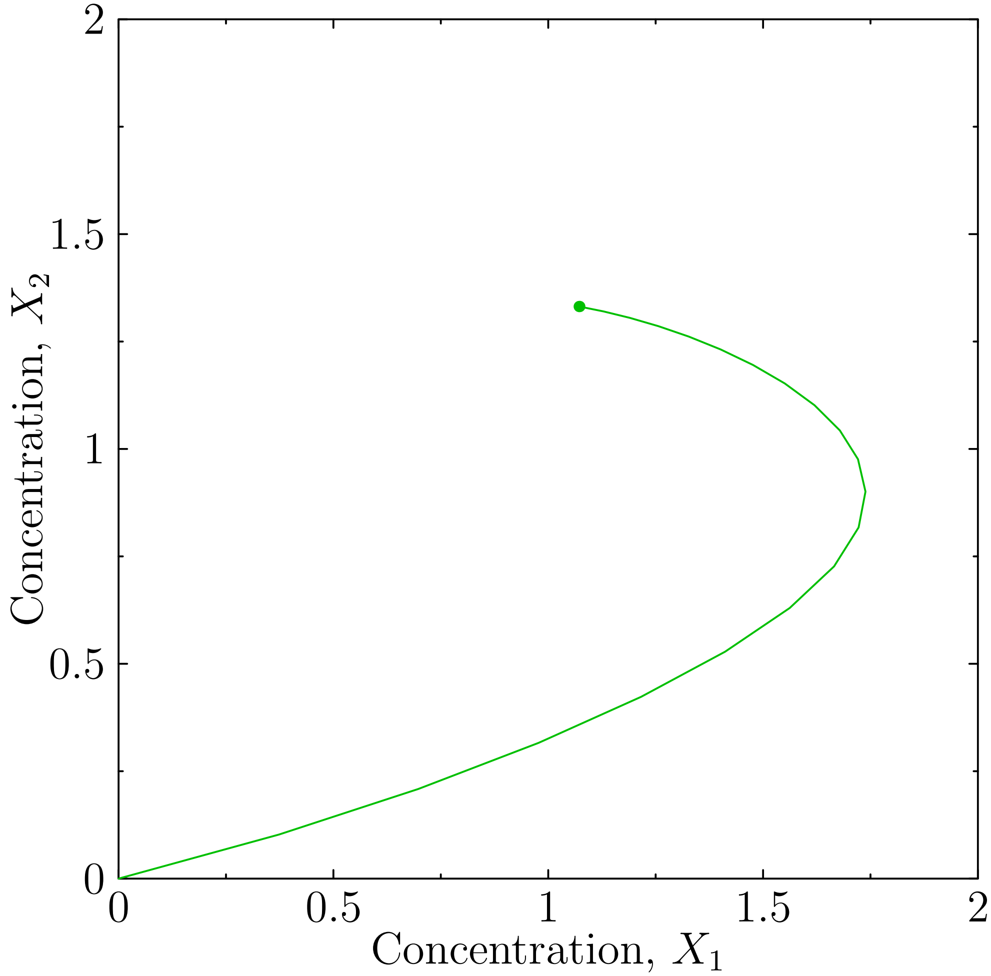




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

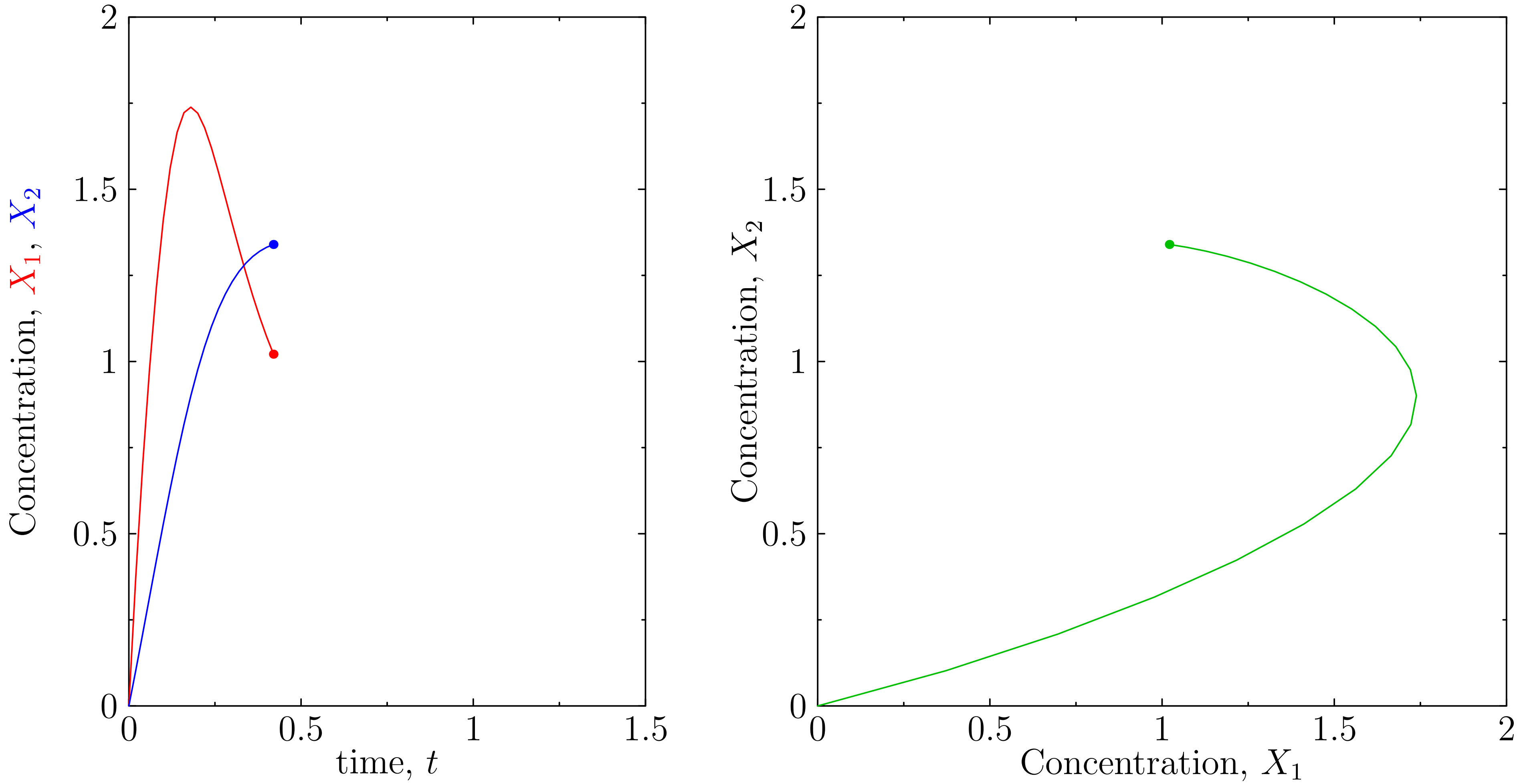
Solving the Dynamics





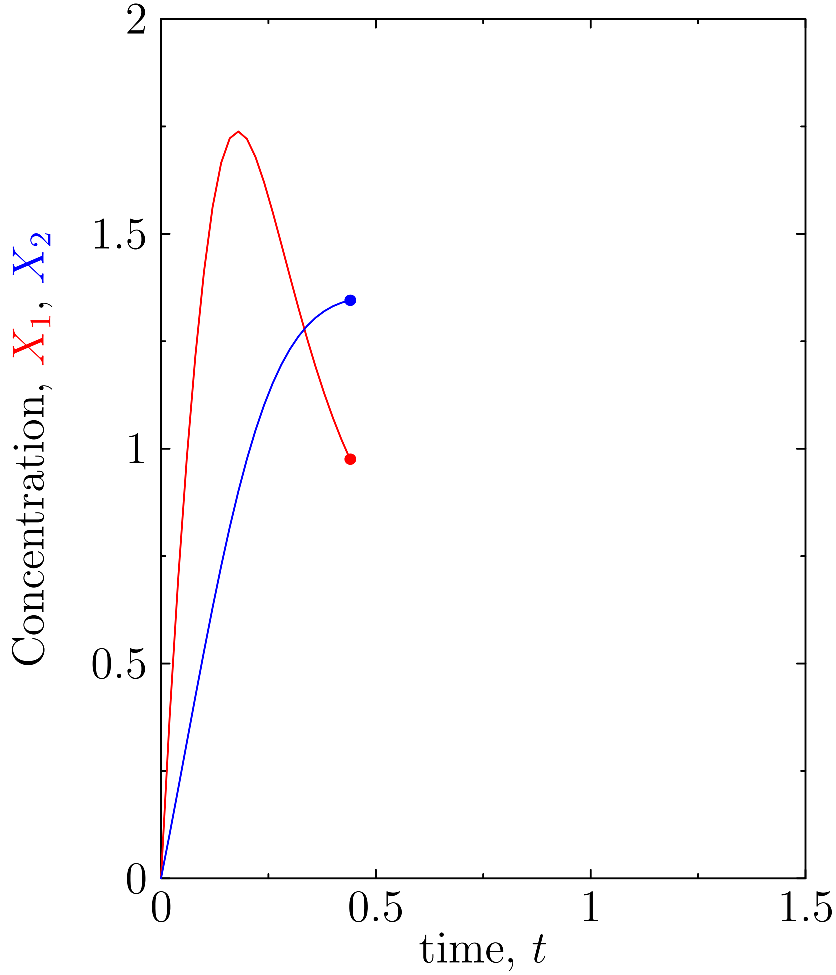
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

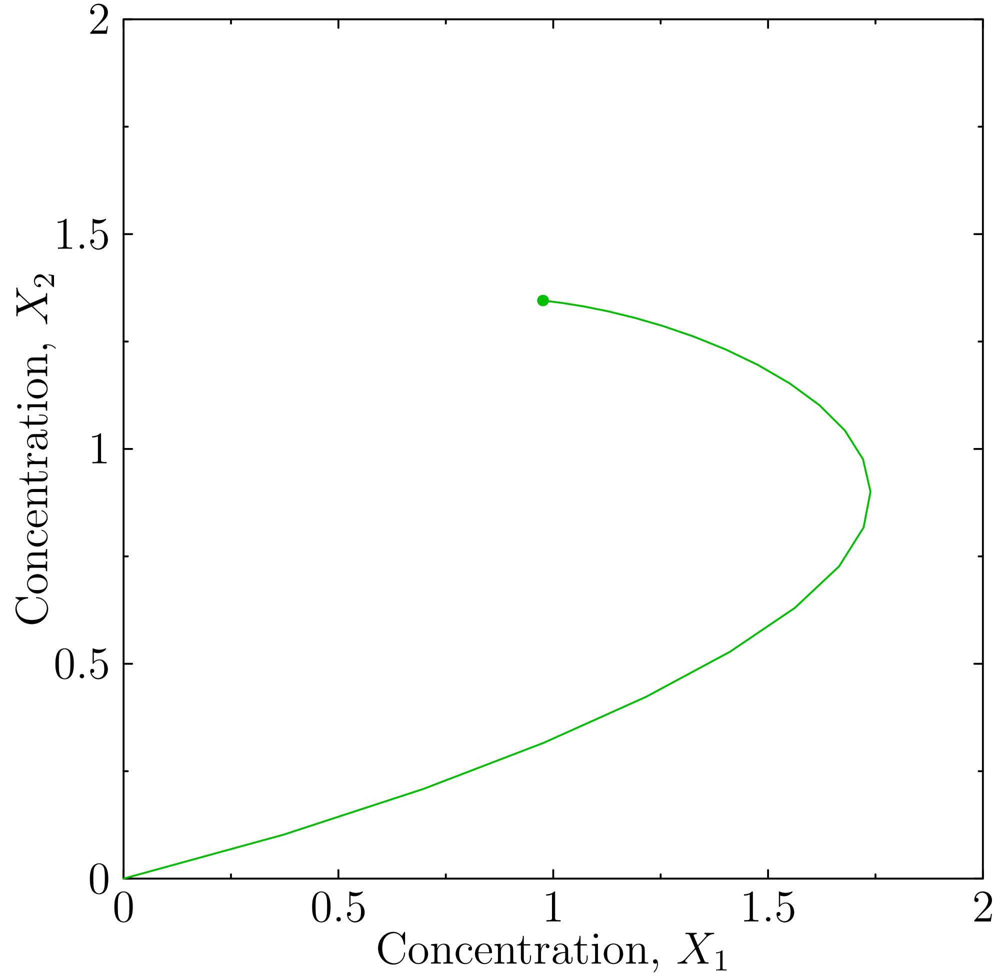
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

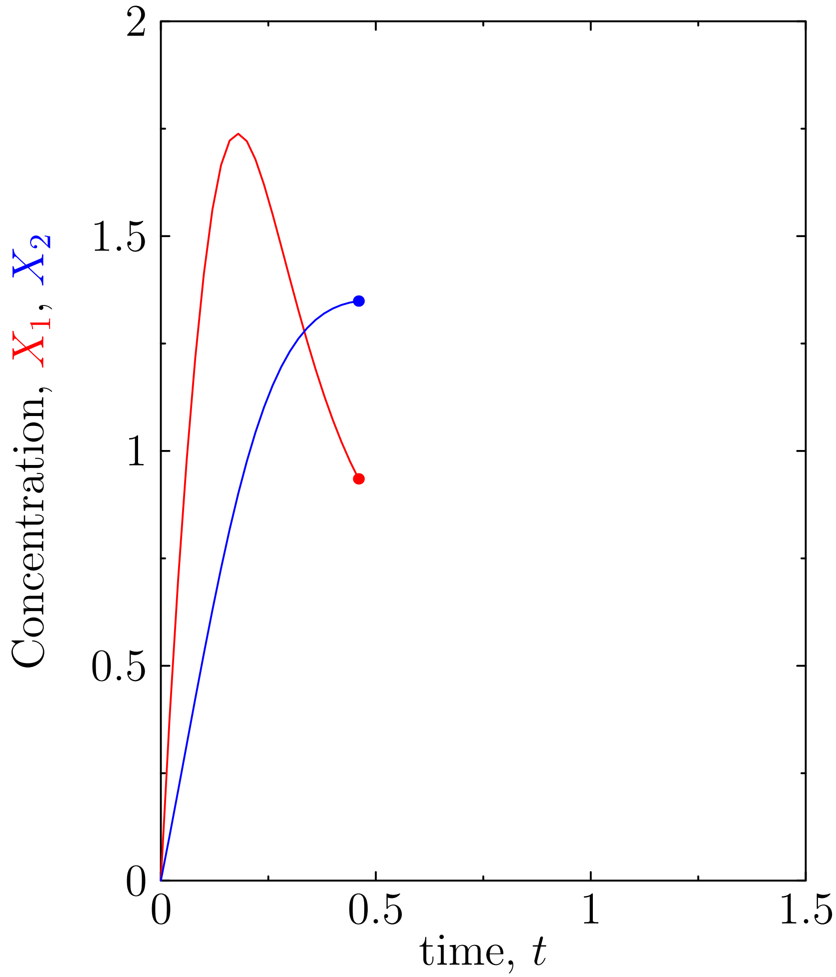
Solving the Dynamics

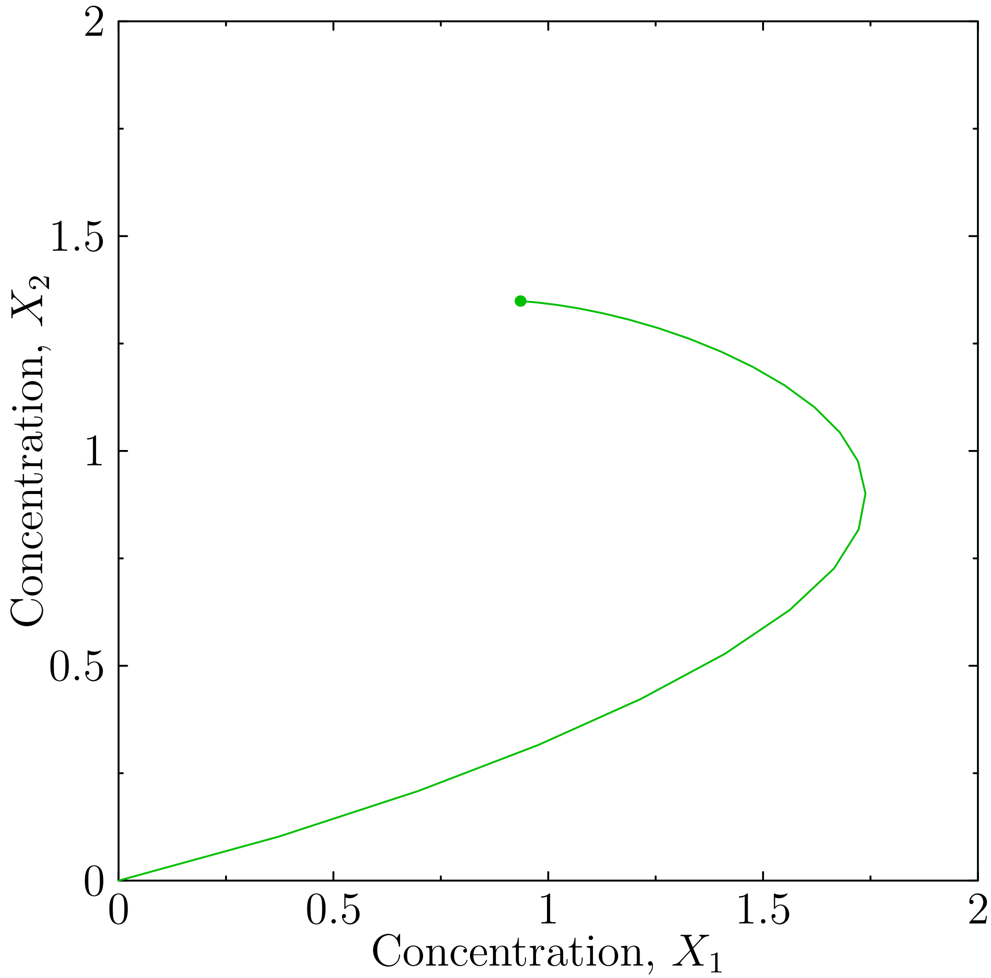




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

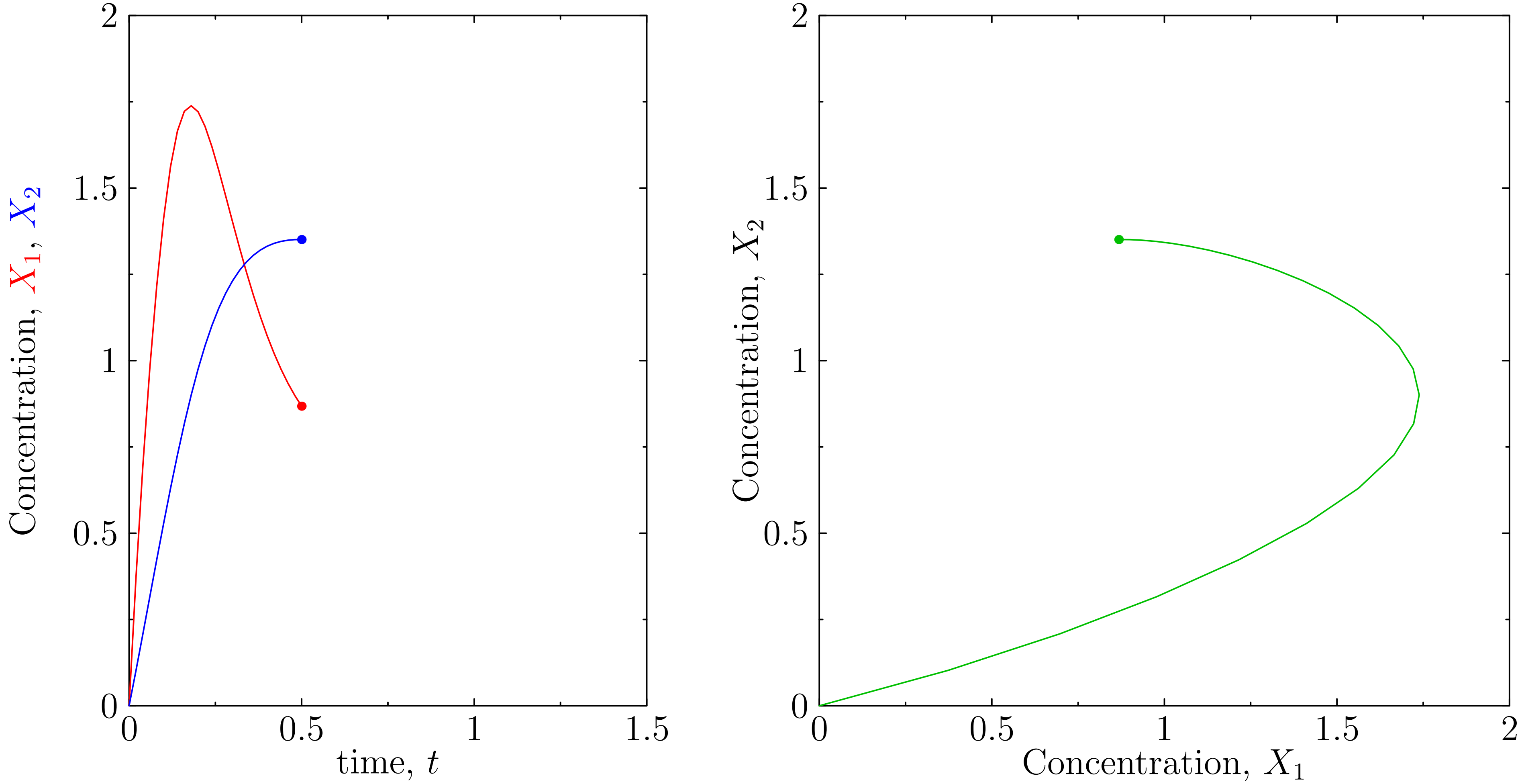
Solving the Dynamics





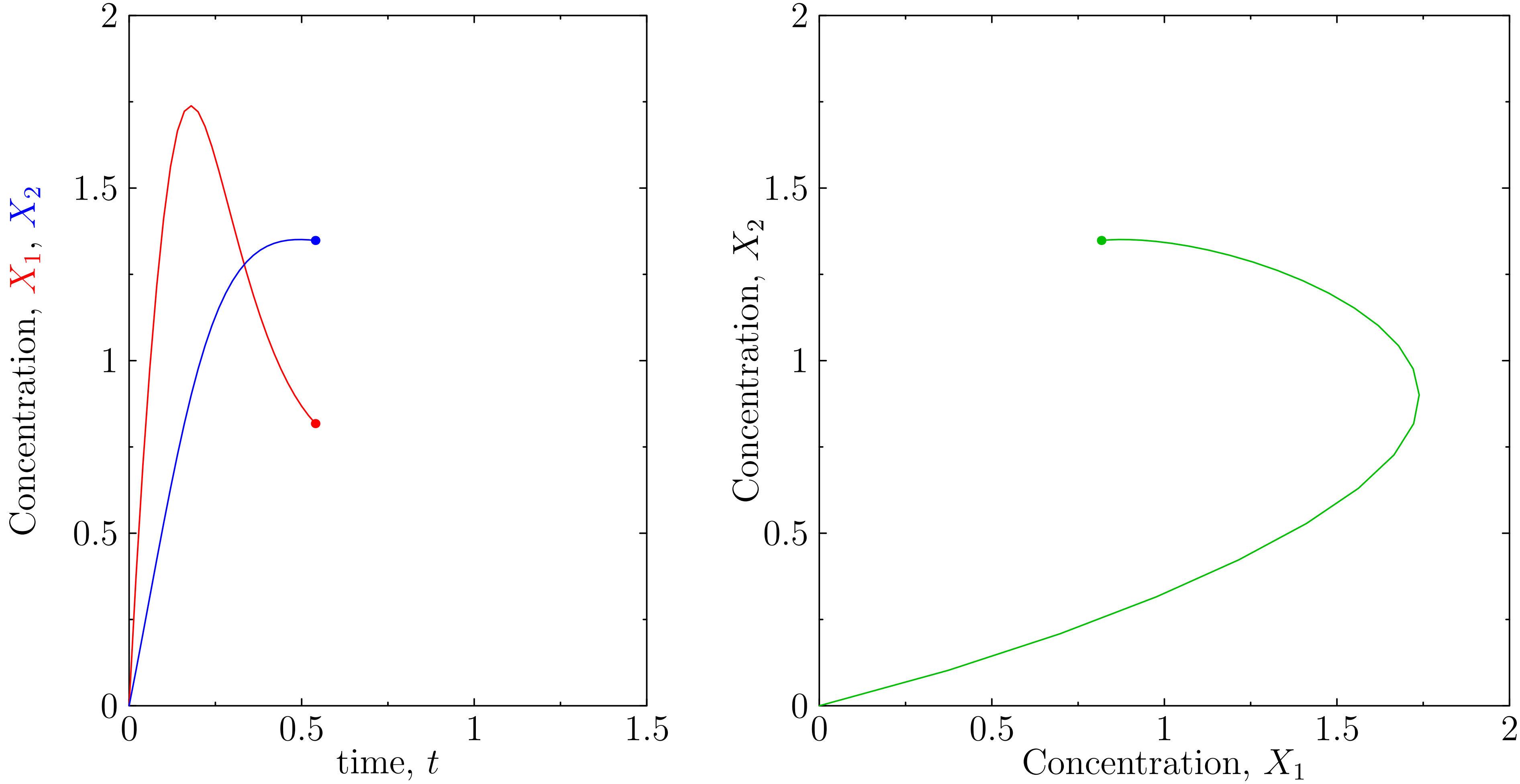
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



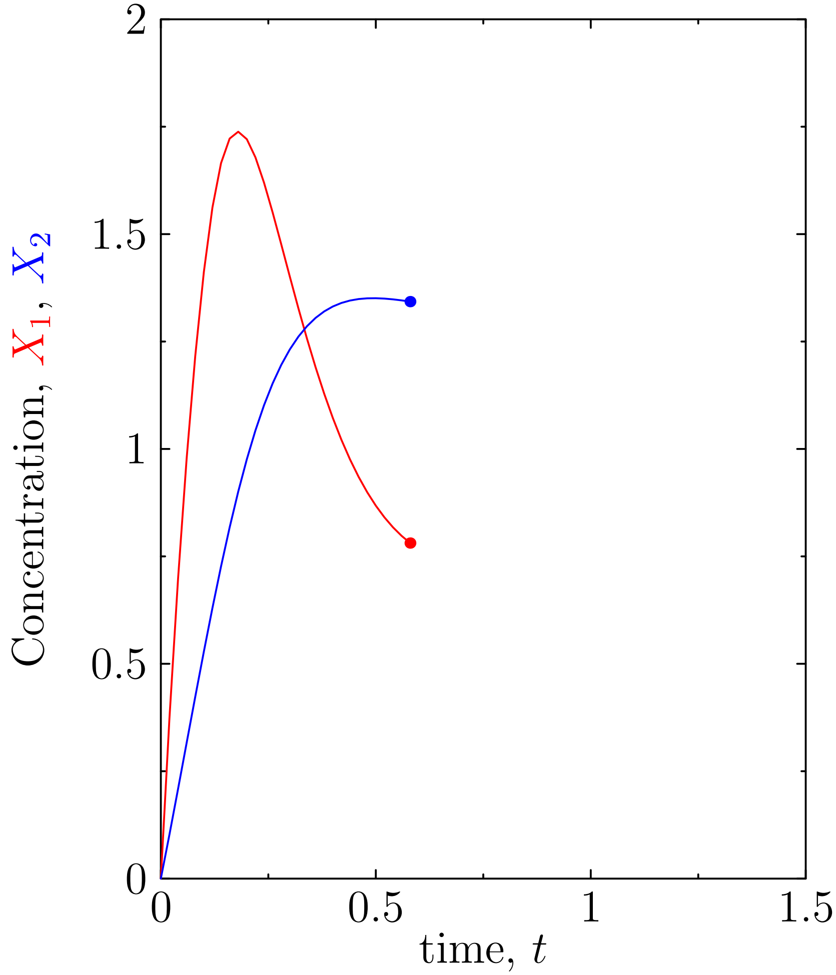
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

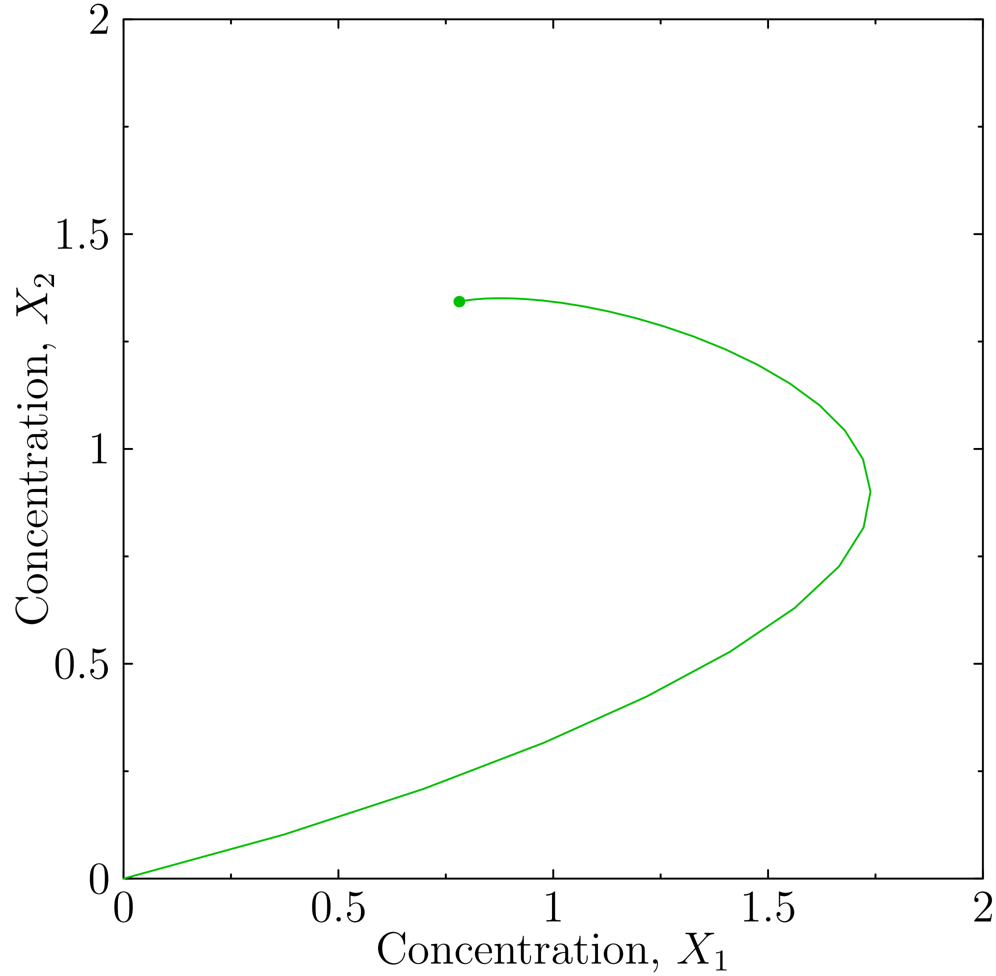
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

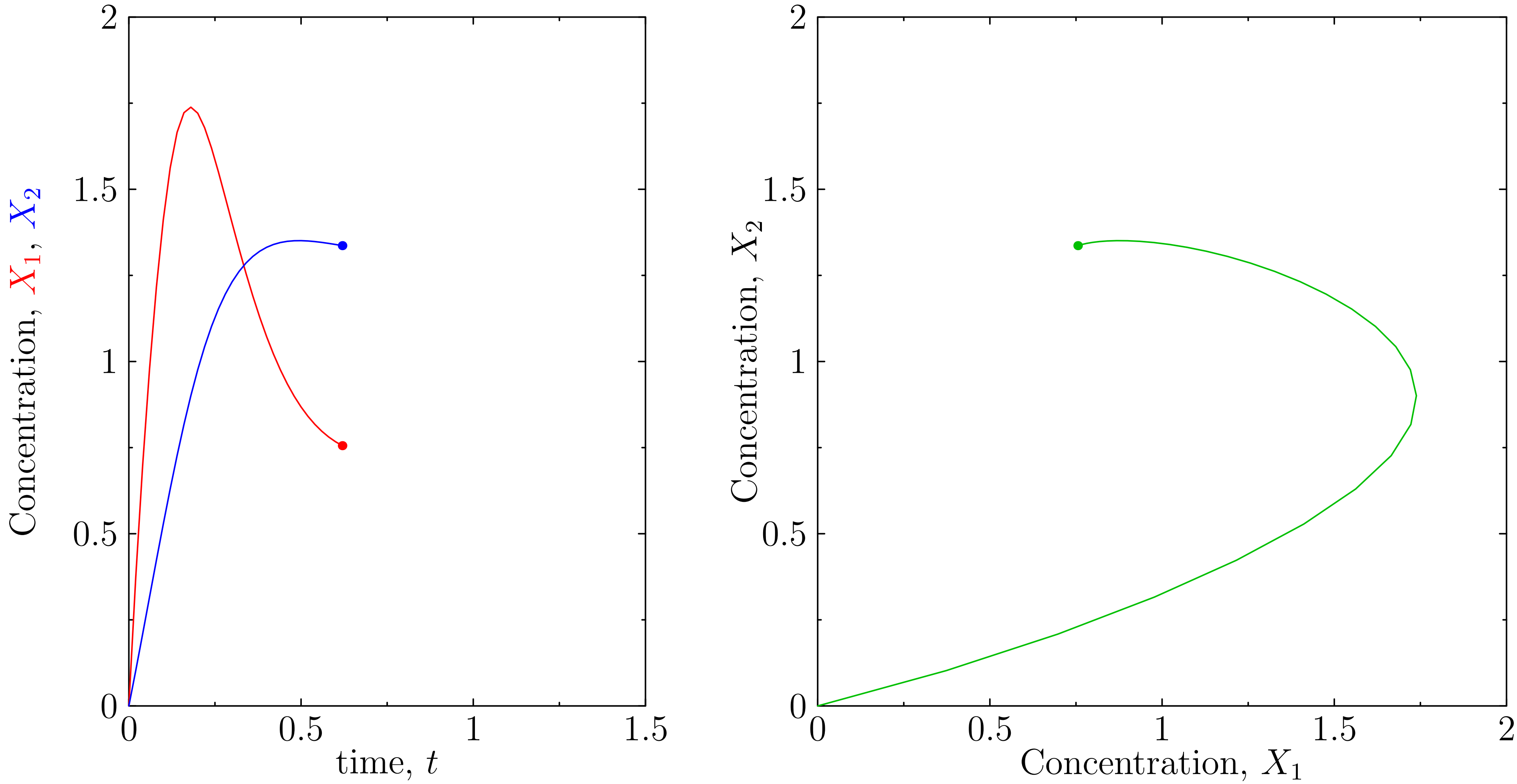
Solving the Dynamics





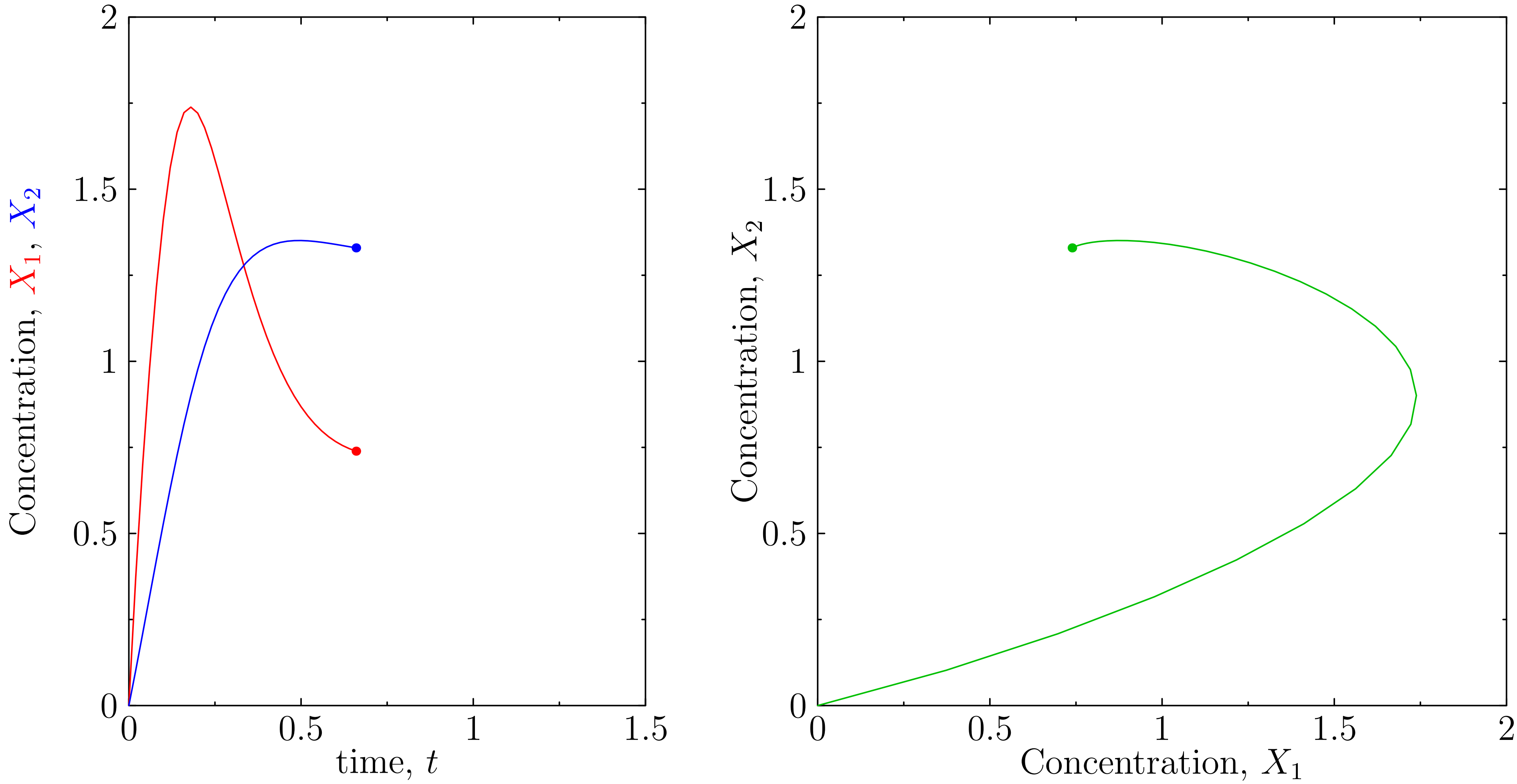
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



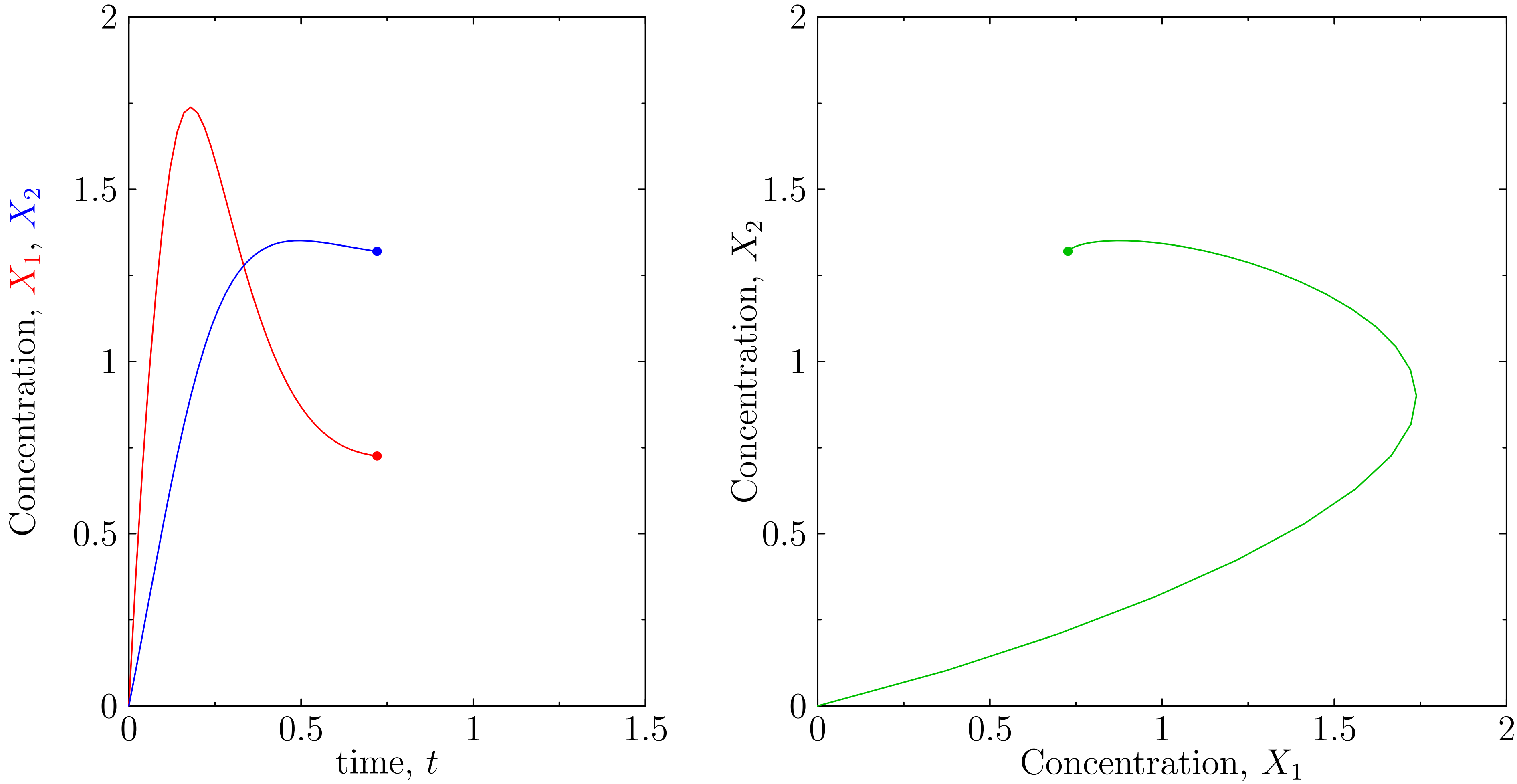
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



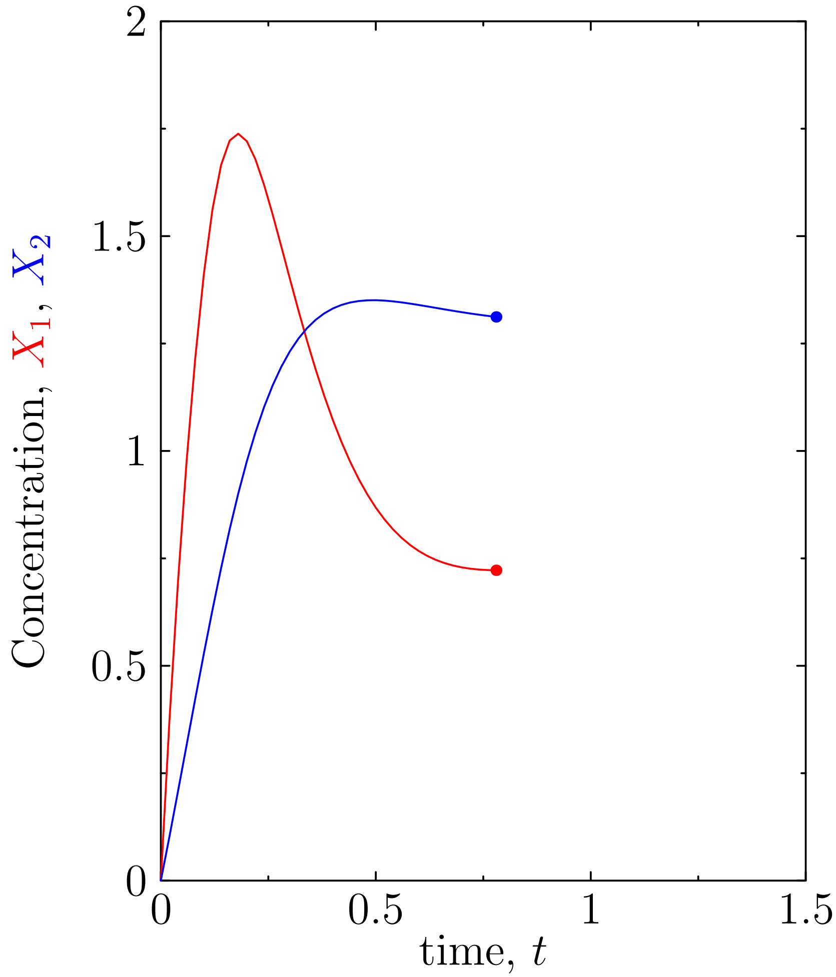
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

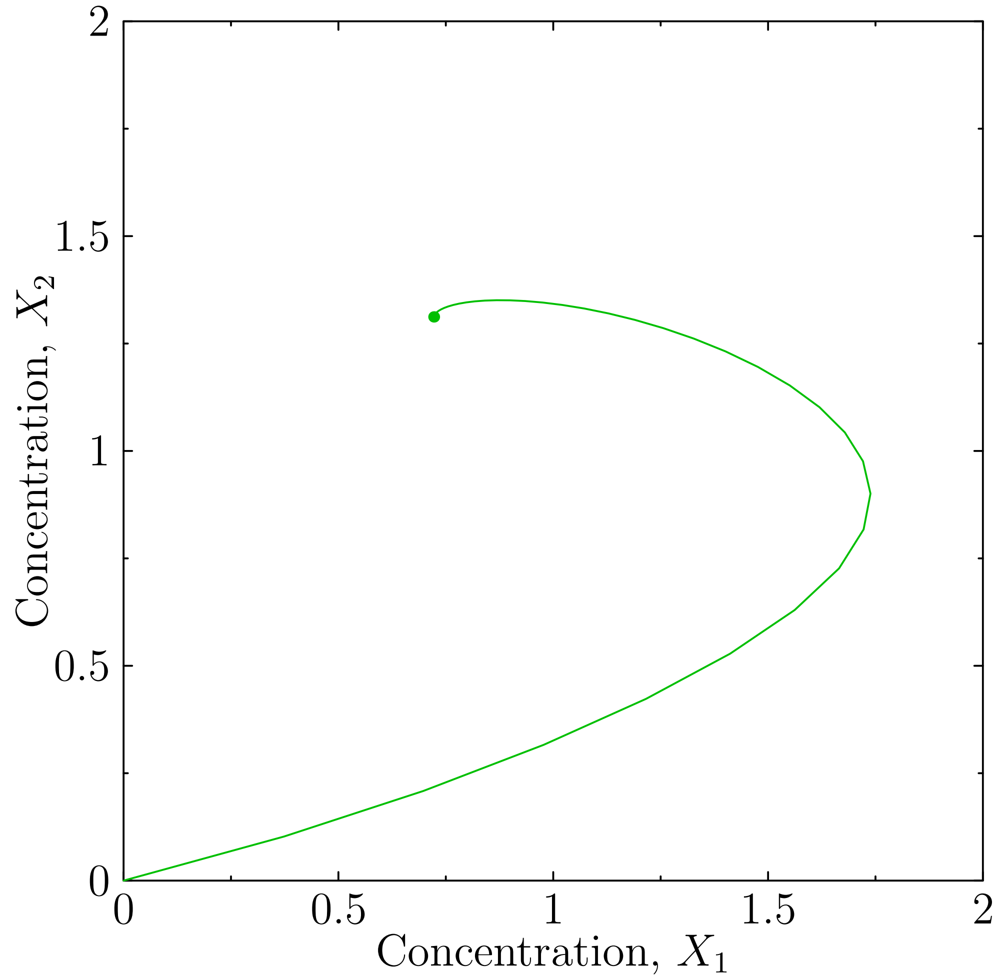
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

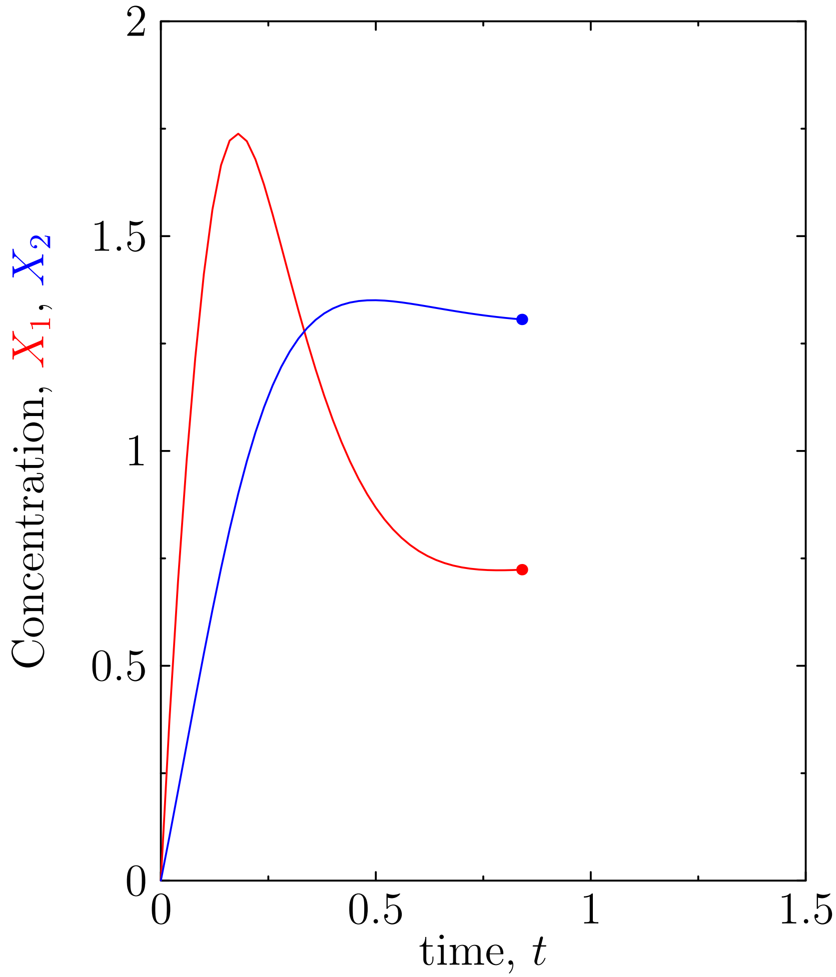
Solving the Dynamics

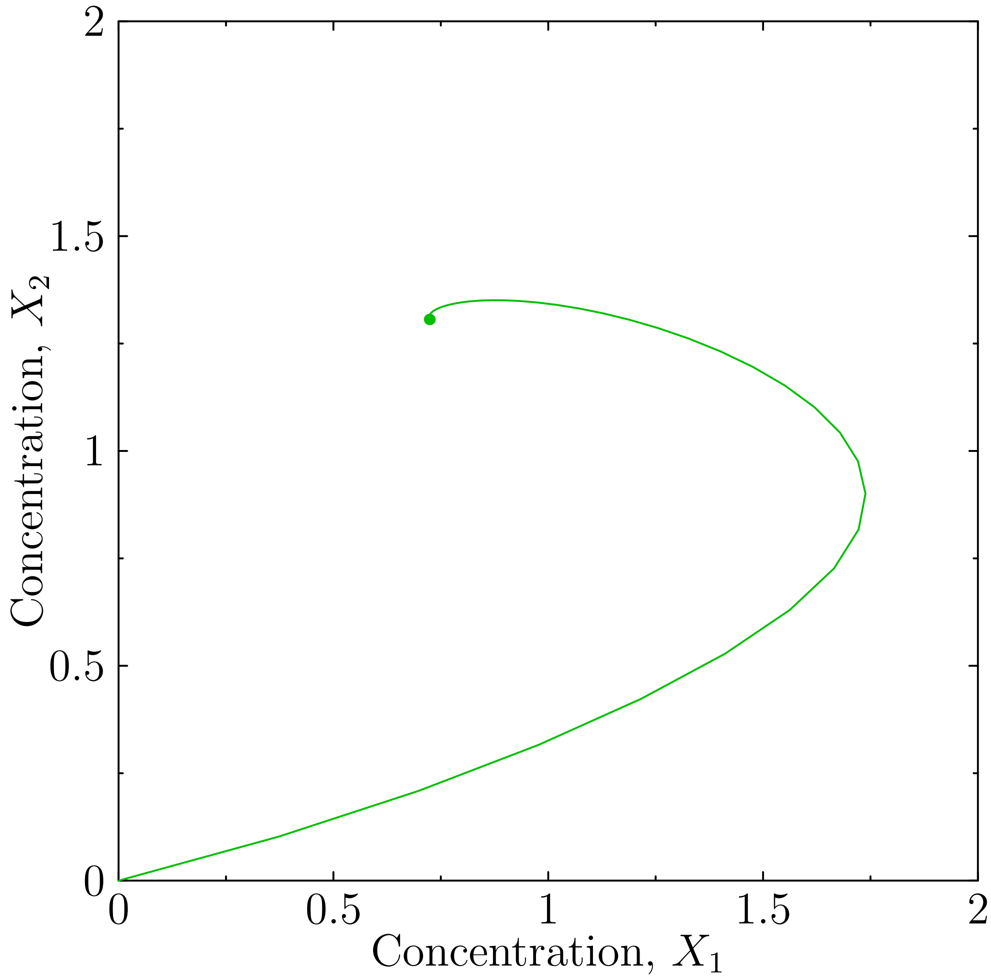




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

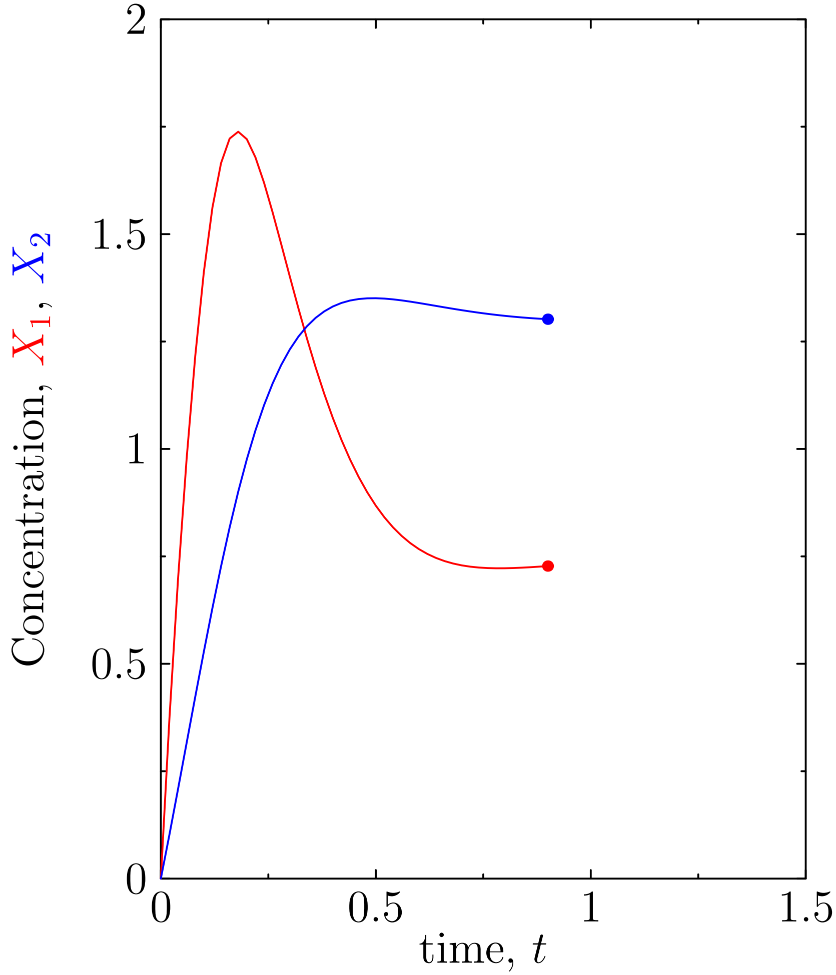
Solving the Dynamics

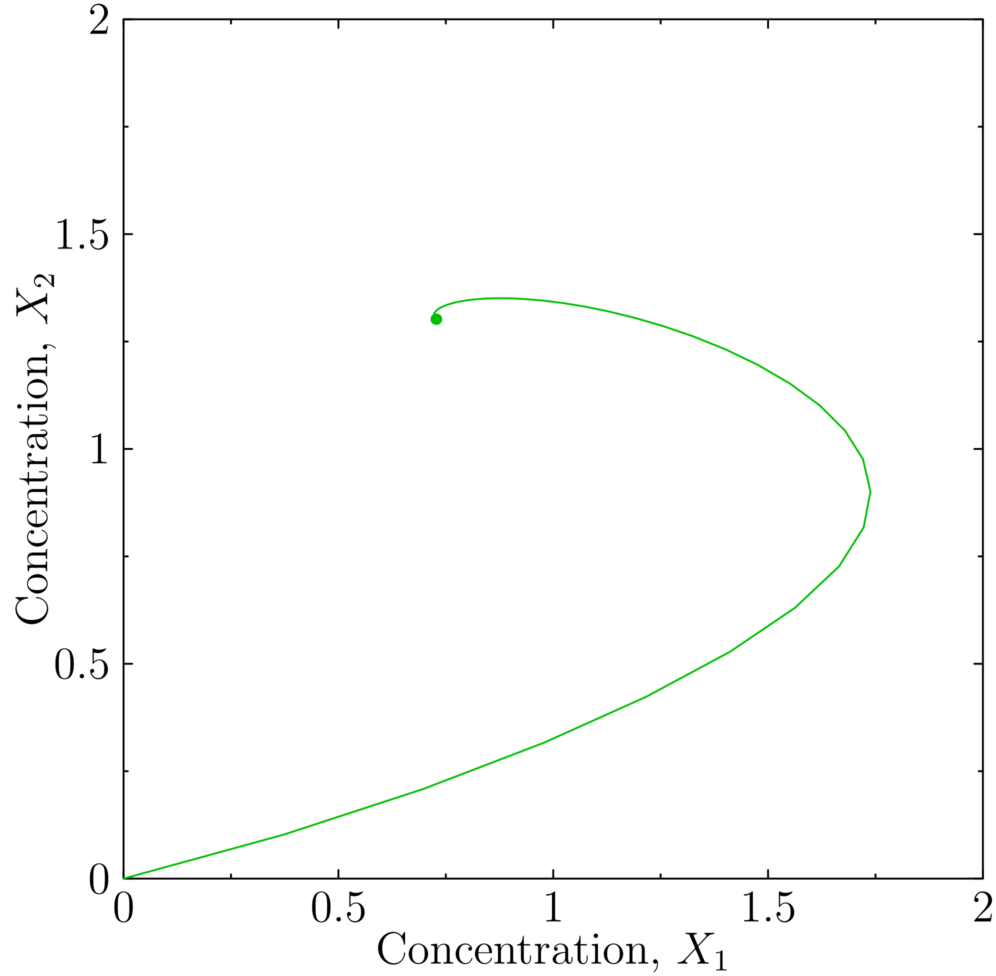




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

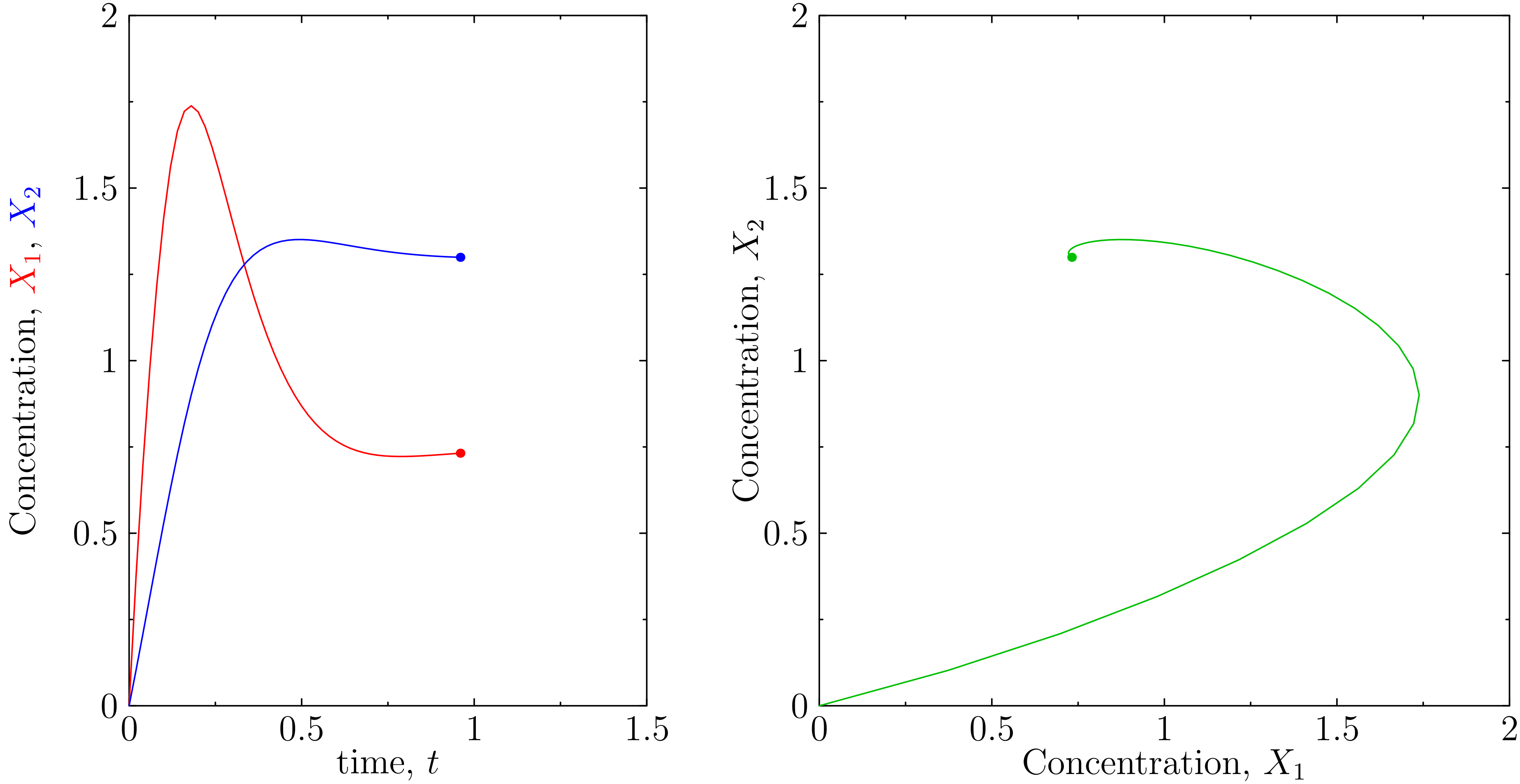
Solving the Dynamics





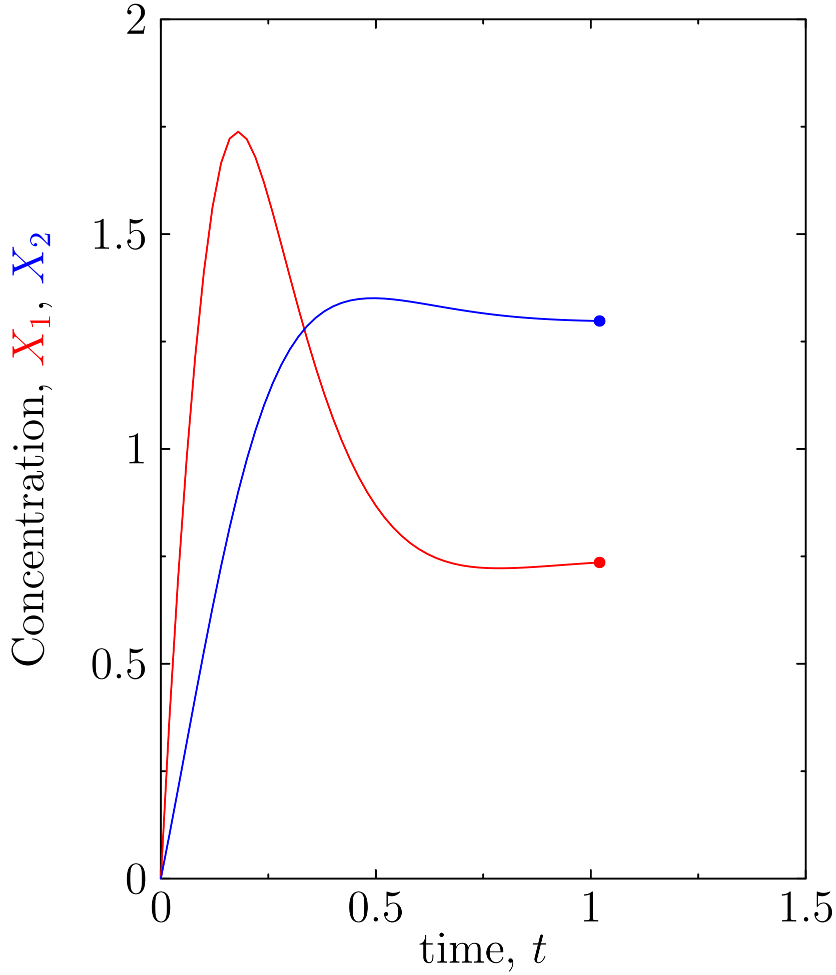
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

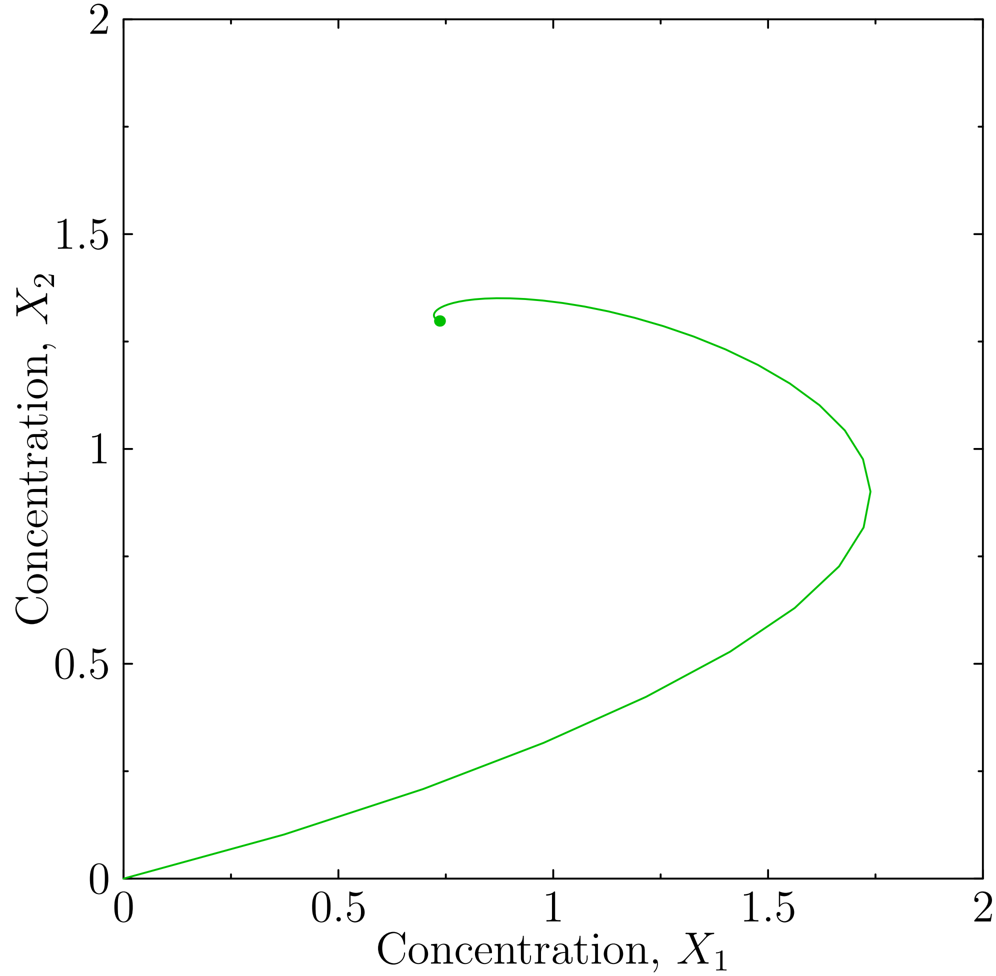
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

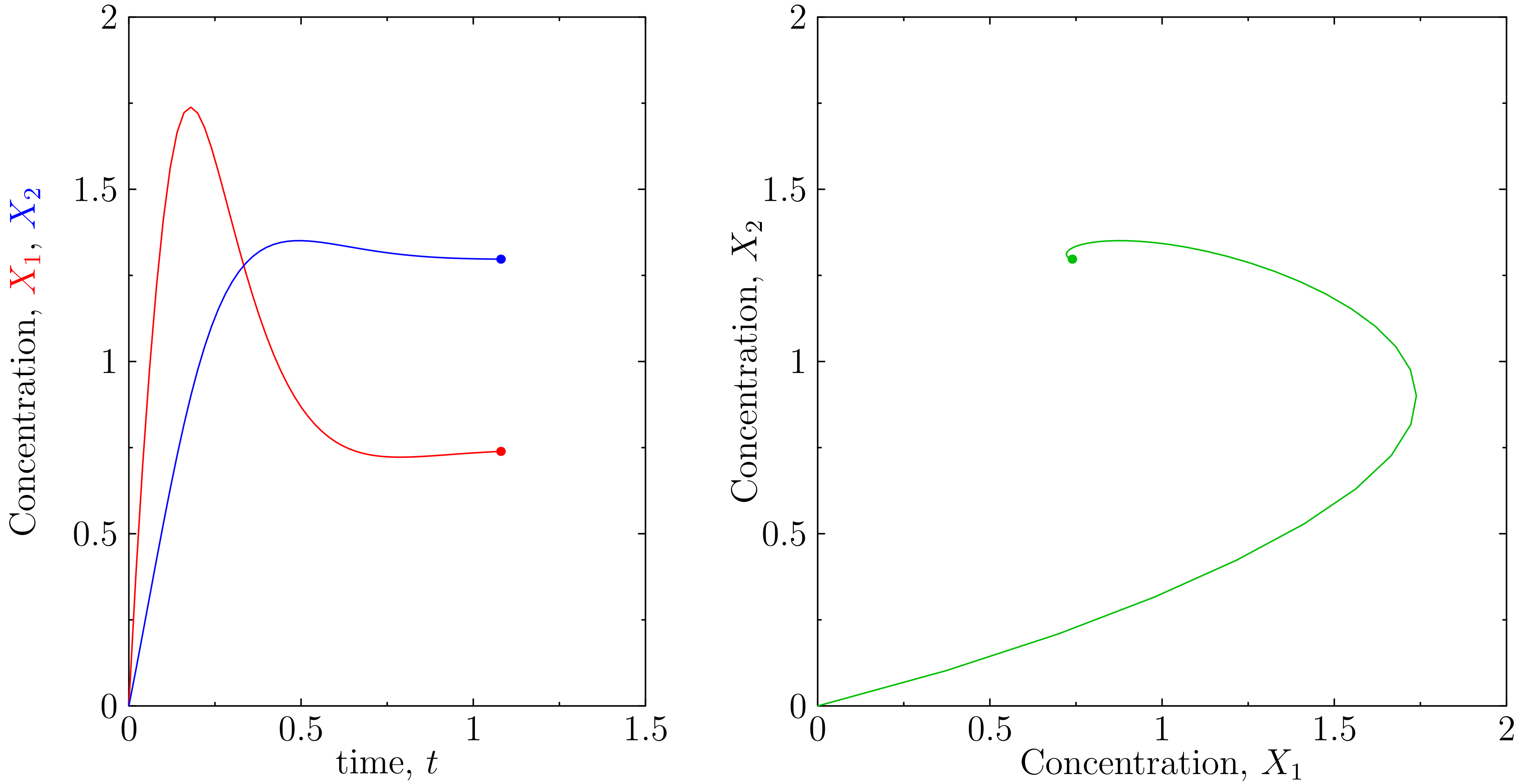
Solving the Dynamics





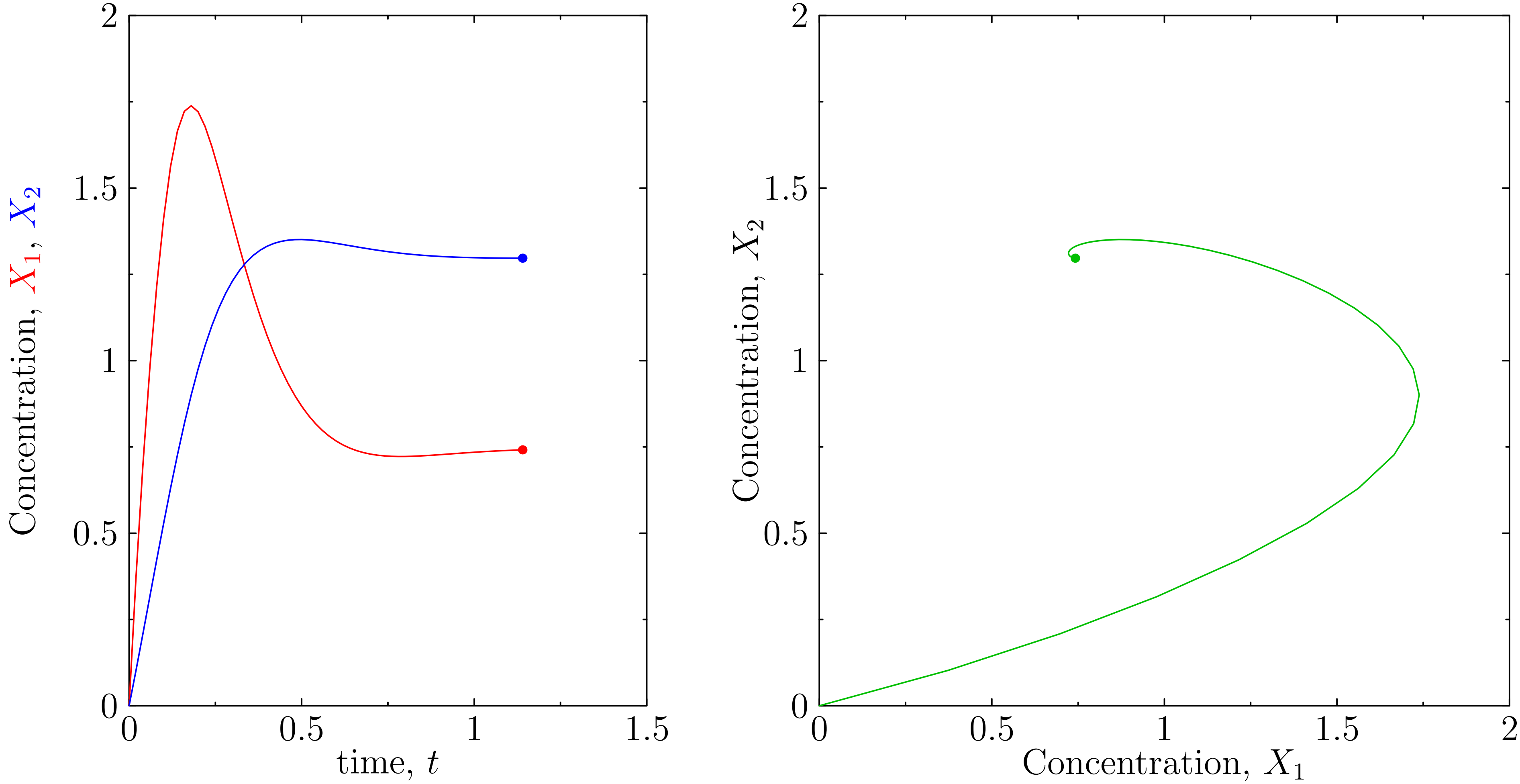
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



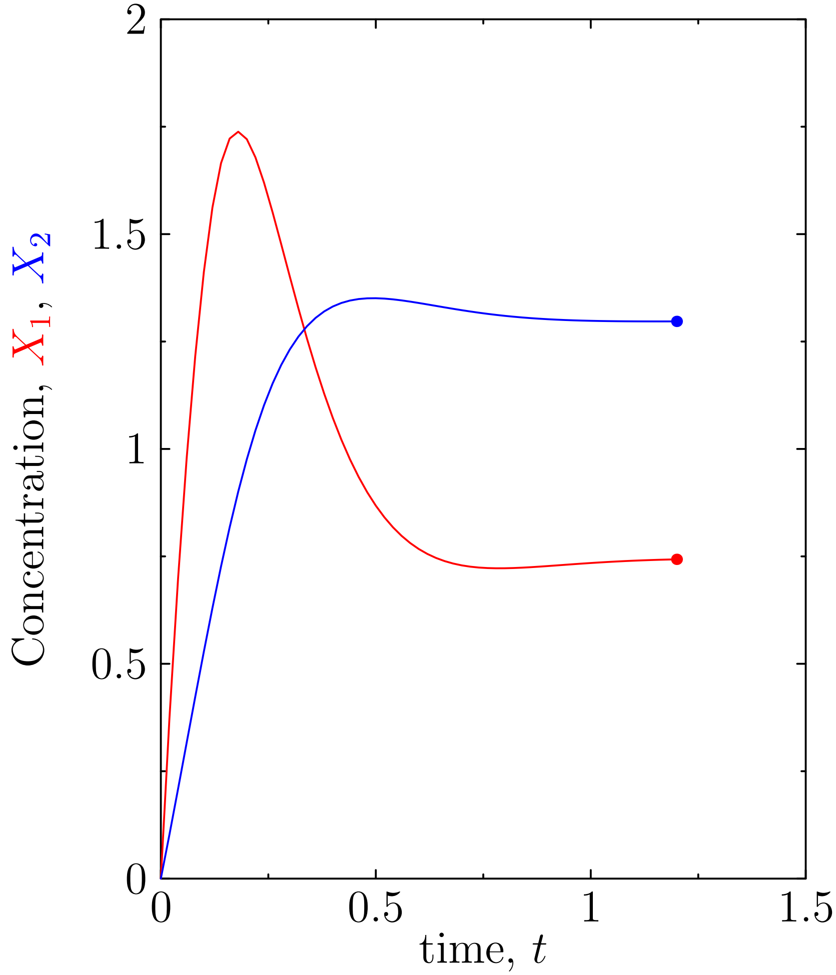
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

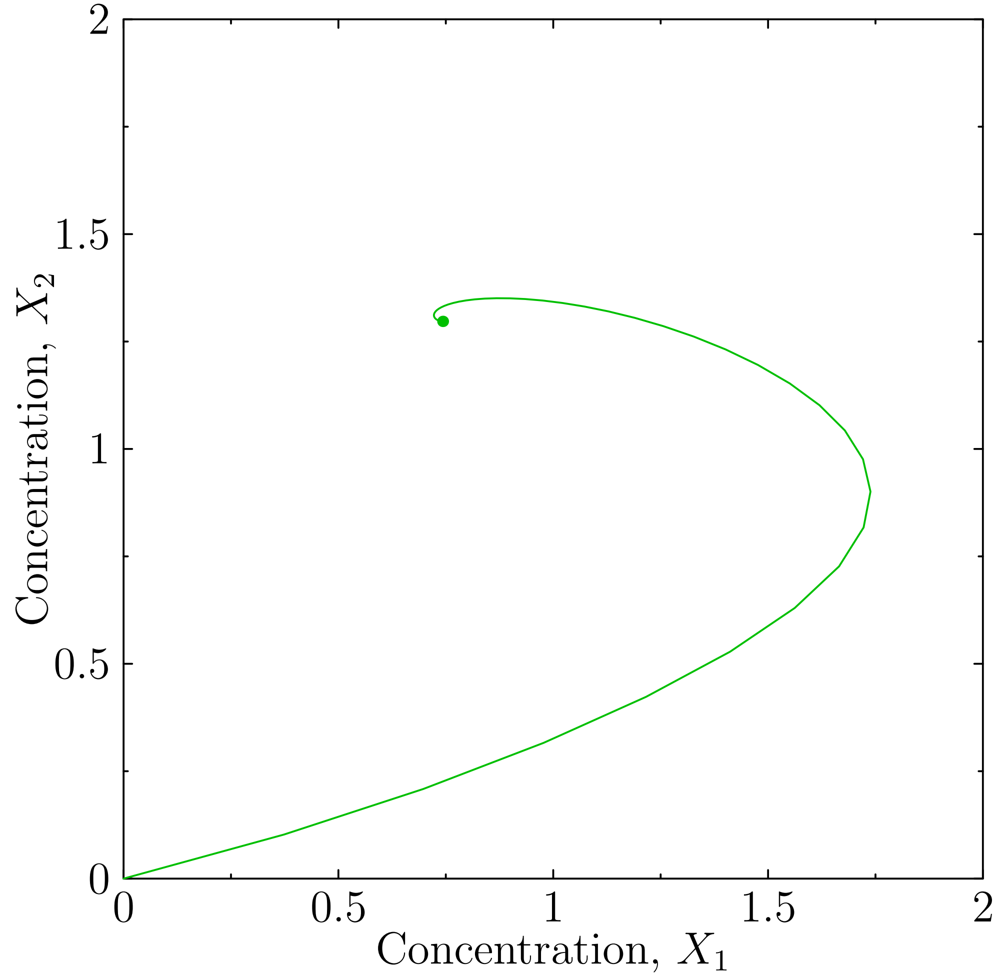
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

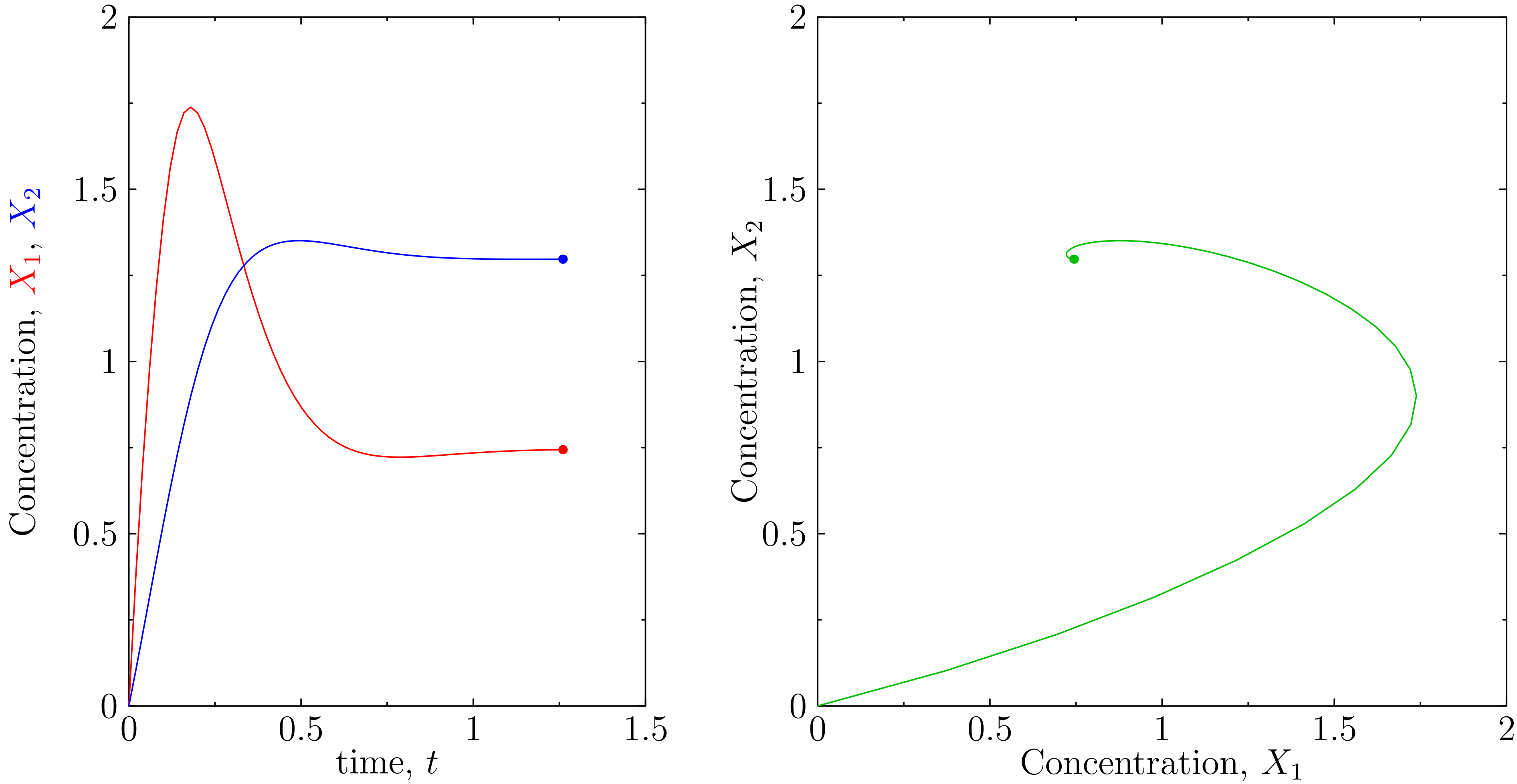
Solving the Dynamics





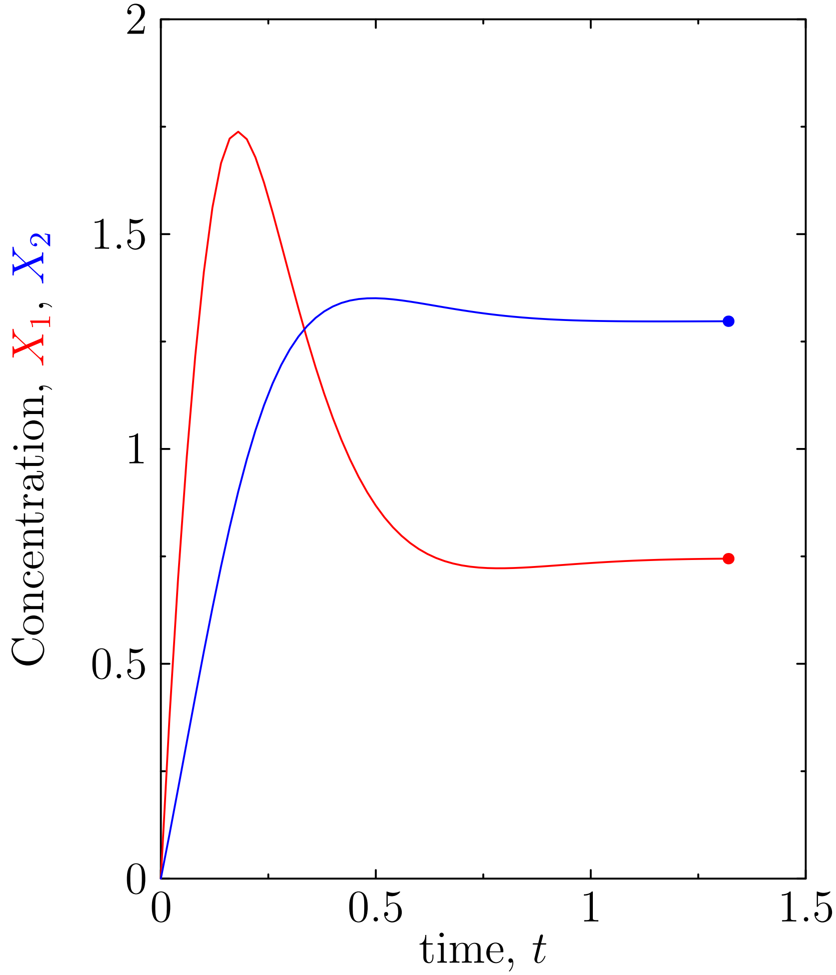
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

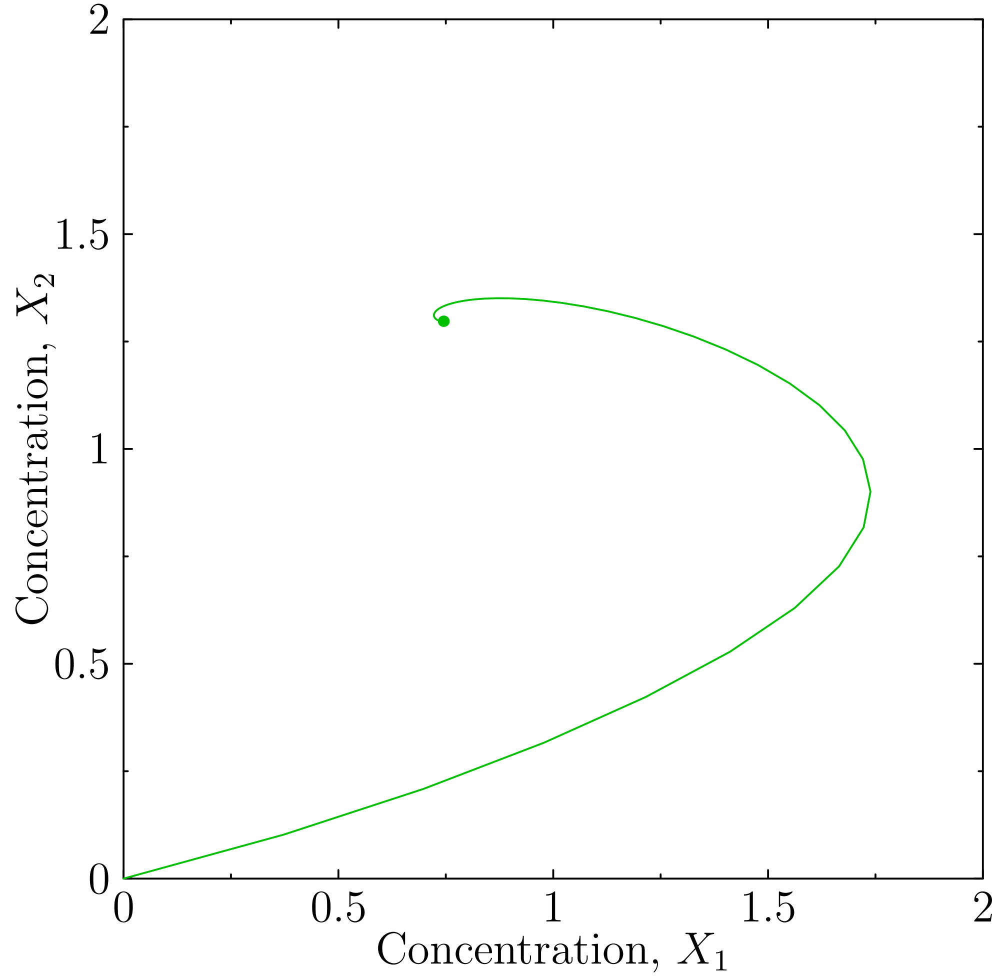
Solving the Dynamics



$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

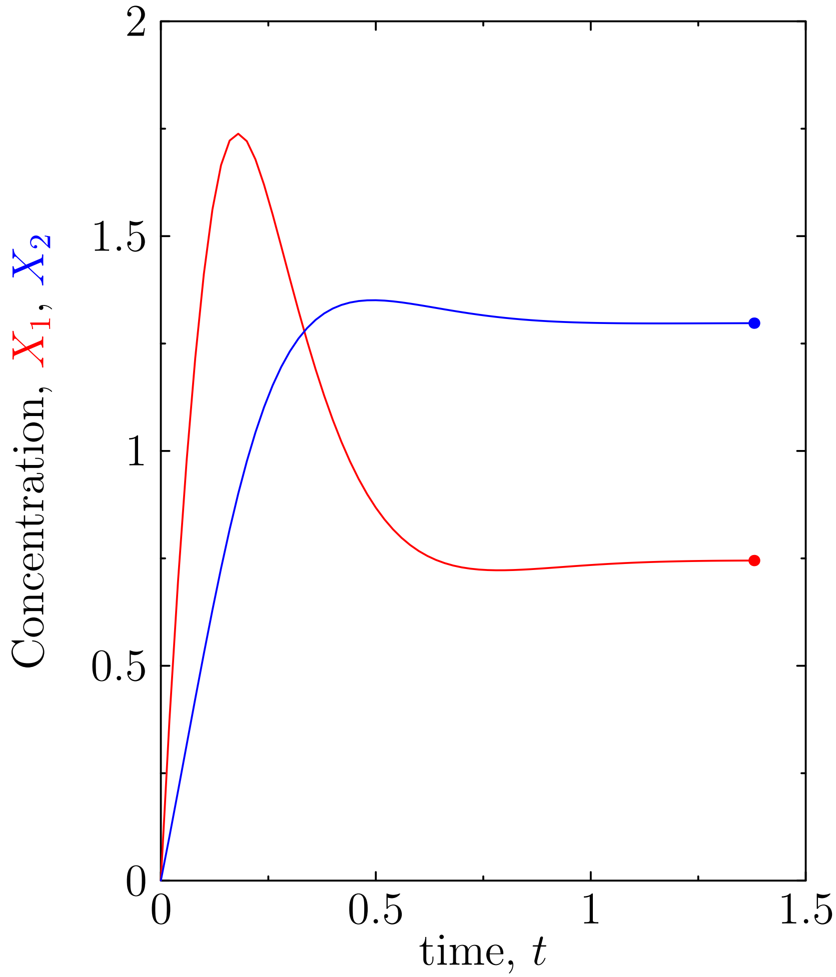
Solving the Dynamics

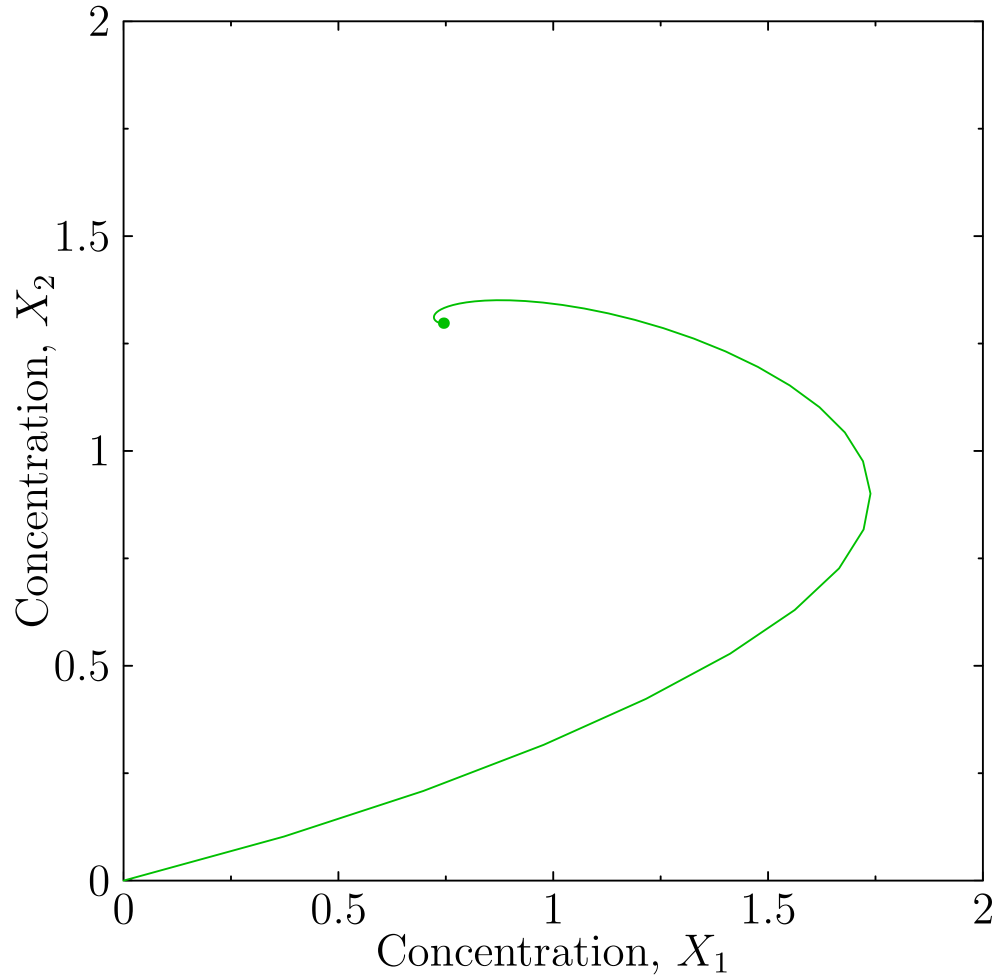




$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

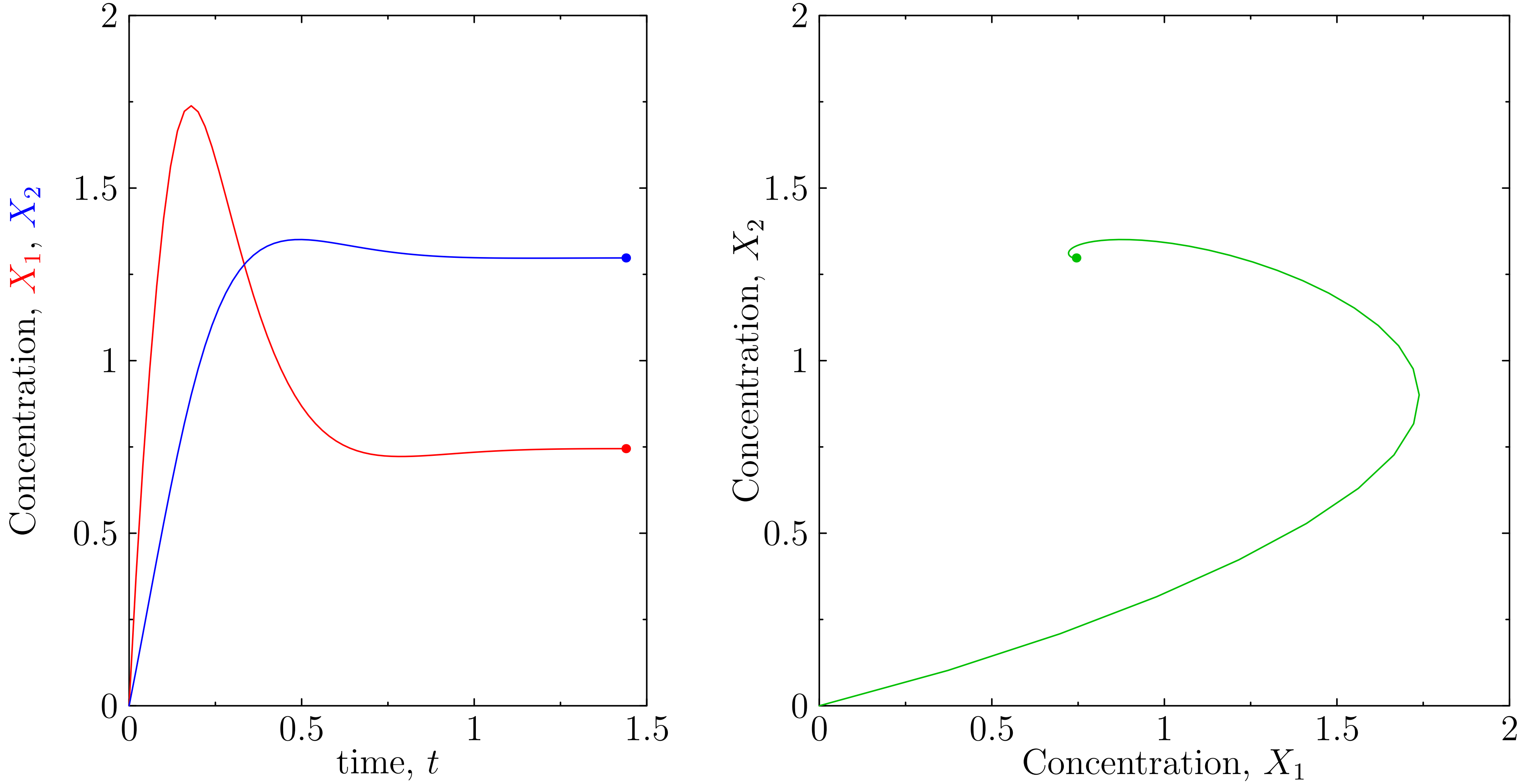
Solving the Dynamics





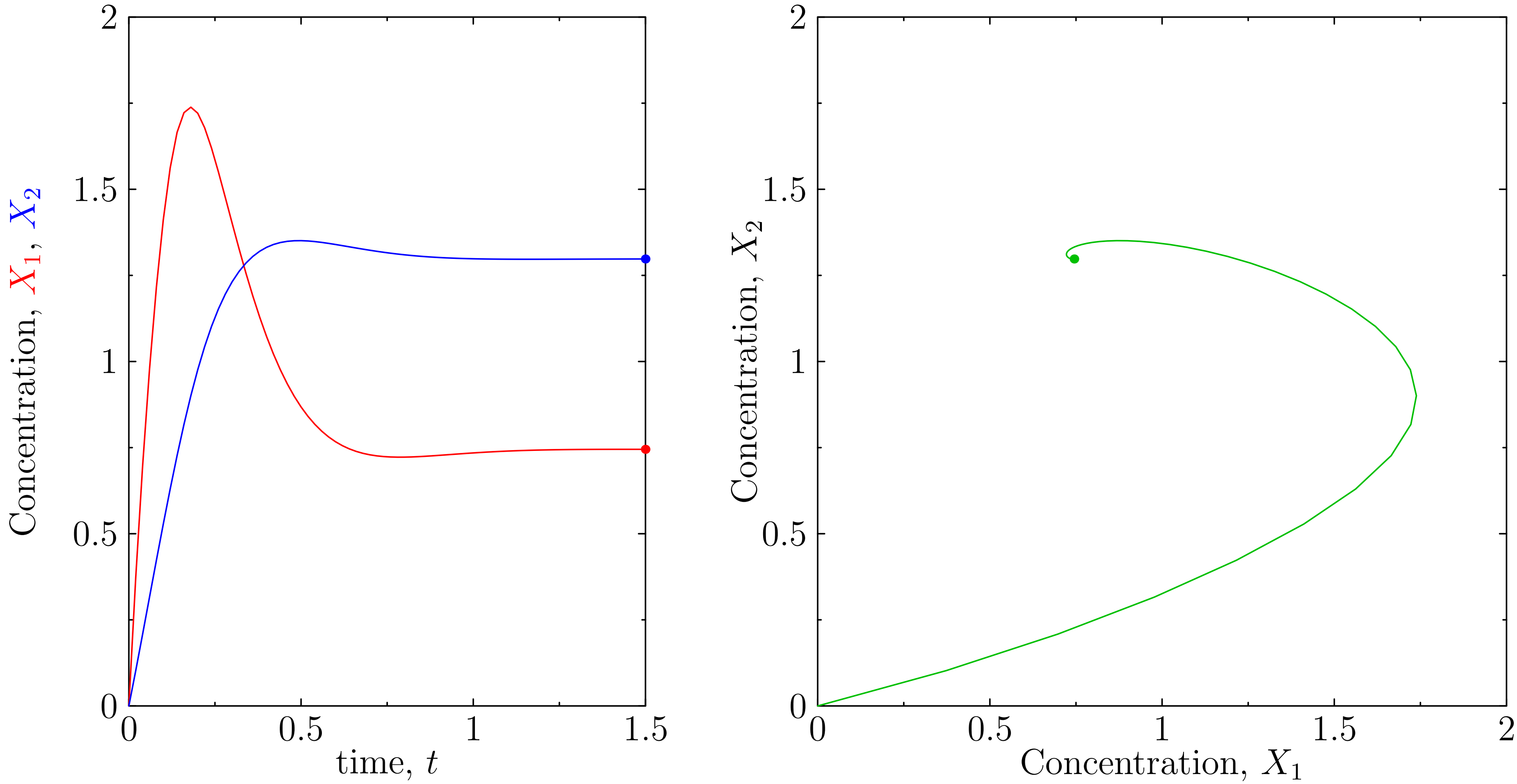
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics



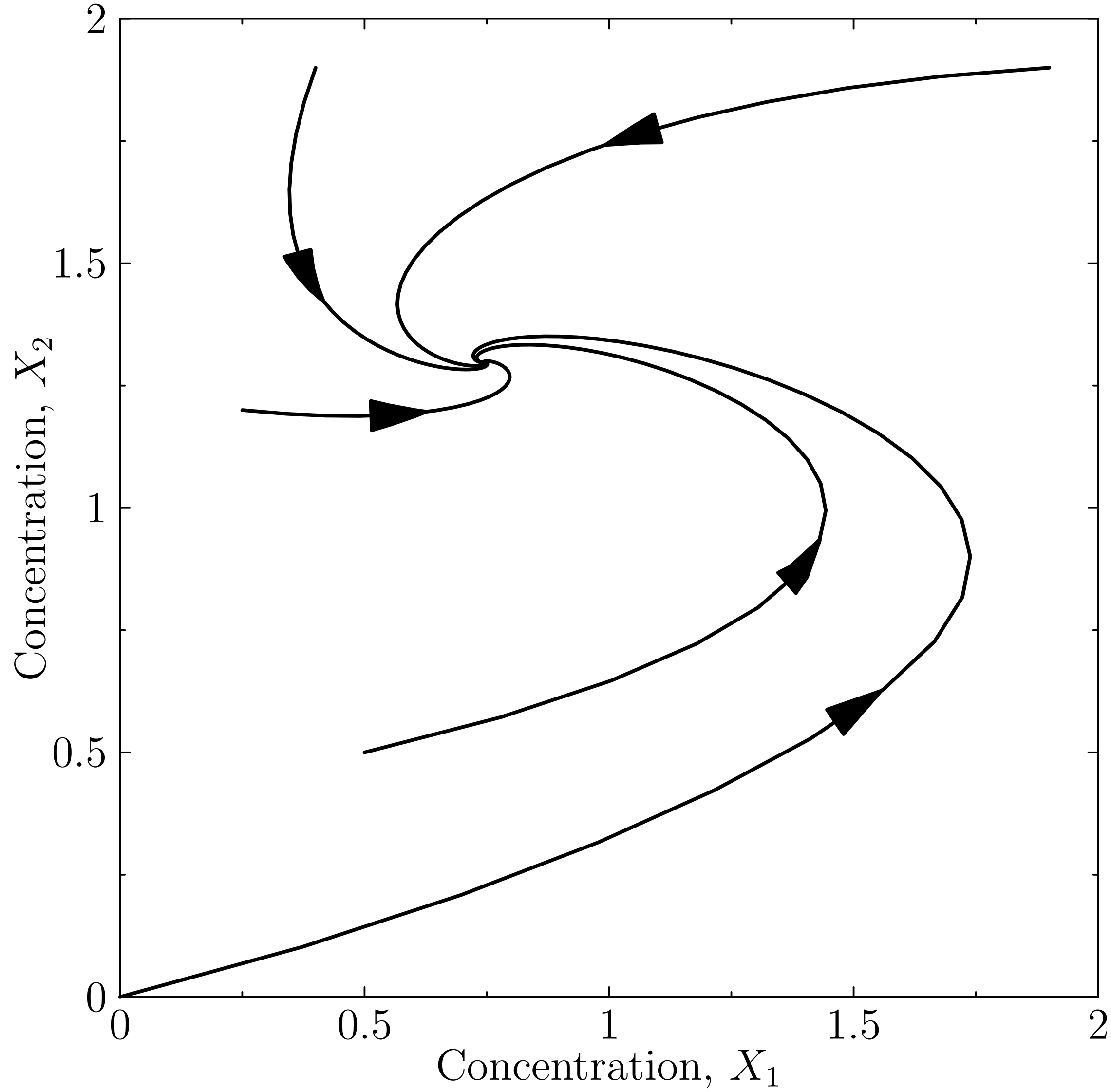
$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Solving the Dynamics

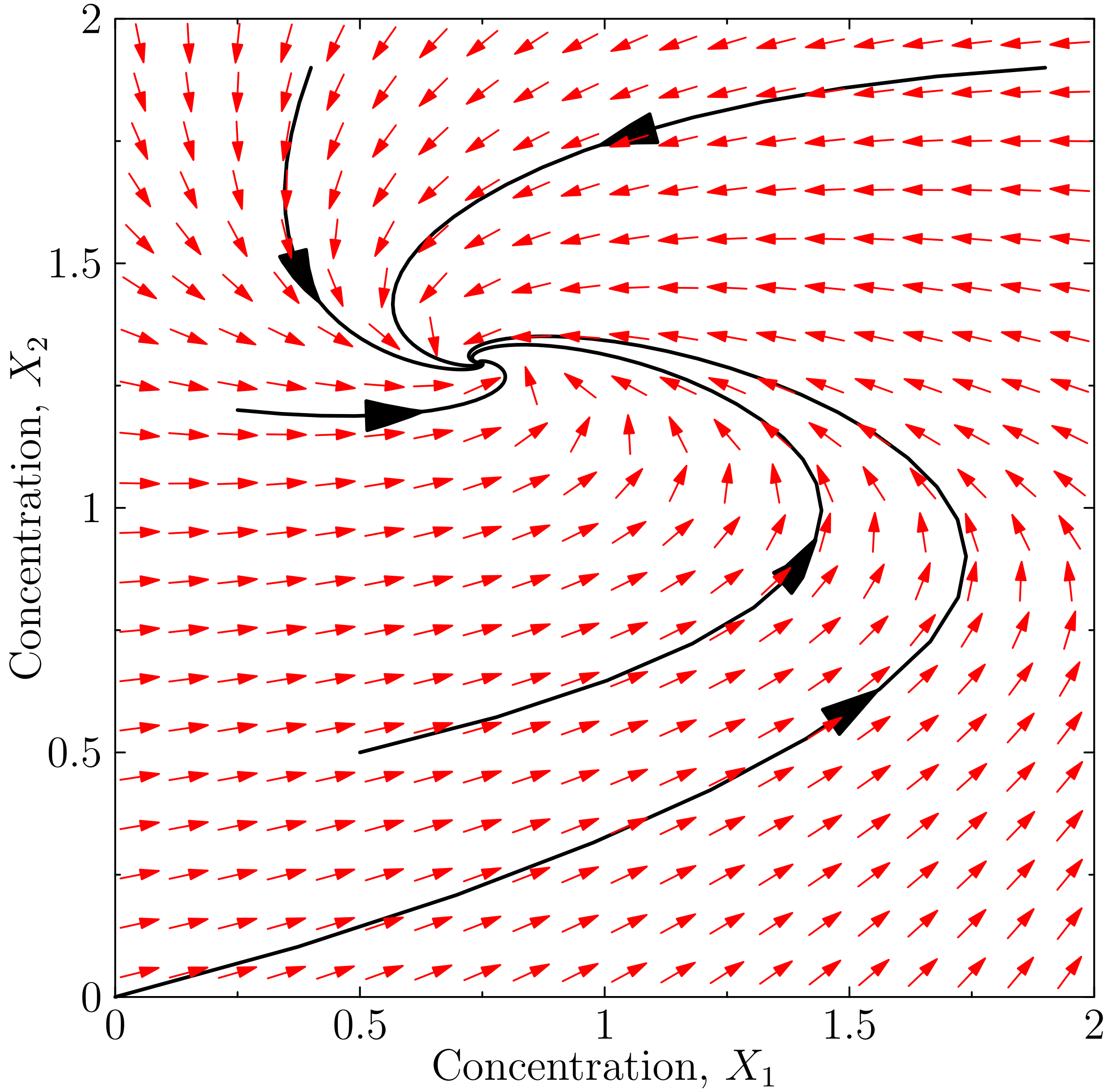


$$K = 1,\ k_1 = 20,\ k_2 = k_3 = k_4 = 5,\ k_5 = 2,\ n = 4$$

Trajectories



Trajectories



Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$

- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1+(X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \quad \Rightarrow \quad X_1 = \frac{k_1}{(k_3+k_5)(1+(X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \quad \Rightarrow \quad X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

- The intersection of the nullclines defines the stationary points

Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$

- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1+(X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \quad \Rightarrow \quad X_1 = \frac{k_1}{(k_3+k_5)(1+(X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \quad \Rightarrow \quad X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

- The intersection of the nullclines defines the stationary points

Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$

- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1+(X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \quad \Rightarrow \quad X_1 = \frac{k_1}{(k_3+k_5)(1+(X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \quad \Rightarrow \quad X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

- The intersection of the nullclines defines the stationary points

Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $\mathrm{d}X_i/\mathrm{d}t = 0$

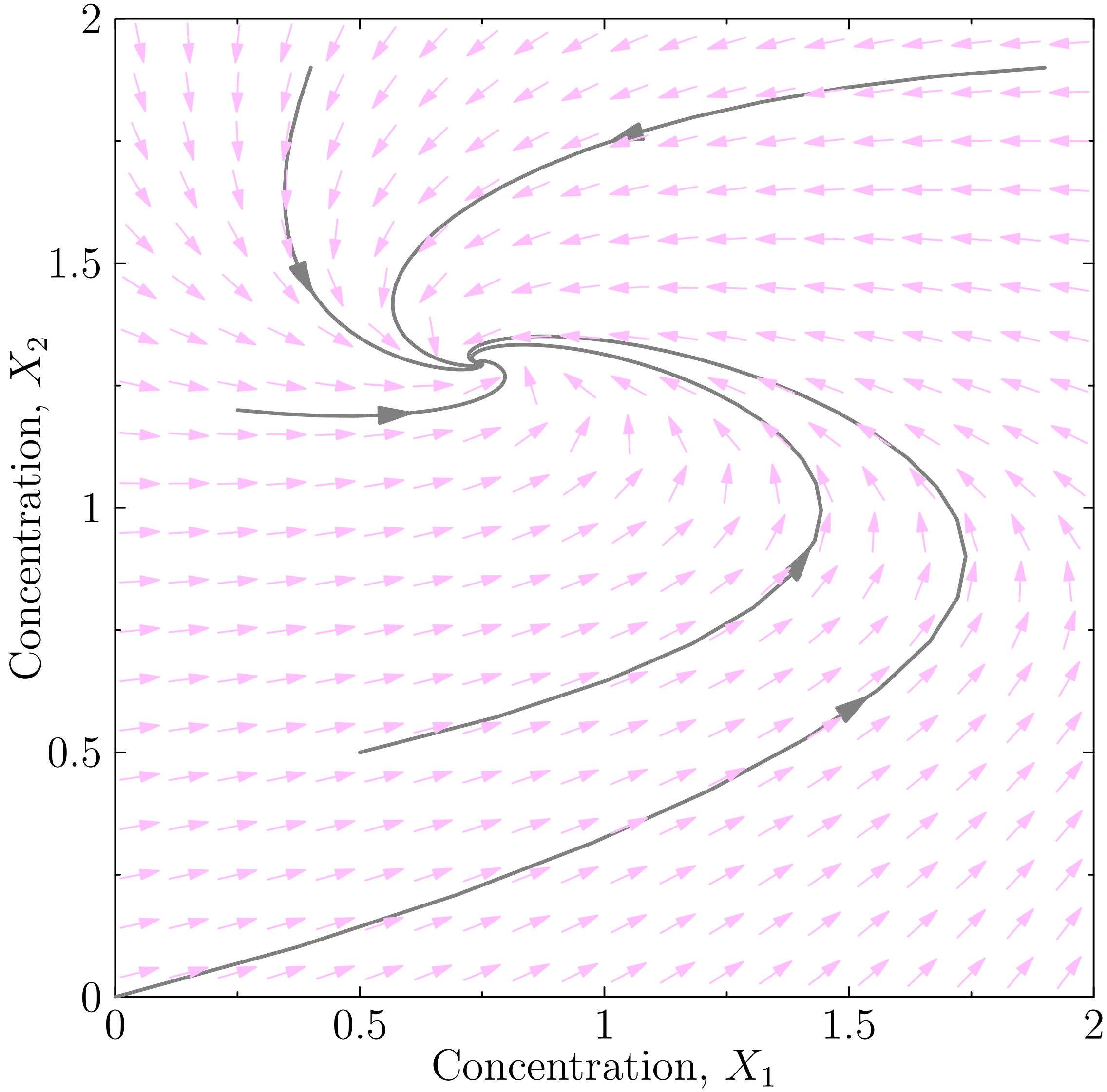
- In our system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1 + (X_2/K)^n} - k_3\, X_1 - k_5\, X_1 = 0 \quad \Rightarrow \quad X_1 = \frac{k_1}{(k_3 + k_5)(1 + (X_2/K)^n)}$$

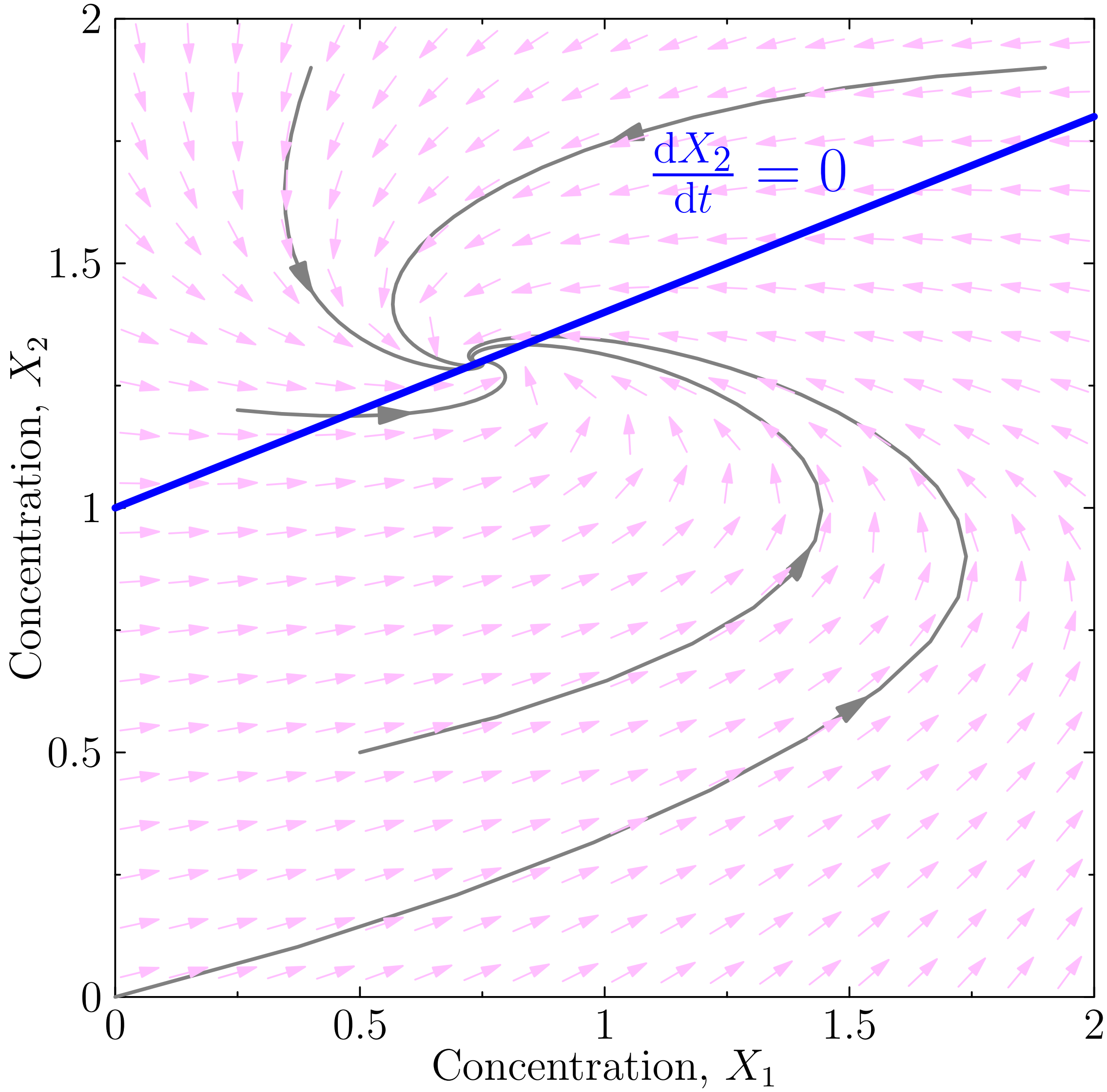
$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\, X_1 - k_4\, X_2 = 0 \quad \Rightarrow \quad X_2 = \frac{k_2 + k_5\, X_1}{k_2}$$

- The intersection of the nullclines defines the stationary points

Nullclines

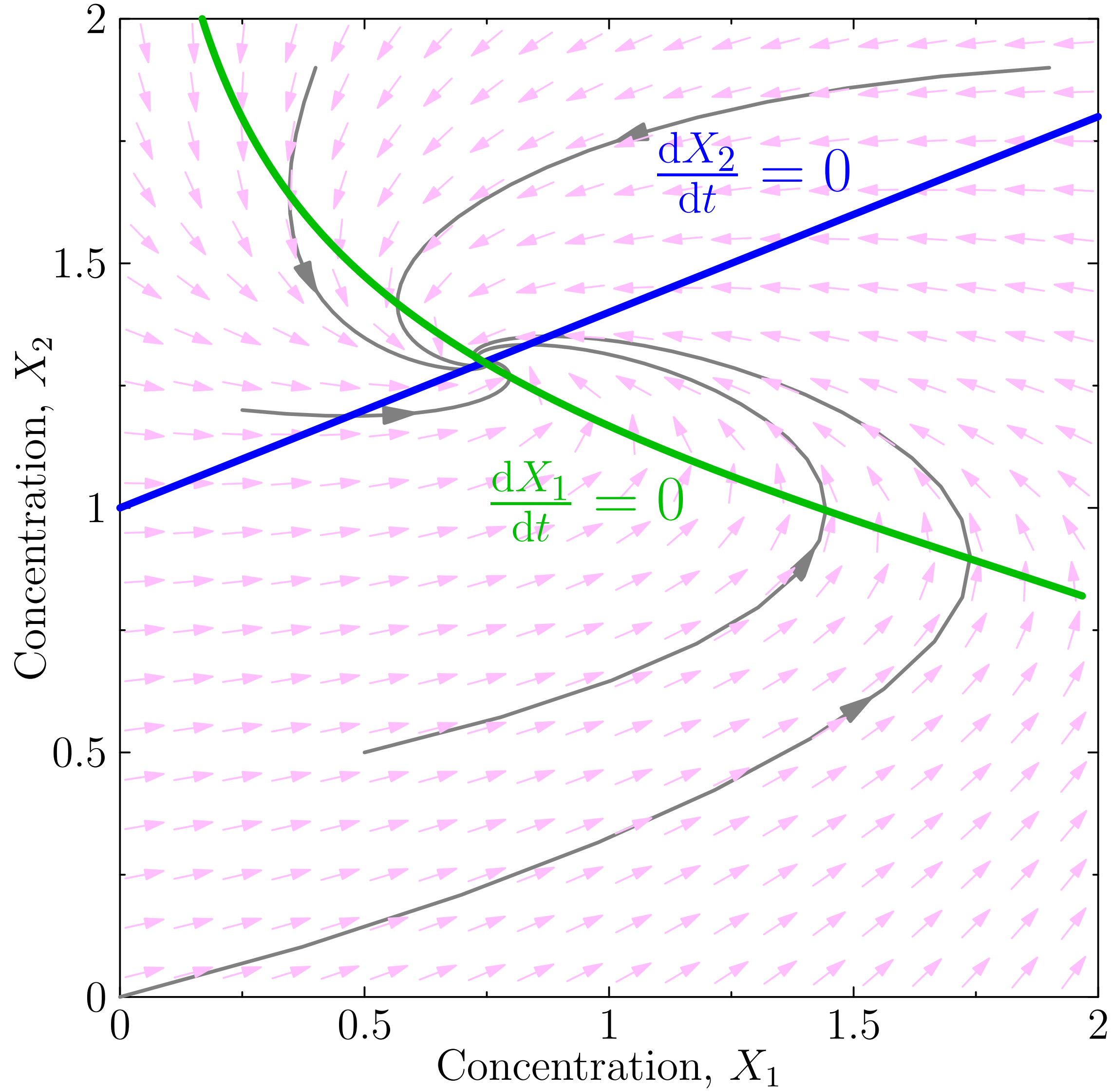


Nullclines



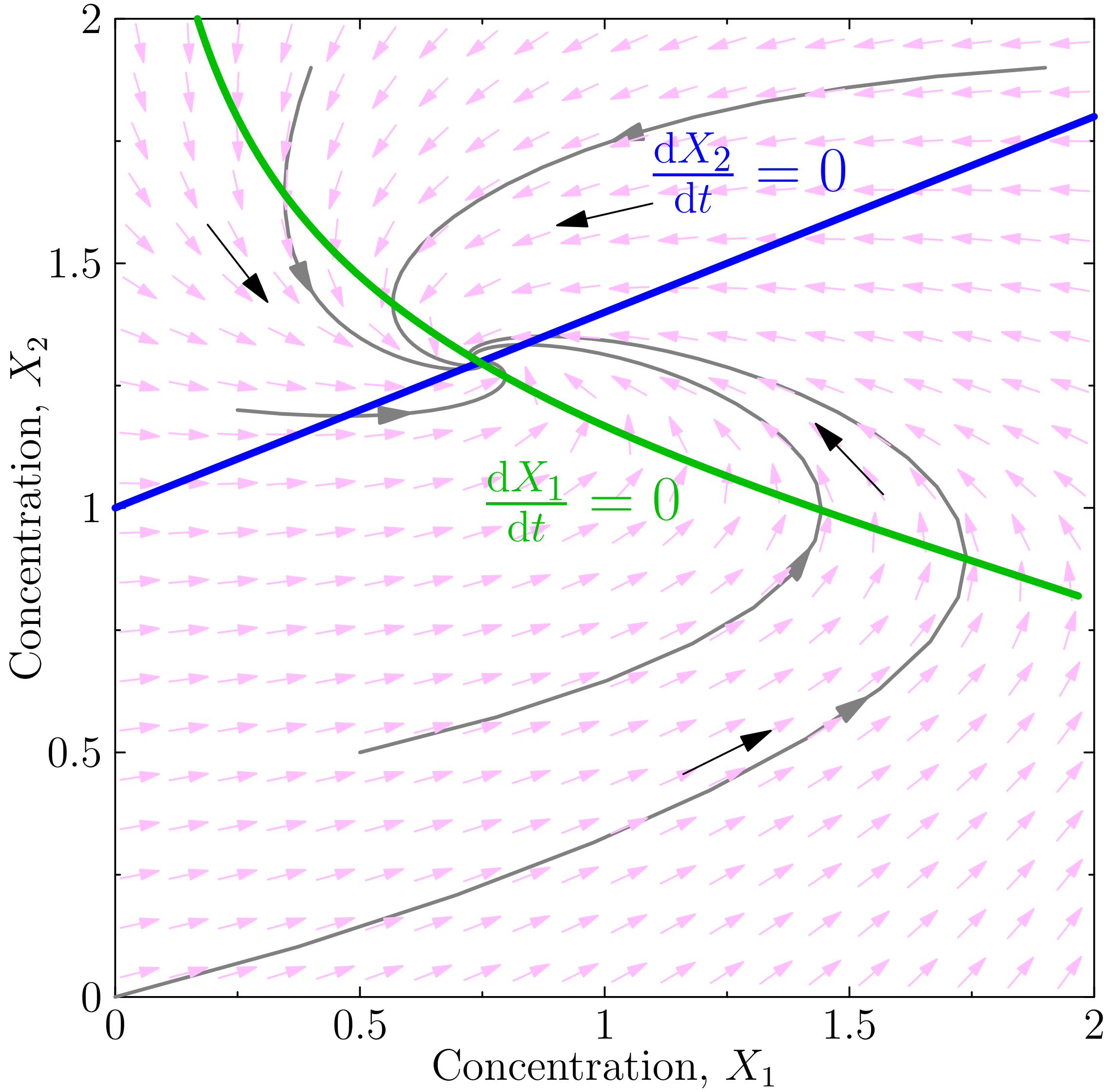
$\frac{\mathrm{d}X_2}{\mathrm{d}t} = 0$
Concentration, X_2
Concentration, X_1
0
0.5
1
1.5
2

Nullclines



$\frac{dX_2}{dt} = 0$
$\frac{dX_1}{dt} = 0$
Concentration, X_2
Concentration, X_1
0
0.5
1
1.5
2
0
0.5
1
1.5
2

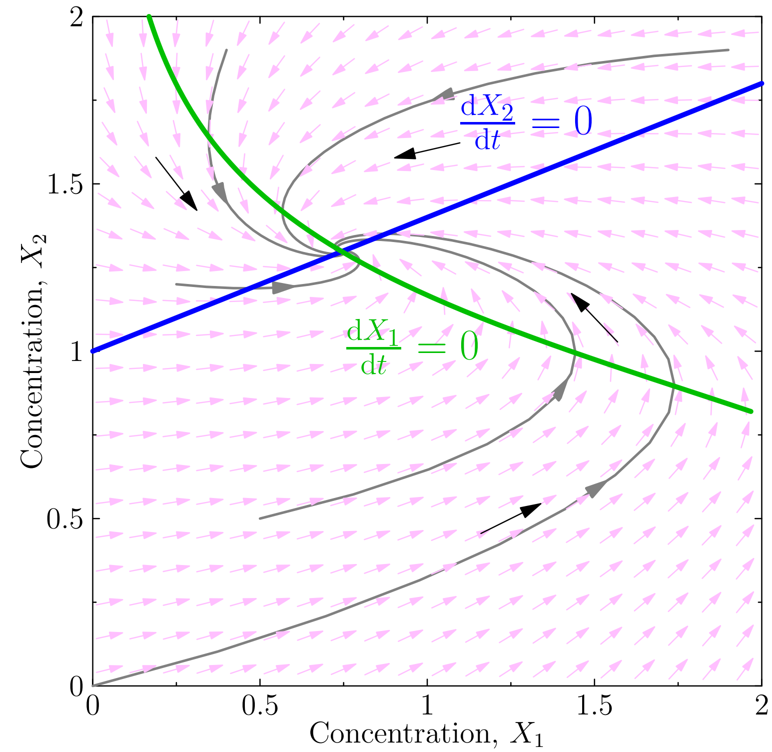
Nullclines



$\frac{dX_2}{dt} = 0$
$\frac{dX_1}{dt} = 0$
Concentration, X_1
Concentration, X_2
0
0.5
1
1.5
2

Outline

1. Understanding ODEs
2. **Stability**
3. Bifications



Bistability

- Let's consider a slightly different system where X_1 and X_2 inhibits each others' expression

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}) = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}) = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ we end up with three fixed points

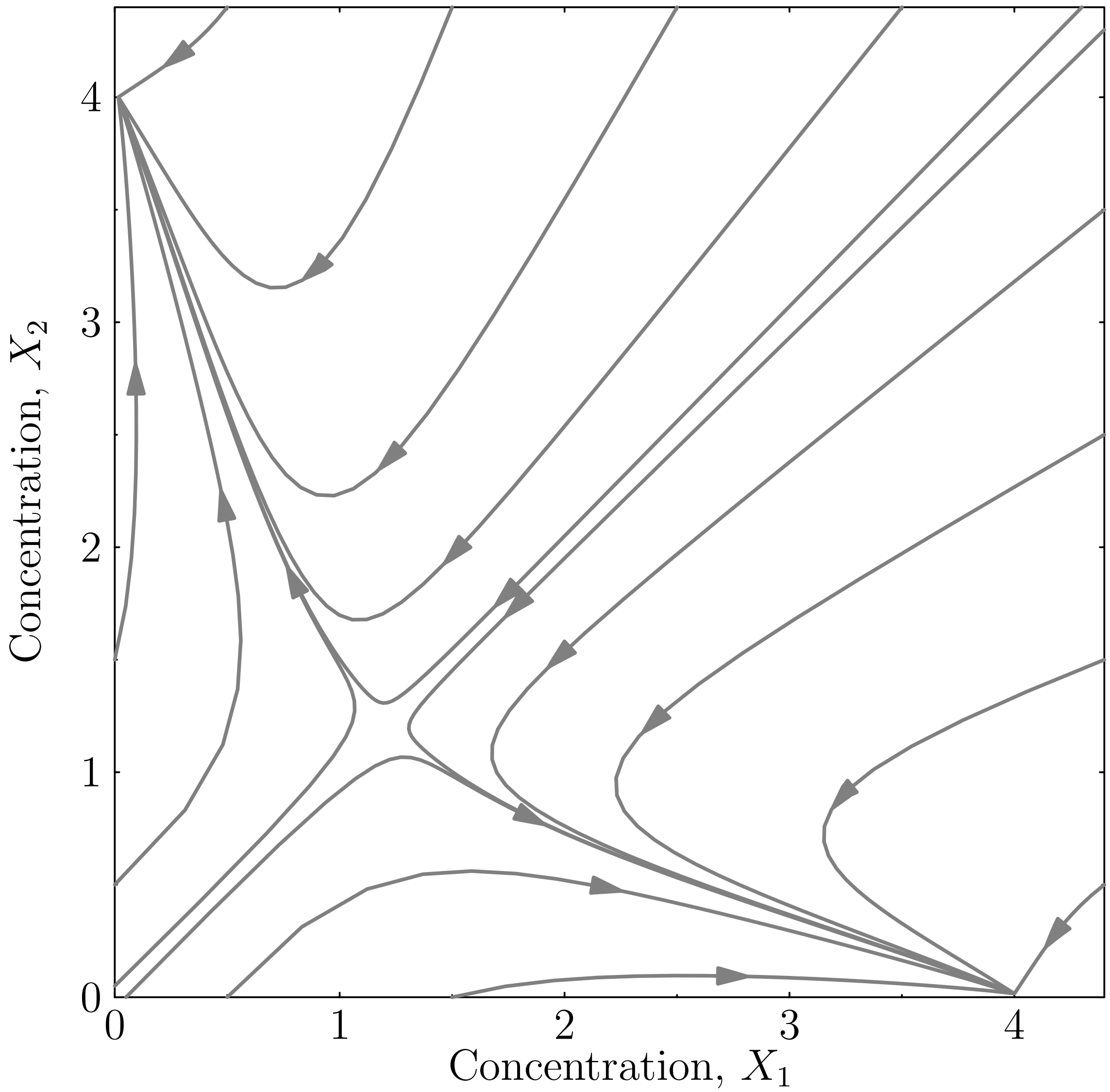
Bistability

- Let's consider a slightly different system where X_1 and X_2 inhibits each others' expression

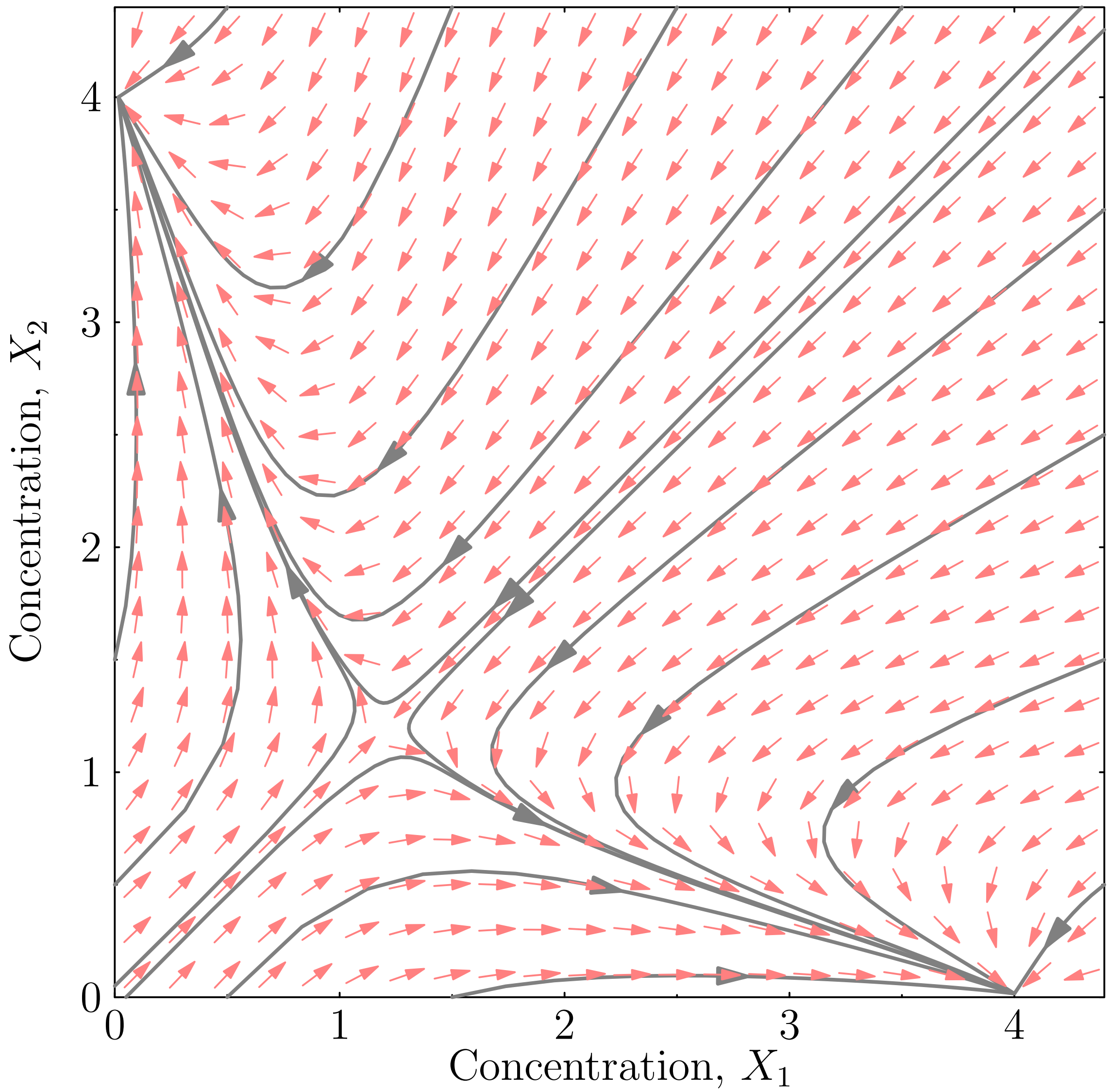
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}) = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}) = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ we end up with three fixed points

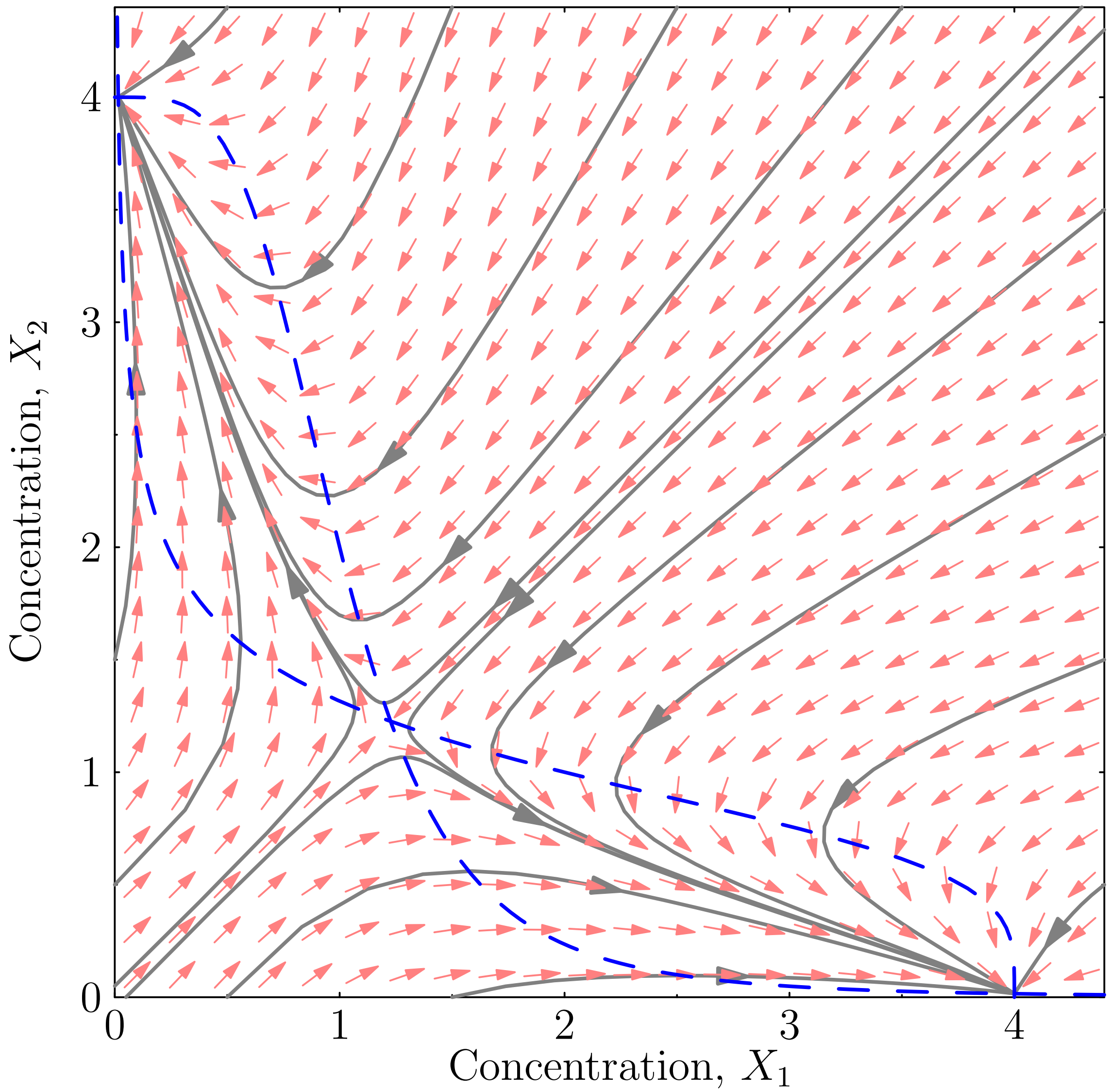
Nullclines and Bistablity



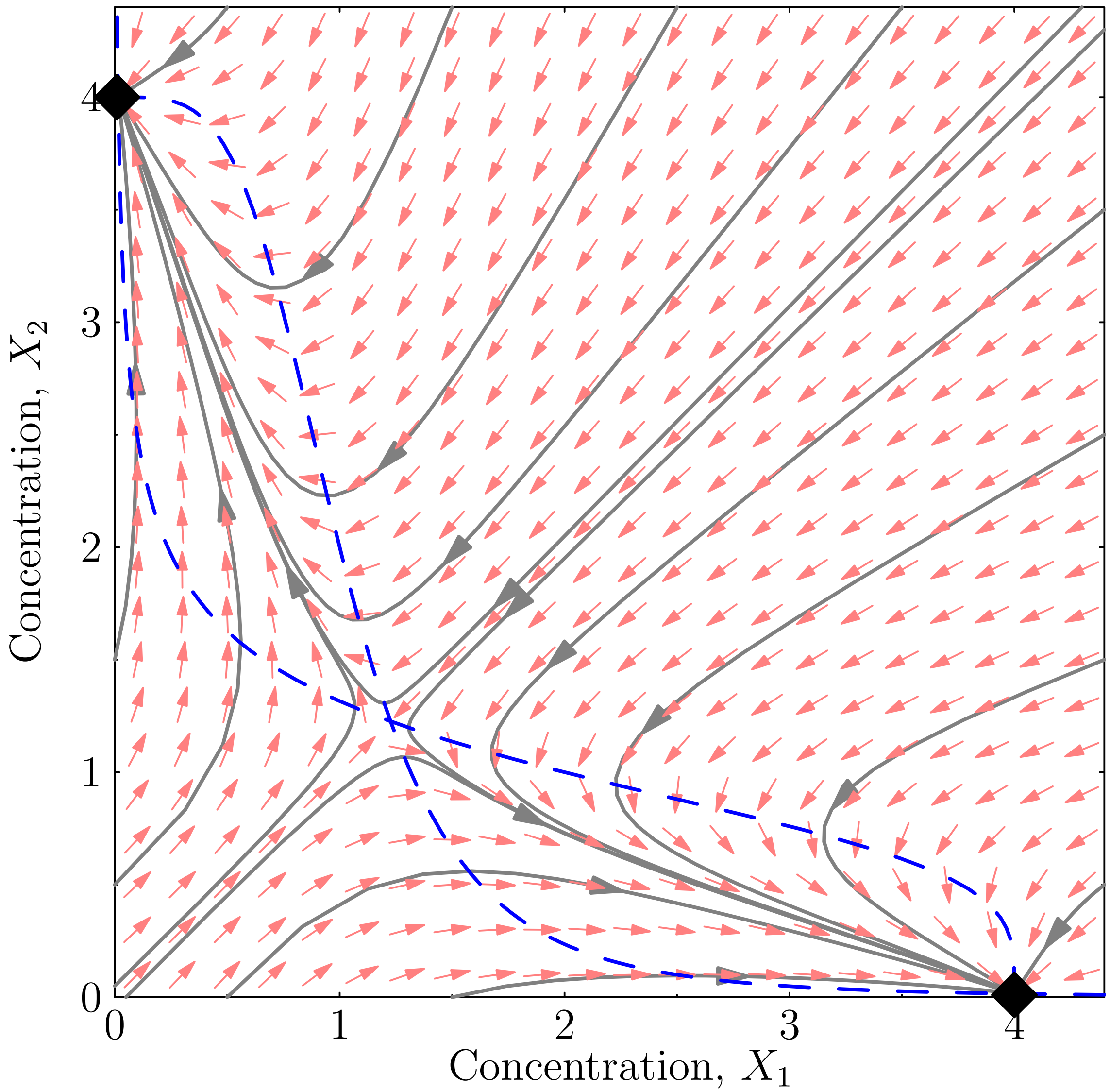
Nullclines and Bistablity



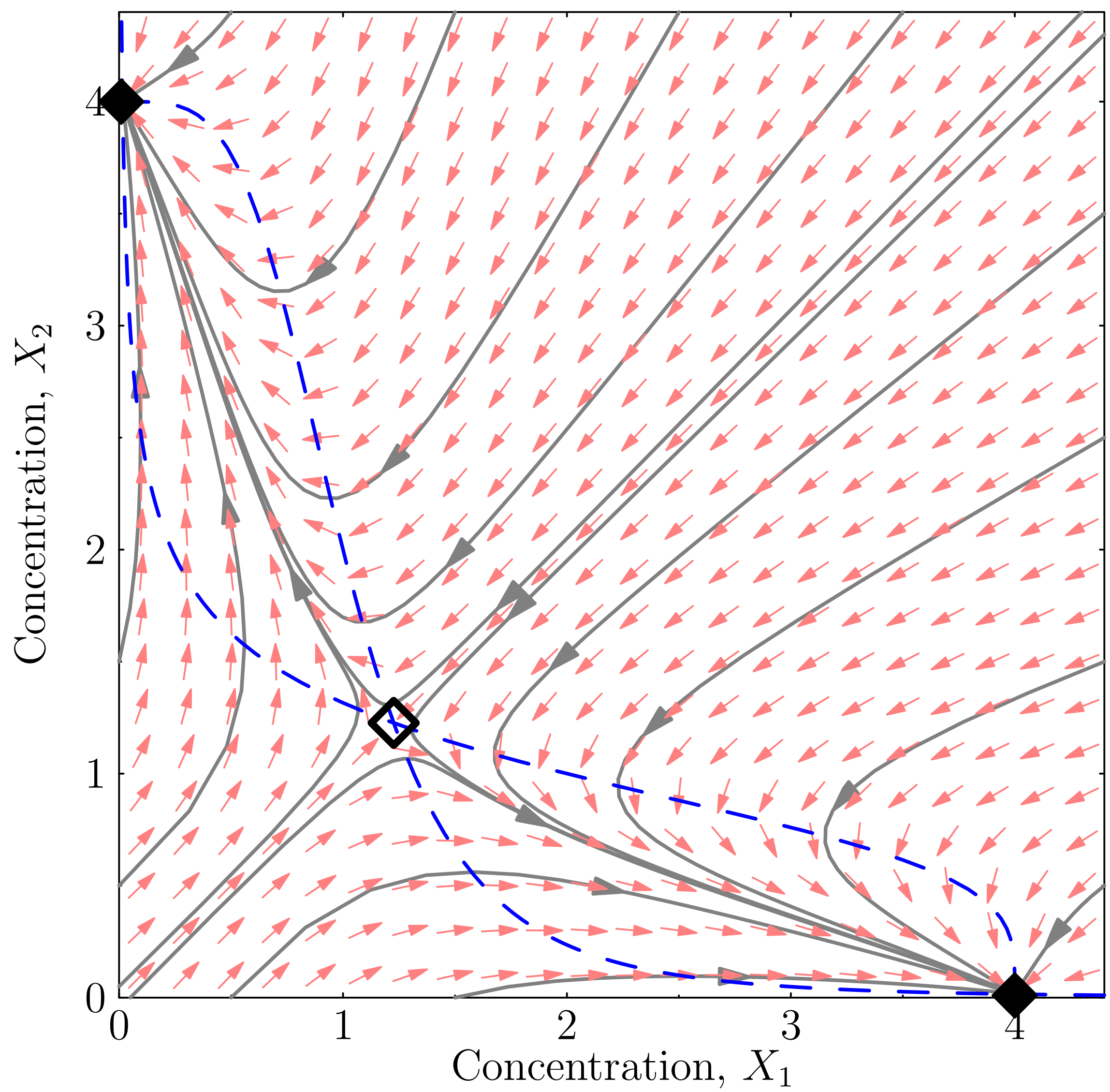
Nullclines and Bistablity



Nullclines and Bistablity

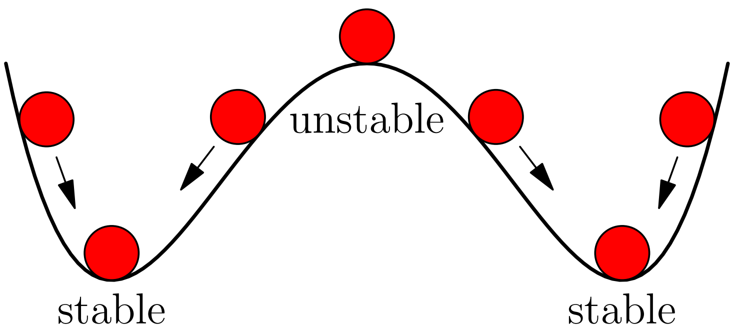


Nullclines and Bistablity

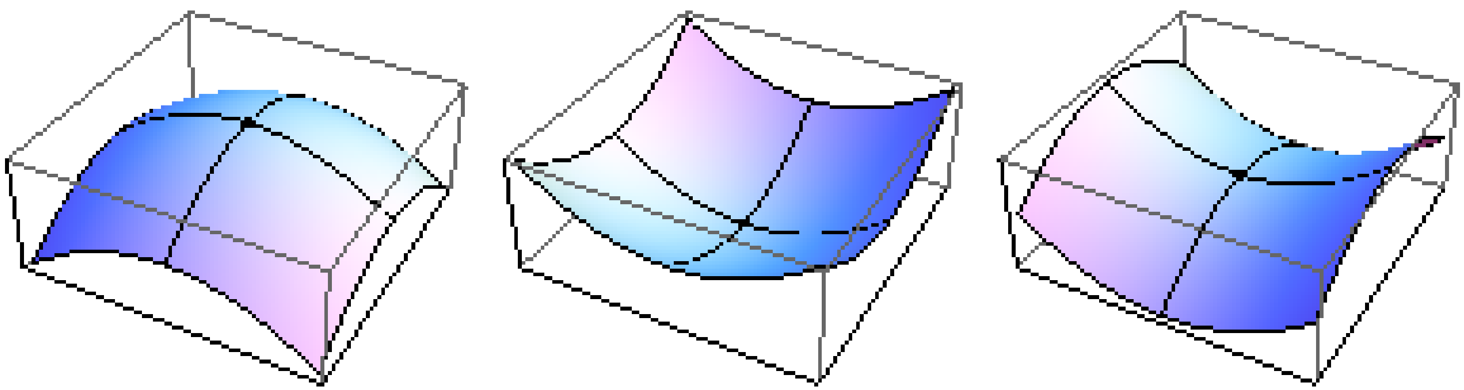


Stability

- Points can be stable or unstable

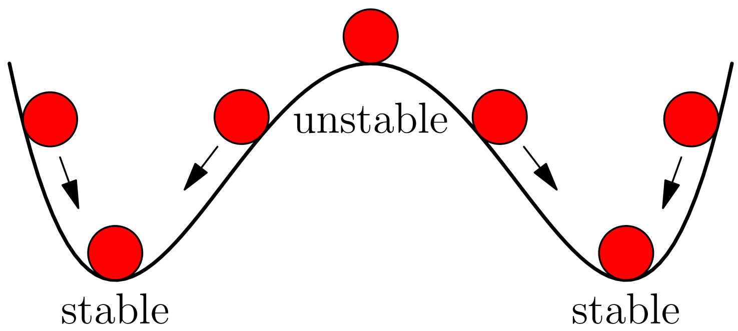


- Above 1D we can also have saddle-points

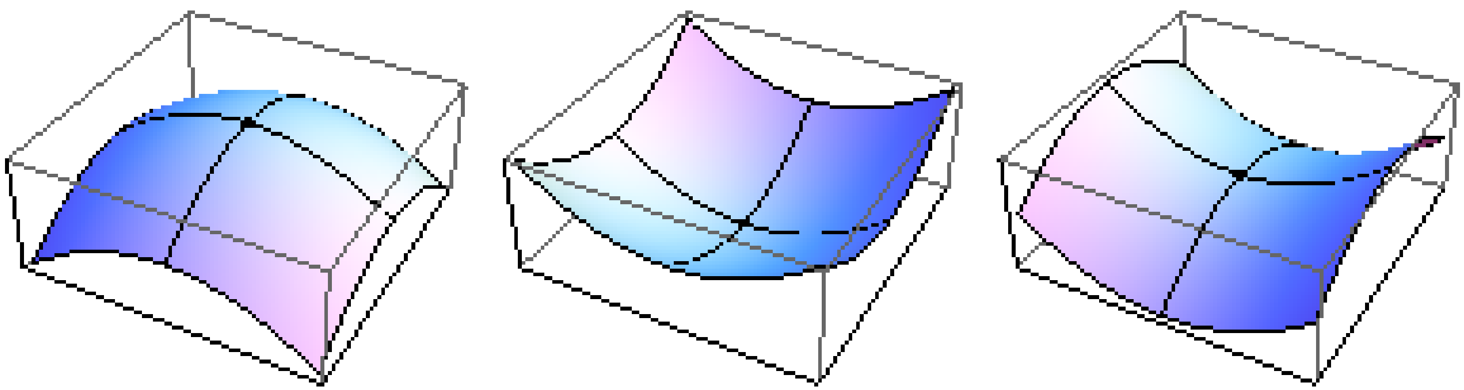


Stability

- Points can be stable or unstable



- Above 1D we can also have saddle-points



Expand Around Fixed-Point

- To study stability at a fixed-point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

since $f_i(\boldsymbol{X}^*) = 0$ (we are at a fixed-point)

- Where

$$\boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d} X_n}$$

Expand Around Fixed-Point

- To study stability at a fixed-point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

since $f_i(\boldsymbol{X}^*) = 0$ (we are at a fixed-point)

- Where

$$\boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Expand Around Fixed-Point

- To study stability at a fixed-point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^{\mathsf{T}} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^{\mathsf{T}} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

since $f_i(\boldsymbol{X}^*) = 0$ (we are at a fixed-point)

- Where

$$\boldsymbol{\epsilon}^{\mathsf{T}} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d} X_n}$$

Expand Around Fixed-Point

- To study stability at a fixed-point $\boldsymbol{X}^*$ we look at what happens when we make a small perturbation $\boldsymbol{\epsilon}$

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*)$$

since $f_i(\boldsymbol{X}^*) = 0$ (we are at a fixed-point)

- Where

$$\boldsymbol{\epsilon}^\mathsf{T} \boldsymbol{\nabla} f_i(\boldsymbol{X}^*) = \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d} X_n}$$

Jacobian

- In vector form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where $\mathbf{J}$ is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

- Note that this is just saying

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} \approx \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Jacobian

- In vector form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where **J** is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

- Note that this is just saying

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} \approx \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Jacobian

- In vector form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where **J** is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

- Note that this is just saying

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} \approx \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Jacobian

- In vector form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$

- where $\mathbf{J}$ is the **Jacobian** matrix with elements

$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$

- Note that this is just saying

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} \approx \epsilon_1 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\boldsymbol{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\boldsymbol{X}^*)}{\mathrm{d}X_n}$$

Linear Approximation

- Let $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then (since $\mathrm{d}\boldsymbol{\epsilon}(t) = \mathrm{d}\boldsymbol{X}(t)$)

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$

- To solve this we note that we can write most square matrices $\mathbf{J}$ as

$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Where $\boldsymbol{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors

Linear Approximation

- Let $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then (since $\mathrm{d}\boldsymbol{\epsilon}(t) = \mathrm{d}\boldsymbol{X}(t)$)

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$

- To solve this we note that we can write most square matrices $\mathbf{J}$ as

$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Where $\boldsymbol{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors

Linear Approximation

- Let $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then (since $\mathrm{d}\boldsymbol{\epsilon}(t) = \mathrm{d}\boldsymbol{X}(t)$)

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$

- To solve this we note that we can write most square matrices **J** as

$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Where **Λ** is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of **J** and the columns of **V** are the eigenvectors

Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\boldsymbol{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- There are also left-handed eigenvectors $\boldsymbol{u}_i$ with the same eigenvalues, λ_i, such that

$$\boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

- In general $\boldsymbol{v}_i \neq \boldsymbol{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{v}_j = 0$

- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\boldsymbol{v}_j = \boldsymbol{v}_i^*$

Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\boldsymbol{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- There are also left-handed eigenvectors $\boldsymbol{u}_i$ with the same eigenvalues, λ_i, such that

$$\boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

- In general $\boldsymbol{v}_i \neq \boldsymbol{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{v}_j = 0$

- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\boldsymbol{v}_j = \boldsymbol{v}_i^*$

Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\boldsymbol{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- There are also left-handed eigenvectors $\boldsymbol{u}_i$ with the same eigenvalues, λ_i, such that

$$\boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

- In general $\boldsymbol{v}_i \neq \boldsymbol{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{v}_j = 0$

- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\boldsymbol{v}_j = \boldsymbol{v}_i^*$

Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\boldsymbol{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- There are also left-handed eigenvectors $\boldsymbol{u}_i$ with the same eigenvalues, λ_i, such that

$$\boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

- In general $\boldsymbol{v}_i \neq \boldsymbol{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{v}_j = 0$

- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\boldsymbol{v}_j = \boldsymbol{v}_i^*$

Matrix Decomposition

- If we write

$$\mathbf{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2, & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Then $\mathbf{M} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{U} = \mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}$

- where $\mathbf{U} = \mathbf{V}^{-1}$ has rows equal to $\boldsymbol{u}_i^{\mathsf{T}}$

- Multiplying $\mathbf{M}$ on the right by $\mathbf{V}$ we find

$$\mathbf{M}\,\mathbf{V} = (\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1})\,\mathbf{V} = \mathbf{V}\,\boldsymbol{\Lambda} \qquad \Rightarrow \qquad \mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- Multiplying $\mathbf{M}$ on the left by $\mathbf{U} = \mathbf{V}^{-1}$

$$\mathbf{U}\,\mathbf{M} = \mathbf{U}\,(\mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}) = \boldsymbol{\Lambda}\,\mathbf{U} \qquad \Rightarrow \qquad \boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

Matrix Decomposition

- If we write

$$\mathbf{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2, & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Then $\mathbf{M} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{U} = \mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}$

- where $\mathbf{U} = \mathbf{V}^{-1}$ has rows equal to $\boldsymbol{u}_i^{\mathsf{T}}$

- Multiplying $\mathbf{M}$ on the right by $\mathbf{V}$ we find

$$\mathbf{M}\,\mathbf{V} = (\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1})\,\mathbf{V} = \mathbf{V}\,\boldsymbol{\Lambda} \qquad \Rightarrow \qquad \mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- Multiplying $\mathbf{M}$ on the left by $\mathbf{U} = \mathbf{V}^{-1}$

$$\mathbf{U}\,\mathbf{M} = \mathbf{U}\,(\mathbf{U}^{-1}\,\boldsymbol{\Lambda}\,\mathbf{U}) = \boldsymbol{\Lambda}\,\mathbf{U} \qquad \Rightarrow \qquad \boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

Matrix Decomposition

- If we write

$$\mathbf{V} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix} \qquad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2, & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Then $\mathbf{M} = \mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1} = \mathbf{V}\,\mathbf{\Lambda}\,\mathbf{U} = \mathbf{U}^{-1}\,\mathbf{\Lambda}\,\mathbf{U}$

- where $\mathbf{U} = \mathbf{V}^{-1}$ has rows equal to $\boldsymbol{u}_i^{\mathsf{T}}$

- Multiplying $\mathbf{M}$ on the right by $\mathbf{V}$ we find

$$\mathbf{M}\,\mathbf{V} = (\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1})\,\mathbf{V} = \mathbf{V}\,\mathbf{\Lambda} \qquad \Rightarrow \qquad \mathbf{M}\,\boldsymbol{v}_i = \lambda_i\,\boldsymbol{v}_i$$

- Multiplying $\mathbf{M}$ on the left by $\mathbf{U} = \mathbf{V}^{-1}$

$$\mathbf{U}\,\mathbf{M} = \mathbf{U}\,(\mathbf{U}^{-1}\,\mathbf{\Lambda}\,\mathbf{U}) = \mathbf{\Lambda}\,\mathbf{U} \qquad \Rightarrow \qquad \boldsymbol{u}_i^{\mathsf{T}}\,\mathbf{M} = \lambda_i\,\boldsymbol{u}_i^{\mathsf{T}}$$

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} = \lambda_i\,\hat{\epsilon}_i(t)$$

- With solution

$$\hat{\epsilon}_i(t) = \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} = \lambda_i\,\hat{\epsilon}_i(t)$$

- With solution

$$\hat{\epsilon}_i(t) = \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} = \lambda_i\,\hat{\epsilon}_i(t)$$

- With solution

$$\hat{\epsilon}_i(t) = \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

- Writing $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} = \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} = \lambda_i\,\hat{\epsilon}_i(t)$$

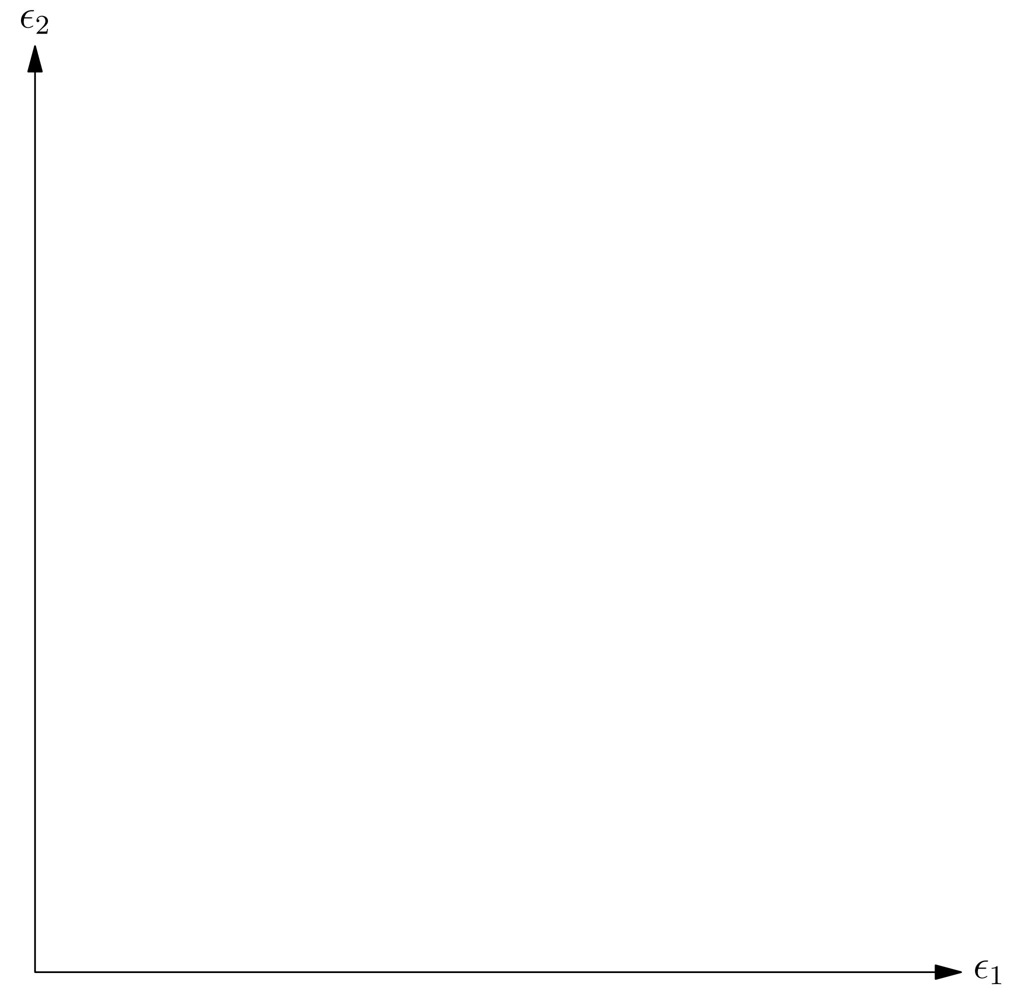
- With solution

$$\hat{\epsilon}_i(t) = \mathrm{e}^{\lambda_i\,t}\,\hat{\epsilon}_i(0)$$

Example

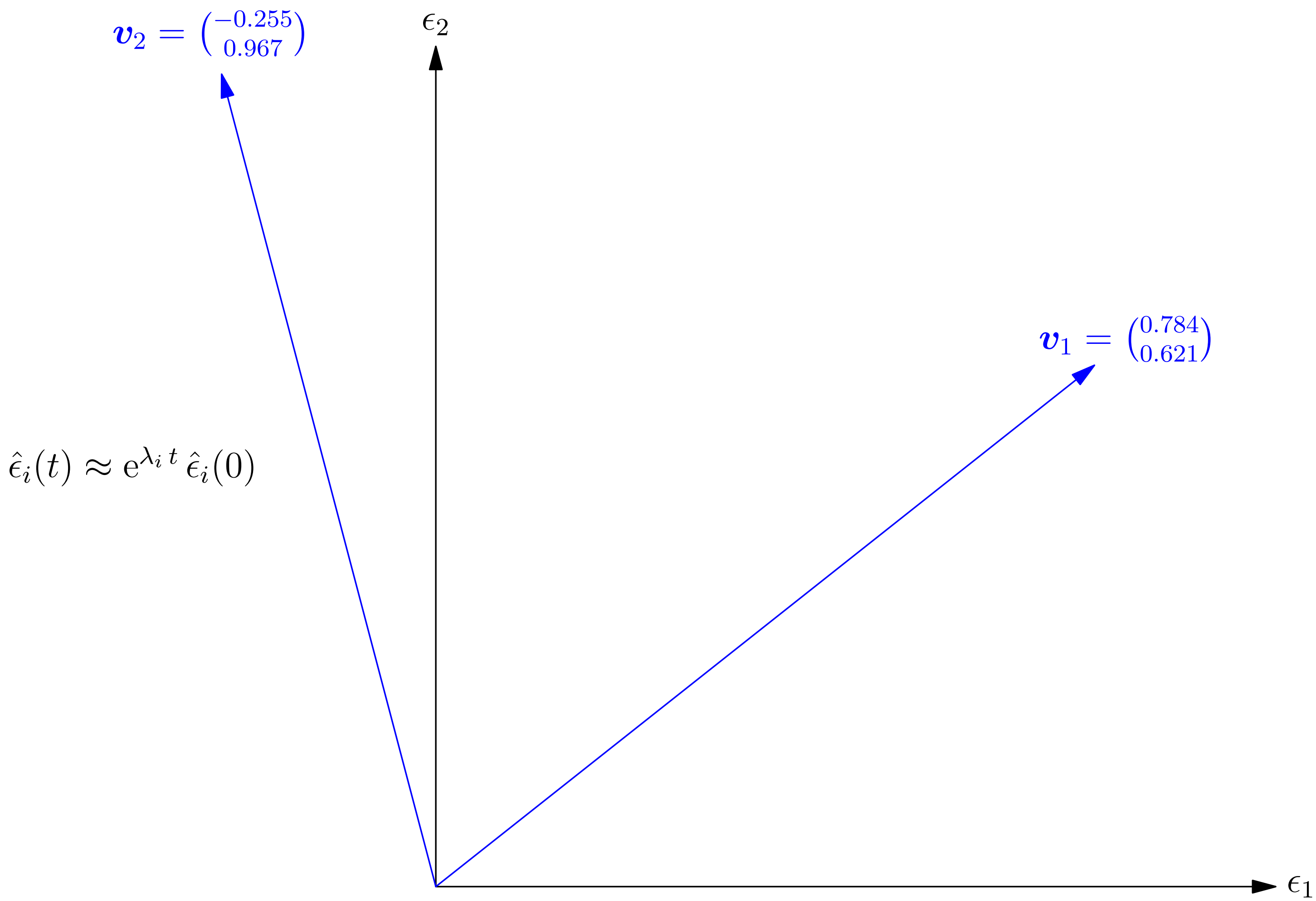
$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$

$$\hat{\epsilon}_i(t) \approx \mathrm{e}^{\lambda_i\, t}\, \hat{\epsilon}_i(0)$$



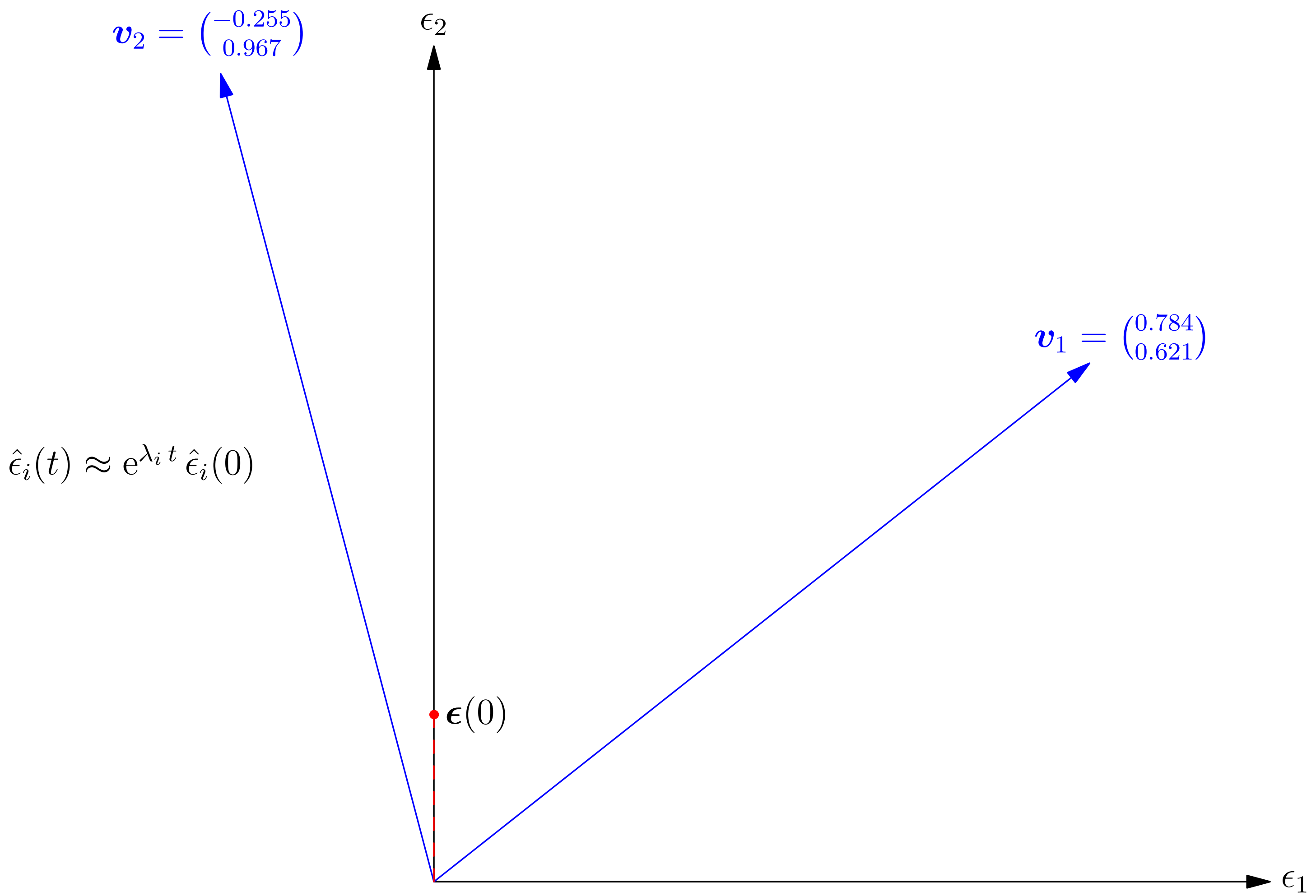
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



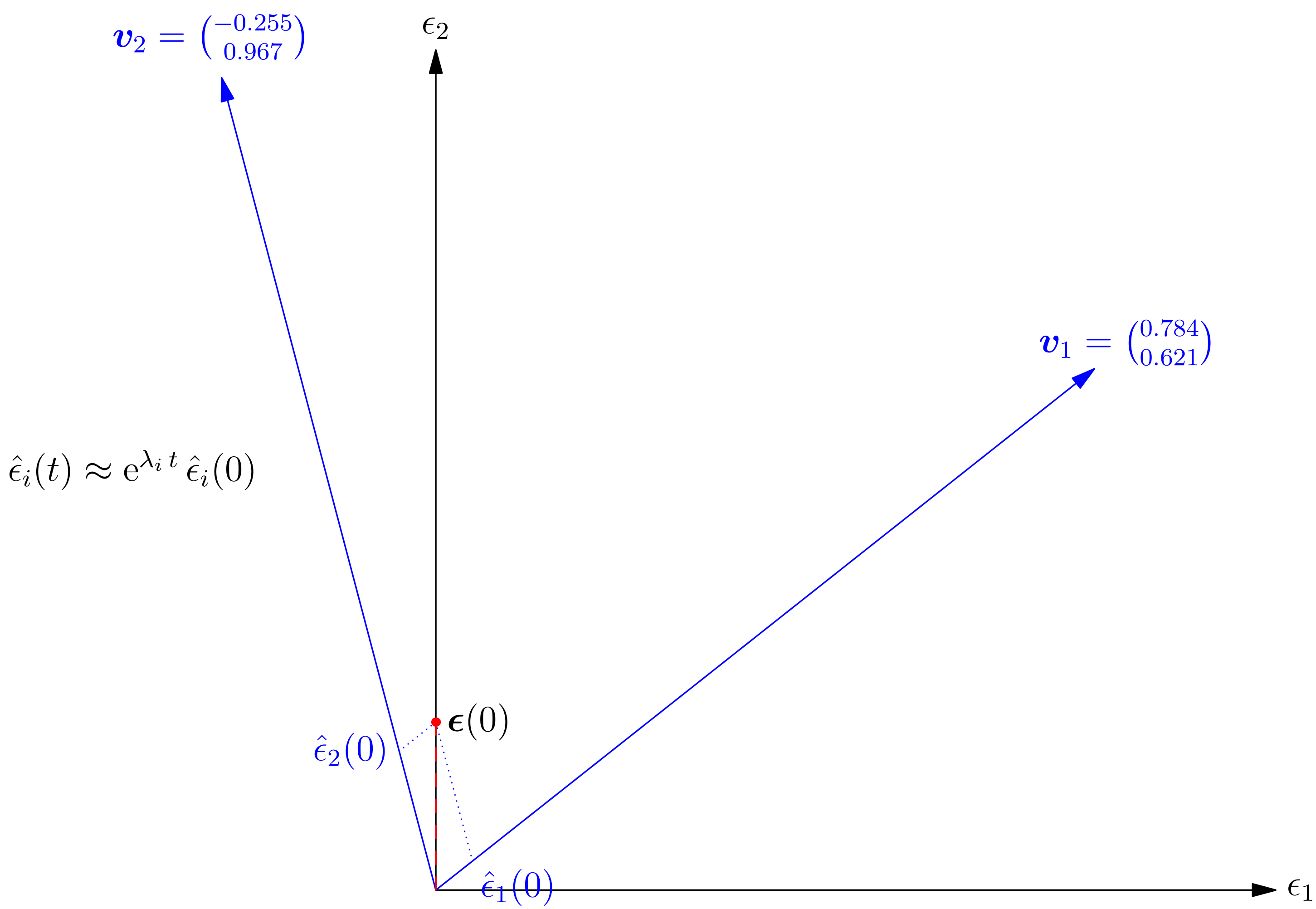
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



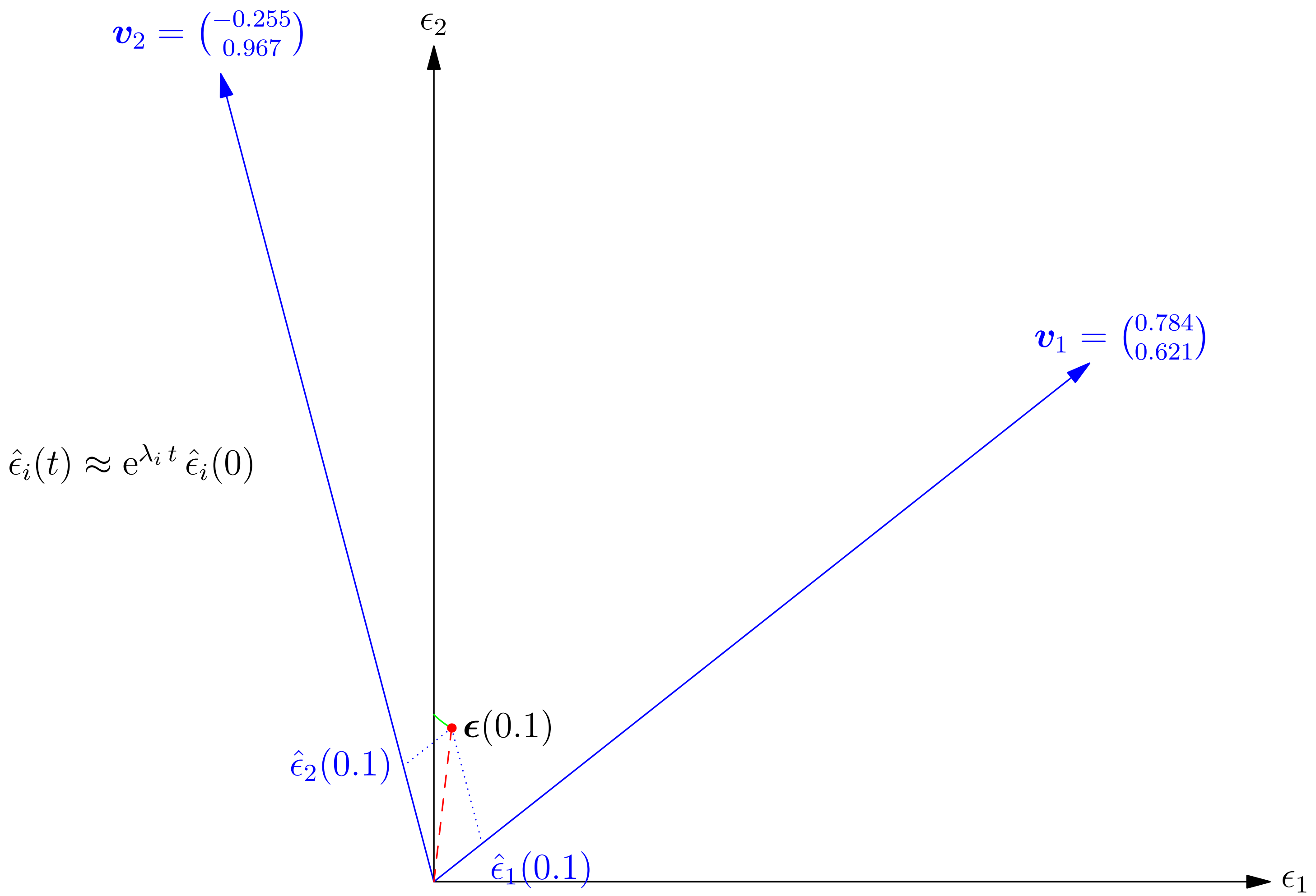
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



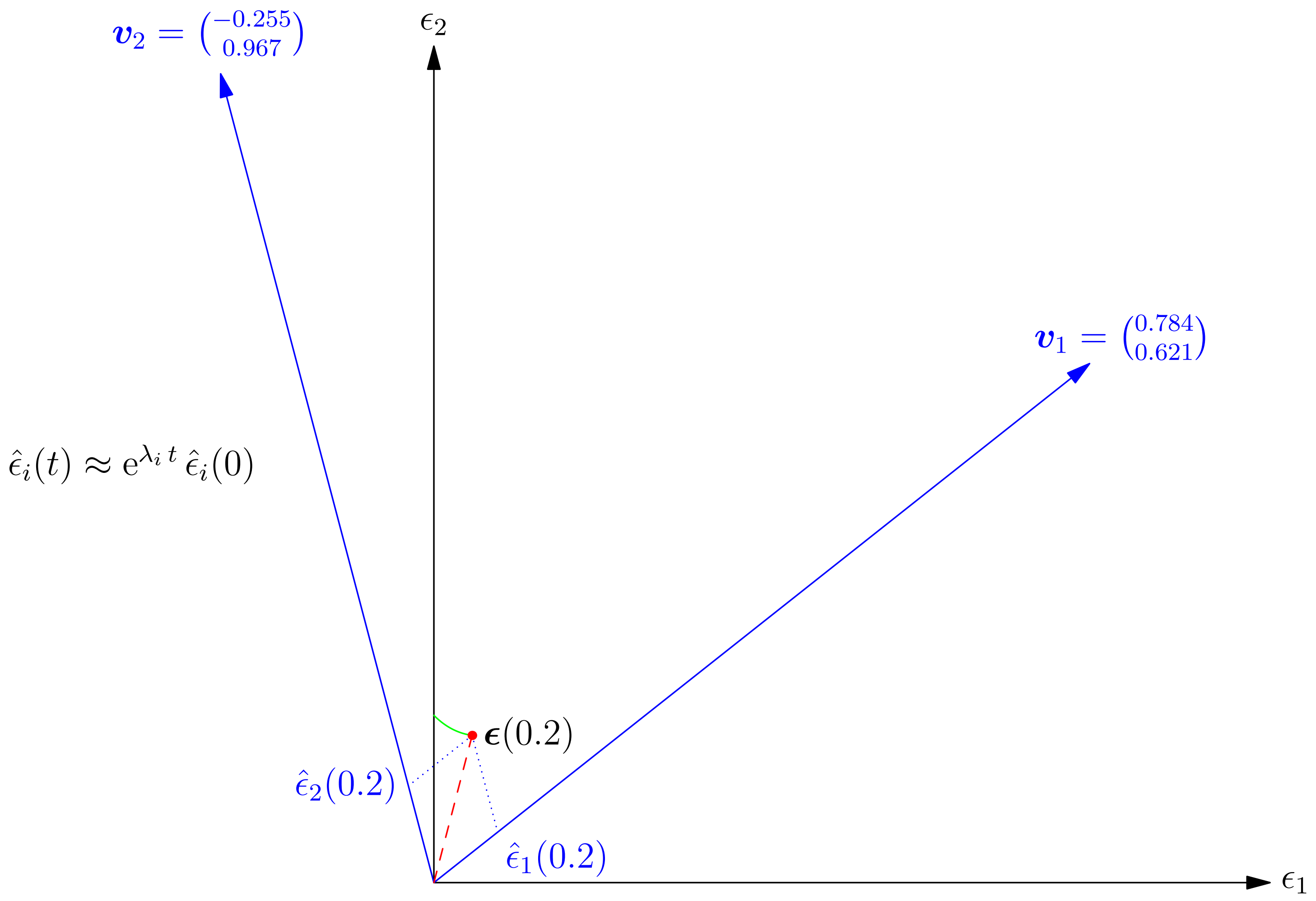
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



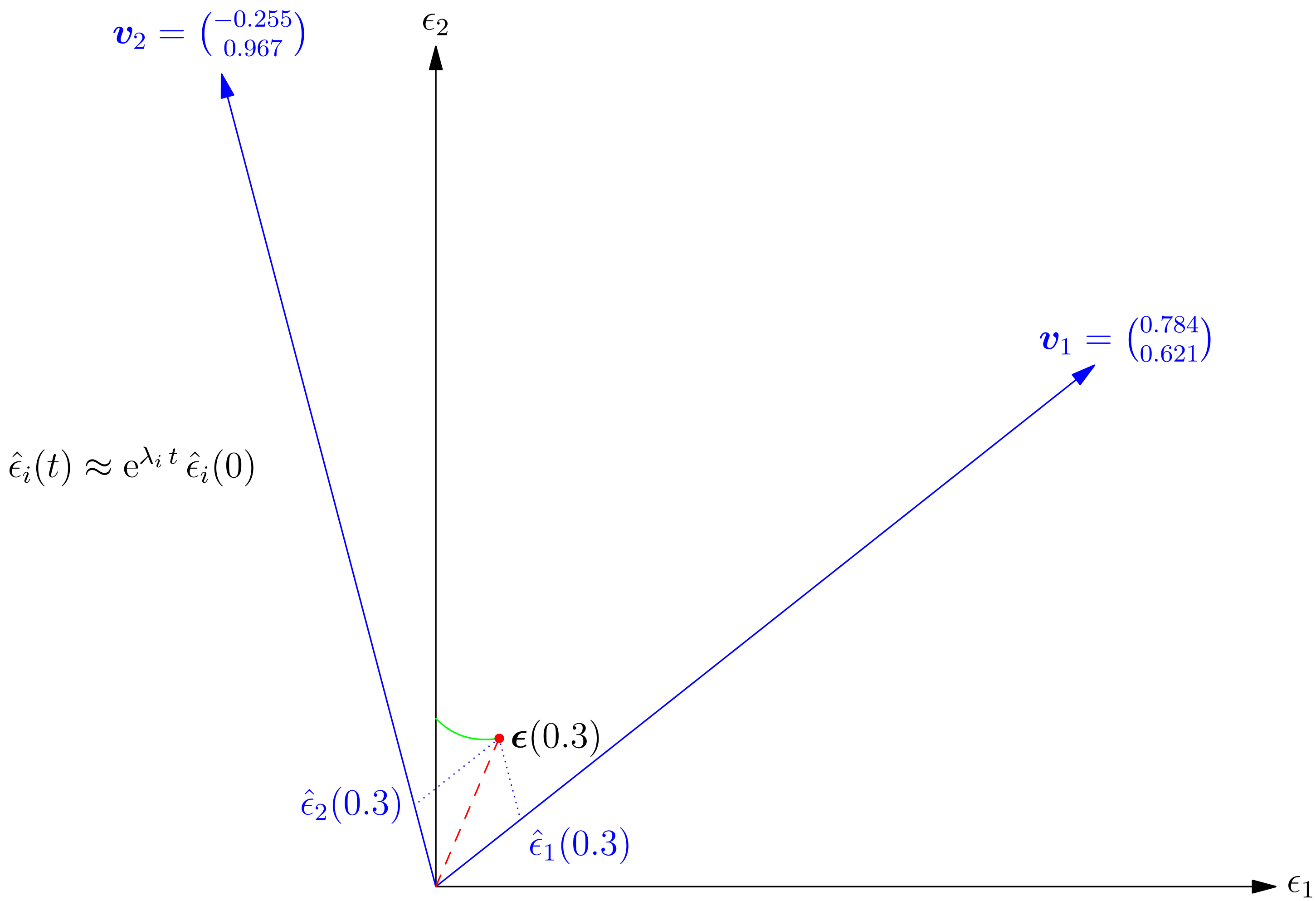
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



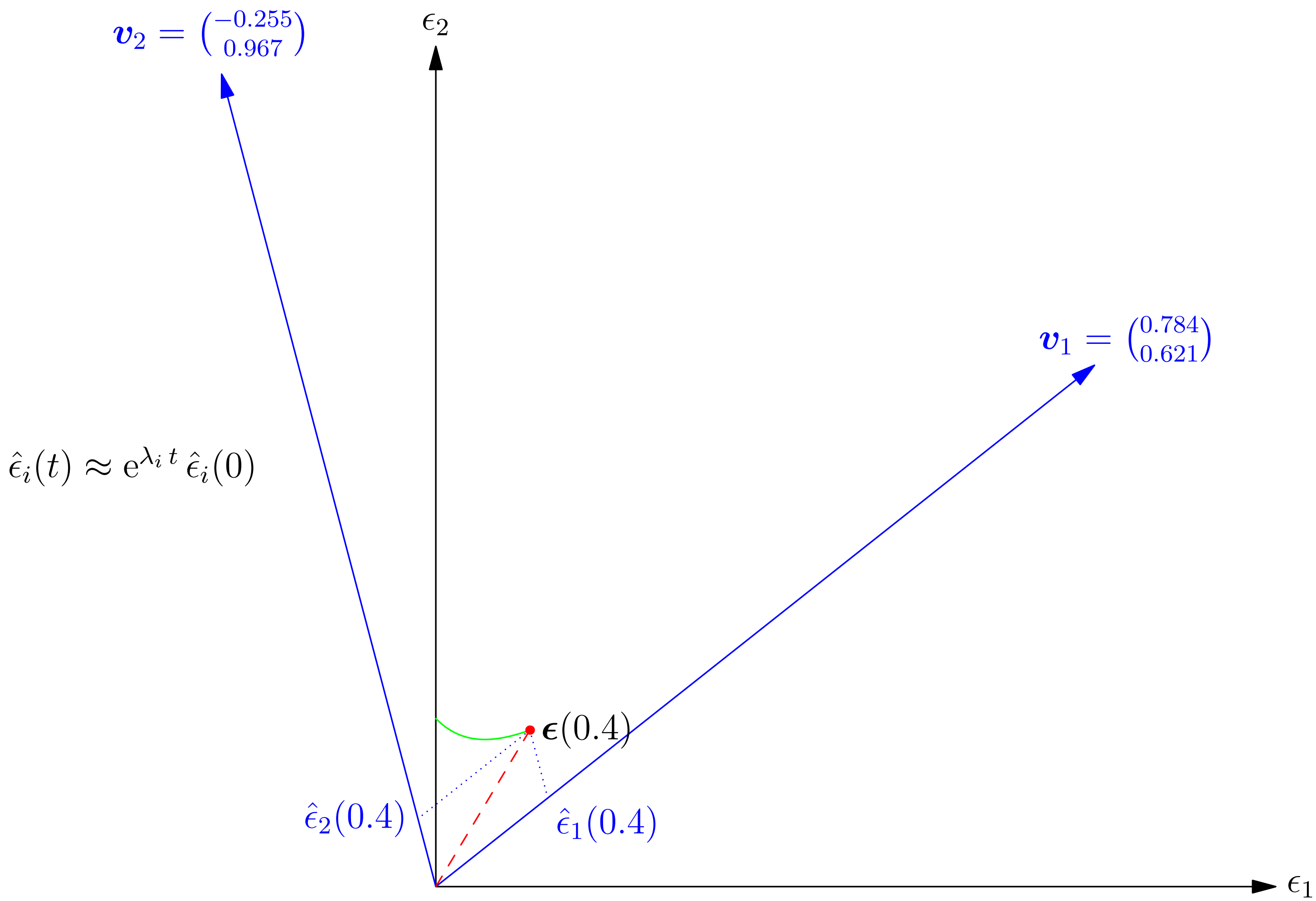
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



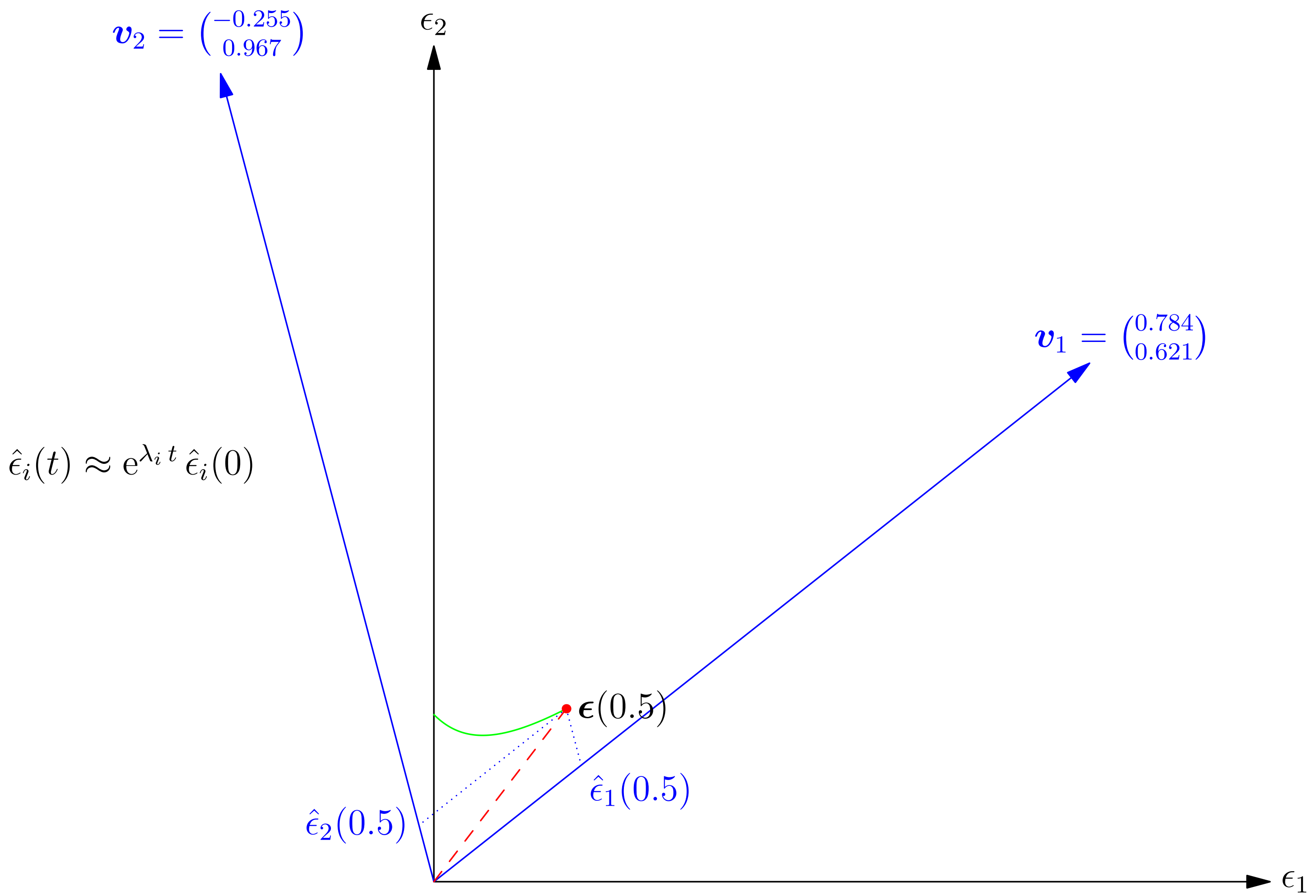
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



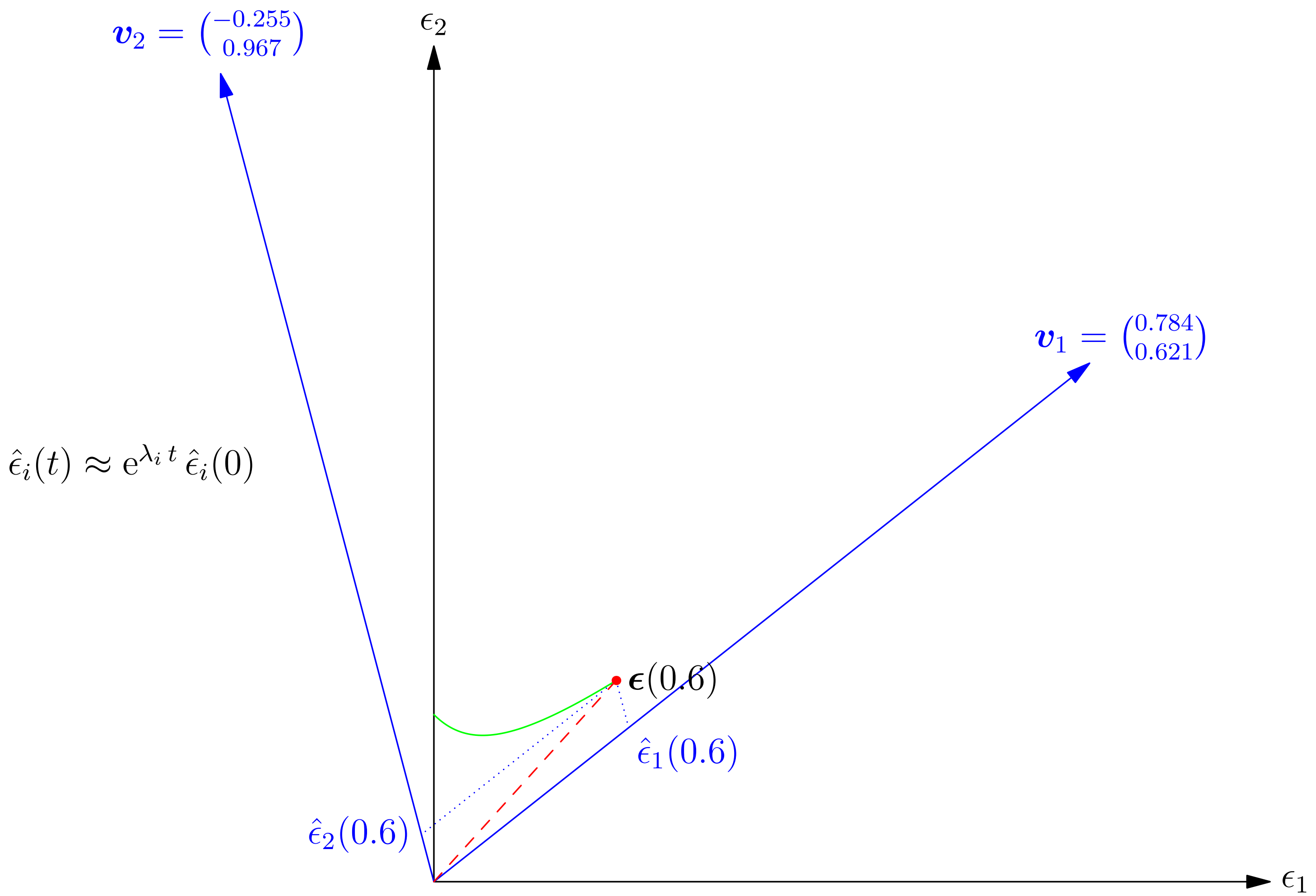
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



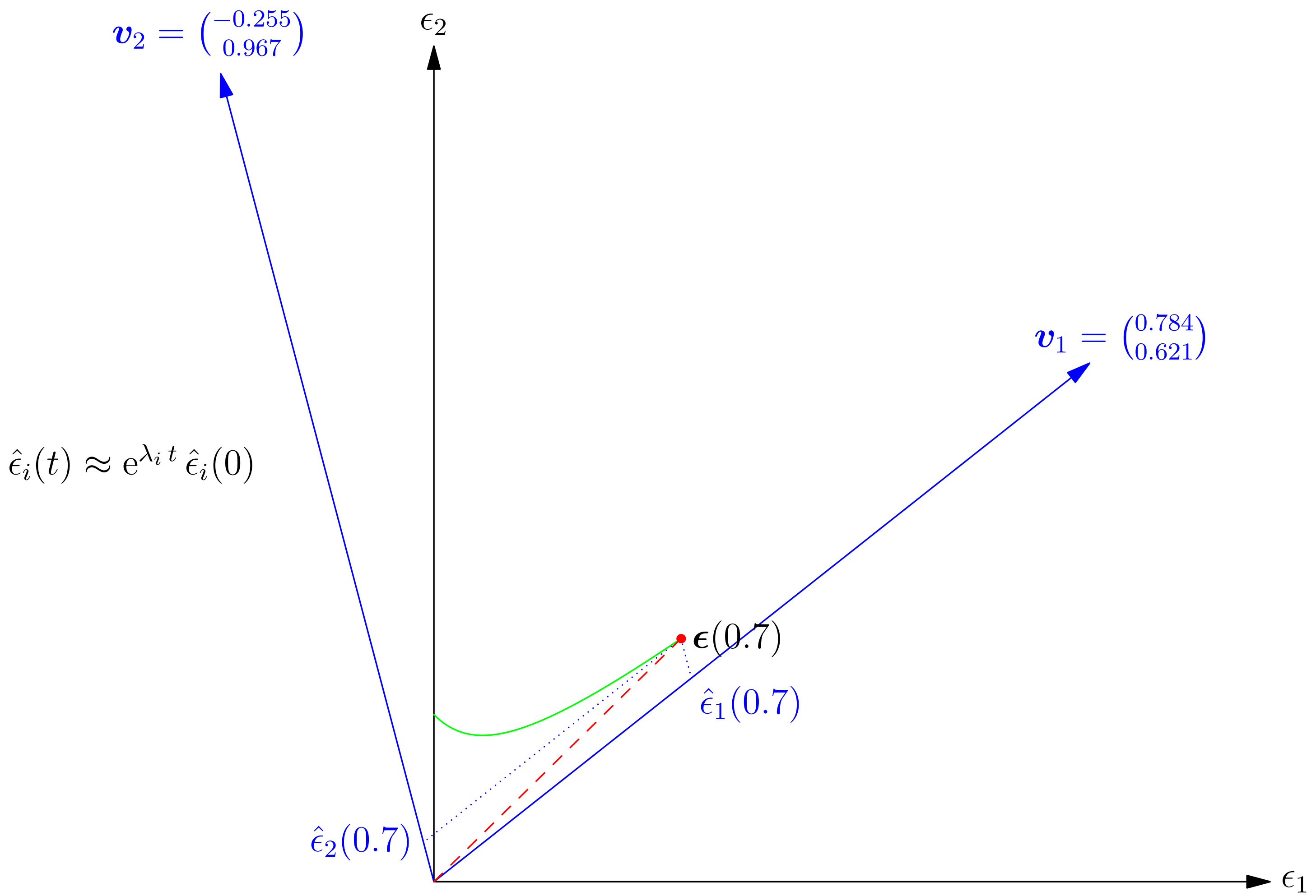
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



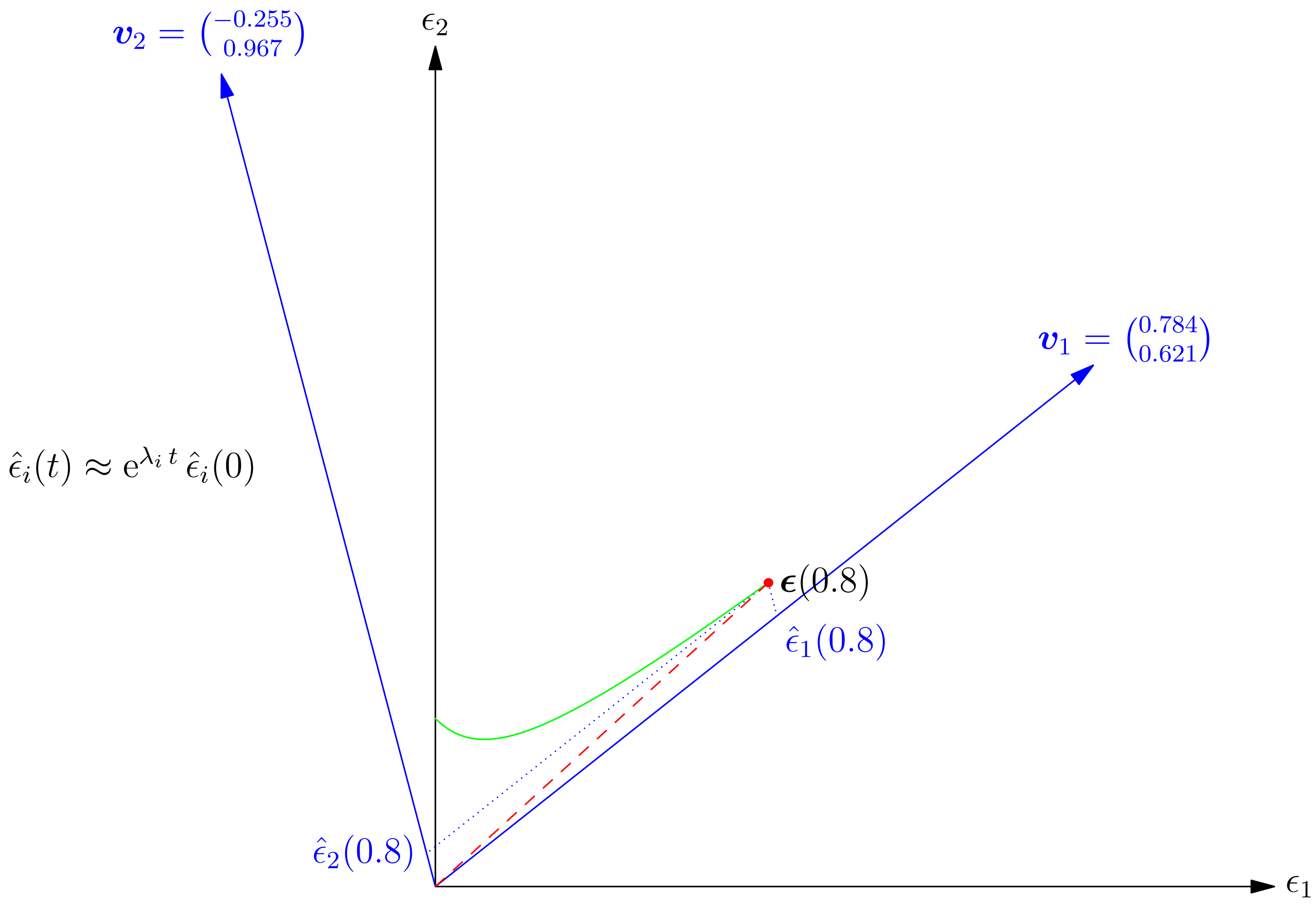
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



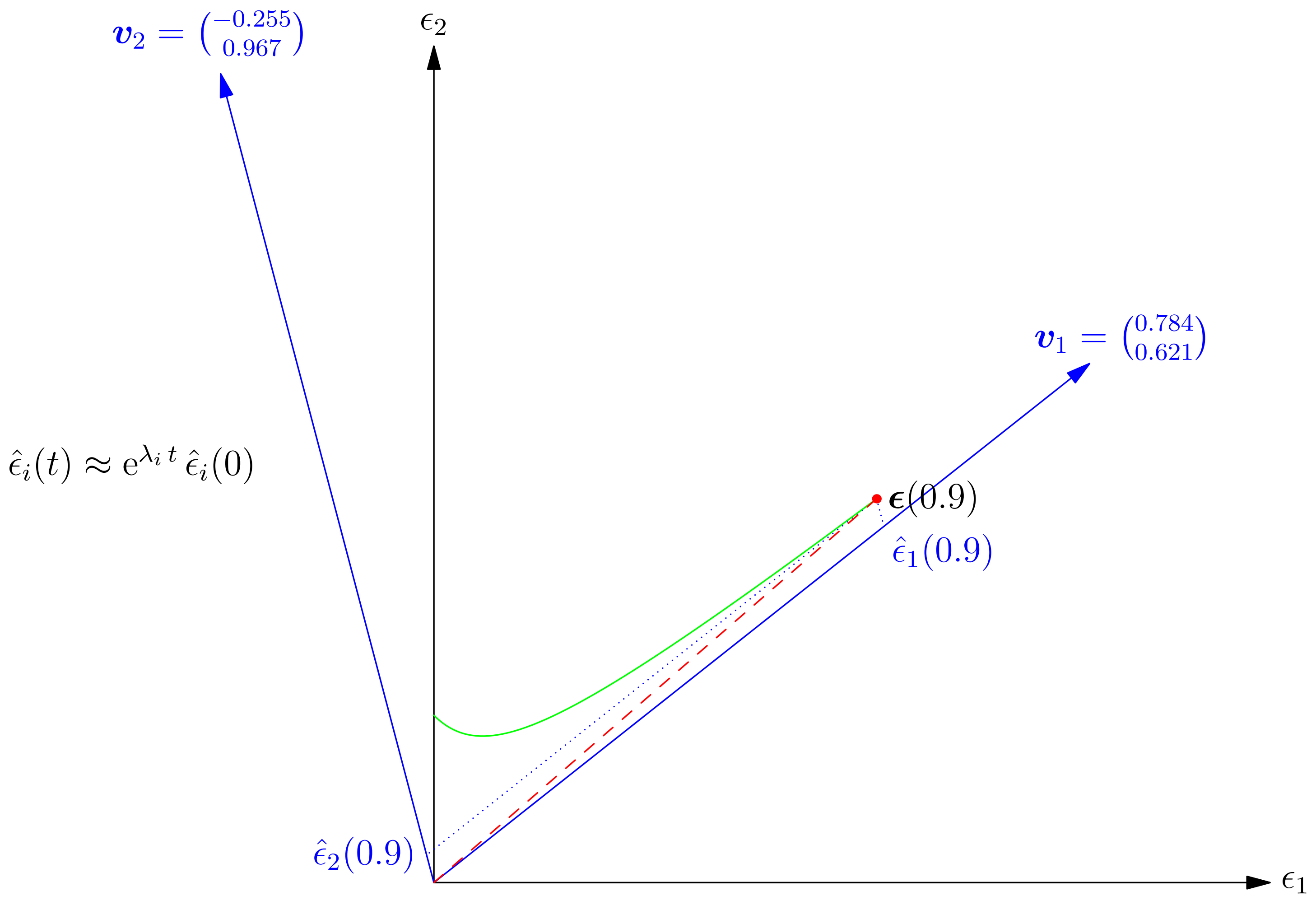
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



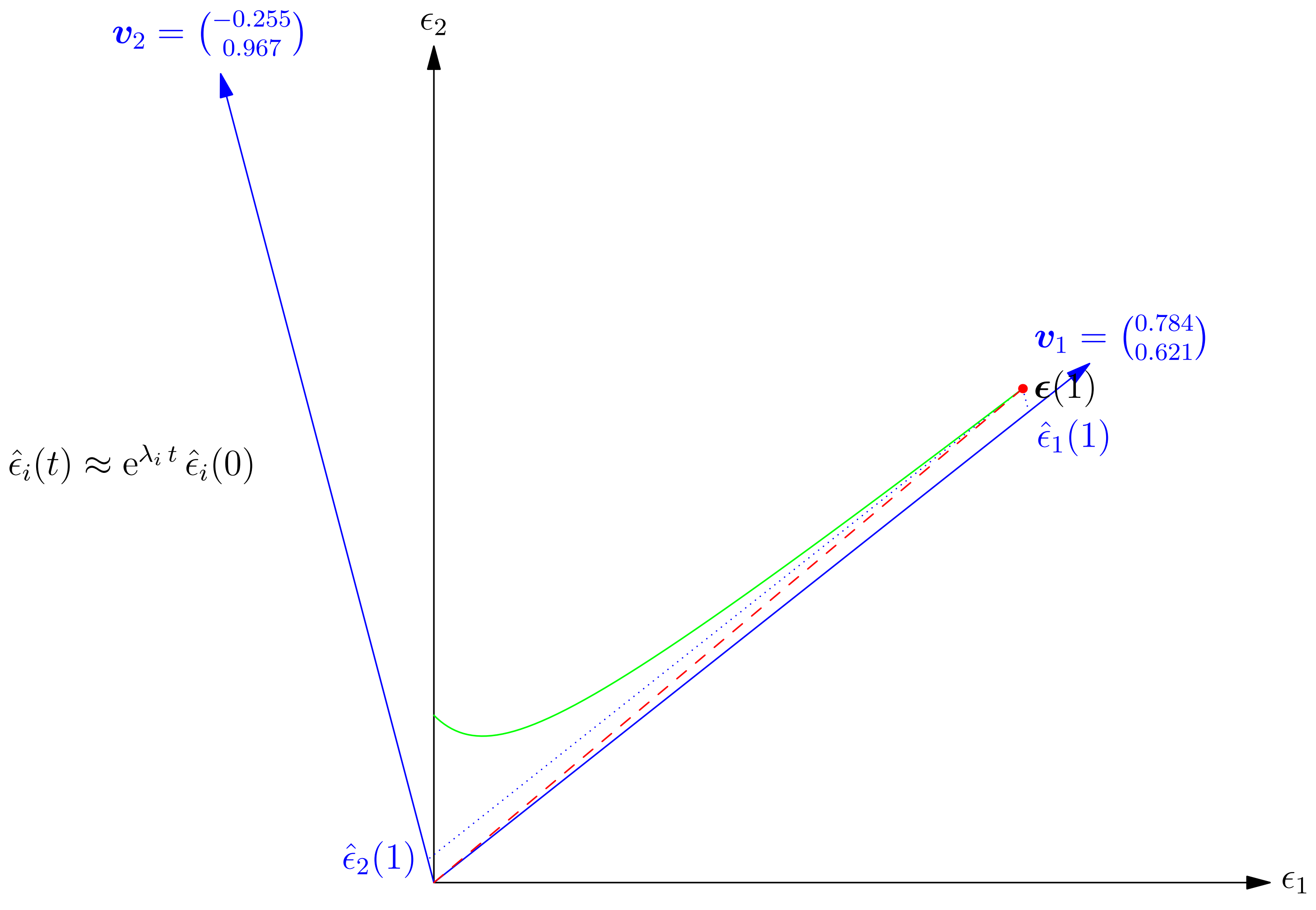
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



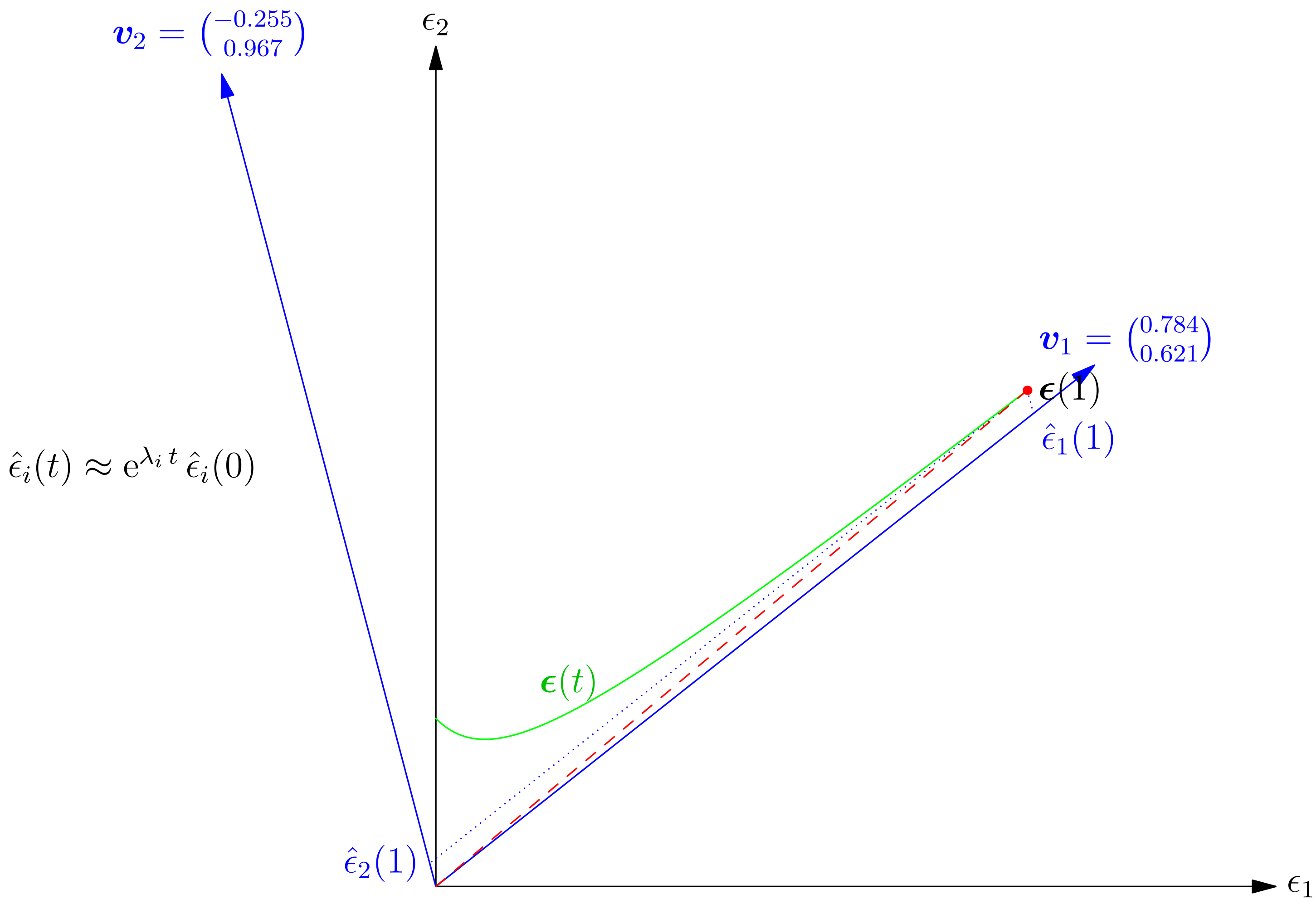
Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



Example

$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$



Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)

- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable

- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point

- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)

- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

Oscillator

- Consider a chain with a positive feedback

$$\varnothing \xrightarrow{k_0} X_1 \qquad k(X_2) = k_1\left(1 + (X_2/K)^n\right)$$

$$X_1 \xrightarrow{k(X_2)} X_2$$

$$X_2 \xrightarrow{k_2} \varnothing$$

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

Oscillator

- Consider a chain with a positive feedback

$$\varnothing \xrightarrow{k_0} X_1$$

$$X_1 \xrightarrow{k(X_2)} X_2$$

$$X_2 \xrightarrow{k_2} \varnothing$$

$$k(X_2) = k_1 \left(1 + (X_2/K)^n\right)$$

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

Oscillator

- Consider a chain with a positive feedback

$$\varnothing \xrightarrow{k_0} X_1 \qquad\qquad k(X_2) = k_1\left(1 + (X_2/K)^n\right)$$

$$X_1 \xrightarrow{k(X_2)} X_2$$

$$X_2 \xrightarrow{k_2} \varnothing$$

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

Oscillator

- Consider a chain with a positive feedback

$$\varnothing \xrightarrow{k_0} X_1$$

$$X_1 \xrightarrow{k(X_2)} X_2$$

$$X_2 \xrightarrow{k_2} \varnothing$$

$$k(X_2) = k_1\,(1 + (X_2/K)^n)$$

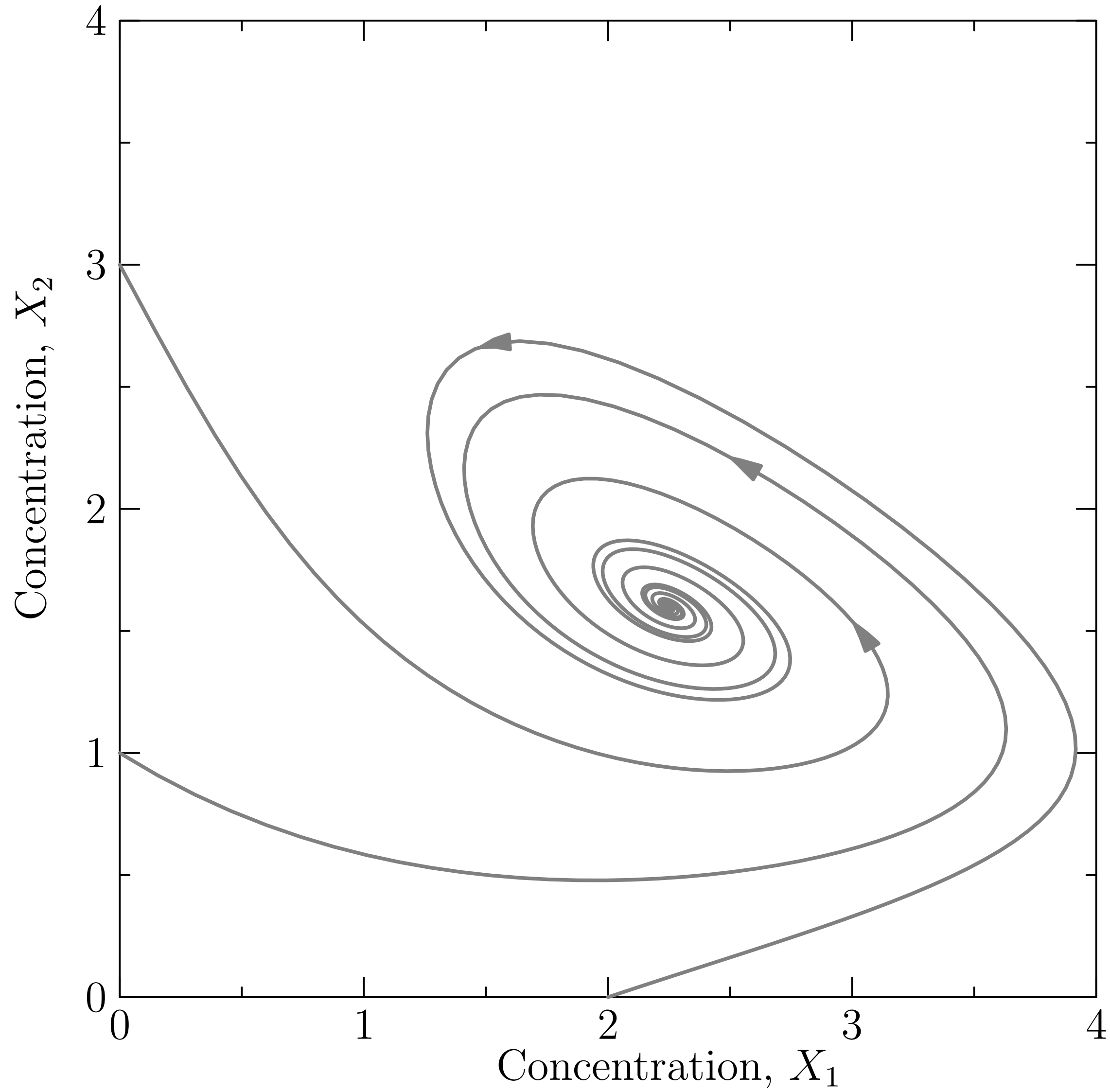
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\,X_2(t)$$

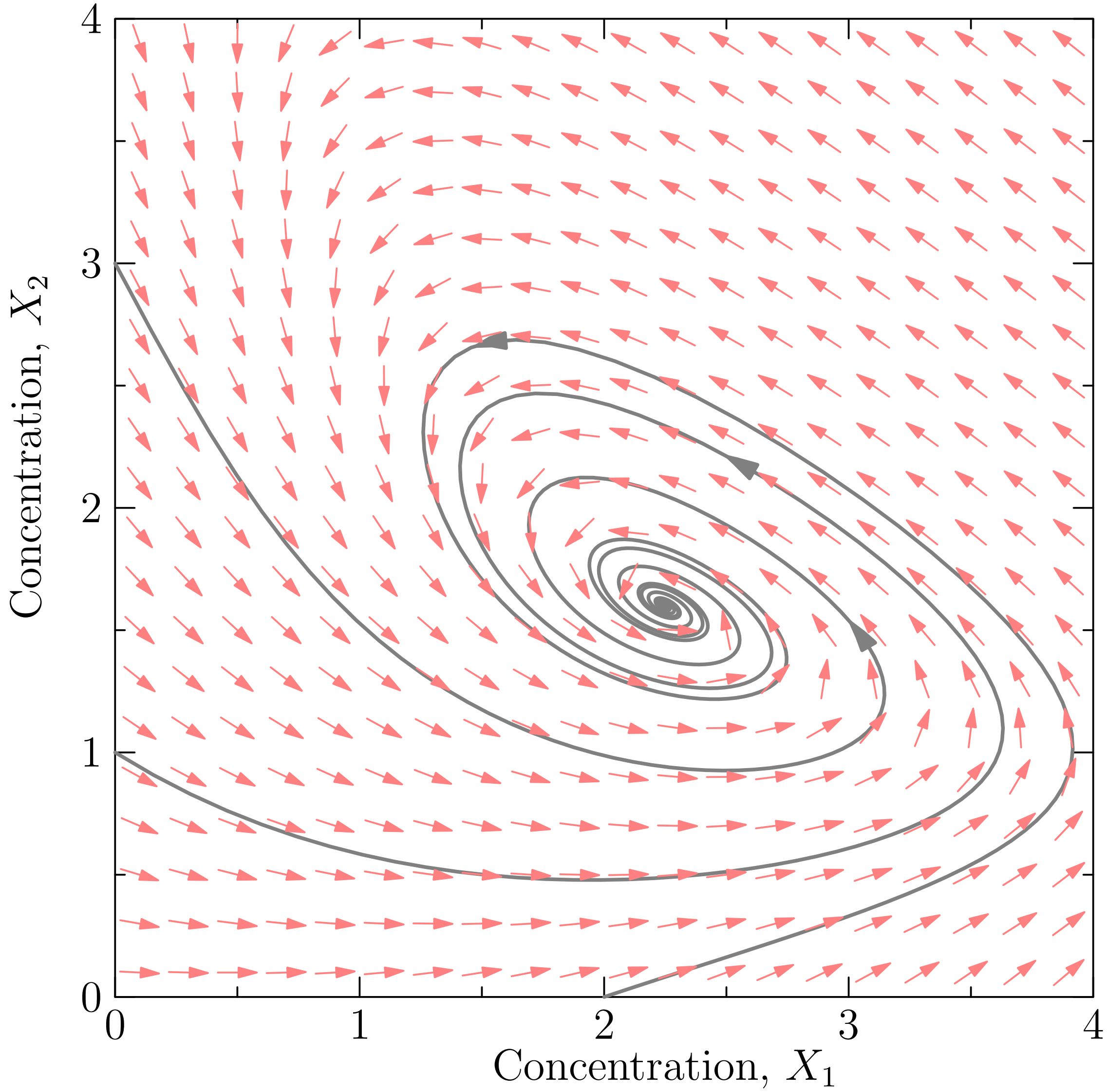
- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- Large values of X_2 will result in most of X_1 being turned into X_2, then X_2 will decay rapidly allowing X_1 to increase

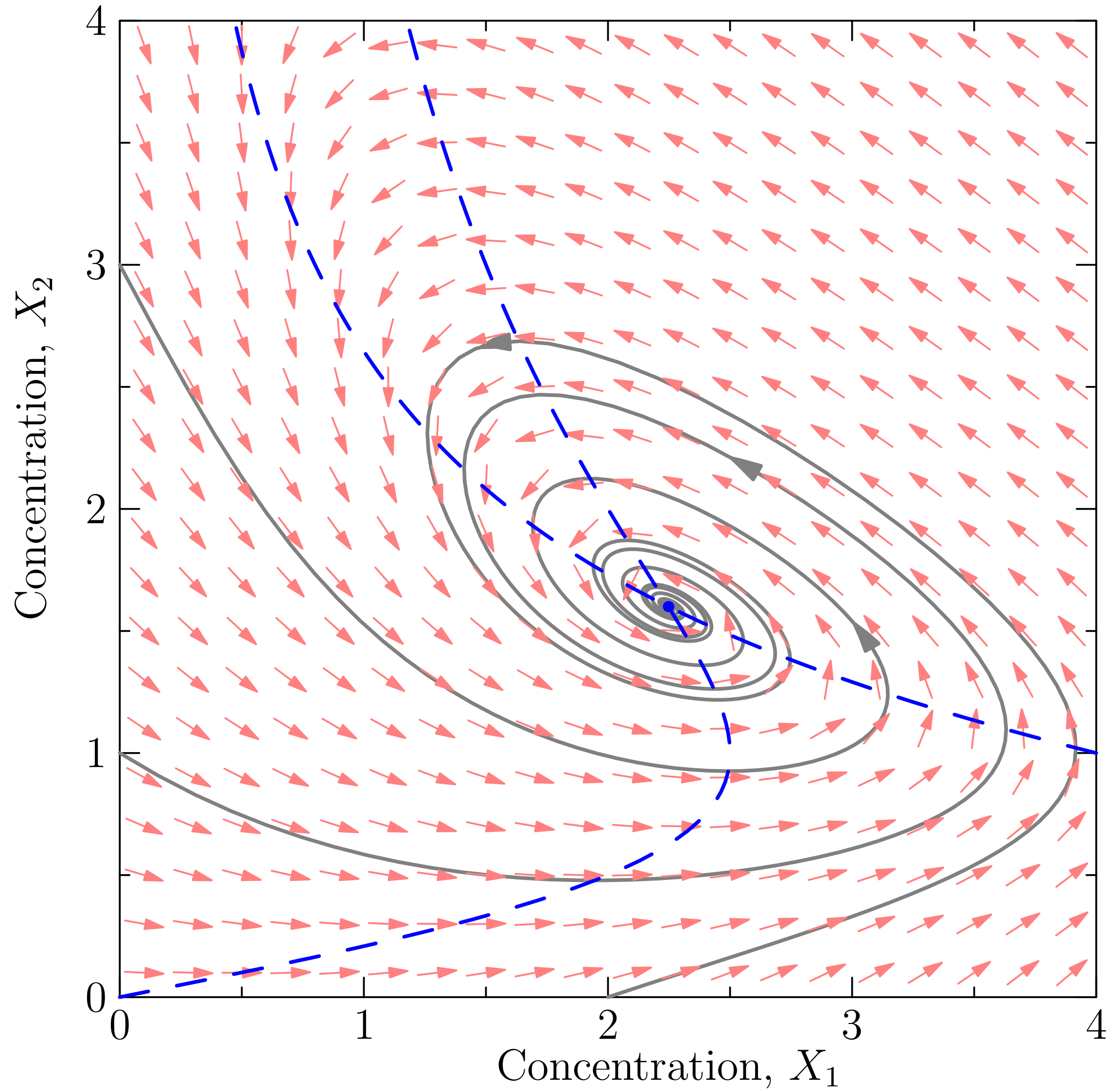
Nullclines and Bistablity



Nullclines and Bistablity



Nullclines and Bistablity



Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$

- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$

- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$
- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Fixed Point Equations

- Our dynamical equations are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = f_1(\boldsymbol{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = f_2(\boldsymbol{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$

- The fixed point is at $\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$

- The Jacobian $J_{ij} = \partial f_i(\boldsymbol{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K} \left(\frac{X_2^*}{K}\right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11} J_{22} - J_{12} J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11} J_{22} - J_{12} J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \mathrm{tr}\,\mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11}\,J_{22} - J_{12}\,J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

Eigen Decompostion

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

- where

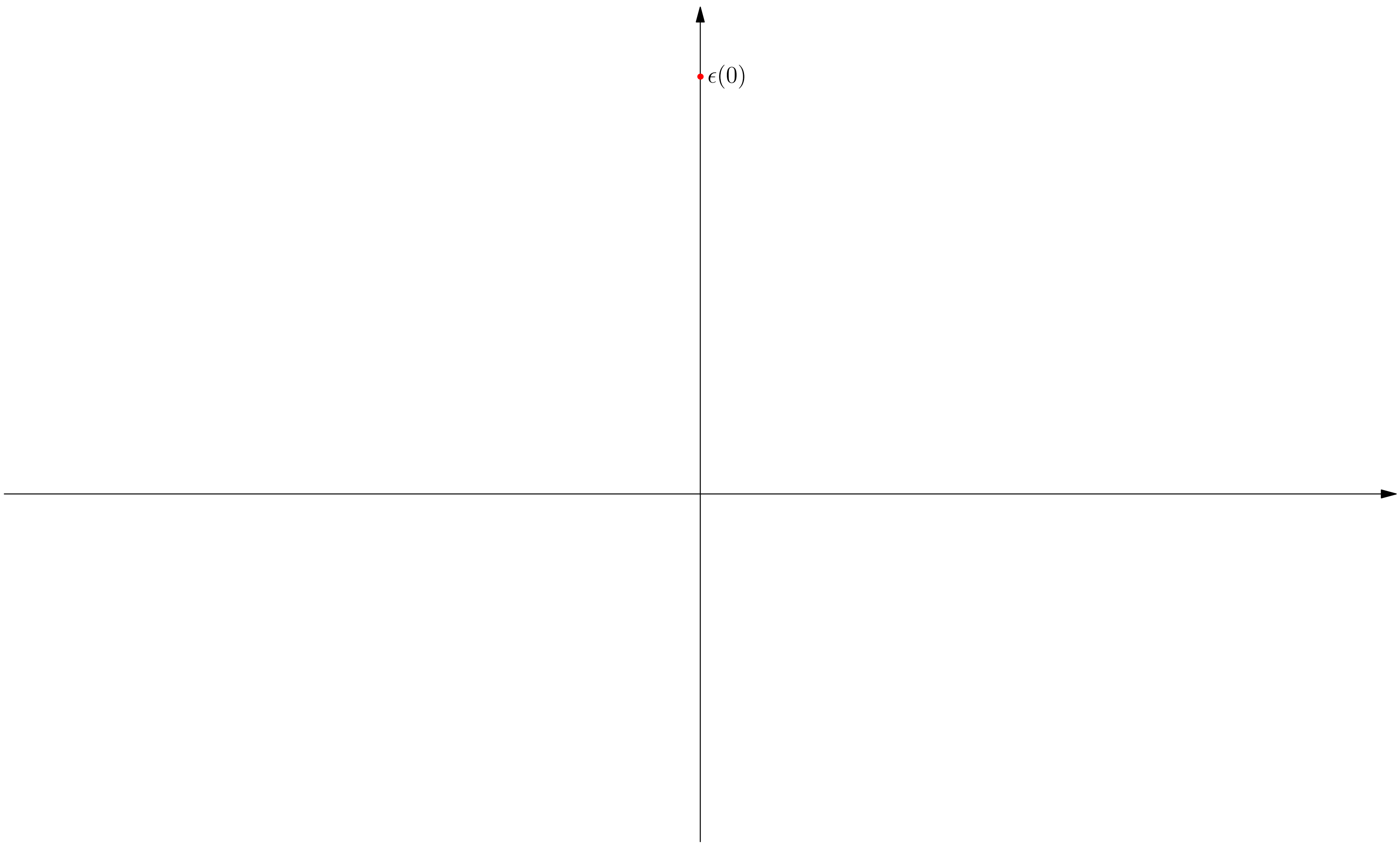
$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- Or $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11}\,J_{22} - J_{12}\,J_{21}$ then $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$

- The eigenvalues have a large complex component which causes a significant spiralling

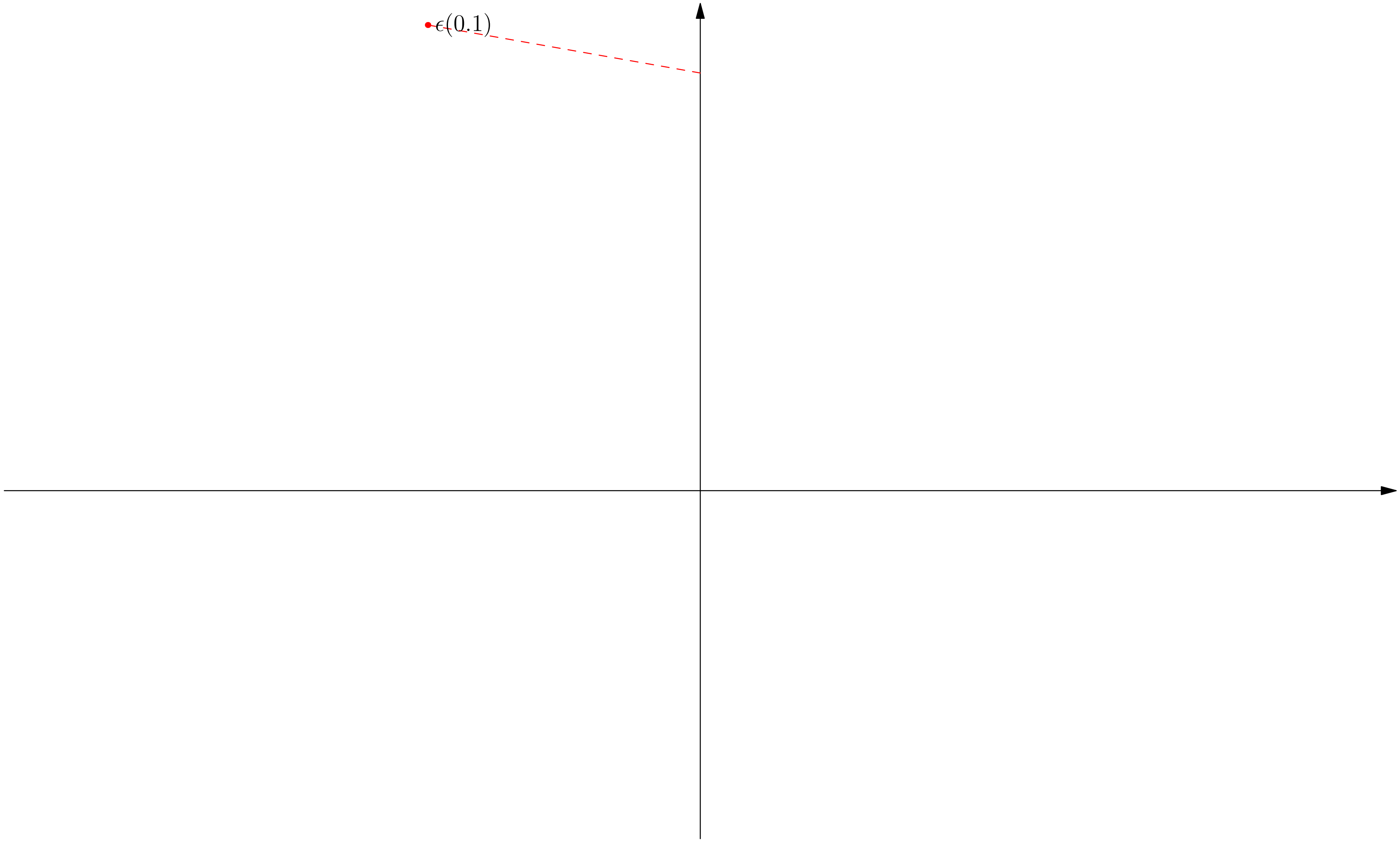
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



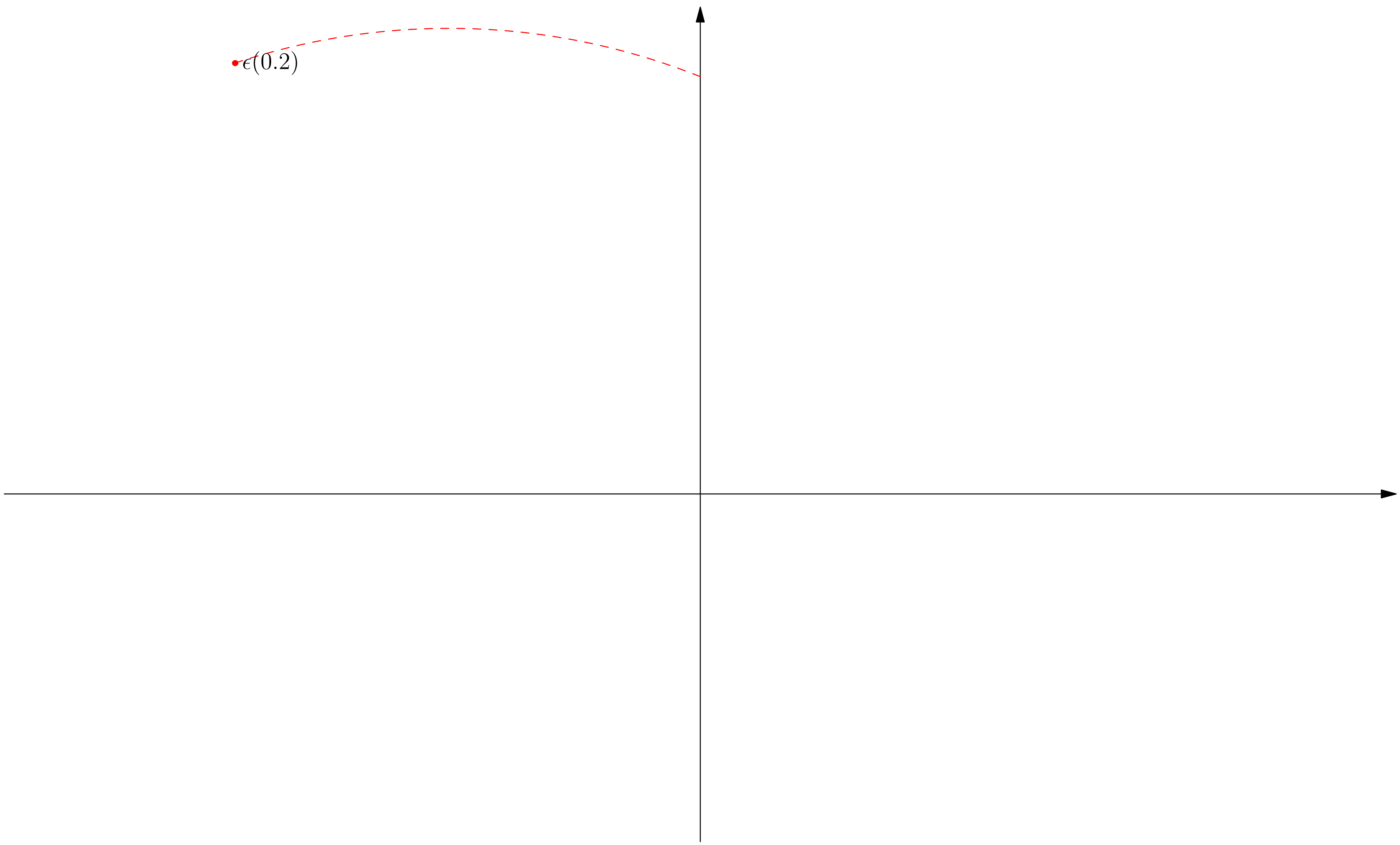
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



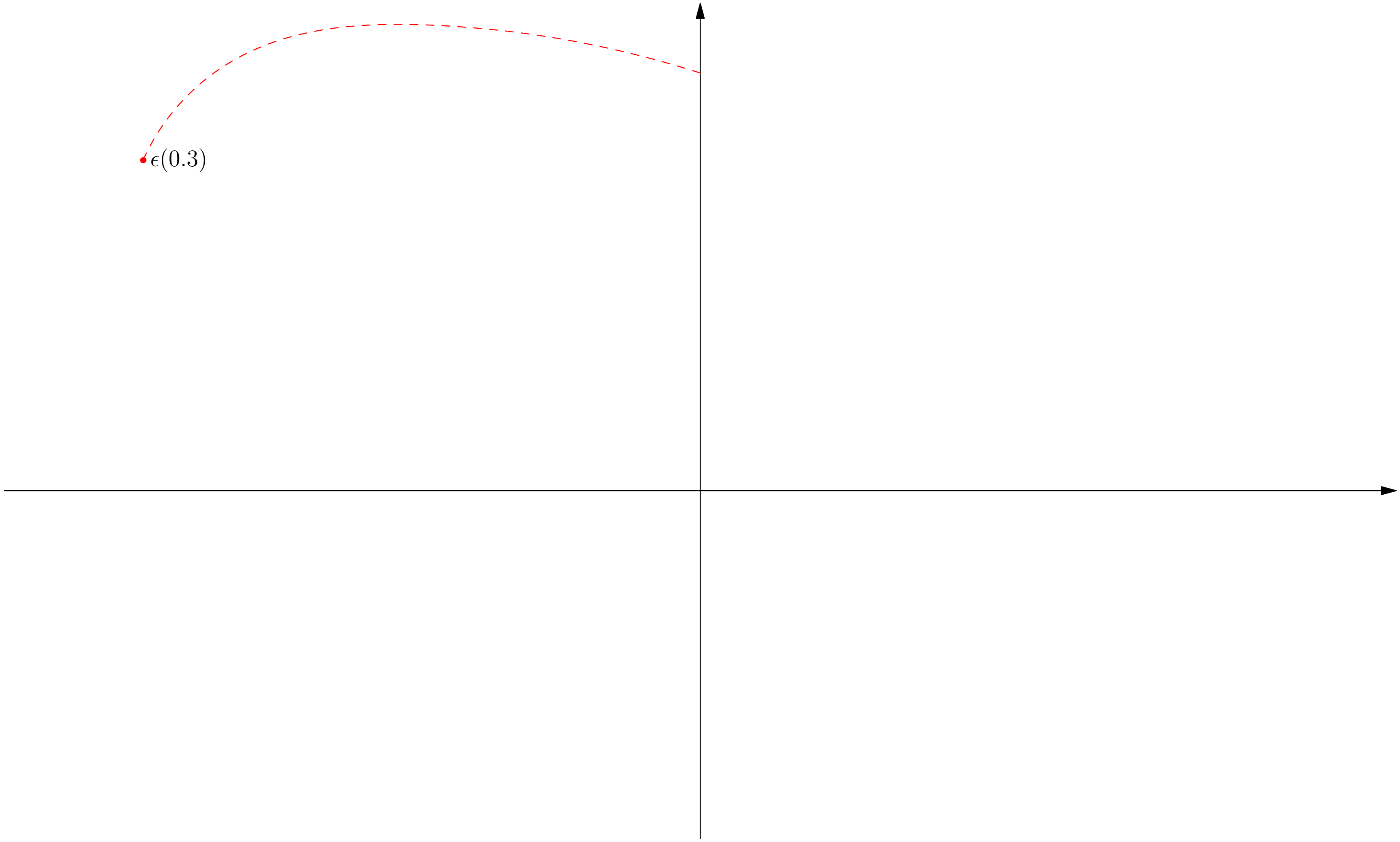
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



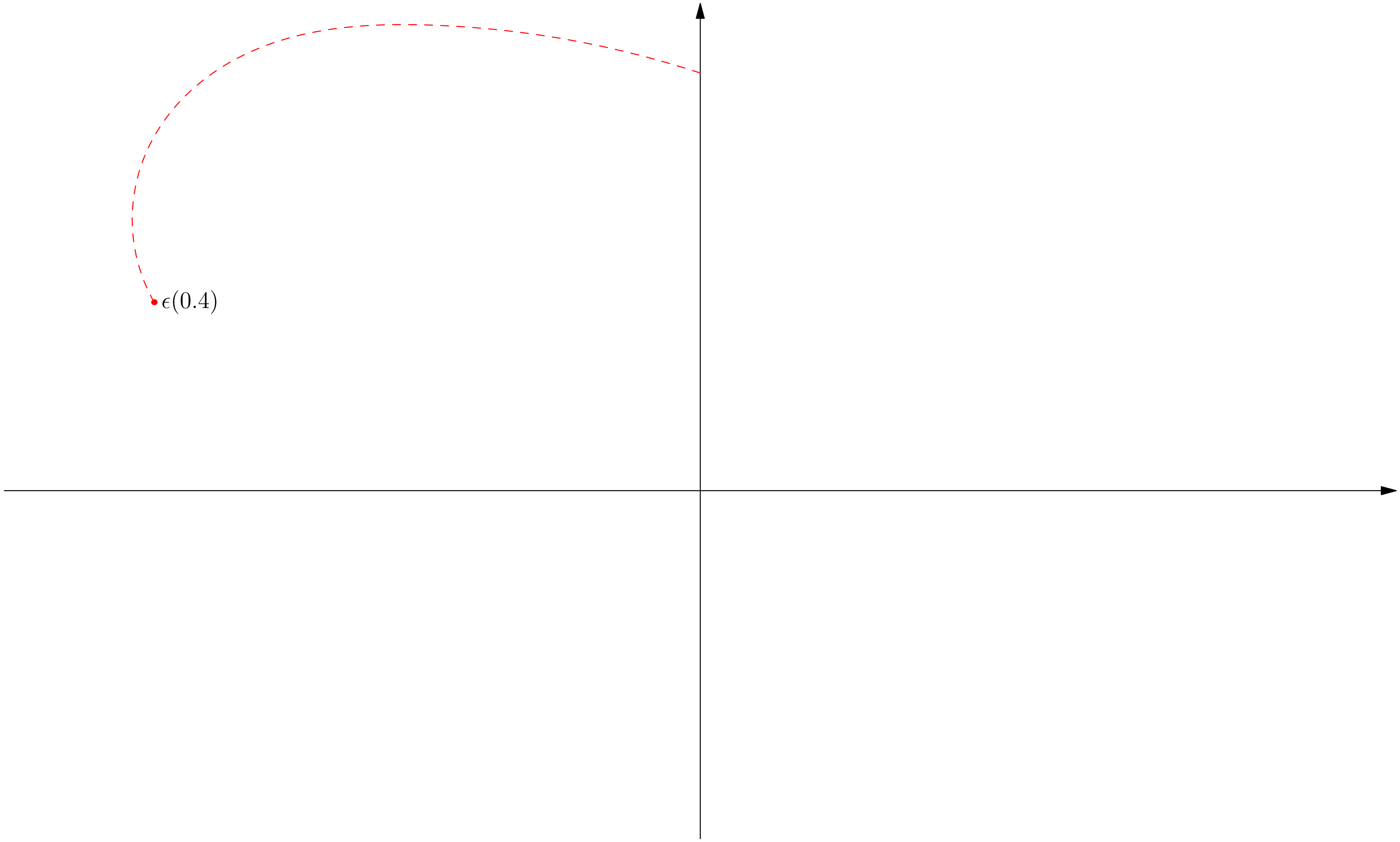
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



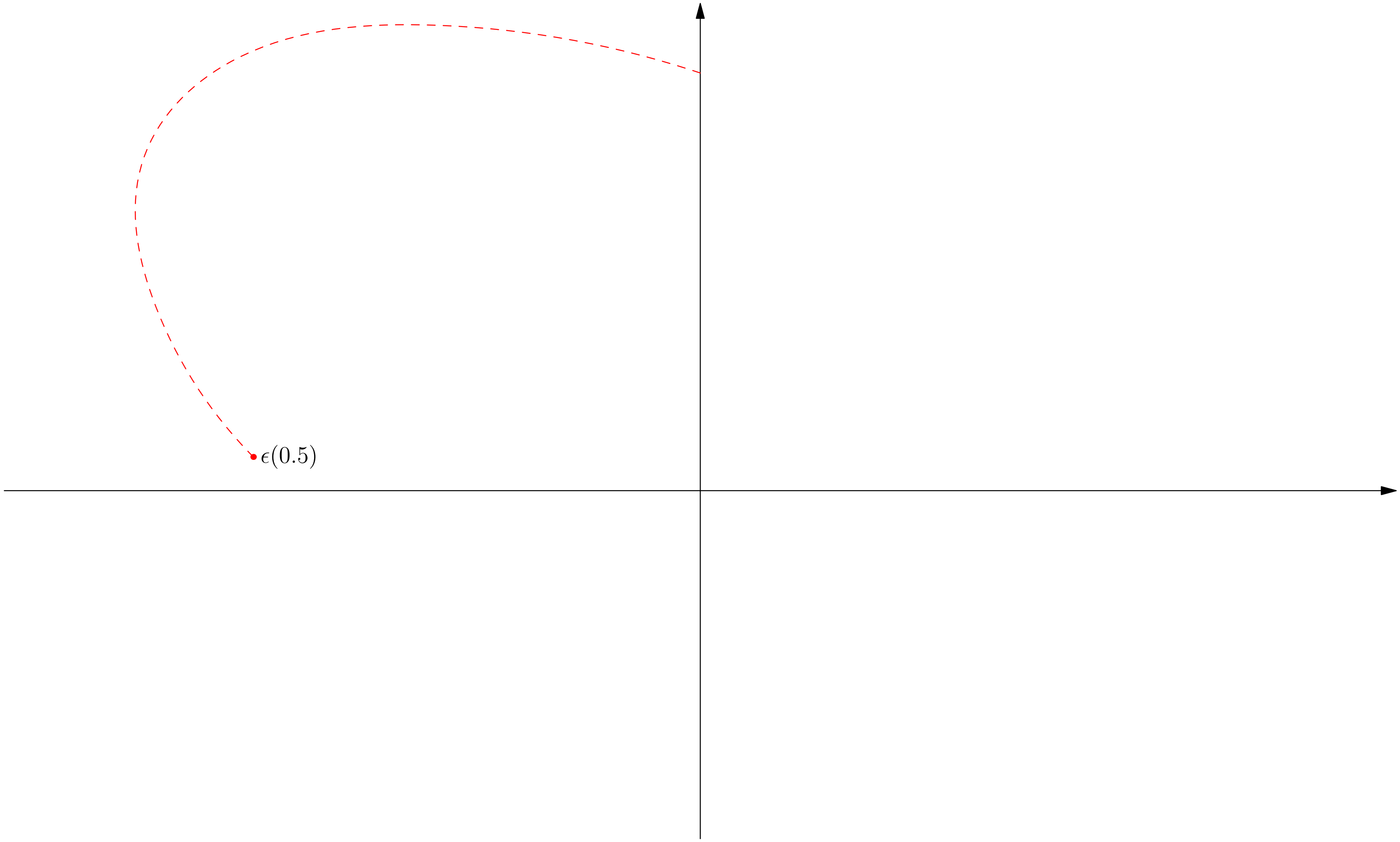
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



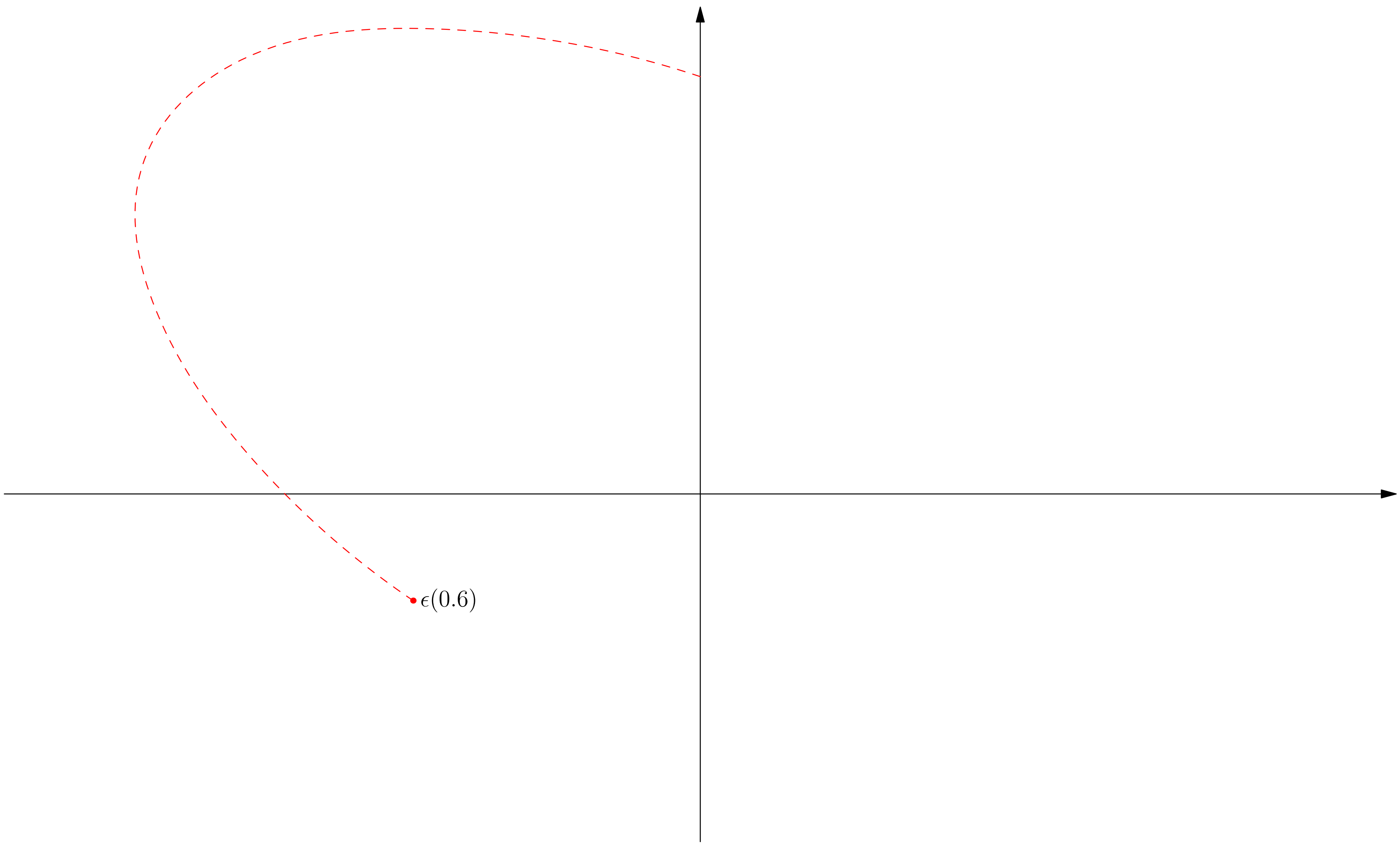
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



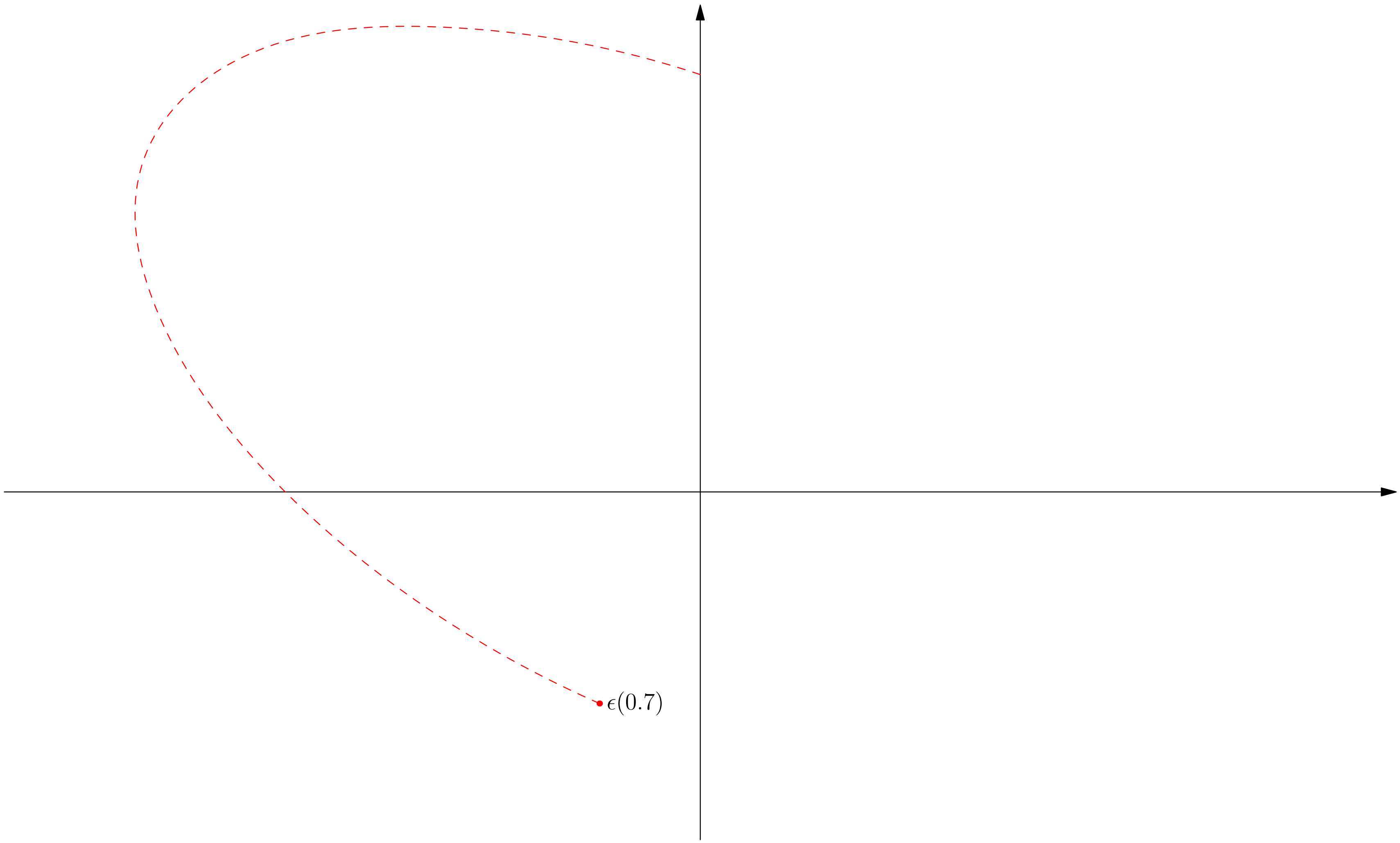
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{\frac{-89}{25}}{\frac{89}{25}} & \frac{\frac{-640}{89}}{\frac{195}{89}} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



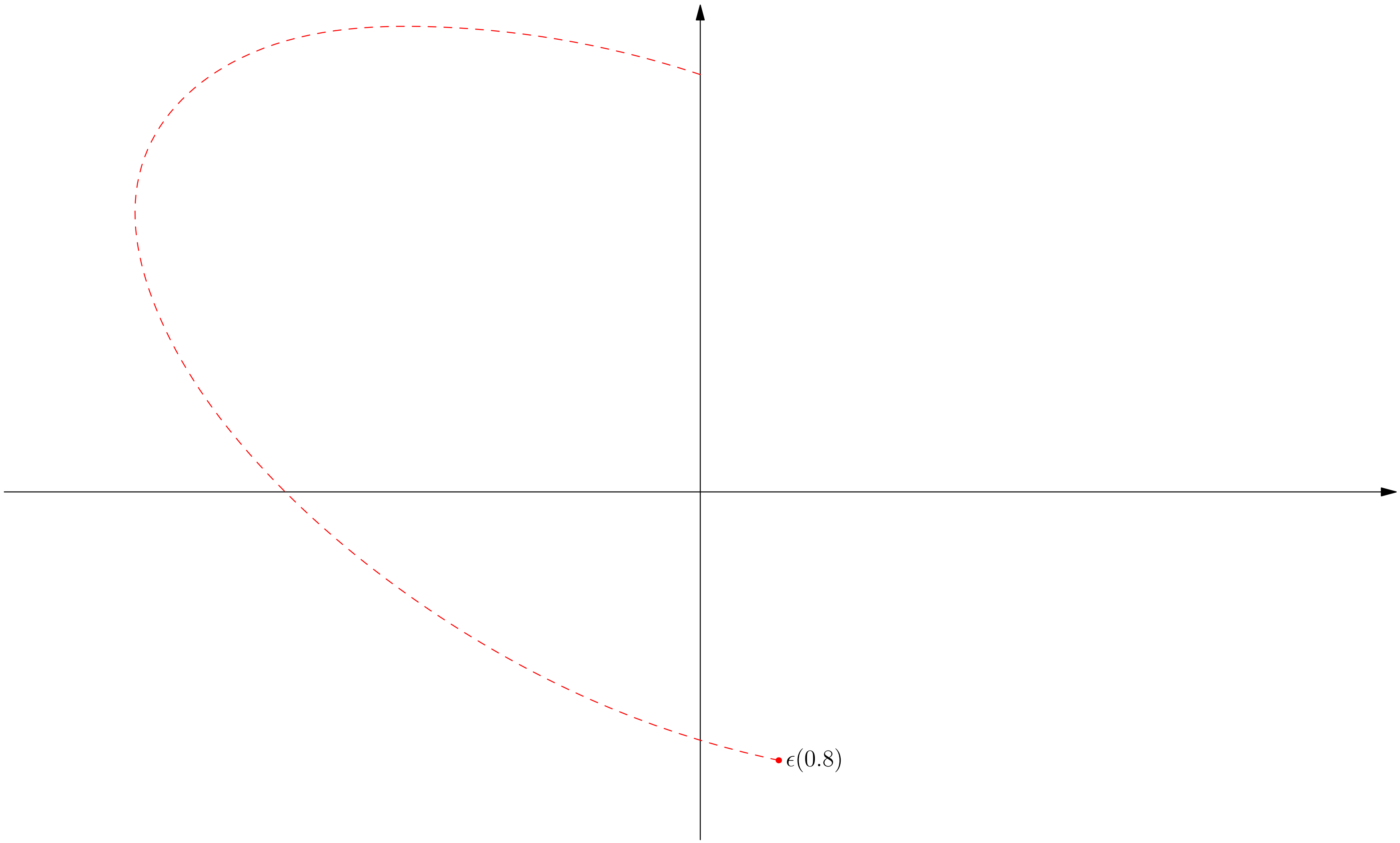
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



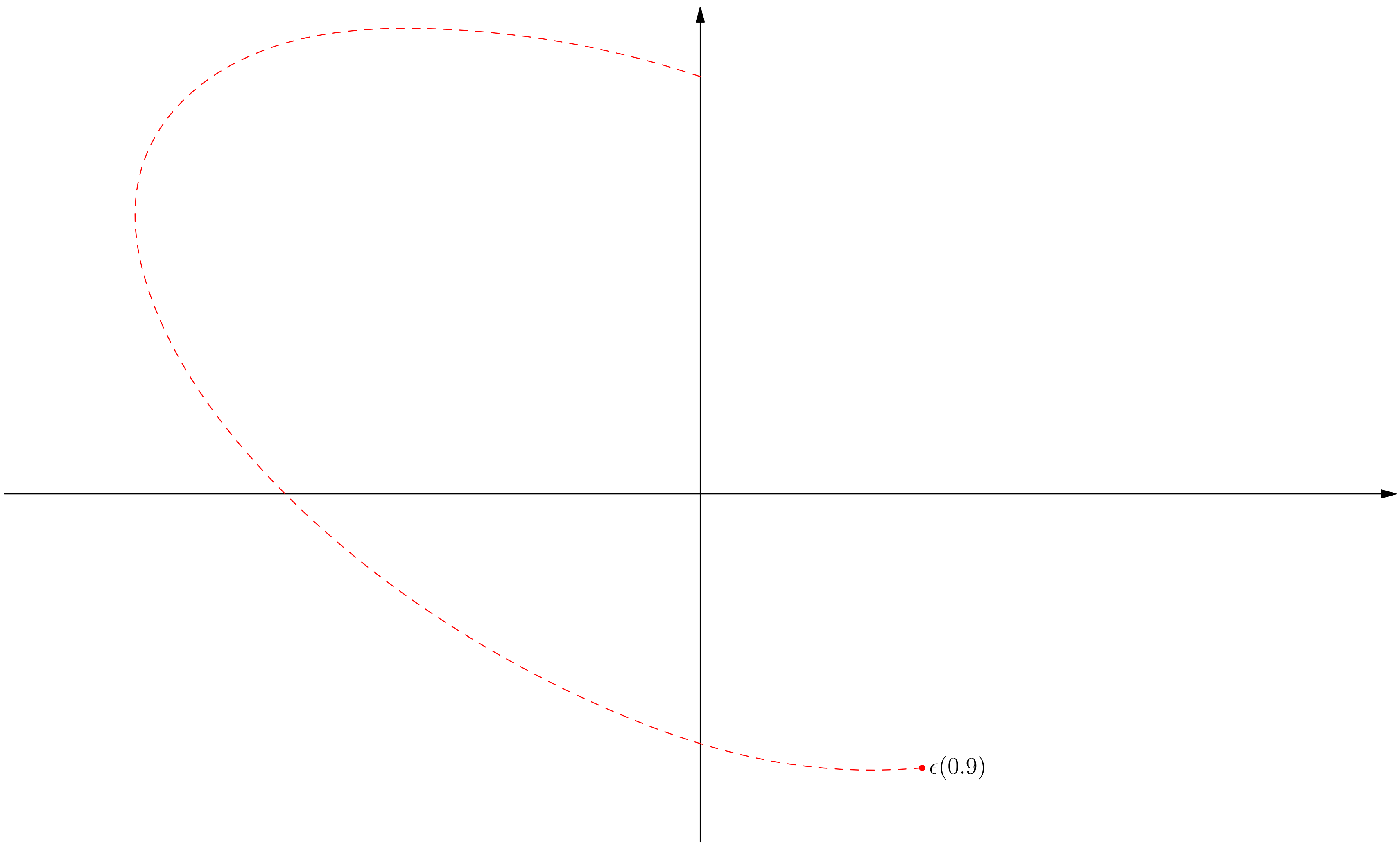
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



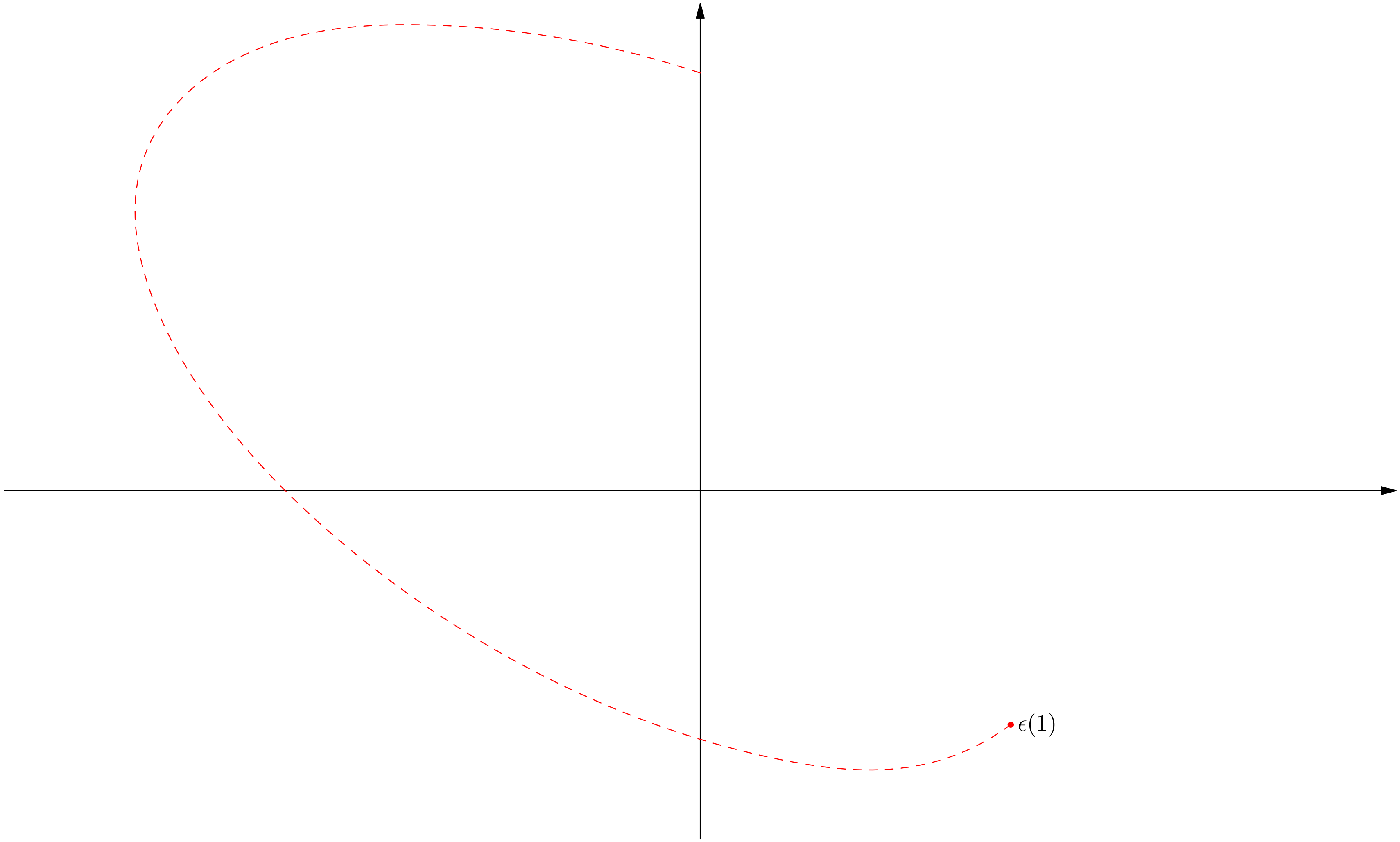
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



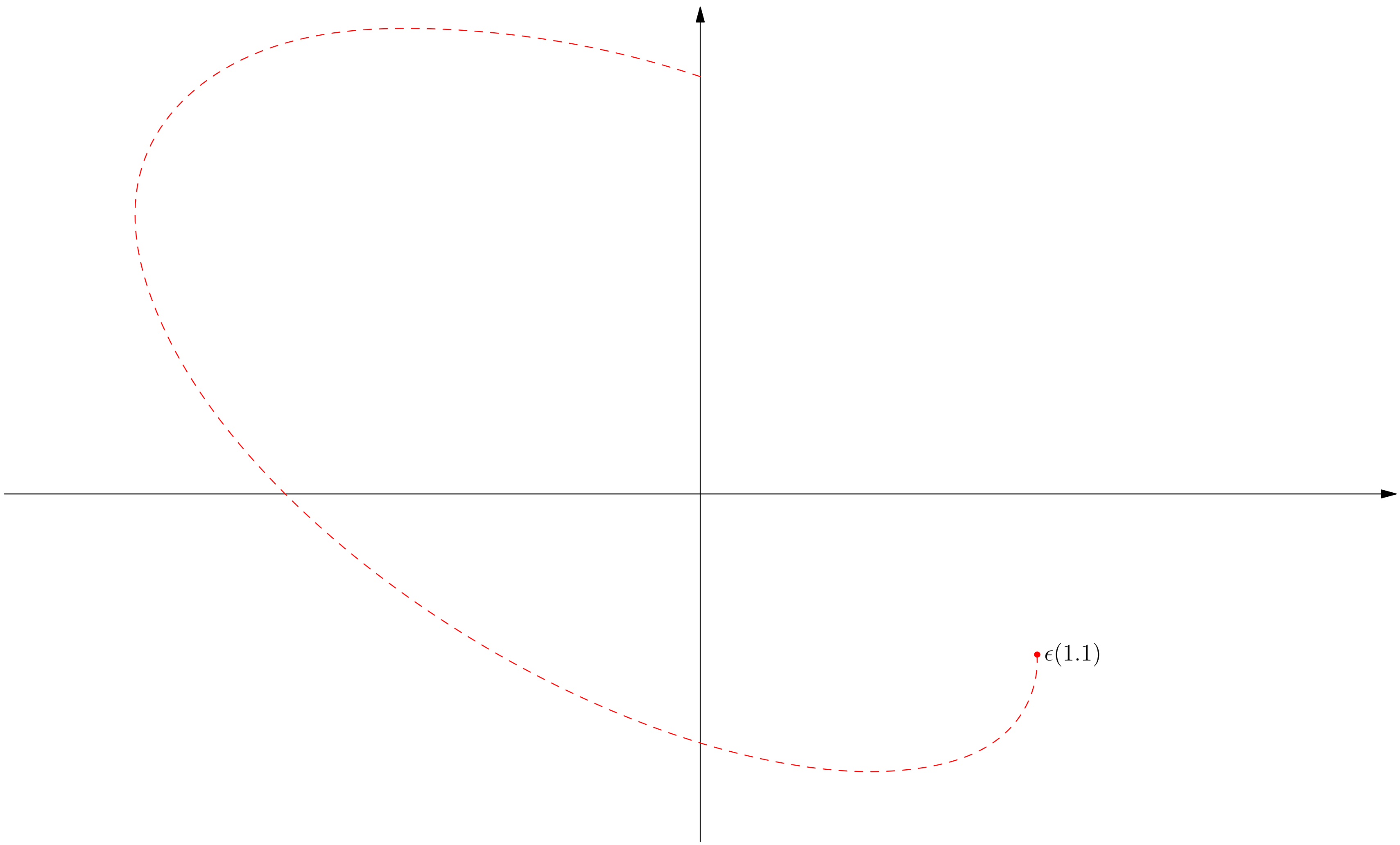
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



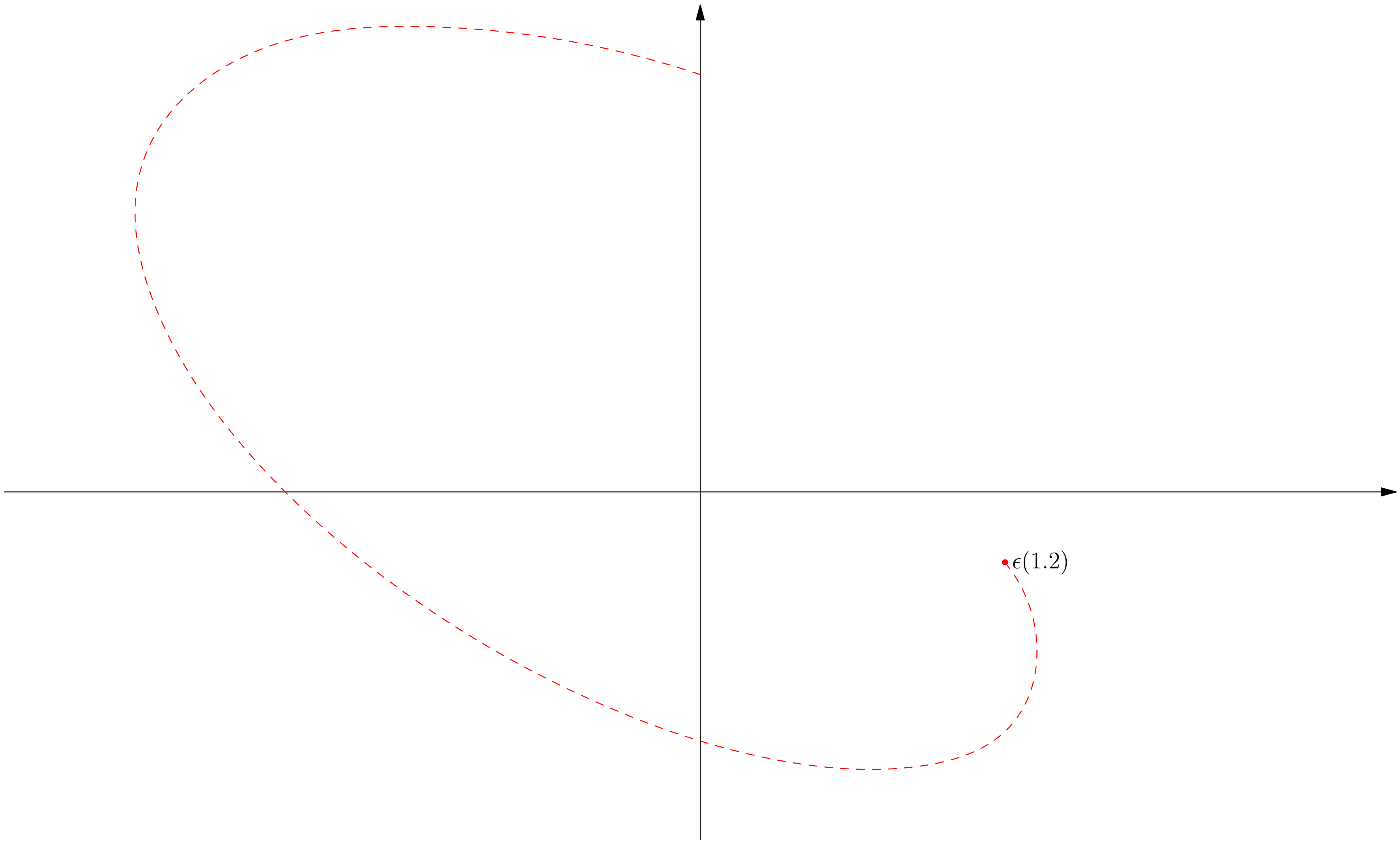
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



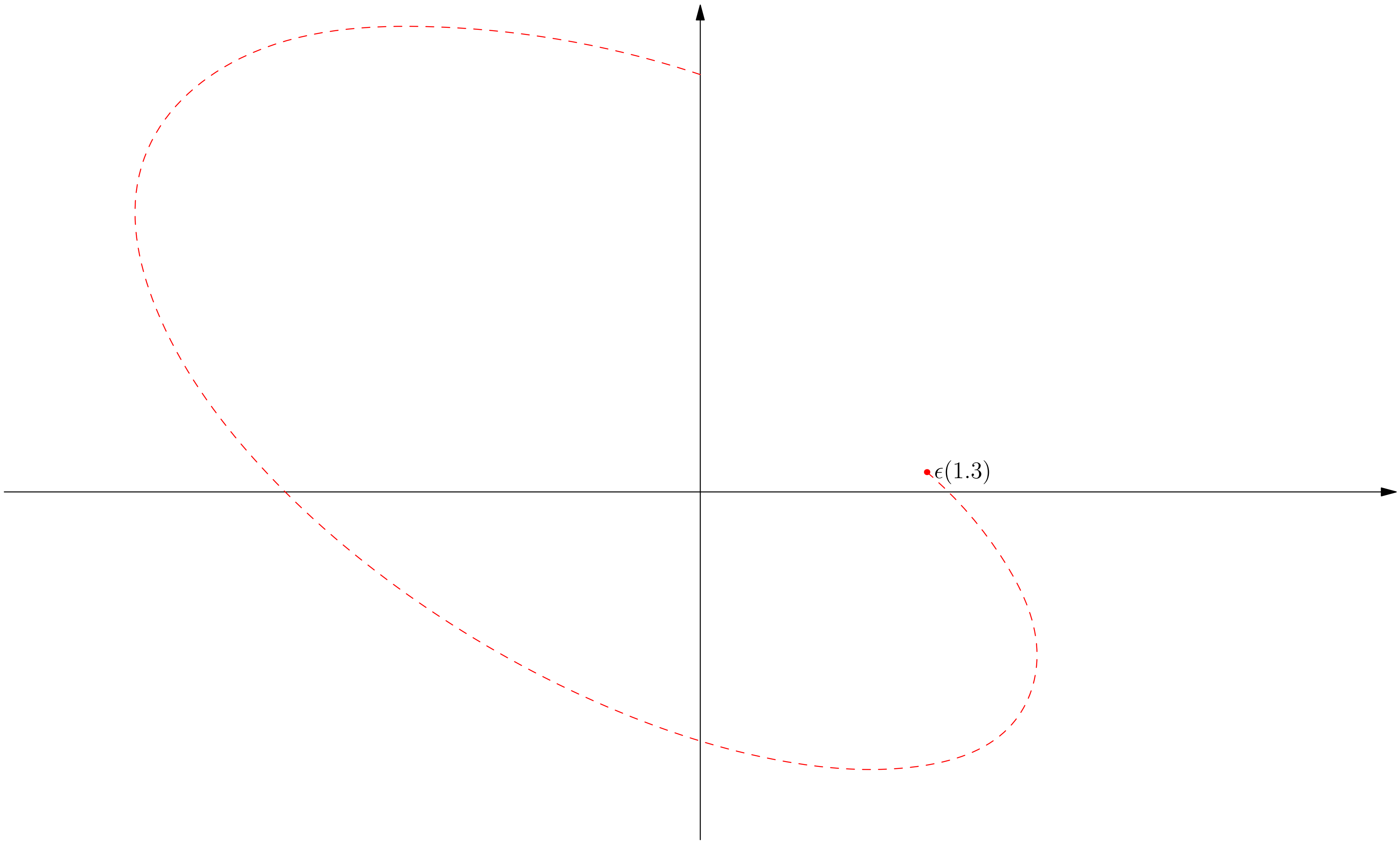
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



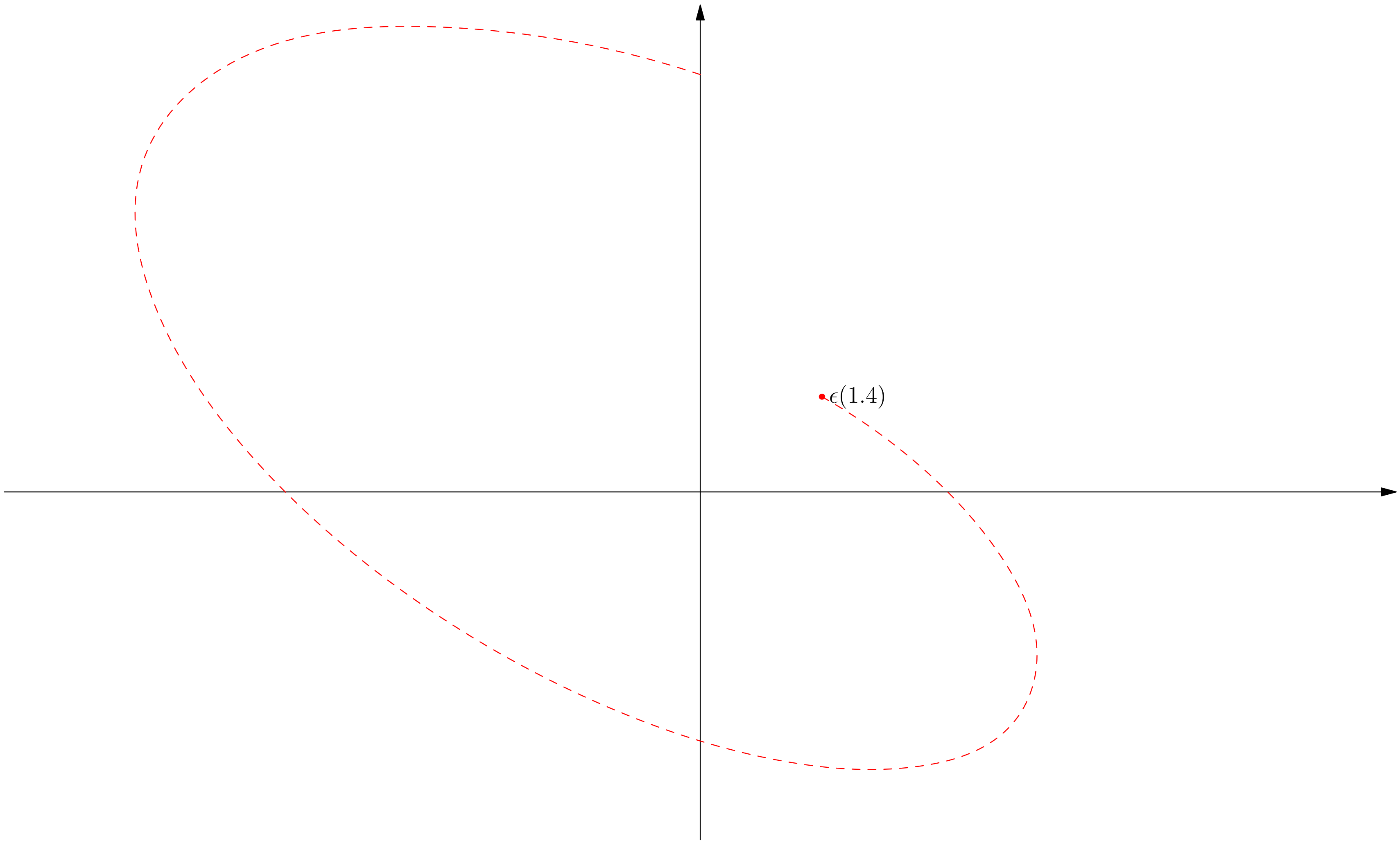
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



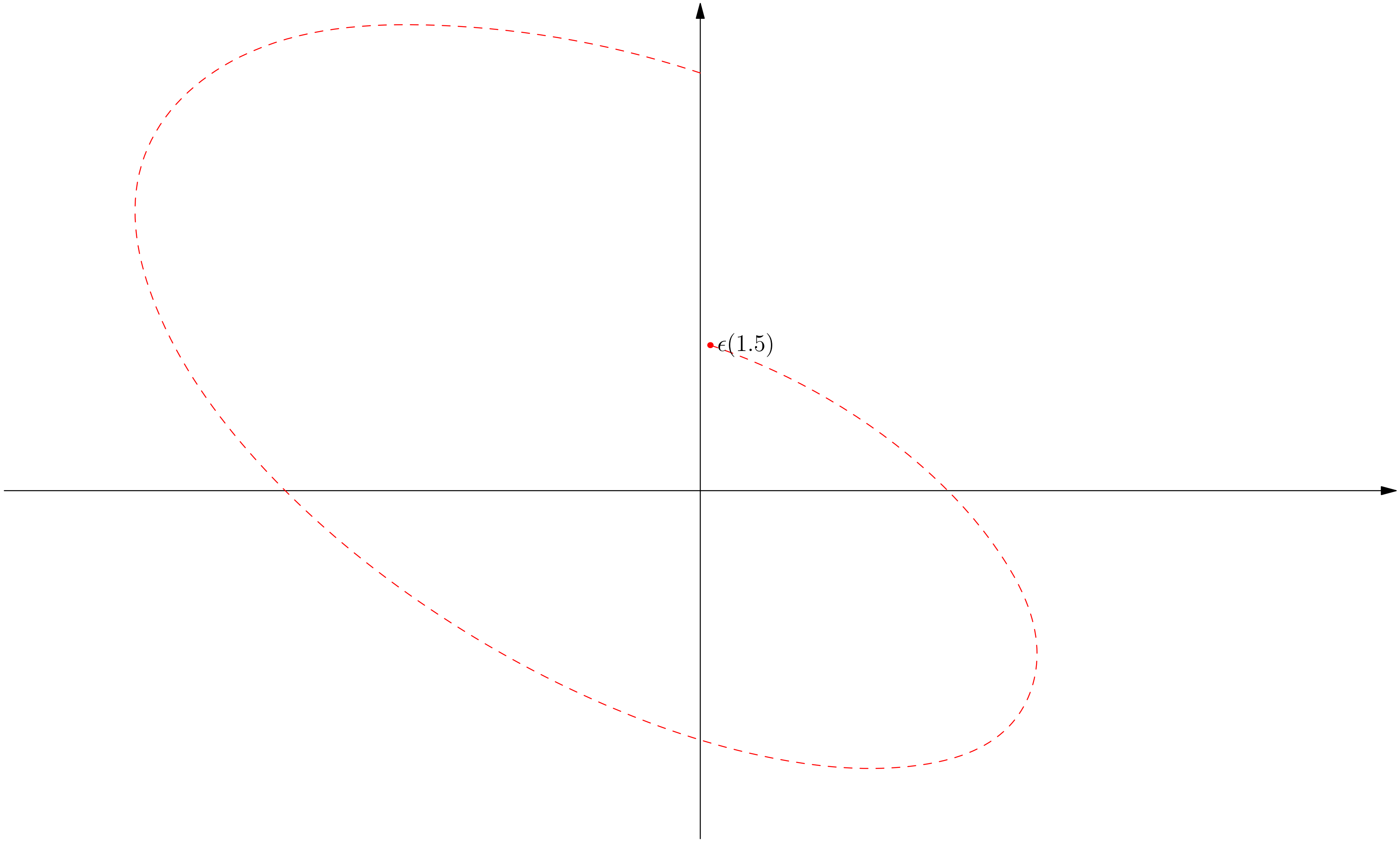
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



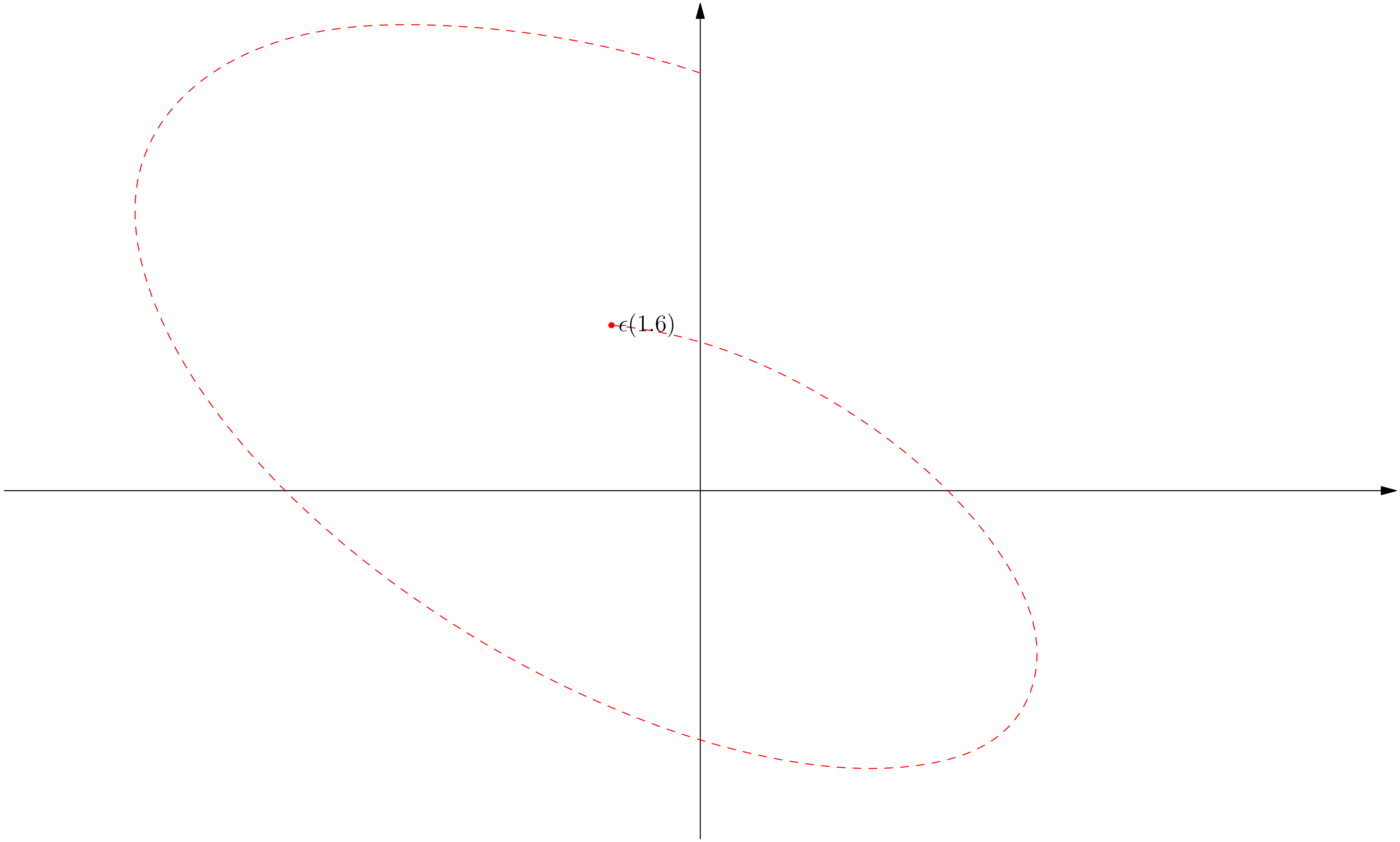
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



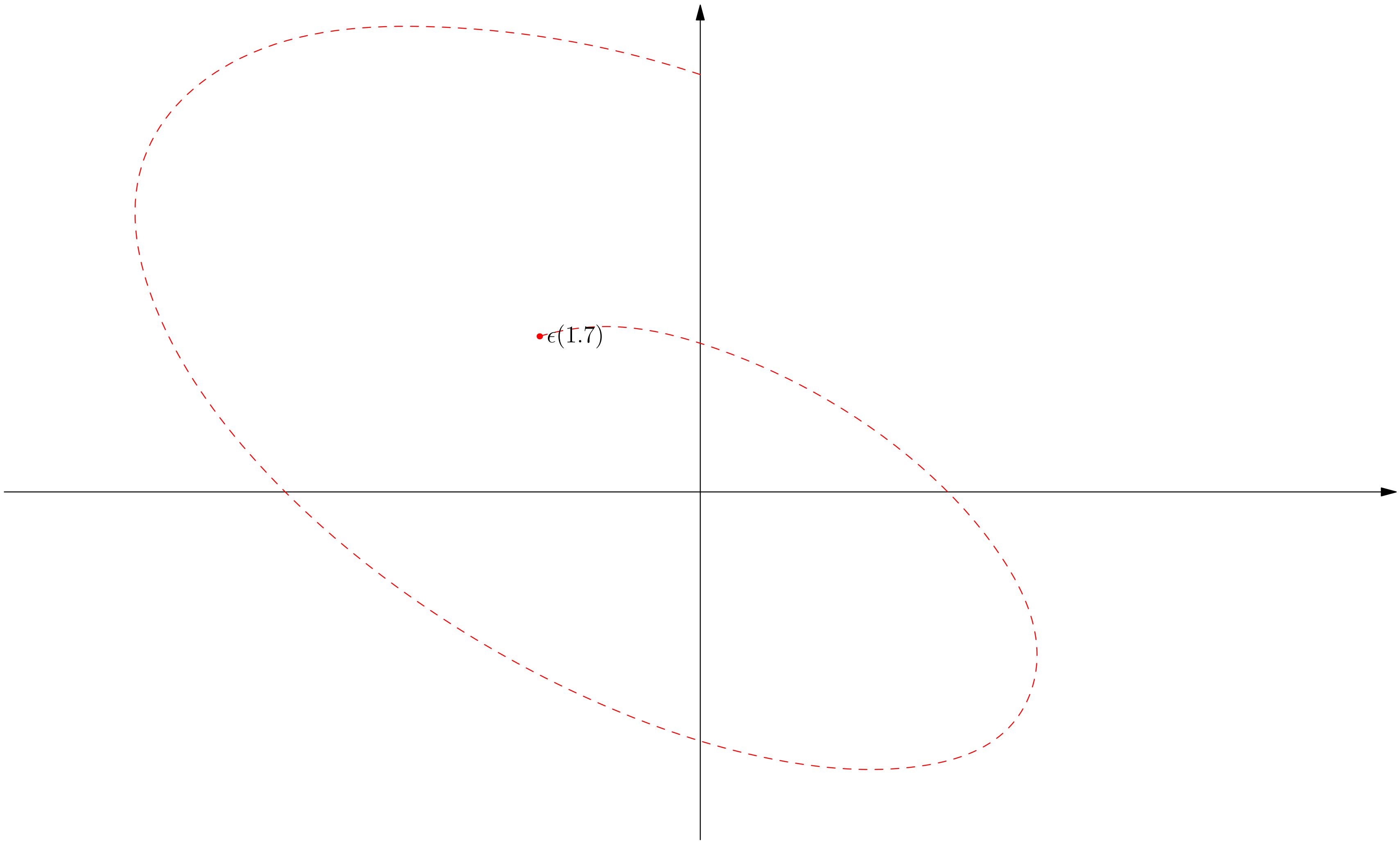
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



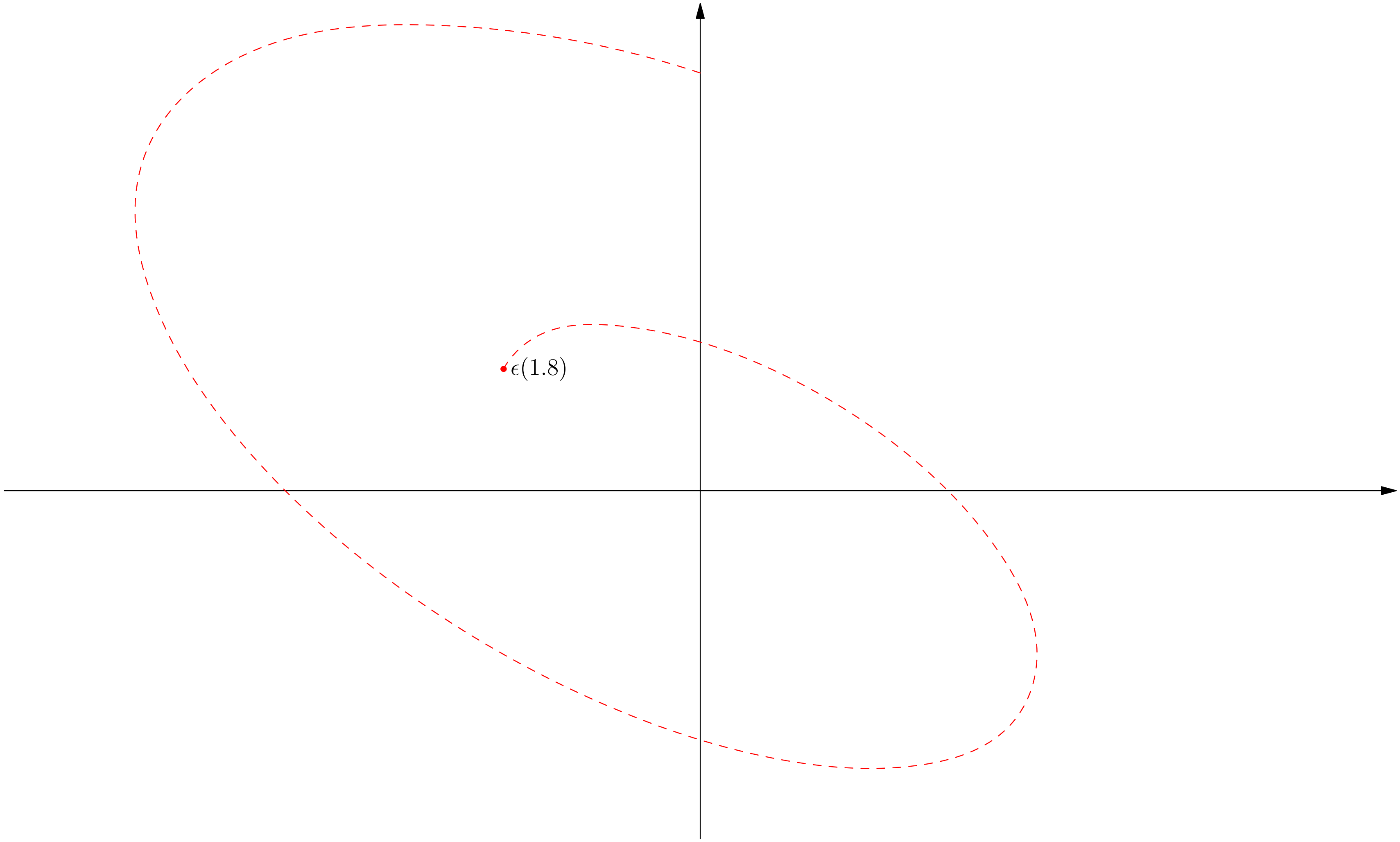
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



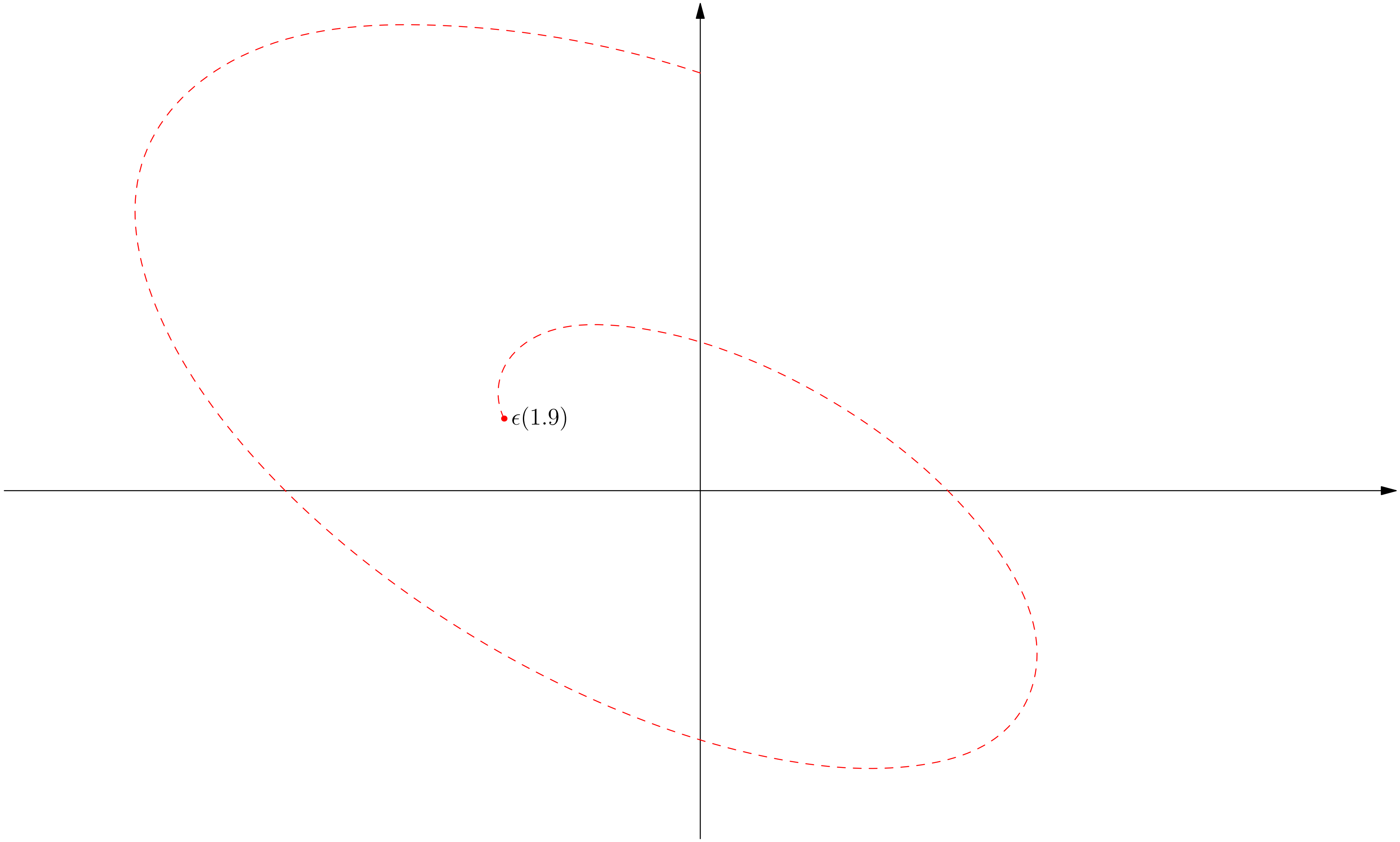
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



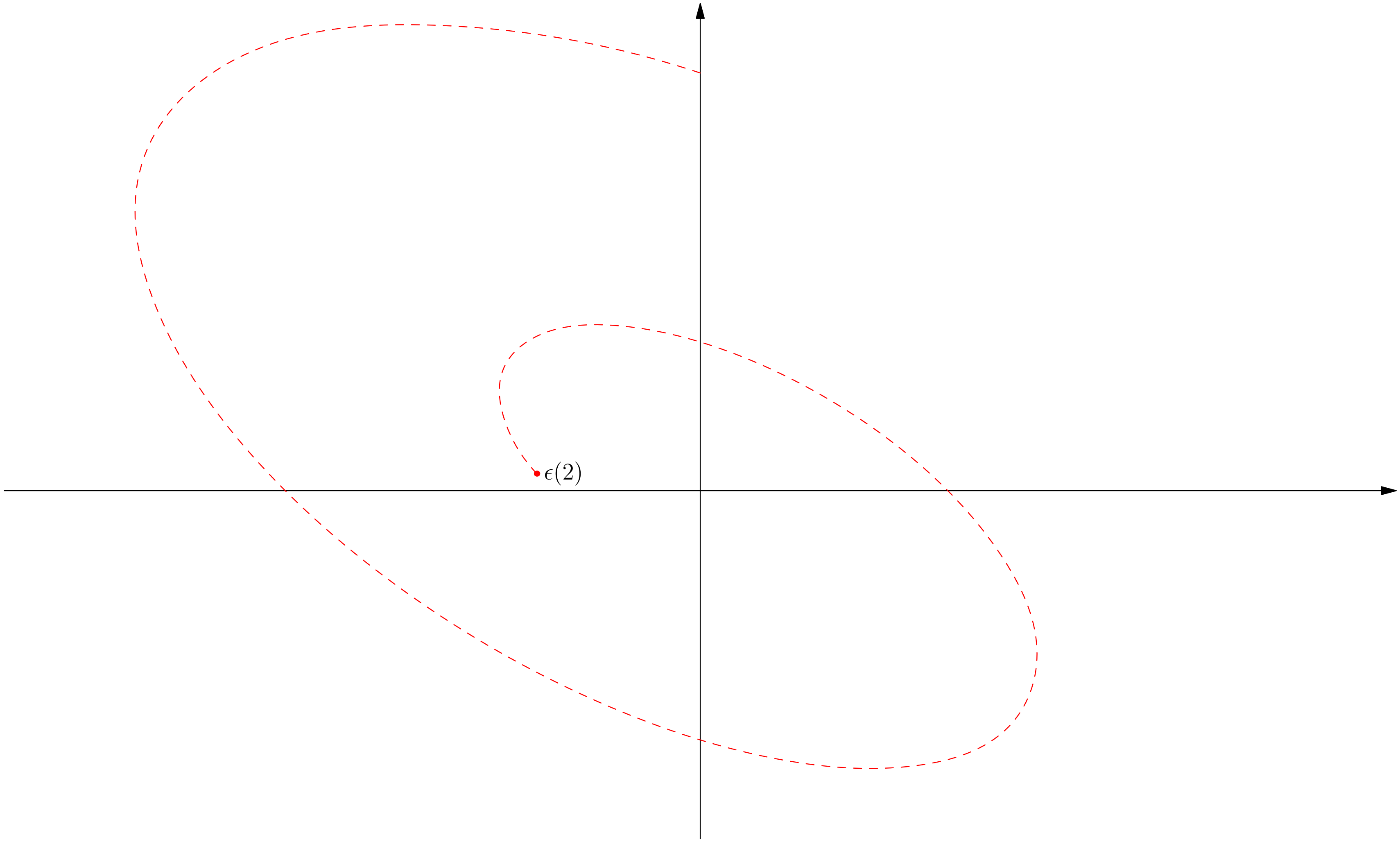
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



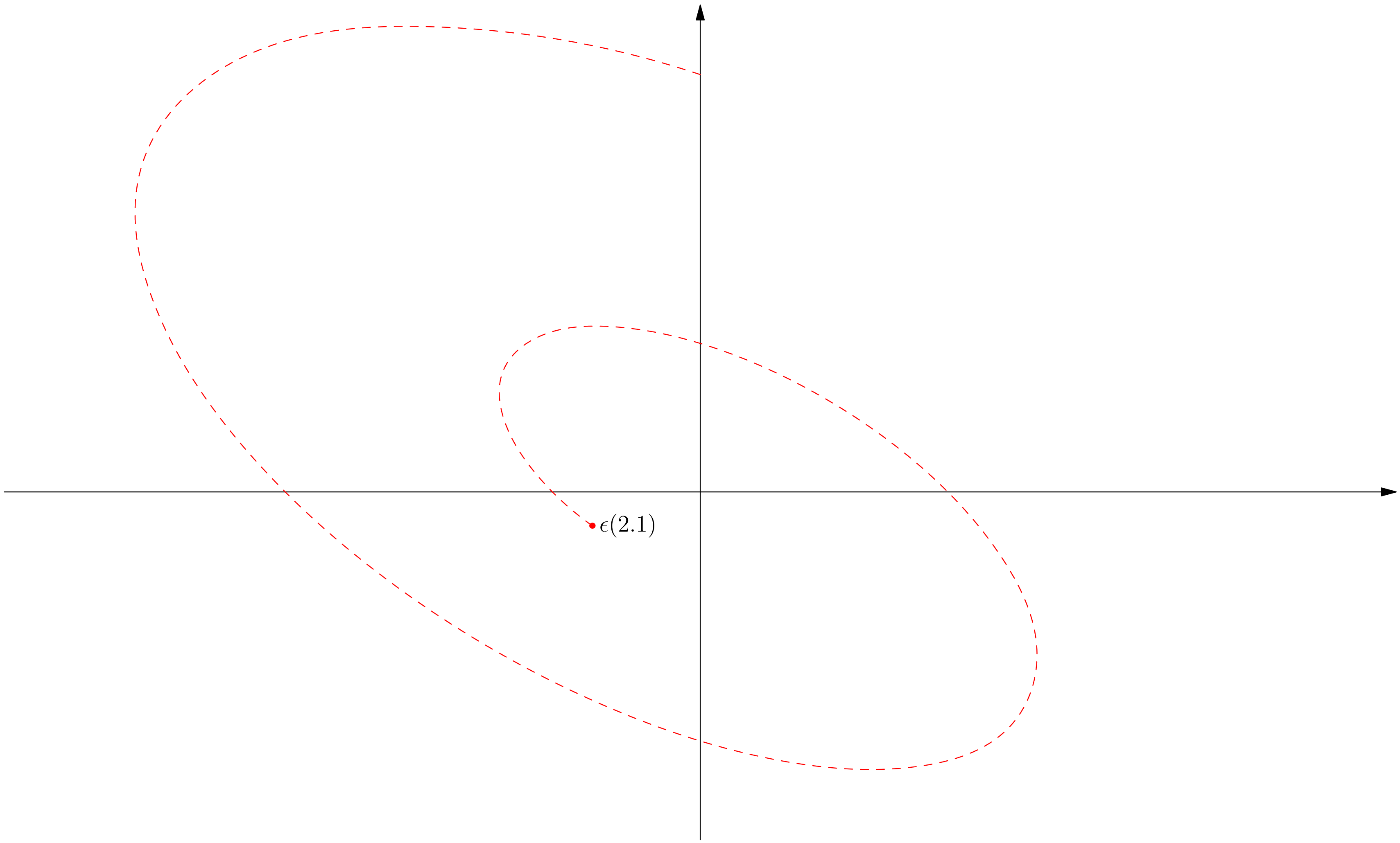
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



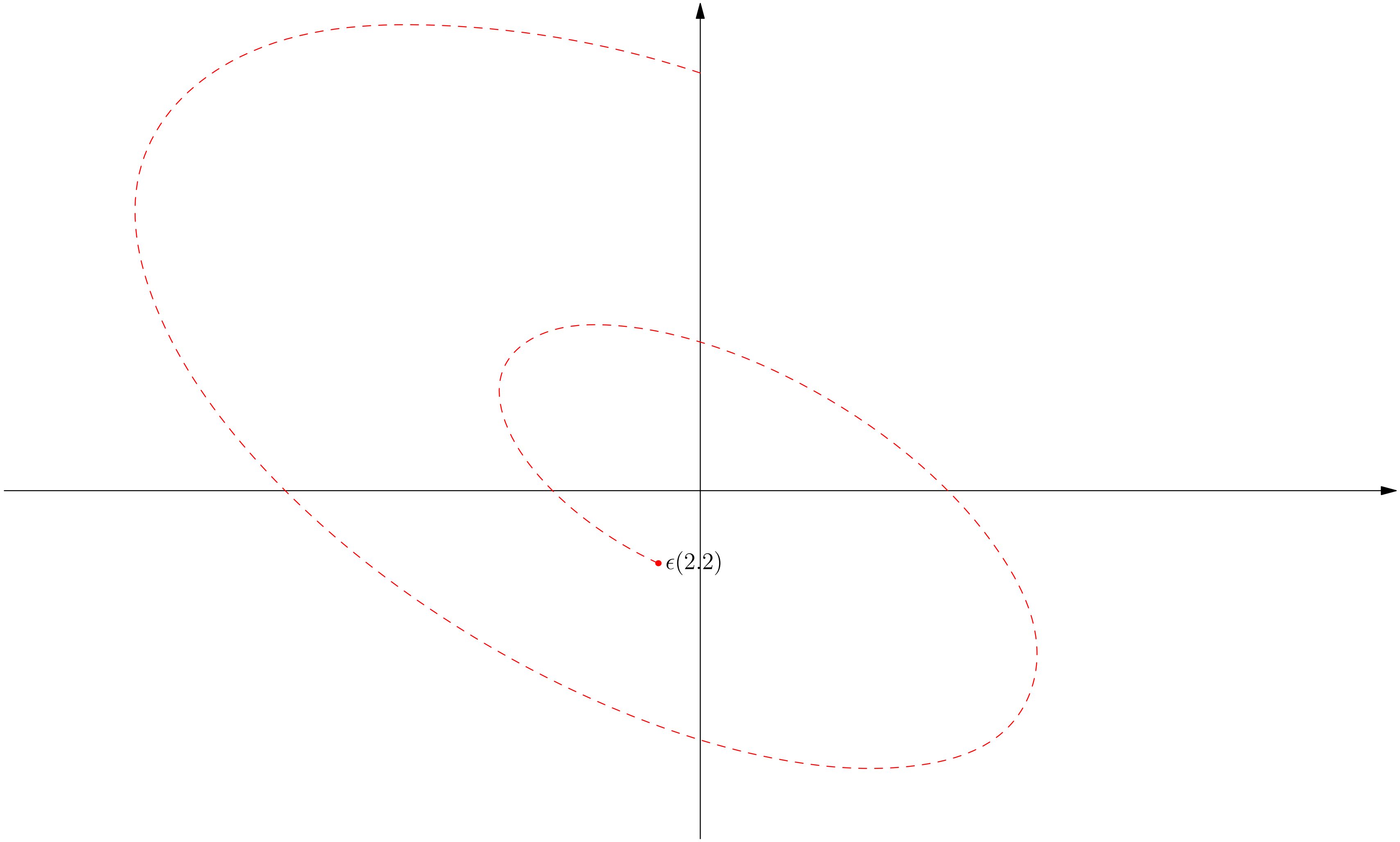
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



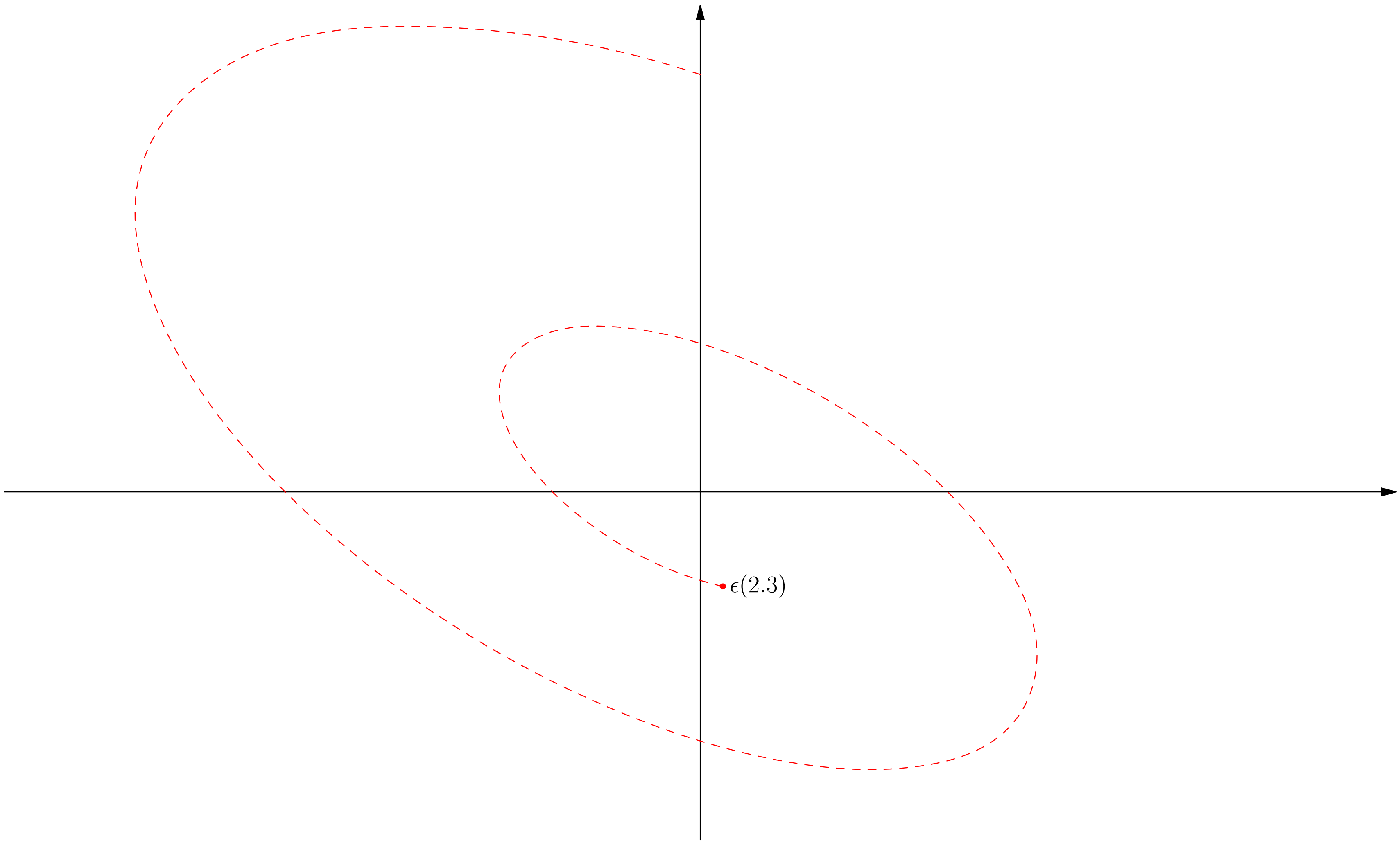
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



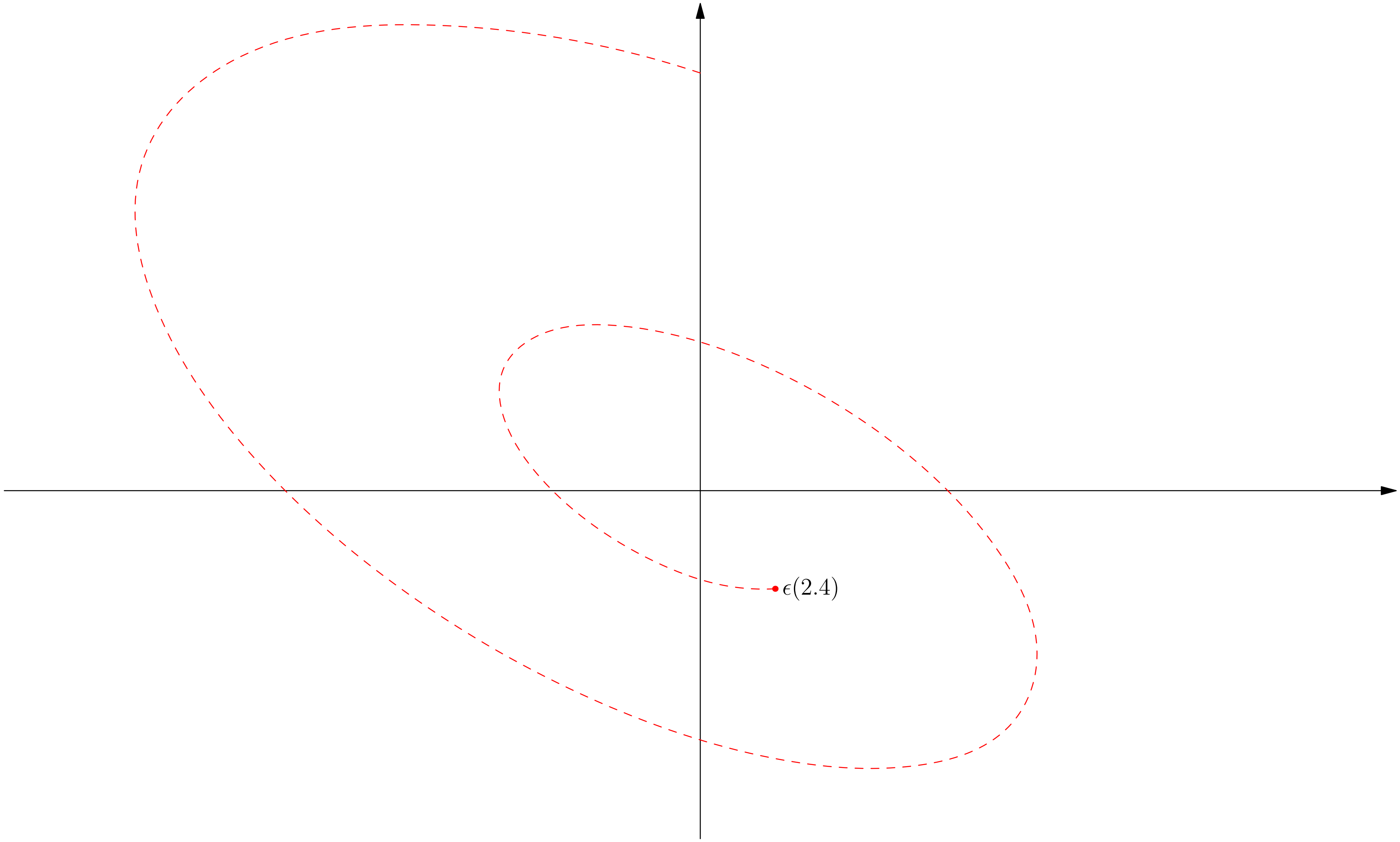
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



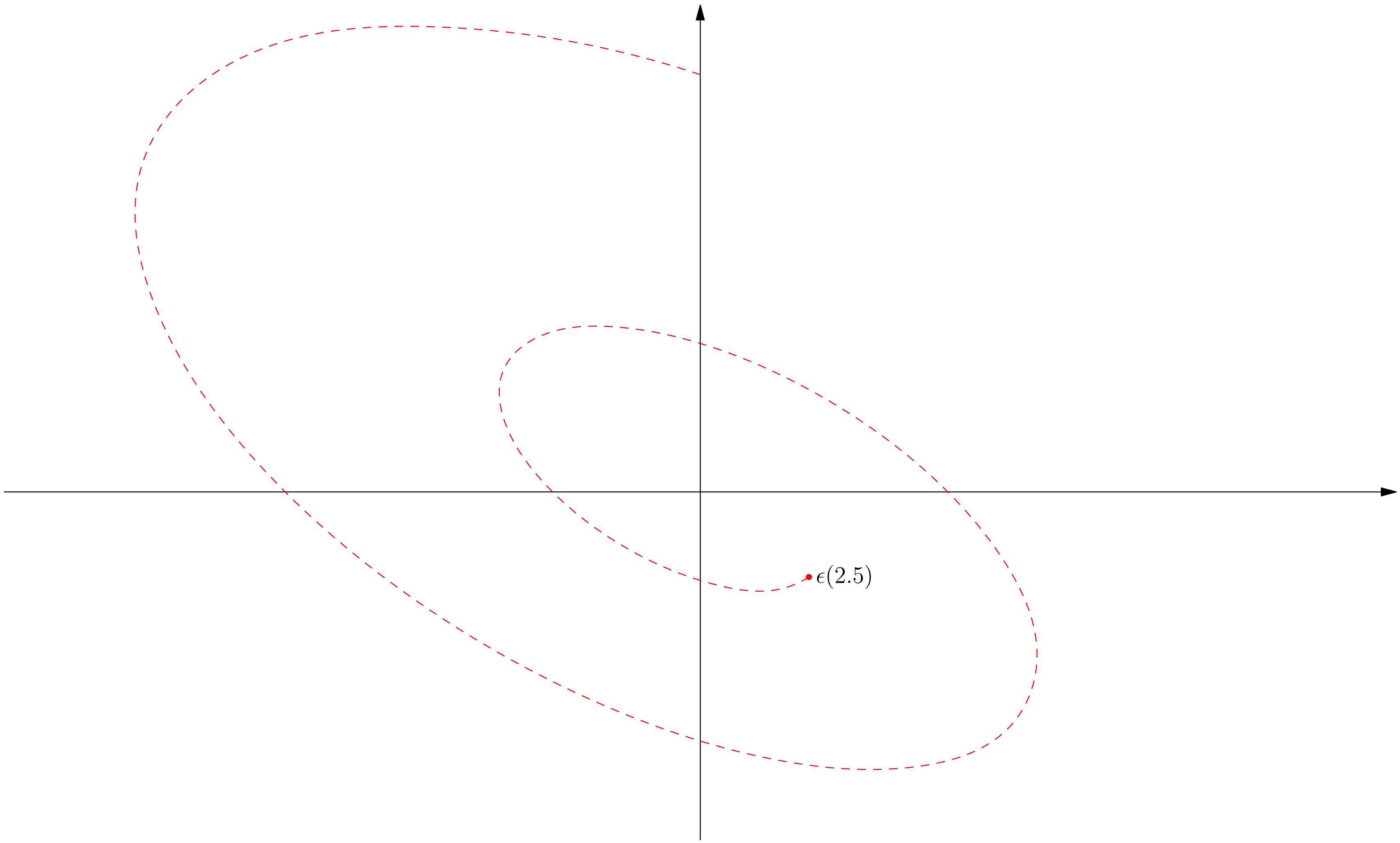
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



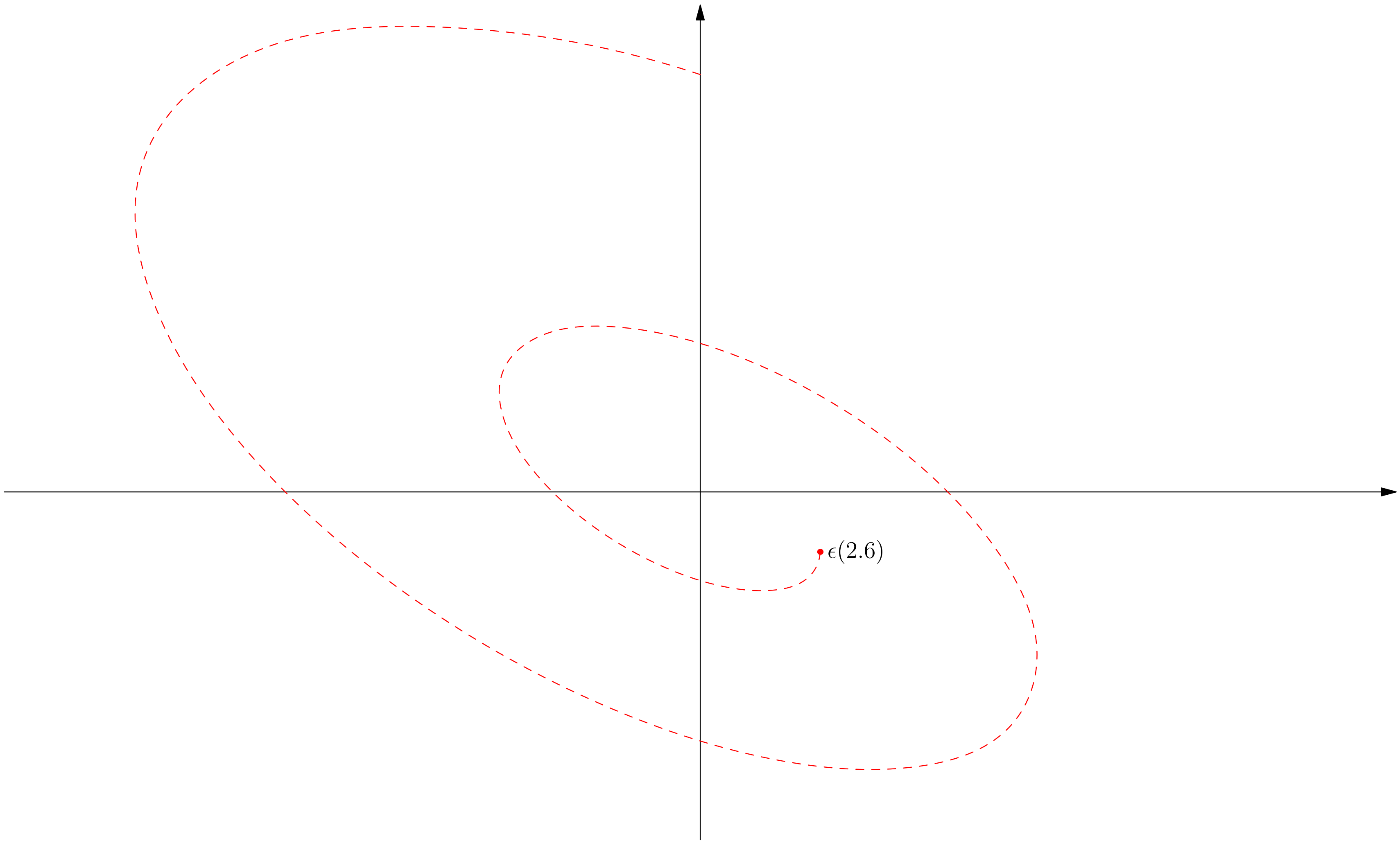
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



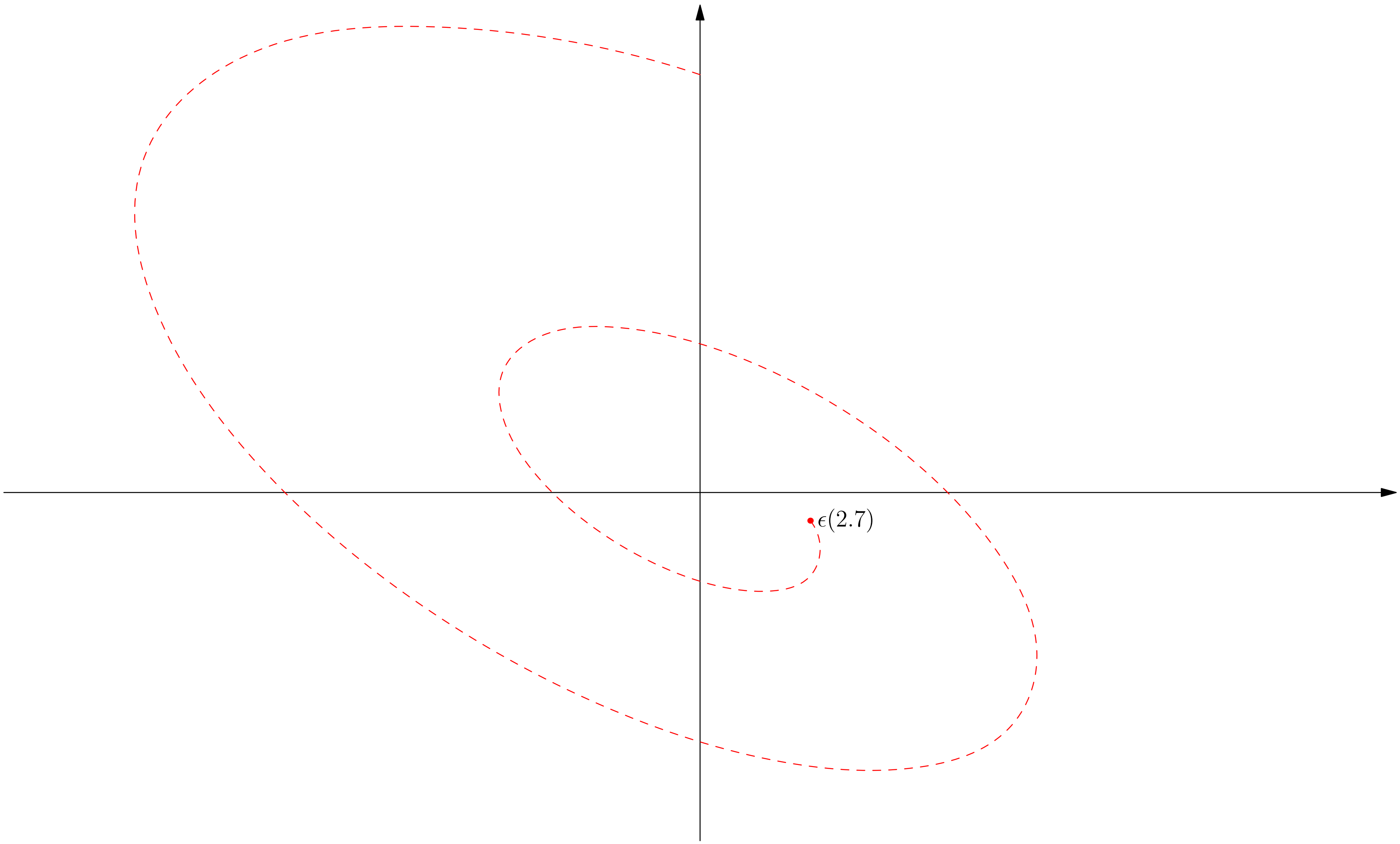
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



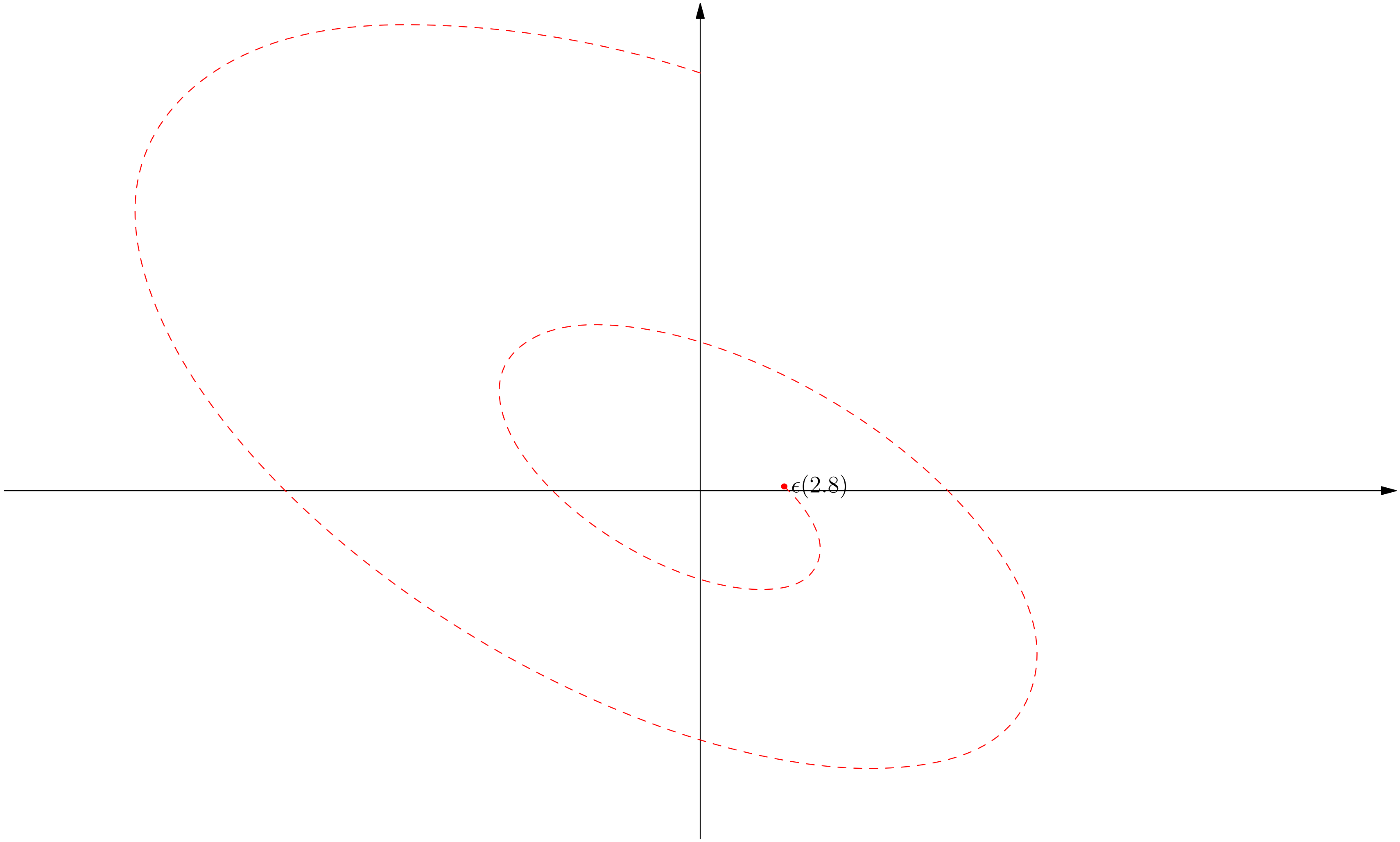
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



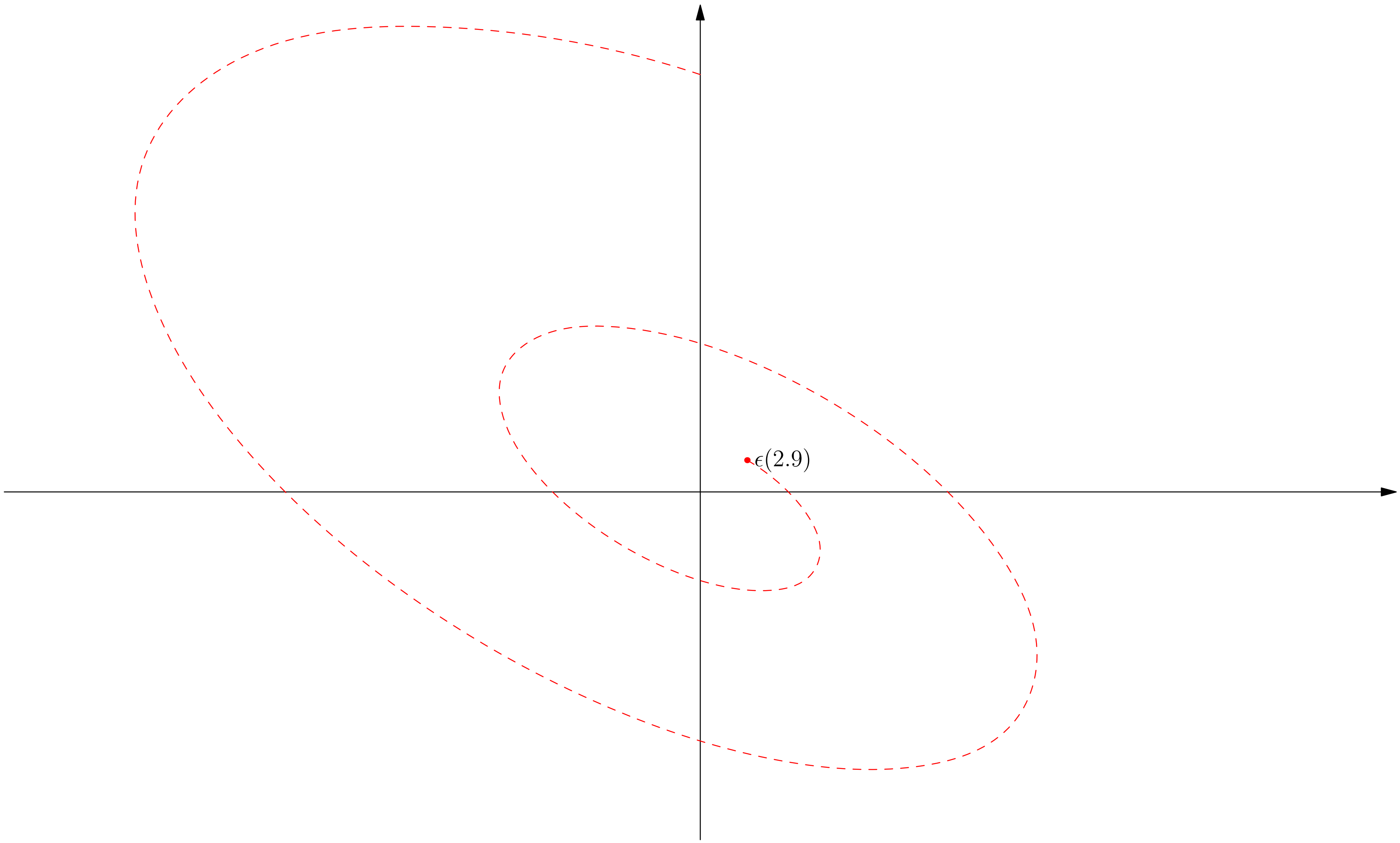
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



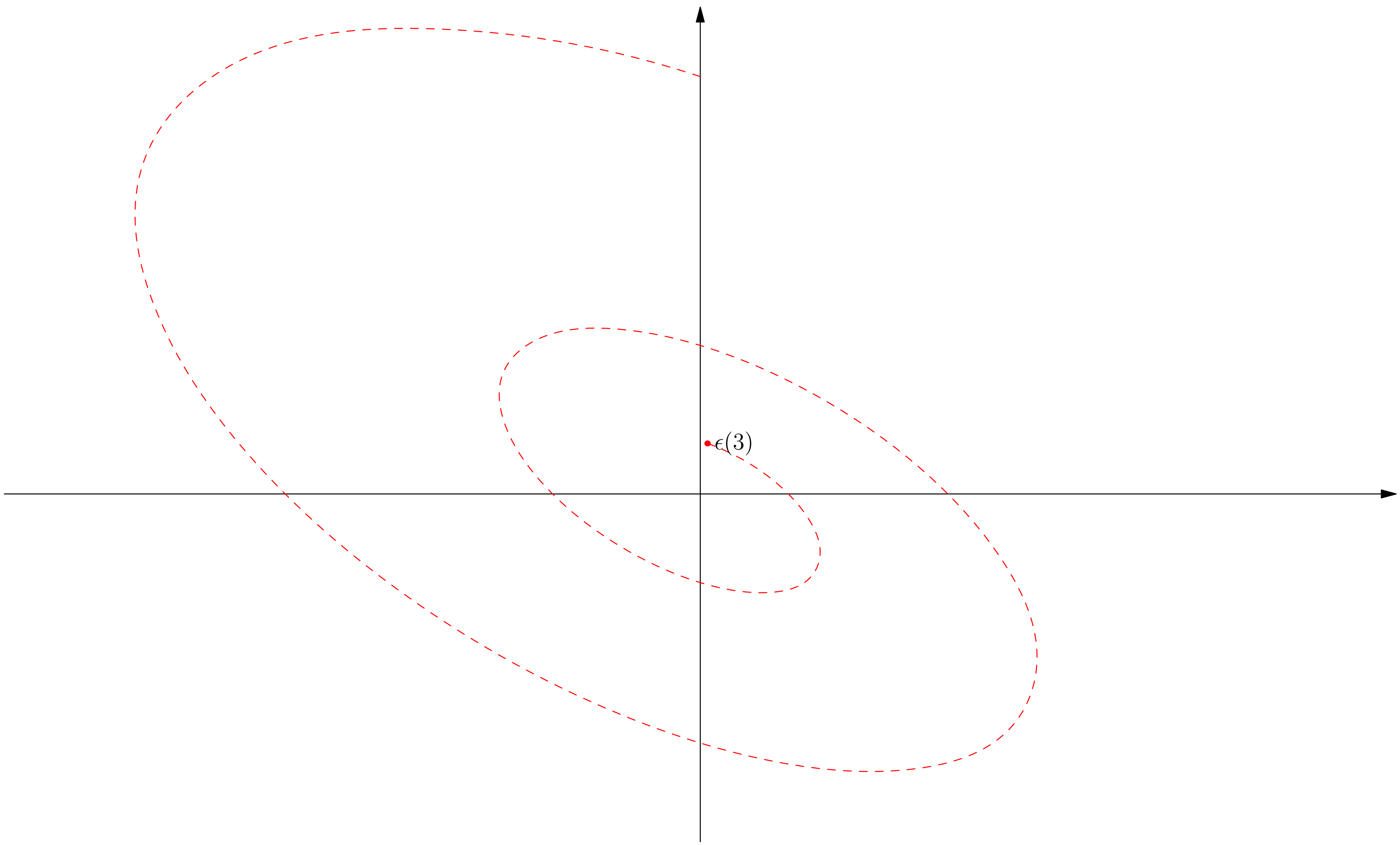
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



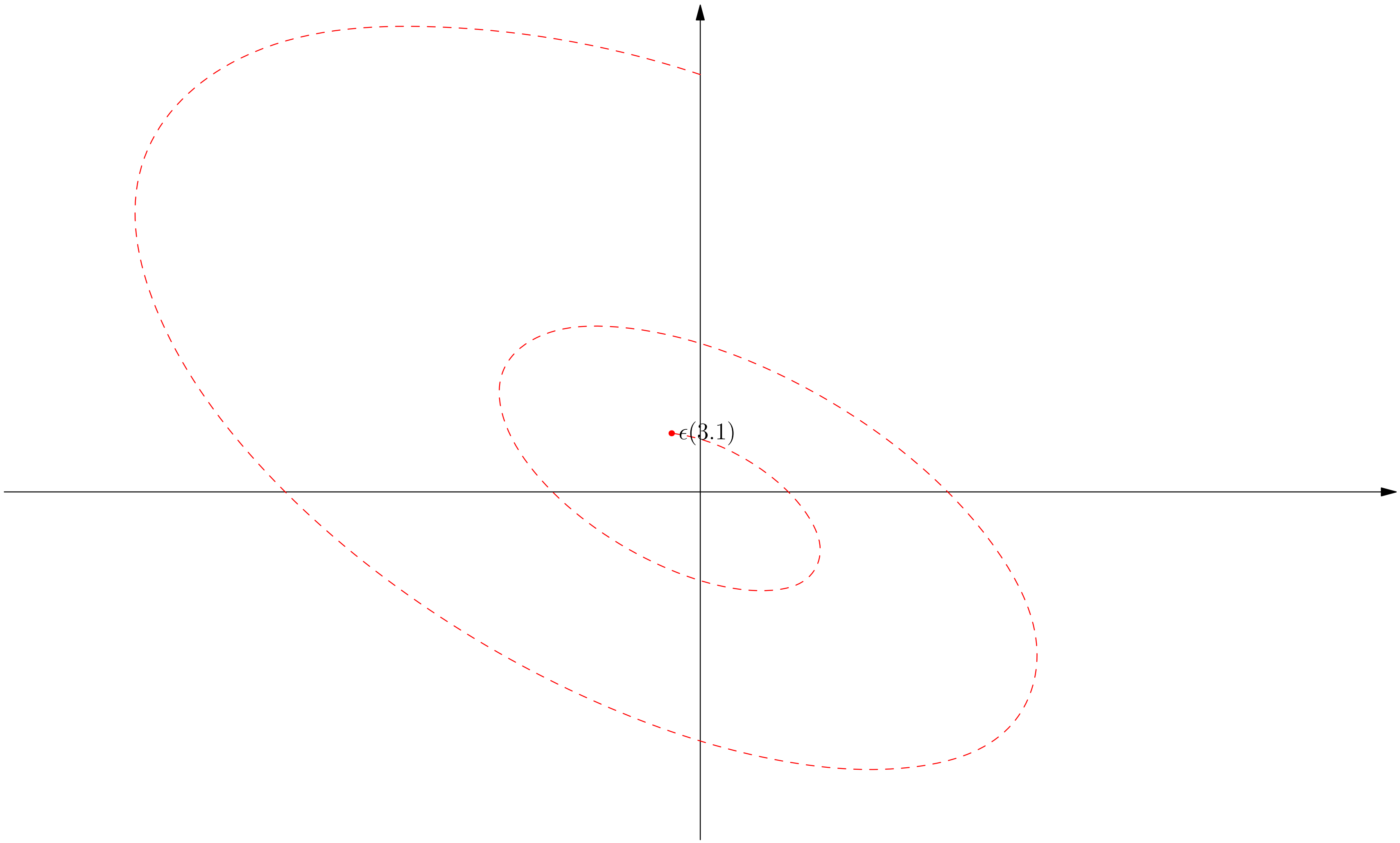
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



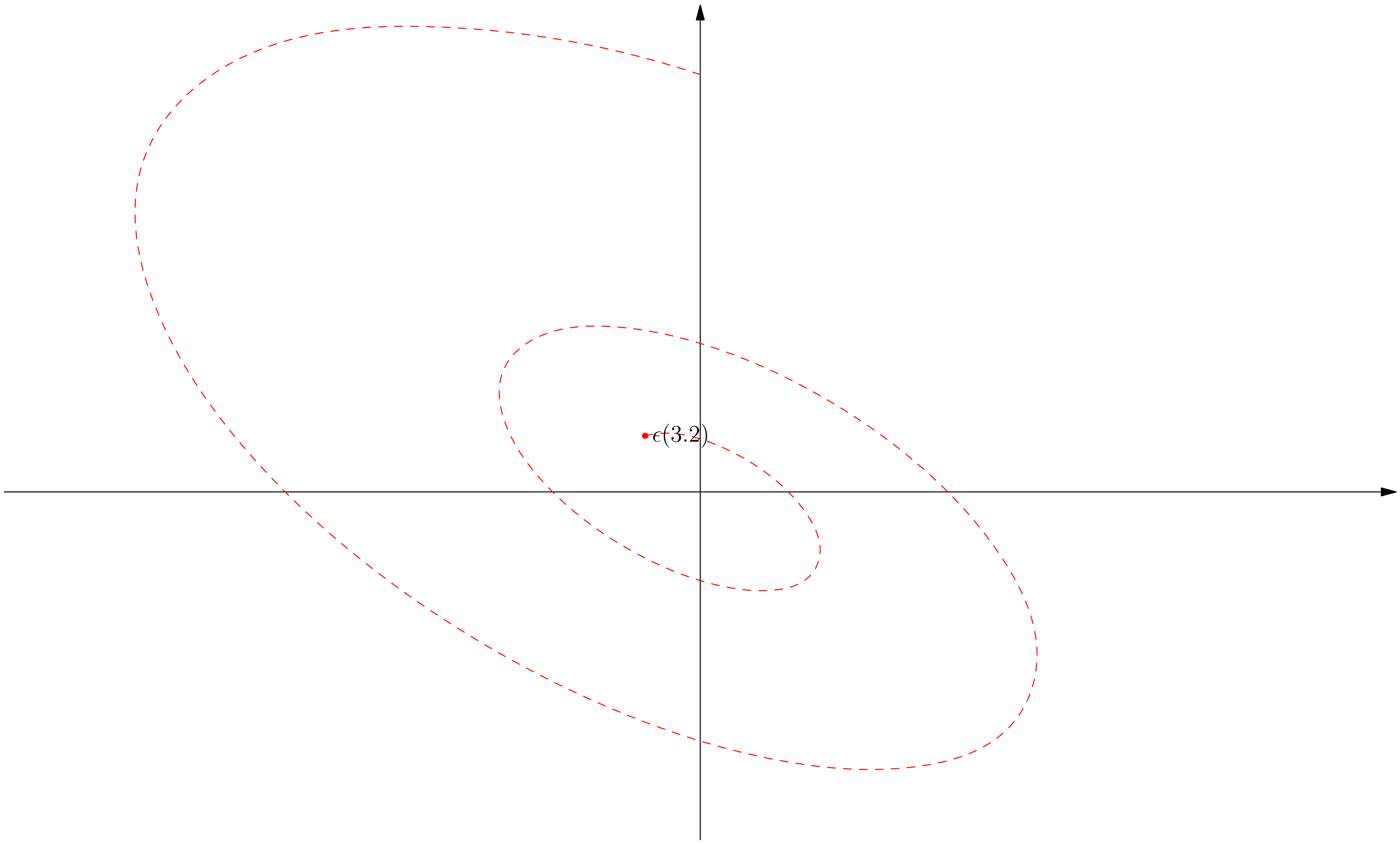
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



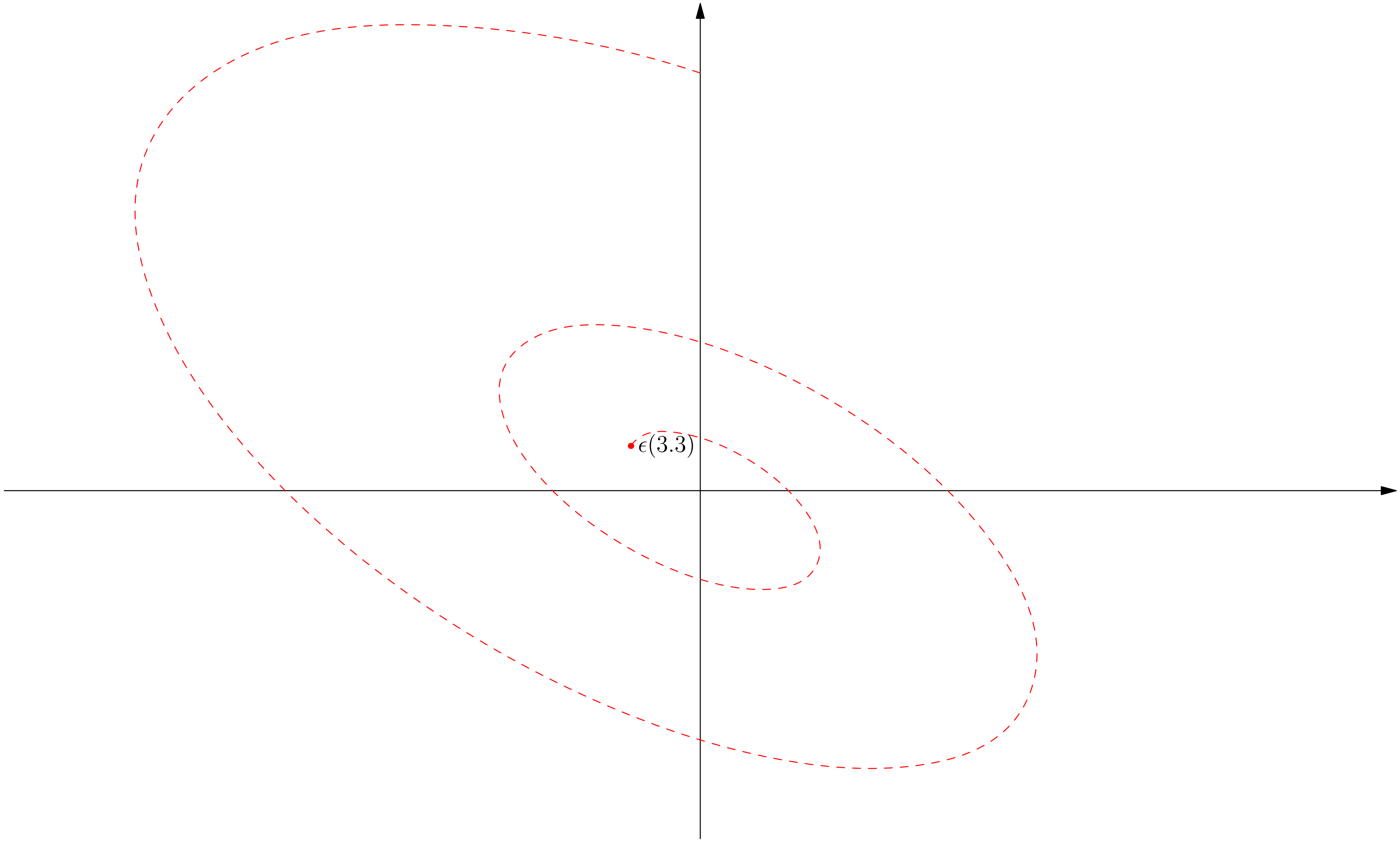
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



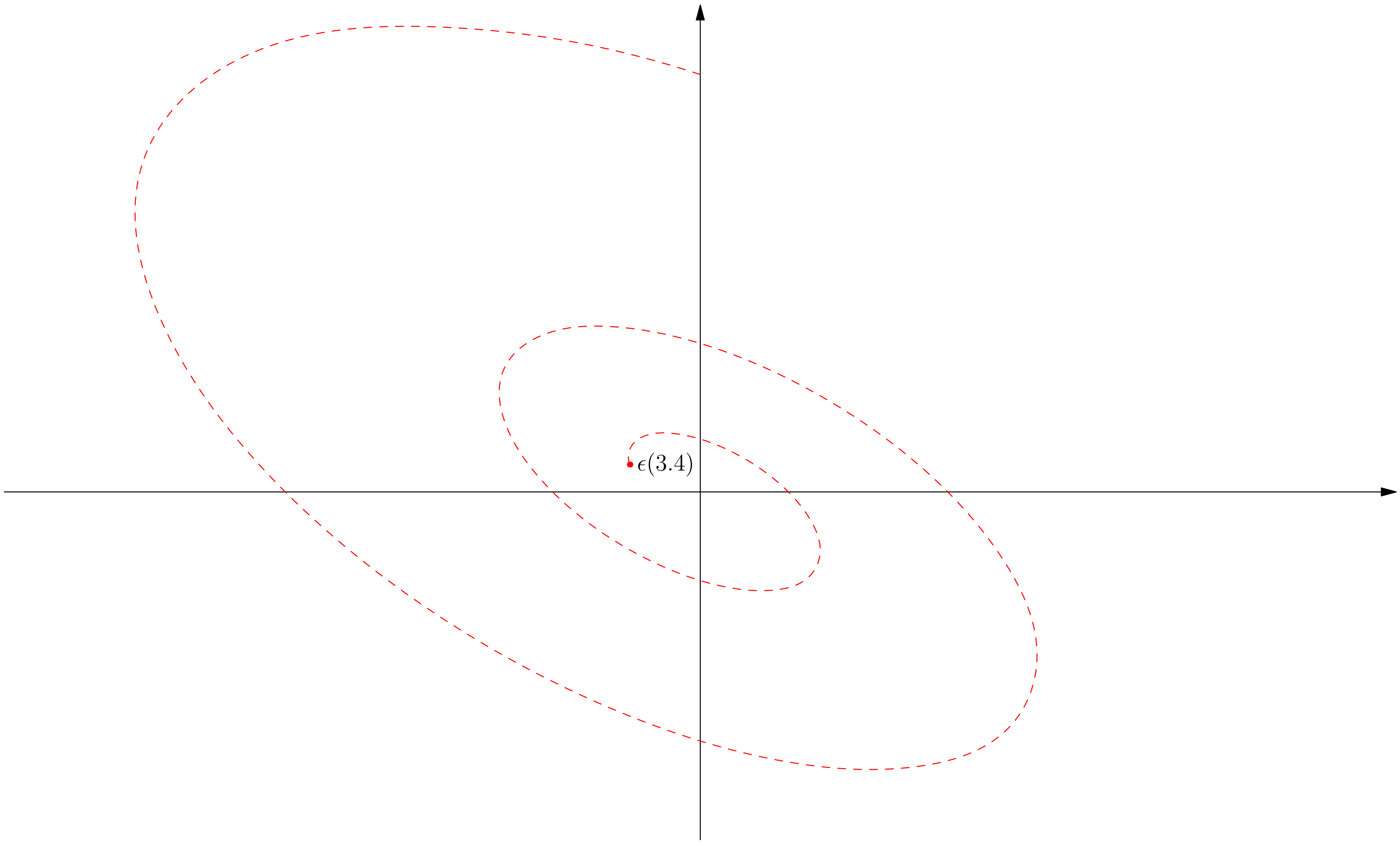
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



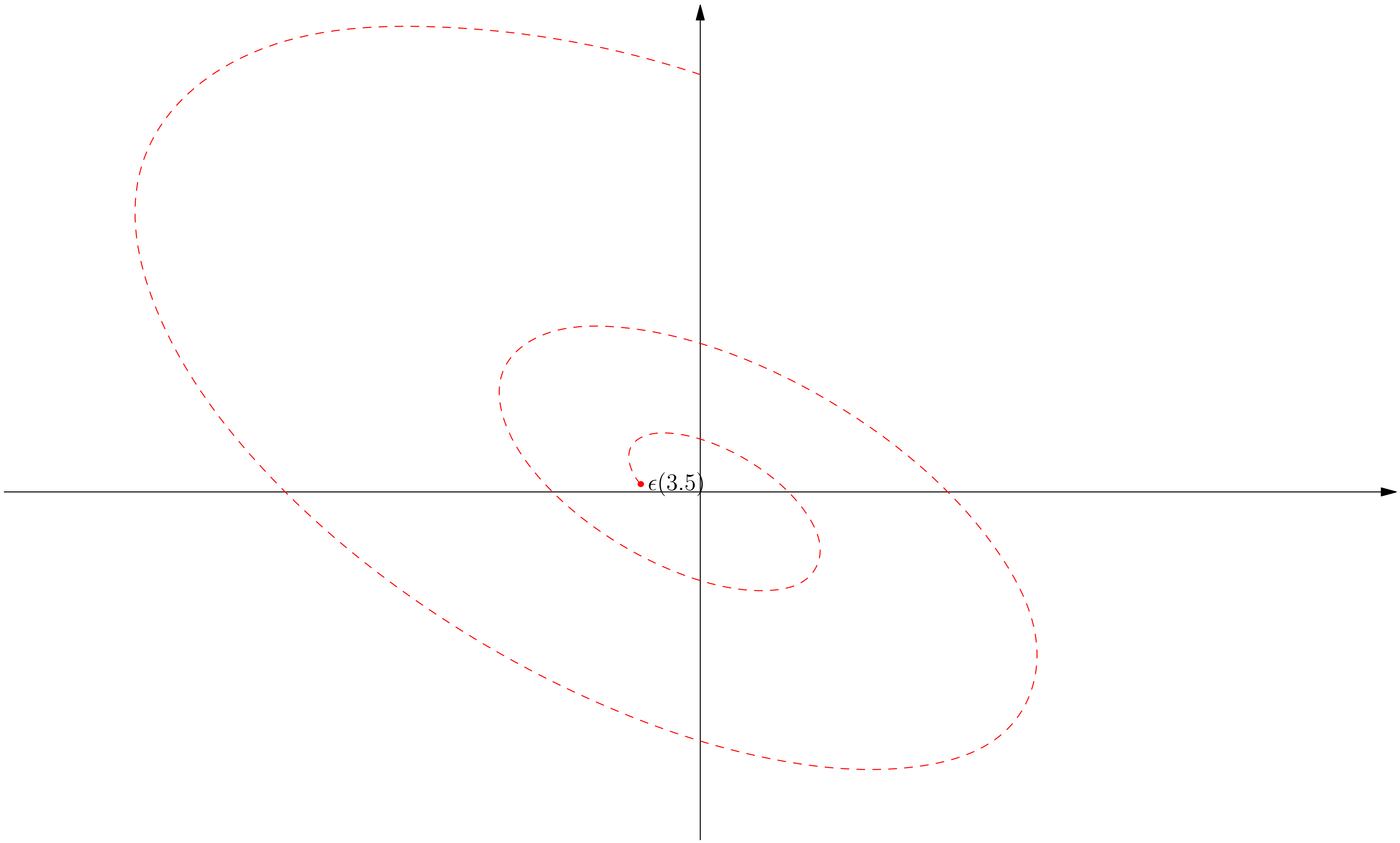
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



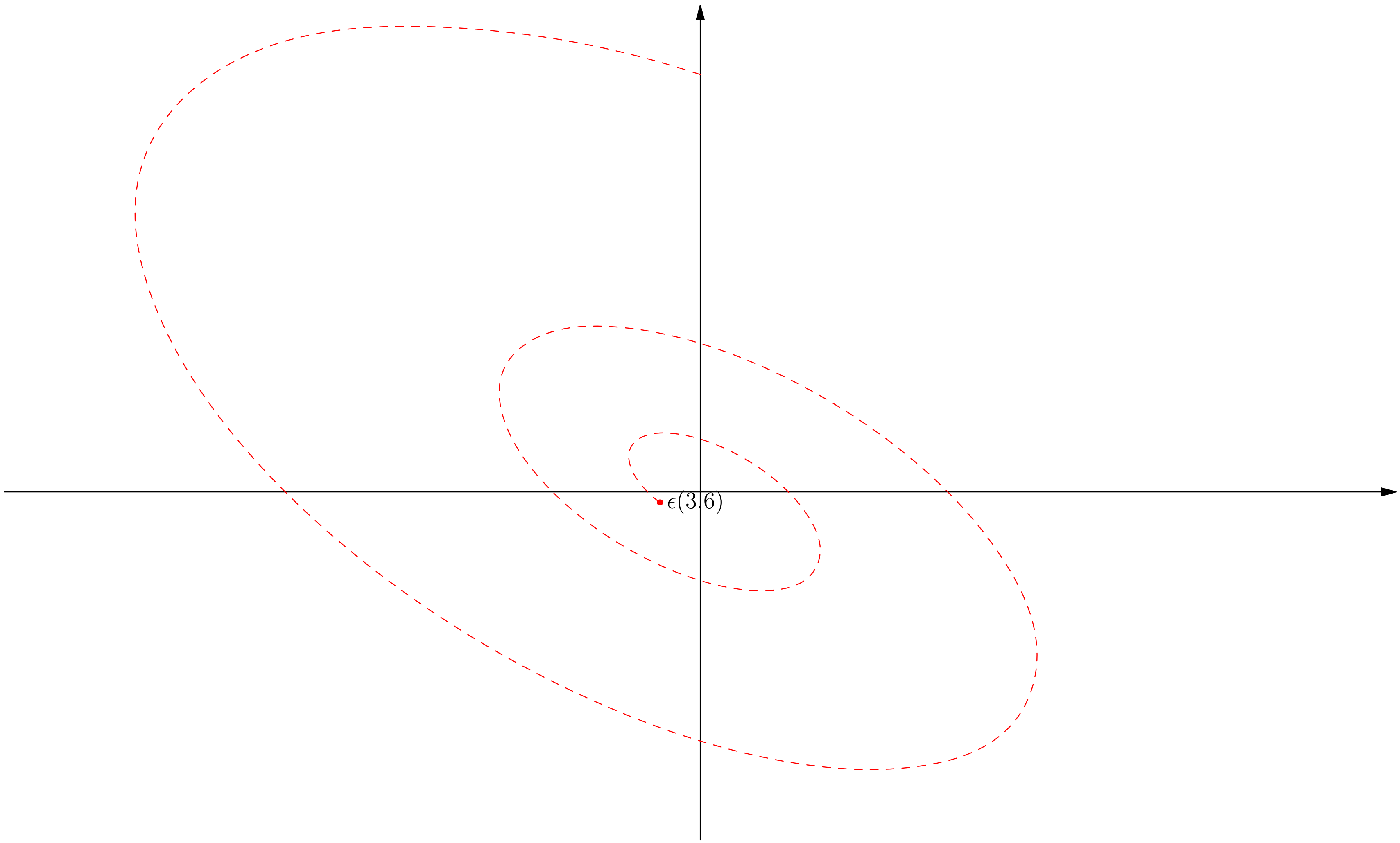
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\Lambda\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



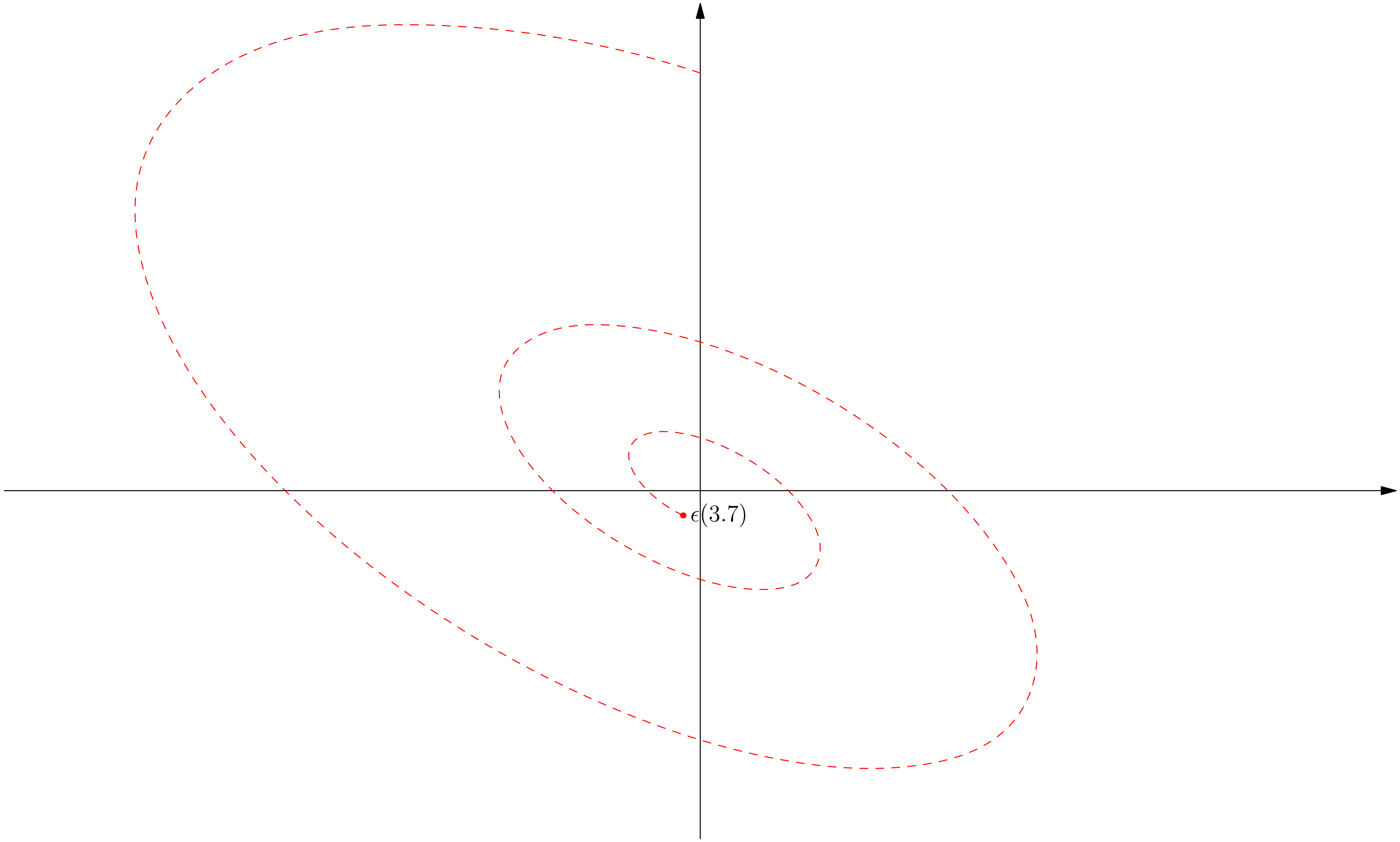
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



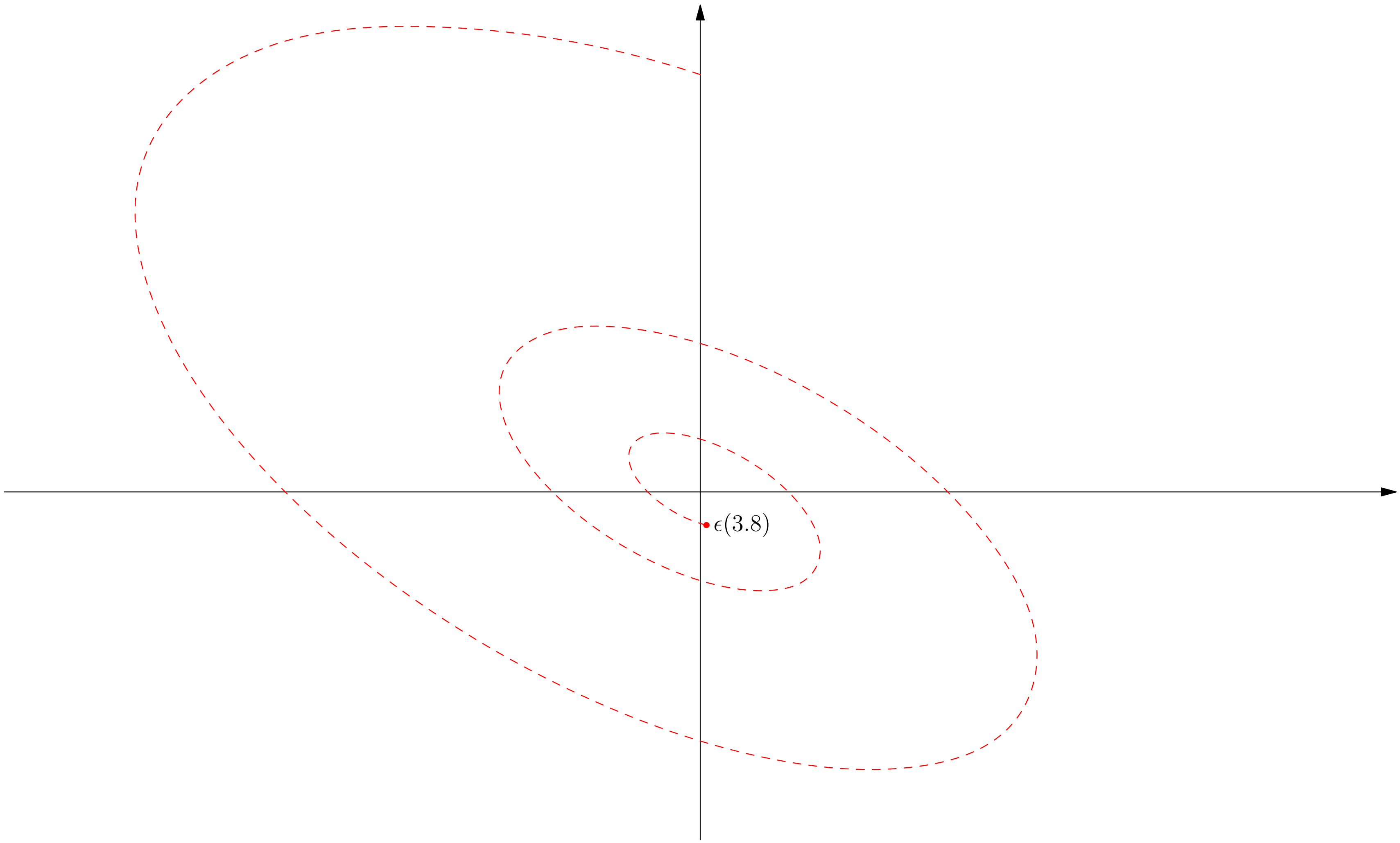
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



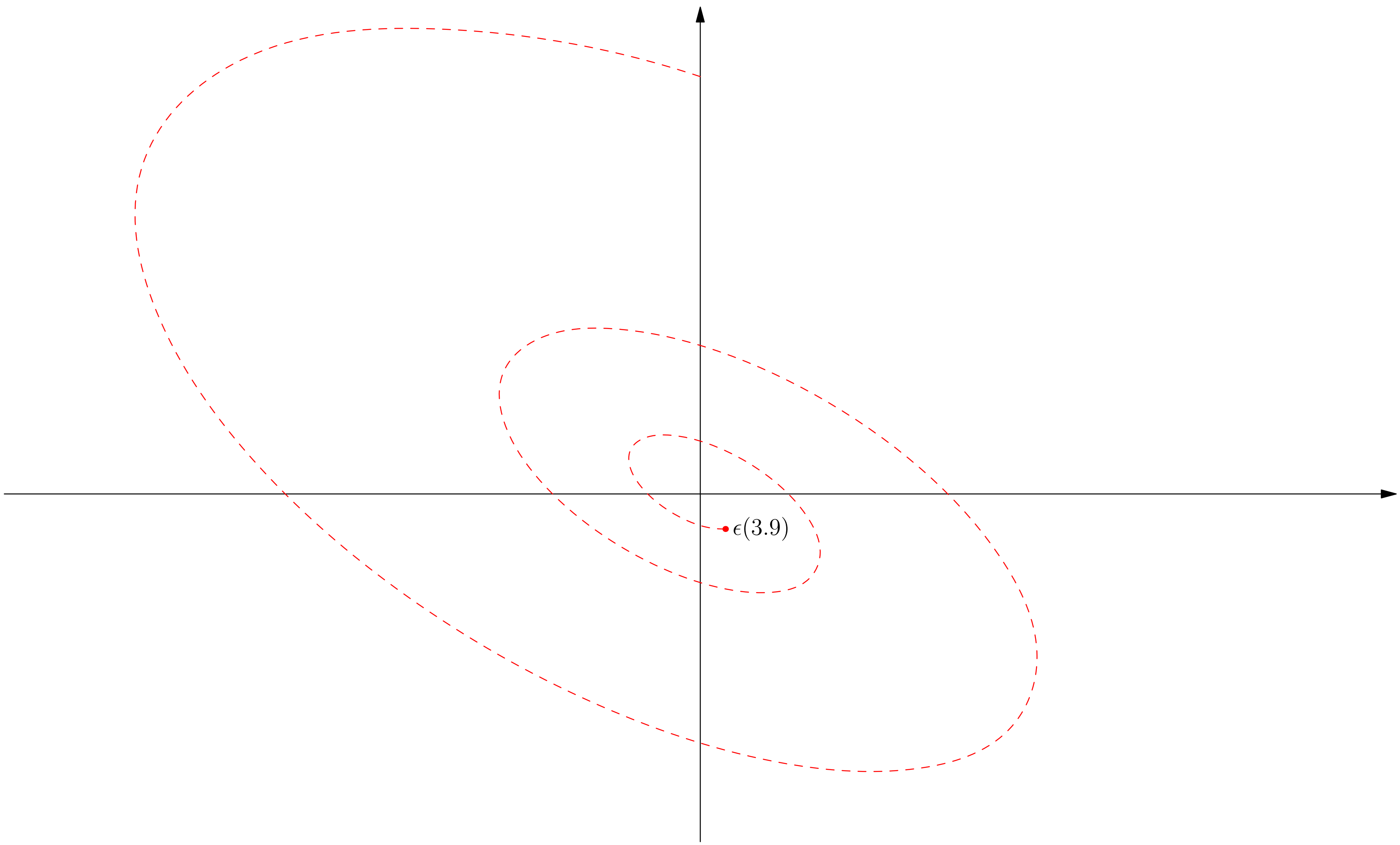
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



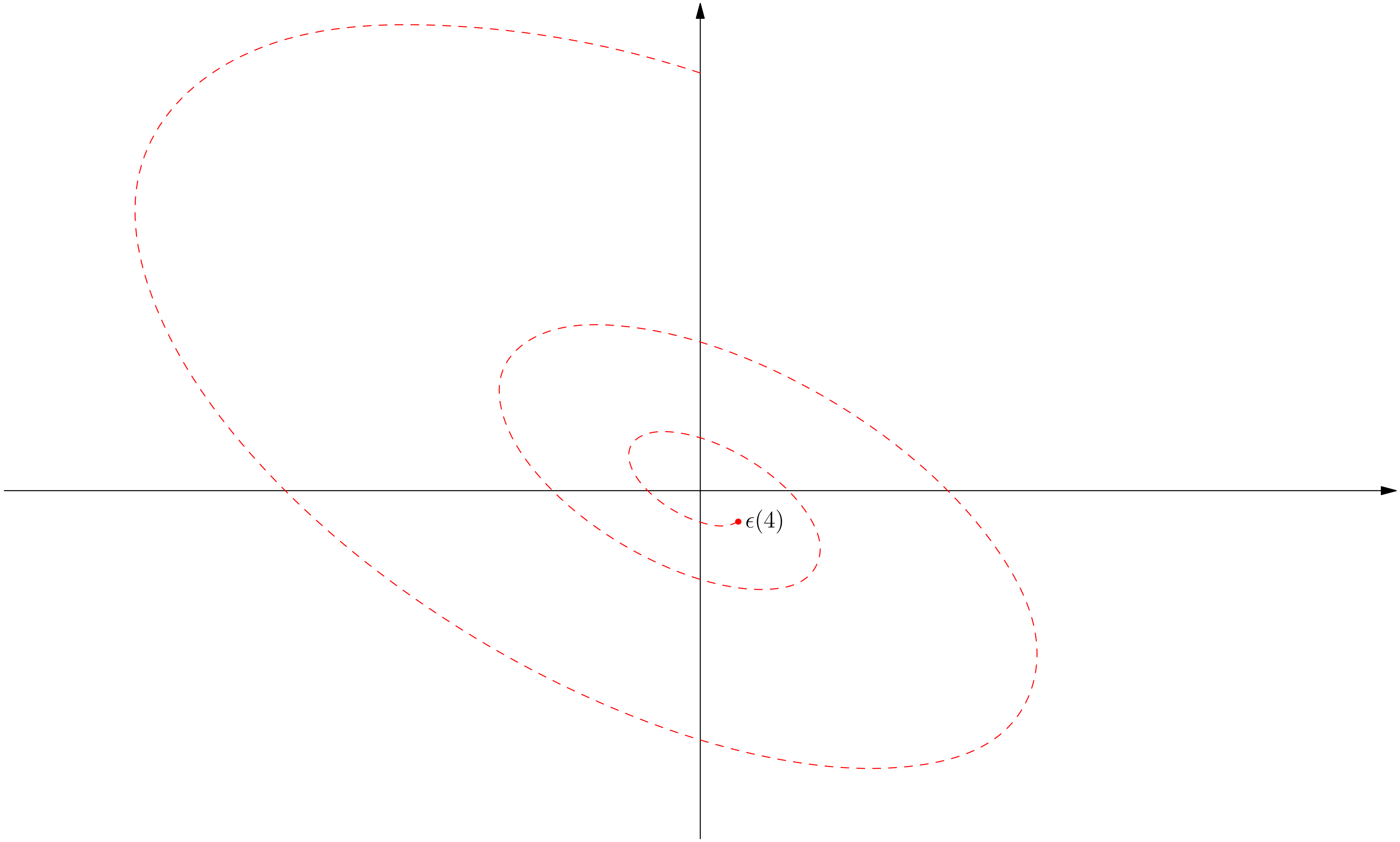
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



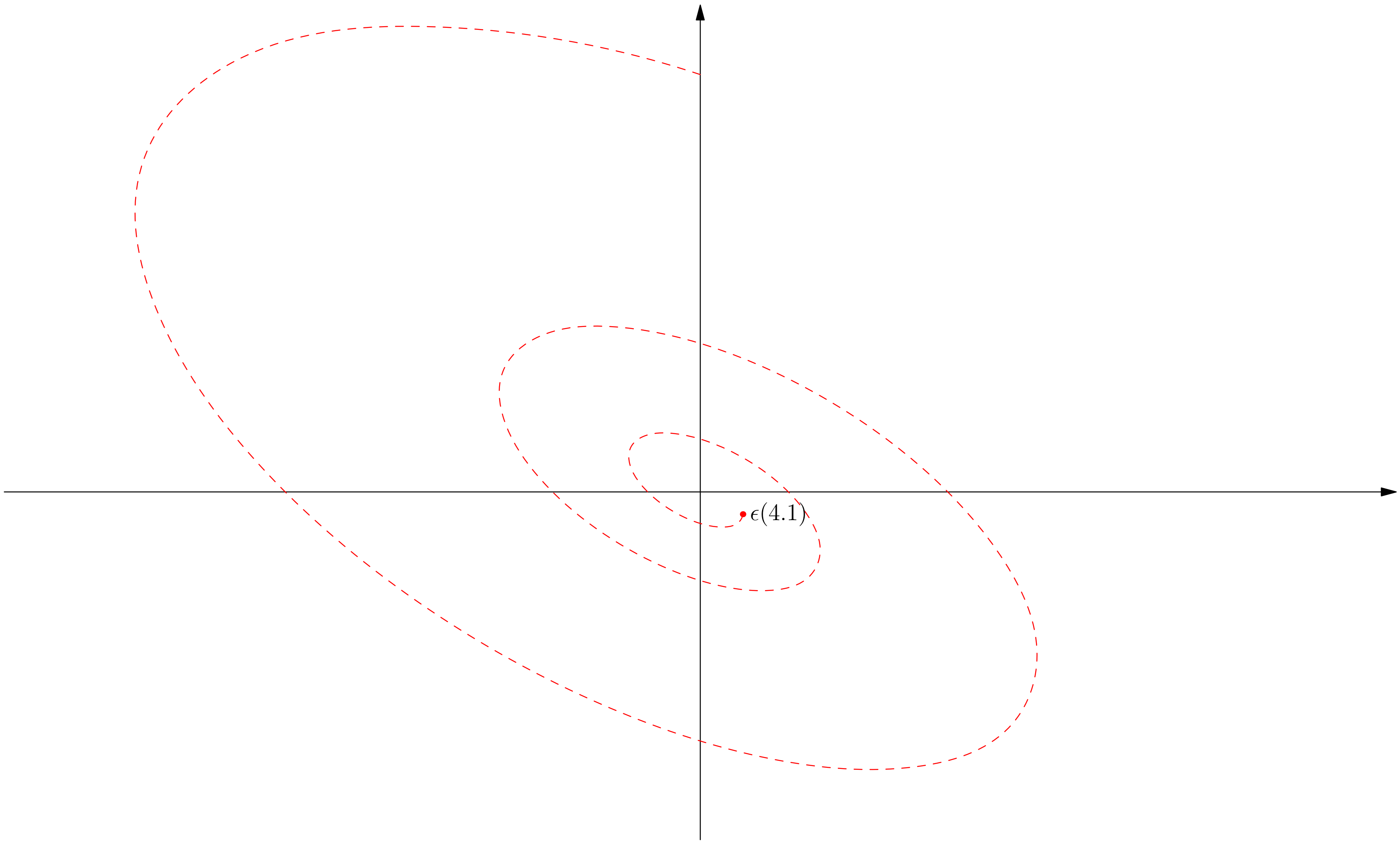
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



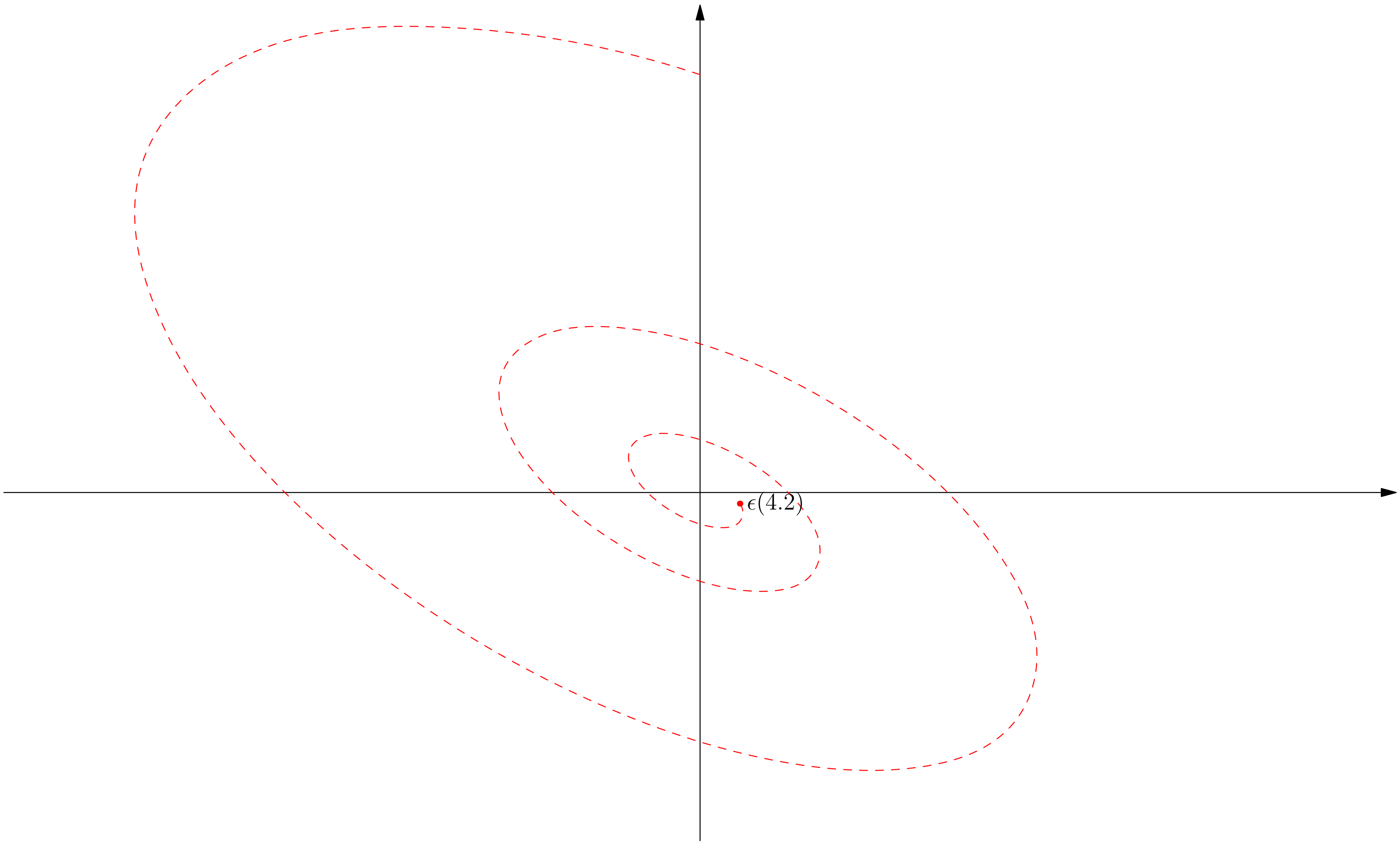
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



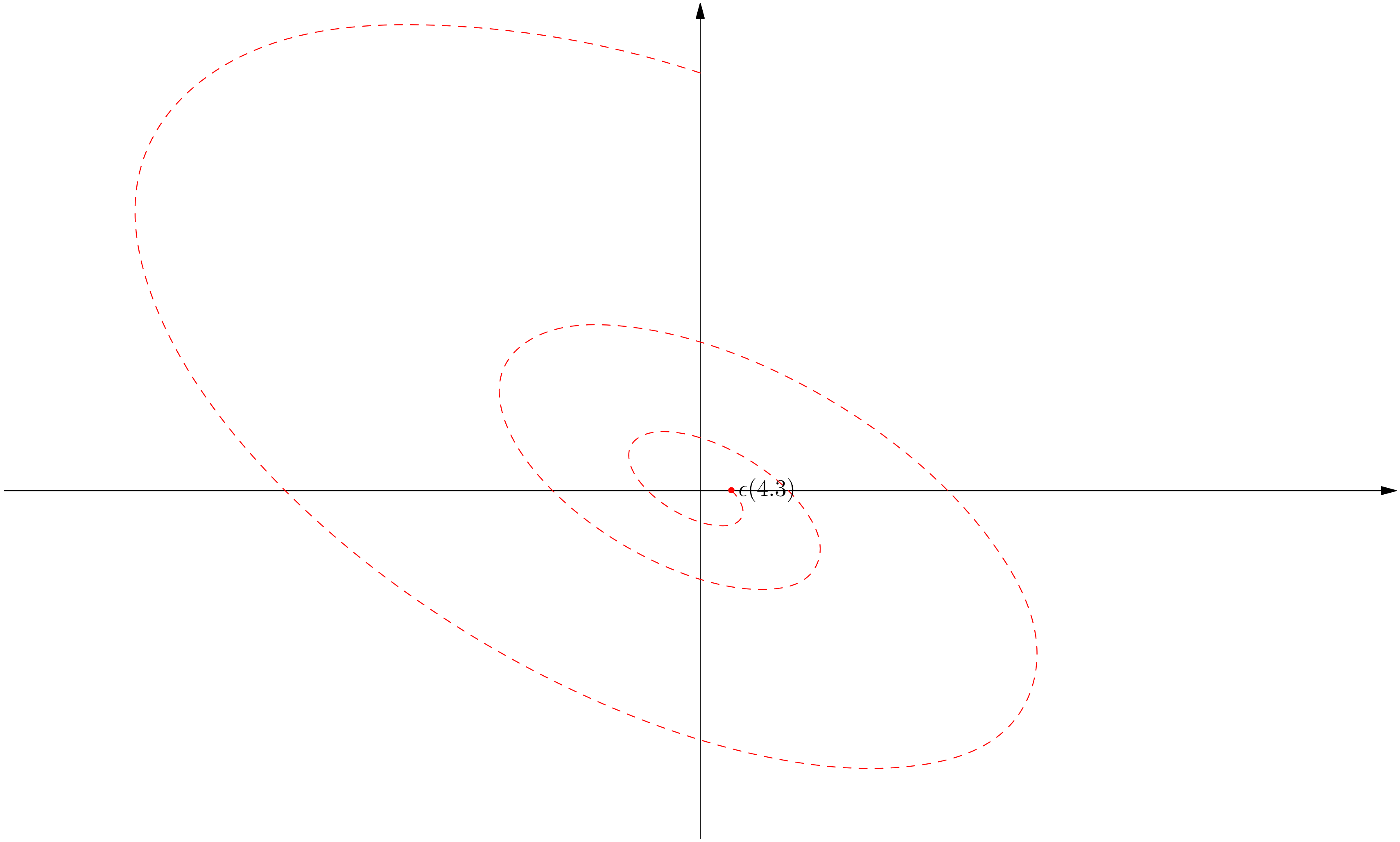
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



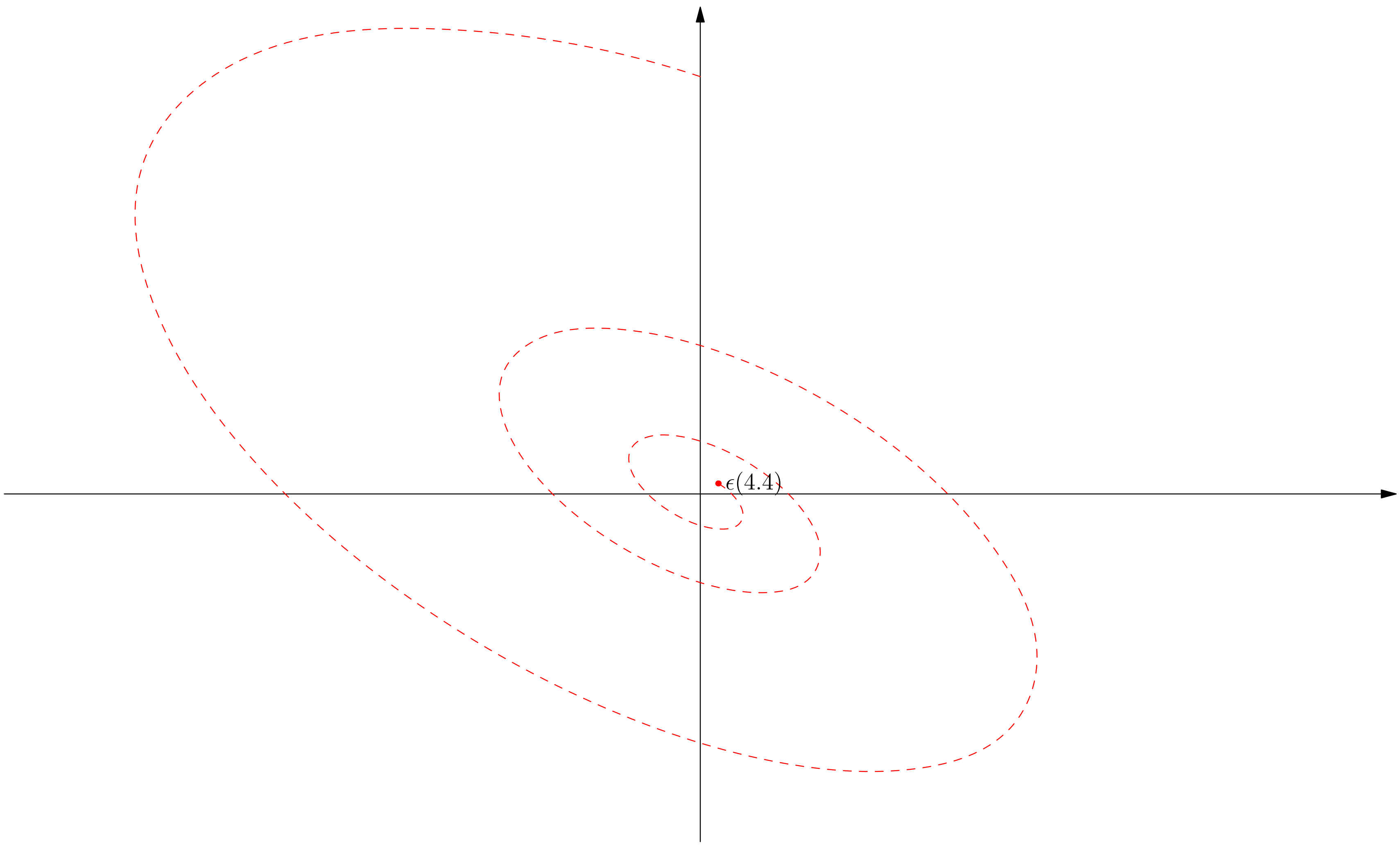
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



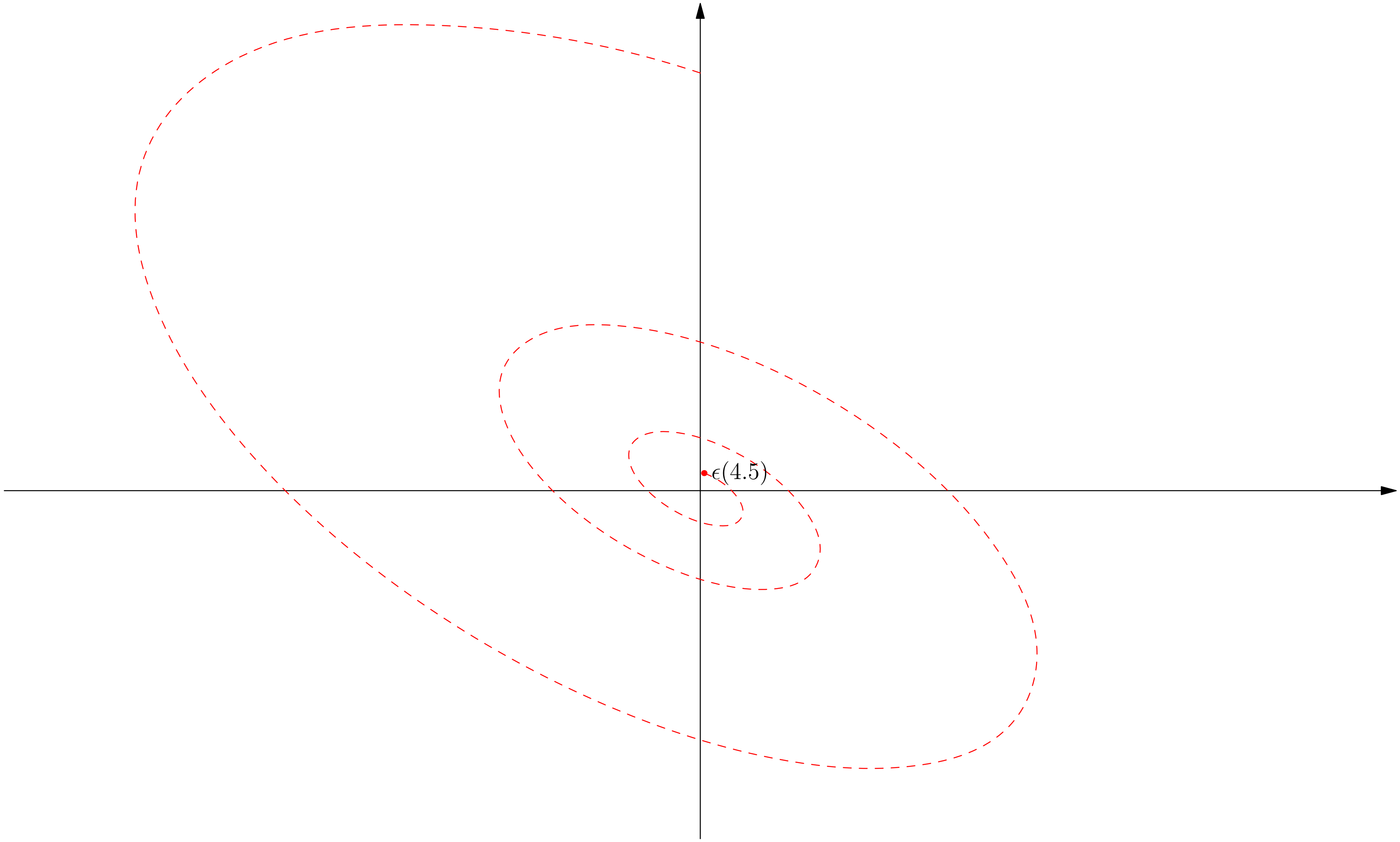
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



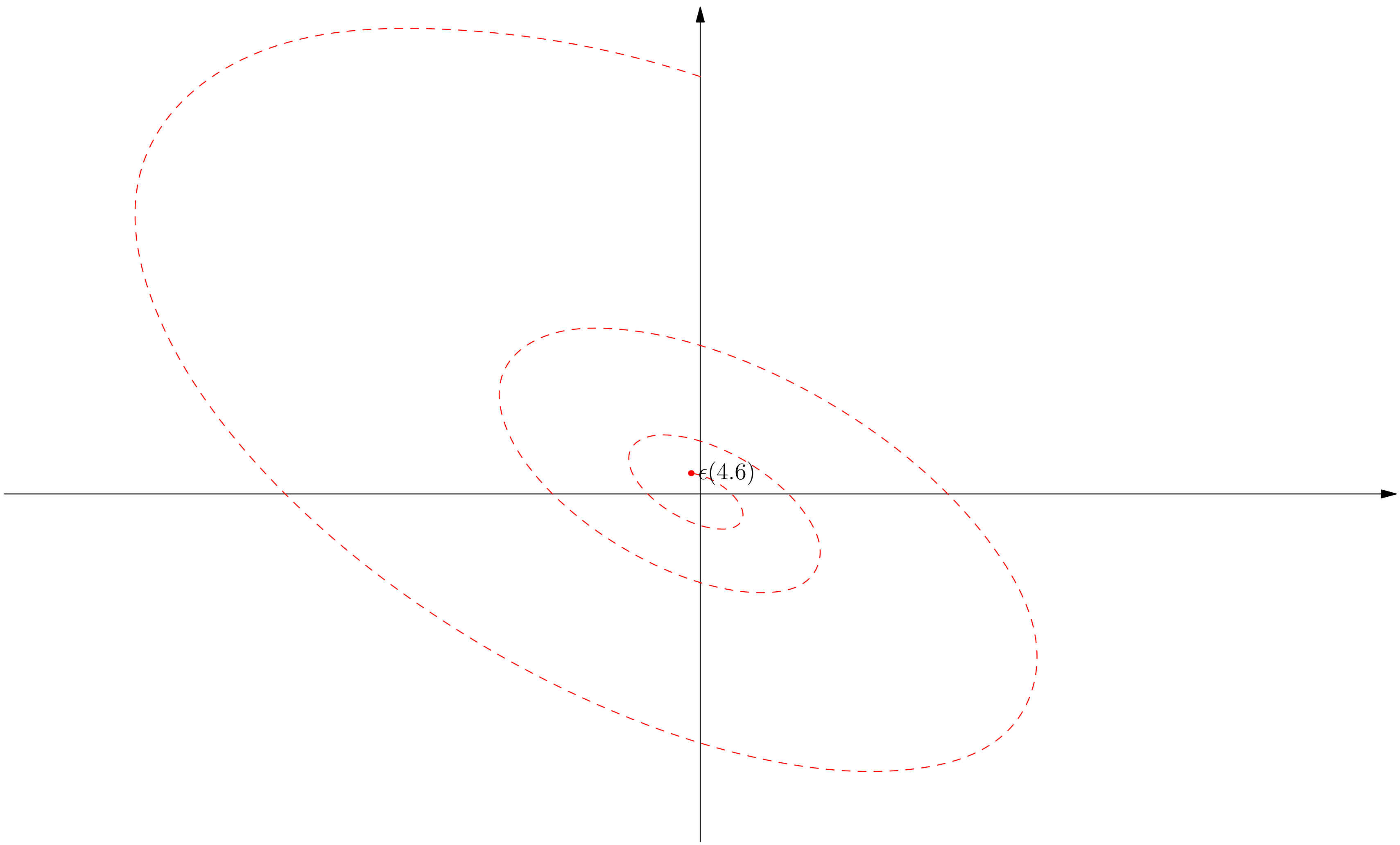
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



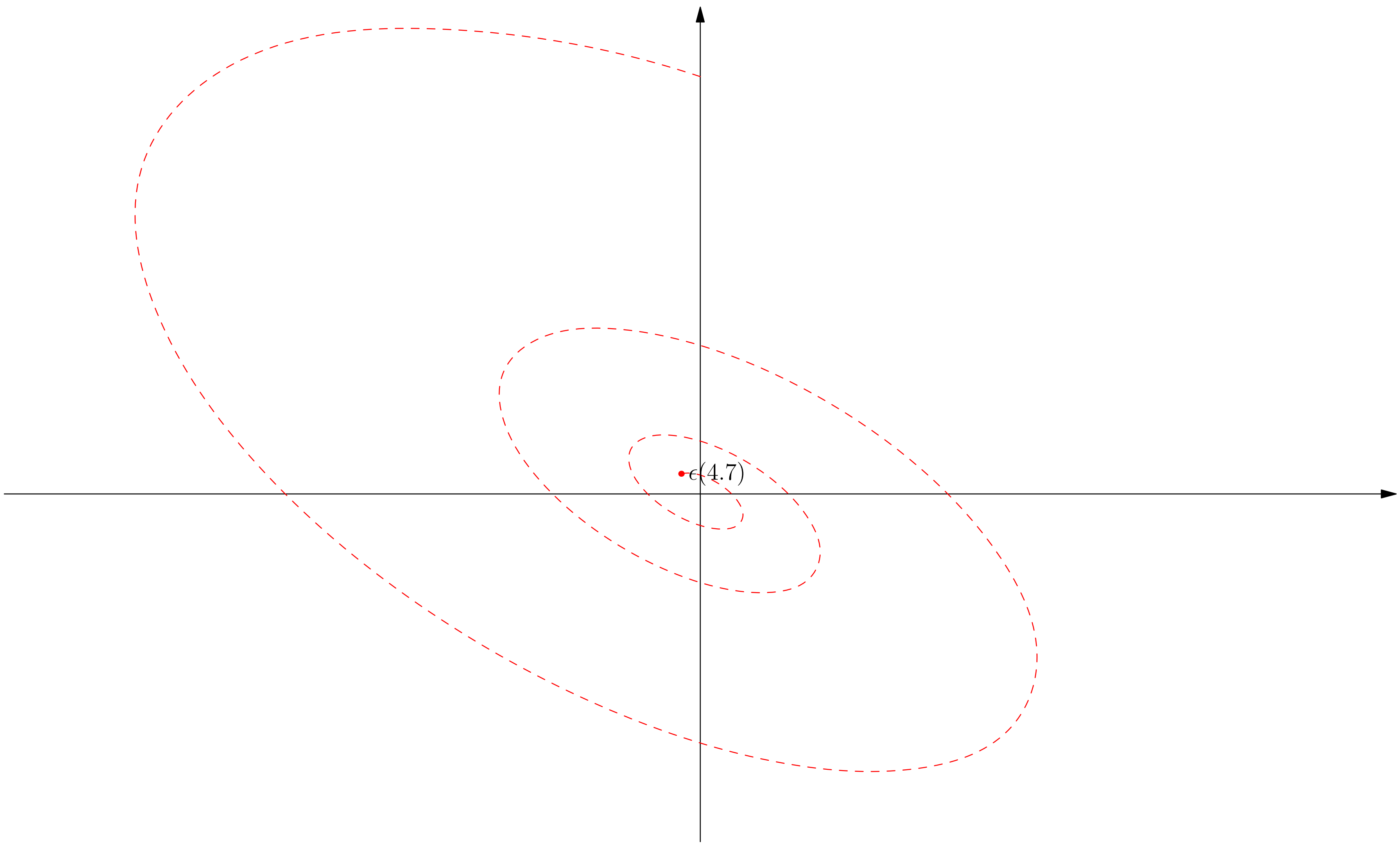
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



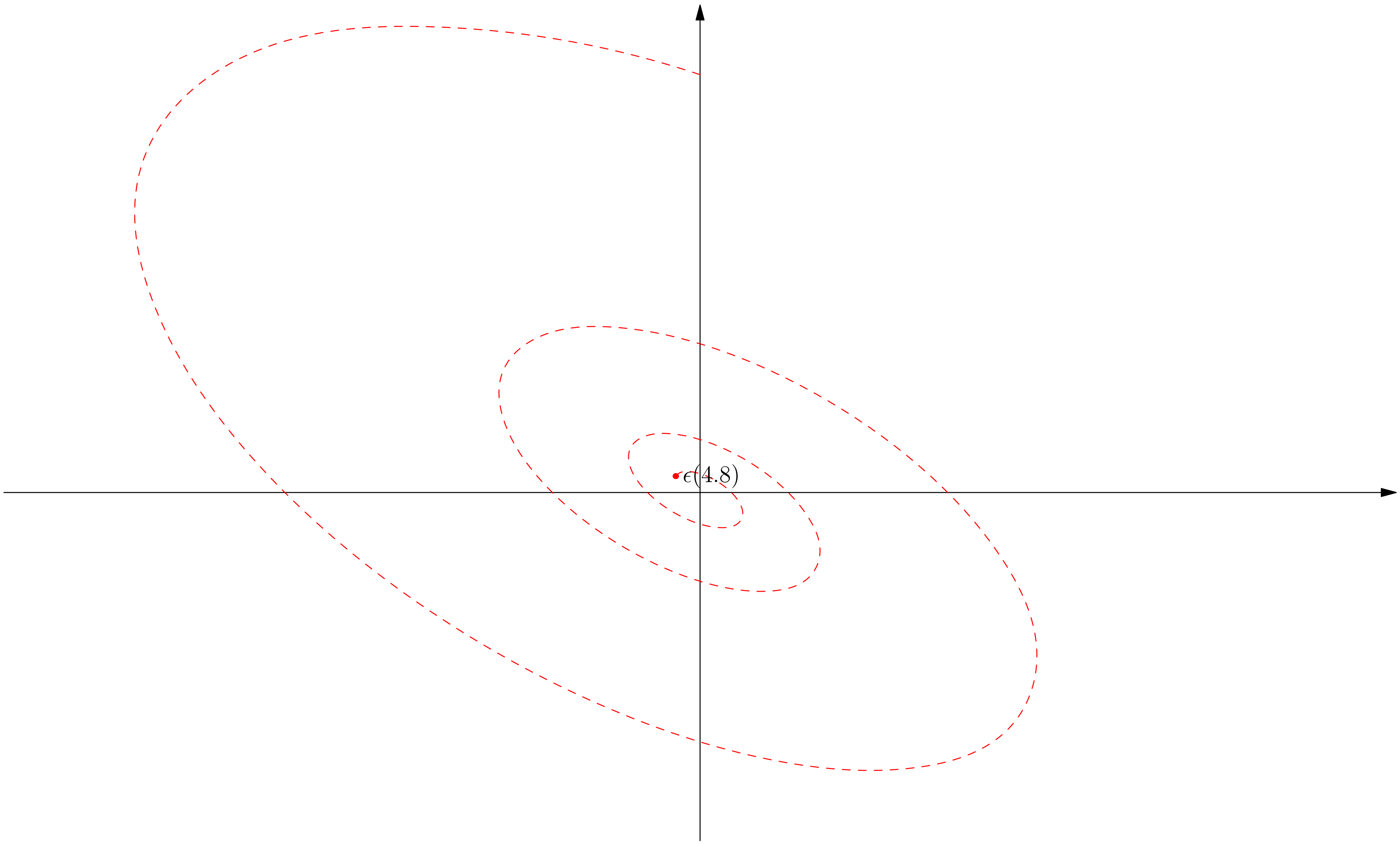
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



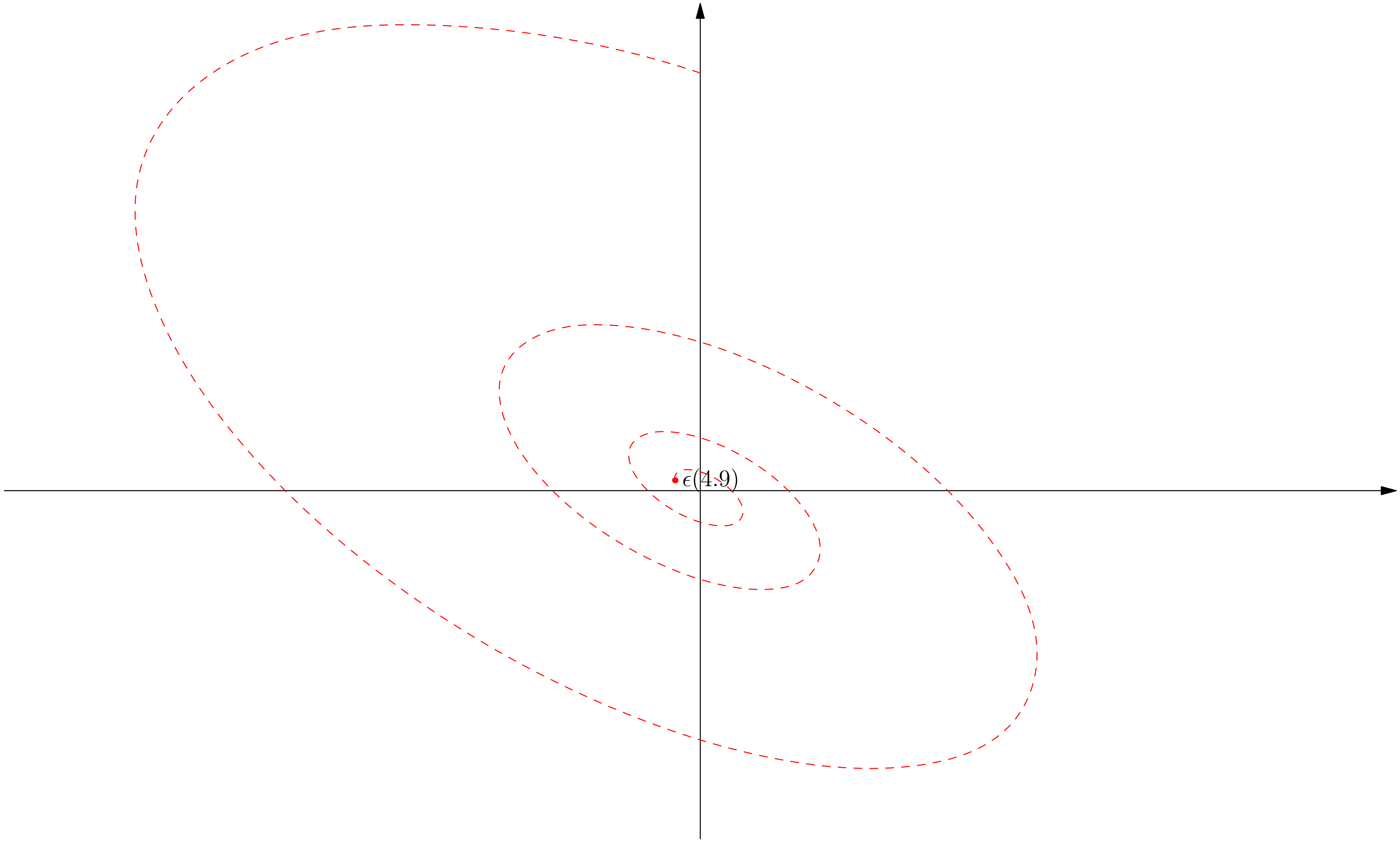
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473\mathrm{i} & -0.327 + 0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685 + 4.163\mathrm{i} & 0 \\ 0 & -0.685 - 4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611 + 0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611 - 0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



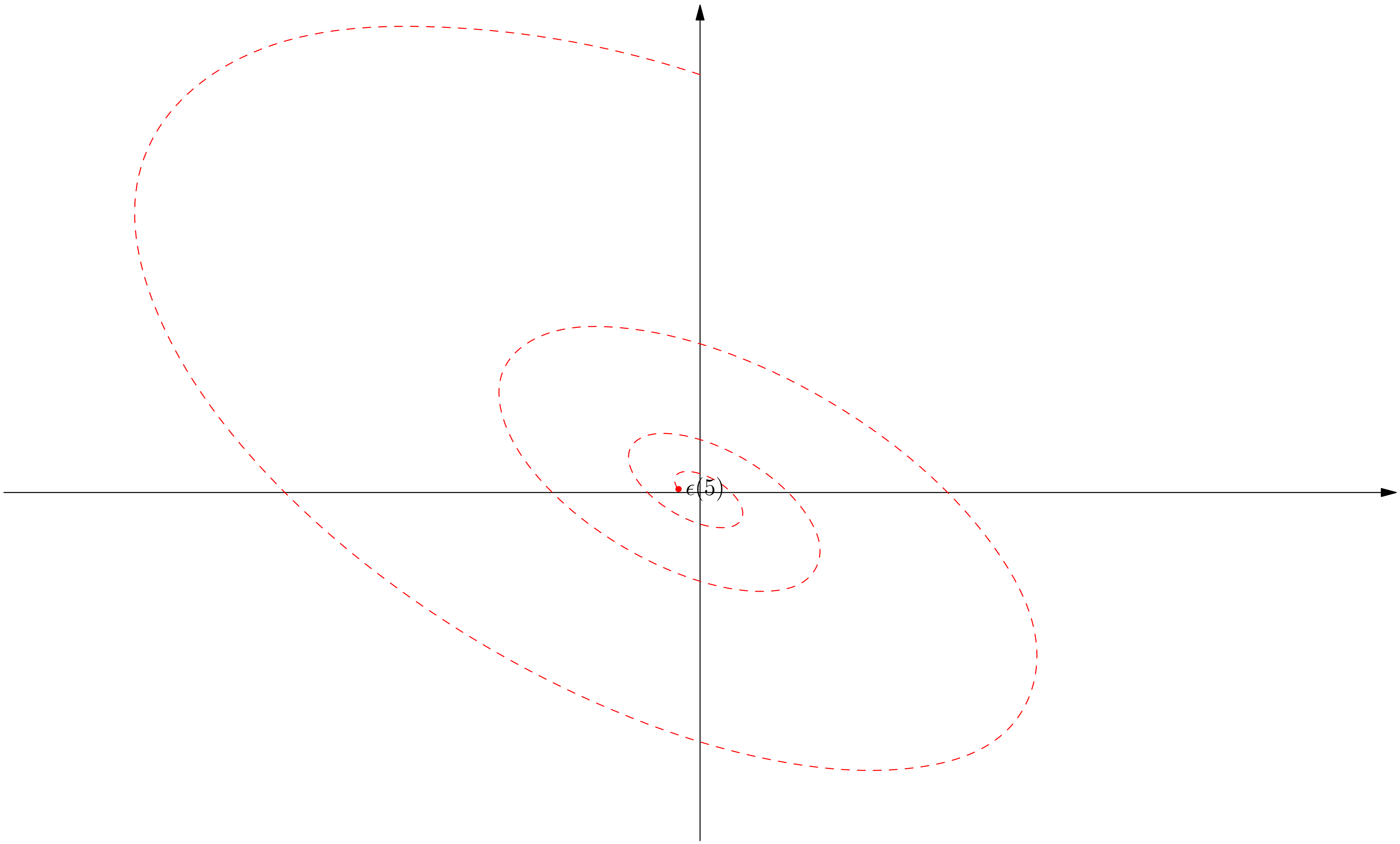
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327-0.473\mathrm{i} & -0.327+0.473\mathrm{i} \end{pmatrix} \begin{pmatrix} -0.685+4.163\mathrm{i} & 0 \\ 0 & -0.685-4.163\mathrm{i} \end{pmatrix} \begin{pmatrix} 0.611+0.422\mathrm{i} & 1.056\mathrm{i} \\ 0.611-0.422\mathrm{i} & -1.056\mathrm{i} \end{pmatrix}$$



Incidentally. . .

- A property of traces ($\operatorname{tr} \mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr} \mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\, B_{ji}$$

- Using $\mathbf{J} = \mathbf{V}\,\boldsymbol{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr} \mathbf{J} = \operatorname{tr} \mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\boldsymbol{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr} \mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr} \mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\, B_{ji} = \sum_{i,j} B_{ji}\, A_{ij}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr} \mathbf{J} = \operatorname{tr} \mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally...

- A property of traces ($\operatorname{tr} \mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr} \mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\, B_{ji} = \sum_{i,j} B_{ji}\, A_{ij} = \operatorname{tr} \mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr} \mathbf{J} = \operatorname{tr} \mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\, B_{ji} = \sum_{i,j} B_{ji}\, A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\mathbf{\Lambda}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\mathbf{\Lambda} = \operatorname{tr}\mathbf{\Lambda}$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\boldsymbol{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\boldsymbol{\Lambda} = \operatorname{tr}\boldsymbol{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\boldsymbol{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\mathbf{\Lambda} = \operatorname{tr}\mathbf{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\mathbf{\Lambda} = \operatorname{tr}\mathbf{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\mathbf{\Lambda}|\,|\mathbf{V}^{-1}| = |\mathbf{\Lambda}|$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\boldsymbol{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\boldsymbol{\Lambda} = \operatorname{tr}\boldsymbol{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\boldsymbol{\Lambda}|\,|\mathbf{V}^{-1}| = |\boldsymbol{\Lambda}| = \prod_i \lambda_i$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

Incidentally. . .

- A property of traces ($\operatorname{tr}\mathbf{M} = \sum_i M_{ii}$) is

$$\operatorname{tr}\mathbf{A}\,\mathbf{B} = \sum_{i,j} A_{ij}\,B_{ji} = \sum_{i,j} B_{ji}\,A_{ij} = \operatorname{tr}\mathbf{B}\,\mathbf{A}$$

- Using $\mathbf{J} = \mathbf{V}\,\boldsymbol{\Lambda}\mathbf{V}^{-1}$

$$\operatorname{tr}\mathbf{J} = \operatorname{tr}\mathbf{V}\,\boldsymbol{\Lambda}\,\mathbf{V}^{-1} = \operatorname{tr}\mathbf{V}^{-1}\,\mathbf{V}\,\boldsymbol{\Lambda} = \operatorname{tr}\boldsymbol{\Lambda} = \sum_i \lambda_i$$

- Two properties of determinants is that $|\mathbf{A}\,\mathbf{B}| = |\mathbf{A}|\,|\mathbf{B}|$ and $|\mathbf{M}^{-1}| = |\mathbf{M}|^{-1}$ thus

$$|\mathbf{J}| = |\mathbf{V}|\,|\boldsymbol{\Lambda}|\,|\mathbf{V}^{-1}| = |\boldsymbol{\Lambda}| = \prod_i \lambda_i$$

- If $\lambda_1 = T/2 - \sqrt{T^2/4 - D}$ and $\lambda_2 = T/2 + \sqrt{T^2/4 - D}$ then

$$\lambda_1 + \lambda_2 = T \qquad\qquad \lambda_1 \times \lambda_2 = D$$

High Dimensional Systems

- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

High Dimensional Systems

- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

High Dimensional Systems

- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

High Dimensional Systems

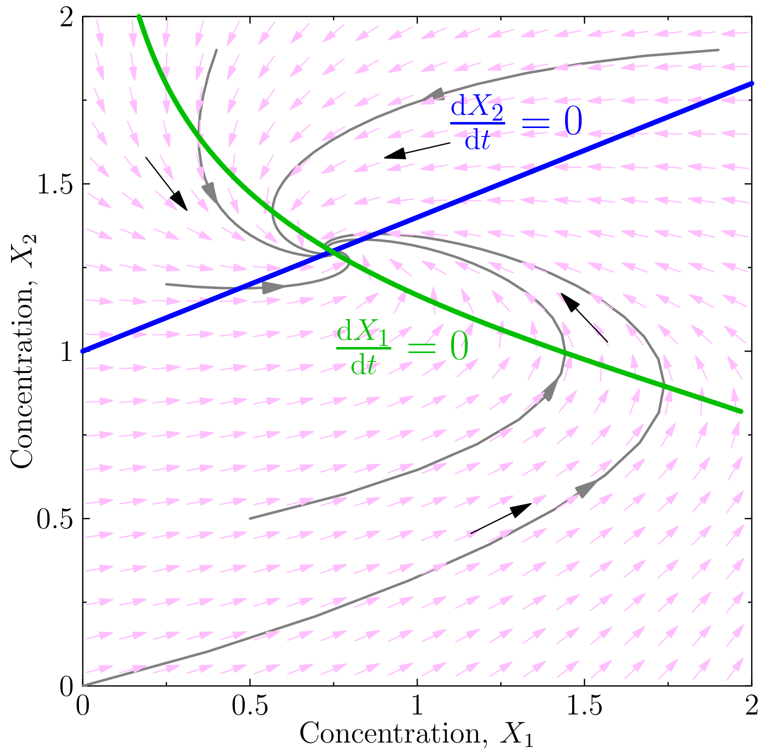
- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

High Dimensional Systems

- We have only considered simple systems with two species of molecules
- Much of the analysis techniques generalise to higher dimensions, but it is far harder to get closed from solutions
- Using programs like Mathematica it is often easy to find fixed point solutions numerically
- However, in biology different reactions happen at very different rates
- We often concentrate on the slowest reactions and assume all the other reactions have reached equilibrium (however, beware this needs to be analysed)

Outline

1. Understanding ODEs
2. Stability
3. **Bifications**



Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\,X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\,X_2(t)$$

- Let $k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$
- But now we look at varying the parameter k_1
- The fixed points are given where the nullclines intersect

Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- Let $k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$

- But now we look at varying the parameter k_1

- The fixed points are given where the nullclines intersect

Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- Let $k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$
- But now we look at varying the parameter k_1
- The fixed points are given where the nullclines intersect

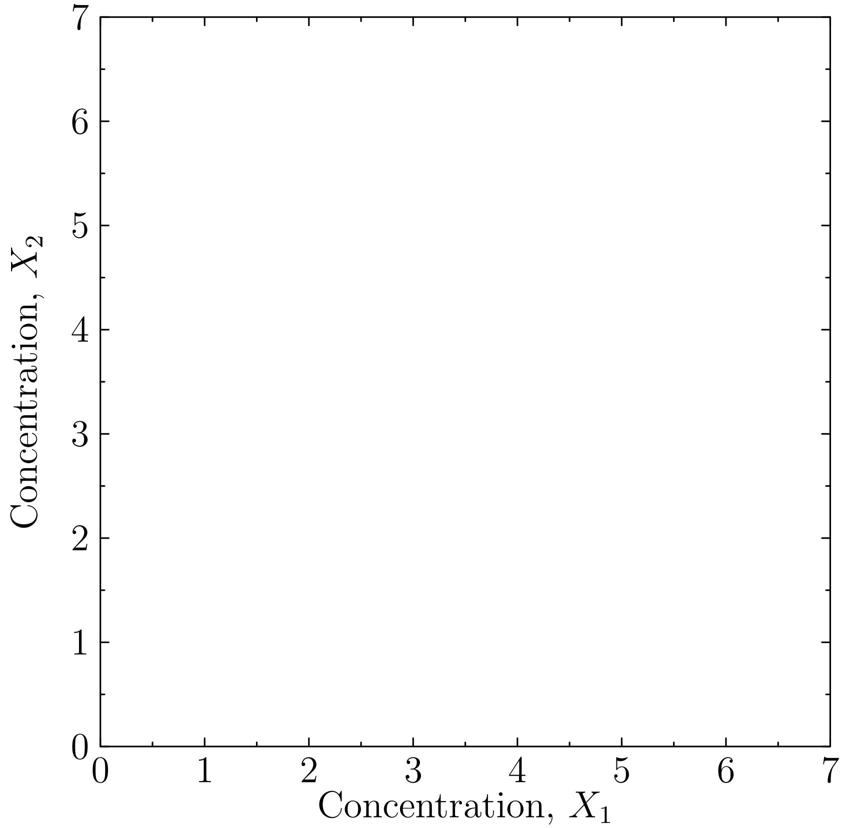
Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

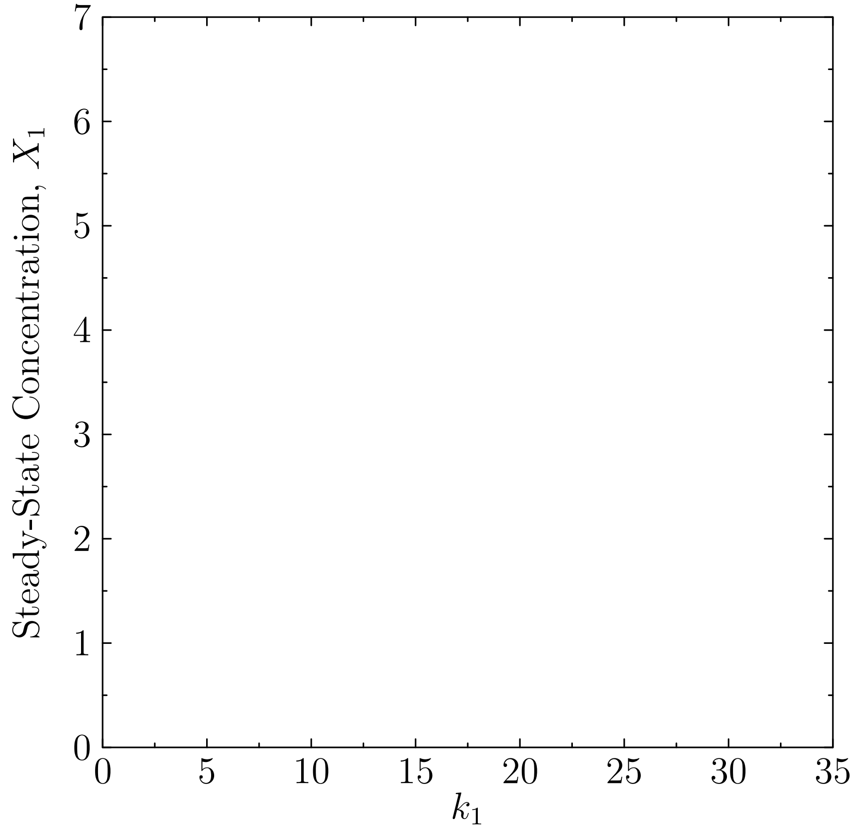
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\,X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\,X_2(t)$$

- Let $k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$
- But now we look at varying the parameter k_1
- The fixed points are given where the nullclines intersect

Varying k_1

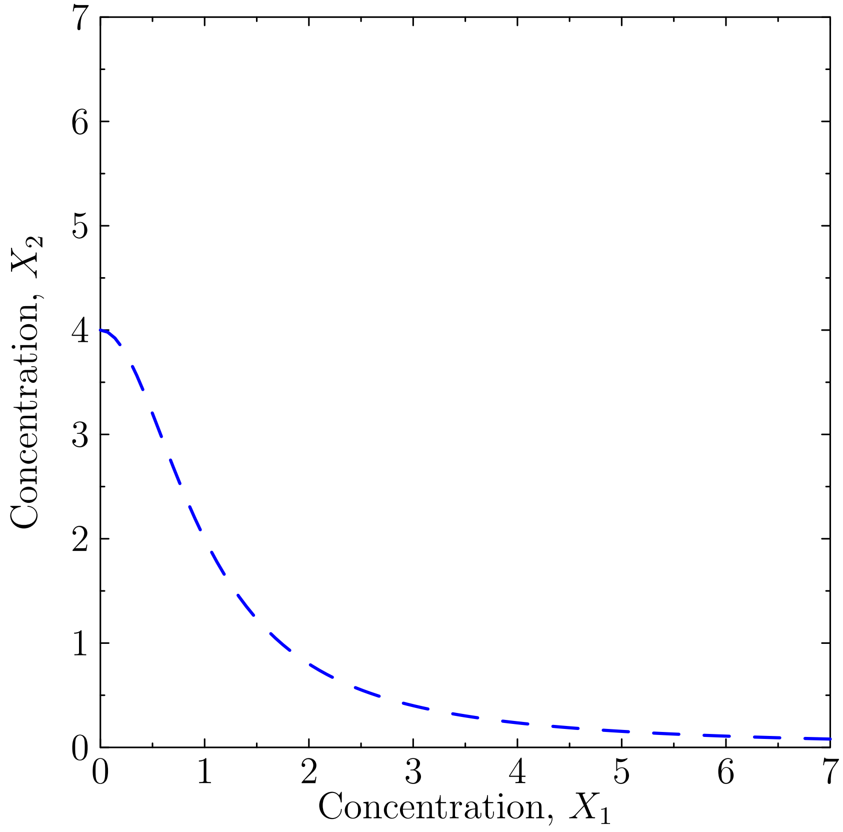


Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

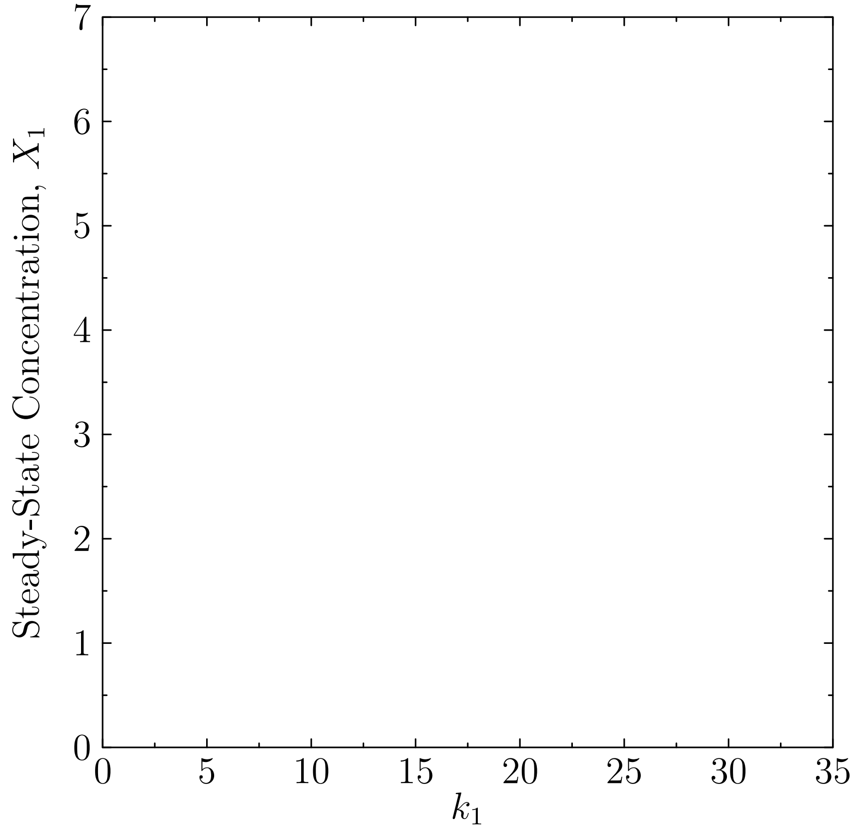


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35
1
2
3
4
6
7

Varying k_1

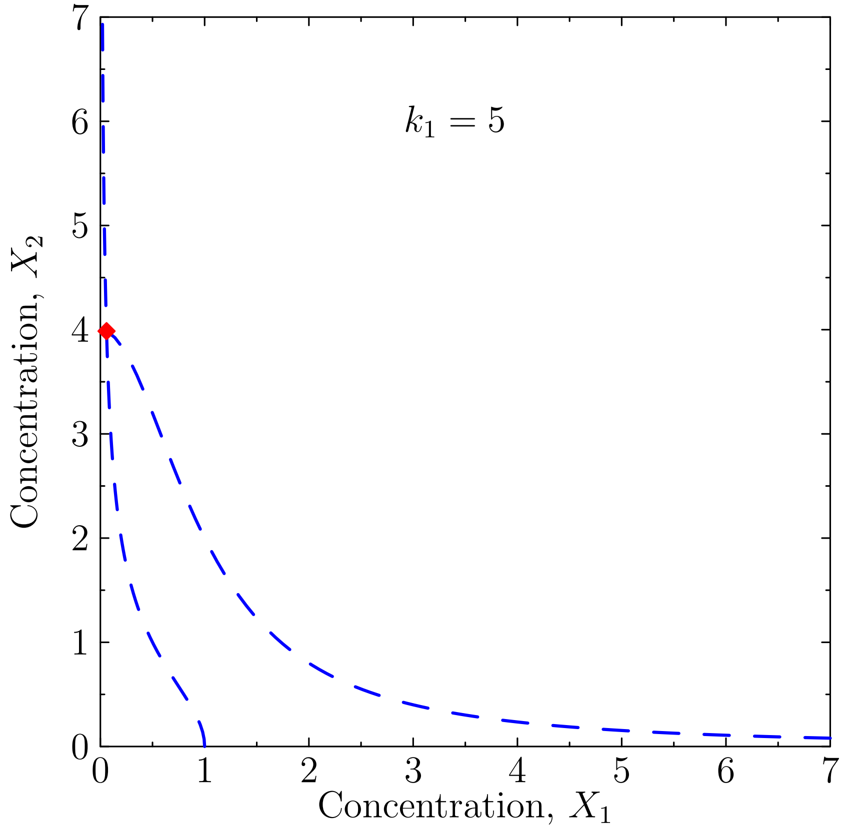


7
6
5
4
3
2
1
0
0
1
2
3
4
5
6
7
Concentration, X_2
Concentration, X_1

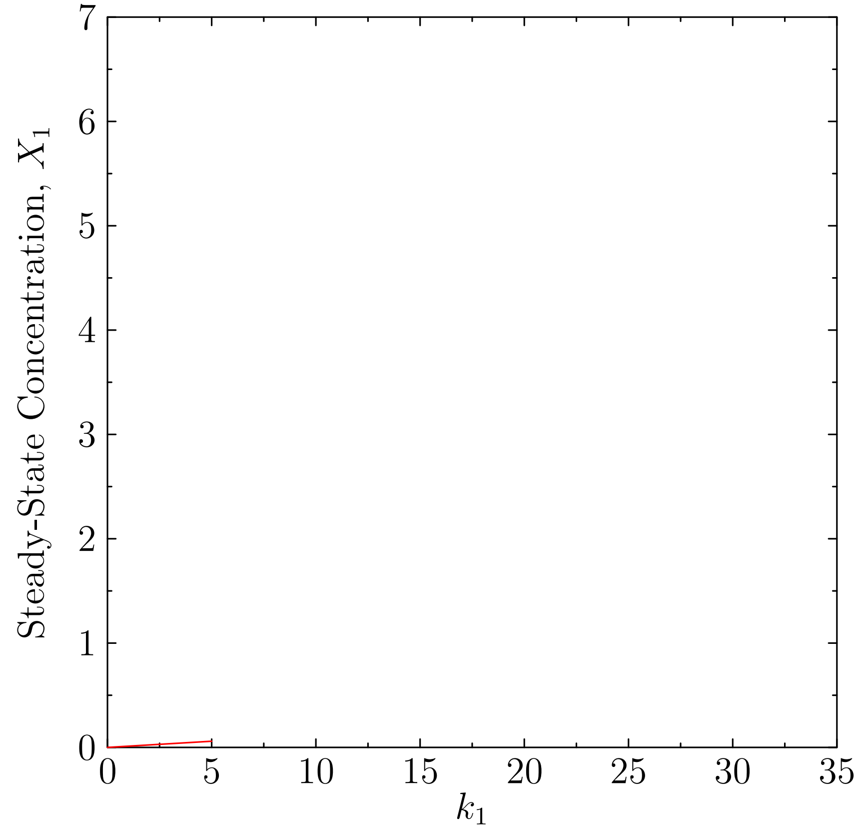


7
6
5
4
3
2
1
0
0
5
10
15
20
25
30
35
Steady-State Concentration, X_1
k_1

Varying k_1

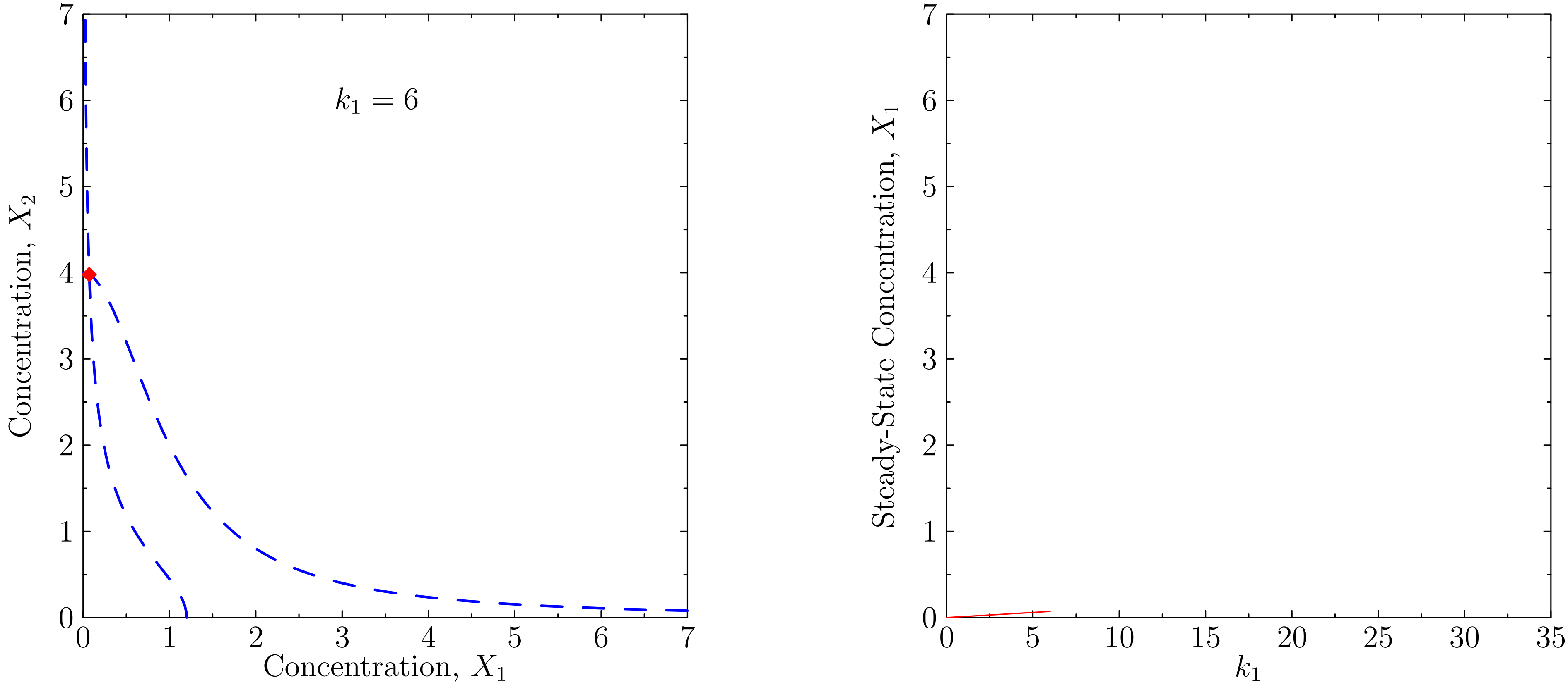


$k_1 = 5$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7



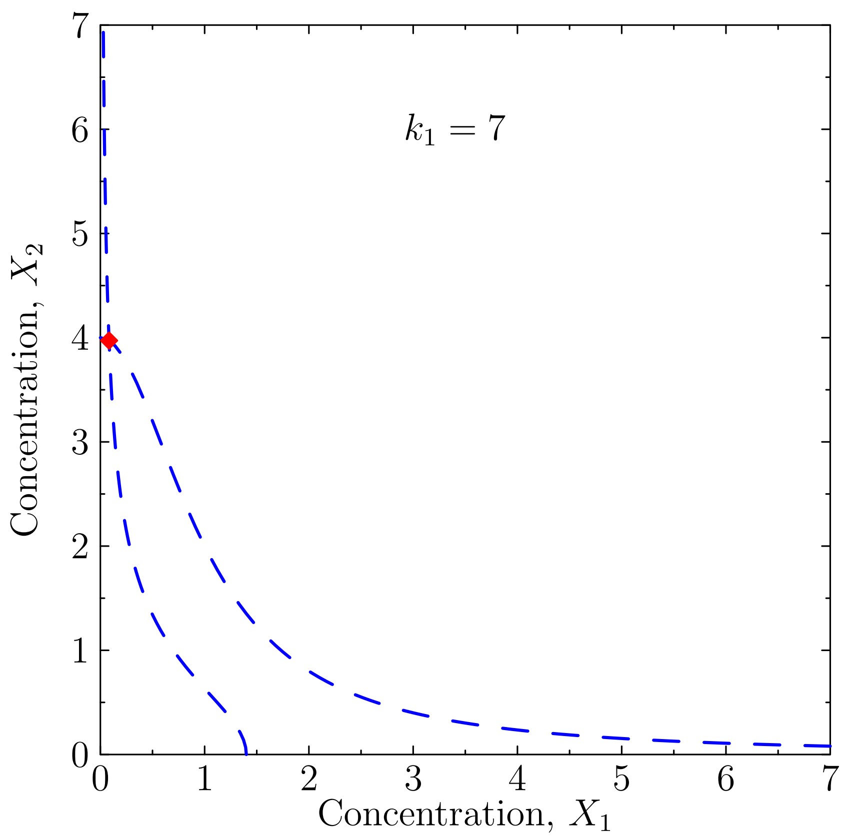
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

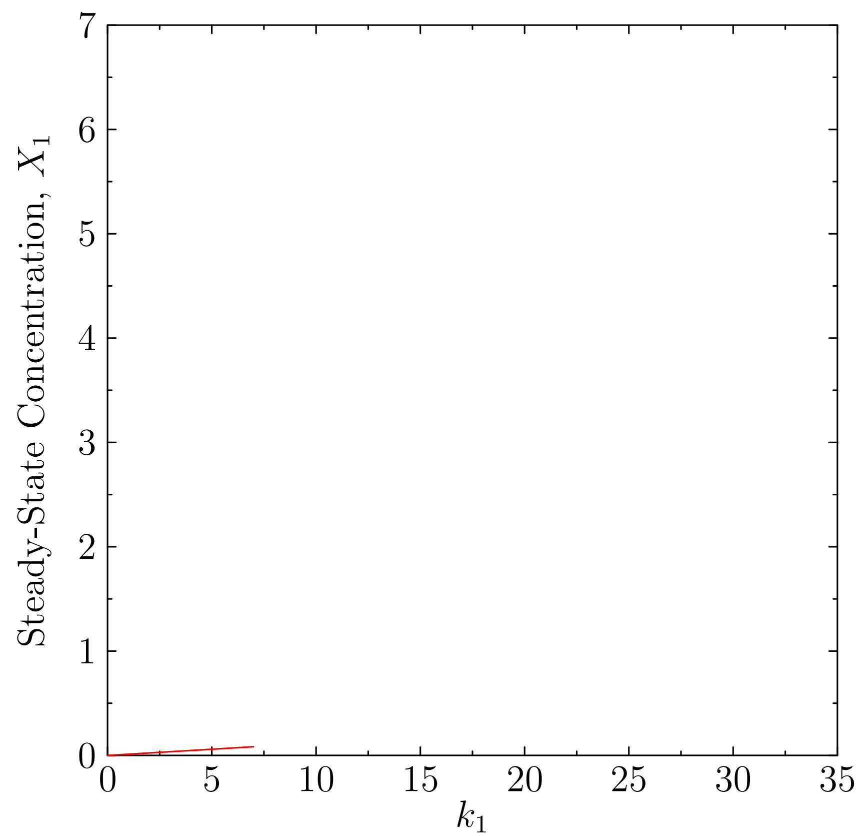


$k_1 = 6$
Concentration, X_1
Concentration, X_2
0
1
2
3
4
5
6
7
Steady-State Concentration, X_1
k_1
10
15
20
25
30
35

Varying k_1

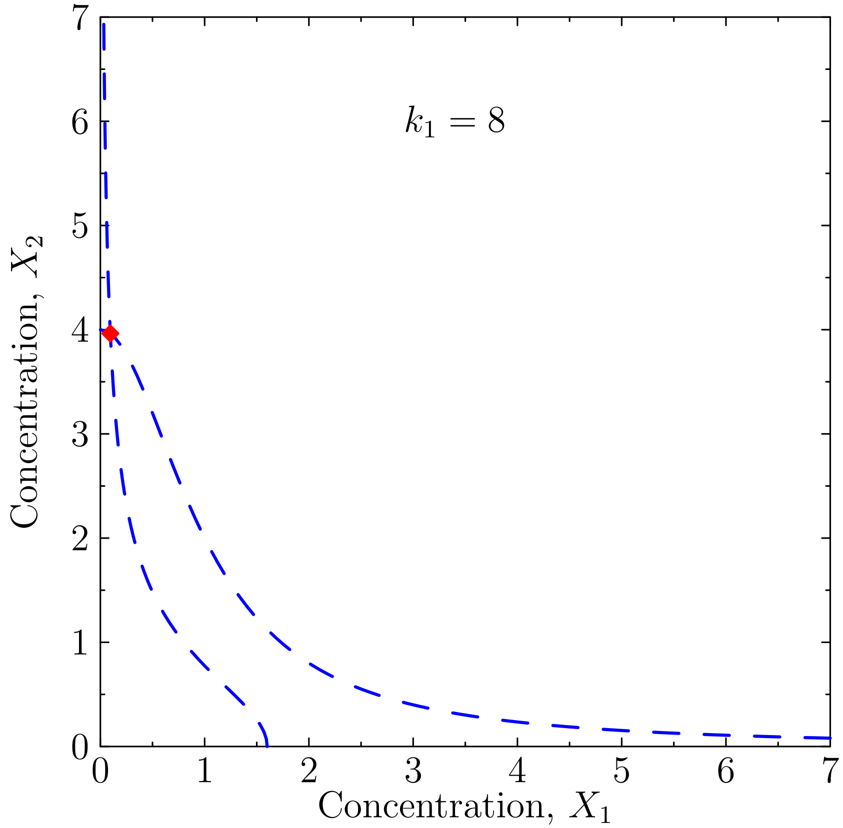


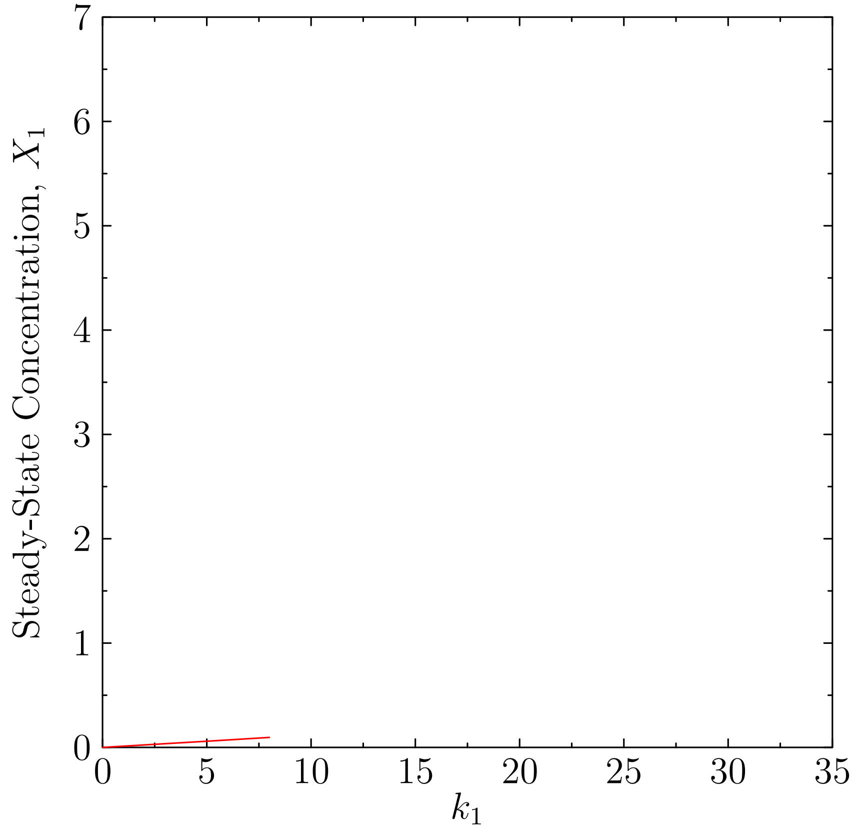
$k_1 = 7$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7



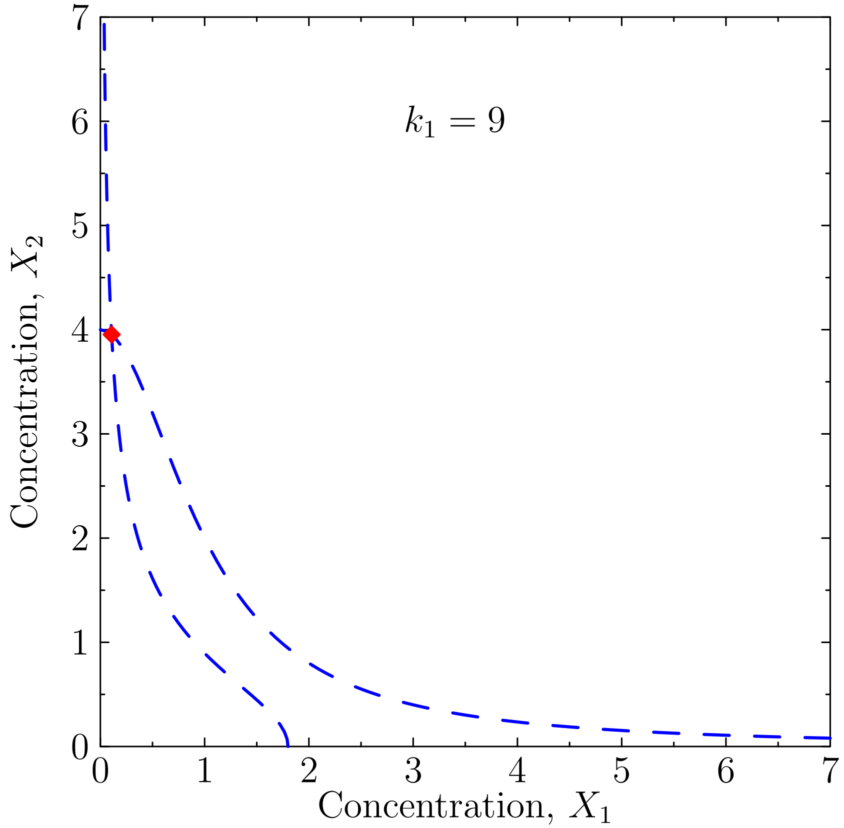
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

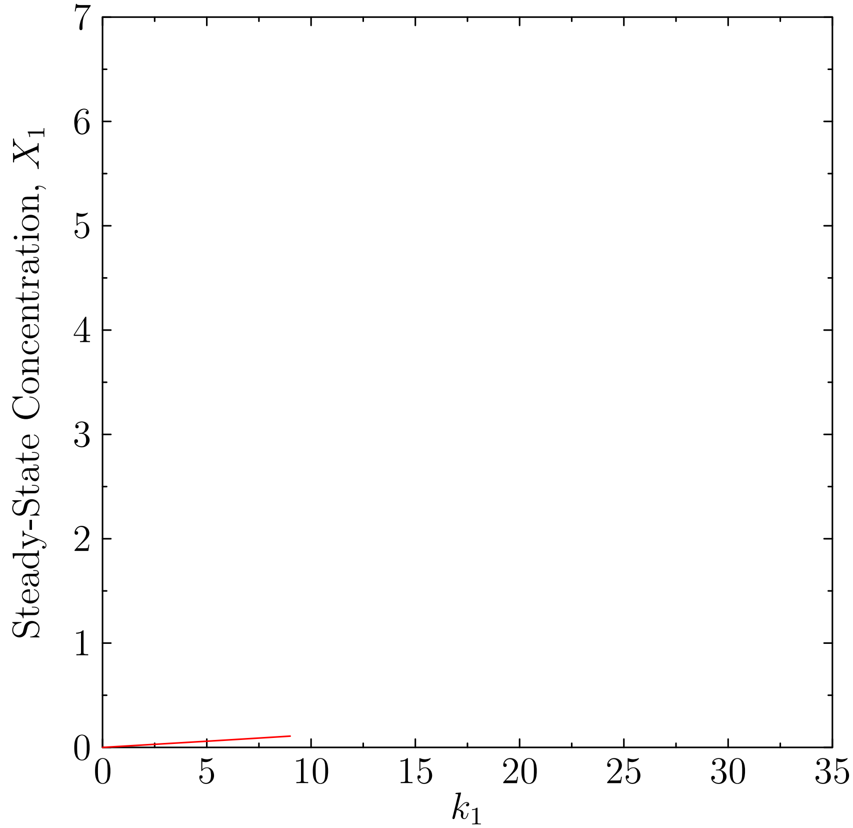

$k_1 = 8$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

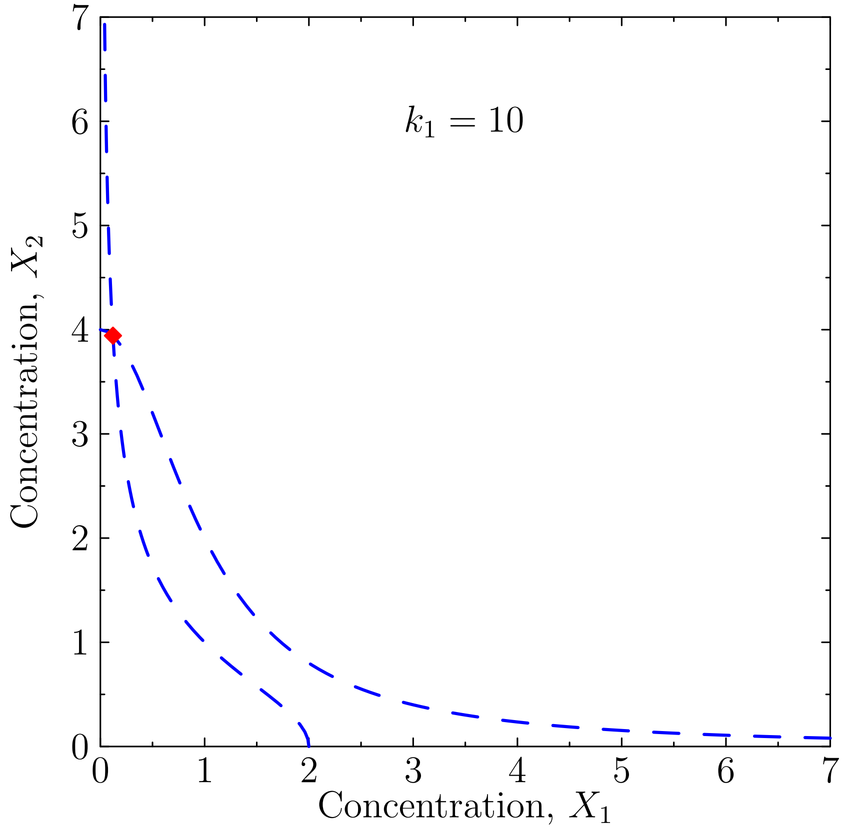


$k_1 = 9$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

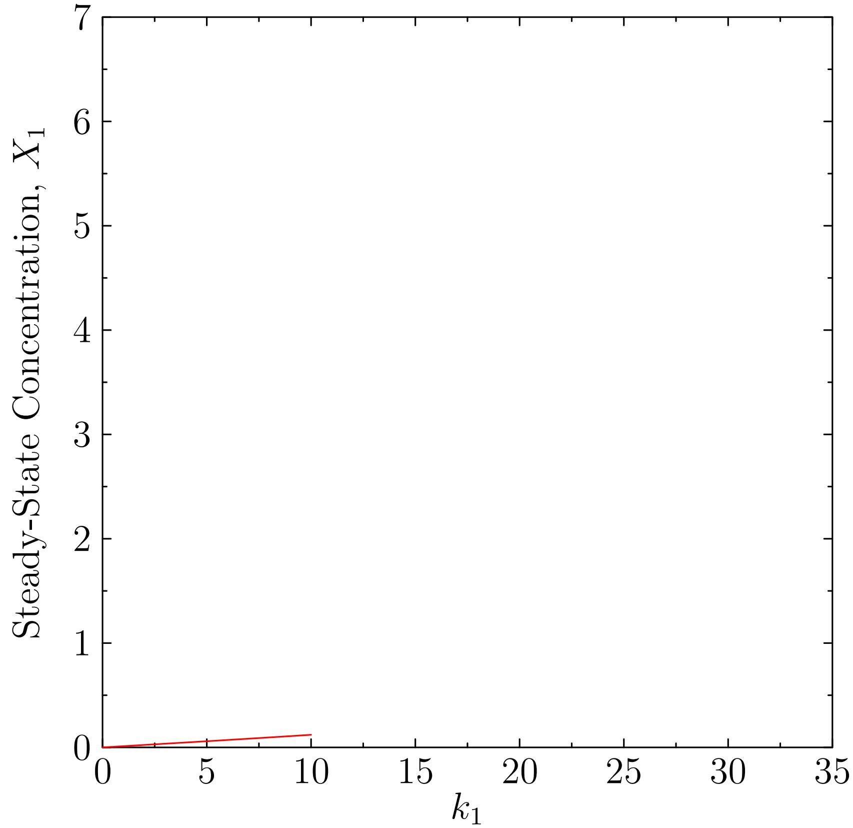


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

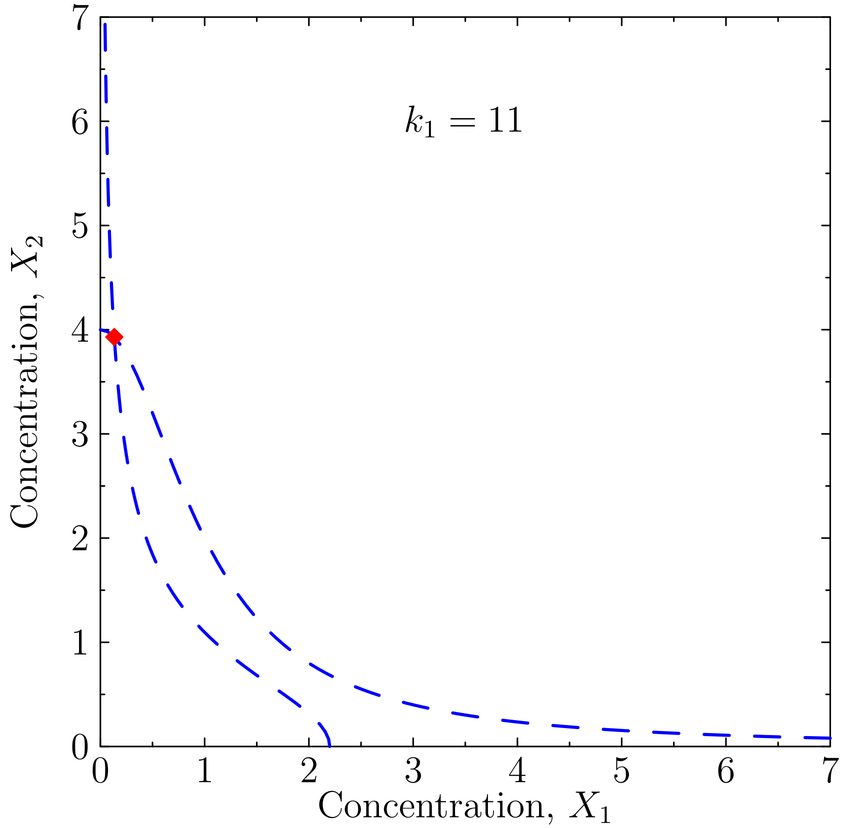


$k_1 = 10$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

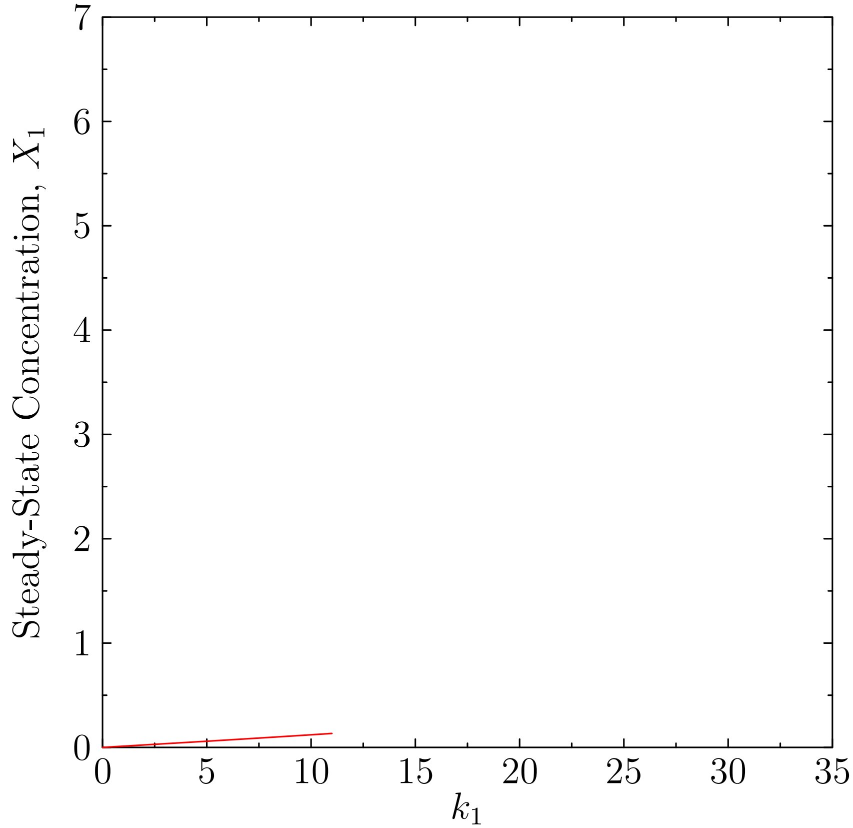


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

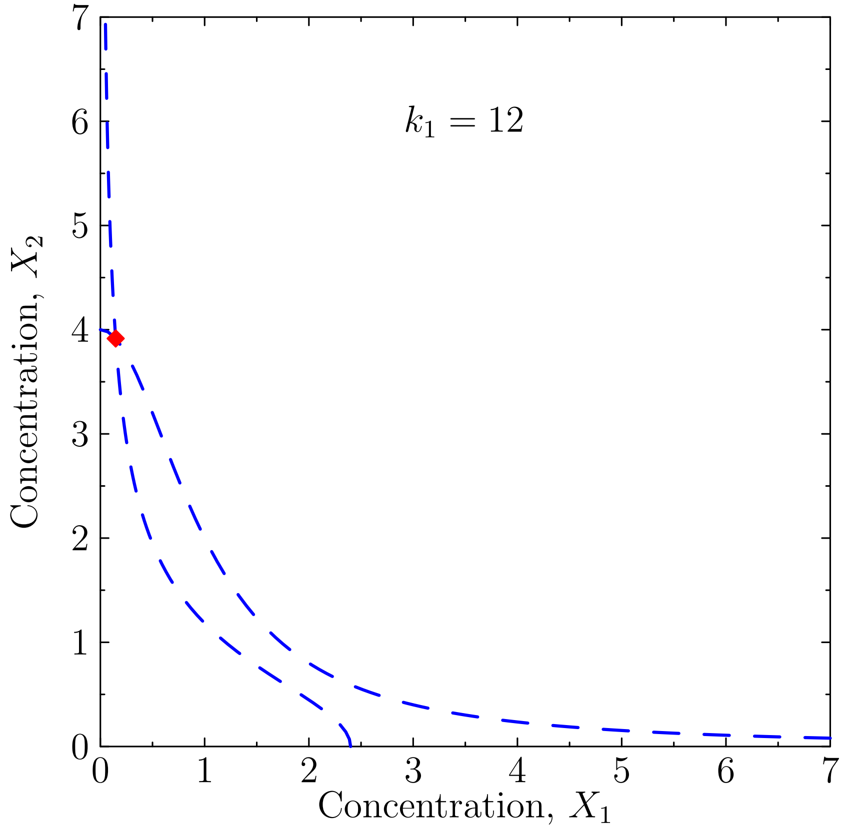


$k_1 = 11$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

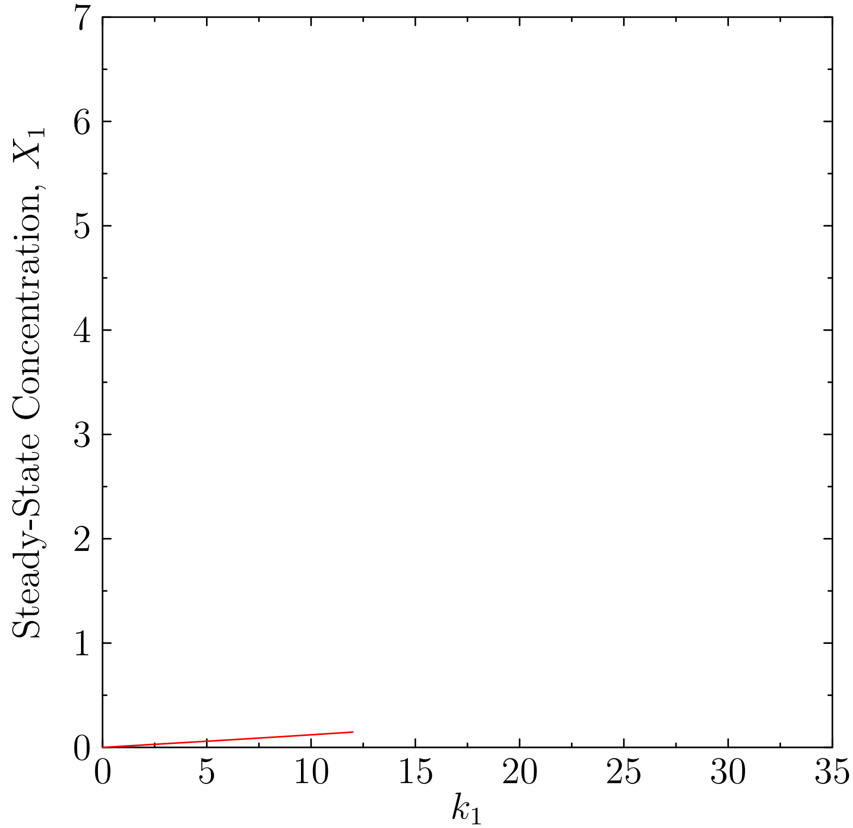


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

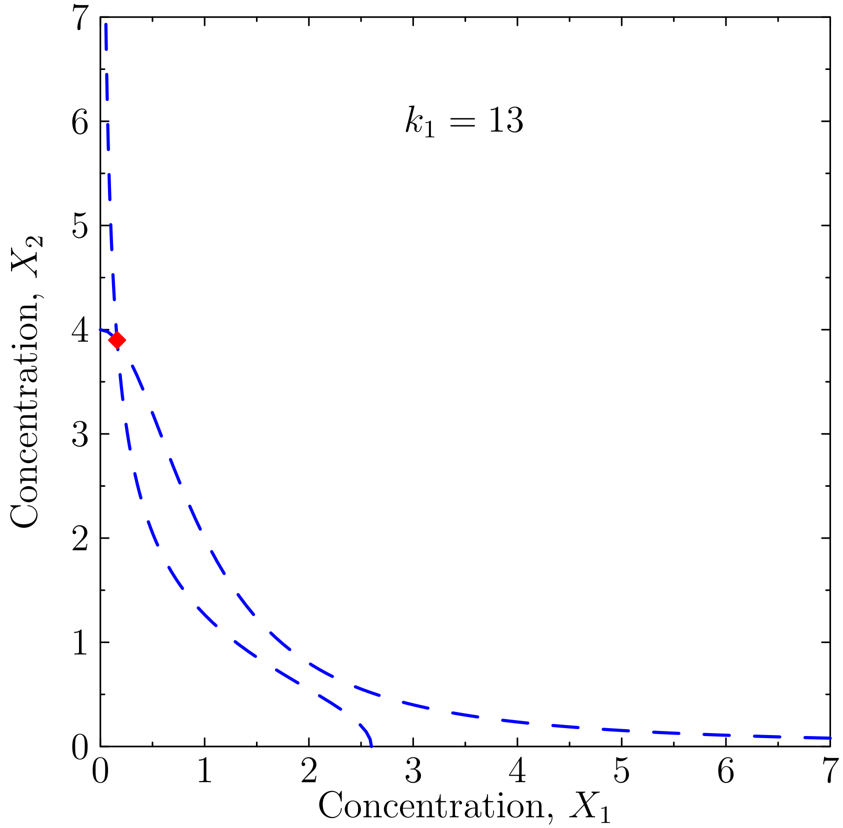


$k_1 = 12$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

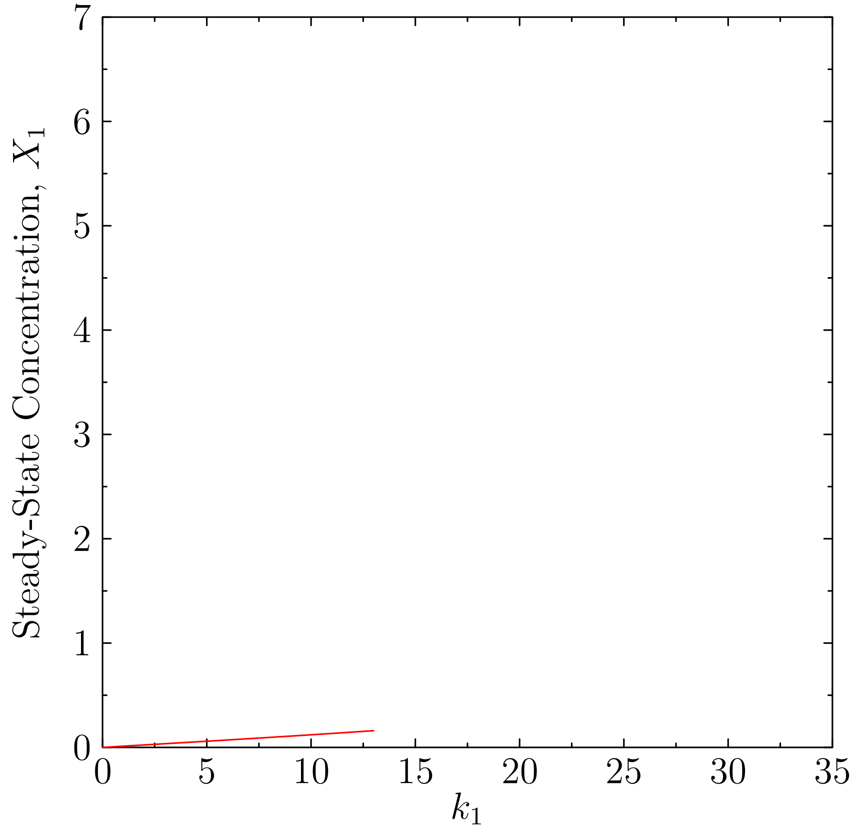


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

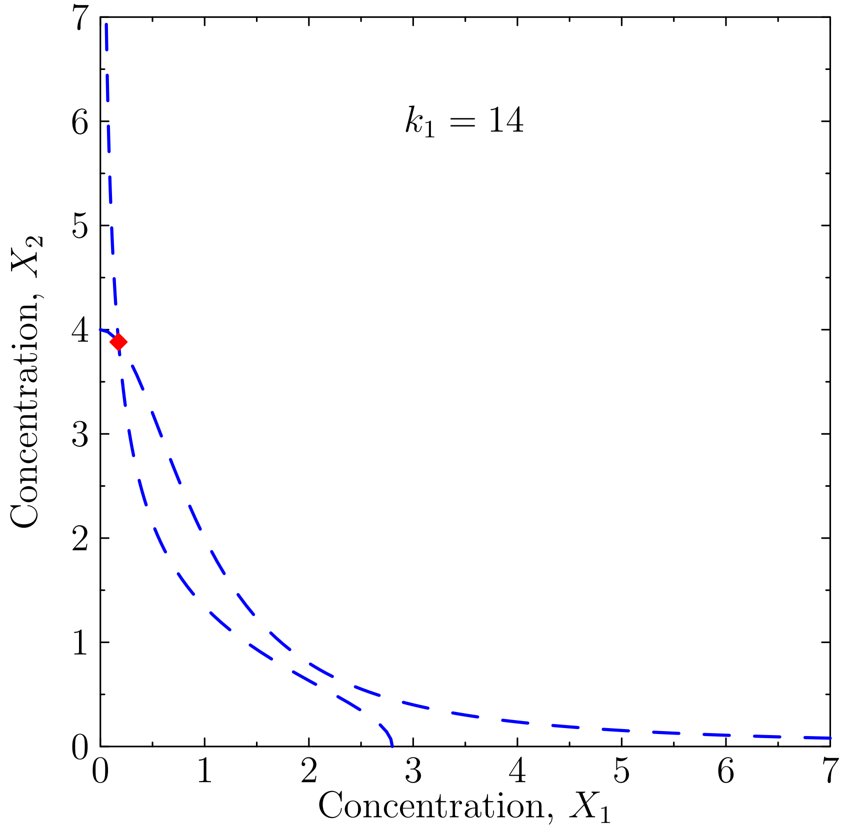


$k_1 = 13$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

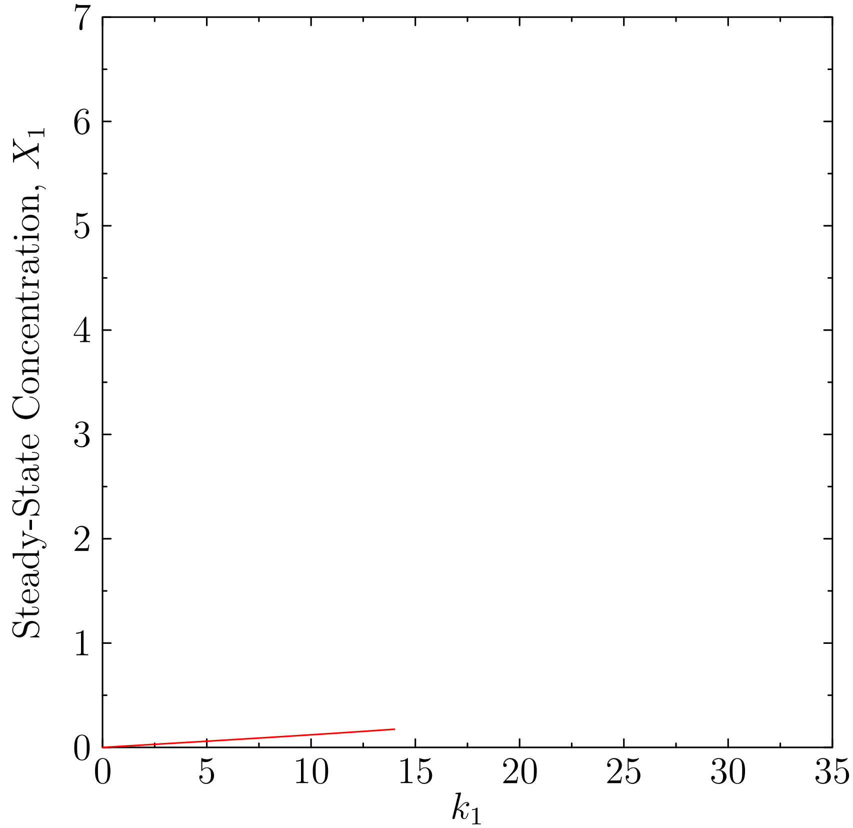


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

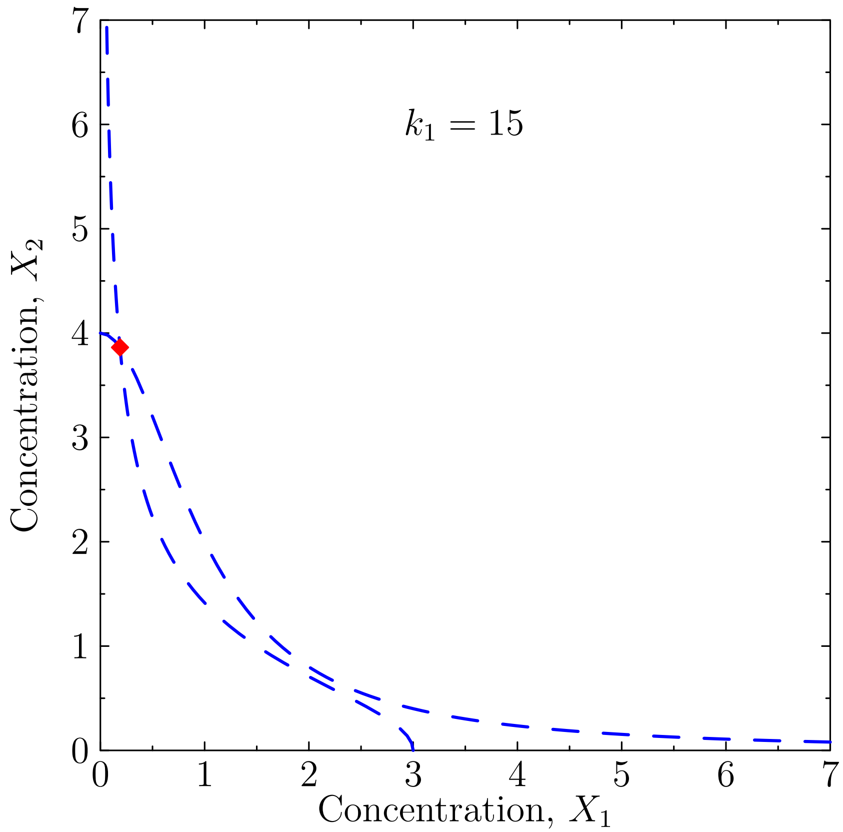


$k_1 = 14$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

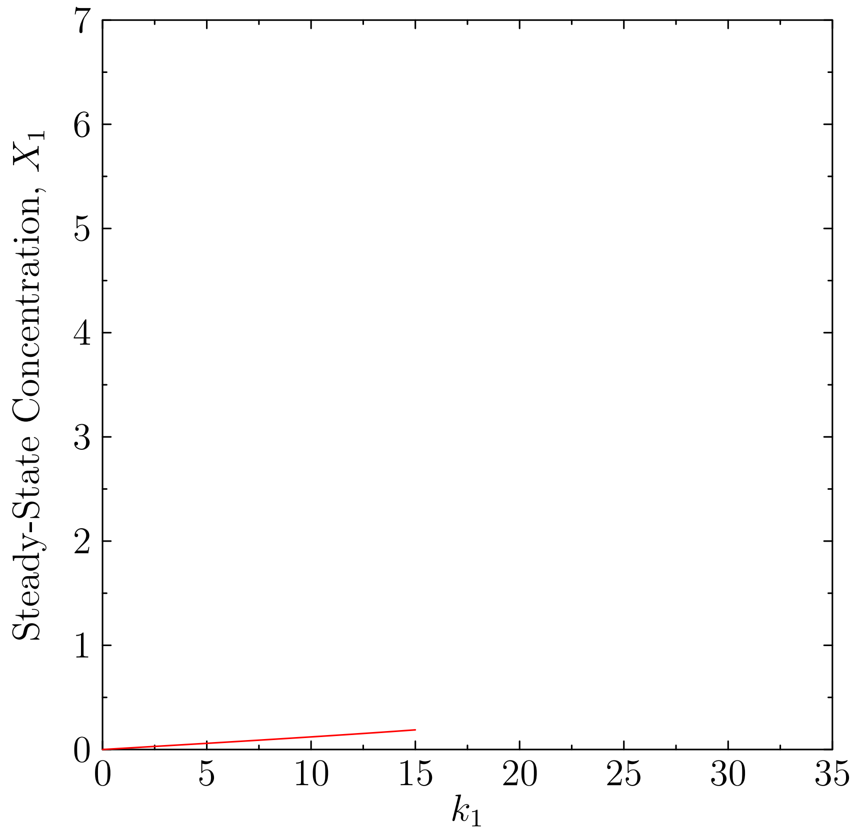


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

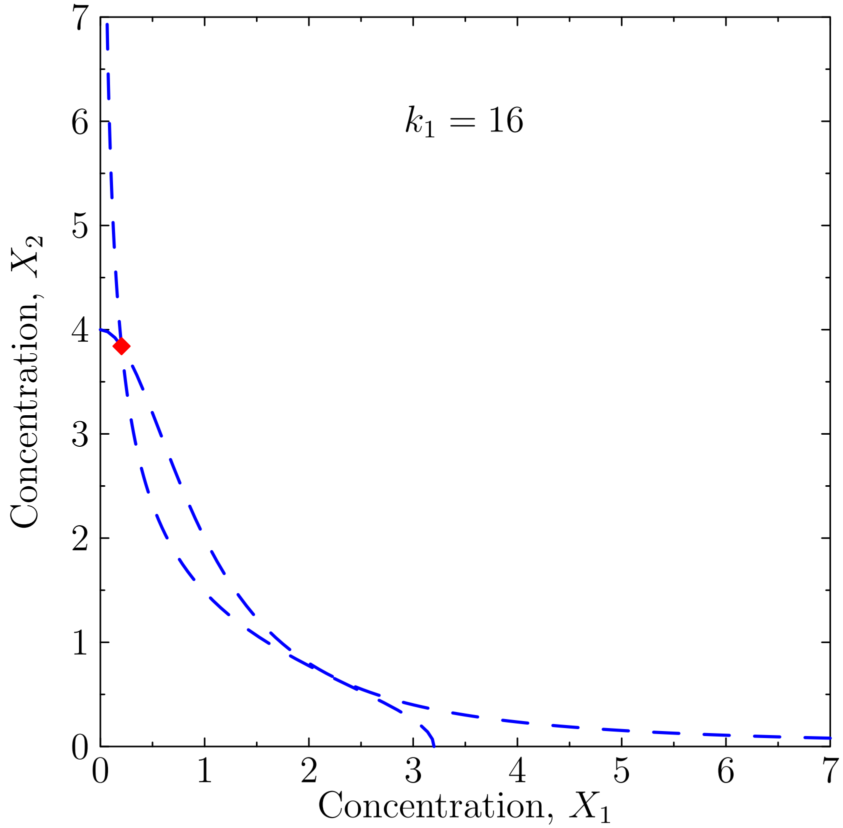


$k_1 = 15$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

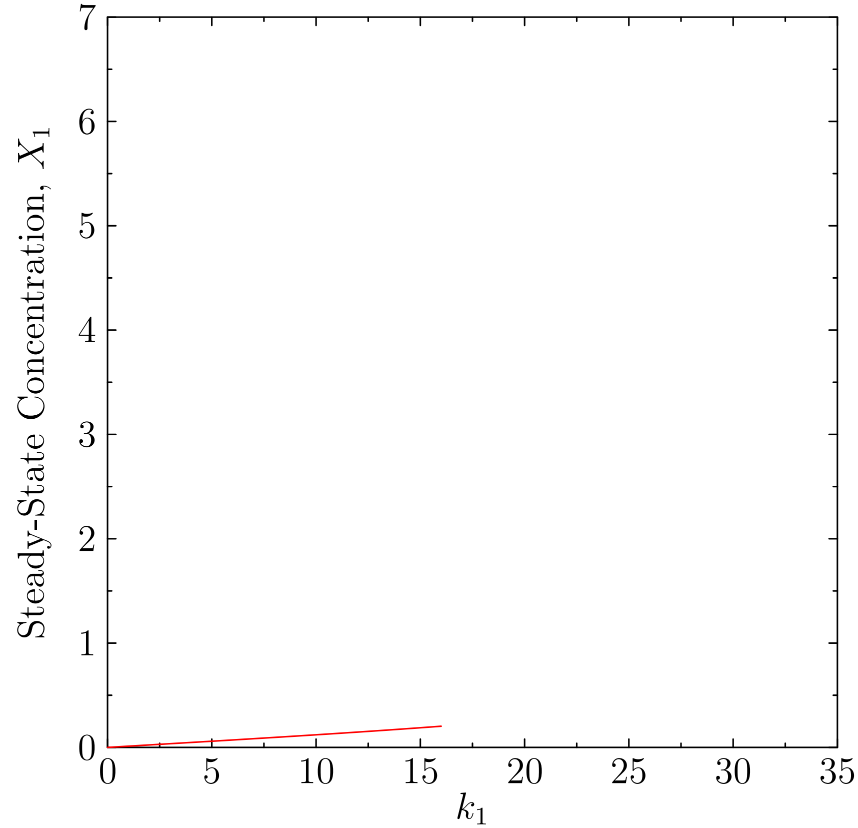


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

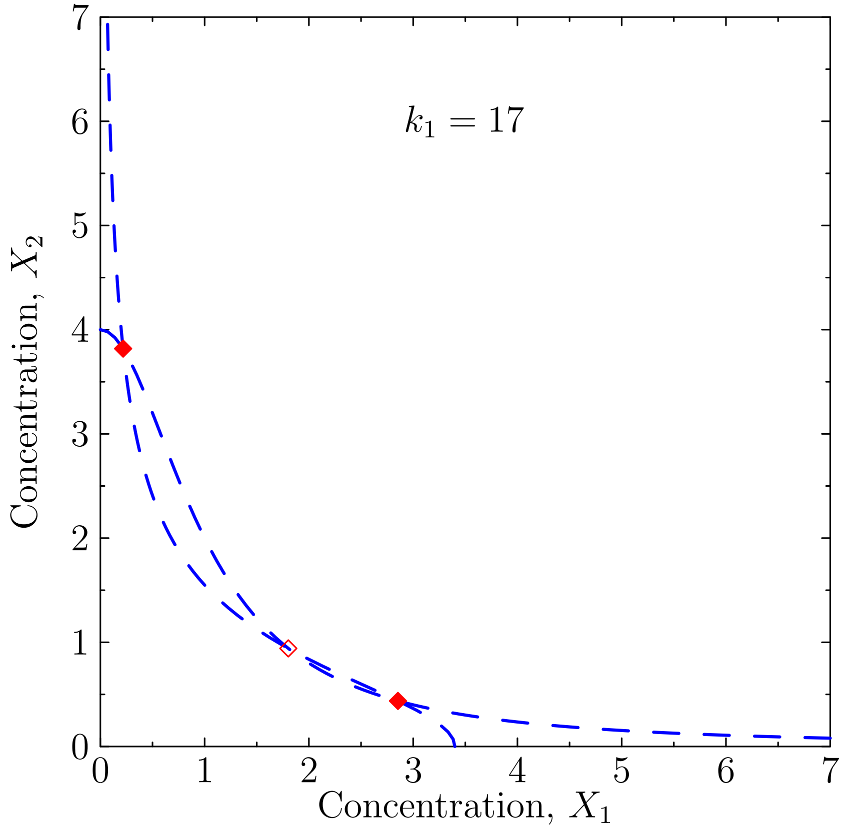


$k_1 = 16$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

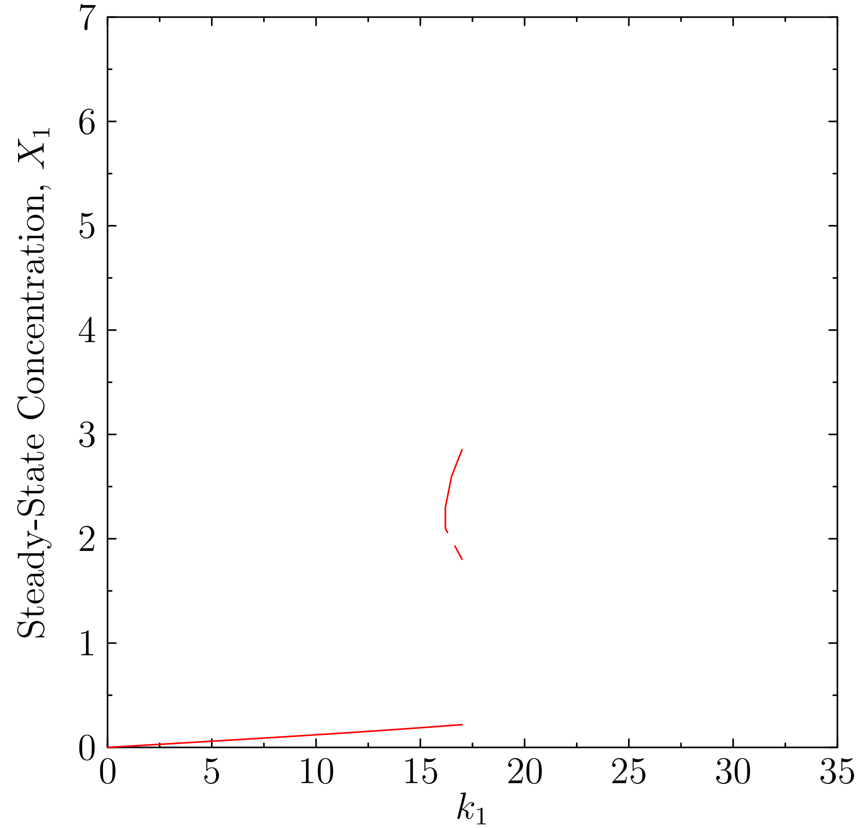


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

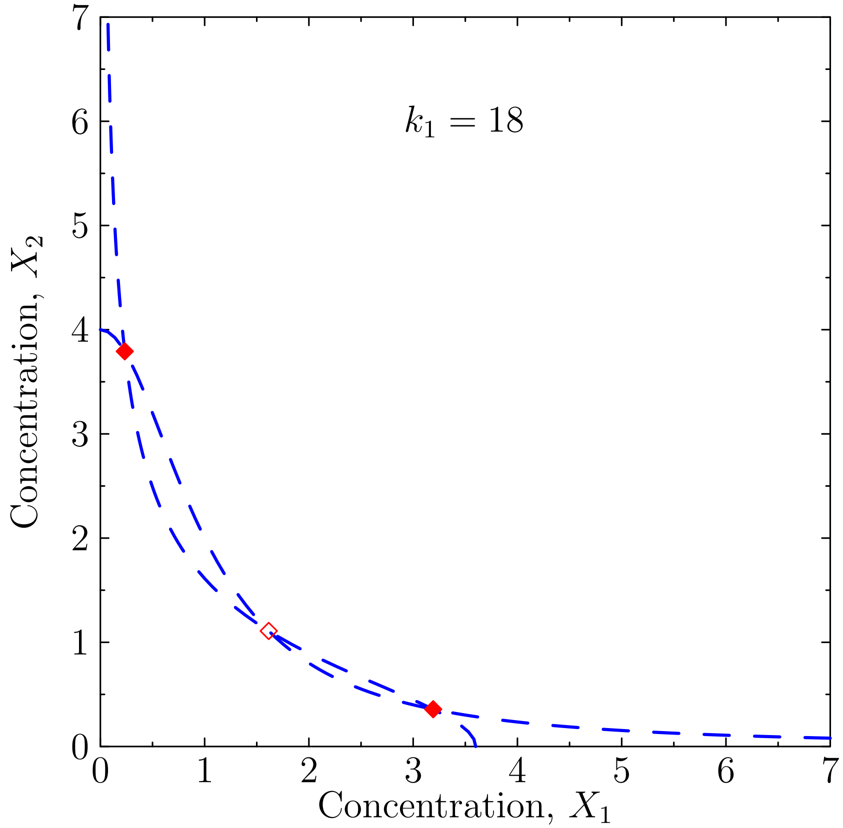


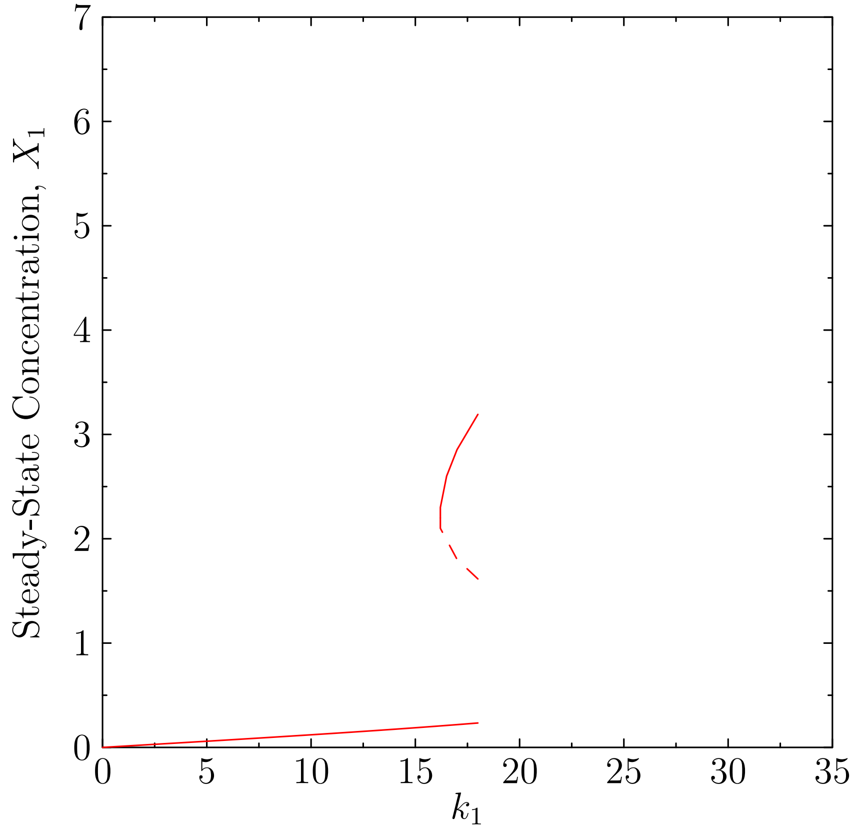
$k_1 = 17$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7



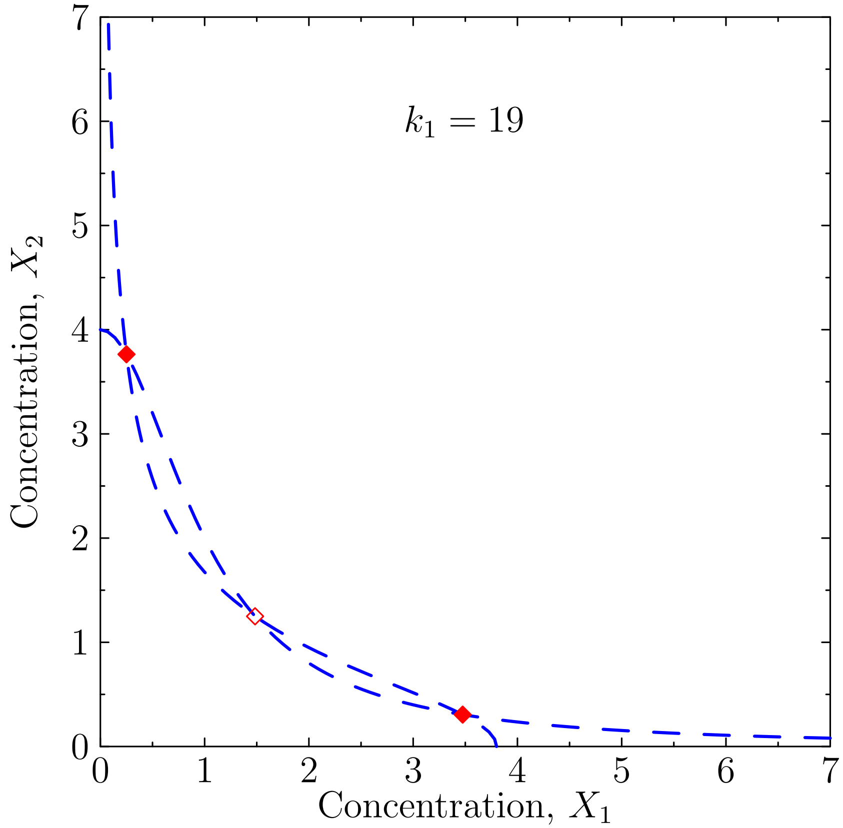
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

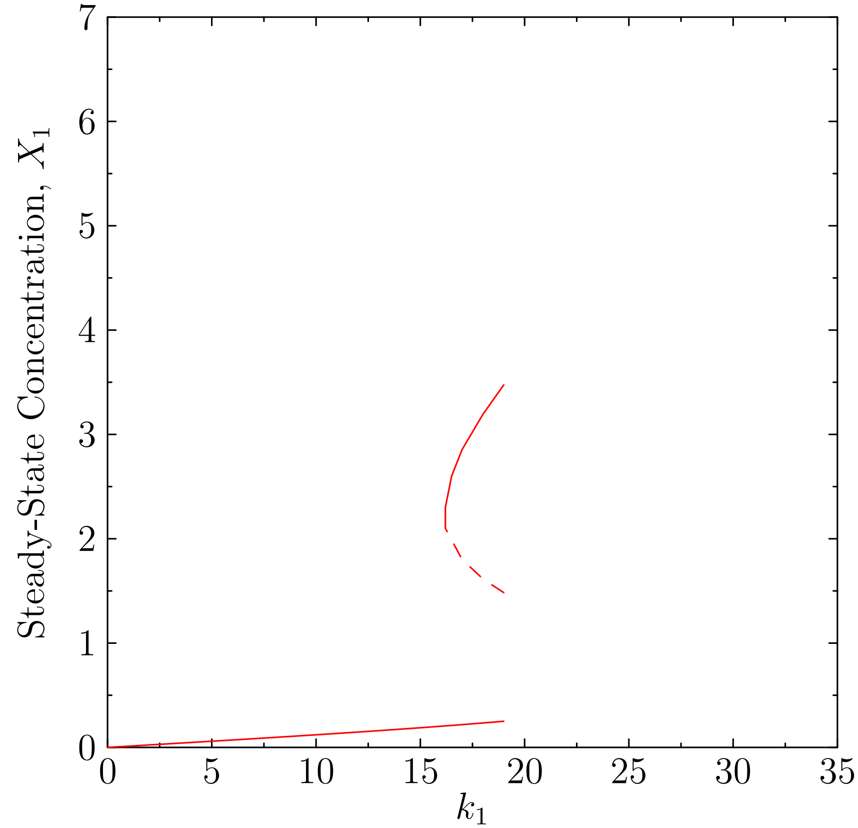

$k_1 = 18$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

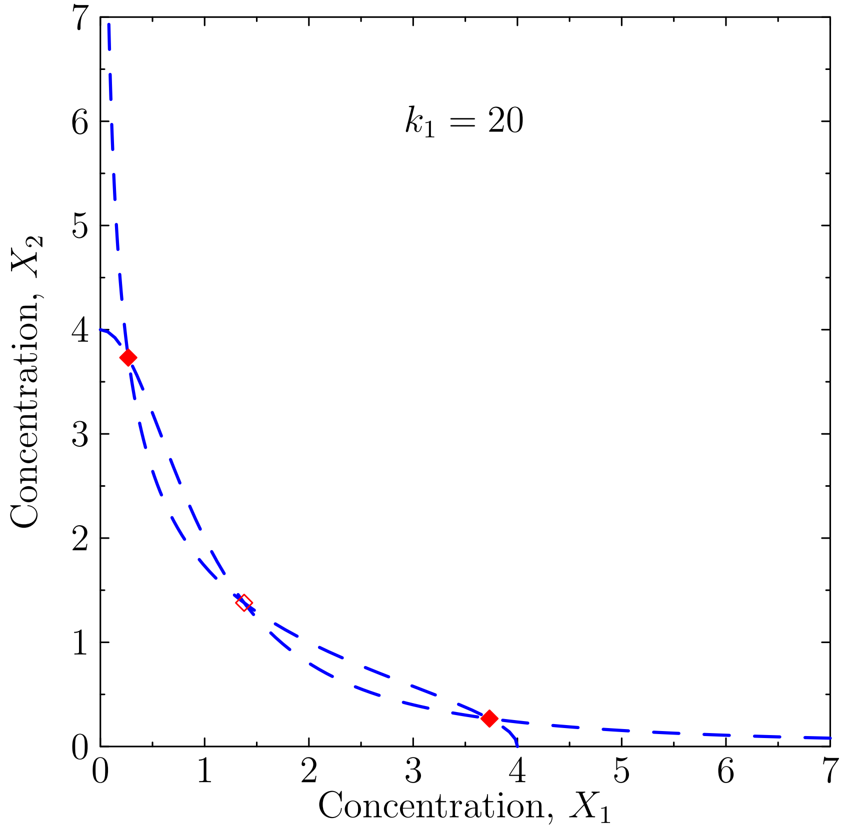


$k_1 = 19$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

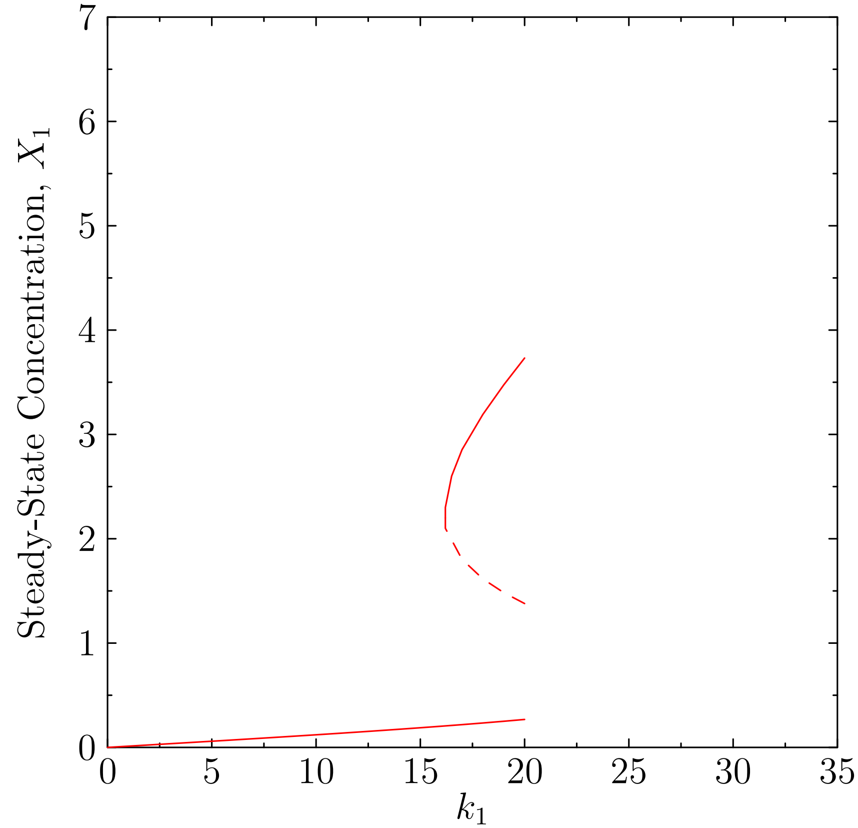


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

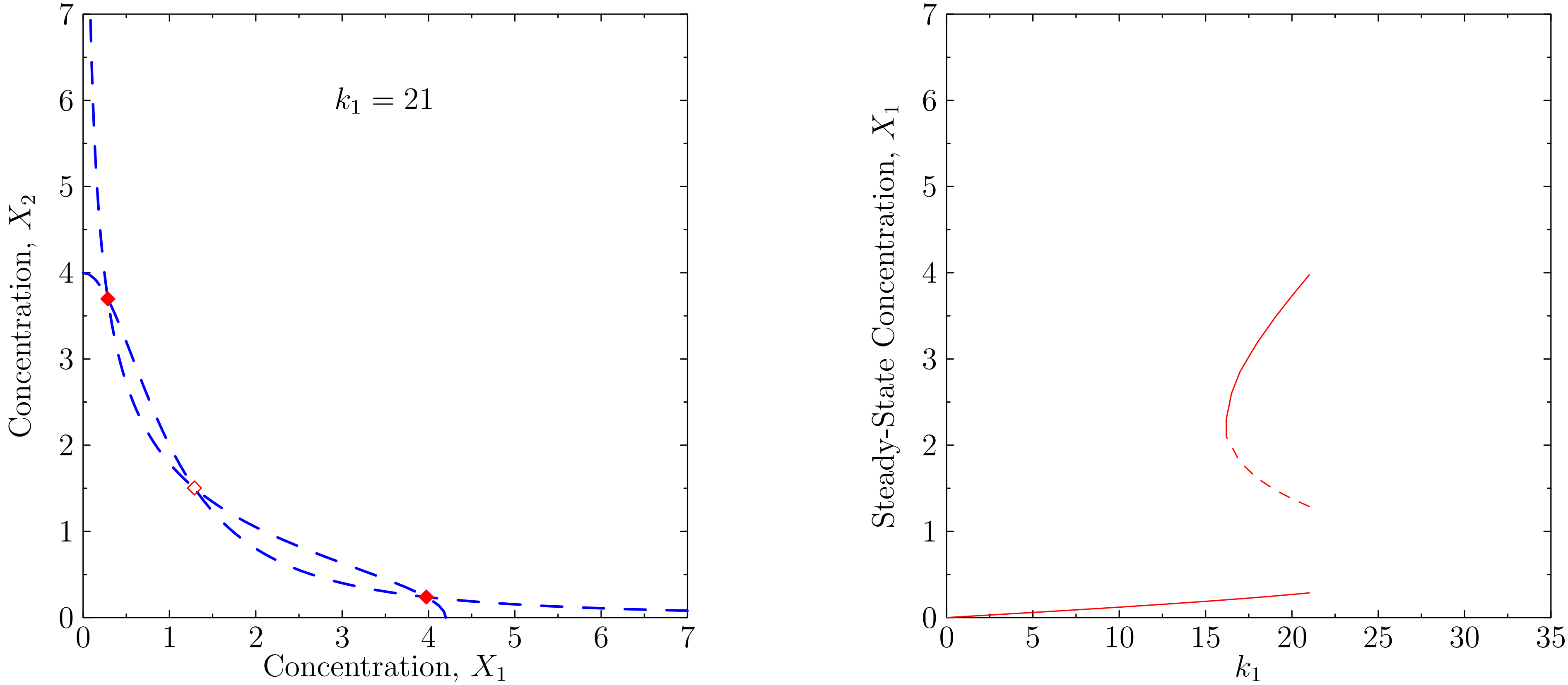


$k_1 = 20$
Concentration, X_1
Concentration, X_2
0
1
2
3
4
5
6
7



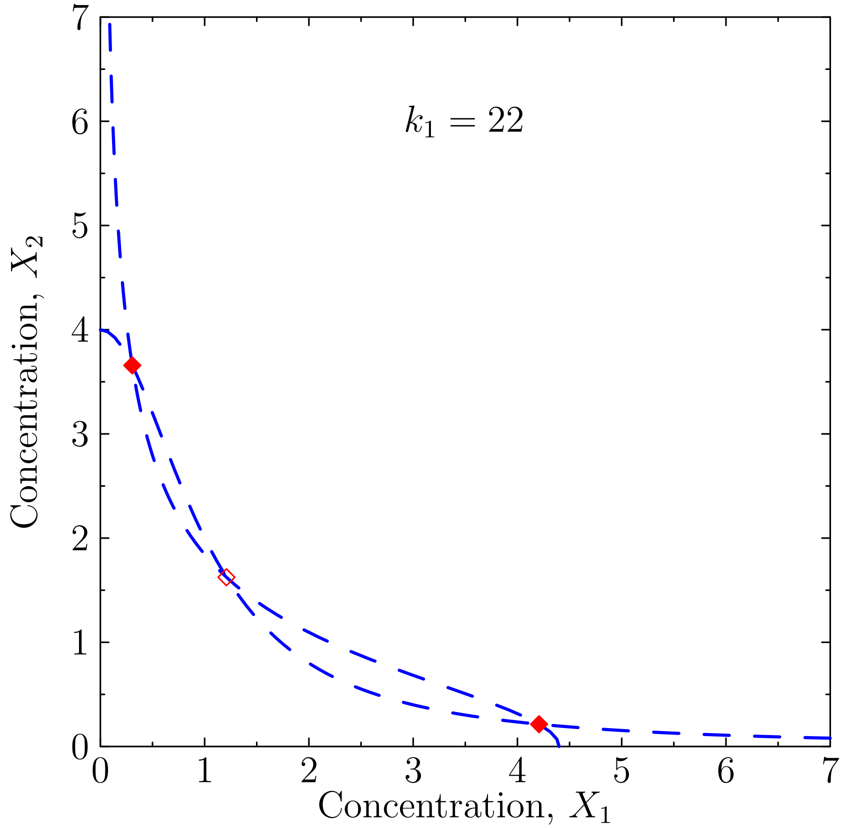
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

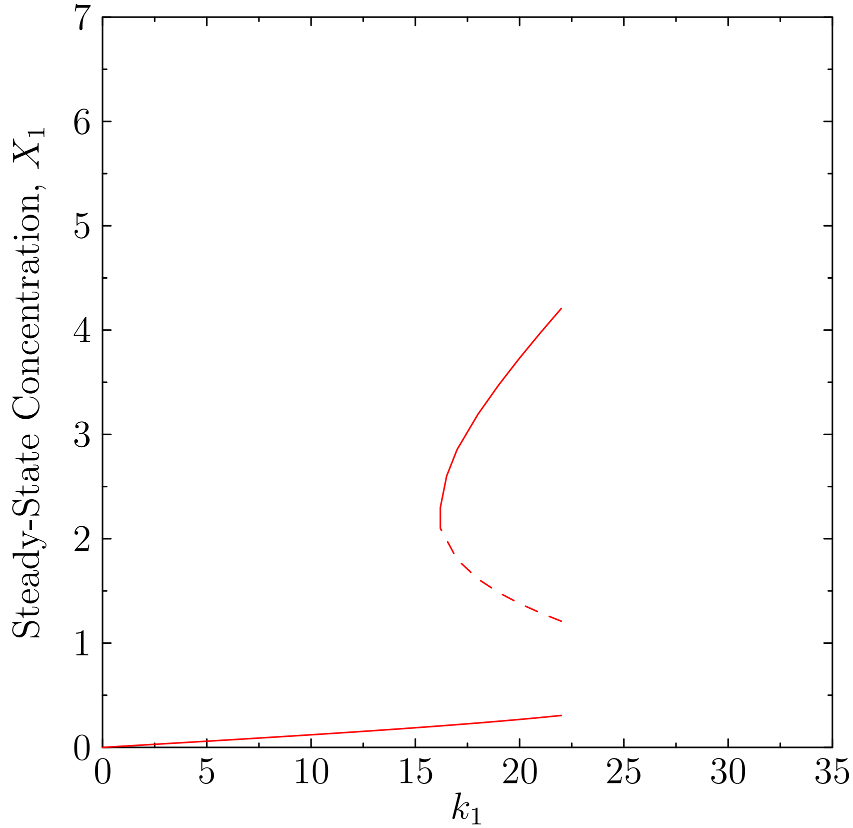


$k_1 = 21$
Concentration, X_2
Concentration, X_1
Steady-State Concentration, X_1
k_1

Varying k_1

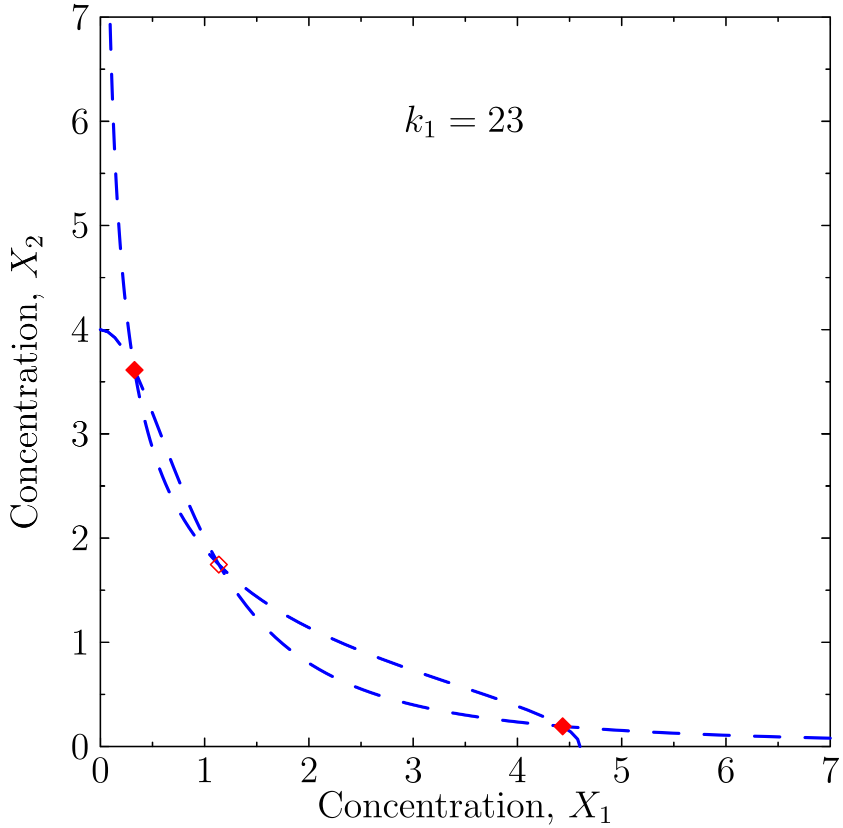


$k_1 = 22$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7
0
1
2
3
4
5
6
7

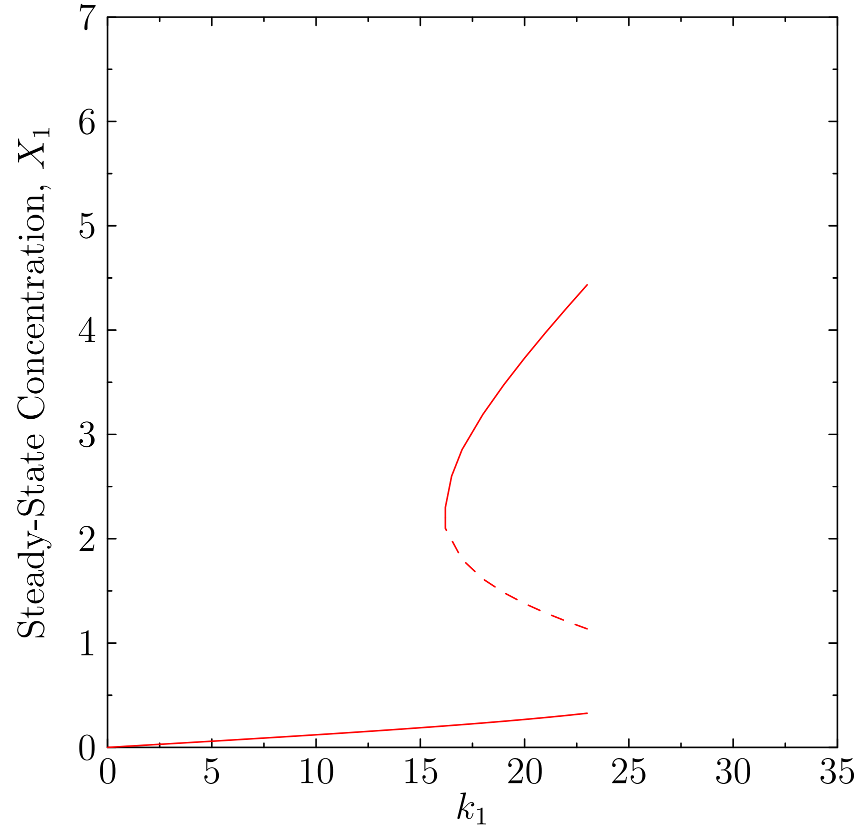


Steady-State Concentration, X_1
k_1
0
1
2
3
4
5
6
7
0
5
10
15
20
25
30
35

Varying k_1

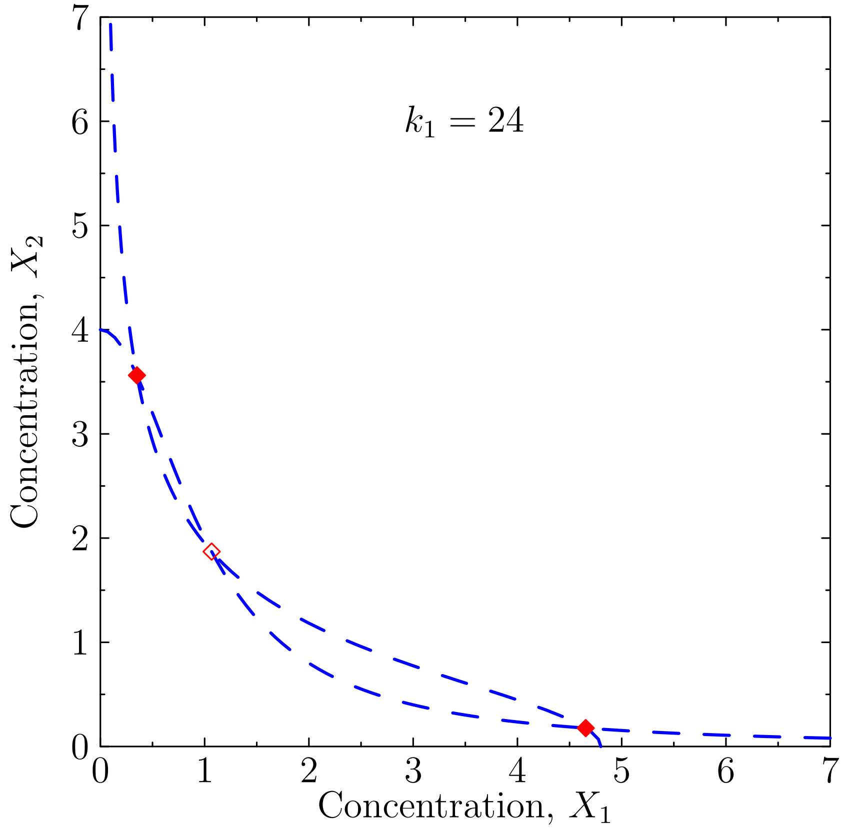


$k_1 = 23$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7
0
1
2
3
4
5
6
7

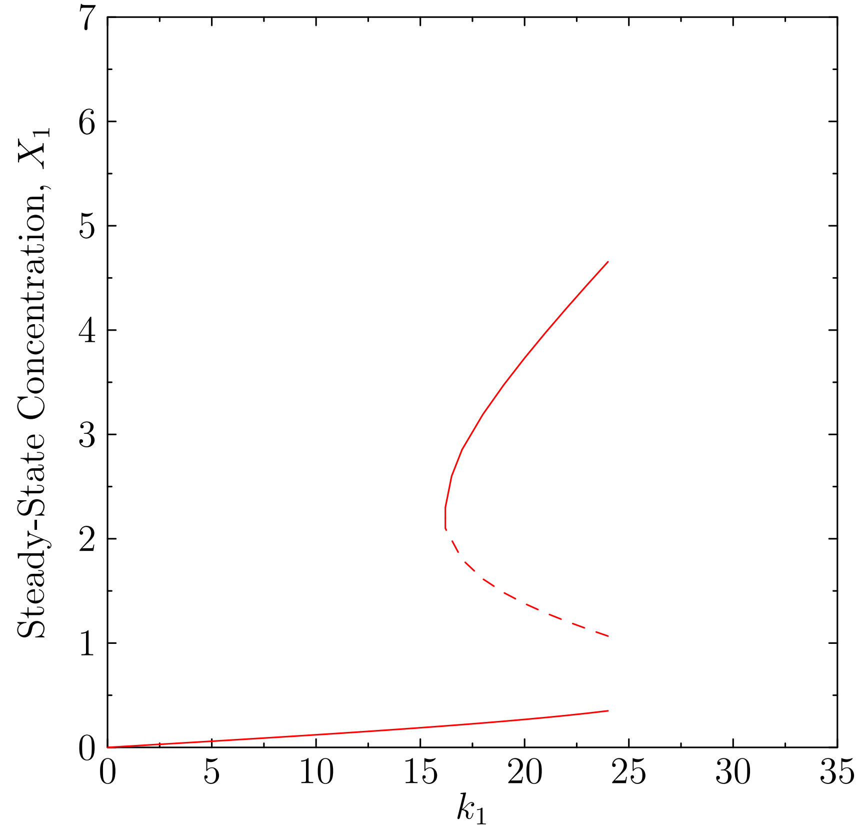


Steady-State Concentration, X_1
k_1
0
1
2
3
4
5
6
7
0
5
10
15
20
25
30
35

Varying k_1

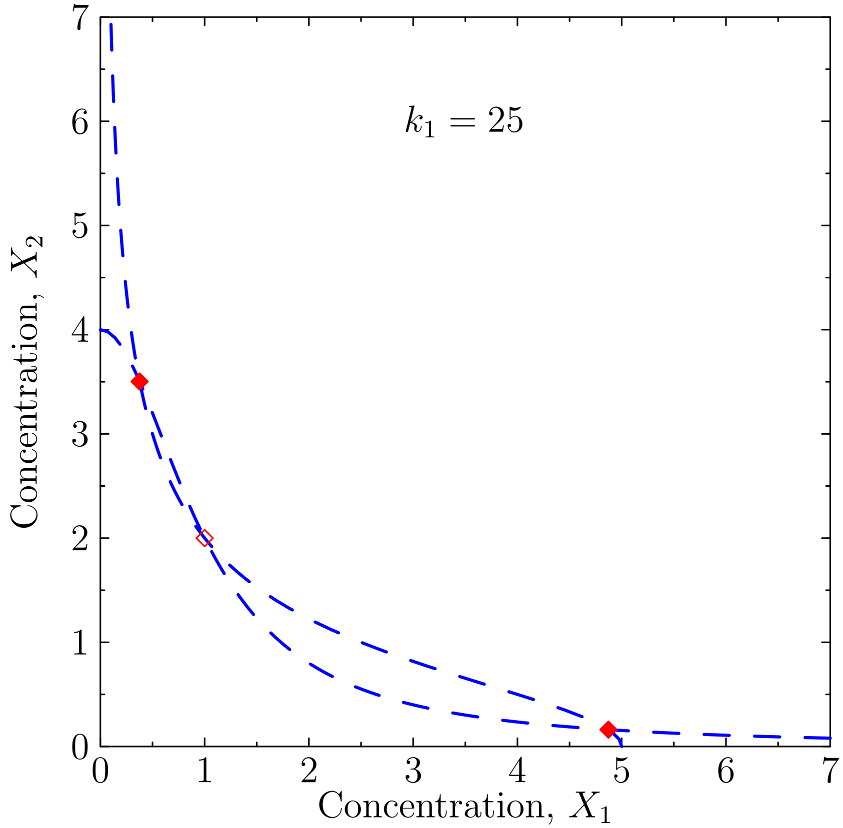


$k_1 = 24$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

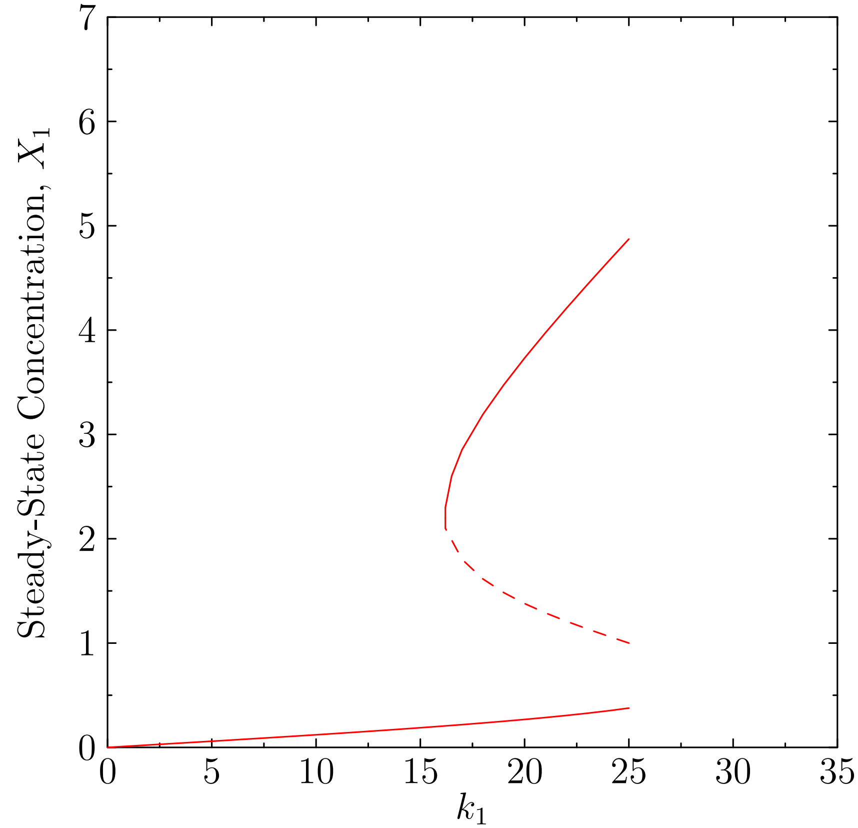


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

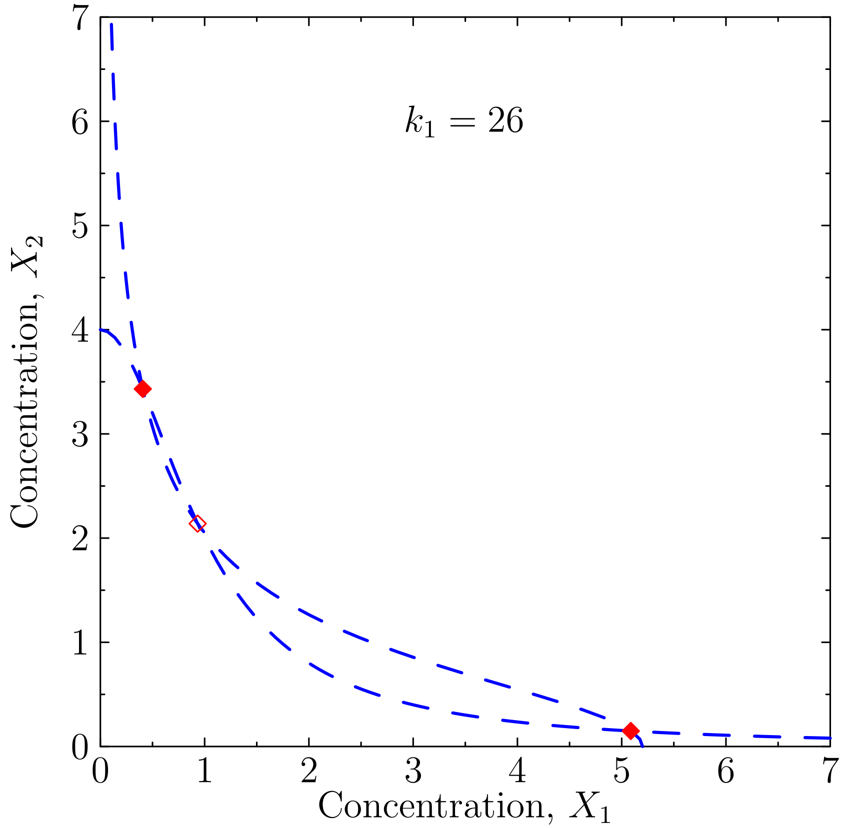


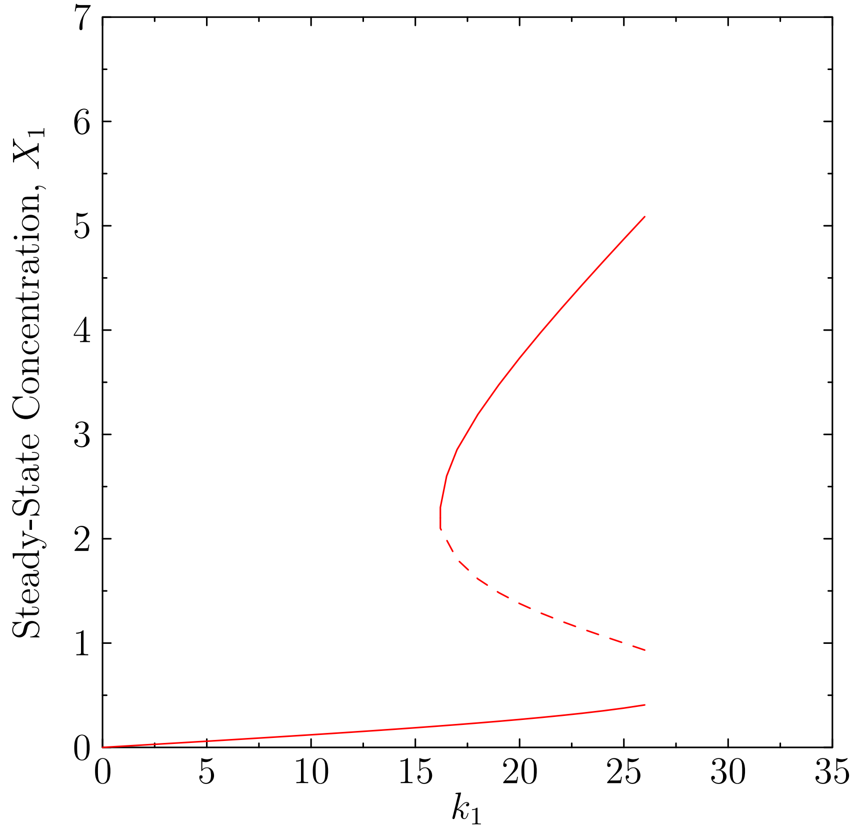
$k_1 = 25$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7



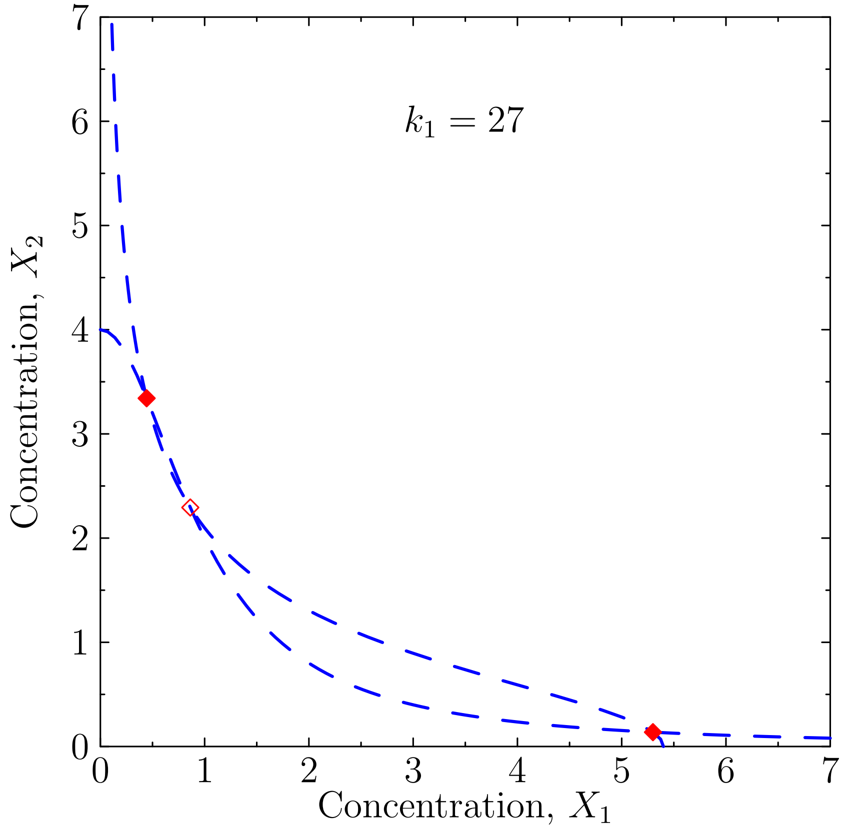
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

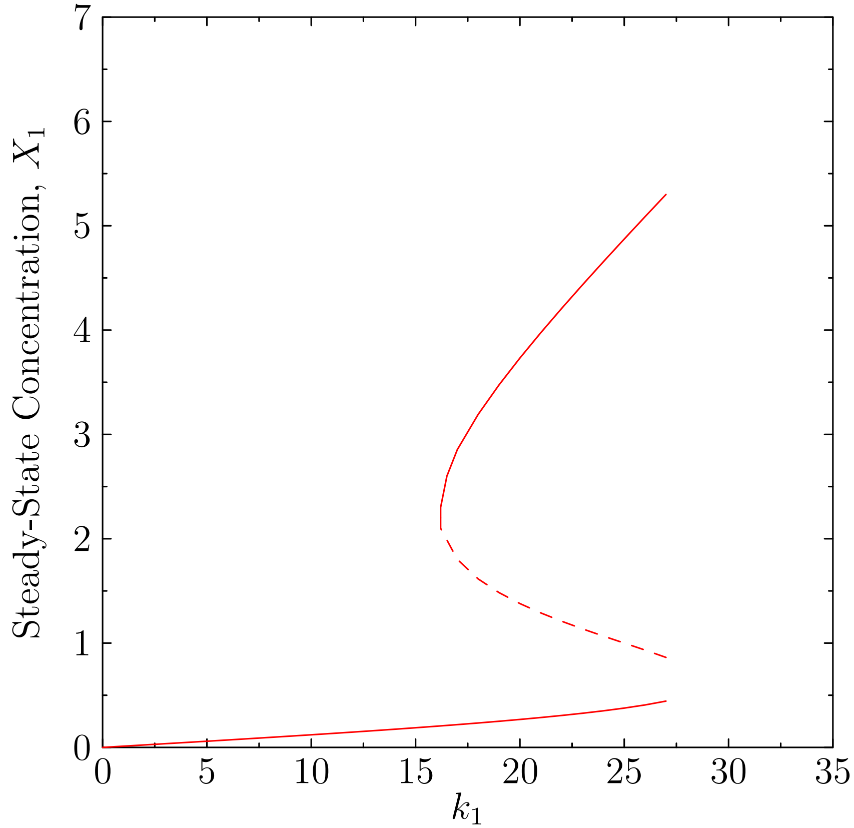

$k_1 = 26$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

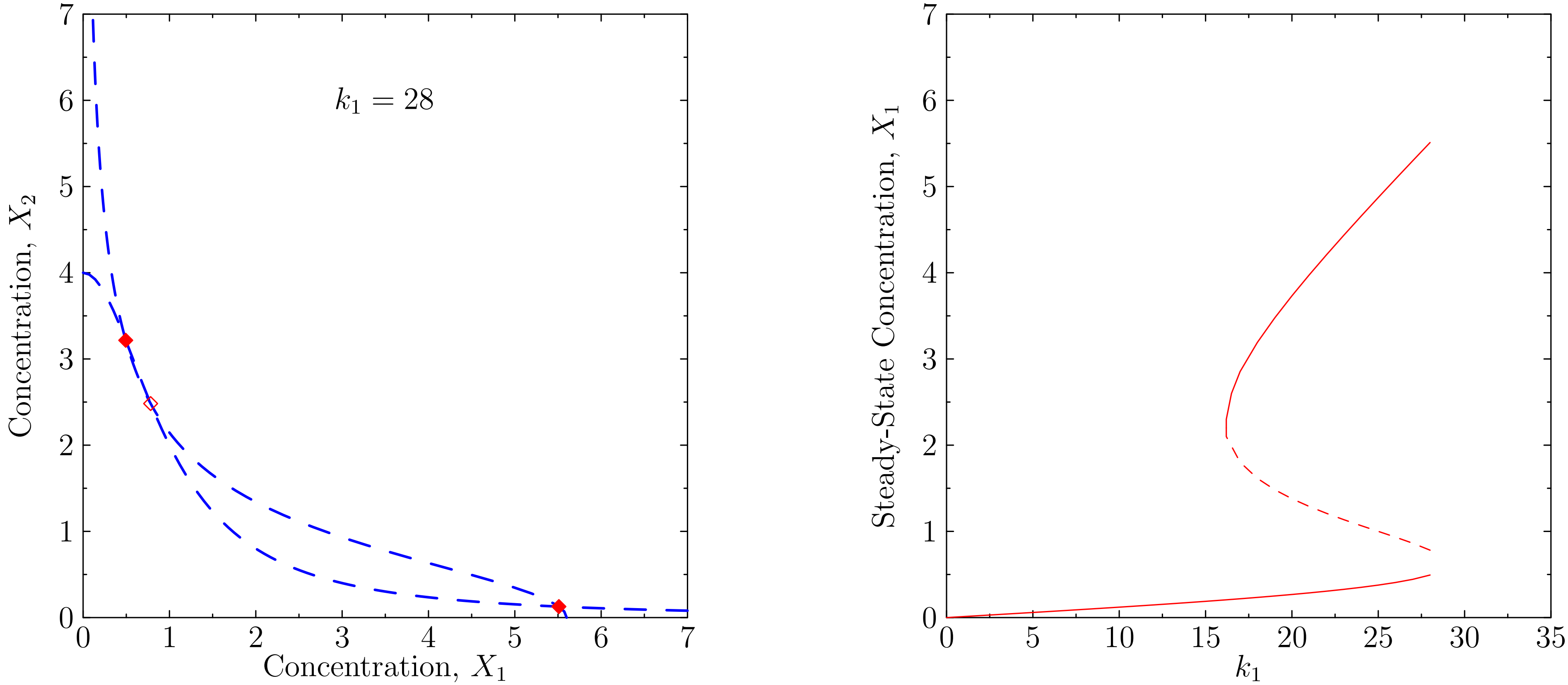


$k_1 = 27$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7



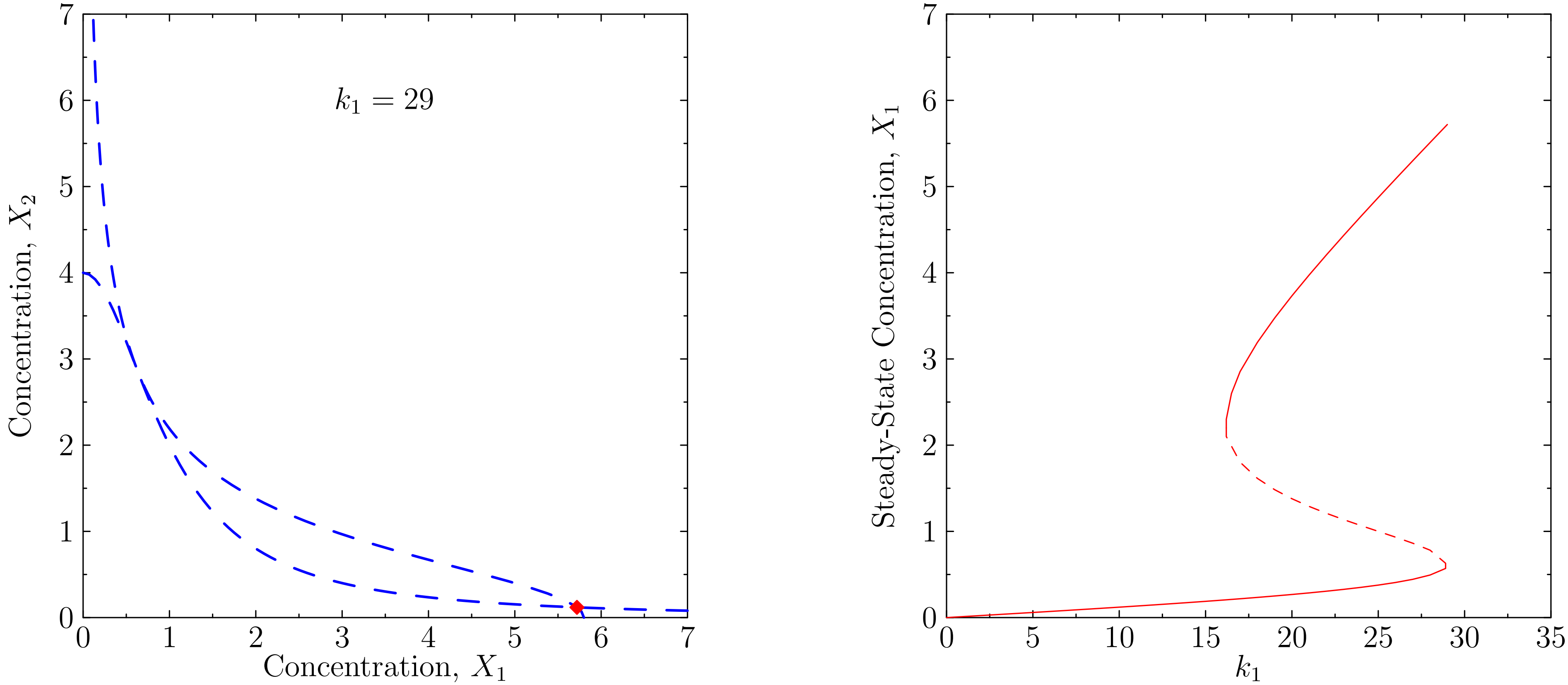
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

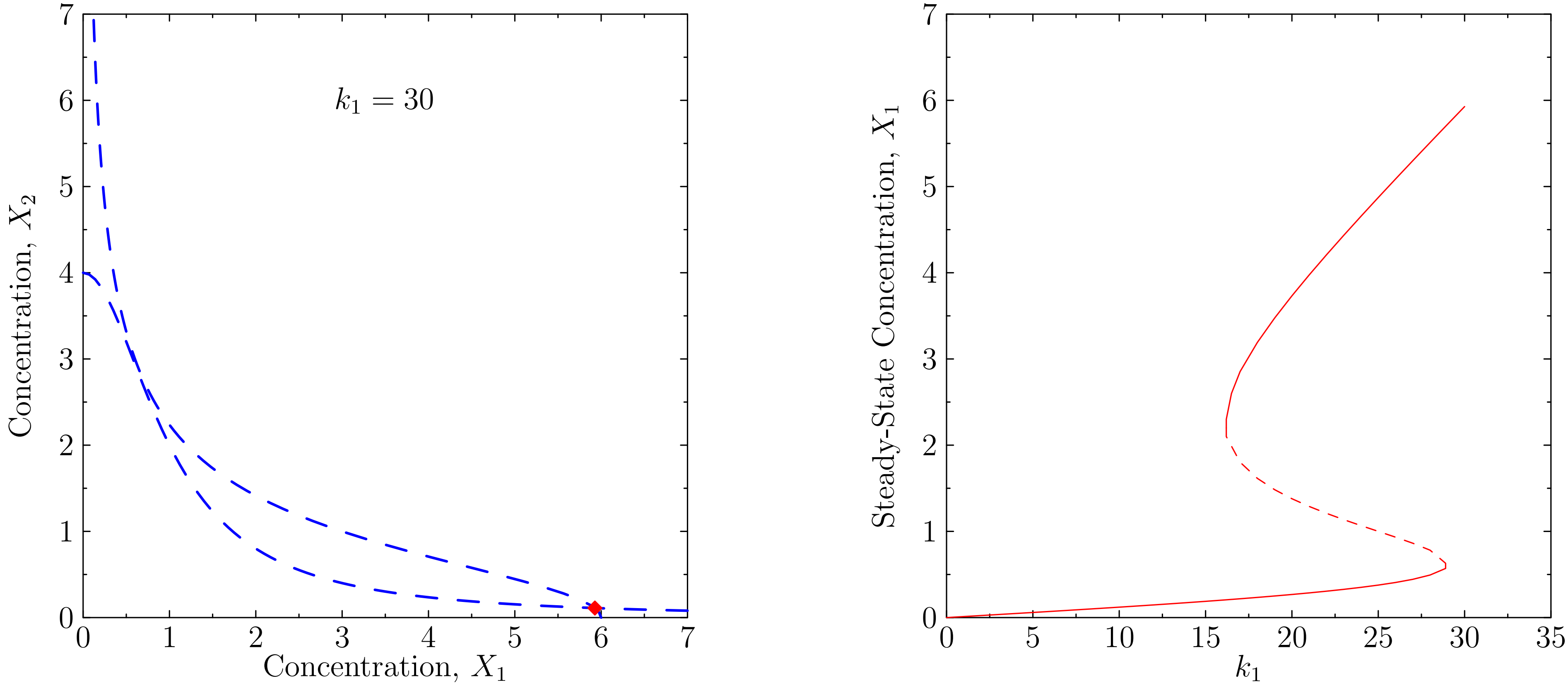


$k_1 = 28$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

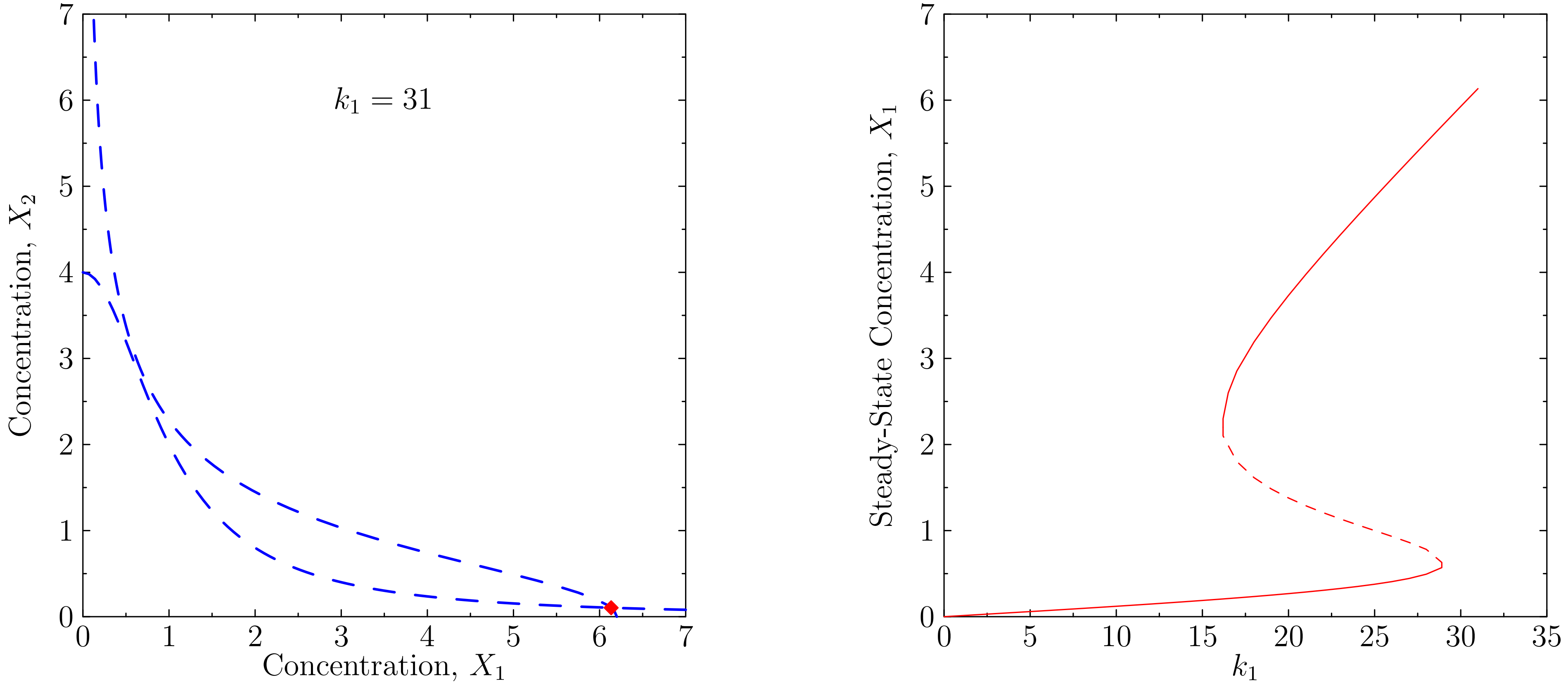

$k_1 = 29$
Concentration, X_1
Concentration, X_2
Steady-State Concentration, X_1
k_1
0
1
2
3
4
5
6
7
0
5
10
15
20
25
30
35

Varying k_1



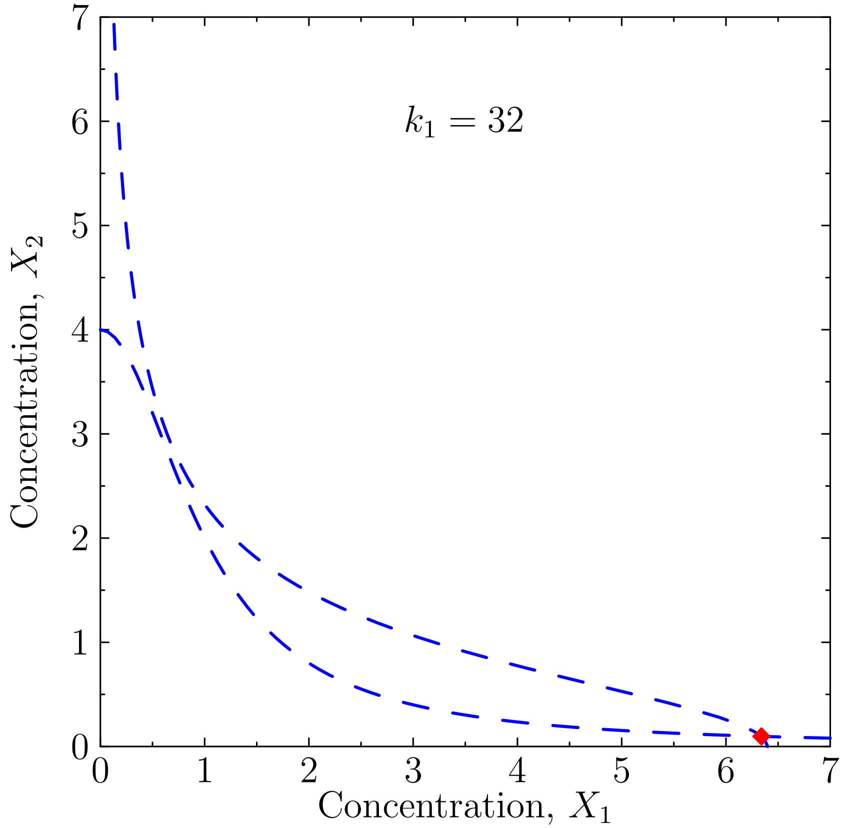
$k_1 = 30$
Concentration, X_2
Concentration, X_1
Steady-State Concentration, X_1
k_1

Varying k_1

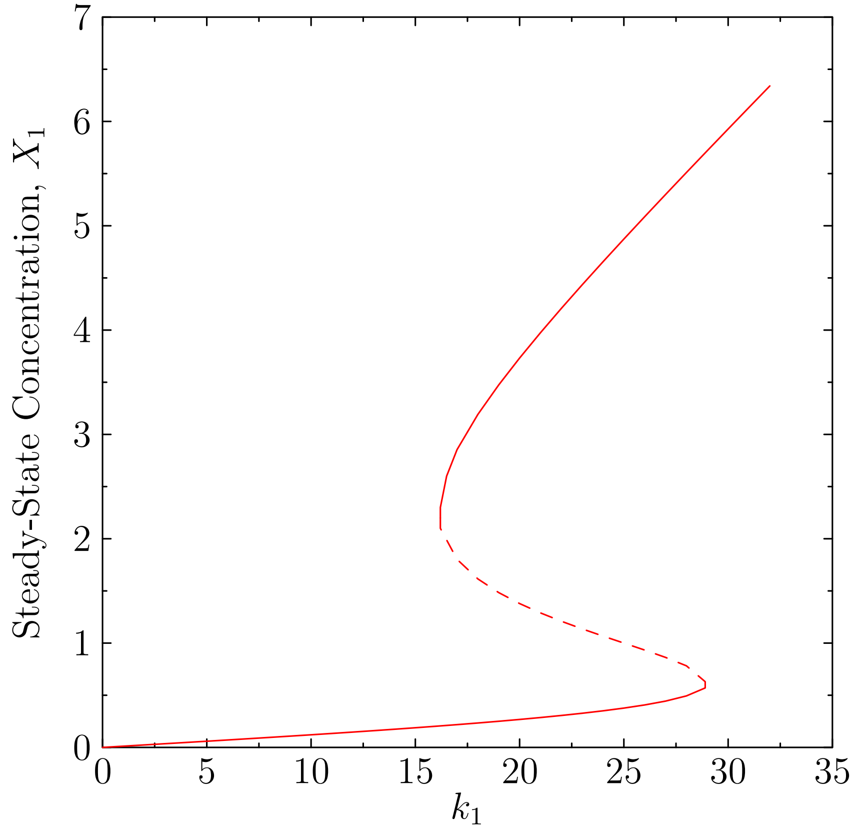


$k_1 = 31$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7
Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

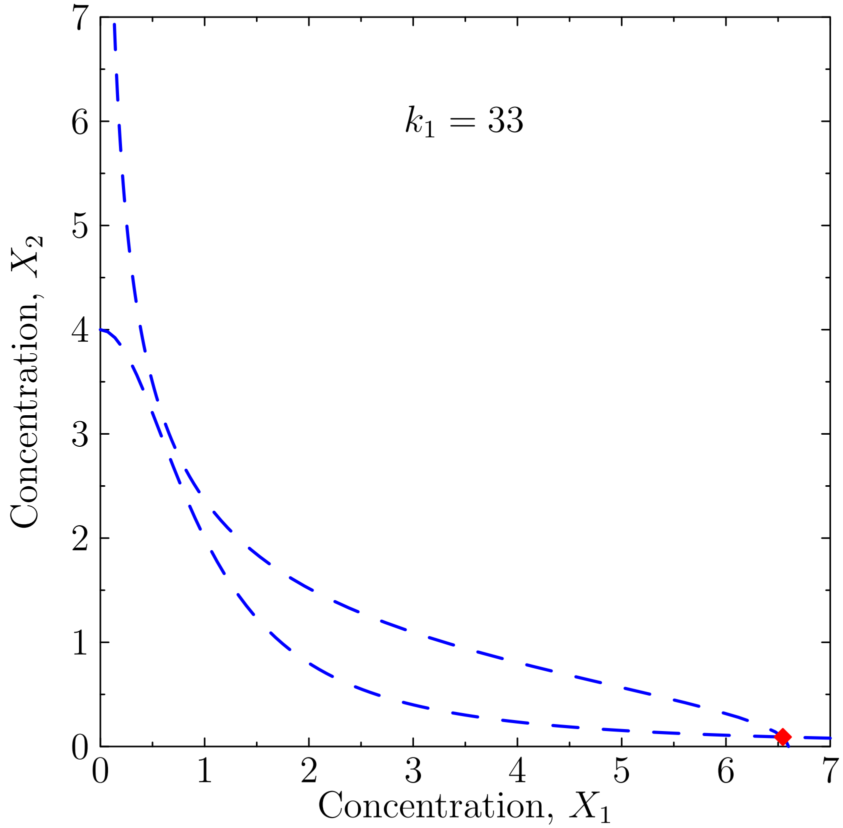


$k_1 = 32$
Concentration, X_1
Concentration, X_2
0
1
2
3
4
5
6
7

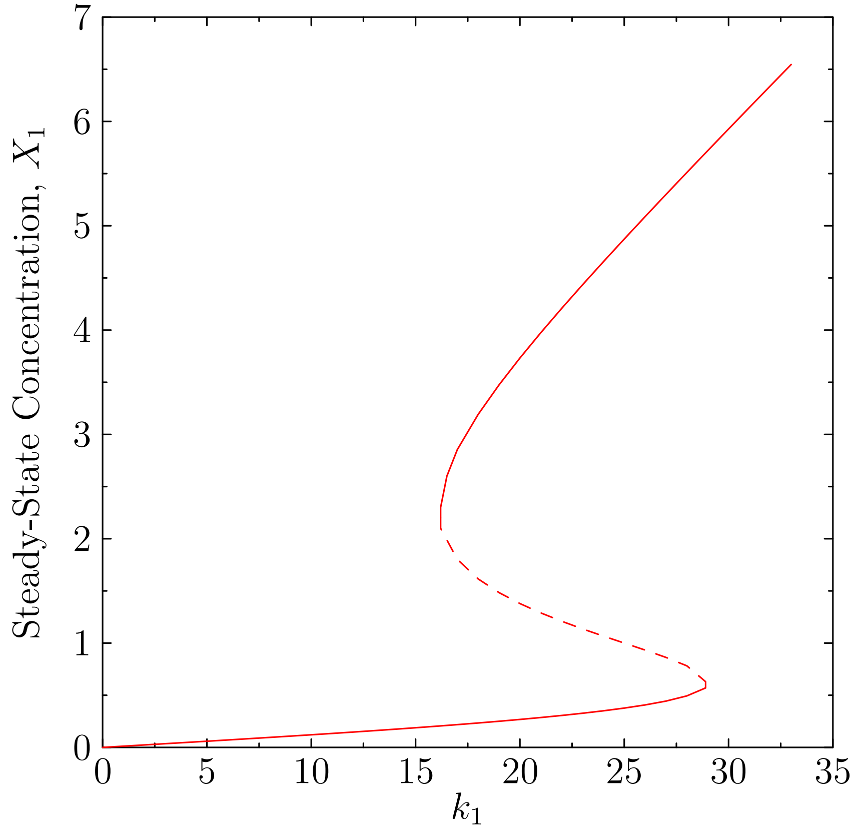


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

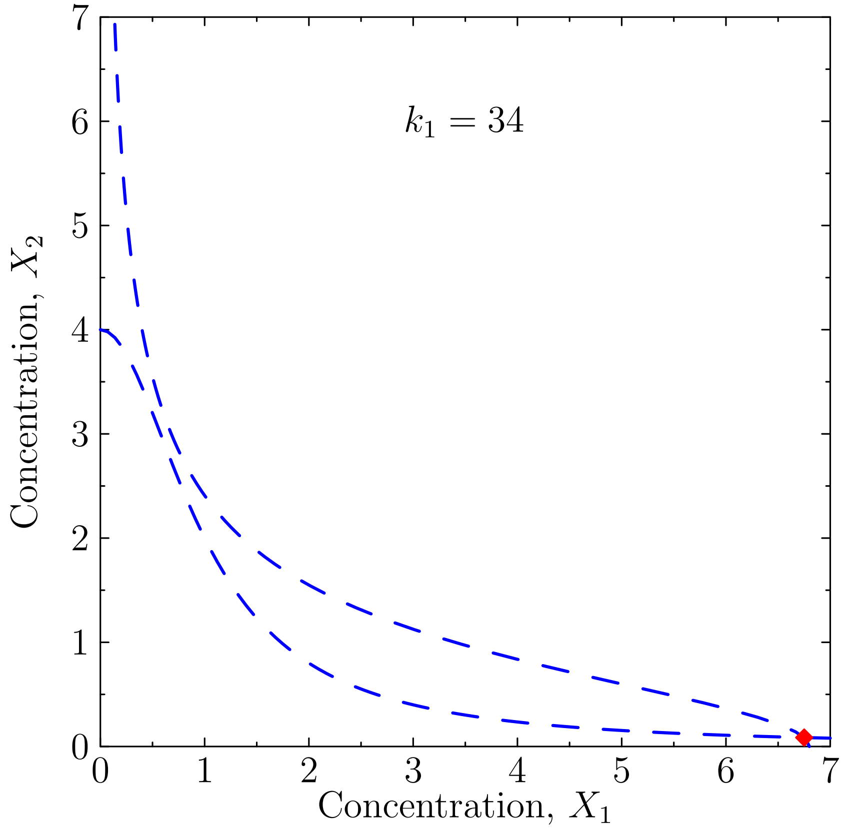


$k_1 = 33$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

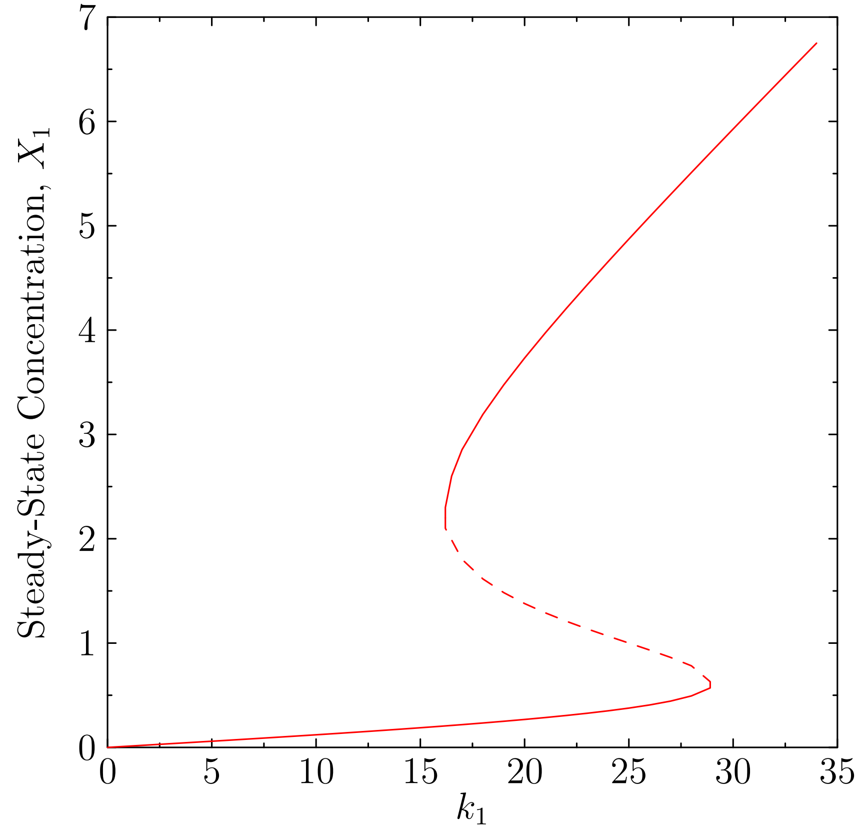


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

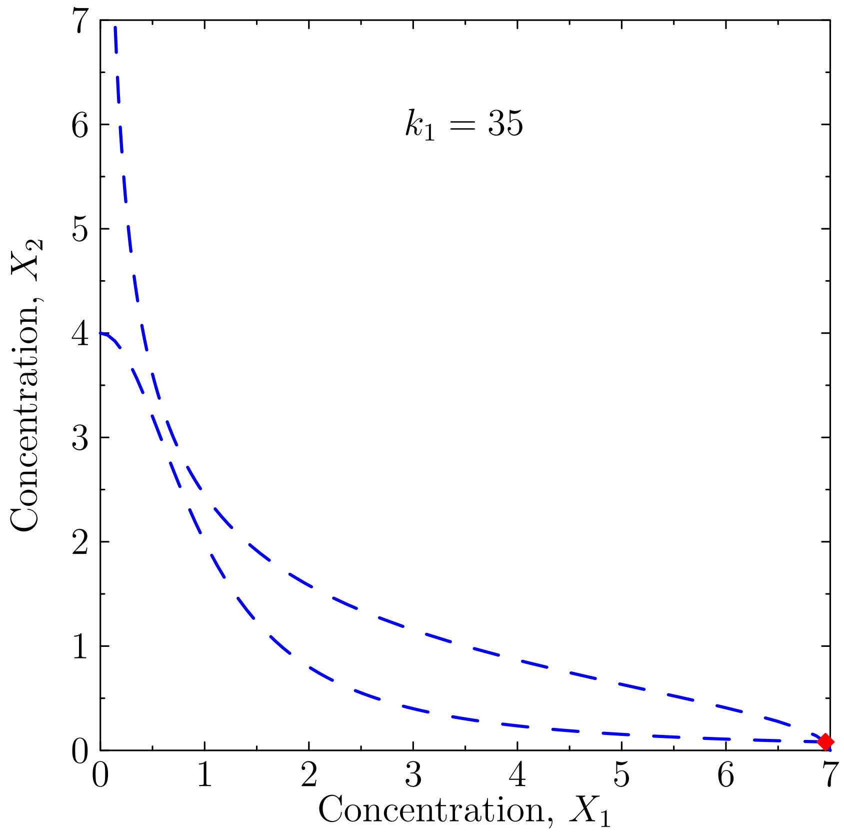


$k_1 = 34$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

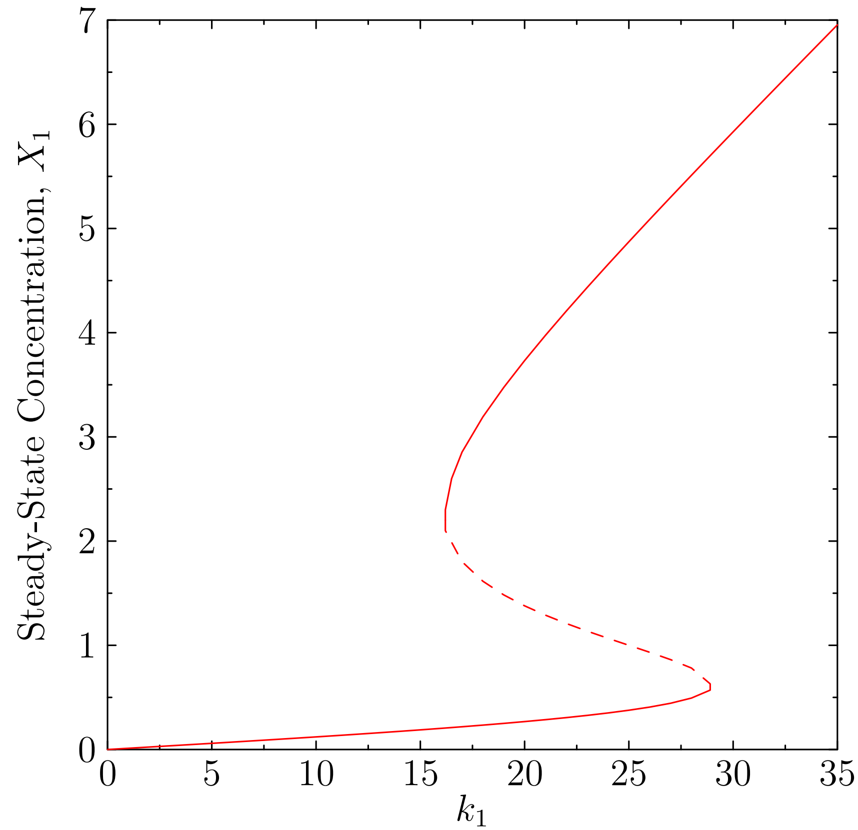


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

Varying k_1

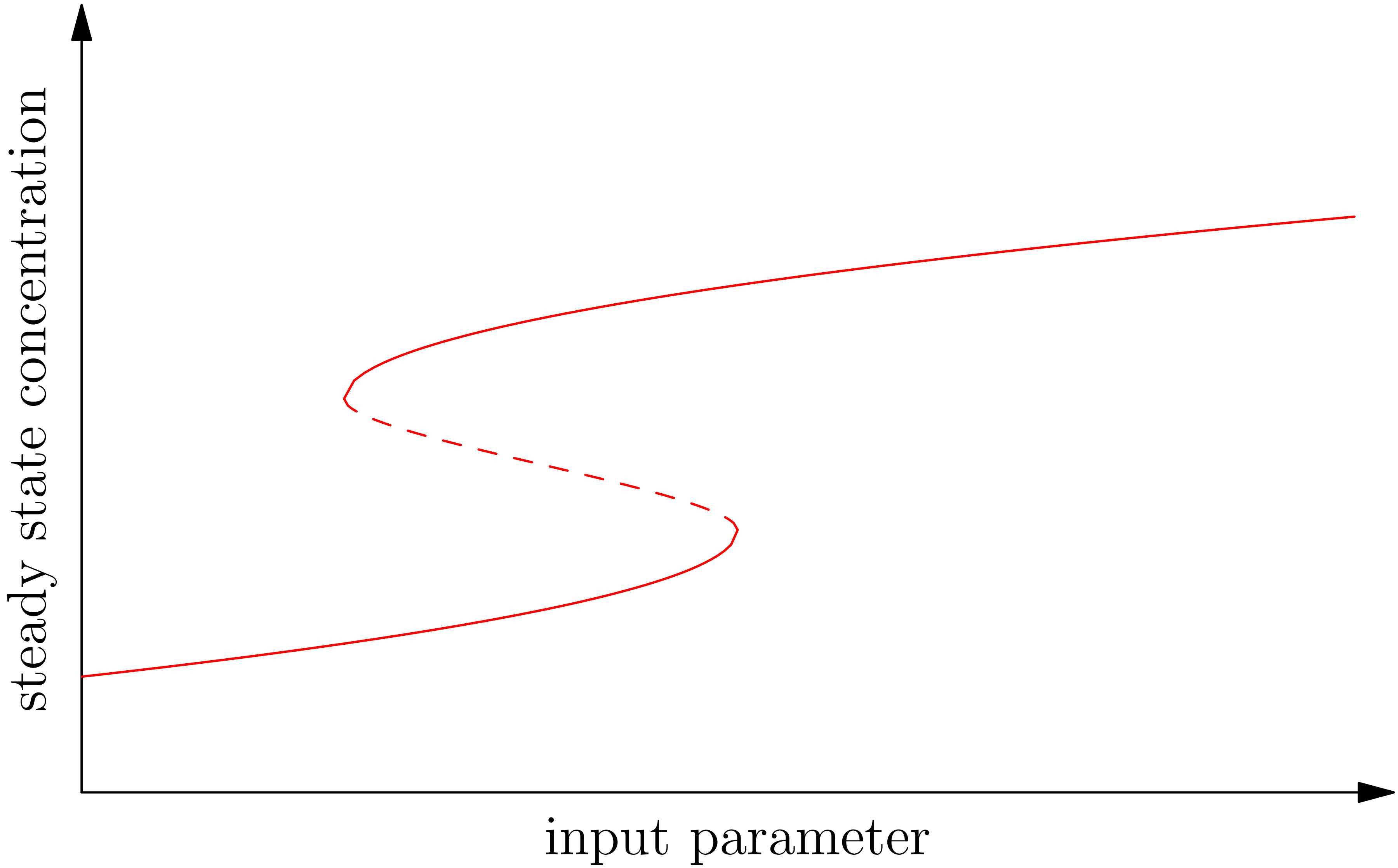


$k_1 = 35$
Concentration, X_2
Concentration, X_1
0
1
2
3
4
5
6
7

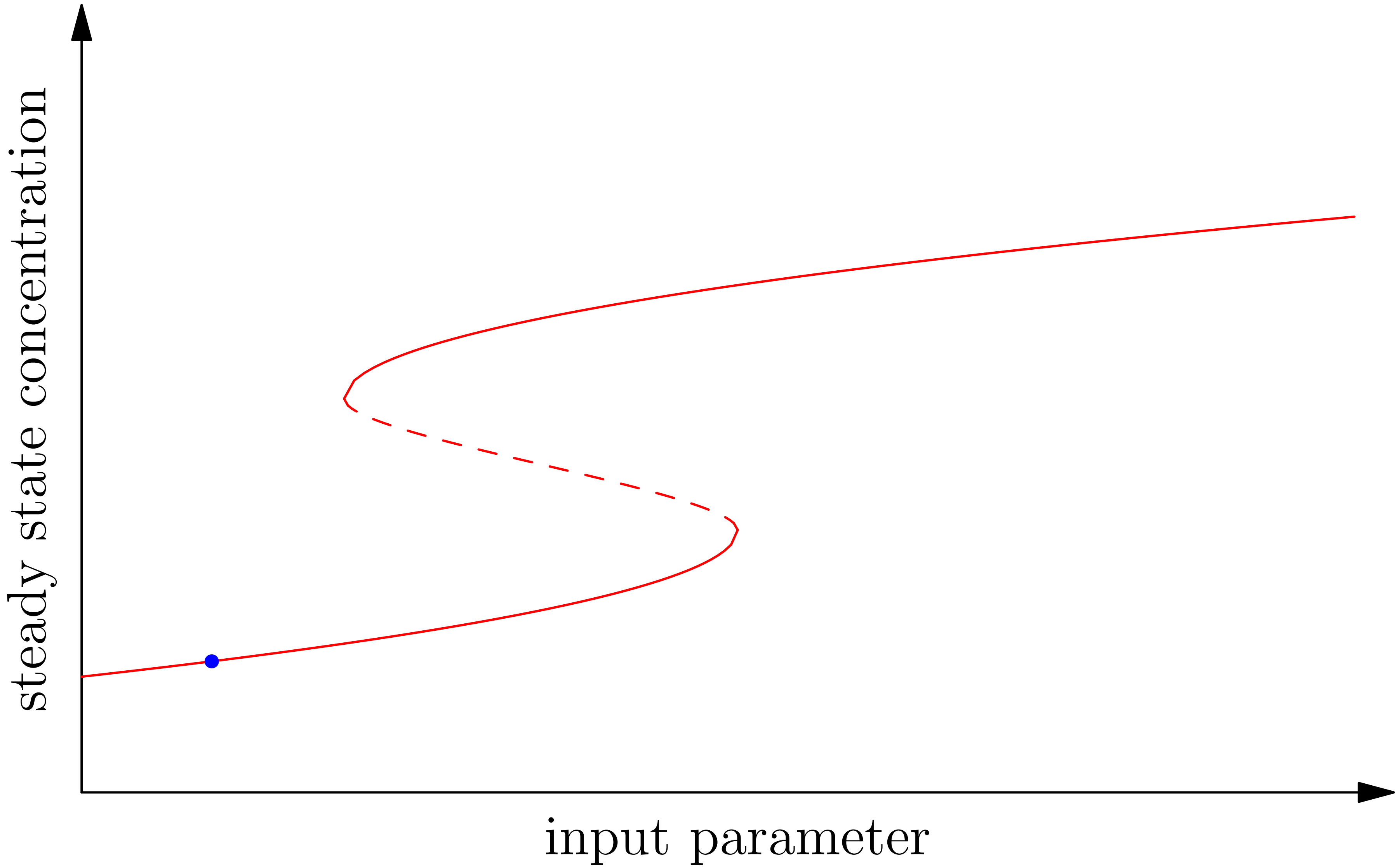


Steady-State Concentration, X_1
k_1
0
5
10
15
20
25
30
35

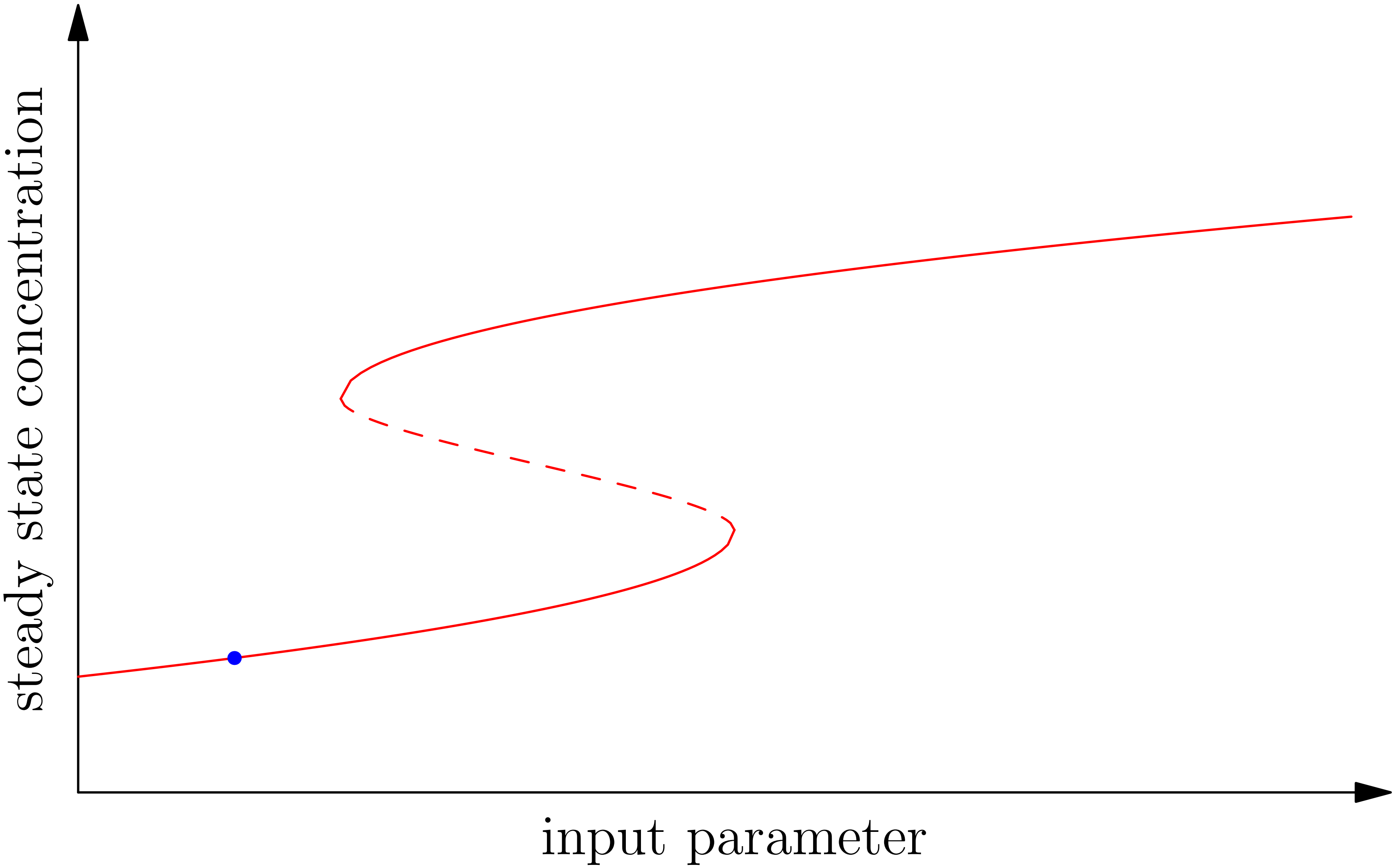
Hysteresis



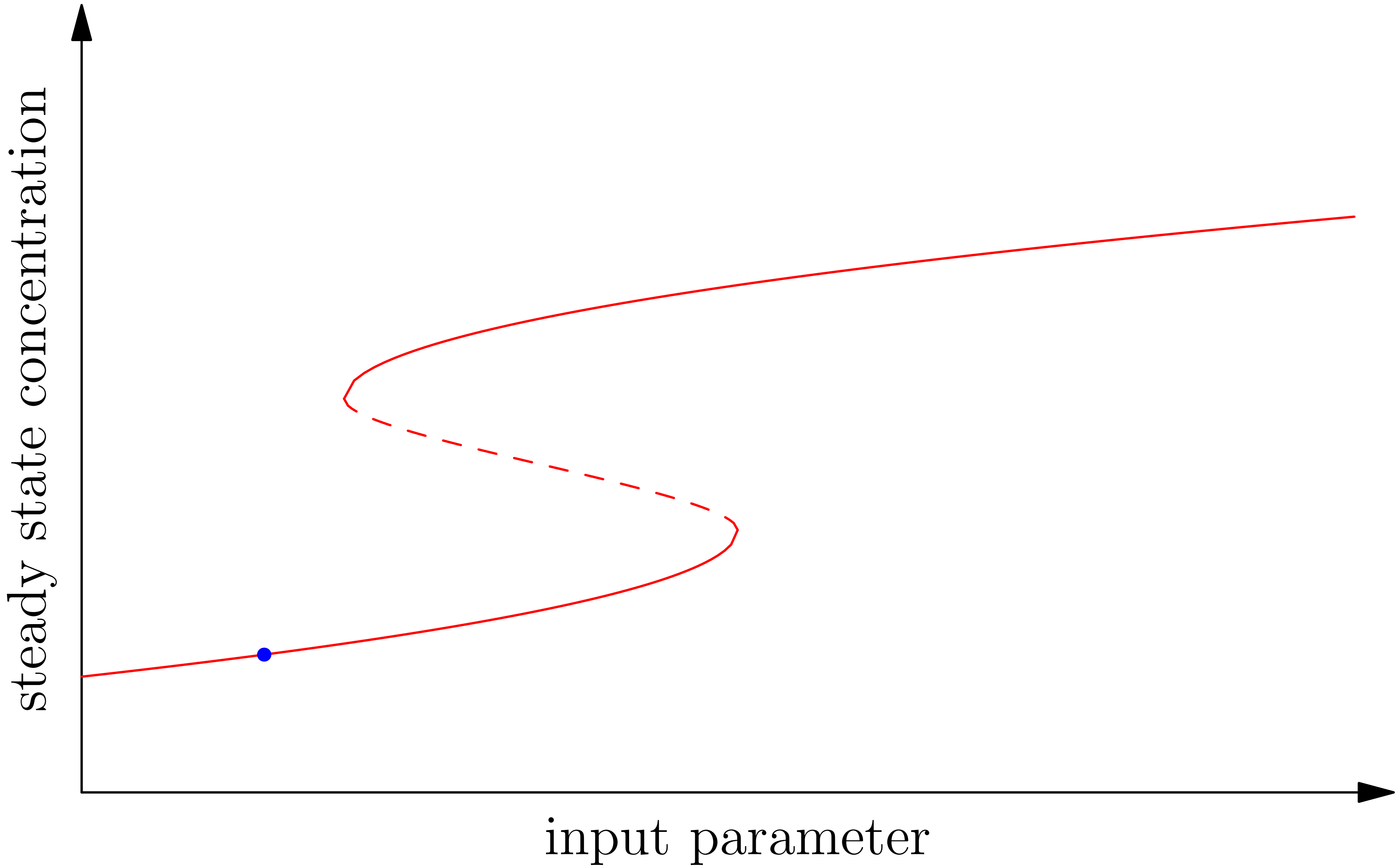
Hysteresis



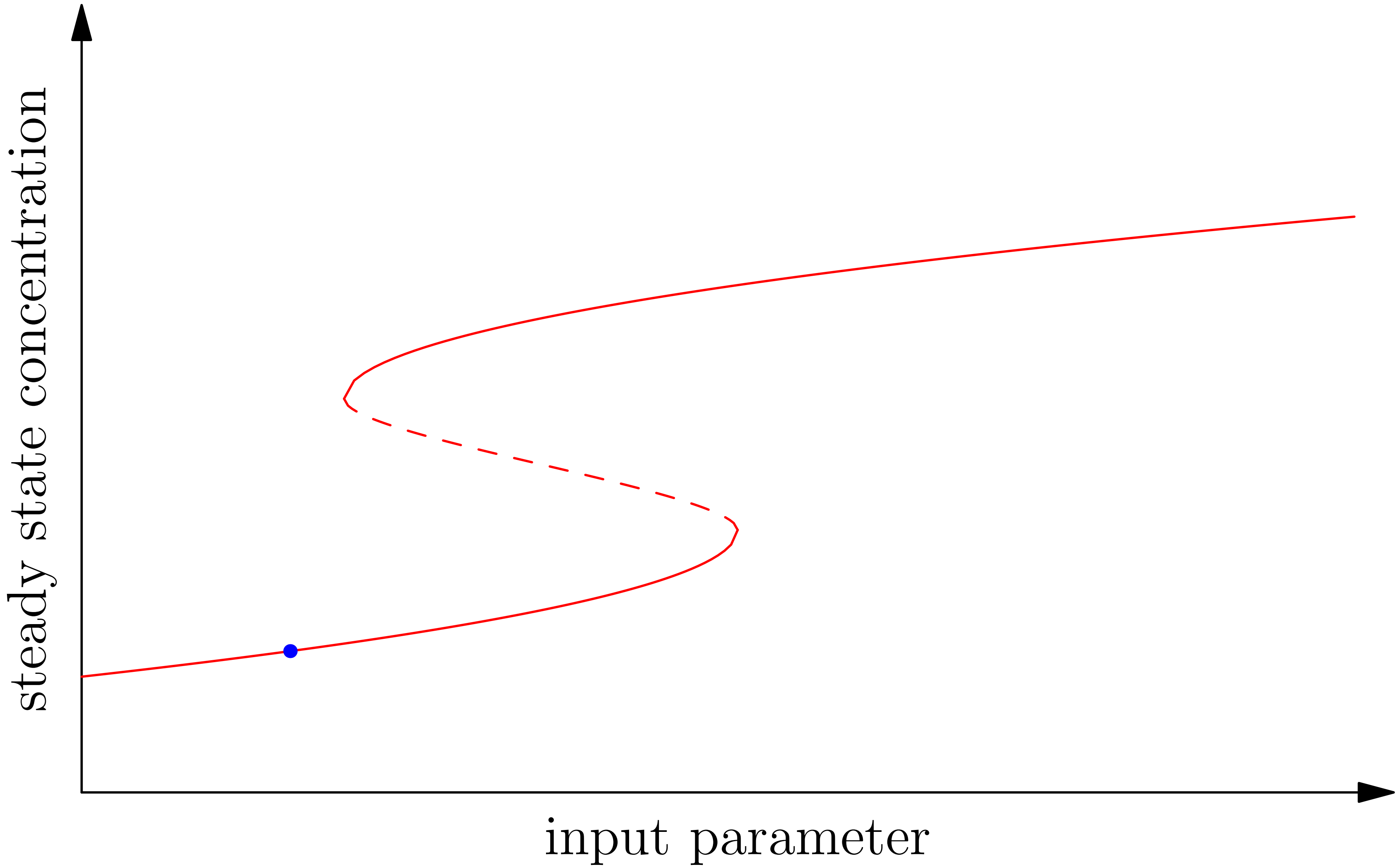
Hysteresis



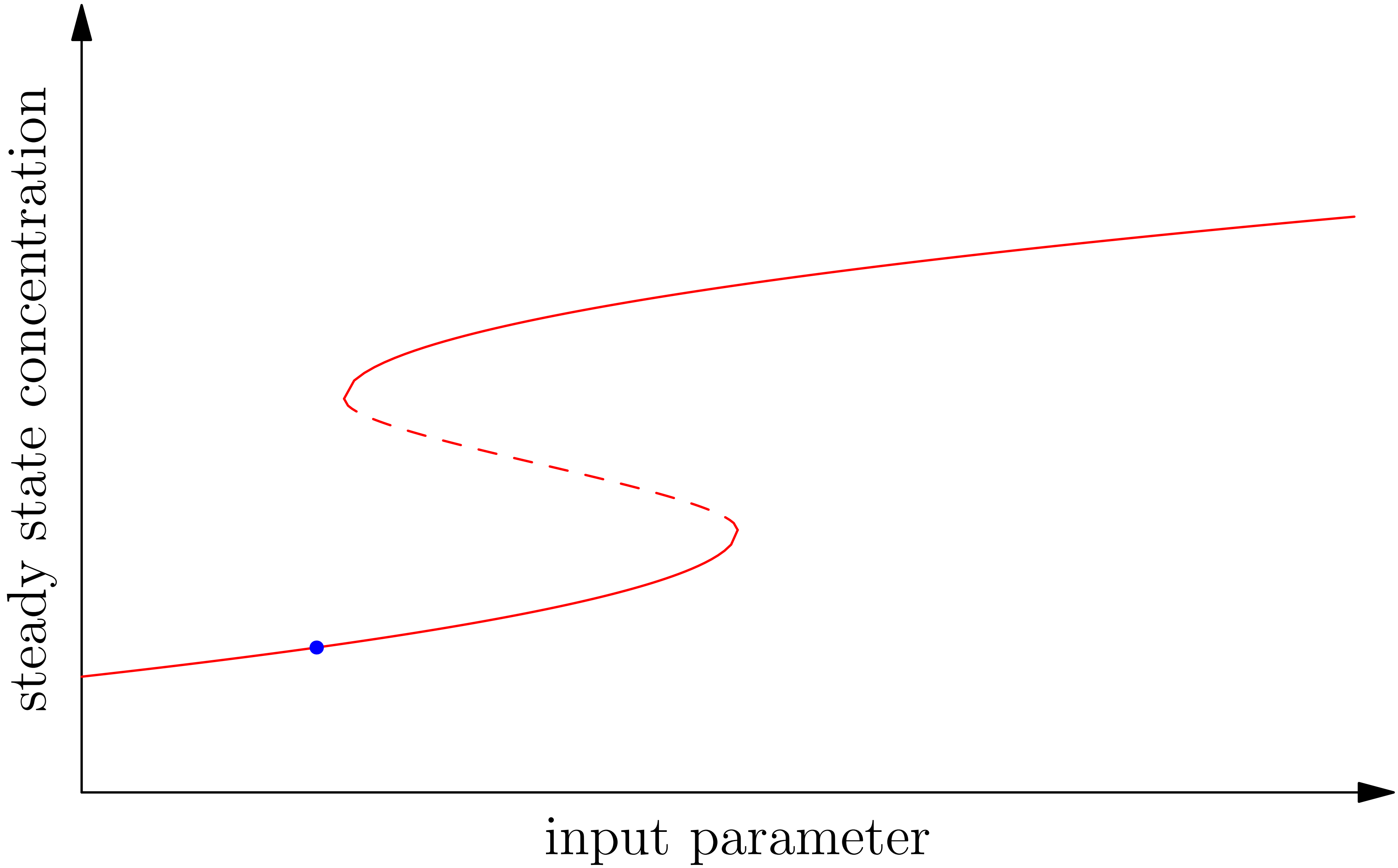
Hysteresis



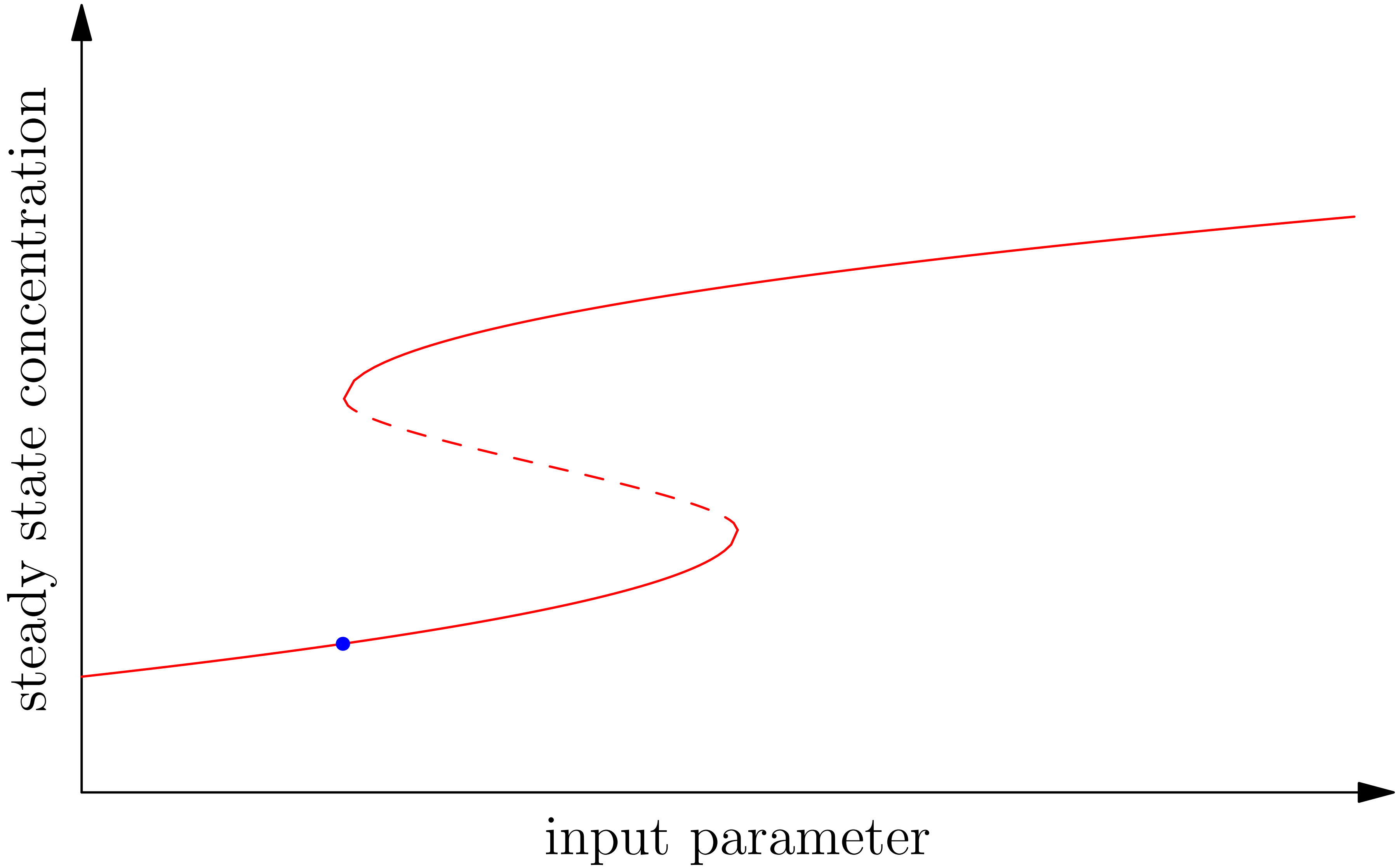
Hysteresis



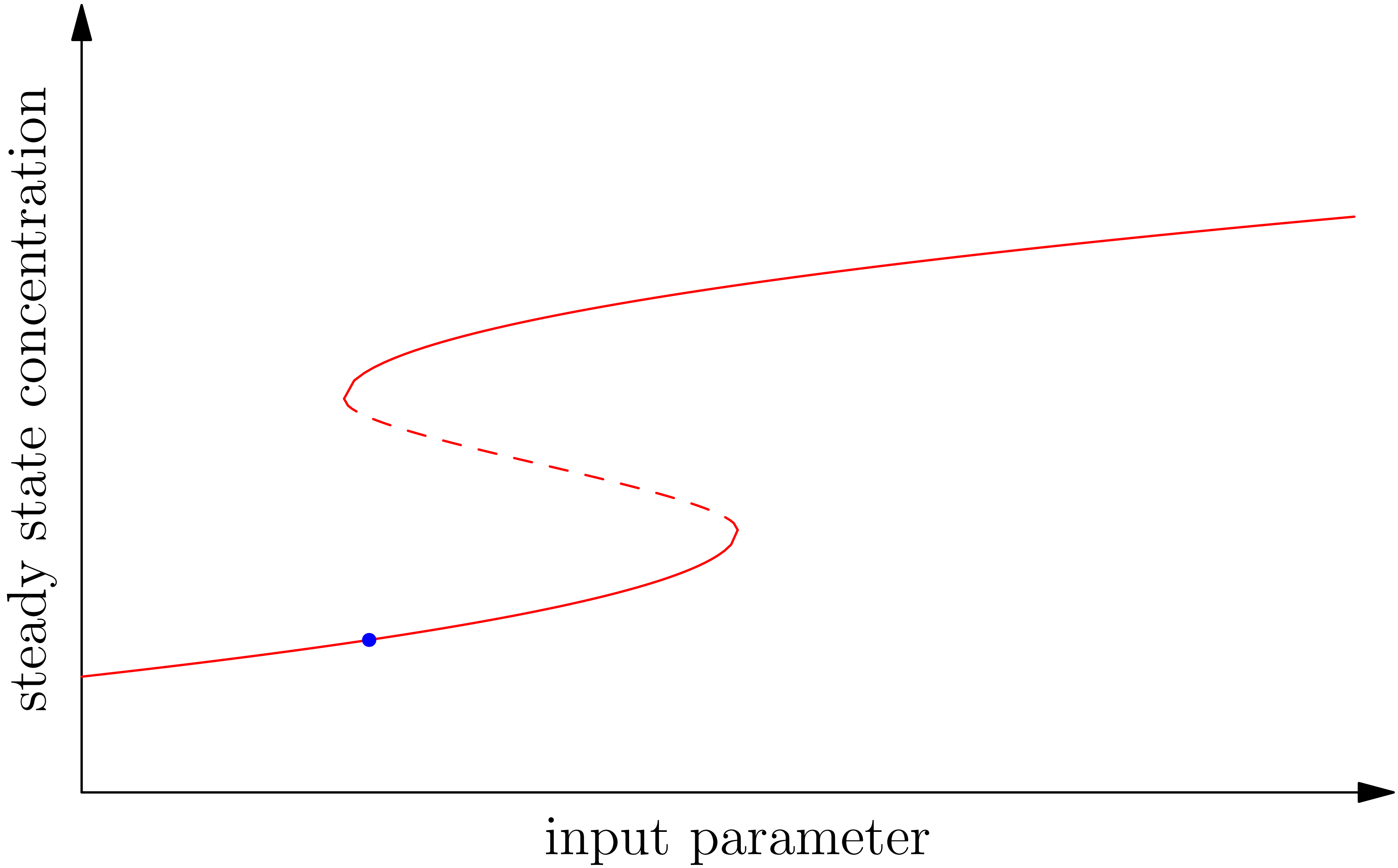
Hysteresis



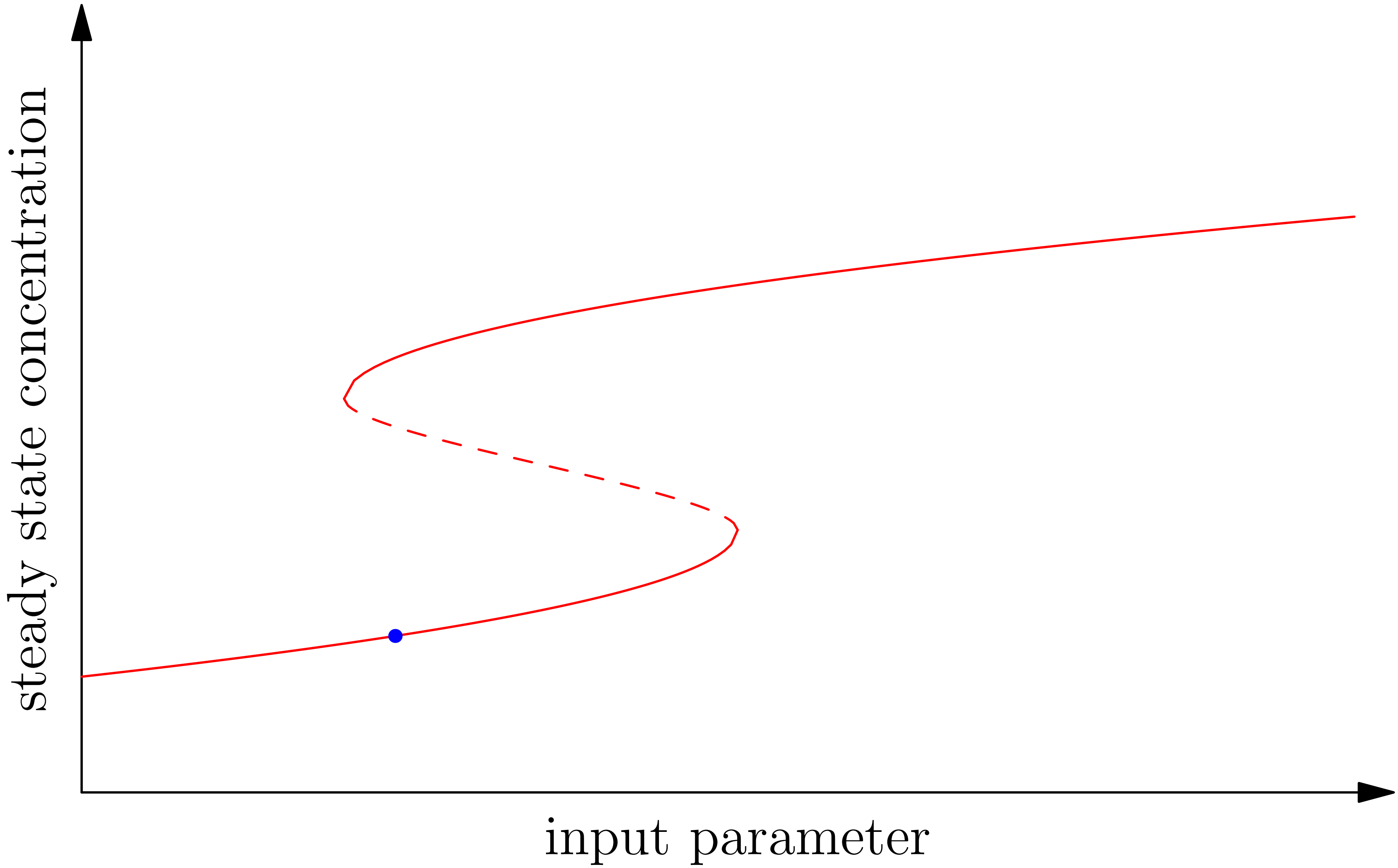
Hysteresis



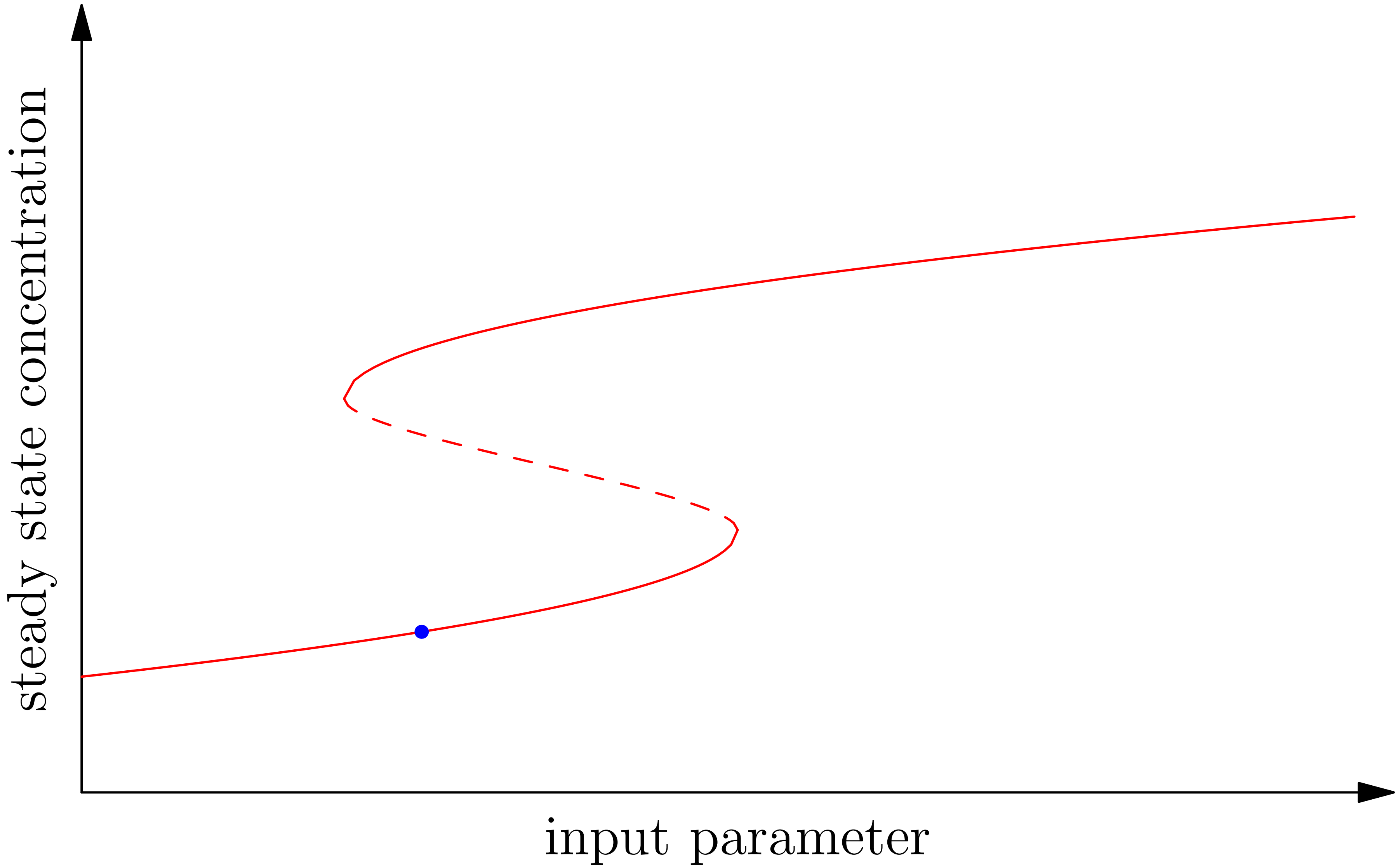
Hysteresis



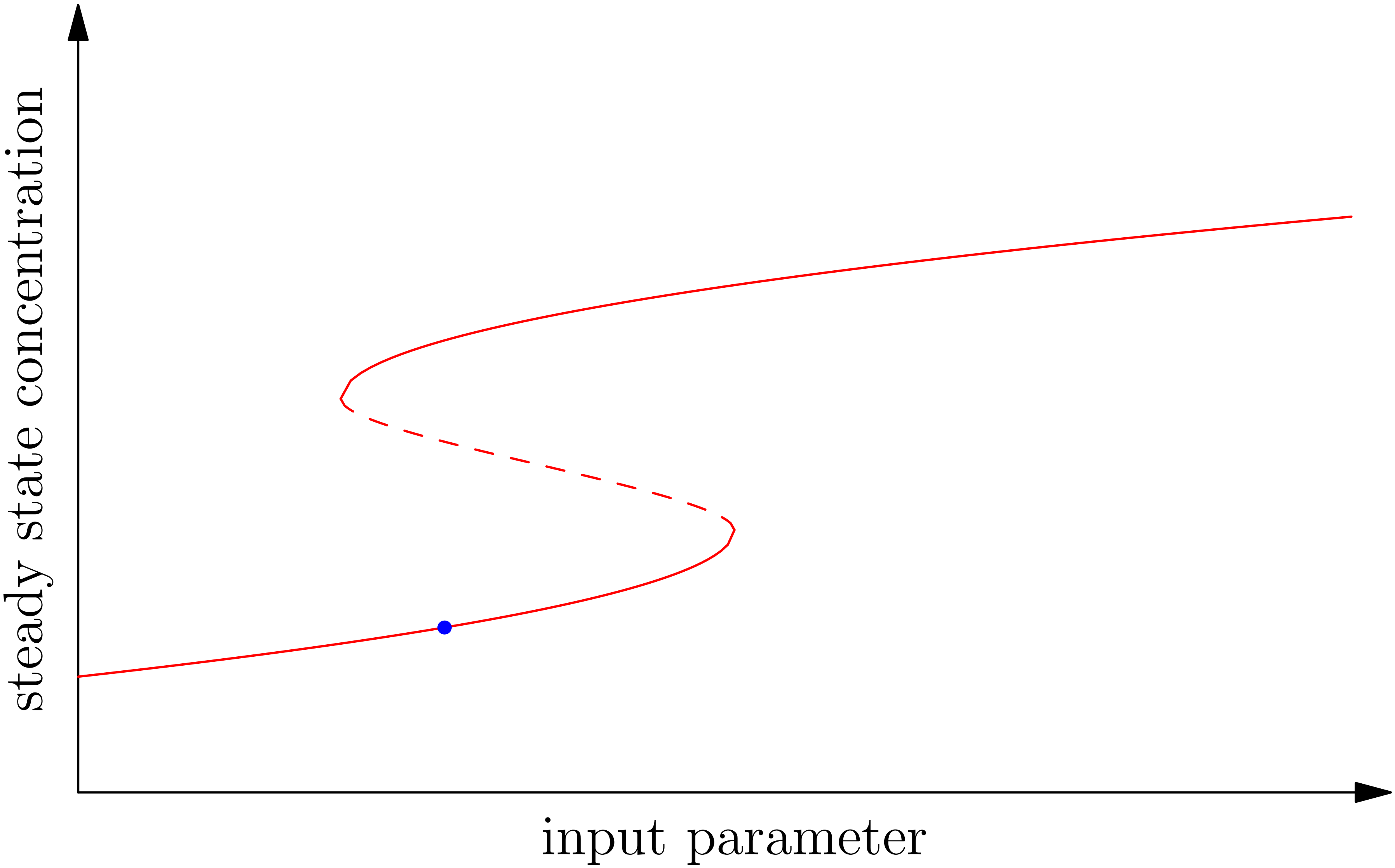
Hysteresis



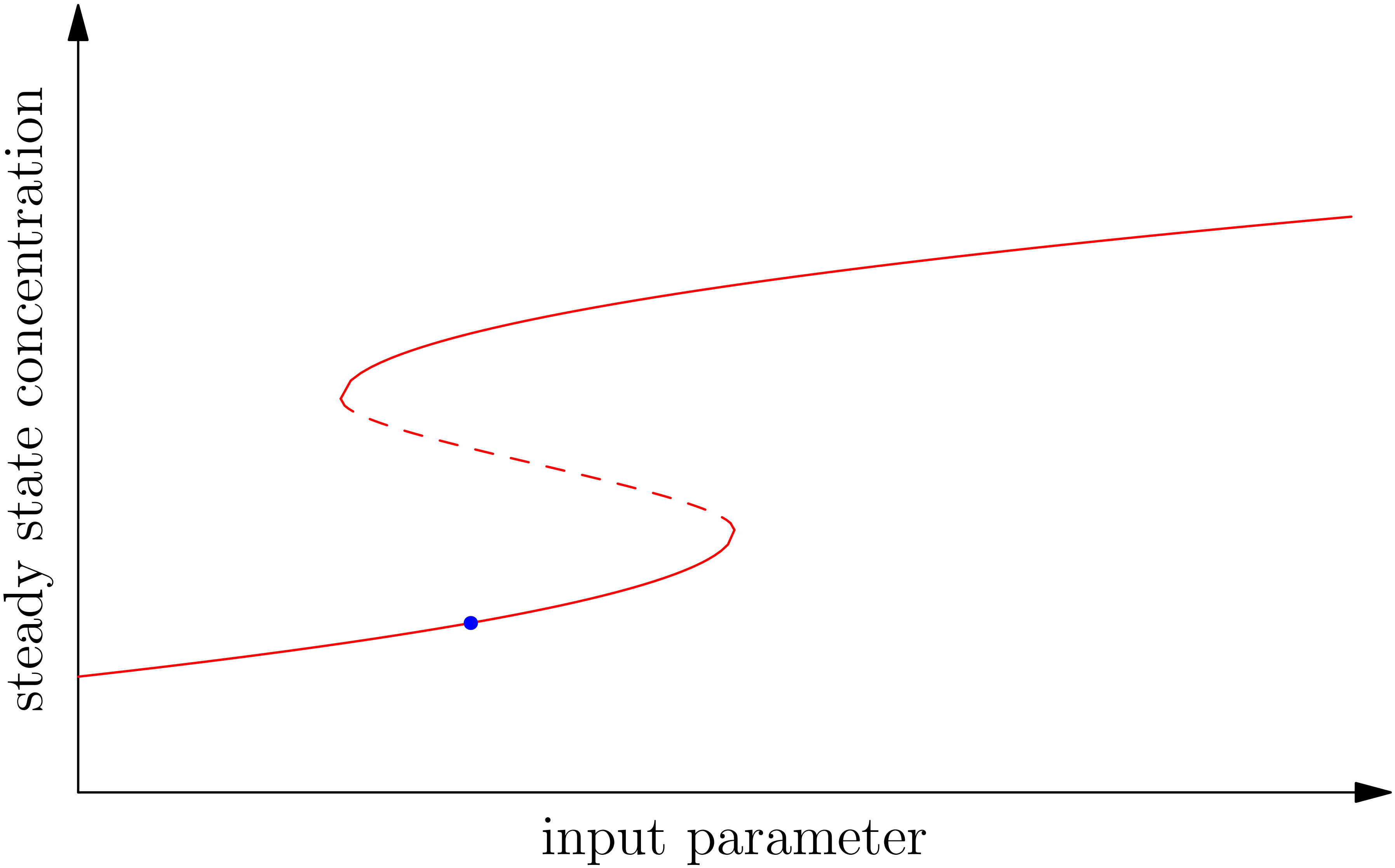
Hysteresis



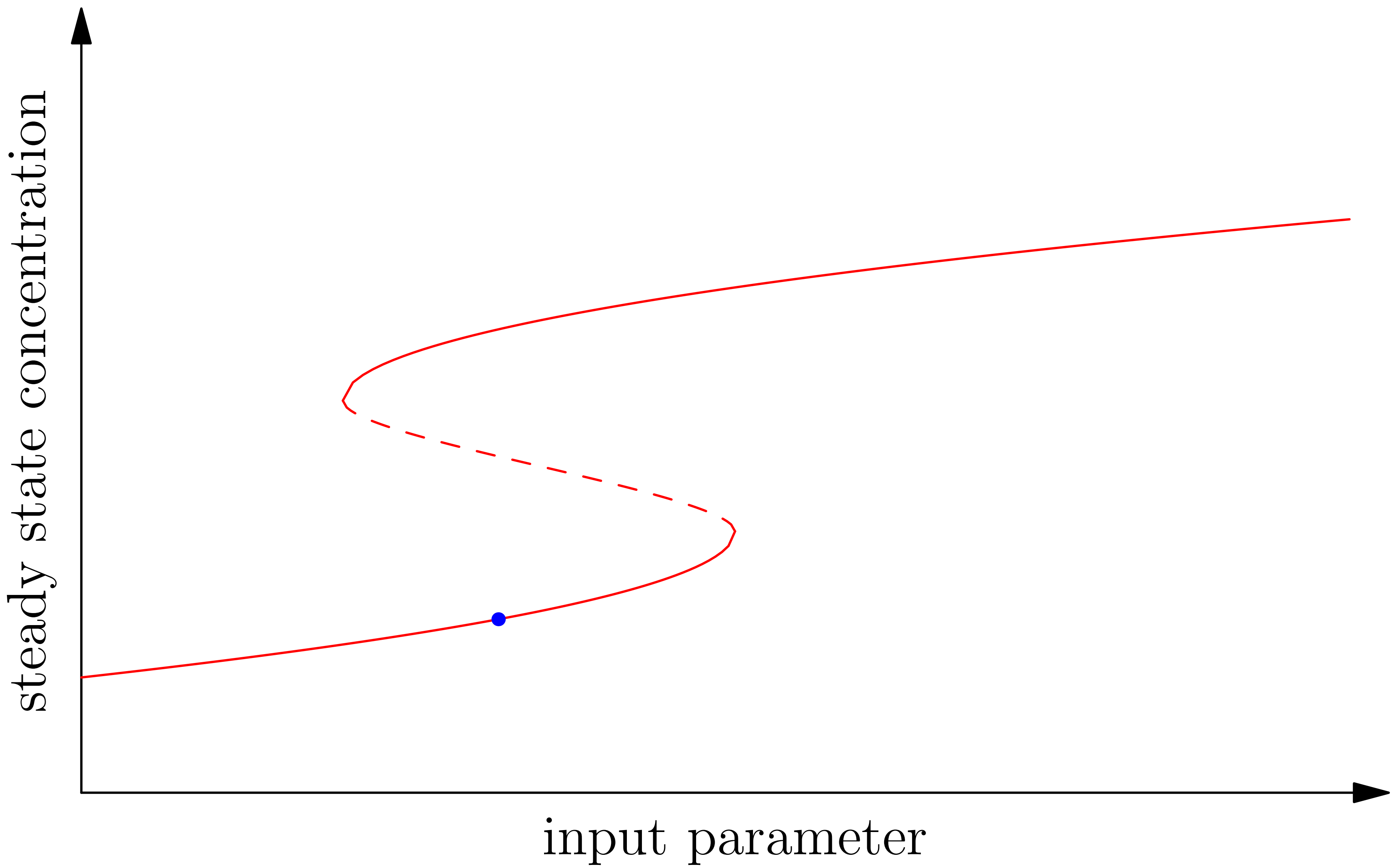
Hysteresis



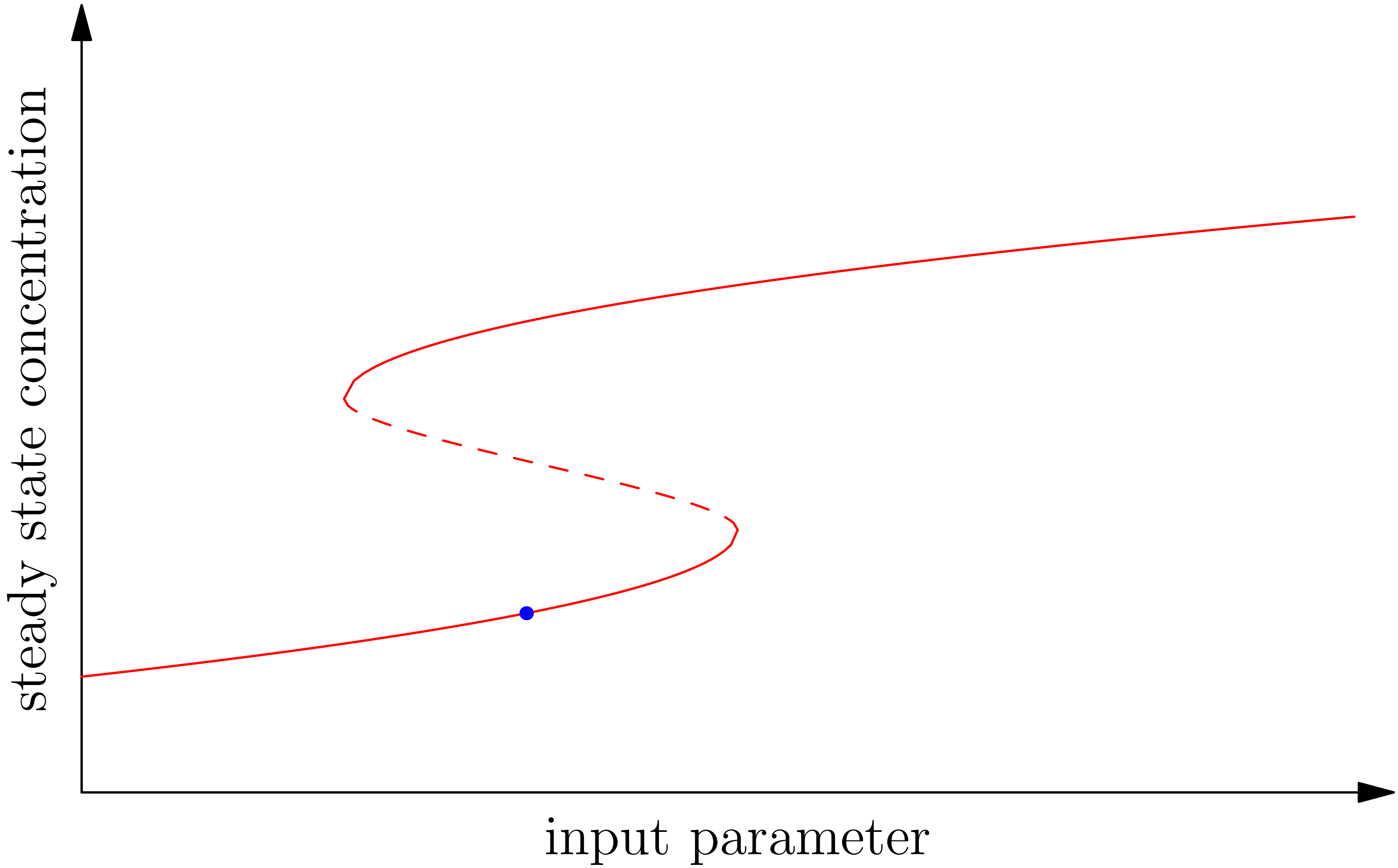
Hysteresis



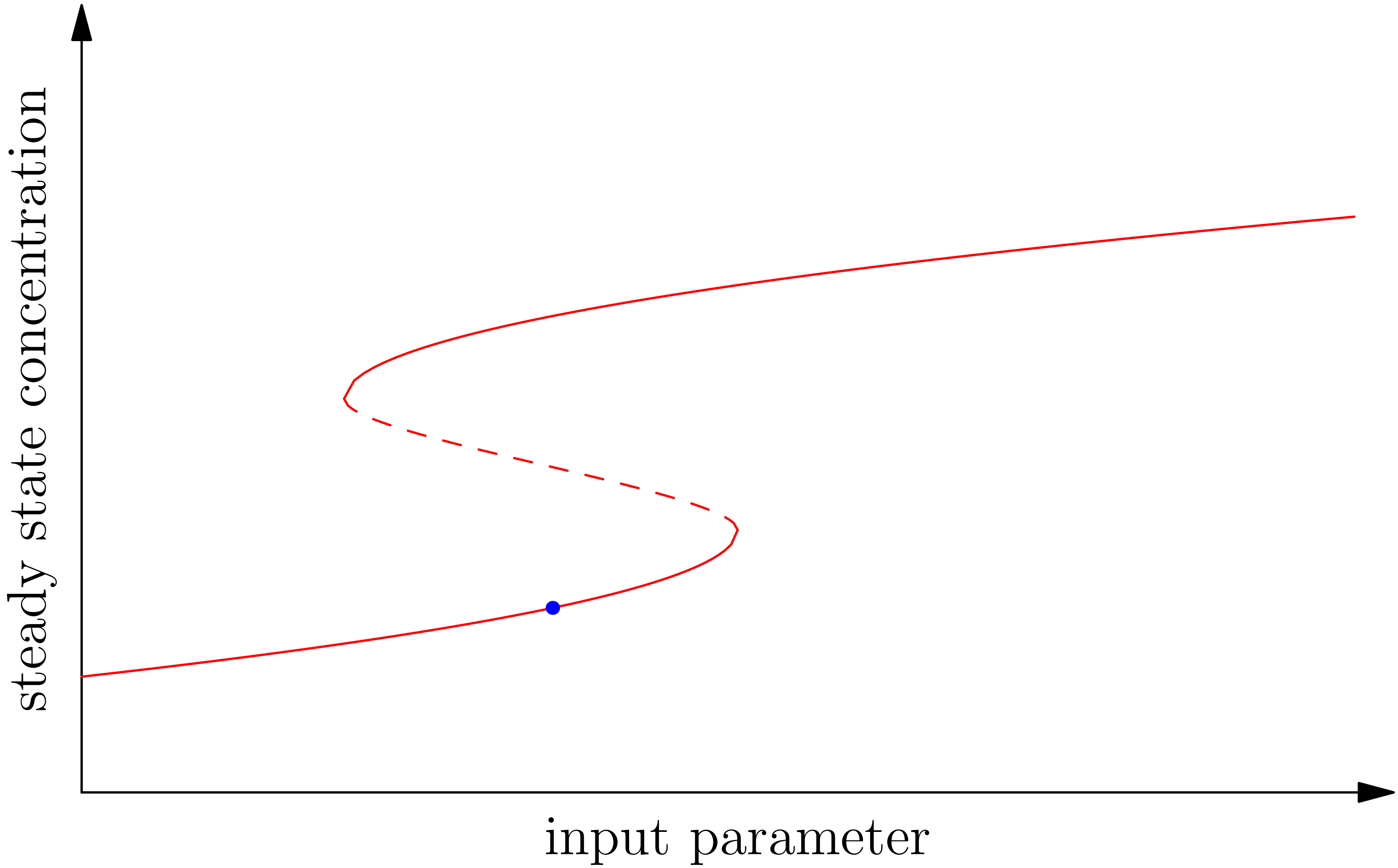
Hysteresis



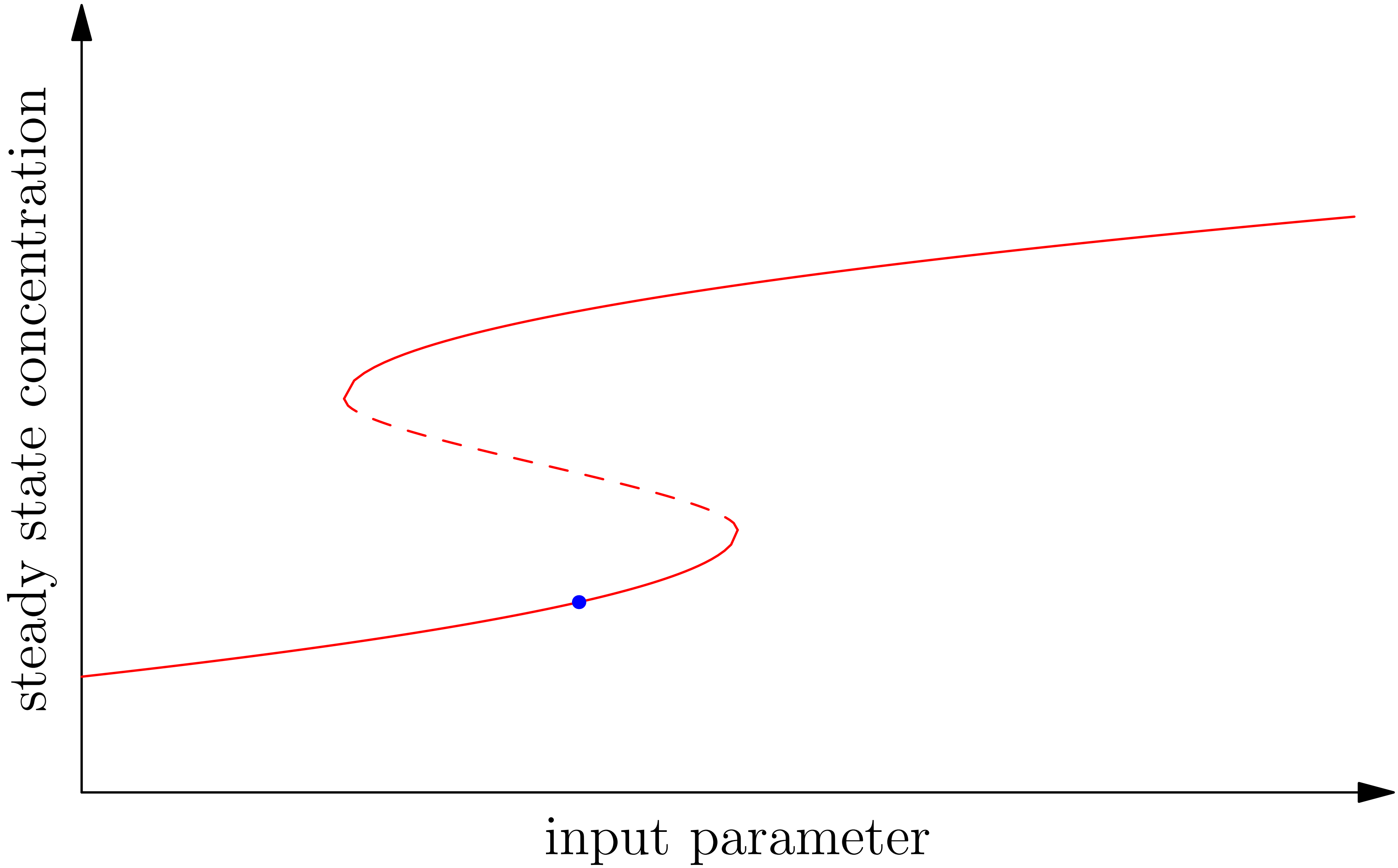
Hysteresis



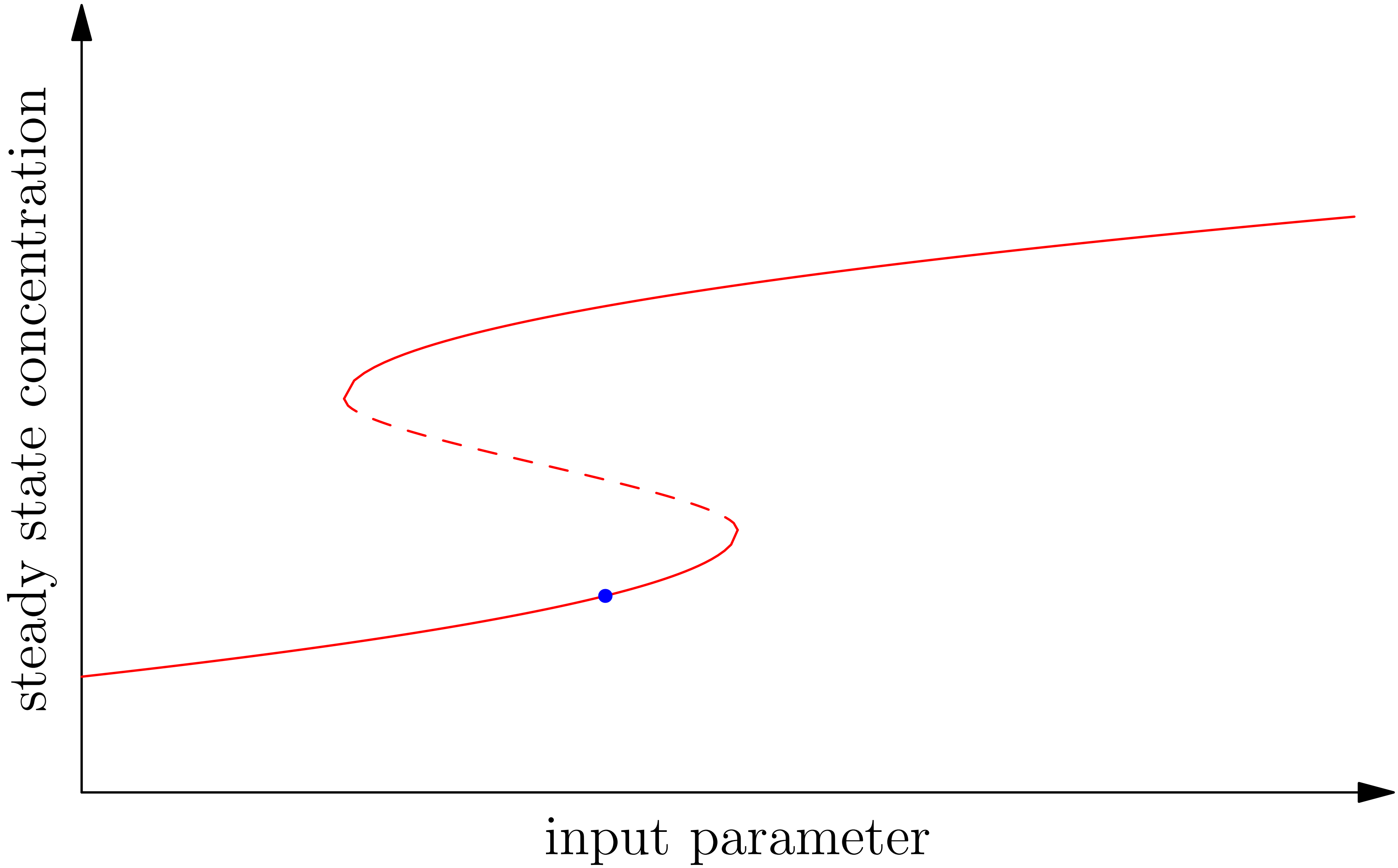
Hysteresis



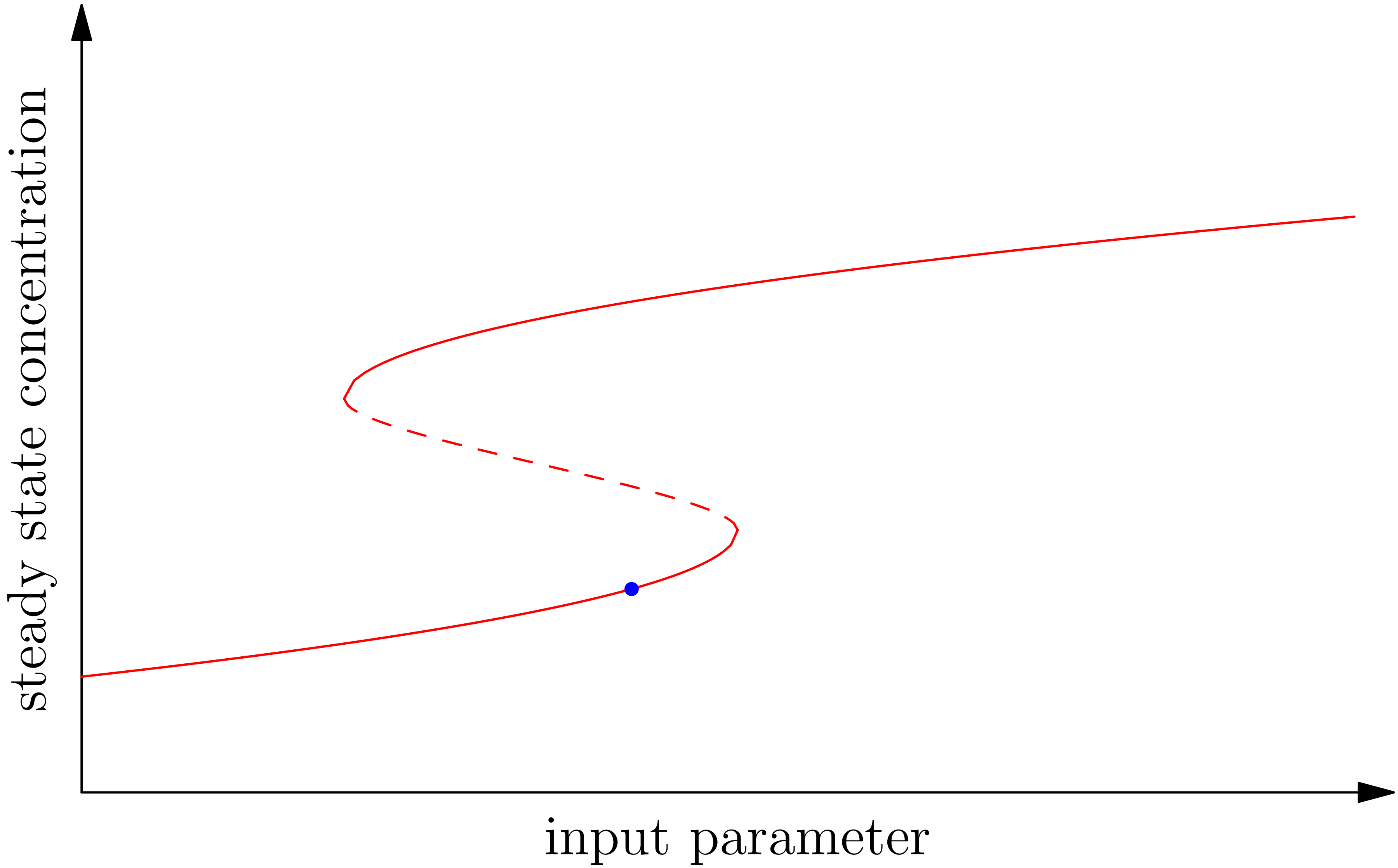
Hysteresis



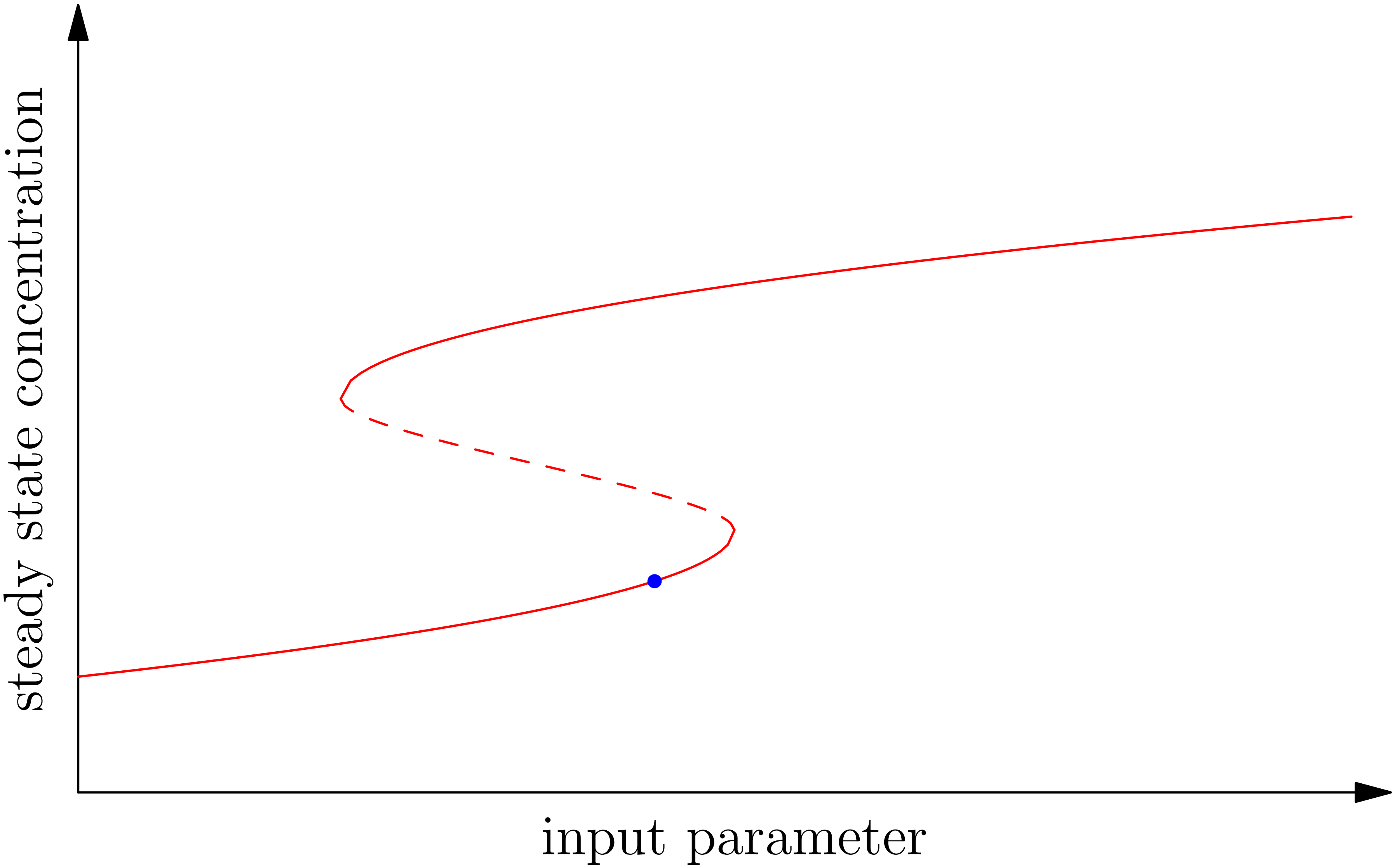
Hysteresis



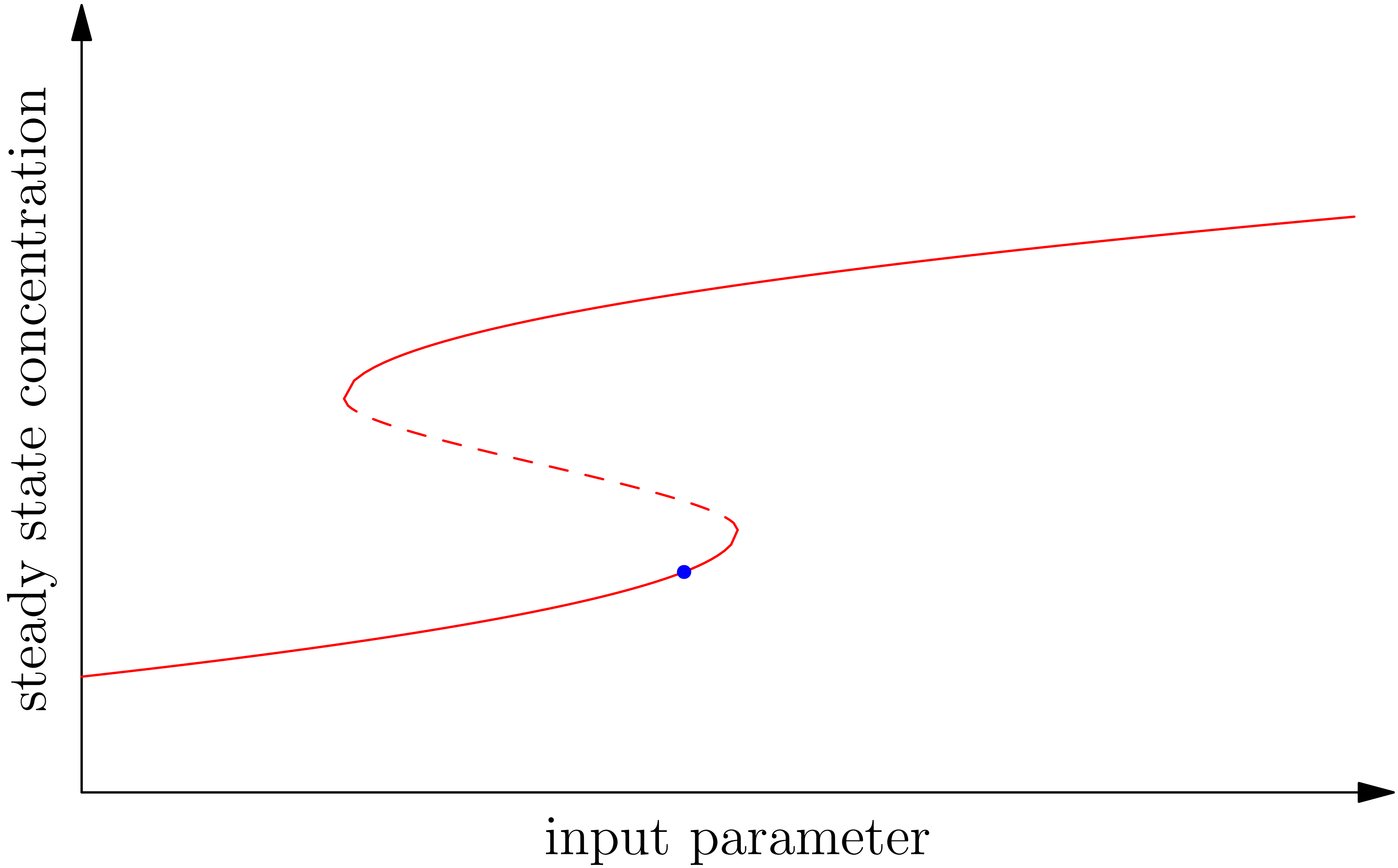
Hysteresis



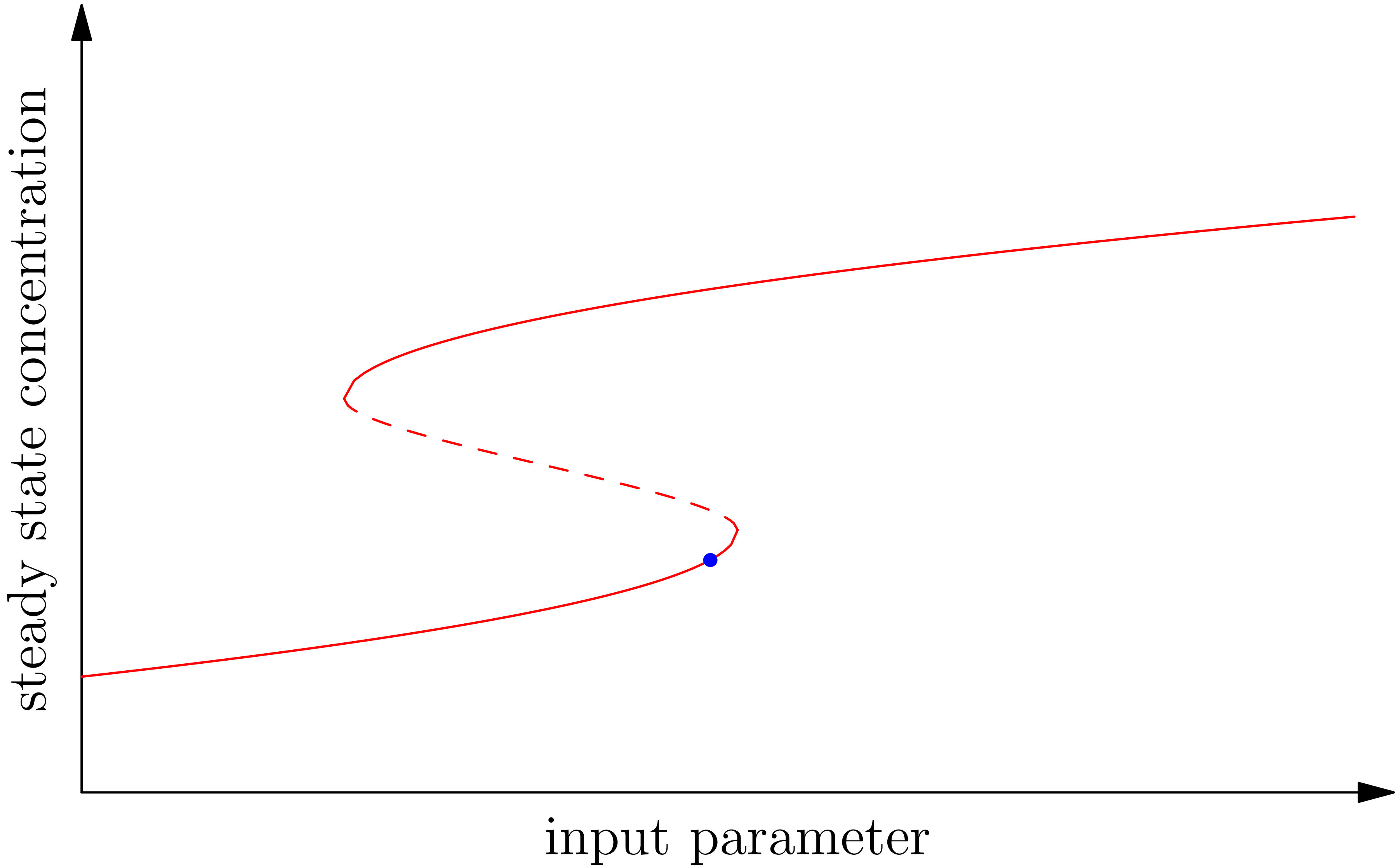
Hysteresis



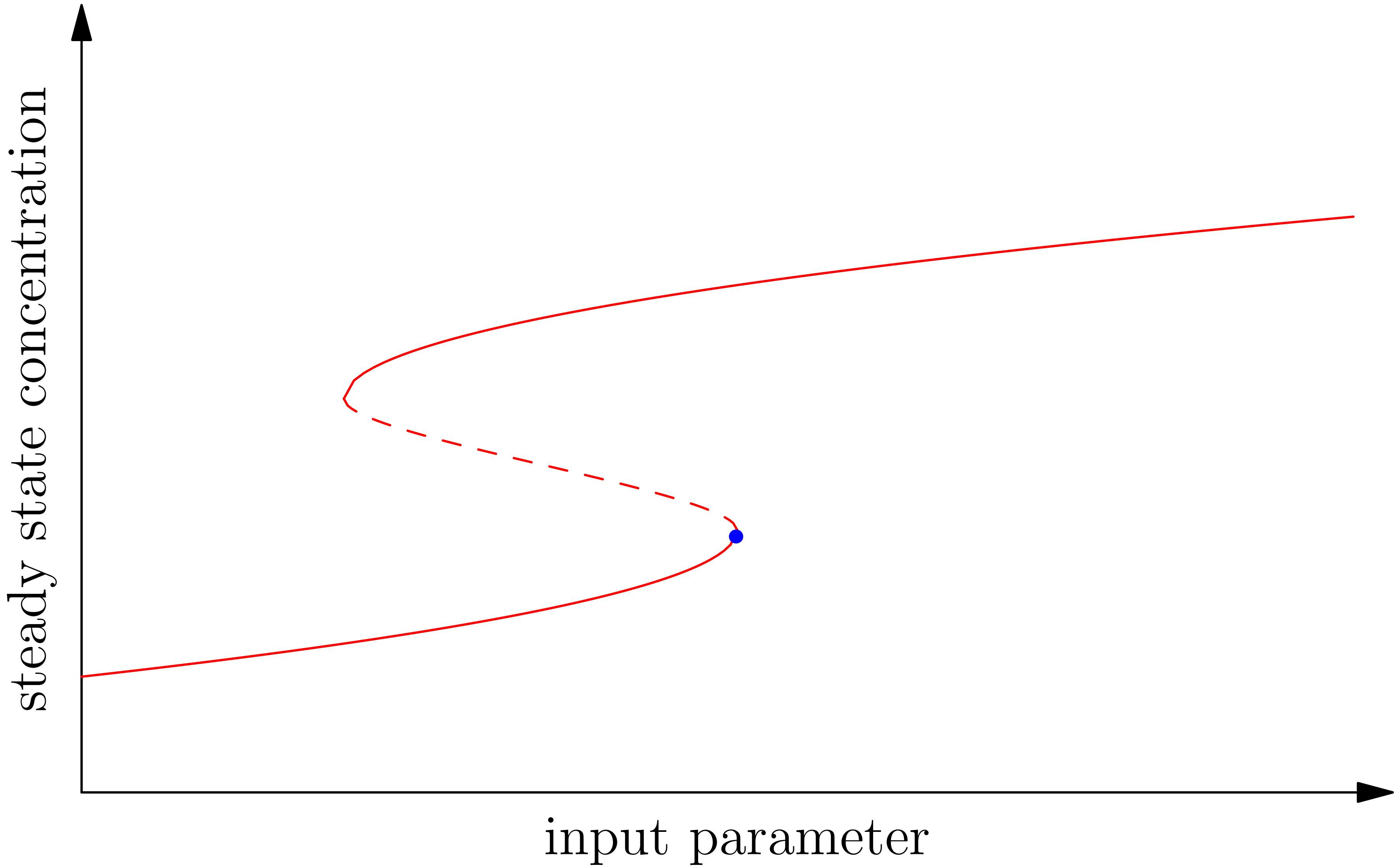
Hysteresis



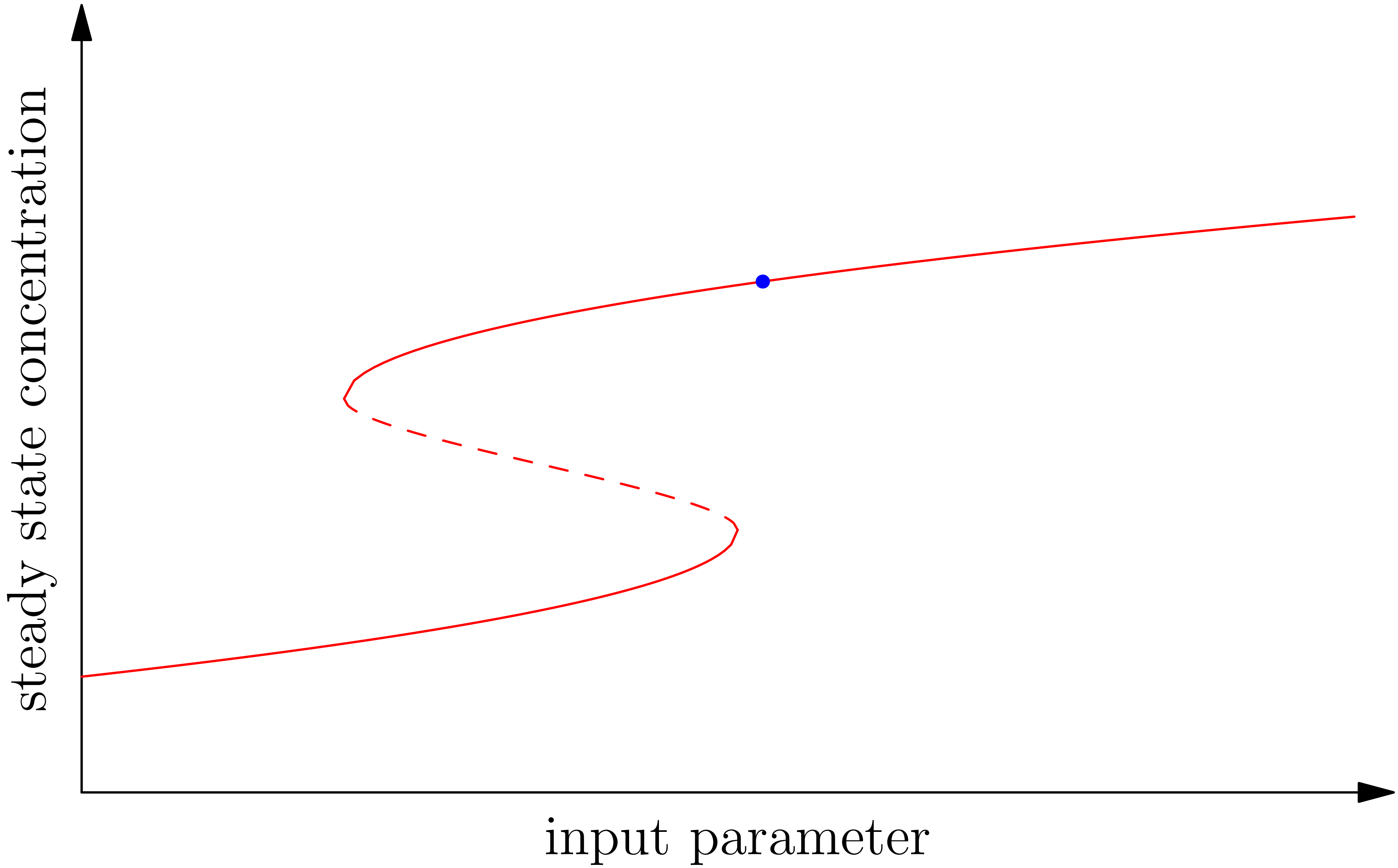
Hysteresis



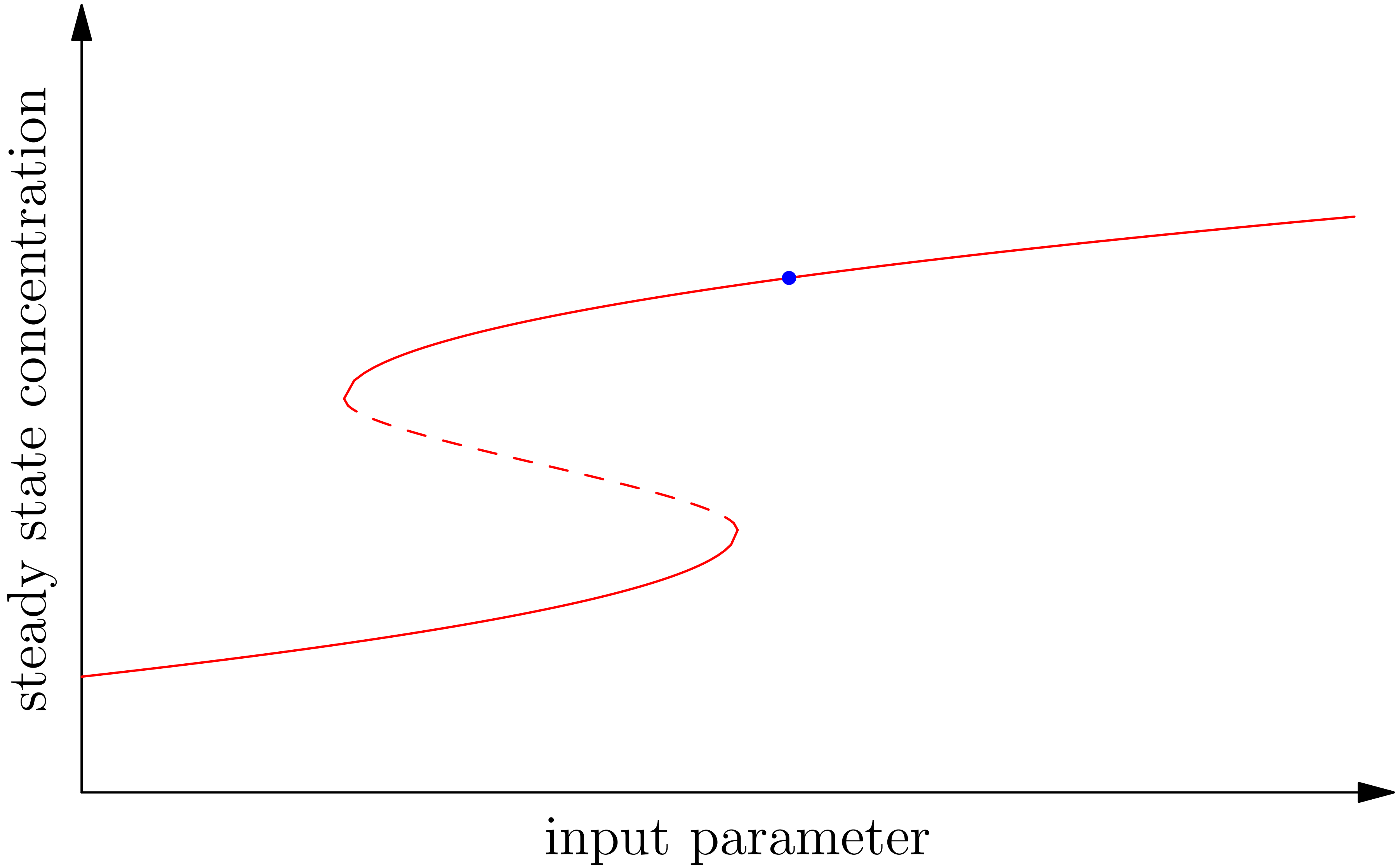
Hysteresis



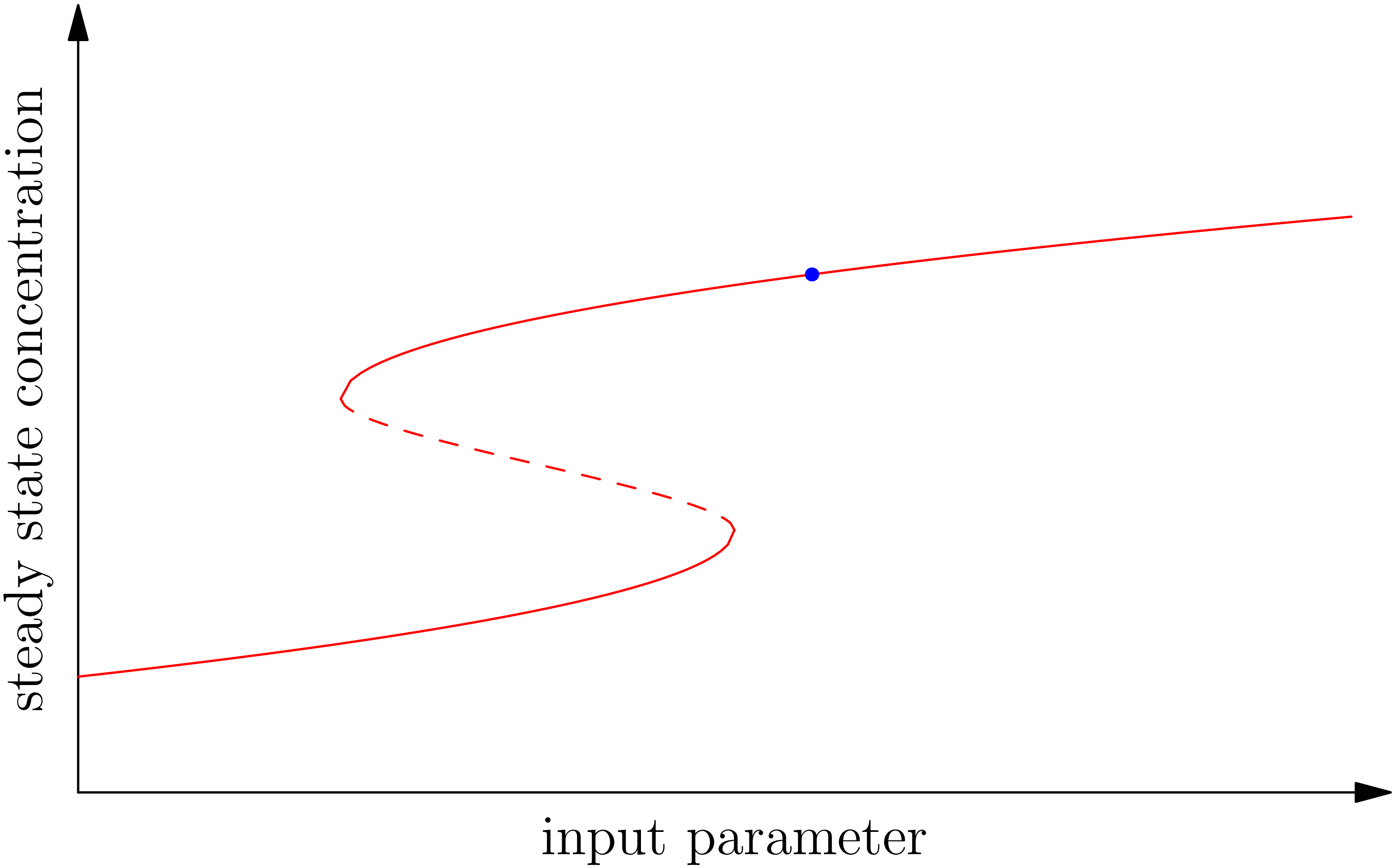
Hysteresis



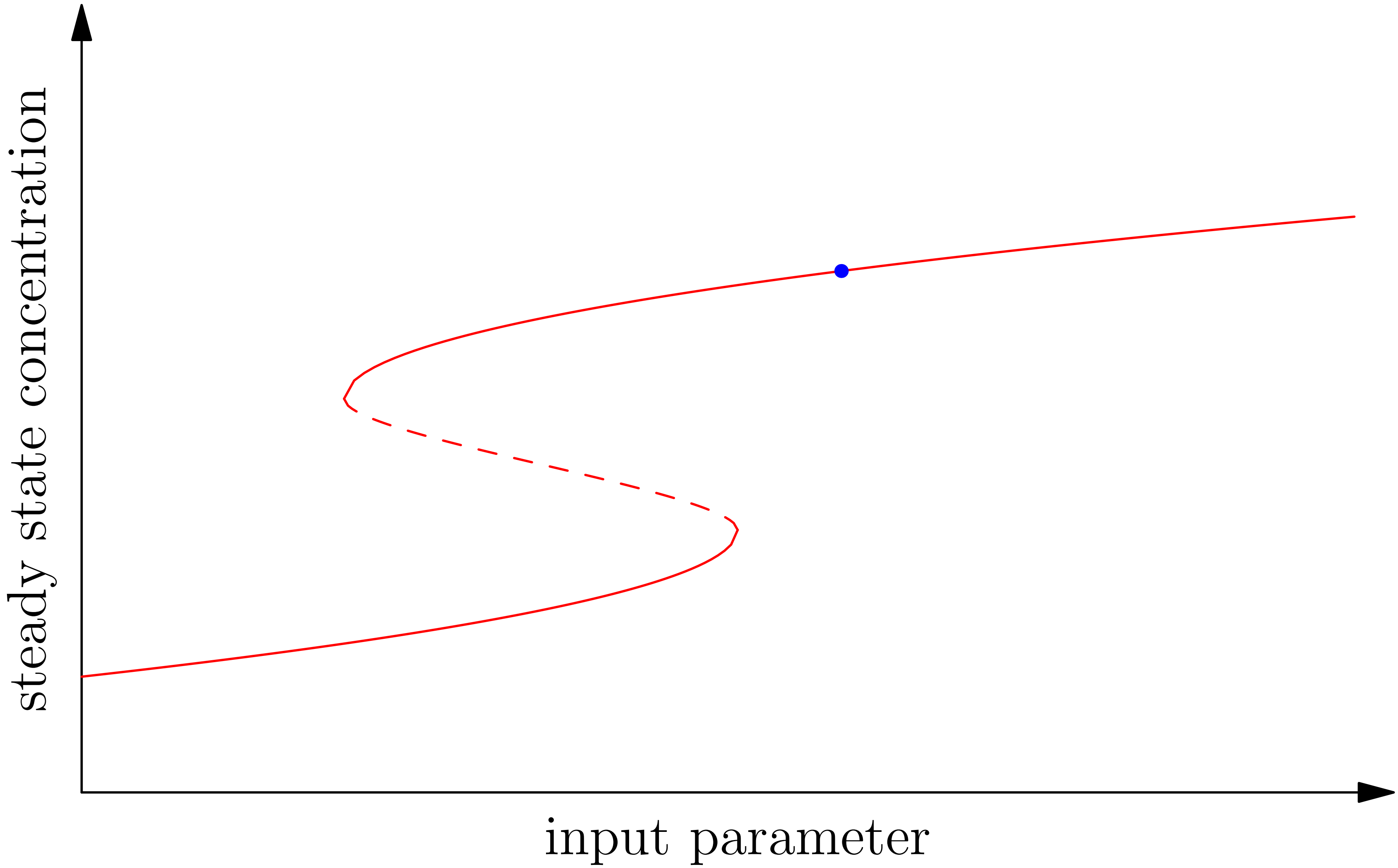
Hysteresis



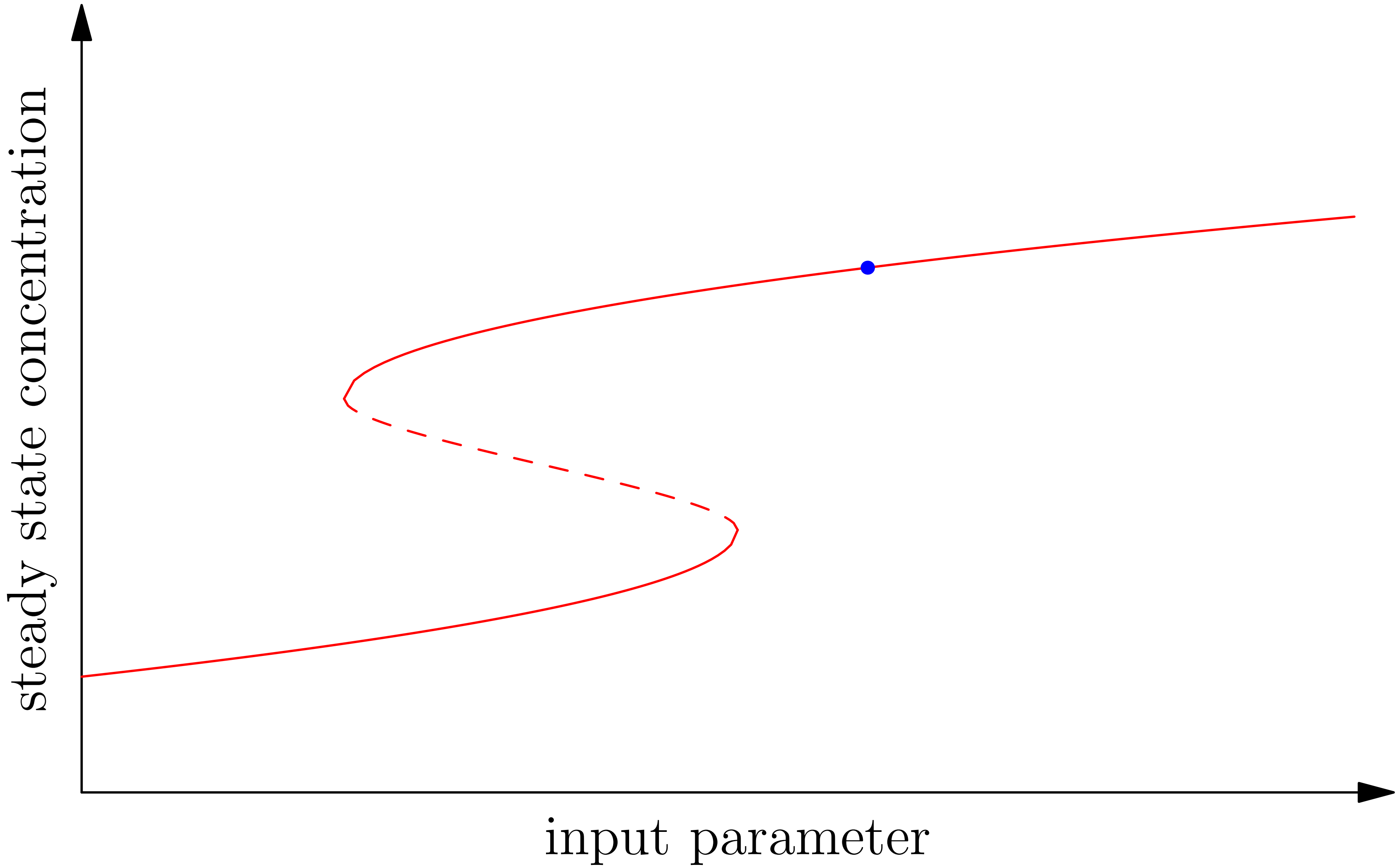
Hysteresis



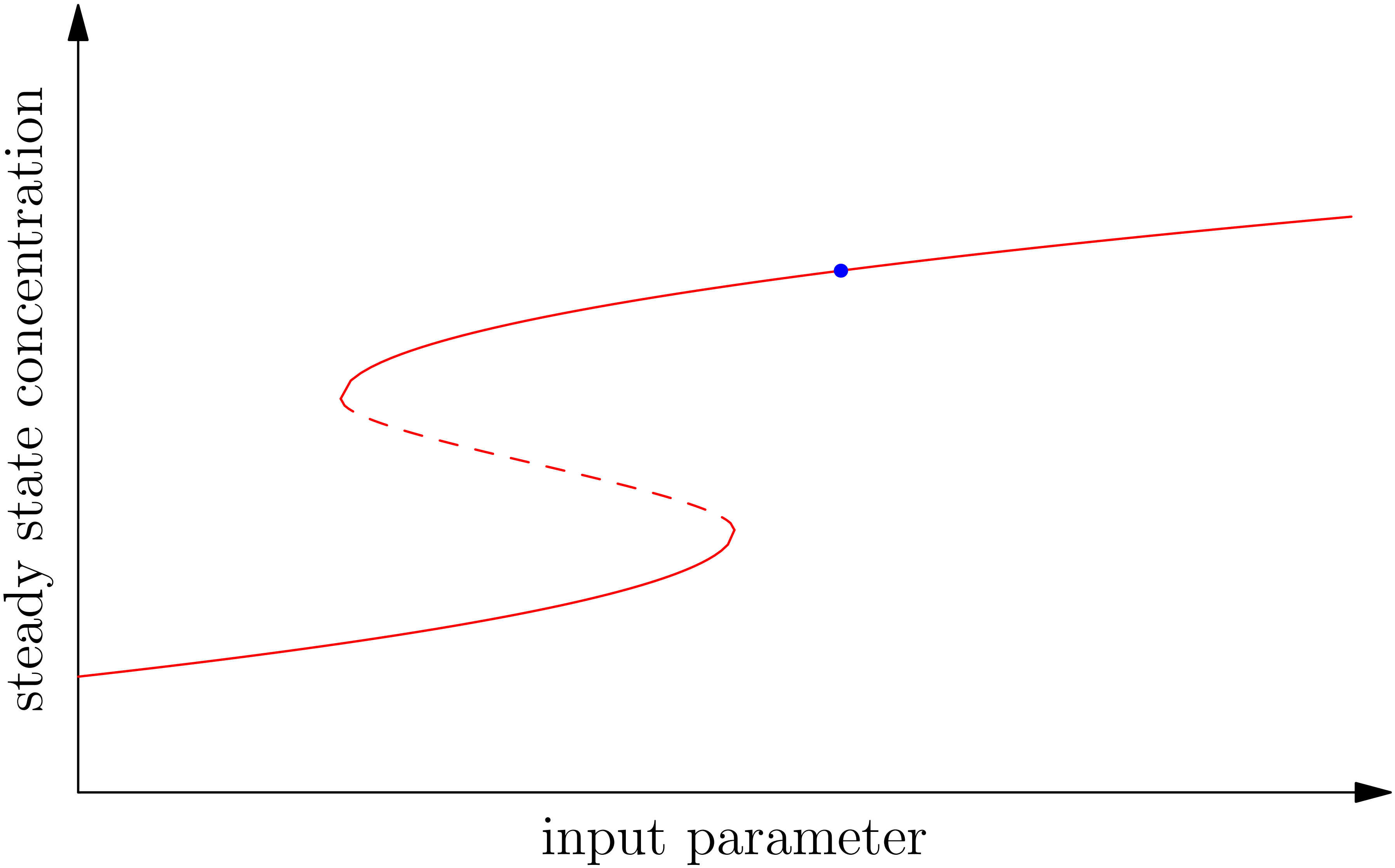
Hysteresis



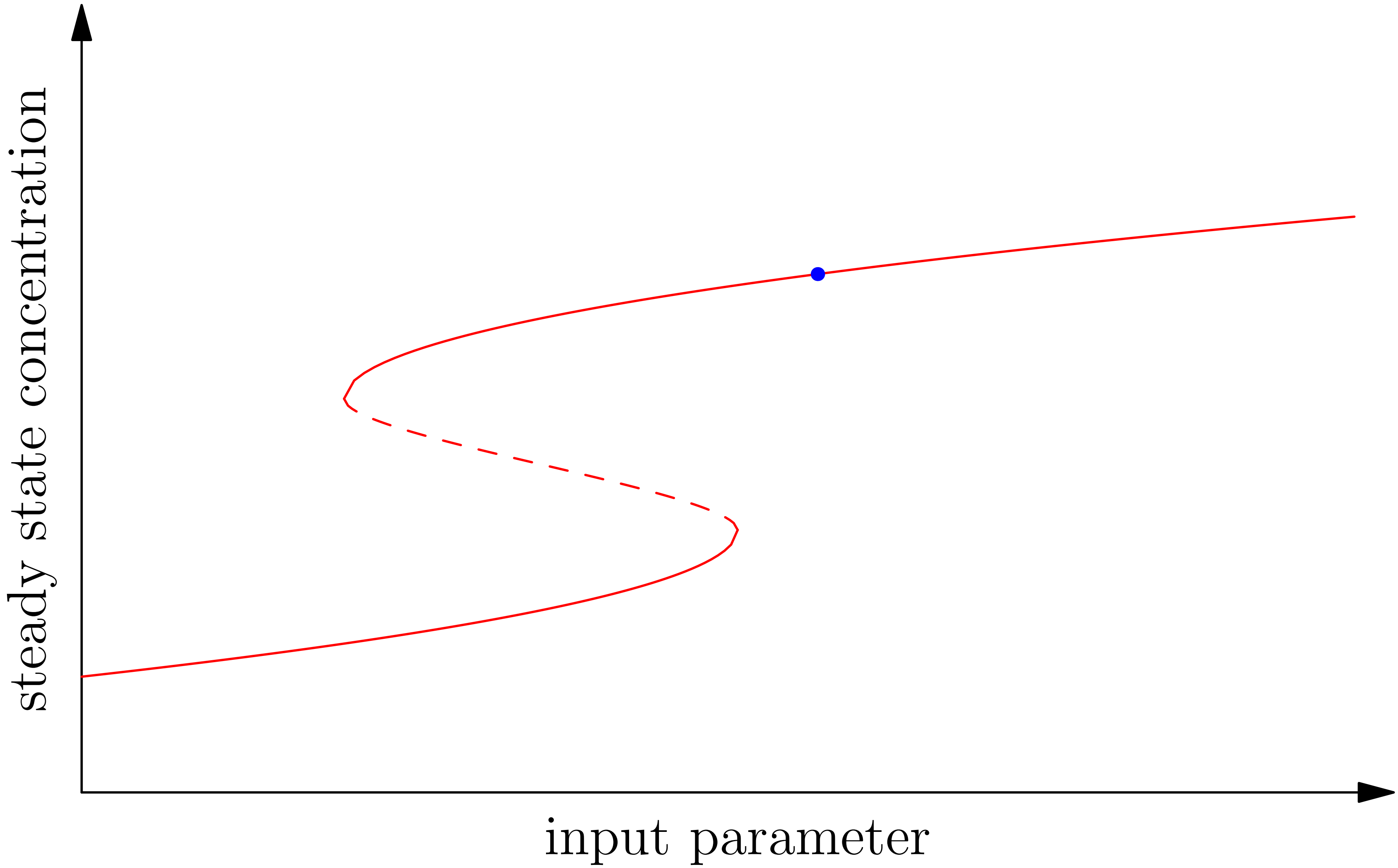
Hysteresis



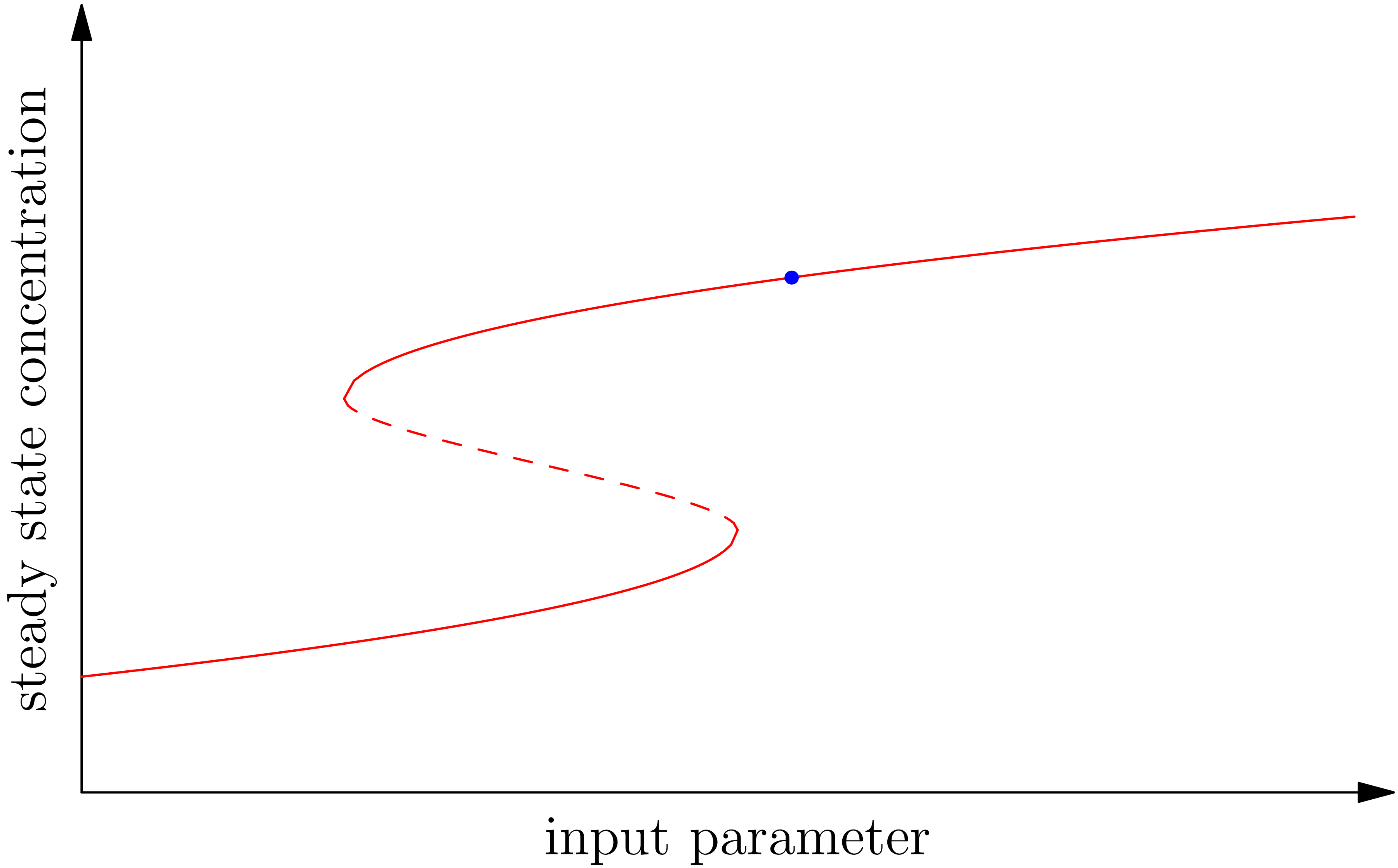
Hysteresis



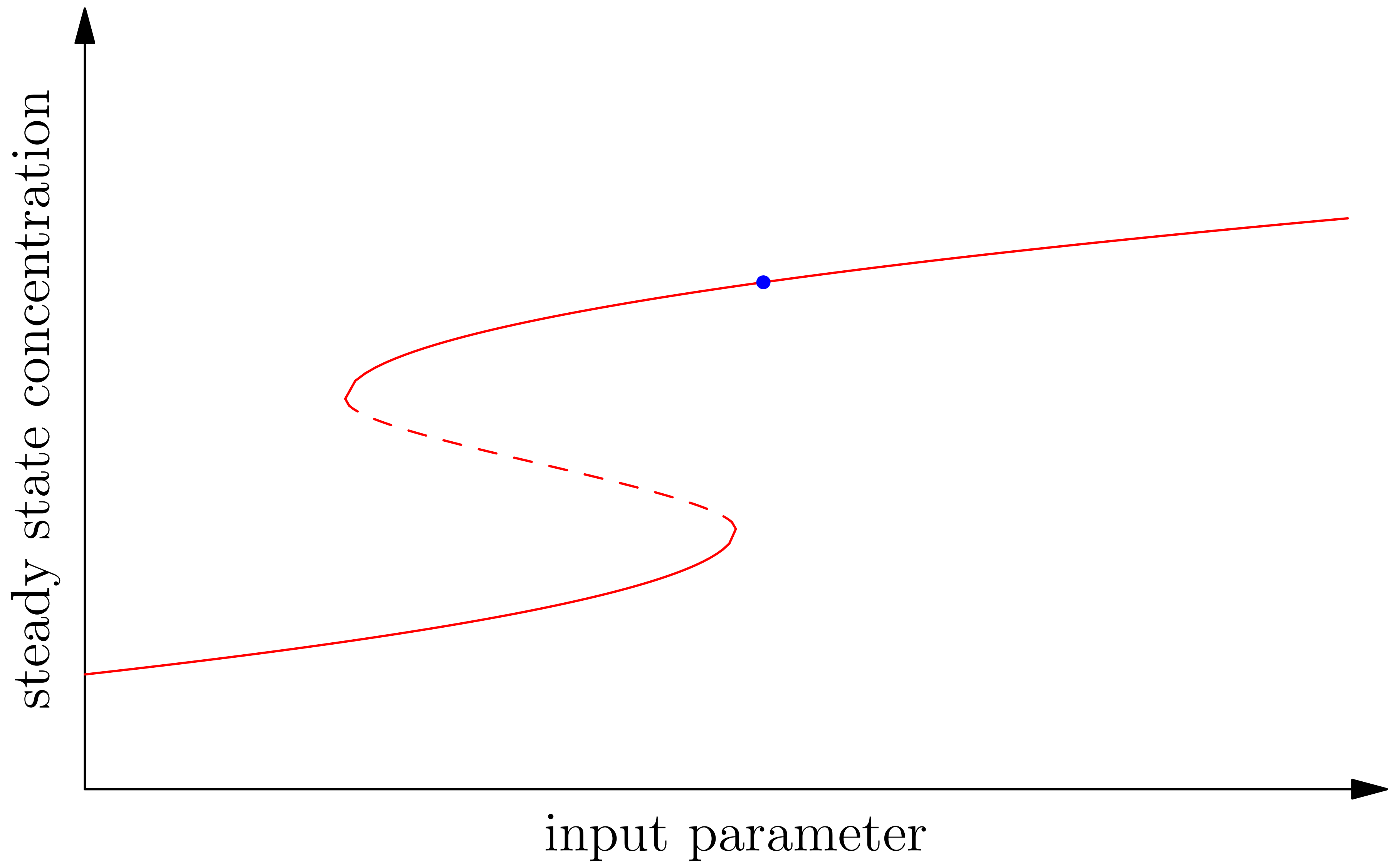
Hysteresis



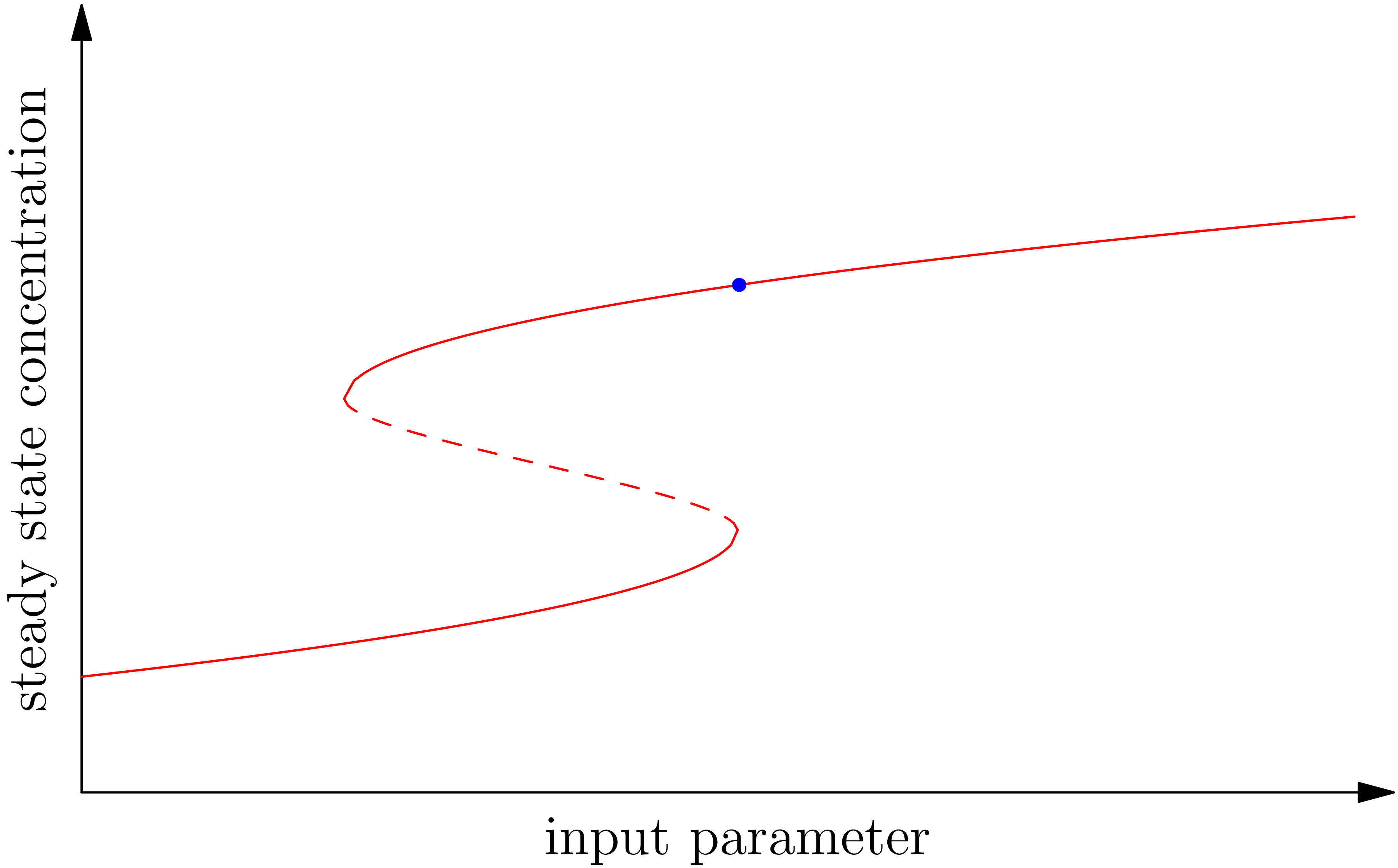
Hysteresis



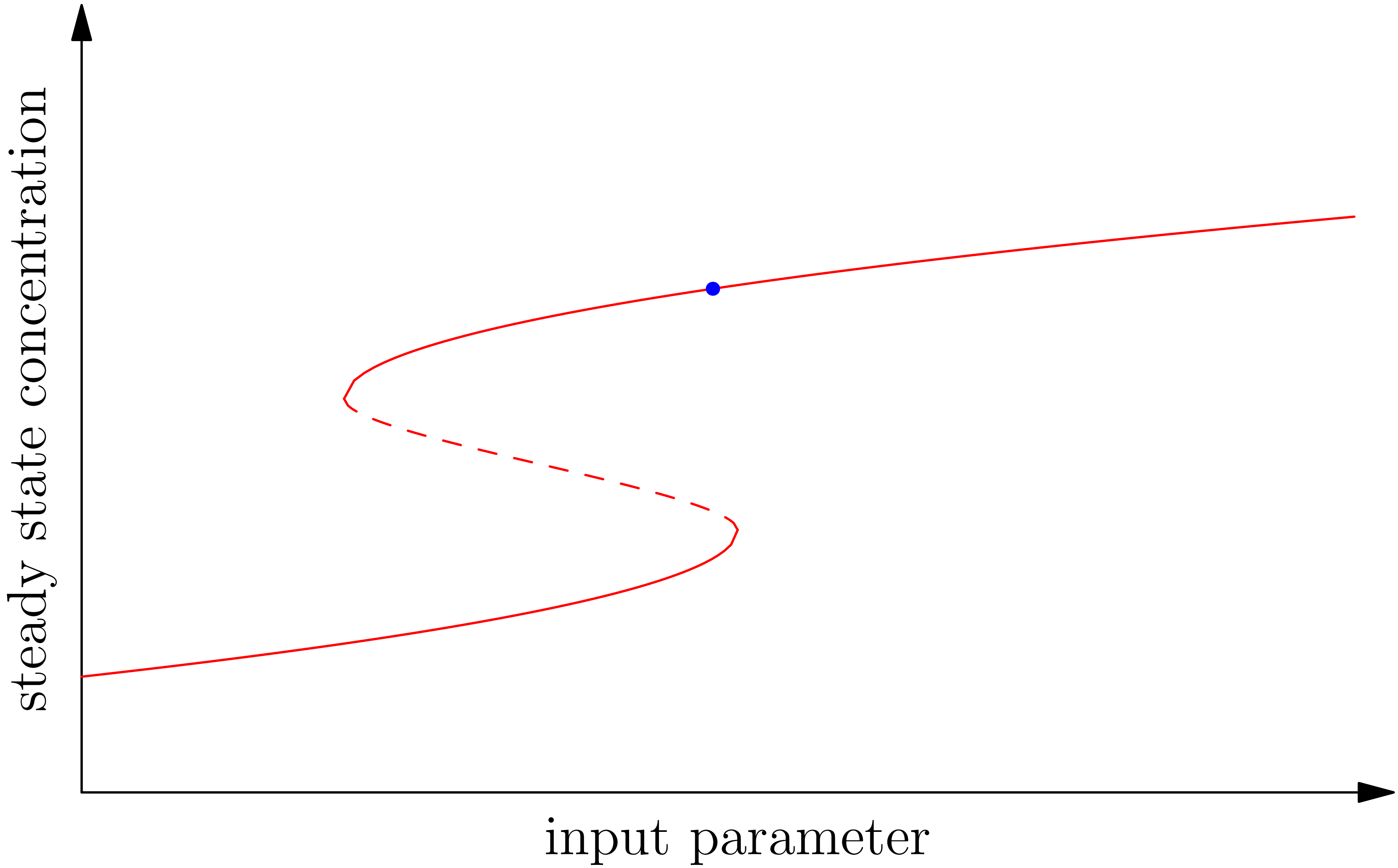
Hysteresis



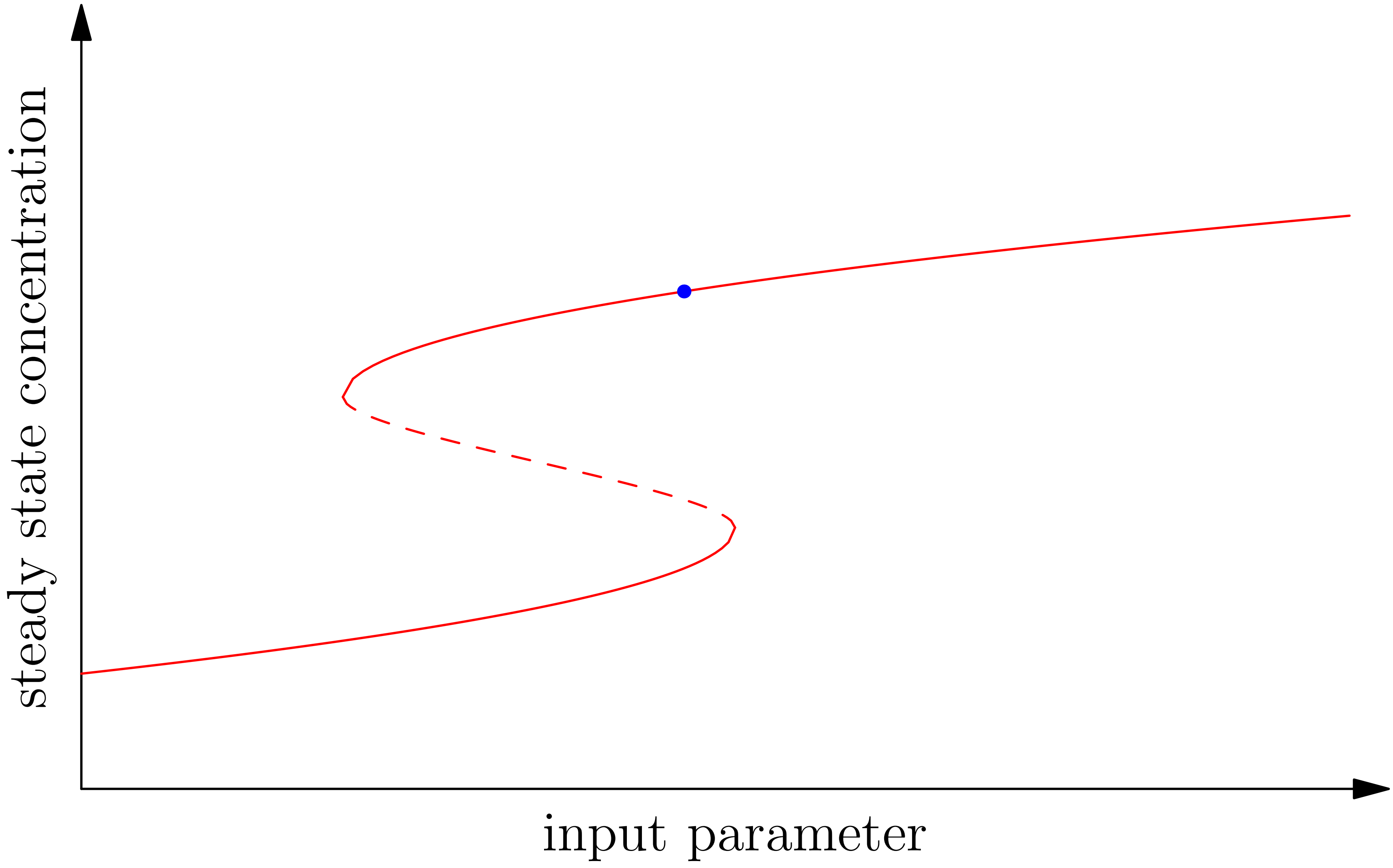
Hysteresis



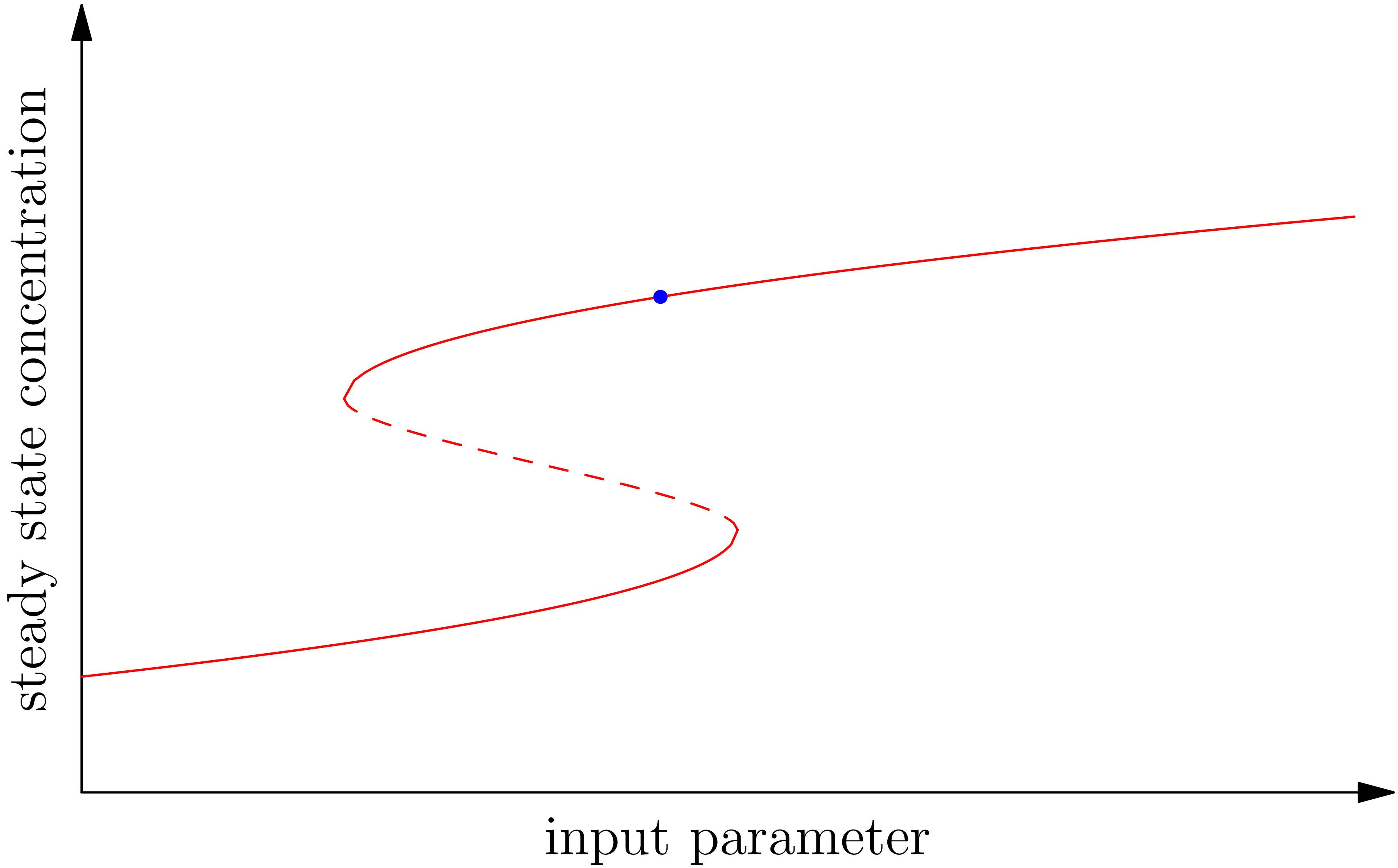
Hysteresis



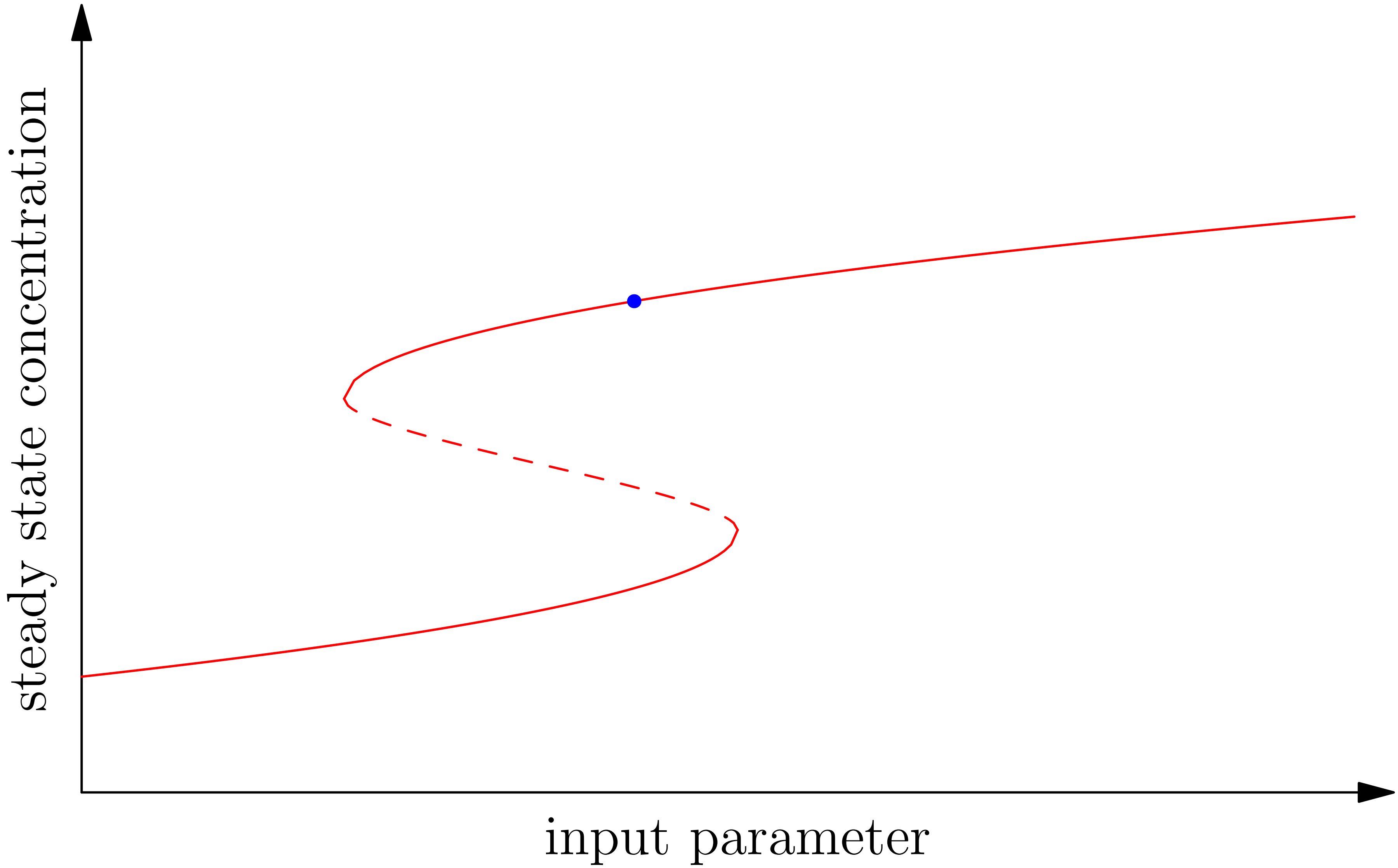
Hysteresis



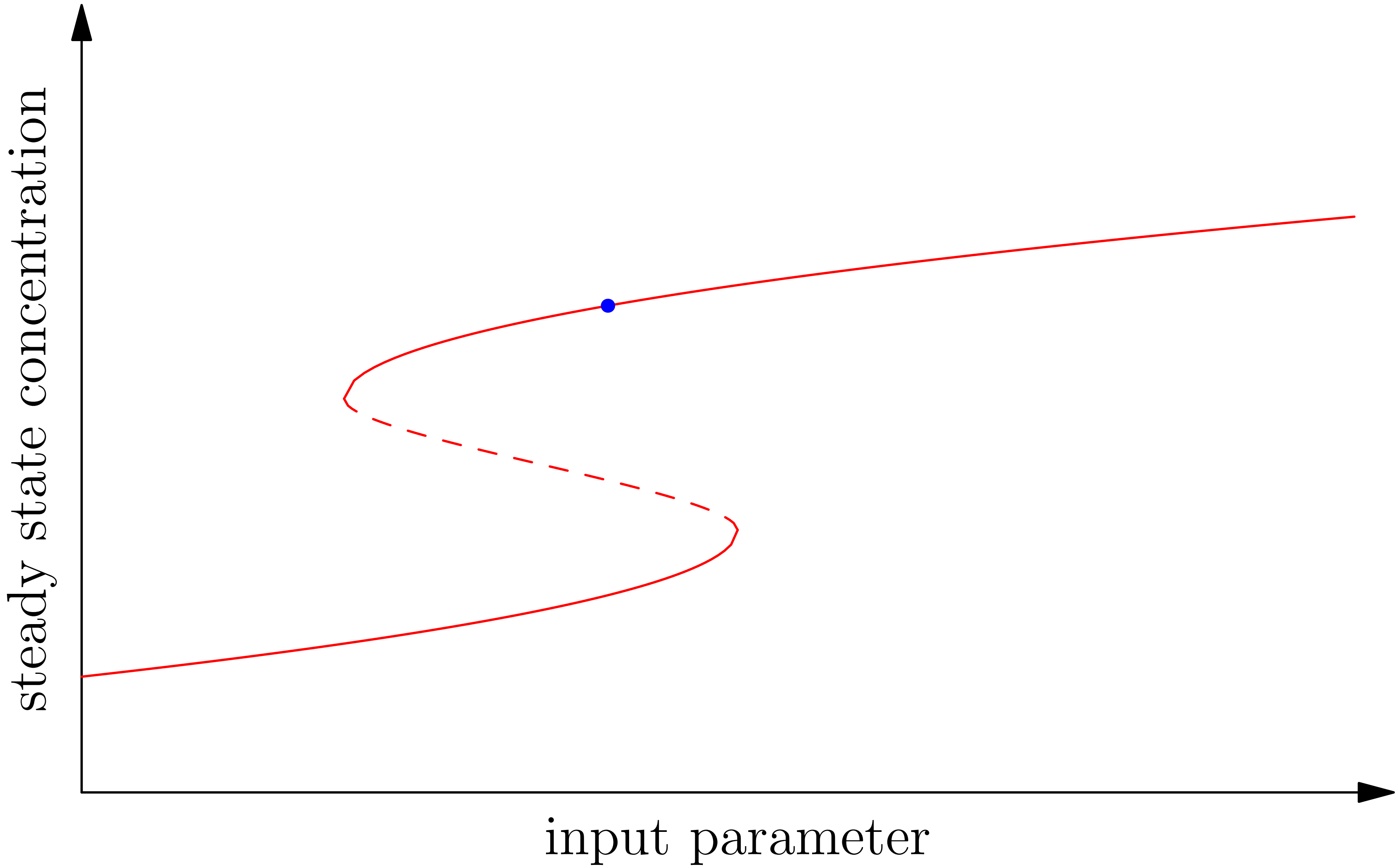
Hysteresis



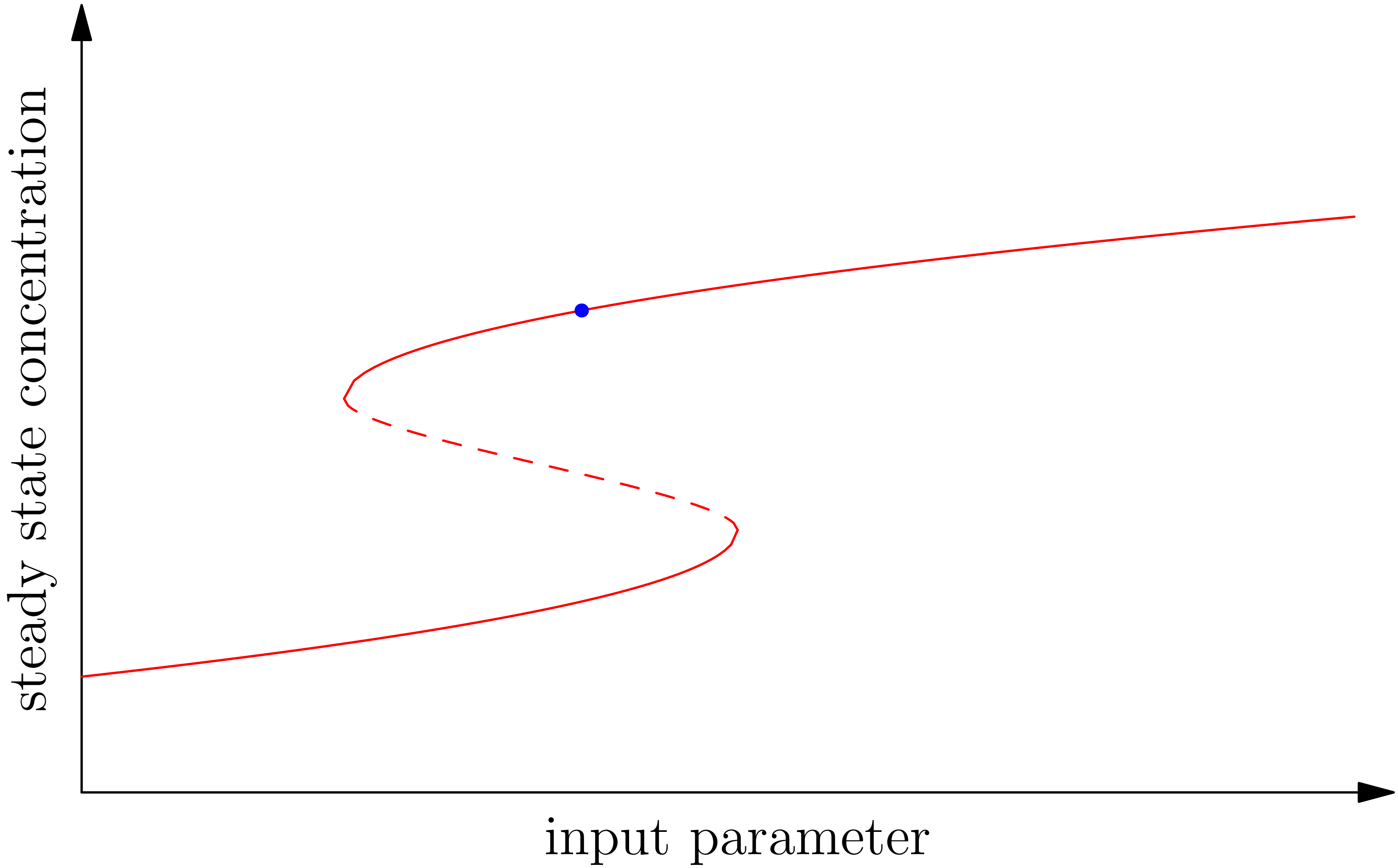
Hysteresis



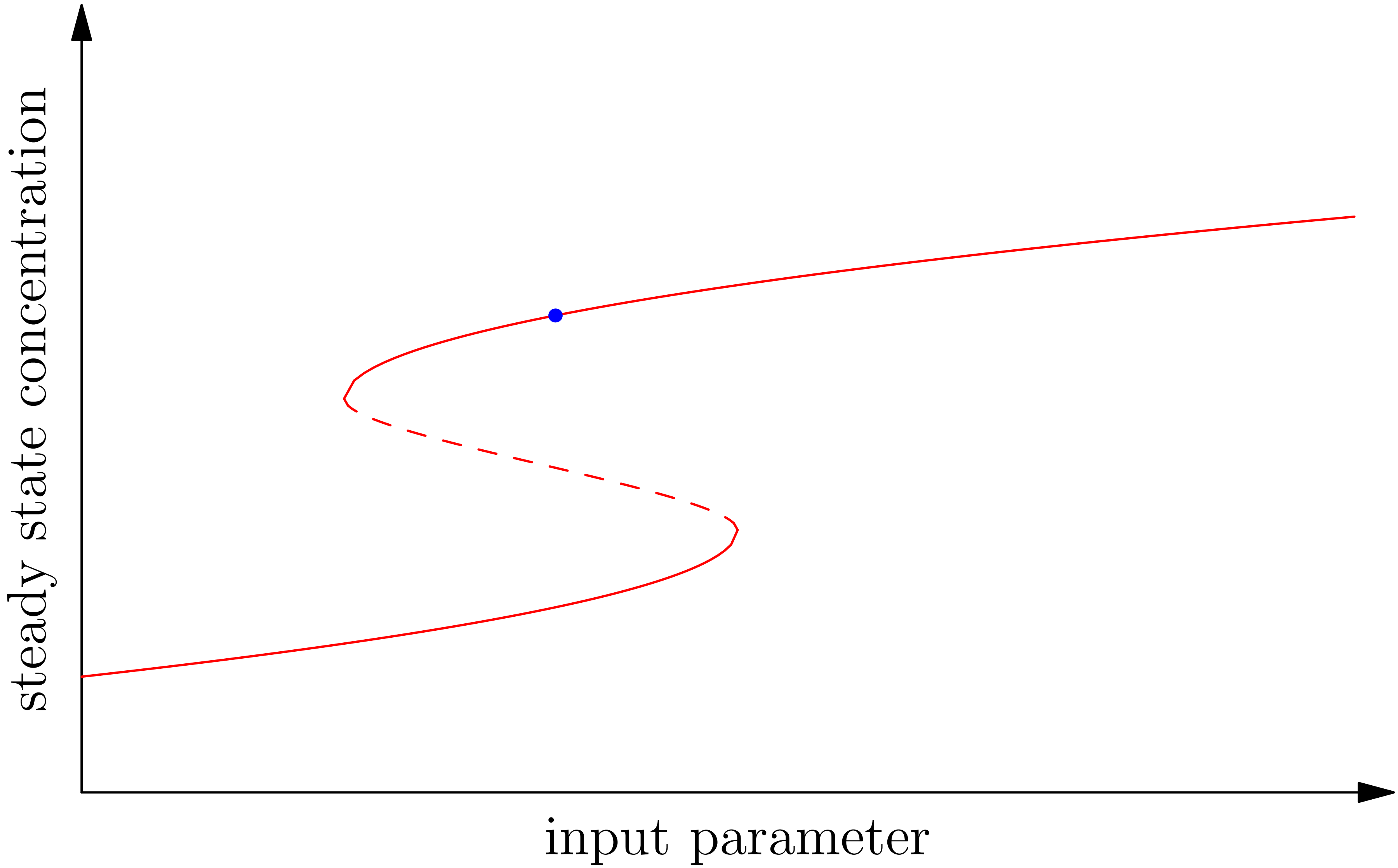
Hysteresis



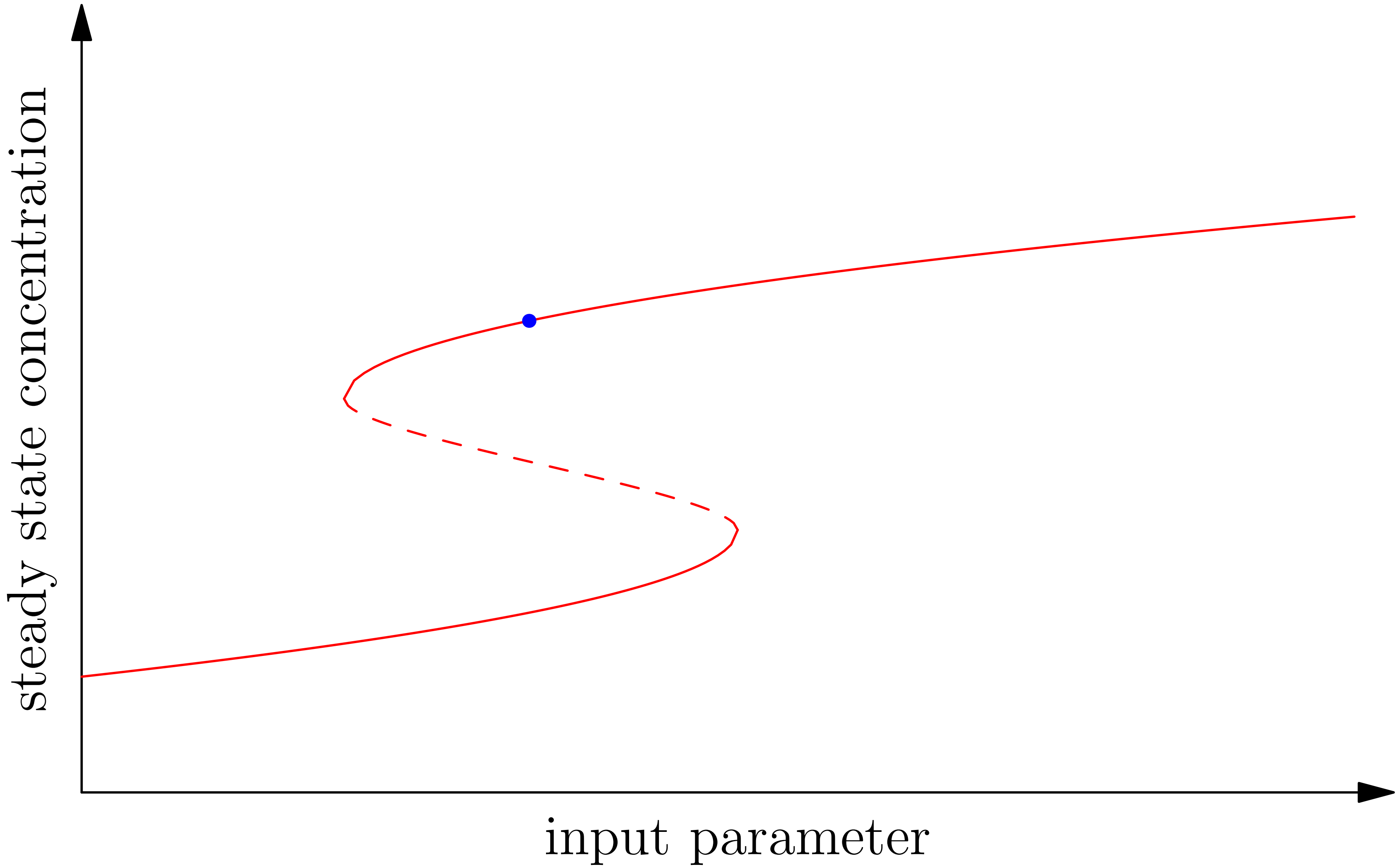
Hysteresis



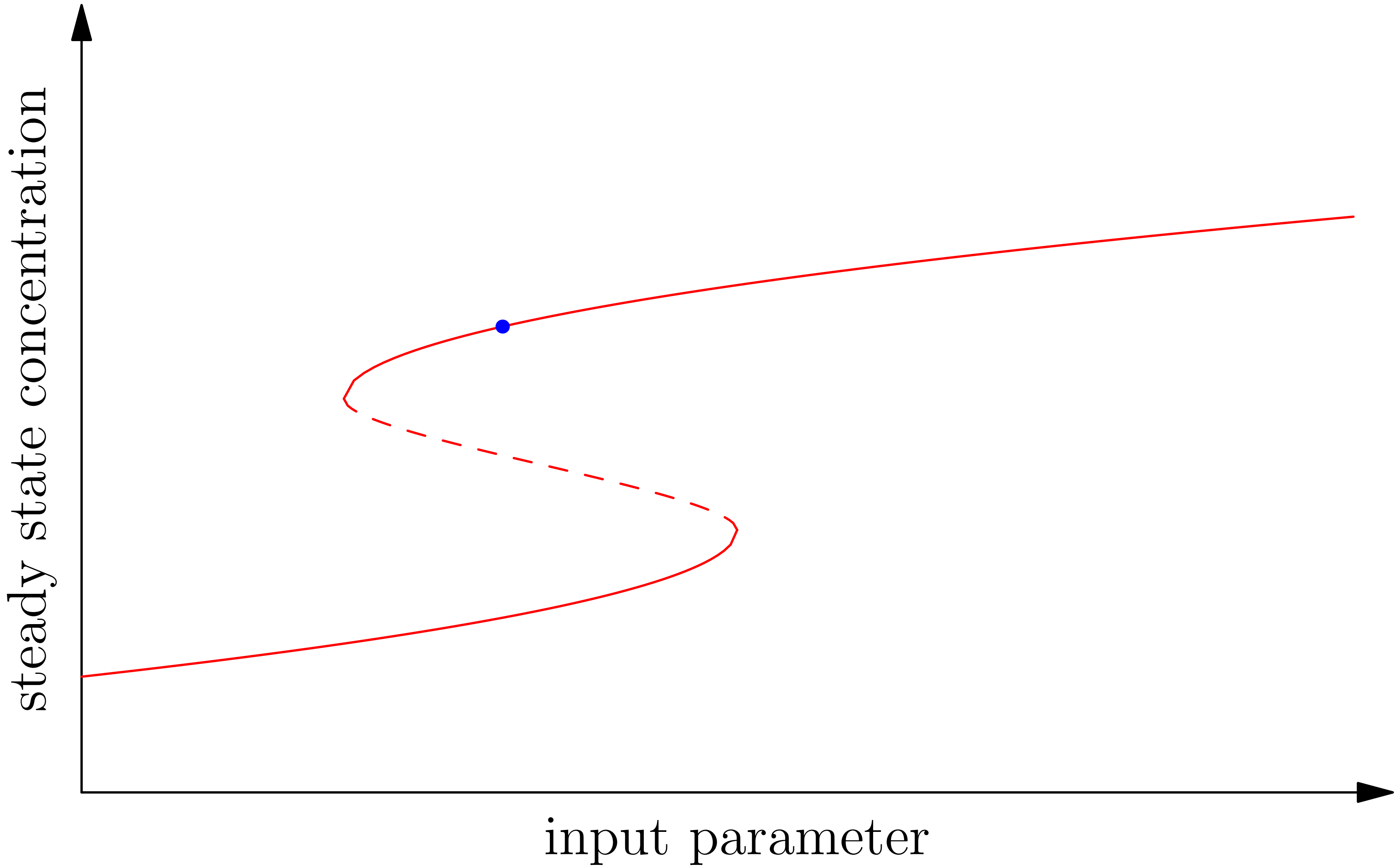
Hysteresis



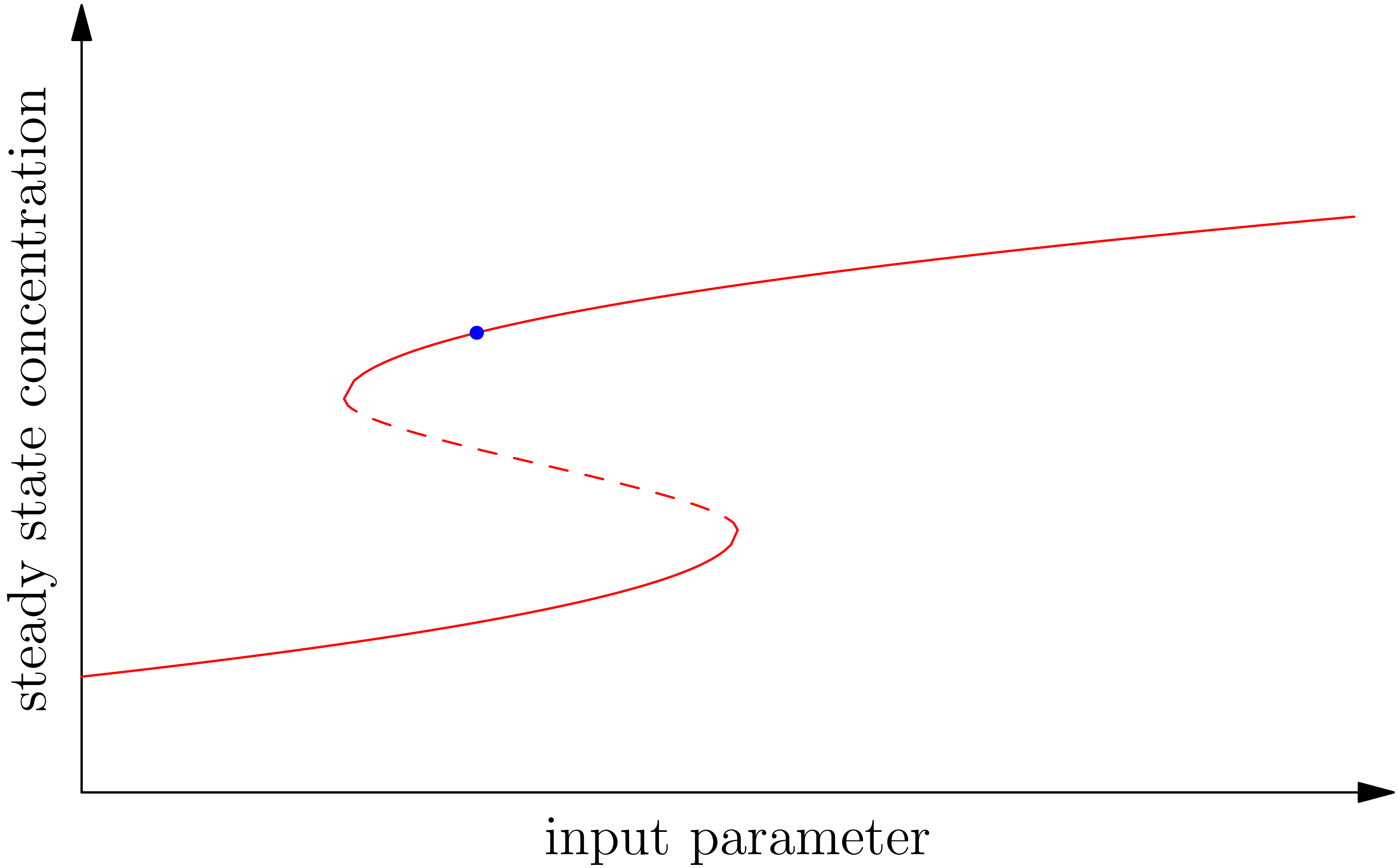
Hysteresis



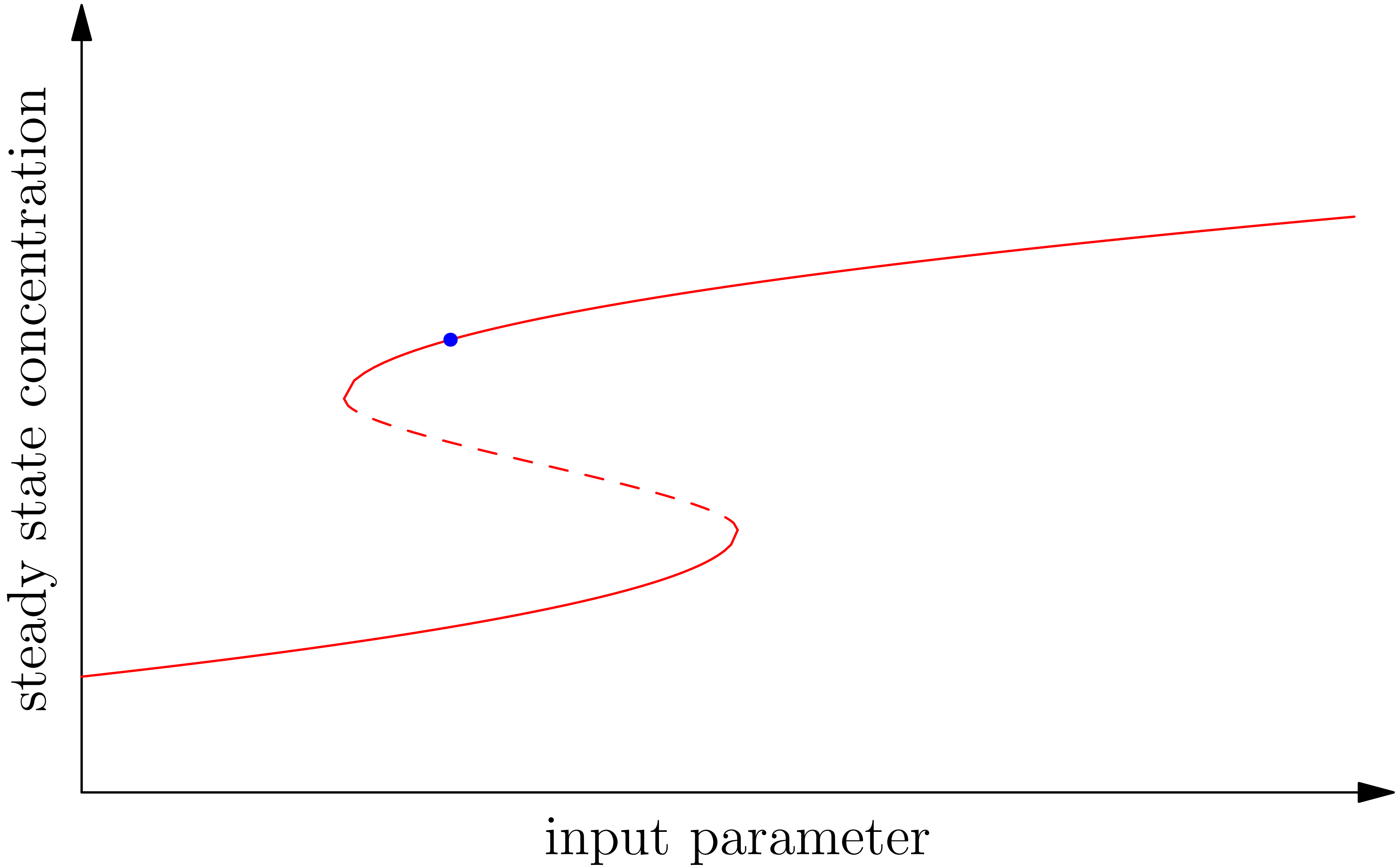
Hysteresis



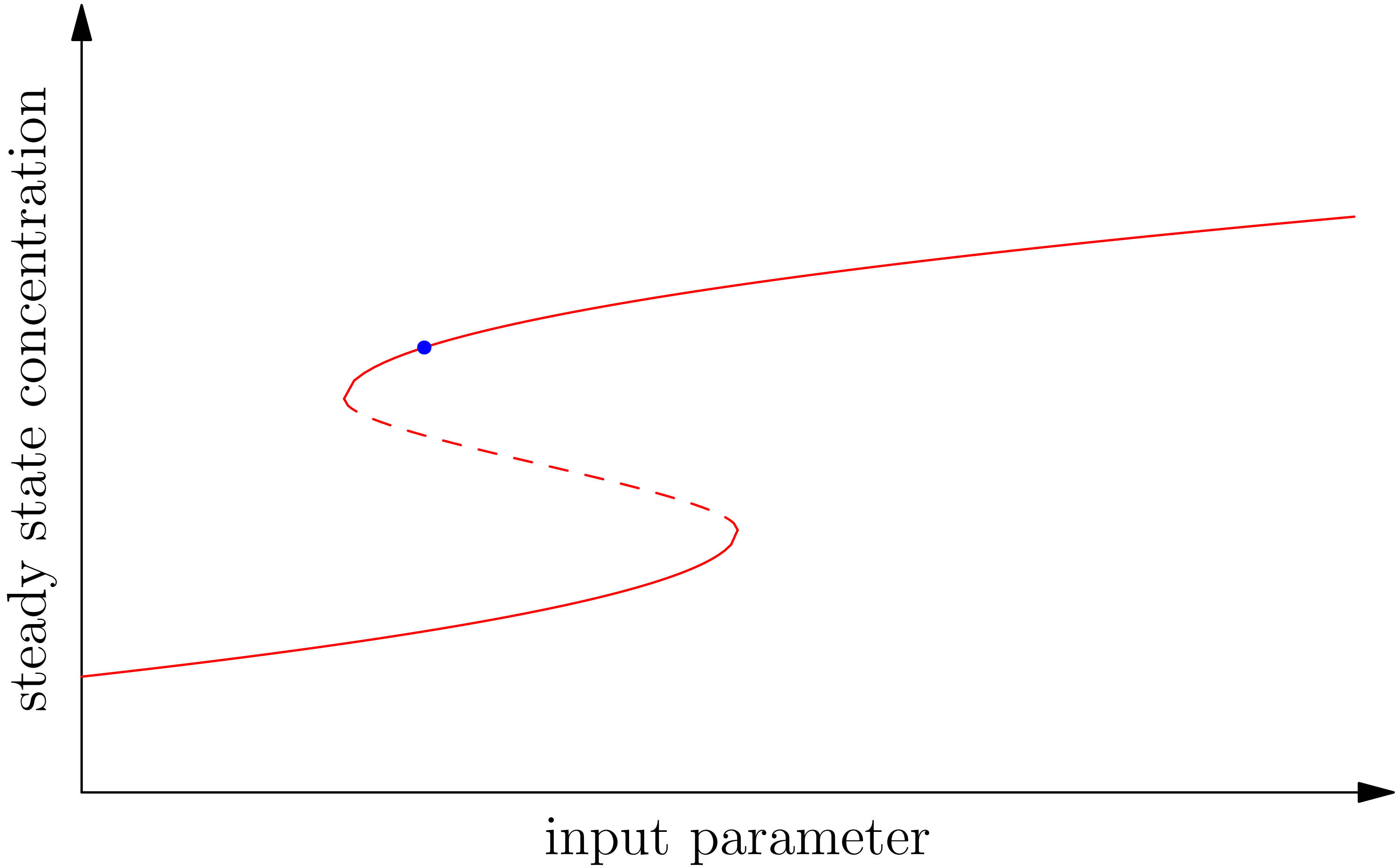
Hysteresis



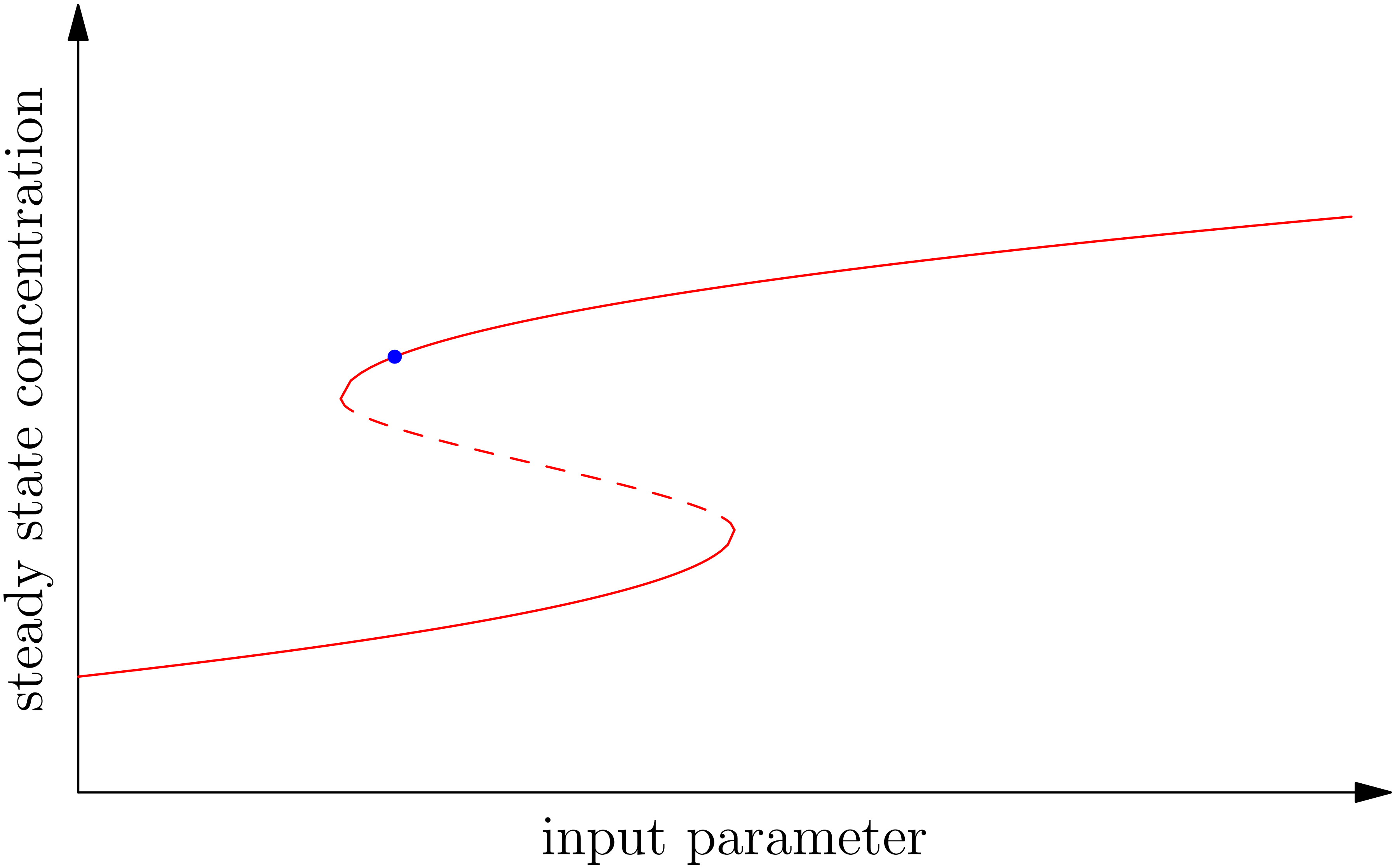
Hysteresis



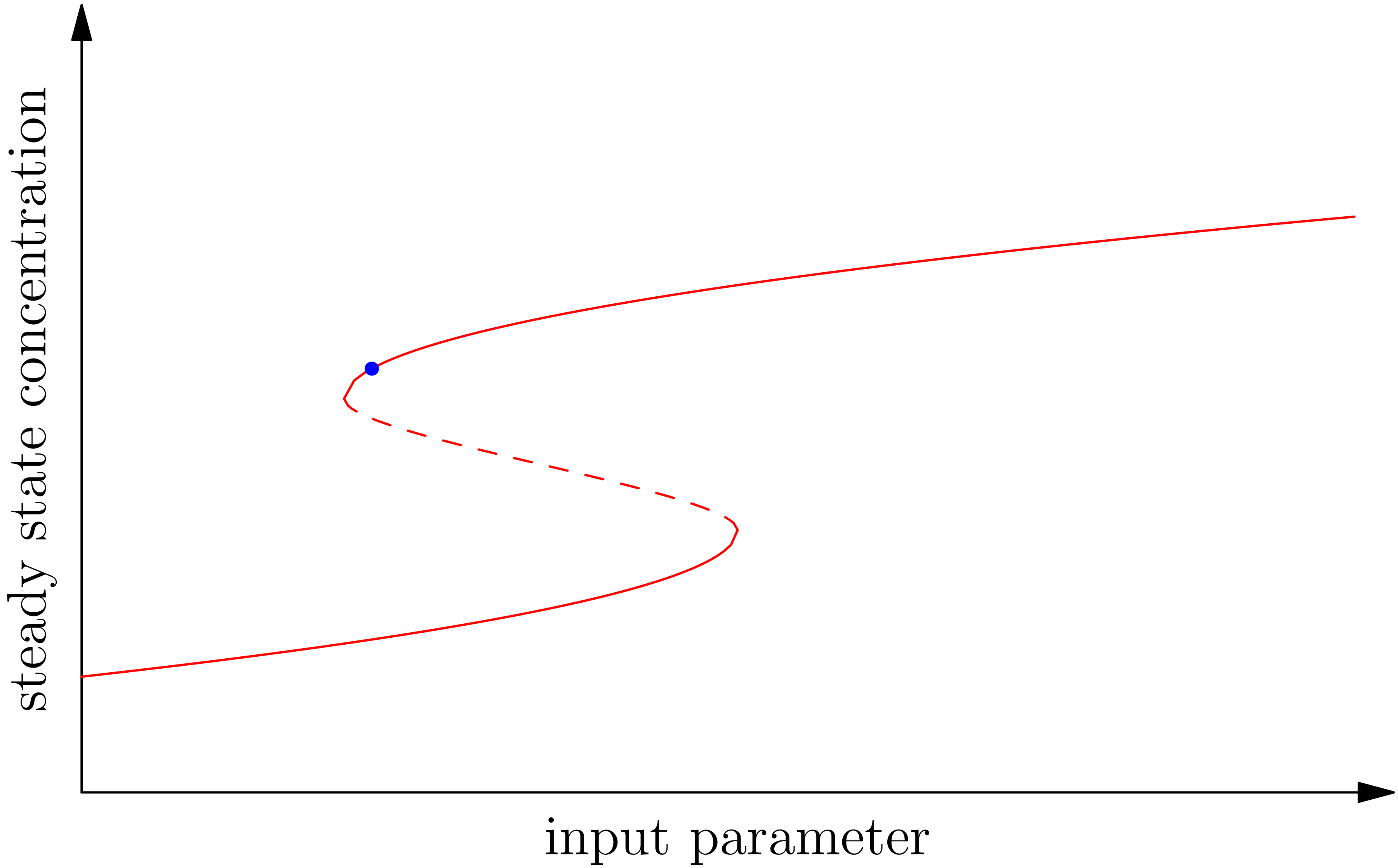
Hysteresis



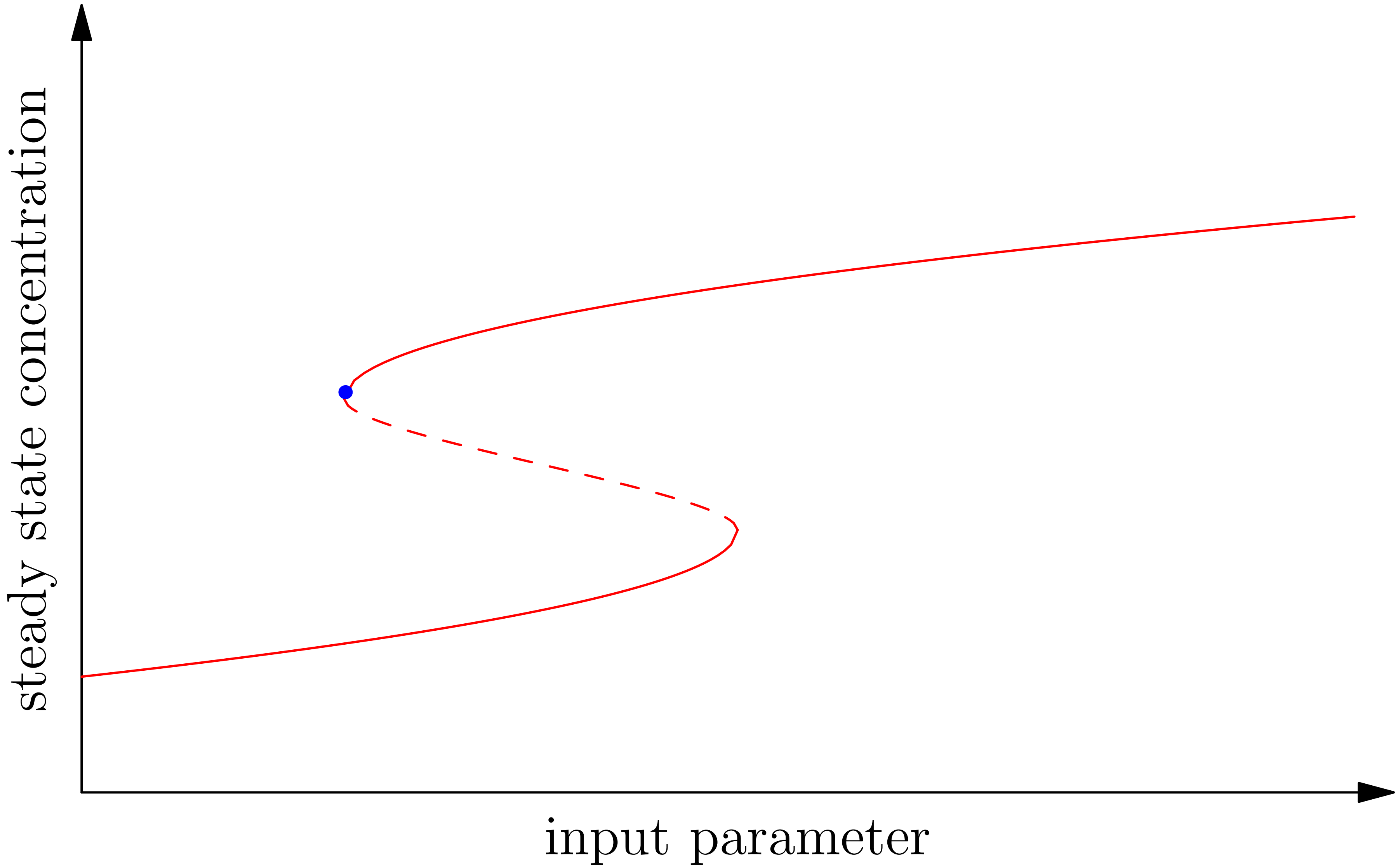
Hysteresis



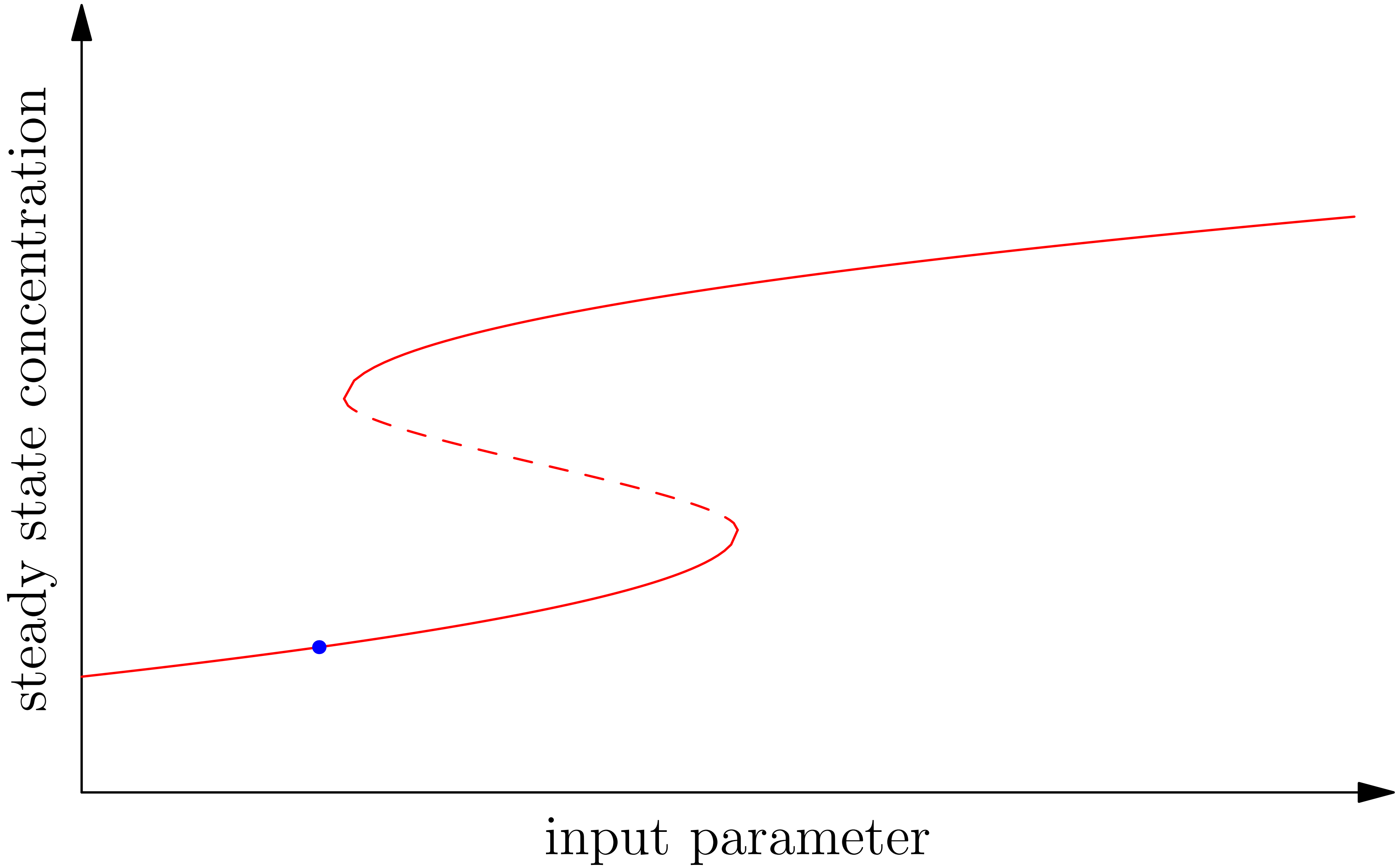
Hysteresis



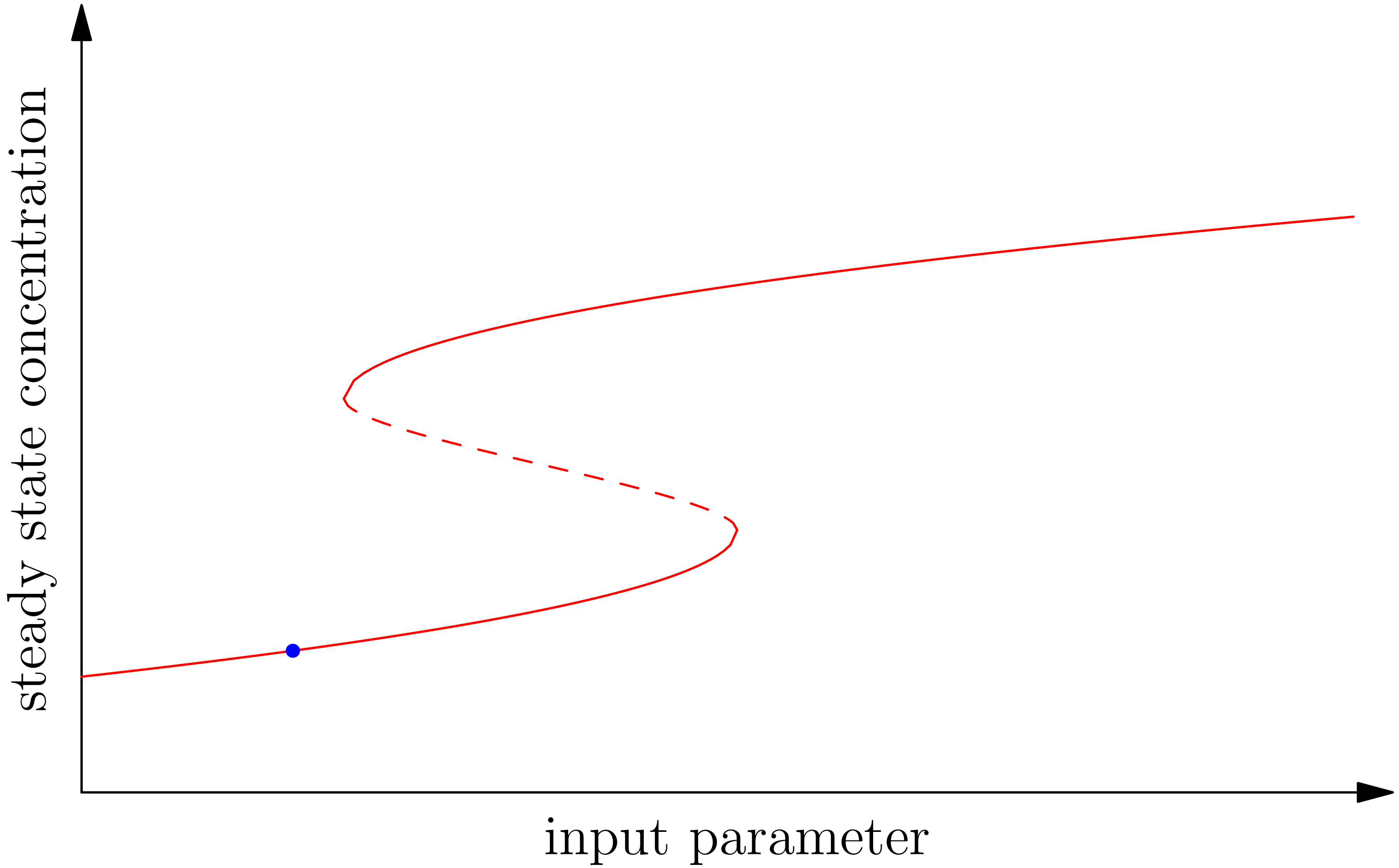
Hysteresis



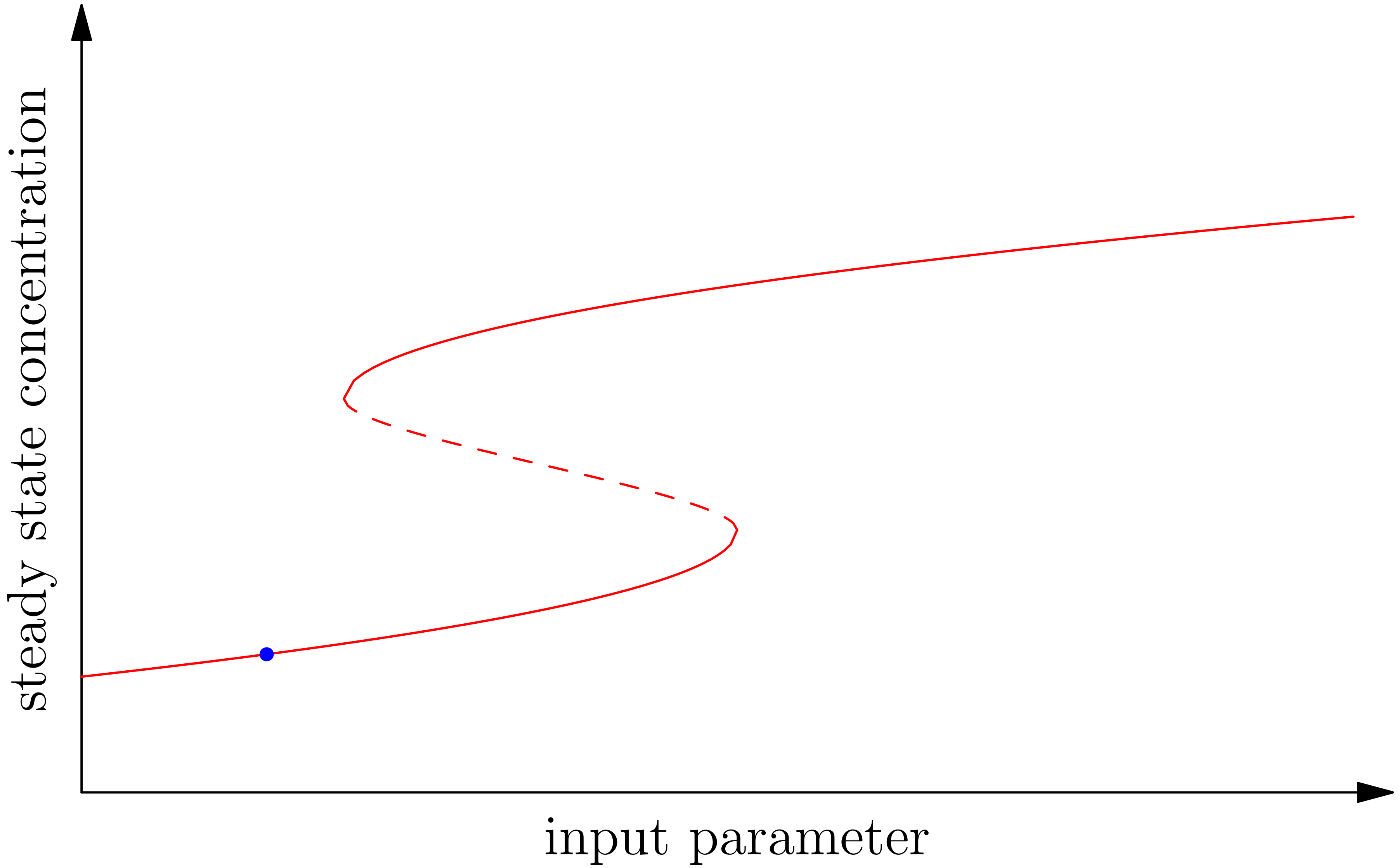
Hysteresis



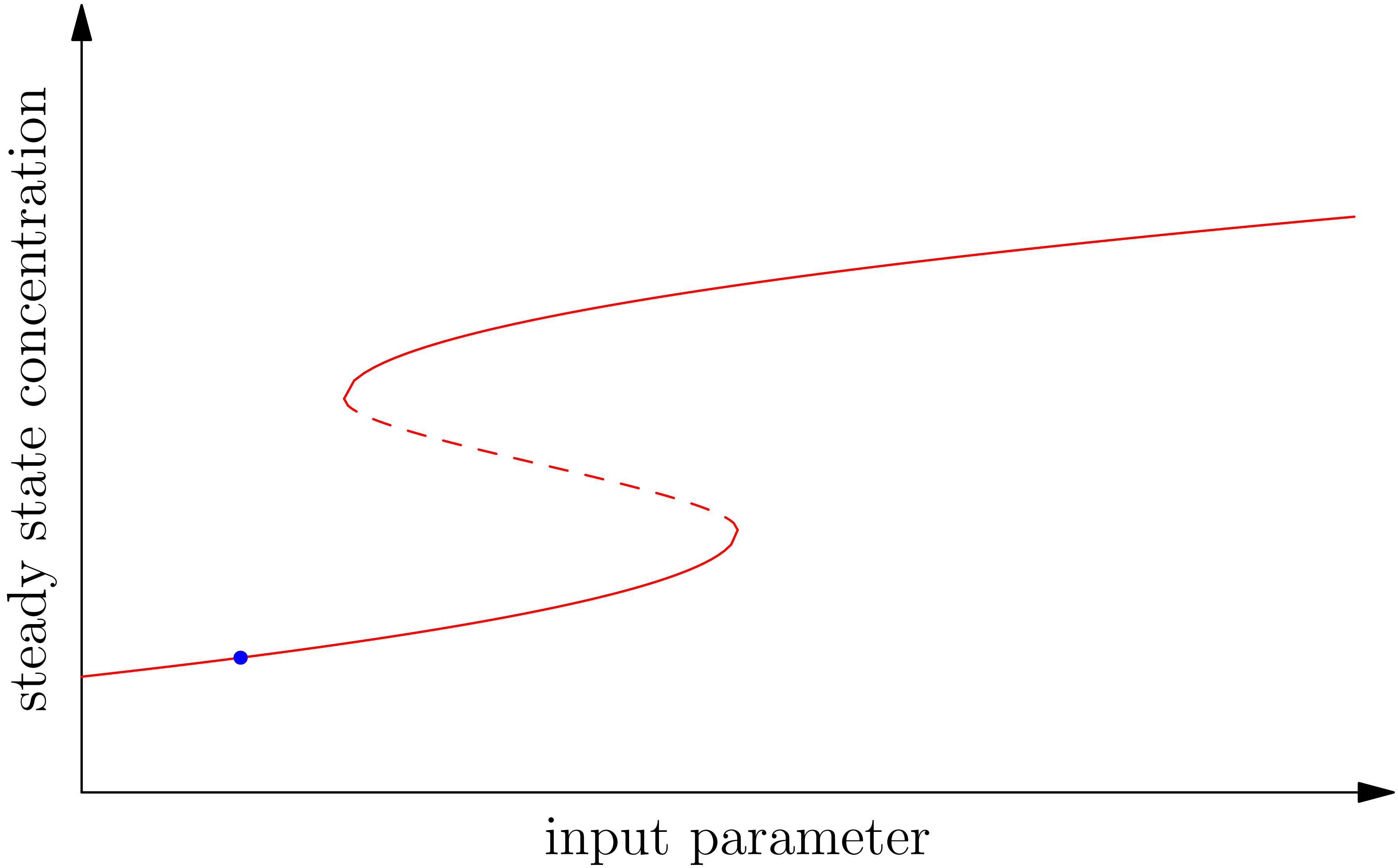
Hysteresis



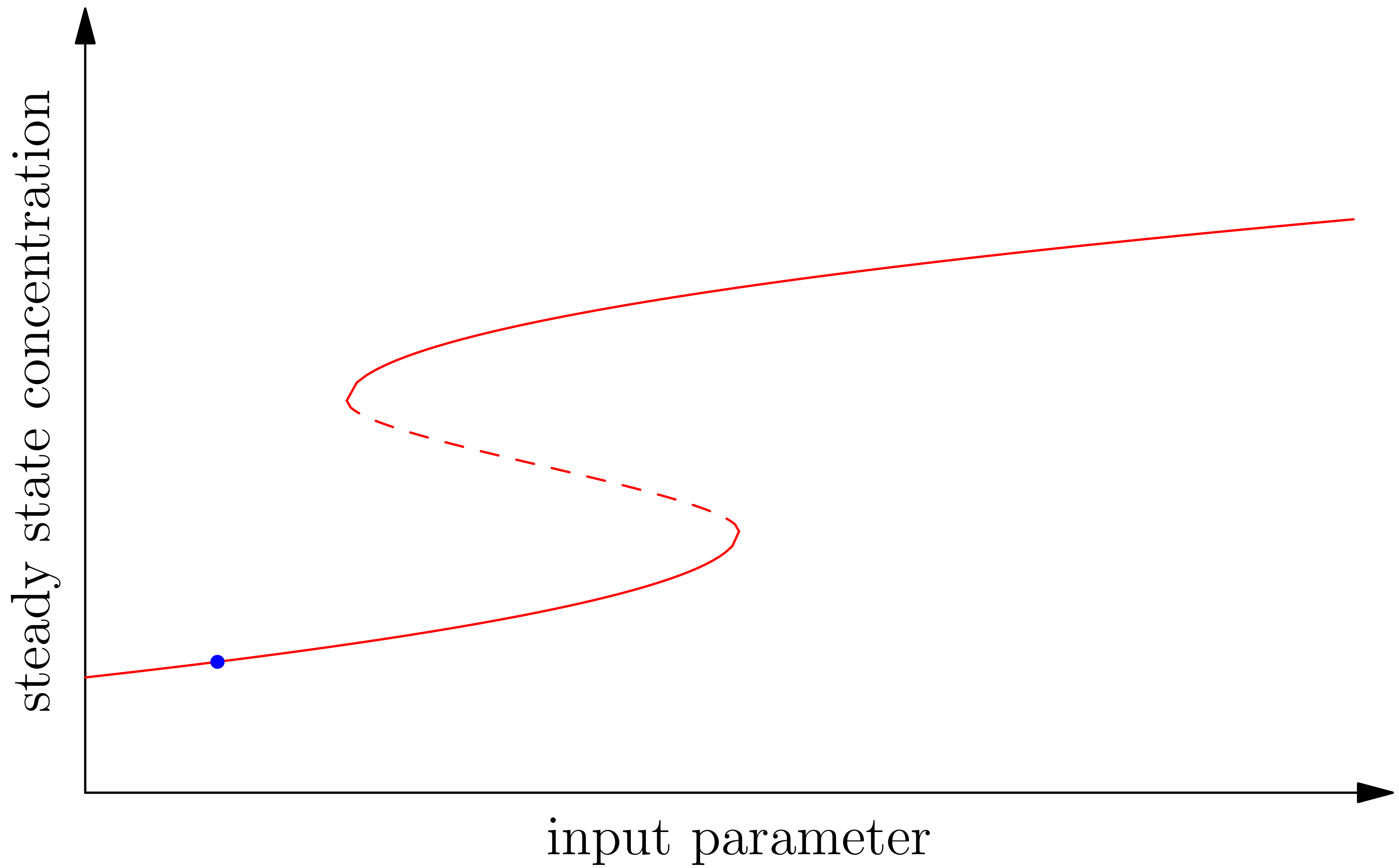
Hysteresis



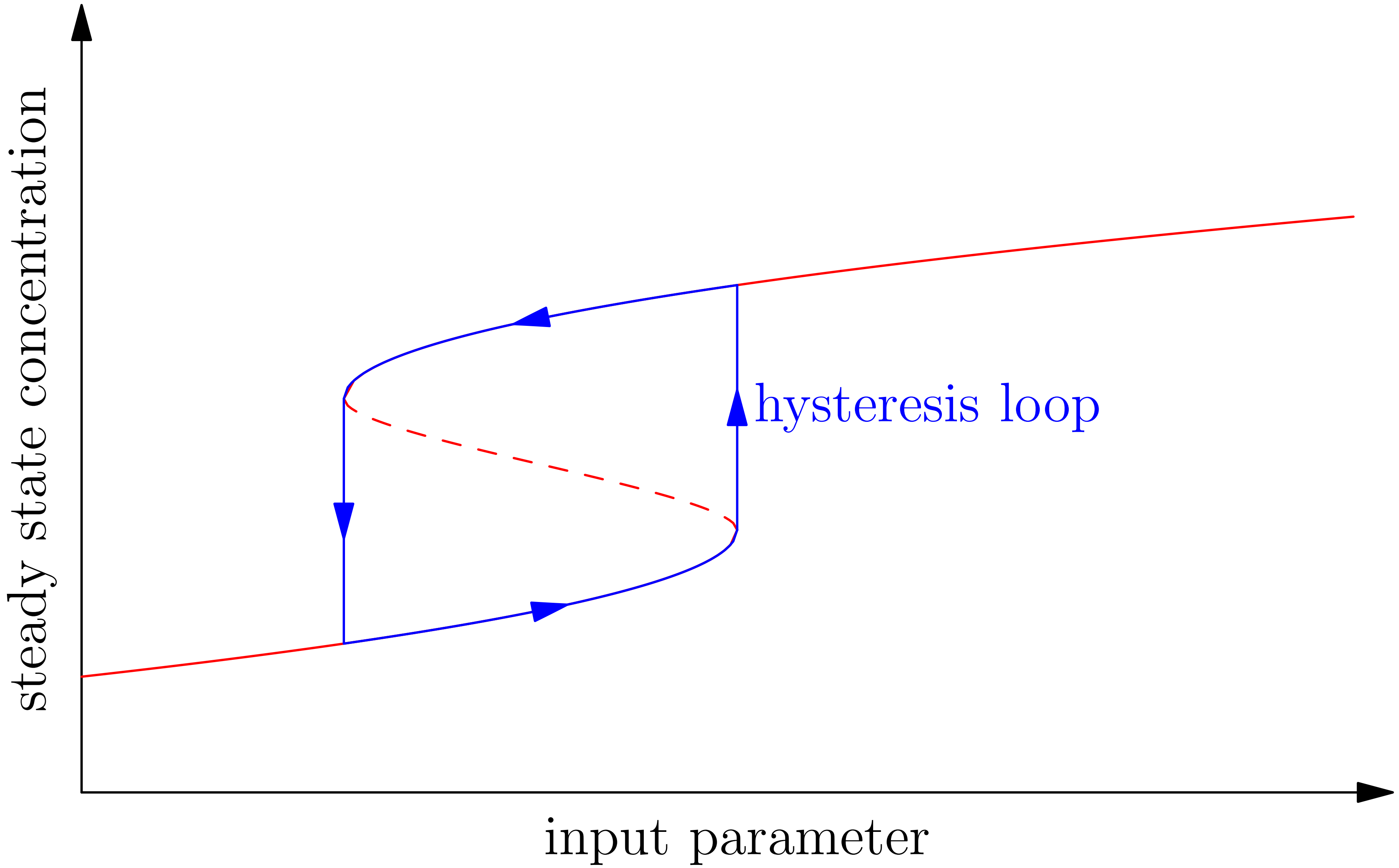
Hysteresis



Hysteresis



Hysteresis



steady state concentration
hysteresis loop
input parameter

Autocatalysis

- Recall the autocatalytic system

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$

- We consider what happens when we vary n

Autocatalysis

- Recall the autocatalytic system

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$

- We consider what happens when we vary n

Autocatalysis

- Recall the autocatalytic system

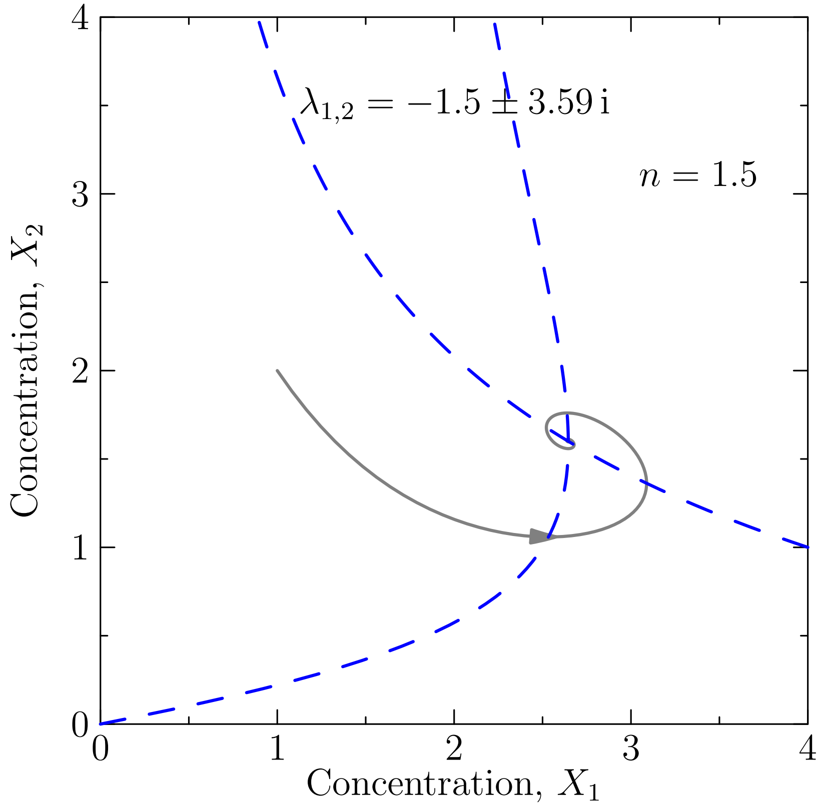
$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$

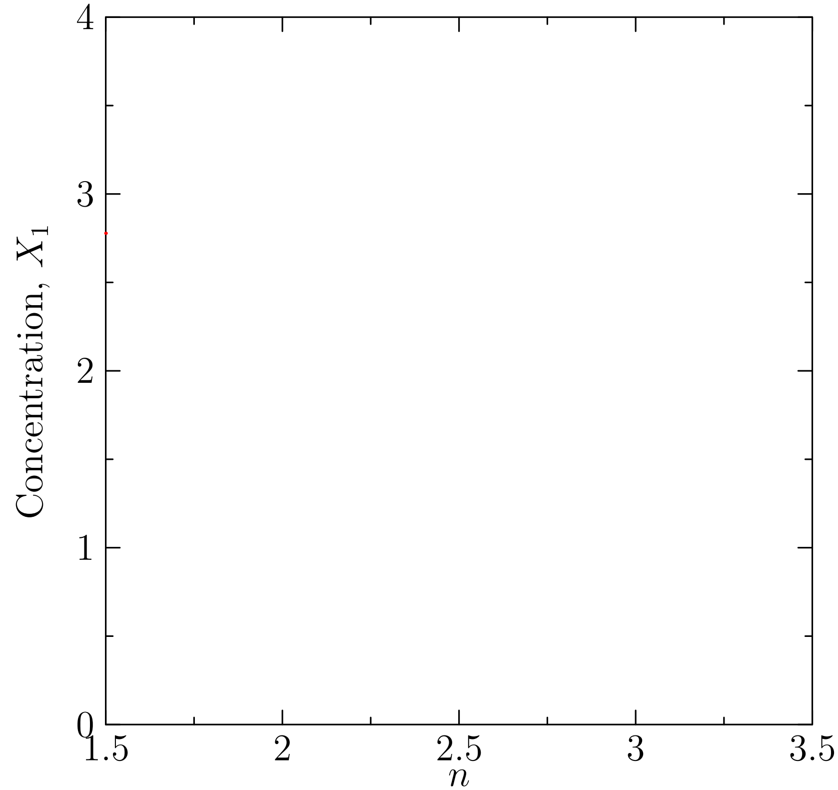
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$

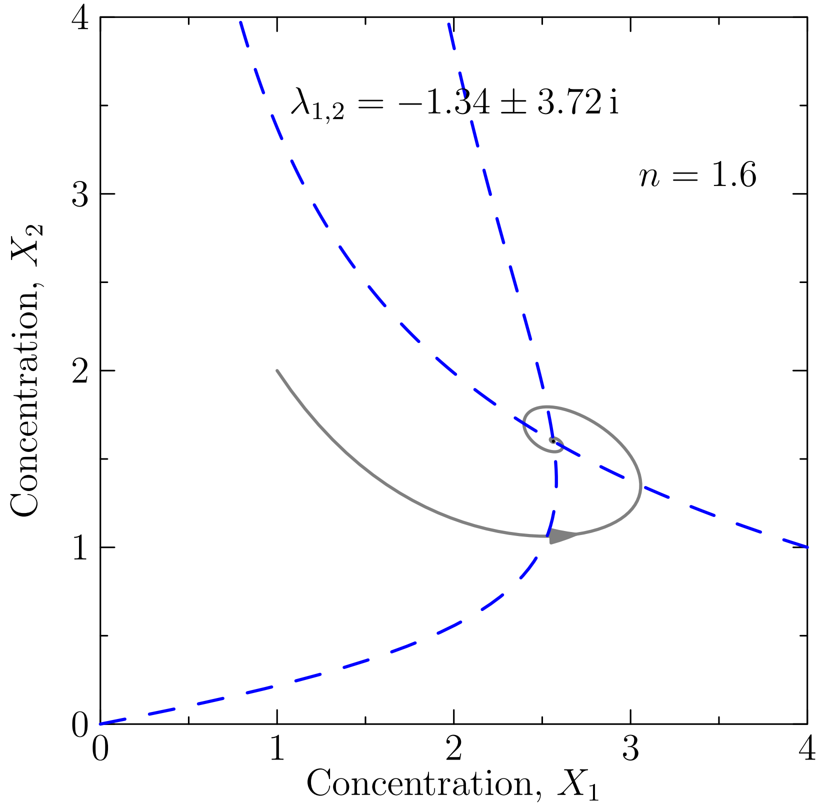
- We consider what happens when we vary n

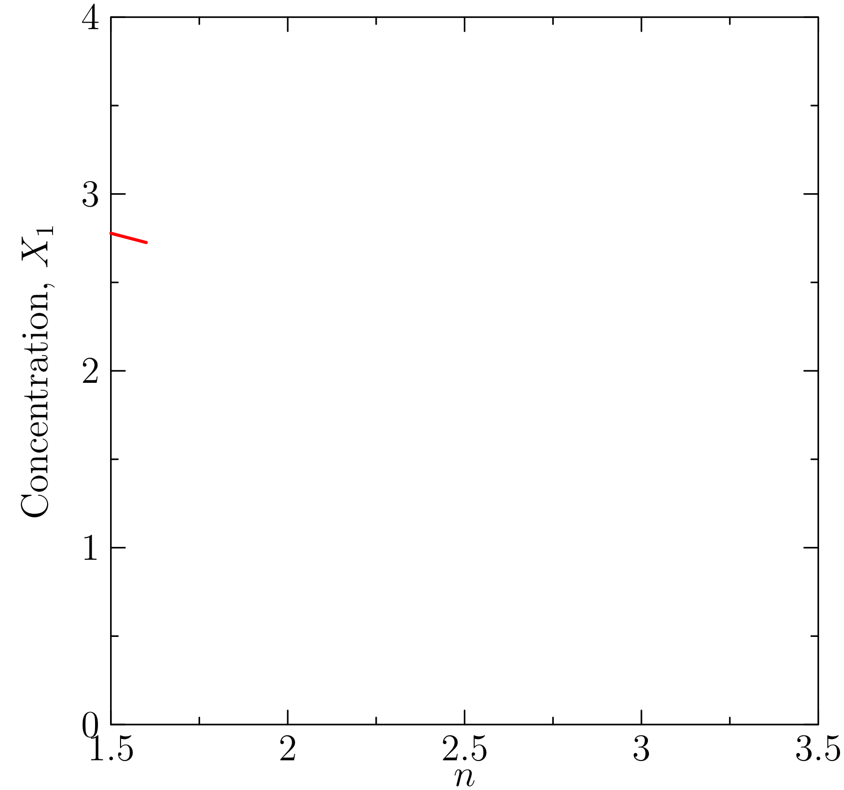
Hopf Bifurcation



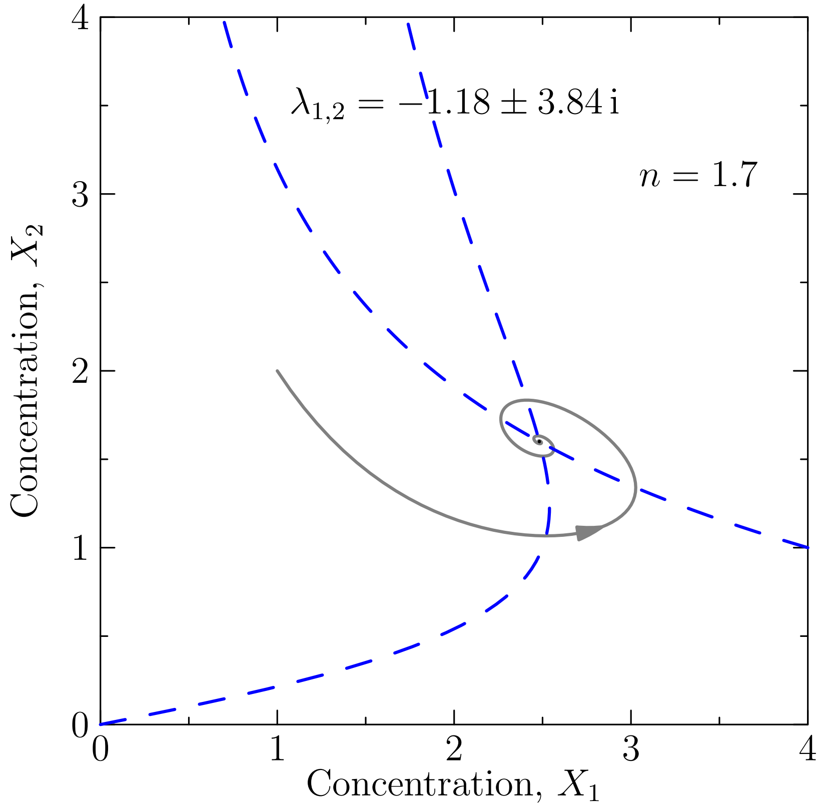


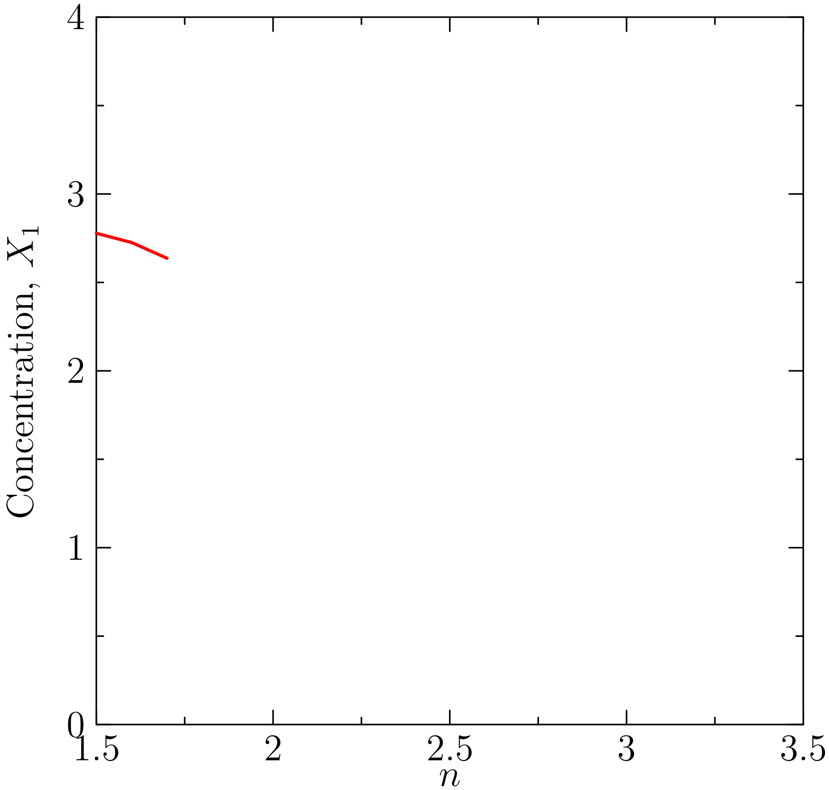
Hopf Bifurcation



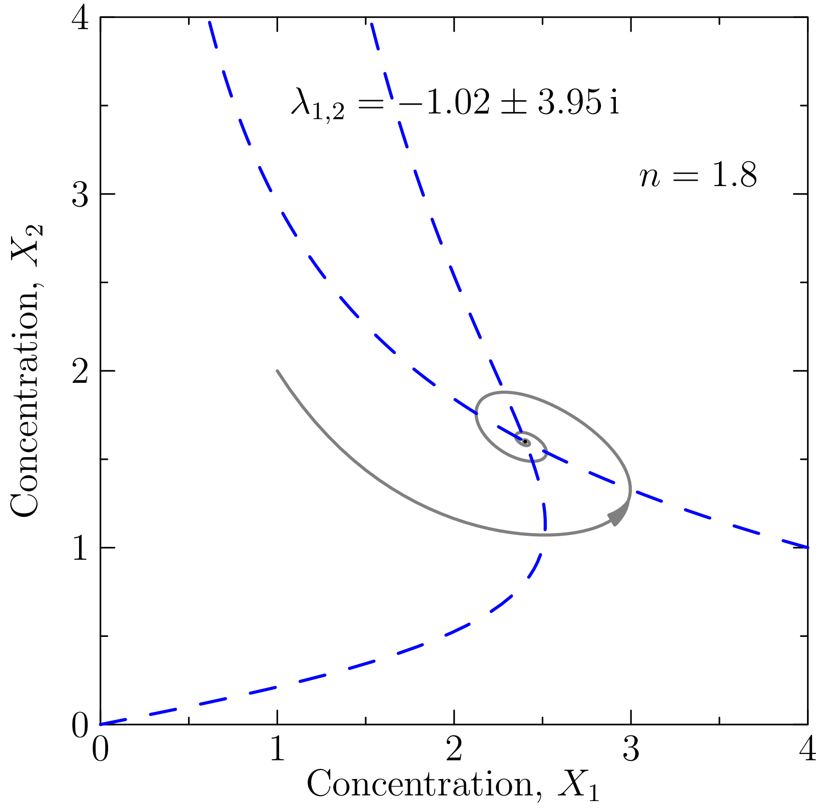


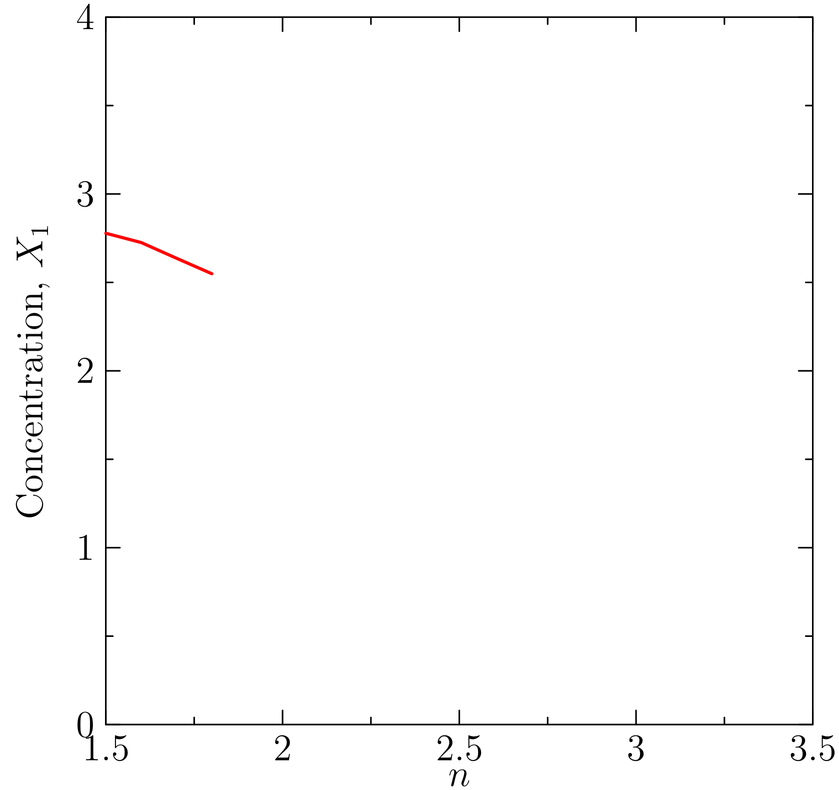
Hopf Bifurcation



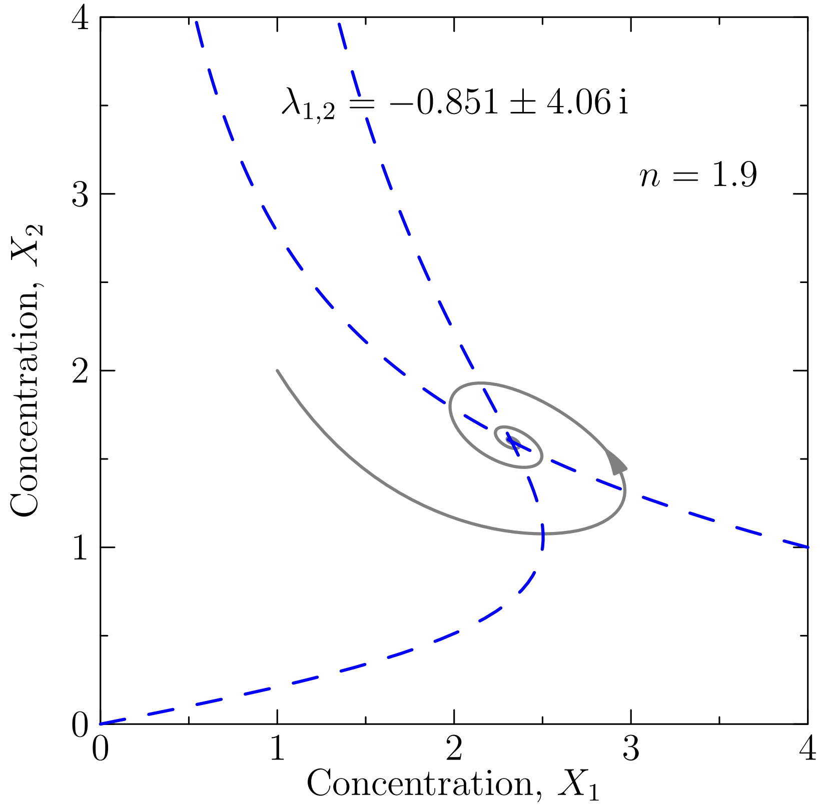


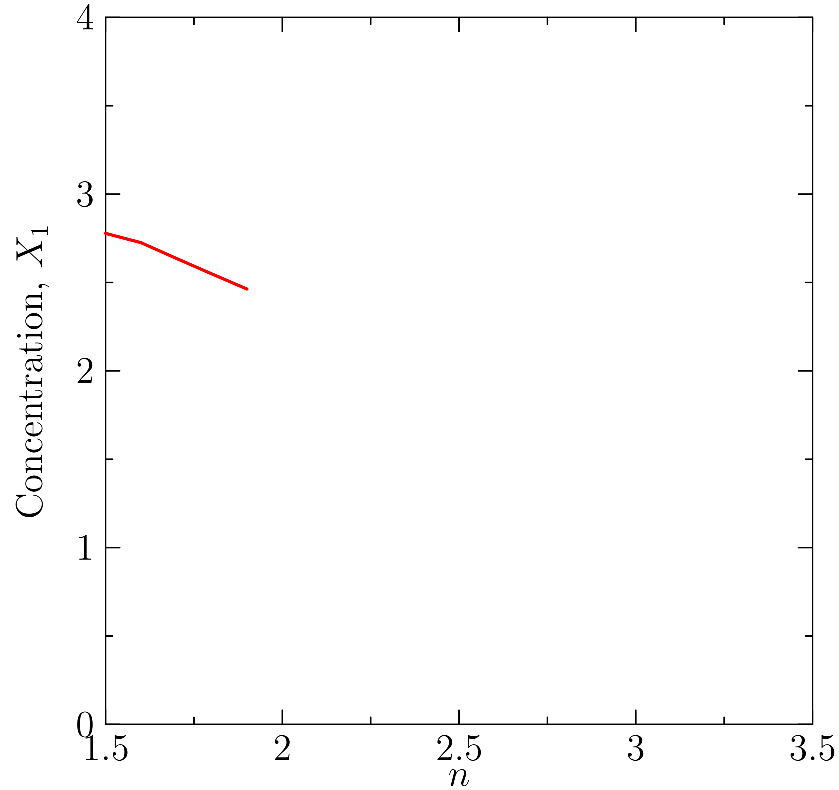
Hopf Bifurcation



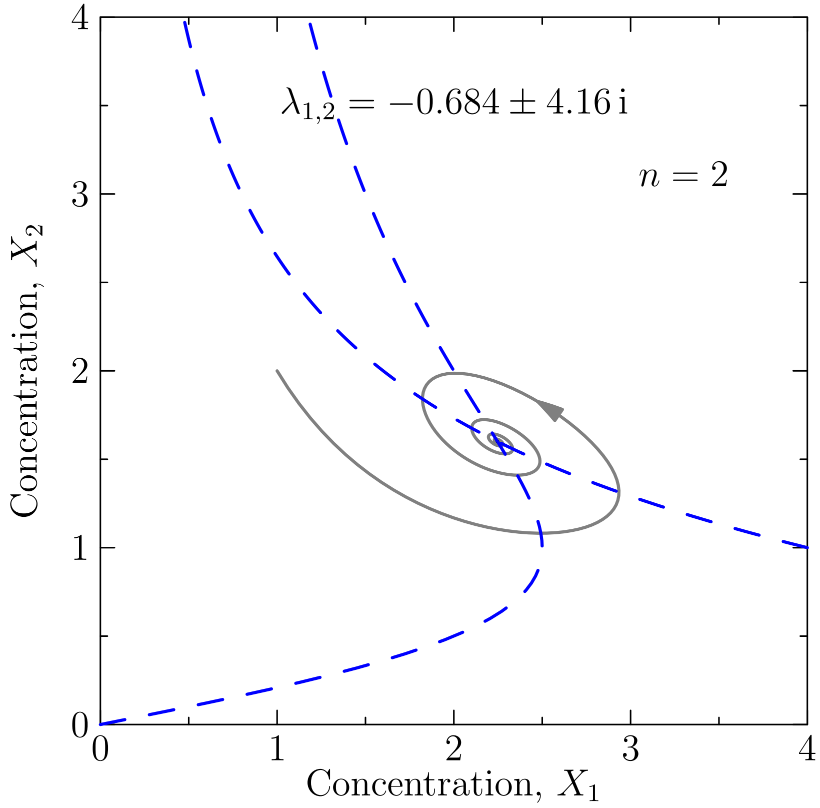


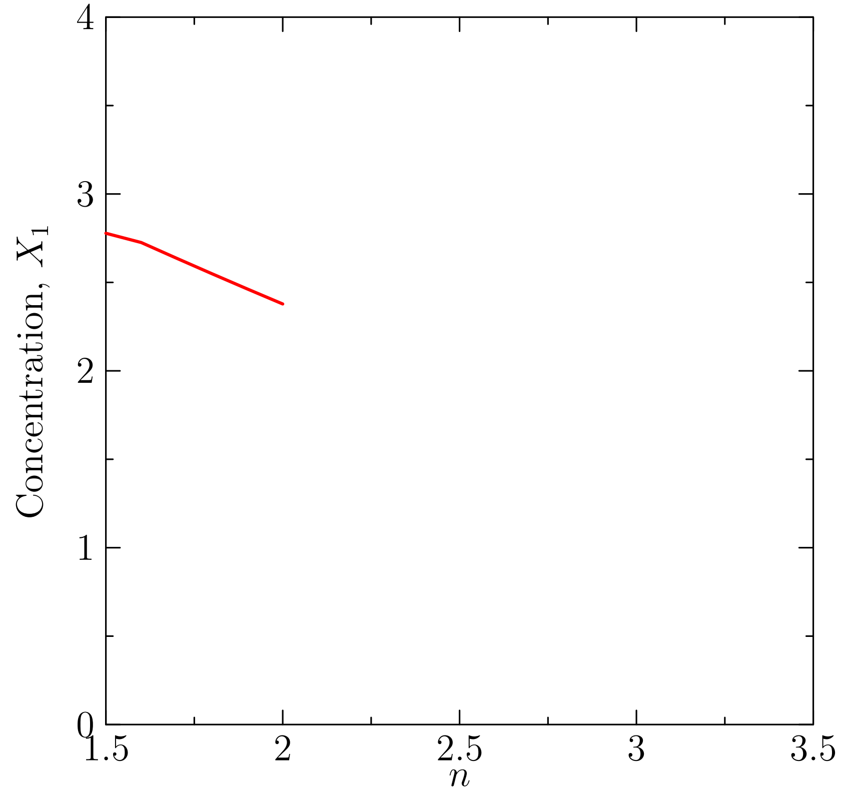
Hopf Bifurcation



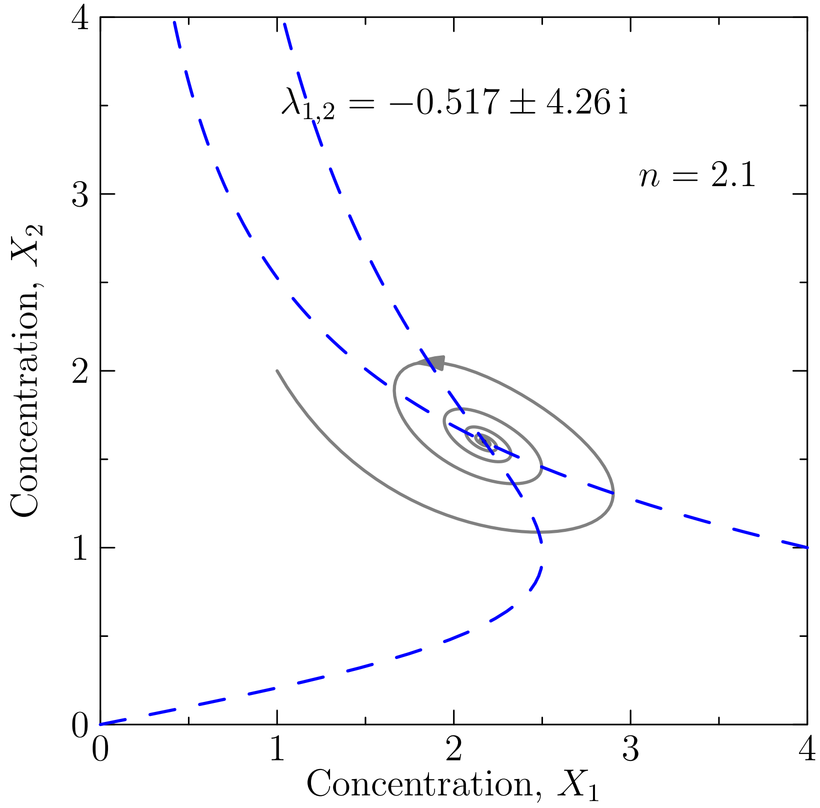


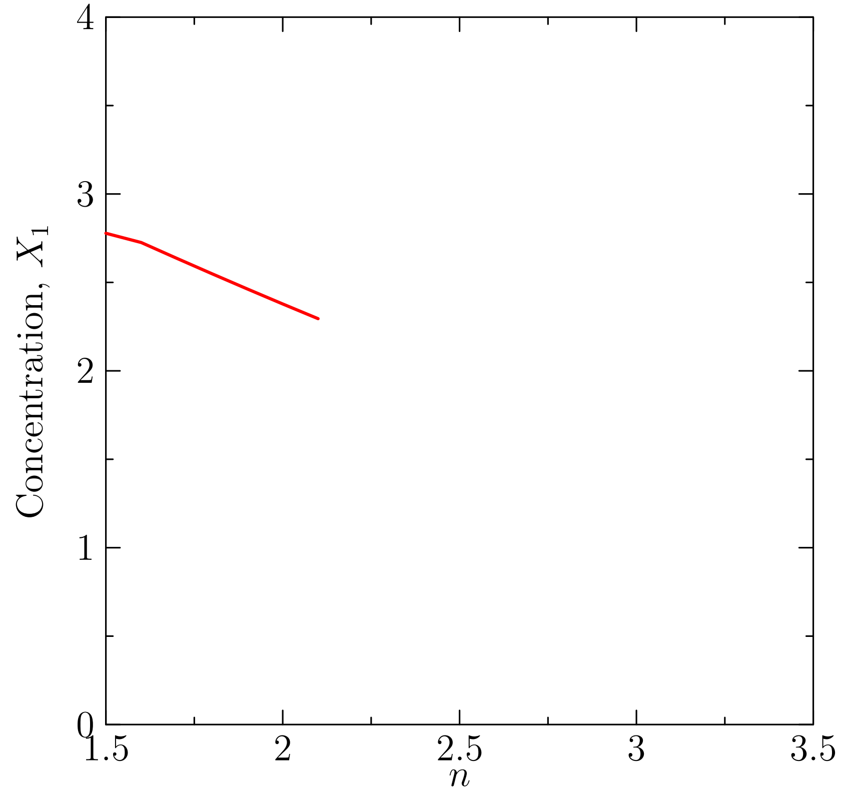
Hopf Bifurcation



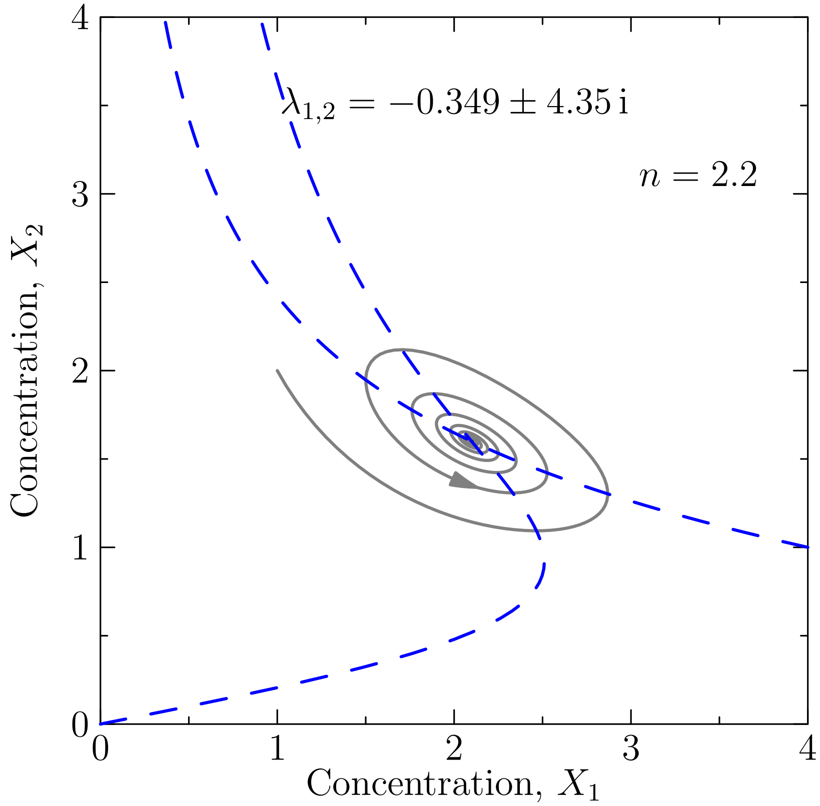


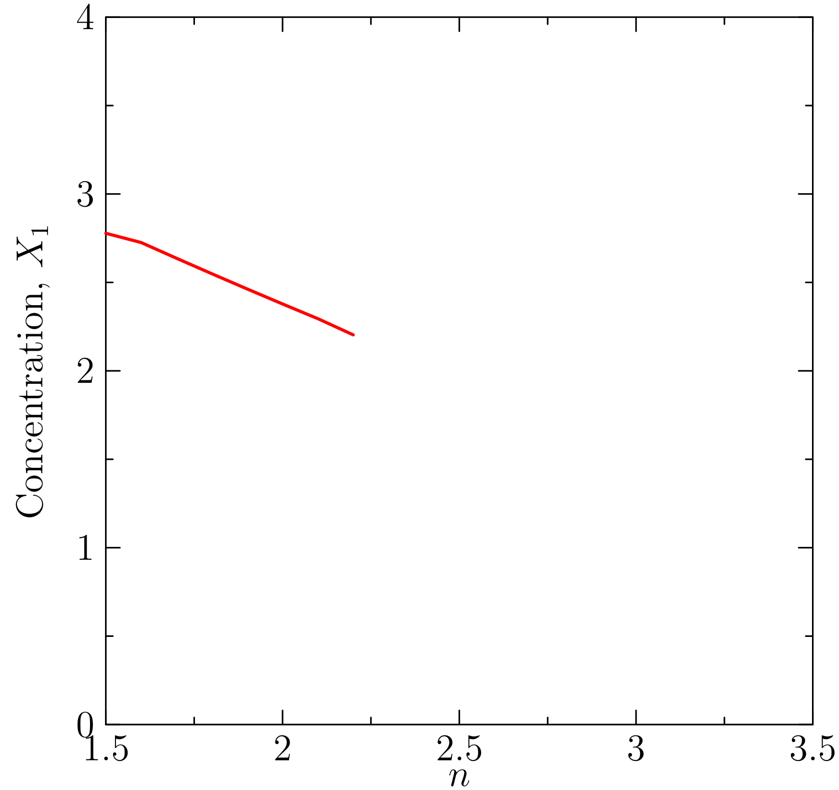
Hopf Bifurcation



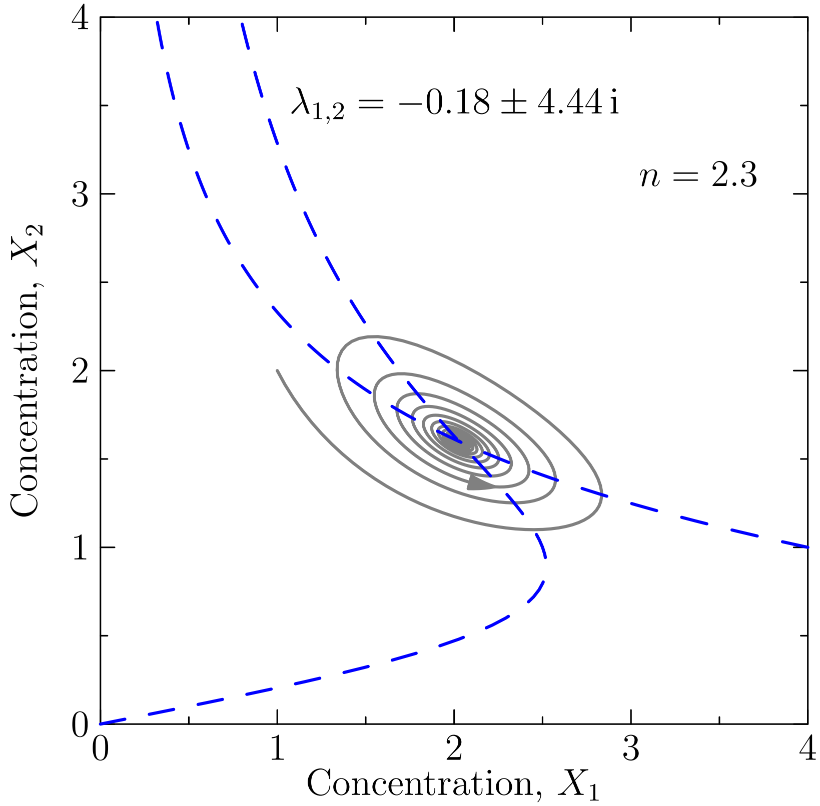


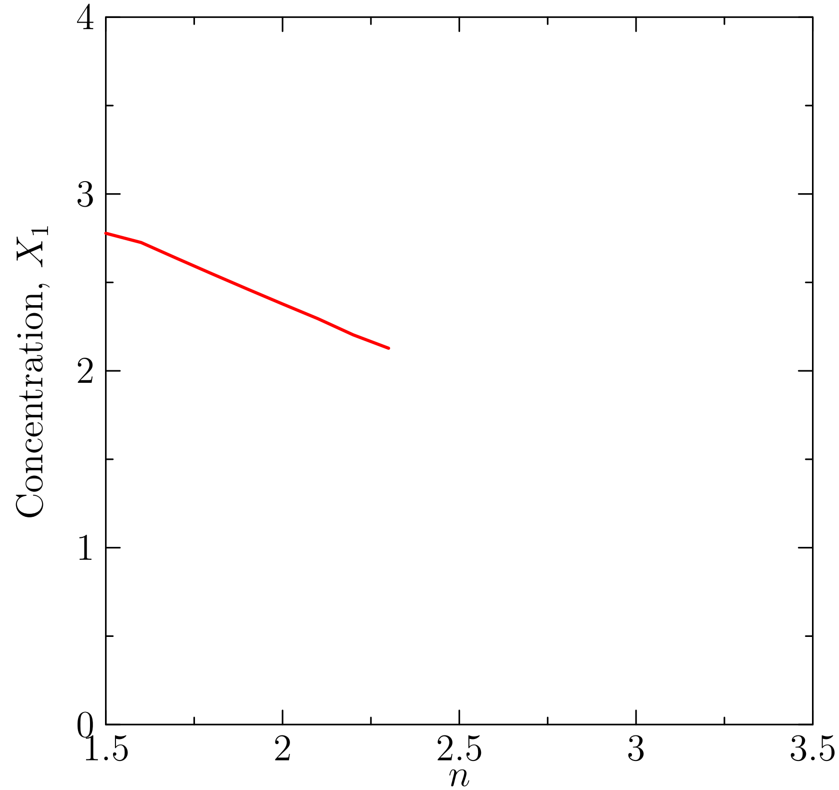
Hopf Bifurcation



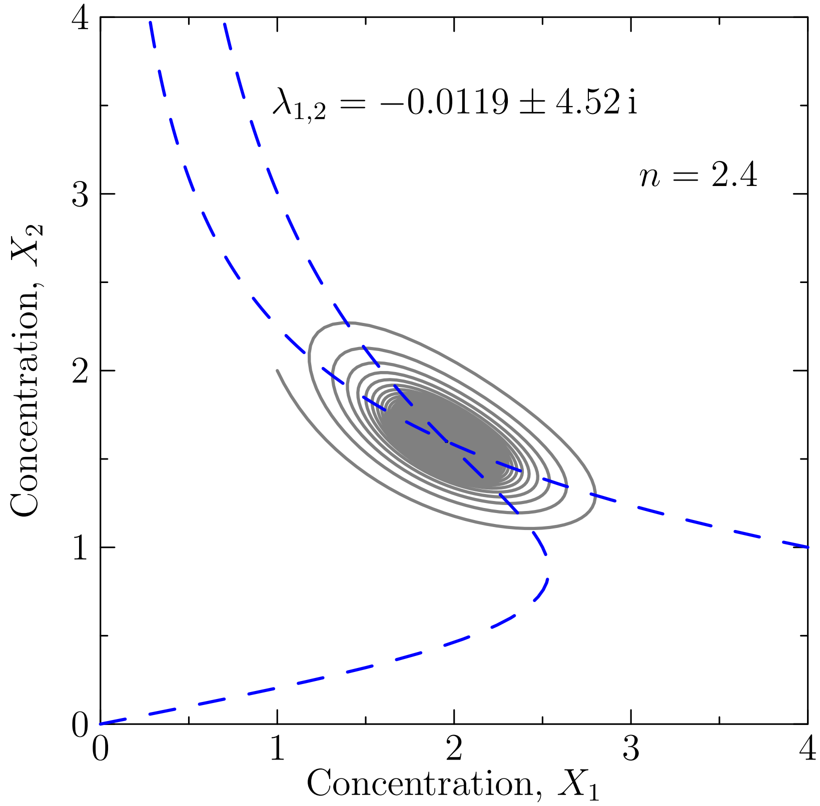


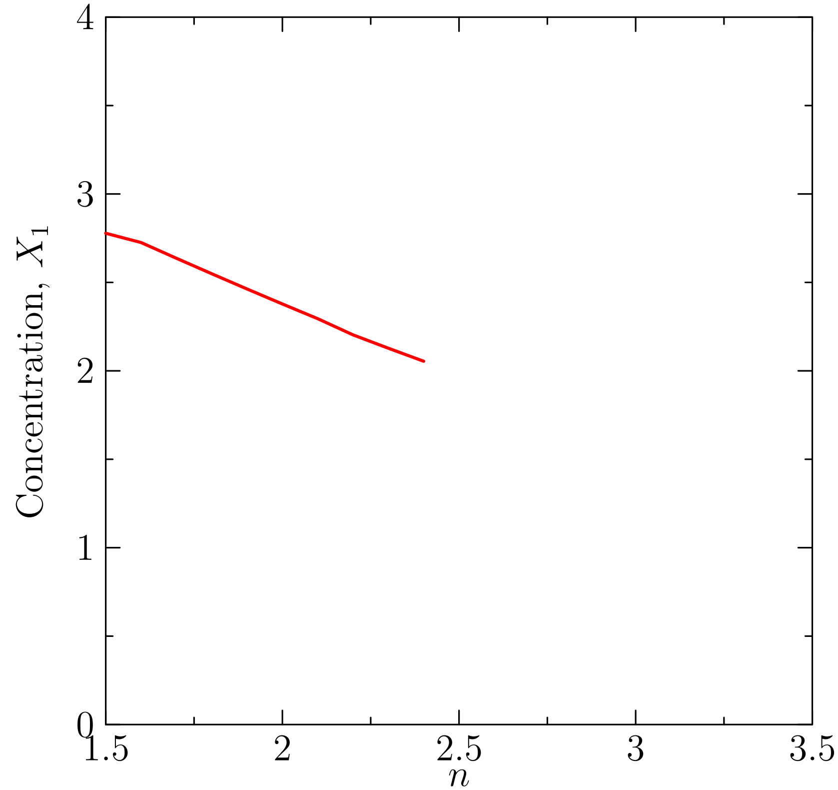
Hopf Bifurcation



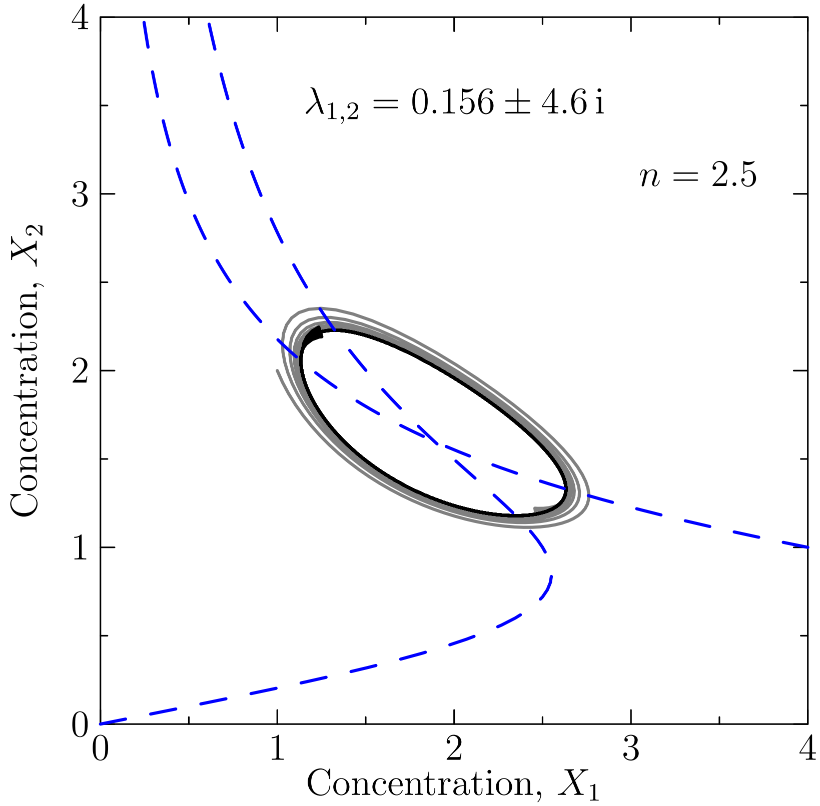


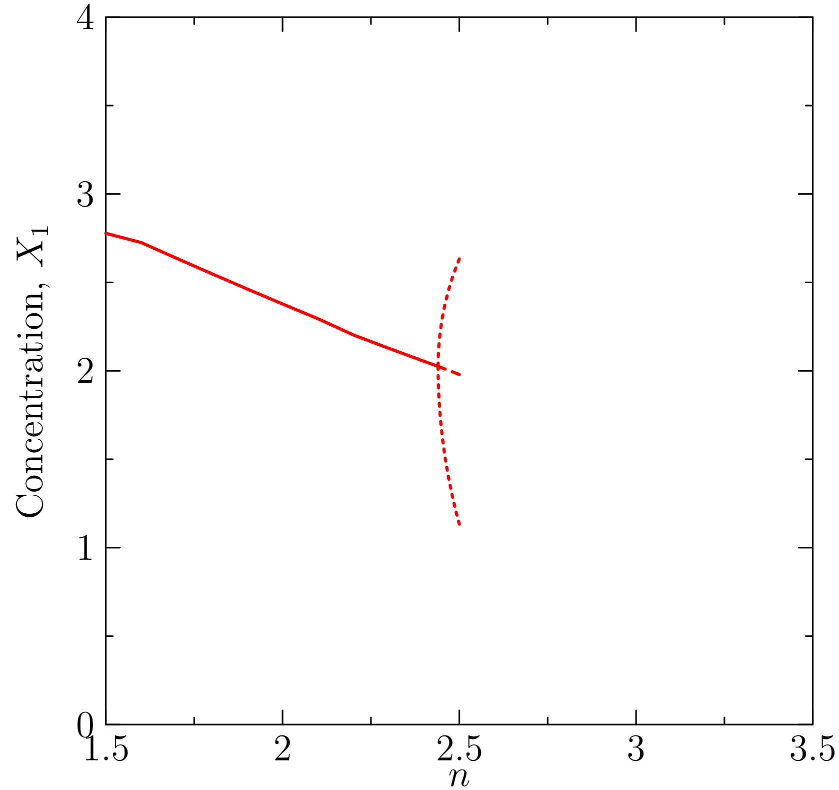
Hopf Bifurcation



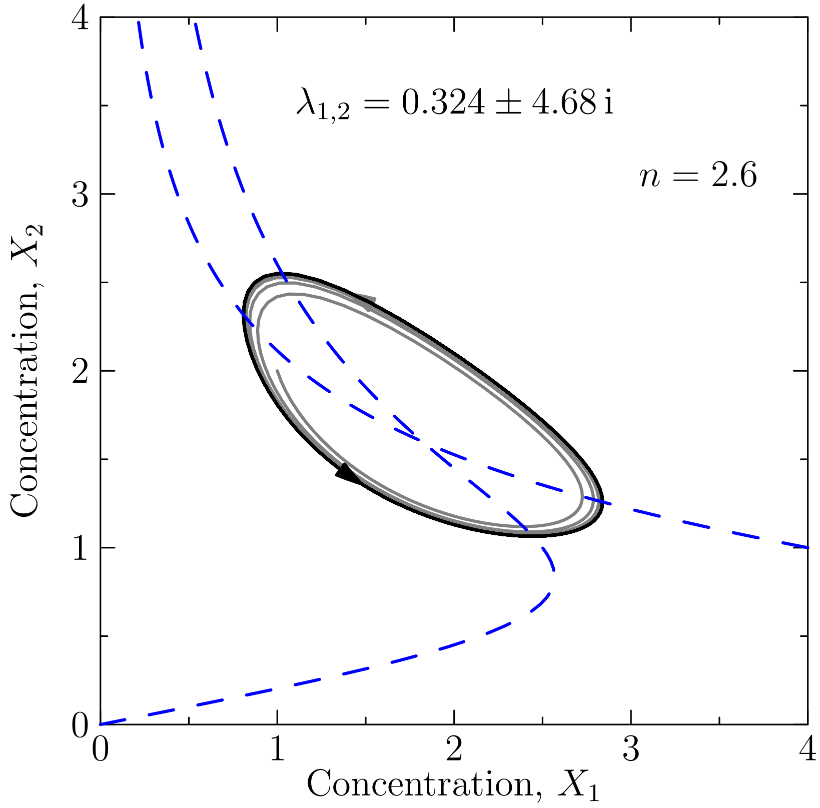


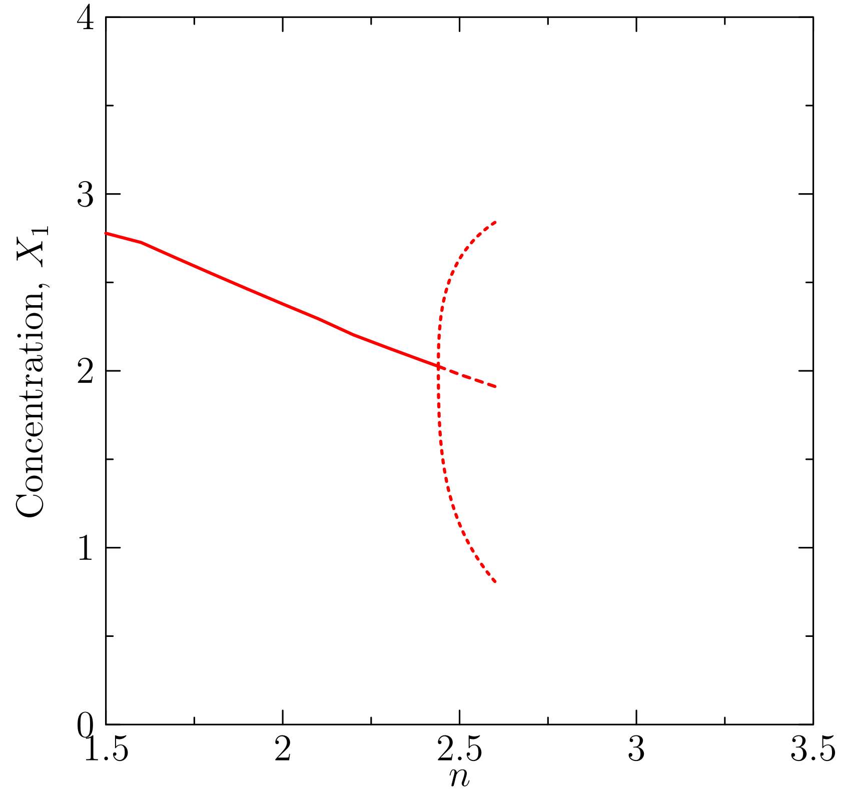
Hopf Bifurcation



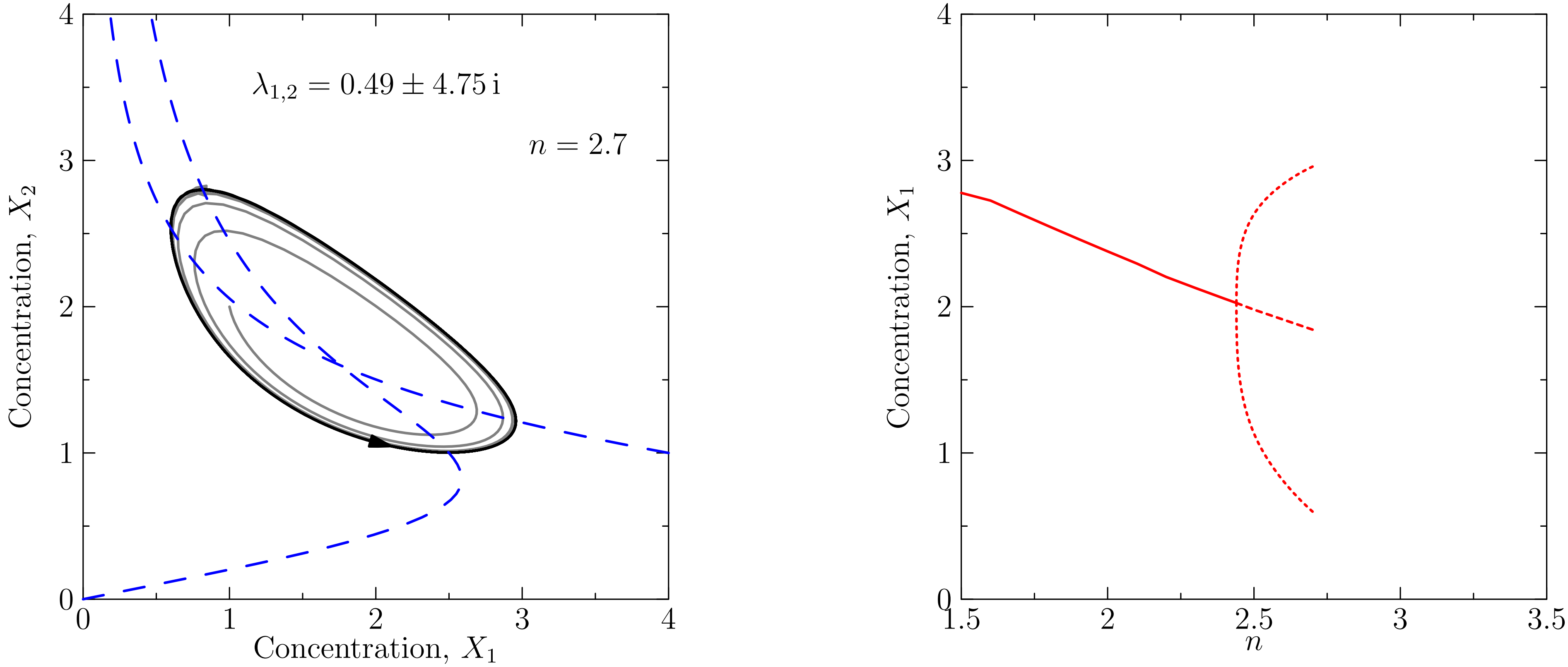


Hopf Bifurcation

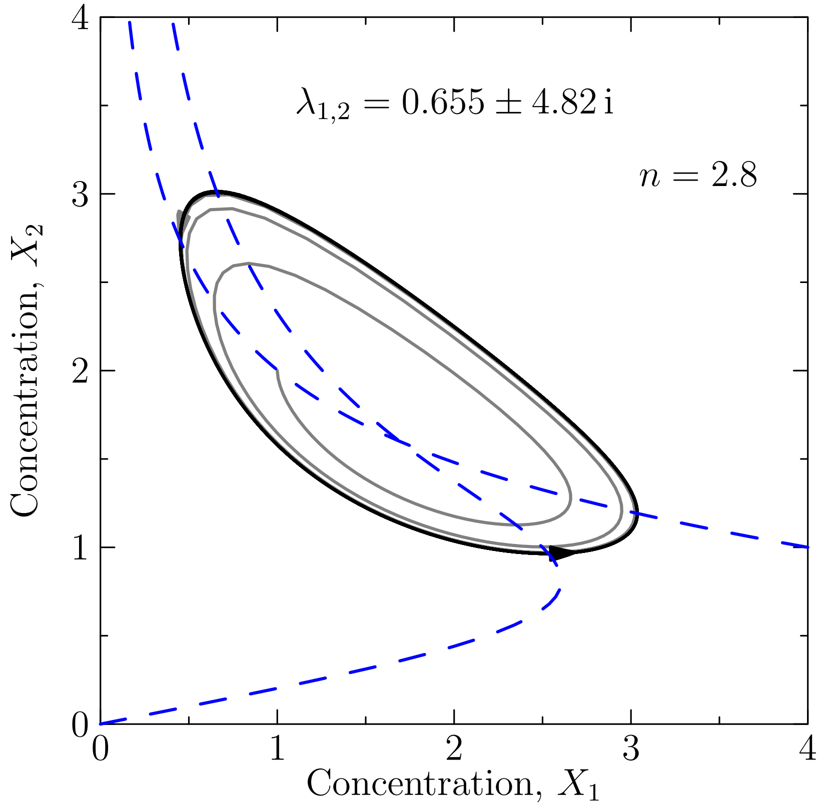


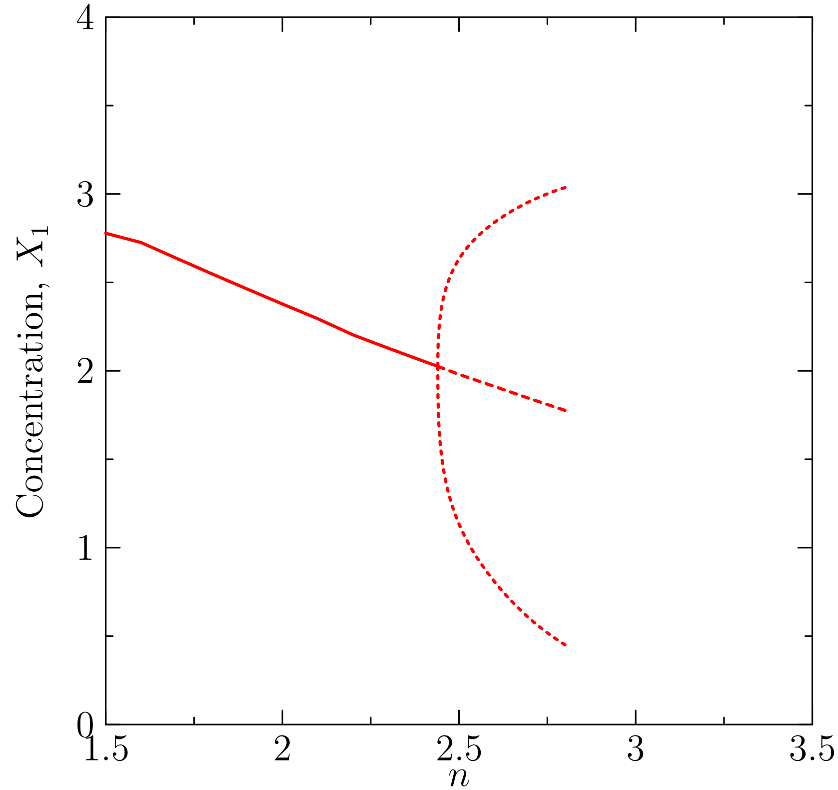


Hopf Bifurcation

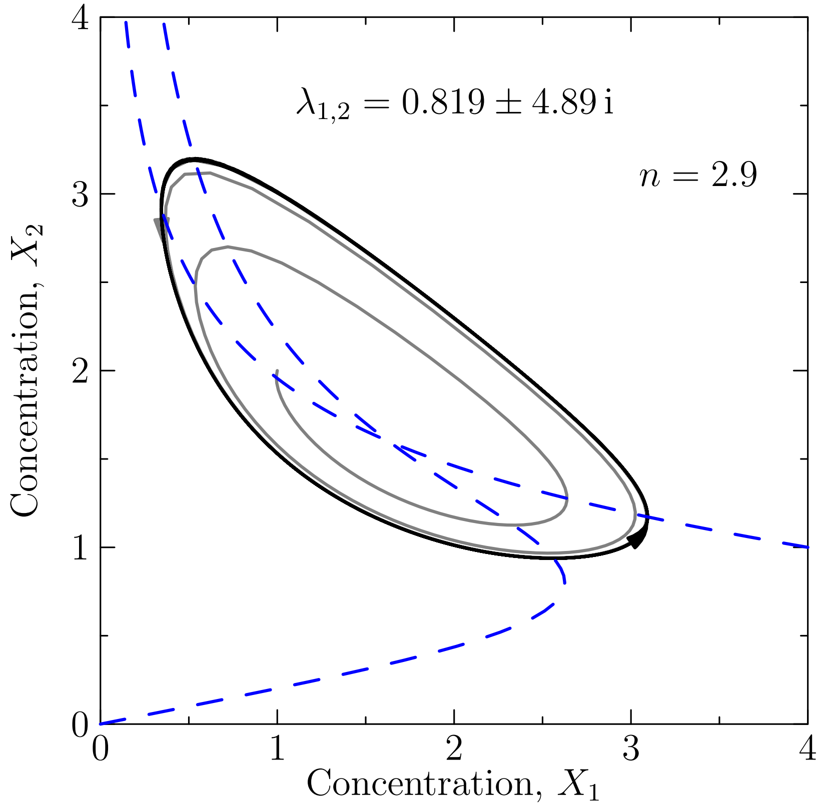


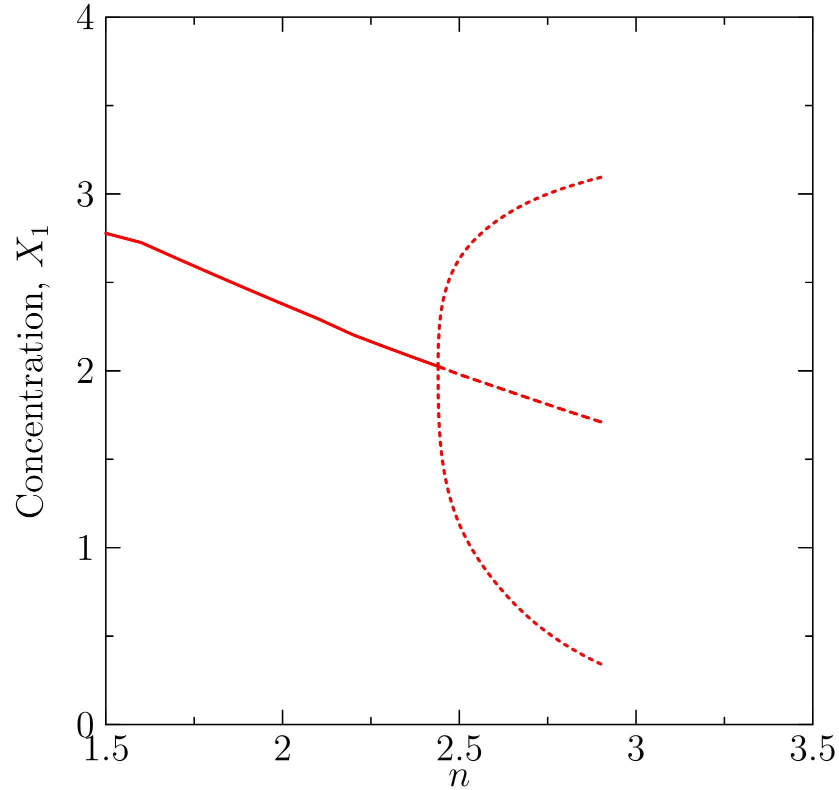
Hopf Bifurcation



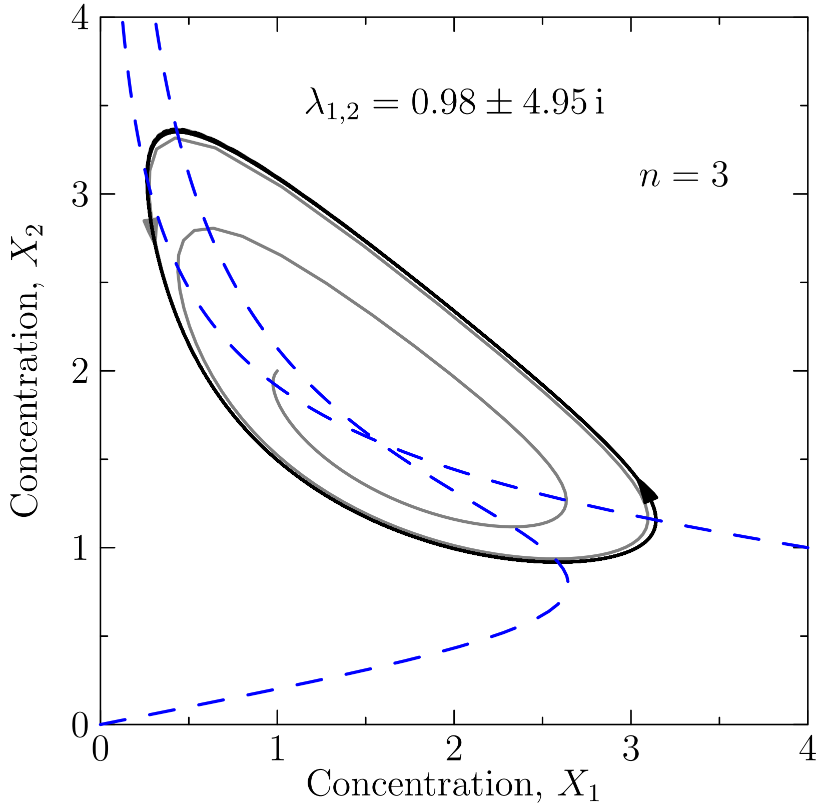


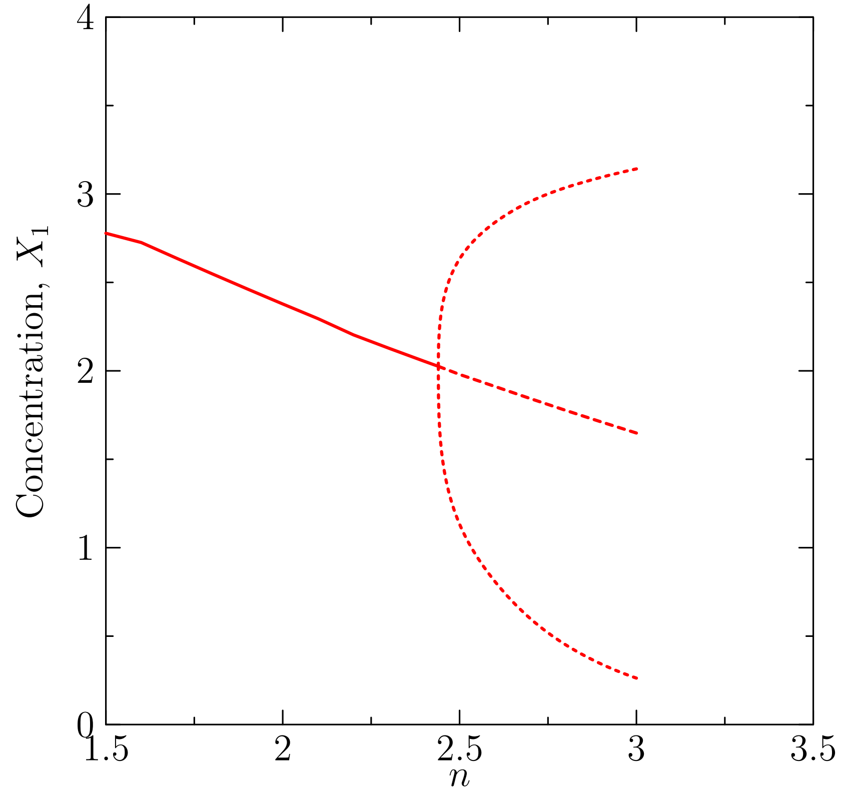
Hopf Bifurcation



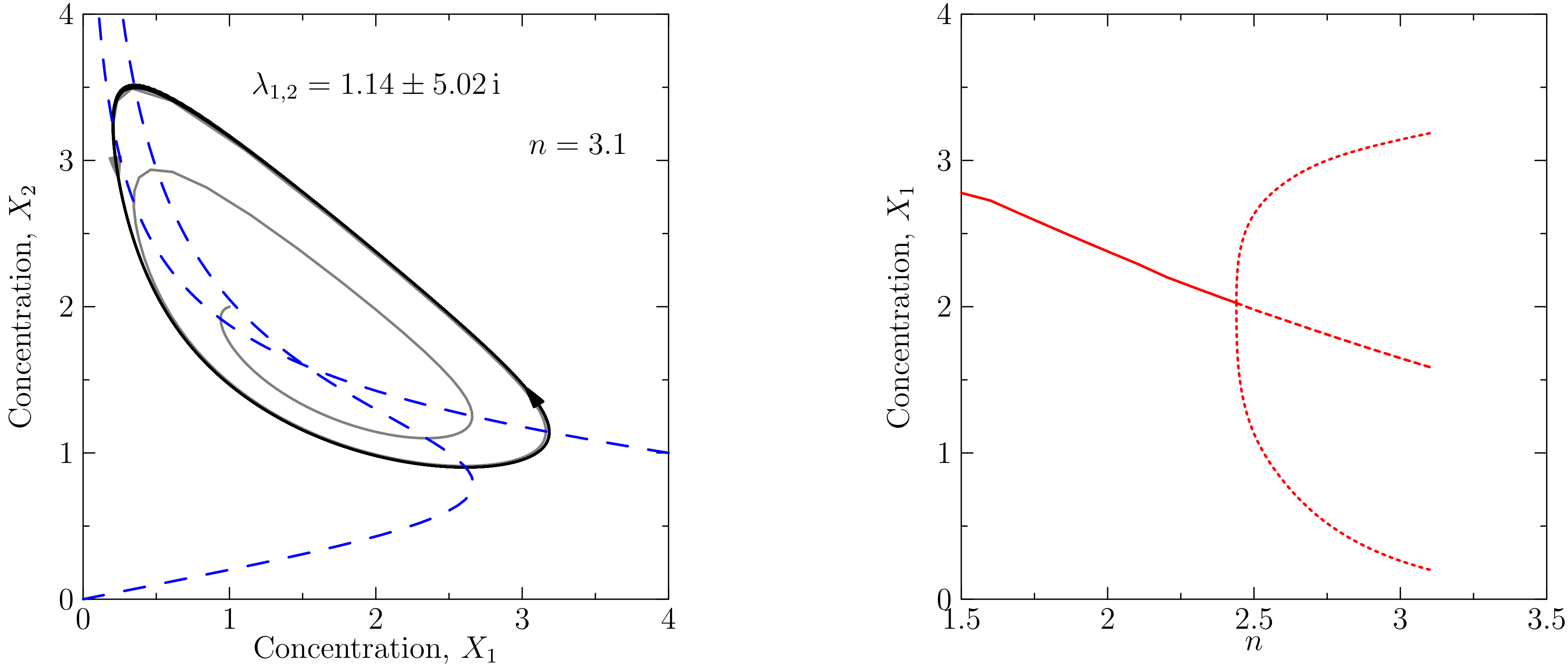


Hopf Bifurcation

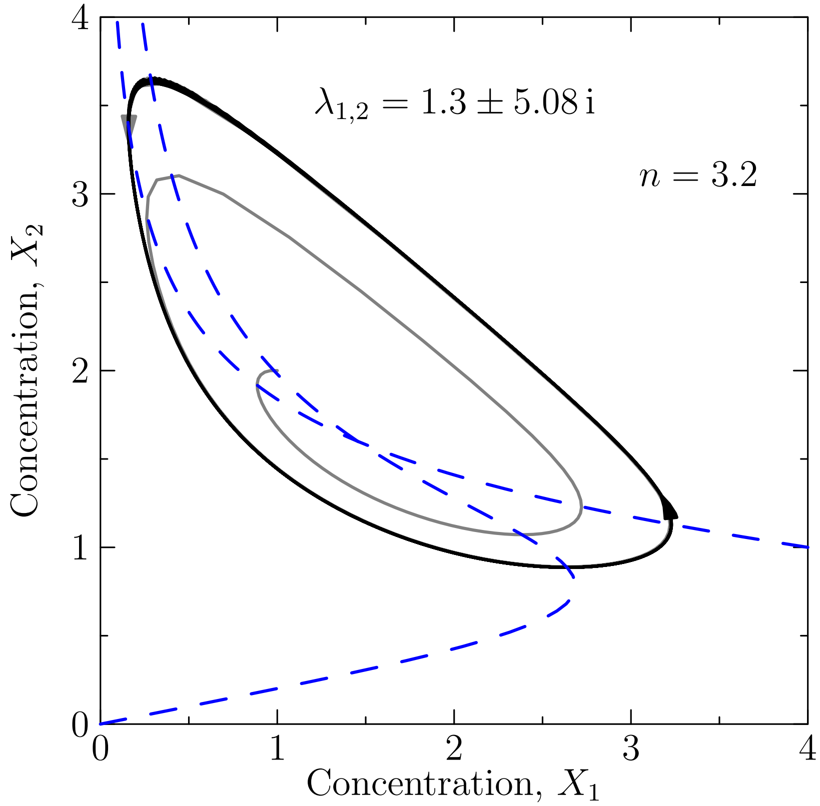


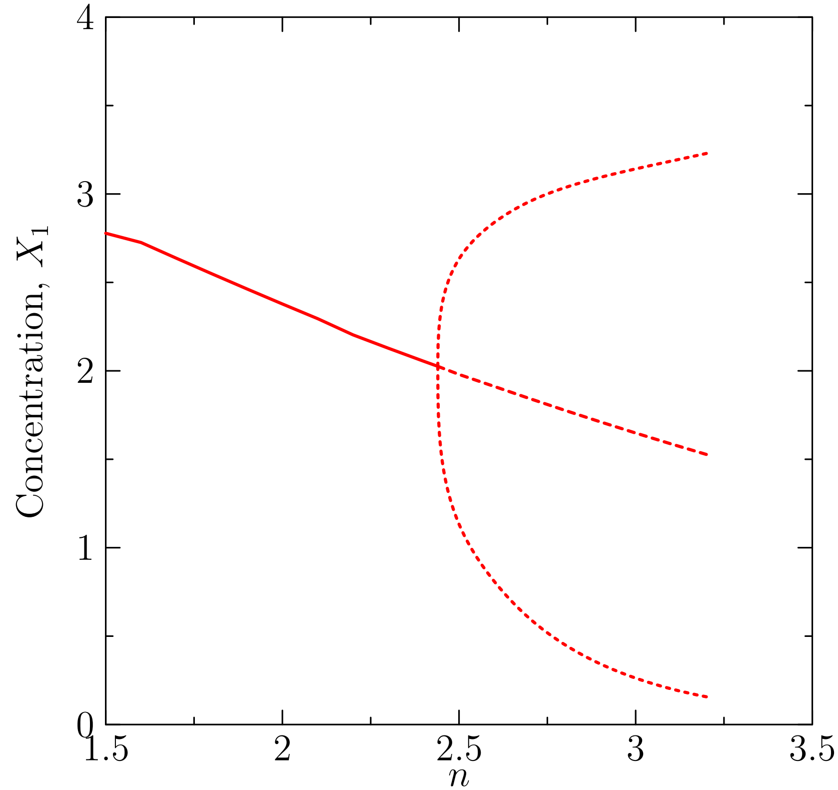


Hopf Bifurcation

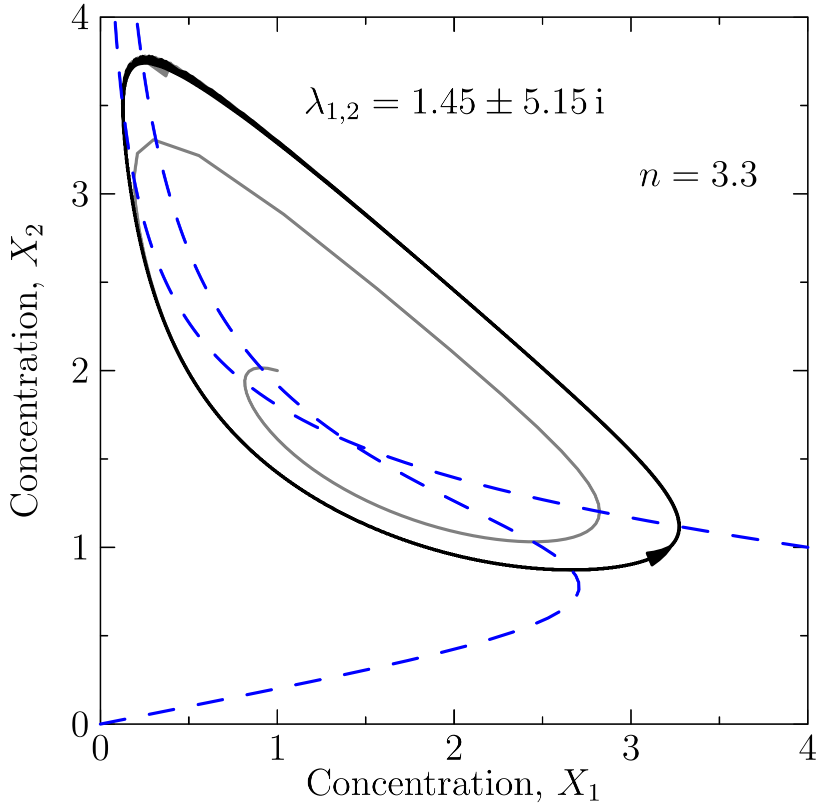


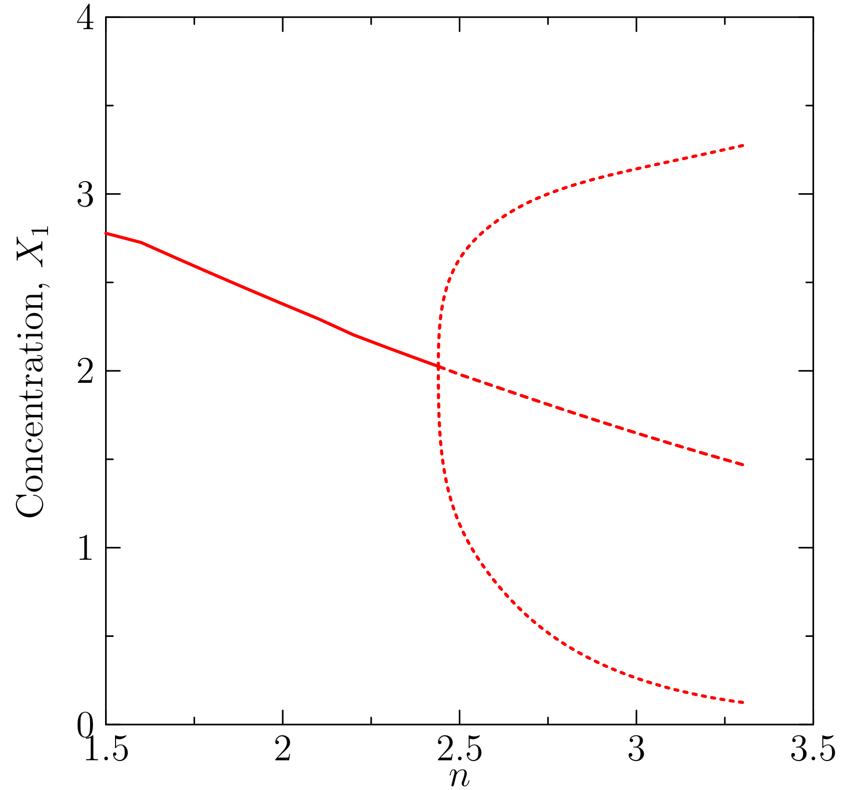
Hopf Bifurcation



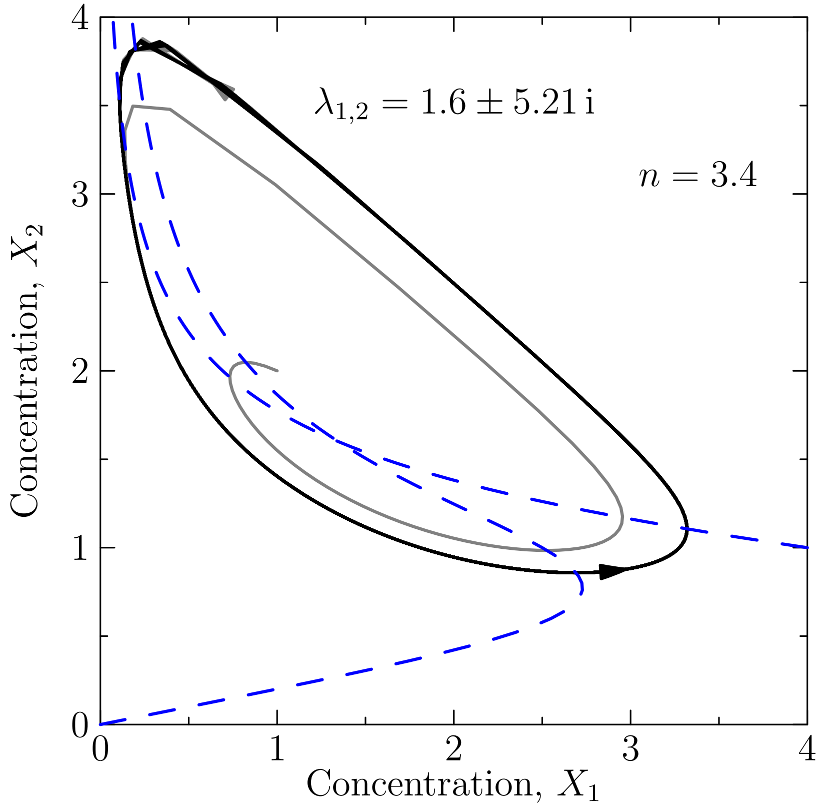


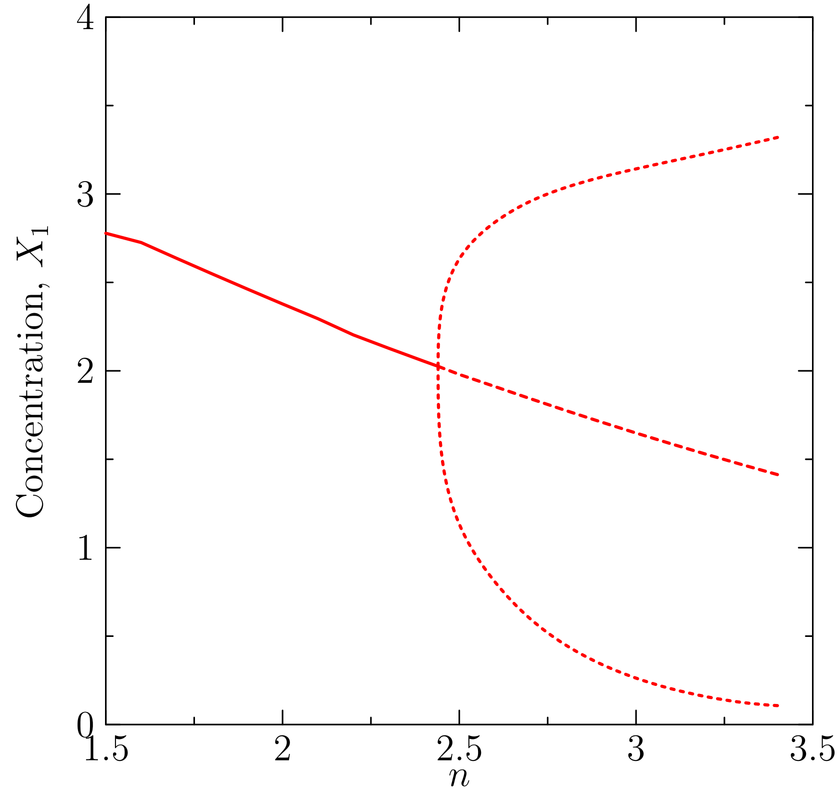
Hopf Bifurcation



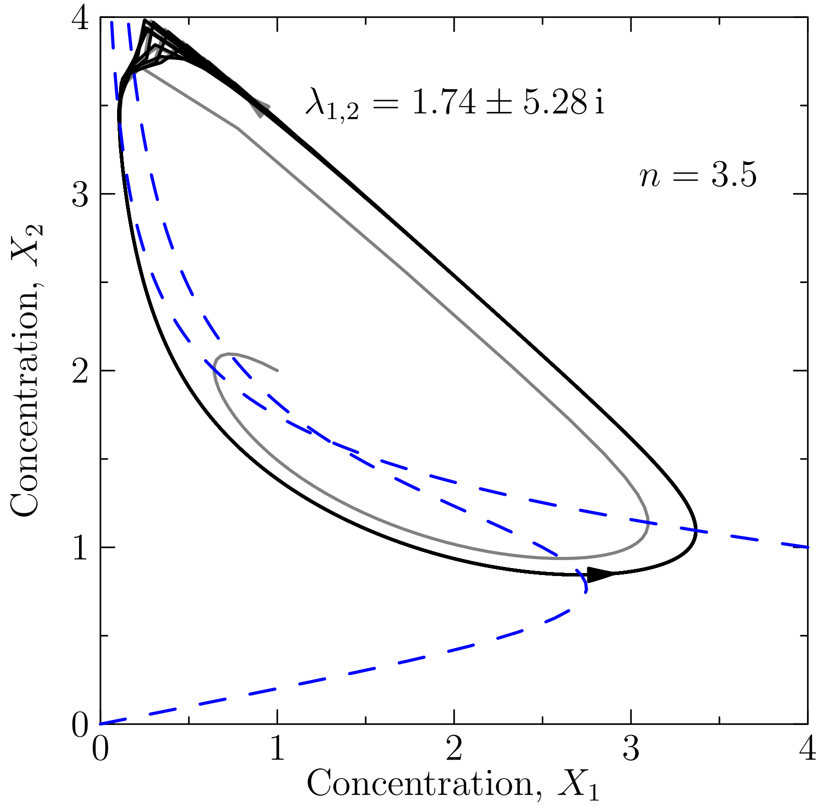


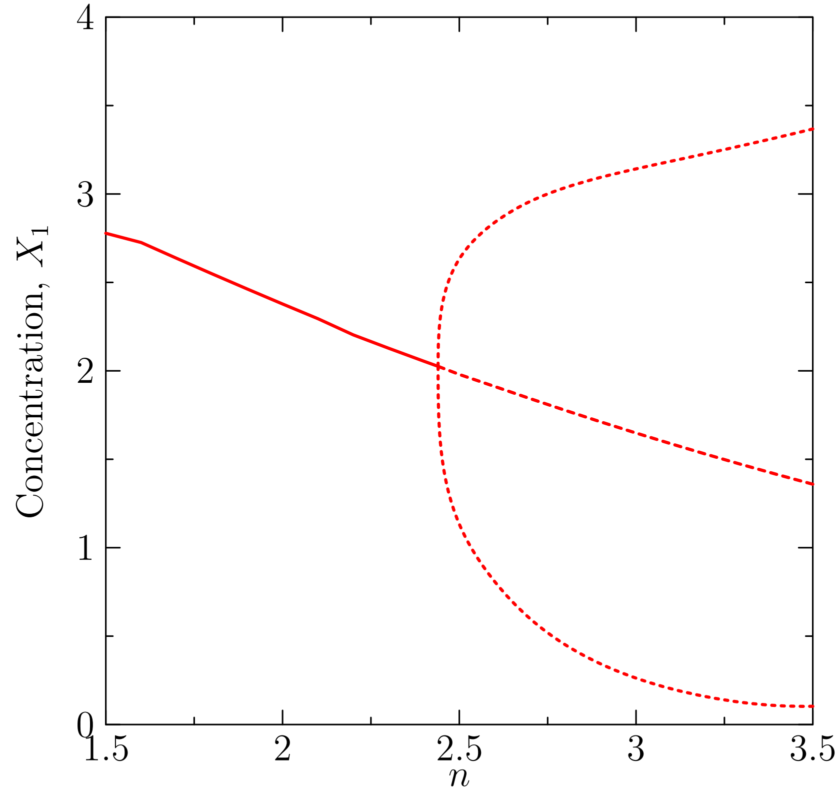
Hopf Bifurcation



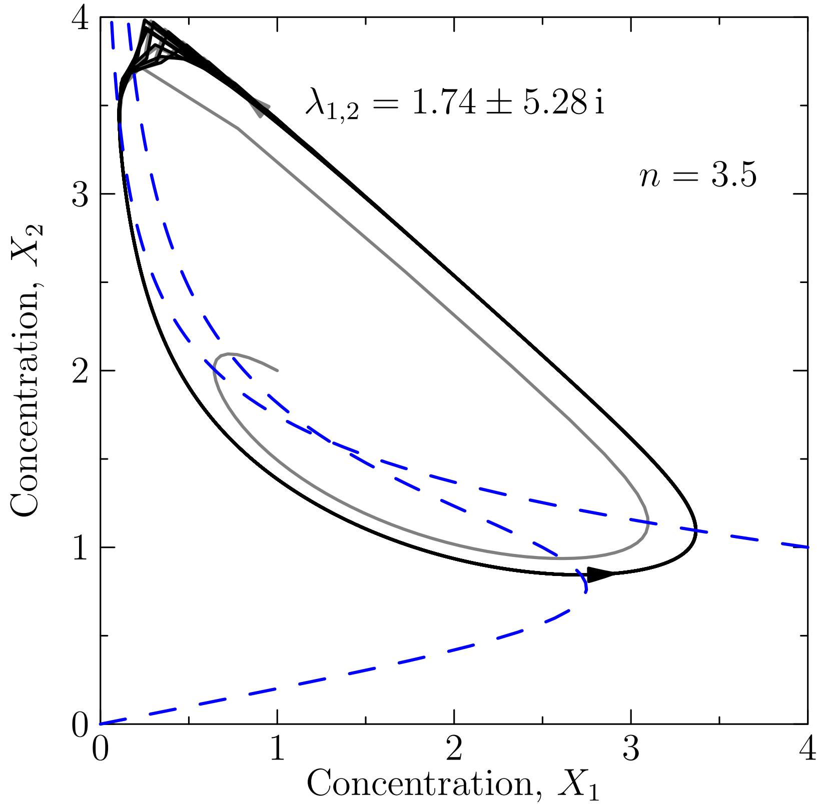


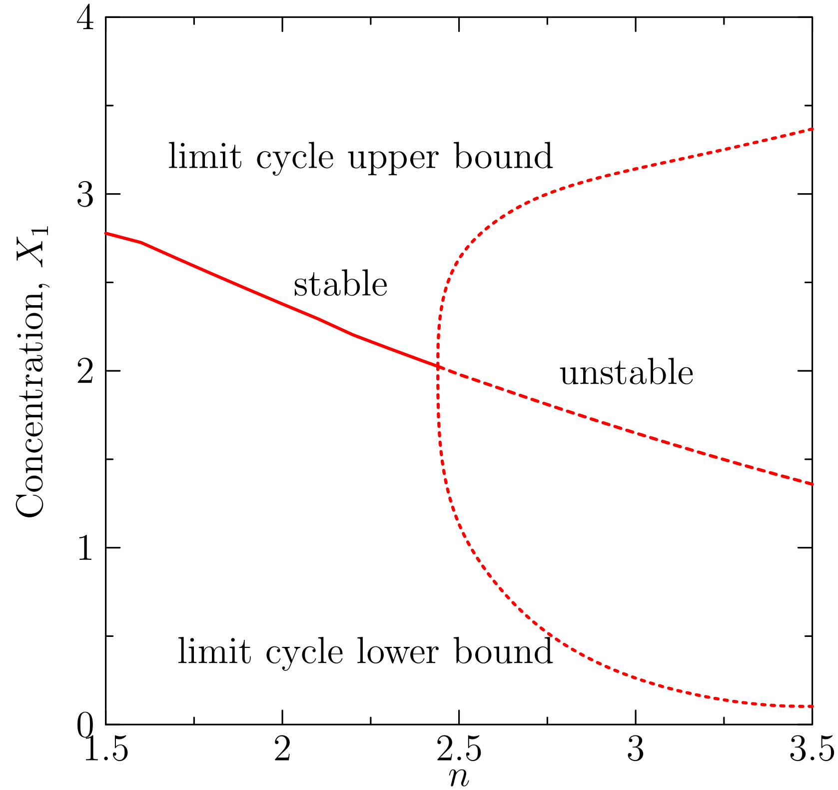
Hopf Bifurcation





Hopf Bifurcation





Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable

Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable

Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable

Summary

- We can understand a lot of behaviours by understanding first order ODEs
- Systems can exhibit fixed points, limit cycles (and even chaotic behaviour in enough dimensions)
- We can study the stability of fixed points
- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable